than -1 the plural of the unit is used and a singular unit is used for values between 1 and -1. (viii) A space is left between the numerical value and unit symbol, for example, 25 kg, but not 25-kg or 25kg. (ix) If the spelled-out name of a unit is used, the normal rules of English are applied. (x) Unit symbols are written in Roman type and quantity symbols are in italic type with superscripts and subscripts in Roman or italic type as appropriate. For numerical notation, some of the rules and conventions include the following: (i) A space should be left between groups of 3 digits on either the right or left hand side of the decimal place, for example, 15 739.012 53. In four digit numbers the space may be omitted. Commas should not be used. (ii) The decimal marker shall be either the point on the line or the comma on the line. The decimal marker chosen should be that which is customary in the context concerned. (iii) Mathematical operations should only be applied to unit symbols (kg/m^3) and not unit names (kilogram/cubic metre). (iv) Values of quantities should be expressed as 2.0 µs or 2.0 x 10^{-6} and not in terms such as parts per million. (v) It should be clear to which unit symbol a numerical value belongs and which mathematical operation applies to the value of a quantity (35 cm x 48 cm, not 35 x 48 cm; or 100 g ± 2 g, not 100 ± 2g). (vi) The value must apply to the whole symbol and not any particular unit within the symbol. NIST has produced a document of more than 80 pages covering all aspects of this which can be downloaded at: http://physics.nist.gov/cuu/Units/rules.html

SABATIER-SENDERENS PROCESS: A method of organic synthesis, typically an unsaturated fat, employing the reaction of hydrogen with carbon dioxide at elevated temperatures and pressures in the presence of a nickel catalyst to produce methane and water. It was named after French chemist, Paul Sabatier (1854-1941) and Jean Baptiste Senderens (1856-1937), French chemists. It is employed commercially for hydrogenating unsaturated vegetable oils to make margarine. Paul Sabatier shared the 1912 Nobel Prize for this discovery with Francois Auguste Victor Grignard for the discovery of the Grignard reaction.

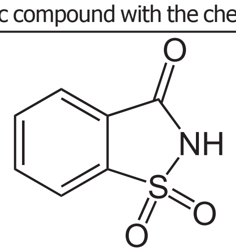
SAC: A bag-shaped part in a living organism, surrounded by a membrane, often containing a special fluid, gas or an organ and usually having a narrow opening or none at all. For example, the yolk in a hen's egg is contained in a sac. **Saccate** means having the form of a sac or pouch.

Saccate

SACCHARIDE: Any of a series of compounds, in which the atoms, carbon and hydrogen are in the ratio of 2:1, especially those containing the group $C_6H_{10}O_5$. They are divided into monosaccharides, disaccharides, trisaccharides and polysaccharides according to the number of saccharide groups present.

SACCHARIMETRY: Measurement of the concentration of sugar in a solution from its optical activity, using a polarimeter.

SACCHARIN: A white crystalline organic compound with the chemical formula $C_6H_4SO_2CONH$, density 0.828 g/cm^3 and melting point about 229 °C (502 K). It was discovered by I. Remsen and C. Fahlberg in 1879. It is about 550 times sweeter than sugar. It is insoluble in water and therefore, used in the form of its soluble sodium salt. Although it has no nutritional value and is excreted unchanged by the body, it is used as an artificial sweetener.



Chemical Structure of Saccharin

SACCHAROMYCES: An industrially important genus in the kingdom of fungi that includes many species of yeast. Many members of this genus are considered very important in food production. An example is *Saccharomyces cerevisiae*, used in baking, brewing and wine making. It is also used in the production of single-cell protein and ergosterol and for experimental studies in cell biology and genetics. Other members of this genus include *Saccharomyces bayanus*, used in making wine and *Saccharomyces boulardii*, used in medicine.

SACCULE (SACCULUS): A chamber of the inner ear from which the cochlea arises in reptiles, birds and mammals. It bears patches of sensory epithelium concerned with balance. The saccule translates head movements into neural impulses which the brain can interpret. It is sensitive to linear translations of the head, specifically movements up and down. When the head moves vertically, the sensory cells of the saccule are disturbed and the neurons connected to them begin transmitting impulses to the brain.

SACHS-WOLFE EFFECT: A phenomenon in which density fluctuations associated with quantum-mechanical effects in the early universe caused 'ripples' in the cosmic microwave background radiation. This effect is the predominant source of fluctuations in the cosmic microwave background for angular scales above about ten degrees.

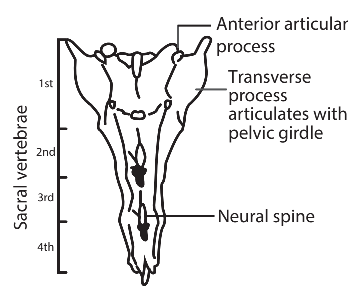
SACHSE REACTION: A reaction of methane at high temperature to produce ethyne: $2CH_4 \rightarrow C_2H_2 + 3H_2$. The reaction occurs at about 1500 °C (1773 K). This high temperature is obtained by burning part of the methane in air.

SACK: A measure of volume of commodities such as cereals and wool. Sacks of different commodities are of different sizes, but a typical measure is 3 bushels, which is about 105.7 litres based on the U.S. bushel or 109.1 litres based on the British Imperial bushel.

SACKUR-TETRODE EQUATION: An equation for the entropy of a perfect monoatomic gas which incorporates quantum considerations which give a more detailed description of its regime of validity. The entropy is given by $S = nRln(\frac{e^{\frac{5}{2}}V}{nN_A\Lambda^3})$, where $\Lambda = h/(2\pi mkT)^{\frac{1}{2}}$, n is the amount of gas, R is the gas constant, e is the base of natural logarithms, V is the volume of the system, N_A is the Avogrado constant, h is the Planck constant, m is the mass of the atom or gas particle, k is the Boltzmann constant and T is the thermodynamic temperature.

SACRAL: Near or relating to the sacrum at the base of the spine.

SACRAL VERTEBRA: Any of the small bones located between the lumbar and the caudal vertebrae in the vertebral column. They articulate securely with the pelvic girdle. They are usually fused to form a single bone to provide a firm support. Sacral vertebra varies from animal to animal. Amphibians have one, reptiles have two and there are three or more in mammals and birds have 10-23 fused in the synsacrum.



Sacral vertebra of a rabbit

SACRIFICIAL ACCEPTOR: Molecular entity that acts as the electron acceptor in a photoinduced electron transfer process and is not restored in a subsequent oxidation process but is destroyed by irreversible chemical conversion.

SACRIFICIAL ANODE (SACRIFICIAL ROD): A metallic anode used in cathodic protection, where it is intended to be dissolved to protect other metallic components. The more active metal is more easily oxidised than the protected metal and corrodes. It generally must oxidise nearly completely before the less active metal will corrode, thus acting as a barrier against corrosion for the protected metal. It is used, for example, to protect a steel structure such as a bridge, a ship, underground or underwater pipeline by wiring to it plates of an electropositive metal such as zinc. The zinc corrodes in preference to the steel, the cathode. Electrons are stripped from the anode and conducted to the protected metal, which becomes the cathode. The cathode is protected from corroding or oxidising, because reduction rather than oxidation takes place on its surface. The anodes are periodically replaced in a technique called **sacrificial protection**.

SACRIFICIAL DONOR: Molecular entity that acts as the electron donor in a photoinduced electron transfer process and is not restored in a subsequent reduction process but is destroyed by irreversible chemical conversion.

SACRIFICIAL PROTECTION (CATHODIC PROTECTION): A technique used to control the corrosion of a metal surface by making it the cathode of an electrochemical cell. The simplest method is by connecting the metal to be protected with another more easily corroded metal, called the **sacrificial anode** or **sacrificial rod** to act as the anode of the electrochemical cell. A typical example is the practice of protecting iron or steel by coating the surface of the iron or steel with another metal which is more reactive than the iron. Galvanisation is a typical example of sacrificial protection whereby the surface of an iron is coated with zinc metal. Cathodic protection systems are most commonly used to protect steel, water or fuel pipelines and storage tanks, steel pier piles, water-based vessels including yachts and ships, offshore oil platforms and onshore oil well casings. Cathodic protection can, in some cases, prevent stress, corrosion and cracking.

SACROILIAC JOINT: A joint that connects the V-shaped sacrum, at the base of the spine and the ilium of the pelvis, which are joined by ligaments. Ligaments and cartilaginous plates connect the bones.

SACRUM (SACRUMS; SACRA): A large, triangular bone at the base of the spine and at the upper and back part of the pelvic cavity, where it is inserted like a wedge between the two hipbones. Its upper part connects with the last lumbar vertebra and bottom part with the coccyx (tailbone). In children, it consists of usually five unfused vertebrae which begin to fuse between ages 16-18 and are usually completely fused into a single bone by age 26.

SADDLE FUNCTION: A function of two vector variables that is convex in one variable and concave in the other variable. In other words, it is a function for which a minimax theorem is obtainable.

SADDLE JOINT: A type of synovial joint in which the articular surfaces are convex in one direction and concave in another. Saddle joints permit flexion, extension, abduction, adduction and rotation. An example is the carpometacarpal joint of the thumb.

SADDLE POINT: 1. In a three-dimensional surface, it is a point that is a minimum in one planar cross-section and a maximum in another plane. At this point, the surface is saddle-shaped. 2. In mathematics, it is a point where all the first partial derivatives of a function vanish but which is not a local maximum or minimum. 3. For a matrix of real numbers, it is an element that is both the smallest element of its row and the largest element of its column or vice versa. 4. For a two-person, zero-sum game, it is an element of the payoff matrix that is the smallest element of its row and the largest element of its column, so that the corresponding strategies are optimal for each player, given the strategy chosen by the other player.

SAFE MODE: On a Windows computer, it refers to a software mode that enables a user to boot the computer with minimal system drivers and features. What makes Safe Mode different from normal mode, is that it uses Windows default drivers and settings. This helps users correct any issues preventing them from getting into normal mode. For example, if a computer is performing sluggishly and becomes slow, booting into Safe Mode helps to diagnose the problem and determine which files are slowing down the computer. Safe Mode is also useful after an unexpected crash and Windows may not fully load and the only way to get the computer to boot is by Safe Mode.

SAFETY GLASS: Glass that does not splinter into sharp pieces when smashed. Examples are **toughened glass** that is made by heating a glass sheet and then rapidly cooling it with a blast of cold air, it shatters into rounded pieces when smashed; and laminated glass that is pressed and heated and only cracks, when struck.

SAFETY LAMP (DAVY LAMP): A safety lamp with a wick and oil vessel burning originally a heavy vegetable oil, with the flame surrounded by fine wire gauze, so that it can be alight in an atmosphere containing firedamp (methane) without causing an explosion. It was devised in 1815 by Sir Humphry Davy for use in coal mines, allowing deep seams to be mined despite the presence of methane and other flammable gases, called **firedamp** or **mine damp**. The lamp also provided a test for the presence of gases. If flammable gas mixtures were present, the flame of the Davy lamp burned higher with a blue tinge as shown by the gauge. Miners could place the safety lamp close to the ground to detect gases such as carbon dioxide that are denser than air and so could collect in depressions in the mine. A flame extinguishes at about 17% oxygen content, air which will still support life, so the lamp gave an early indication of a problem.

SAFETY RING (WRITE RING): A plastic ring that fits within the reel of magnetic tape that helps prevent data from being overwritten accidentally or erased.

SAFFLOWER (AMERICAN SAFFRON; FALSE SAFFRON): *Carthamus tinctorius* of division Magnoliophyta, class Magnoliopsida order Asterales, family Asteraceae) is an oilseed thistle-like herb grown mainly in India and used in the manufacture of margarine and other residual oilseed cake. Safflower has long been cultivated in South Asia and Egypt for food and medicine and as a costly but inferior substitute for the true saffron dye. In the United States, it is more important as the source of safflower oil, which has recently come into wide use as cooking oil.

SAFFRON: Golden-coloured, pungent substance obtained from the dried stigmas of flowers of the saffron crocus (*Crocus sativus*), a bulbous perennial of the iris family. The colour and flavour are essential ingredients for certain dishes, as well as for some baked goods. Used as a dye, it has been the official colour for the robes of Buddhist priests and for royal garments in several cultures.

SAFRANIN (SEFRANINE; SAFRANIN O; BASIC RED 2): A biological stain used in optical microscopy that colours lignified and cutinised tissues nuclei red and chloroplast pink. It is used mainly for plant tissues, in conjunction with a blue or green counterstain. This is the classic counterstain in a Gram stain. It can also be used for the detection of cartilage and mast cell granules.

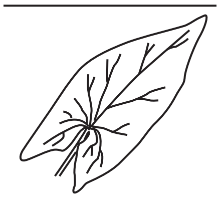
SAGE: An aromatic perennial herb, *Salvia officinalis*, of the mint family, native to the Mediterranean. Its leaves are used fresh or dried as flavouring in many foods. The flowers may be purple, pink, white or red. Since the middle ages, sage tea has been brewed as a spring tonic and a stimulant believed to strengthen the memory and promote wisdom.

SAGITTAL: 1. Shaped like an arrow. 2. A section through an organism that is bisected in the longitudinal plane. 3. Situated in the direction of the sagittal suture; said of an anteroposterior plane or section parallel to the median plane of the body.

SAGITTAL CREST: A ridge bone found in the skulls of some mammals (excluding humans) and reptiles that projects up, from front to back, along the top midline of the skull to which the jaw muscles are attached. The most prominent sagittal crests are found on adult male gorillas.

SAGITTAL PLANE (PARAMEDIAN PLANE): An imaginary vertical plane that divides the body into right and left parts. A sagittal plane is at right angles to a coronal plane.

SAGITTATE: In botany, shaped like the head of an arrow, with two pointed lobes extending downward from the base, especially as seen in some leaves.



Sagittate leaf

SAHEL: A dry zone, extending from Senegal in the west to Sudan in the east and separating the tropical regions of western and central Africa from the Sahara. The Sahel is about 5,400 km from the Atlantic Ocean in the west to the Red Sea in the east. It is mostly covered in grassland and savannah, with areas of woodland and shrubland. Rainfall averages between 102 and 203 mm and falls mostly from June to September. Nomadic herding and limited cultivation of peanuts and millet are possible in most areas.

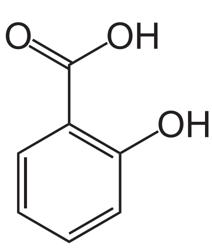
SAINFOIN: Leguminous perennial herb (*Onobrychis viciaefolia*) of the family Leguminosae, indigenous in South Europe and in temperate West Asia, grown mainly in areas with calcareous soil too dry or too barren for clover. It is classified in the division Magnoliophyta, class Magnoliopsida order Rosales and family Leguminosae.

SAINTPAULIAS (AFRICAN VIOLETS): A genus of herbaceous perennial flowering plants in the family Gesneriaceae, native to Tanzania and Kenya which are commonly cultivated as house plants. They can grow from 6-15 centemetres tall with rounded to oval succulent leaves which have hairs and petiolated. The flowers can be white, pale blue, violet or purple and grow in clusters of 3–10 or more, borne on slender peduncles. They can be propagated by leaf cutting or tissue culture.

SAL AMMONIAC (AMMONIUM CHLORIDE): A white crystalline solid with the chemical formula NH_4Cl, density 1.5274 g/cm^3, melting point 338 °C (611 K) and decomposes. Sal ammoniac was the former name of **ammonium chloride**. It typically forms as encrustations formed by sublimation around volcanic vents and found around volcanic fumaroles, guano deposits and burning coal seams. It is commonly used as a flux in the soldering of stained-glass windows, in the refining of precious metals, as expectorant and in dry cell electrolyte.

SALAMANDER: An amphibian of the order Urodela or Caudata. Salamanders have tails and small, weak limbs, but some do not have hind limbs, with not more than four toes on their front limbs and five on their rear ones, but some species have fewer digits. Superficially, they resemble the unrelated lizards, which are reptiles, but they are easily distinguished by their lack of scales and claws and by their moist, usually smooth skins. Salamanders are found in damp regions of the northern temperate zone and are most abundant in North America. Most are under 15 cm long, but the giant salamander of Japan (*Megalobatrachus japonicus*) may reach a length of over 1.5 m. Most salamanders are terrestrial as adults, living near water or in wet vegetation, but some are aquatic and a few are arboreal, burrowing or cave-dwelling. Most are nocturnal and all avoid direct light.

SALICYLIC ACID (2-HYDROXYBENZOIC ACID): A white crystalline organic compound with the chemical formula HOC_6H_4COOH, density 1.443 g/cm^3, melting point 159 °C (432 K) and boiling point 211 °C (484 K). It is a naturally occurring aromatic carboxylic acid found in certain plants. It is soluble in ethanol and ether but is only slightly soluble in water. It is used as an antiseptic in medicine and in the preparation of azo dyes. It is also used in foodstuff industries. Salicylic acid and its derivatives are toxic when consumed in large amounts. Sodium salicylate is used to a small extent as a food preservative and as an antiseptic in mouthwashes and toothpastes. The major use of salicylic acid is in the preparation of its ester derivatives; since it contains both a hydroxyl (–OH) and a carboxyl (–CO_2H) group, it can react with either an acid or an alcohol. The hydroxyl group reacts with acetic acid to form the acetate ester, acetylsalicylic acid. Several useful esters are formed by reaction of the carboxyl group with alcohols.



Chemical Structure of Salicylic Acid

SALINATION: A process by which the salt concentration of soil or water increases, especially as a result of irrigation in hot climate.

SALINE: 1. Relating to or containing common table salt or chemical salt, especially of sodium, potassium, magnesium, etc. 2. A solution of common salt (sodium chloride) and distilled water, especially one having the same concentration as body fluids. It is used as diluents for drugs and as a plasma substitute.

SALINE SOIL: A non-alkali soil or a soil that is not sodic containing enough soluble salts to interfere with the growth of most crop plants. The conductivity of the saturation extract is greater than 4 deciSiemens/metre (dS/m), the exchangeable-sodium percentage is less than 15, and the pH is usually less than 8.5.

SALINE-ALKALI SOIL (SALINE-SODIC SOIL): 1. A soil containing enough exchangeable sodium to interfere with the growth of most crop plants, and containing appreciable quantities of soluble salts. The exchangeable-sodium percentage is greater than 15, the conductivity of the saturation extract is greater than 4 deciSiemens/metre (dS/m) at 25°C, and the pH is usually 8.5 or less in the saturated soil. 2. A saline-alkali soil has a combination of harmful quantities of salts and either a high alkalinity or high content of exchangeable sodium, or both, so distributed in the profile that the growth of most crop plants is reduced.

SALINISATION (SALINIZATION): An increase in the salt content of soil which can lead to retarded plant growth and render the soil infertile. It occurs in hot areas where water readily evaporates from the soil and also due to irrigation. In coastal districts, salinisation can occur when sea water percolates into the soil as a result of the over-pumping of groundwater.

SALINITY: Measurement of the mass of dissolved salt present in a given amount of water. In most circumstances, especially in the context of seawater, the vast majority of dissolved solids and salts can be used interchangeably. Salinity is measured by an instrument called salinometer in grams of salt per kilogram (g/kg), which can also be expressed as parts per million (ppm).

SALINOMETER: An instrument for measuring the salinity of water or solution. There are two types available: the hydrometer type to measure density; and one that measures the electrical conductivity of a solution.

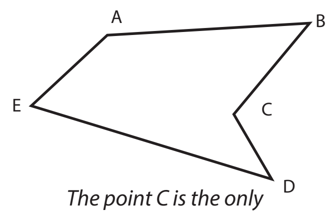
SALIVA: A watery fluid secreted by the salivary glands in the mouth. Human saliva is composed of 98% water, while the other 2% consists of other compounds such as electrolytes, mucus, antibacterial compounds and various enzymes. As part of the initial process of food digestion, the enzymes in the saliva break down some of the starch and fat in the food at the molecular level. Saliva also breaks down food caught in the teeth, protecting them from bacteria that cause decay. It lubricates and protects the teeth, the tongue and the tender tissues inside the mouth. Saliva also plays an important role in tasting food, by trapping thiols produced from odourless food compounds by anaerobic bacteria living in the mouth.

SALIVARY AMYLASE: An enzyme secreted by the salivary glands that catalyses the hydrolysis of starch into maltose and dextrin.

SALIVARY DIGESTION: The process of converting starch into sugars by the action of salivary amylase in an animal's mouth.

SALIVARY GLANDS: Glands in many terrestrial animals that secrete saliva in the mouth. In man there are three pairs: the sublingual, sub-mandibular and sub-maxillary glands, which also secrete amylase, an enzyme that breaks down starch into maltose. In other organisms such as insects, salivary glands are often used to produce biologically important proteins like silk or glues and fly salivary glands contain polytene chromosomes that have been useful in genetic research. **Salivation** is the act or process of secreting saliva or an abnormally abundant flow of saliva.

SALIENT: 1. In mathematics, a salient angle is an angle pointing outward at less than 180°; an interior angle of a polygon is salient if its vertex points outwards. 2. A salient point on curve is a point at which two branches of the curve meet with different tangents.



The point C is the only point that is ***not salient***

SALMON: Common name for marine fish characterised by an elongated body covered with small, rounded scales and a fleshy fin between the dorsal fin and tail. They occur in cold and temperate waters. They are members of the family Salmonidae in the Salmoniformes order. They include the Atlantic salmon, *Salmo salar* and similar fishes of the same family (Salmonidae), especially the Pacific salmon, which constitute the genus Oncorhynchus with six species. The average size of adult Atlantic salmon is 71-76 cm long and weigh 3.6-5.4 kg. Some uncommon adults can grow to 13.6 kg. The Chinook salmon or King salmon, *Oncorhynchus tshawytscha*, is the largest species of Pacific salmon with an adult average weight of 10 kg, although some uncommon individuals can grow to 22–36 kg. Salmons live most of their life in the ocean, but as adults, return to the stream where they hatched in order to spawn. Adult Pacific salmon die soon after spawning, but many Atlantic salmon return to the sea and after one or two years in open waters may spawn again, some up to three or four times. Most members of the salmon family are valuable as food.

SALMONELLA: A varied group of bacteria or a rod-shaped gram-negative non-spore forming bacterium found in the intestine that can cause food poisoning, gastroenteritis and typhoid fever. They are aerobic or facultatively anaerobic and most are motile. They can exist for long period outside their host. They can be divided into three broad groups. One of them causes typhoid and paratyphoid fevers while a second group causes salmonella food poisoning which is characterised by stomach pains, vomiting, diarrhoea and headache. They are found in sewage and surface water. Humans are infected by consuming contaminated foods like egg, meat, etc.

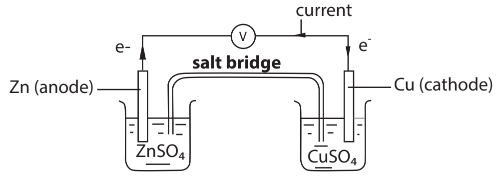
SALMONELLOSIS: A food poisoning disease caused by salmonella bacteria. It is usually characterised by gastrointestinal upset, diarrhoea, fever, vomiting and abdominal cramps and sometimes results in death. It often results from eating contaminated food which is not well cooked.

SALSIFY: A temperate biennial flowering plant with long white roots which are used as vegetable. They belong to the family Asteraceae in the genus *Tragopogon*.

SALSODA (SODIUM CARBONATE; WASHING SODA; SODA ASH): A white, crystalline, slightly alkaline salt or sodium salt of carbonic acid with the chemical formula Na_2CO_3, density 2.20 g/cm^{-3}, melting point 50 °C (323 K) and a boiling point of 851 °C (1124 K). It most commonly occurs as a crystalline heptahydrate, which readily effloresces to form a white powder, the monohydrate. It is domestically well known for its everyday use as a water softener. It has a cooling alkaline taste and can be extracted from the ashes of many plants and synthetically produced in large quantities from salt (sodium chloride) and limestone in a process known as the Solvay process. It is used in photography, in cleaning, in pH control of water, in textile treatment, glasses and glazes, as a food additive and a volumetric reagent. For aquatic organisms it is a very important or only source of carbon. It is used as a general cleanser, in the manufacture of glass, as an additive in municipal pools to neutralise the acidic effects of chlorine and raise pH.

SALT (ROCK SALT): A chemical compound formed when an acid and a base react or when the hydrogen of an acid is replaced by a metal or its equivalent, such as ammonium (NH_4). Examples are zinc sulfate ($ZnSO_4$), sodium chloride (NaCl), potassium chloride (KCl) and sodium carbonate (Na_2CO_3). Salts are typically crystalline ionic compounds, having many uses. For example, sodium chloride serves as a preservative, food seasoning and also used in the petroleum industry.

SALT BRIDGE: A bridge of a salt solution, usually potassium chloride, potassium nitrate or an agar gel placed between the two half-cells of a galvanic cell, either to reduce to a minimum the potential of the liquid junction between the solutions of the two half-cells or to isolate a solution under study from a reference half-cell and prevent chemical precipitations.



Salt bridge of an electrochemical cell

SALT FLATS (ALKALI FLATS): Level, barren areas in dry regions covered with evaporites, mainly salts of the alkali metals and alkaline earth metals. They are formed by the evaporation of water in a depression and deposition of its fine sediment and dissolved minerals.

SALT HYDROLYSIS: A reaction in which a salt of a weak acid or weak base or both is dissolved in water, with the water autoionising into negative hydroxyl ions and positive hydrogen ions; and the salt breaking down into positive and negative ions. In other words, it is a process in which a cation or an ion of a salt dissolved in water accepts H^+ ions resulting in excess hydroxyl ions or donates H^+ to water resulting in excess hydroxonium ions. For example, sodium acetate dissociates in water into sodium and acetate ions. Sodium ions react very little with hydroxyl ions, whereas acetate ions combine with hydrogen ions to produce neutral acetic acid and the net result is a relative excess of hydroxyl ions, causing a basic solution.

SALT POISONING (SALT TOXICITY): A disease of pigs, usually caused by inadequate supply of water or increase in salt ration. The animals become constipated before twitching, fits and death.

SALTATORIAL: Describes limbs adapted for leaping, for example, as in grasshoppers.

SALTATORY: Desribes animals such as kangaroos, anurans (tail-less amphibians) and rabbits that are adapted for leaping.

SALTATORY CONDUCTION: A method of rapid axon neuronal transmission in vertebrate nerves resulting from the action potential jumping from one node of Ranvier to another, skipping the myelin-sheathed regions of membrane. This reduces the capacitance of the neuron, allowing much faster transmission.

SALTING: 1. A low-lying area of land regularly flooded with salt water. 2. A method of addition of salt for the purpose of preserving food.

SALTING IN: The increase in solubility that is displayed by typical globular proteins on the addition of small amounts of certain salts such as ammonium sulfate.

SALTING OUT: 1. A method of separating proteins based on the principle that proteins are less soluble at high salt concentrations. The salt concentration needed for the protein to precipitate out of the solution differs from protein to protein. This process is also used to concentrate dilute solutions of proteins. Dialysis can be used to remove the salt if needed. 2. Precipitation of a colloid such as gelatine by the addition of large amounts of a salt. In other words, it is the process of adding brine to the product of saponification to give soap and glycerol.

SALTLICKS: 1. A natural deposit of salt which animals go to lick to supplement their salt intake. Salt licks often occur naturally, providing the sodium, calcium, iron, phosphorus and zinc required in the springtime for bone, muscle and other growth in some animals such as deer moose, elephants, cattle and sheep. 2. A block of salt or other preparation, which may also contain minerals such as magnesium, phosphorus, calcium, potassium, sulfur or iodine that livestock lick to supplement their salt intake.

SALTPETRE [POTASSIUM NITRATE; NITRATE OF POTASH; POTASSIUM TRIOXONITRATE(V)]: A transparent, colourless or white powder or crystals of potassium nitrate (KNO_3) occurring naturally in deposits. It is a strong oxidising agent used in fireworks, explosives, matches, fertilisers, glassmaking, steel tempering, food curing, as a reagent and as an oxidiser in solid rocket propellants. The term is also used for sodium nitrate (Chile saltpetre) and calcium nitrate (lime saltpetre), both of which are used in the nitric acid industry and as fertilisers and for ammonium nitrate (Norway saltpetre), a high explosive and fertiliser.

SALTWATER INTRUSION/ENCROACHMENT: Displacement of fresh surfacewater or groundwater by the advance of saltwater due to its greater density. This occurs usually in coastal and estuarine areas.

SALTWATER MARSH (SALT MARSH): The ecological system of plants and animals influenced by tidal waters with salinity. It is an environment in the upper coastal intertidal zone between land and salt water or brackish water. It is dominated by dense stands of halophytic (salt-tolerant) plants such as herbs, grasses or low shrubs. These plants are terrestrial in origin and are essential to the stability of the salt water marsh in trapping and binding sediments. Saltwater marshes play a large role in the aquatic food web and the exporting of nutrients to coastal waters. They also provide support to terrestrial animals such as migrating birds, as well as providing coastal protection.

SALVAGE VALUE: The amount that somebody may pay for something no longer useful in itself but because of potentially useful materials contained in it.

SALVERFORM: A gamopetalous corolla, which possesses a narrow tube and an abruptly spreading and flattened limb, as seen in flowers such as Phlox, Primula and primrose.

Salverform corolla

SAMARA: A dry single-seeded winged indehiscent fruit such as elm and ash fruits. **Samaroid** means resembling a samara.

SAMARIUM: Metallic element in Group 3 (IIIB) of the Periodic Table, the lanthanides, with the chemical symbol Sm, atomic number 62, atomic weight 150.35, specific gravity at 20°C is 7.54, melting point 1,072 °C (1345 K), boiling point 1,791 °C (2064 K). It has 7 naturally occurring isotopes, which are five stable isotopes, ^{144}Sm, ^{149}Sm, ^{150}Sm, ^{152}Sm and ^{154}Sm and two extremely long-lived radioisotopes, ^{147}Sm (1.06 × 1011y) and ^{148}Sm. The most abundant (26.75% natural abundance) is ^{152}Sm. Samarium is used exclusively in ferromagnetic alloy $SmCO_5$ which produces permanent magnet five times stronger than any other material. It is also used in special alloys for making nuclear reactor parts.

SAME CHANGE SUBTRACTION RULE METHOD: A method of solving subtraction problems by adding or subtracting the same number from both numbers without changing the difference. It is based on the assumption that a subtraction problem is easier if the smaller number ends in one or more zeros. For example, in the equation 98 - 8, add 2 to both numbers to change the equation to 100 -10. Now, because of the zeroes, it is easier to solve the problem:

$$\begin{array}{r} 100 \\ -10 \\ \hline 90 \end{array}.$$

SAMPLE: 1. A representative selection of a population that is examined to gain statistical information about the whole. 2. A portion of material selected from a larger quantity of material, which can be qualified, for example, bulk sample, representative sample, primary sample, bulked sample and test sample. The term 'sample' implies the existence of a sampling error, i.e. the results obtained on the portions taken are only estimates of the concentration of a constituent or the quantity of a property present in the parent material. If there is no or negligible sampling error, the portion removed is a test portion, aliquot or specimen. The term **specimen** is used to denote a portion taken under conditions such that the sampling variability cannot be assessed, usually because the population is changing and is assumed, for convenience, to be zero. The manner of selection of the sample is prescribed in a sampling plan.

SAMPLE COVARIANCE: A statistical measure of how much two variables change together. This measure is equal to the sum of the product of the deviations of corresponding values of the two variables from their respective means. For n pairs of values of two random variables x and y, this is given by $Cov(X,Y)=\frac{\sum_{i=1}(X_i-\bar{X})(Y_i-\bar{Y})}{n-1}$, where $\bar{X}$ and $\bar{Y}$ are the means of the populations of x and y, respectively.

SAMPLE ERROR: An error which arises when the absorbance of some samples changes with time, for example, as a result of photochemical reaction, formation of aggregates or adsorption on the cell wall.

SAMPLE HANDLING: Any action applied to a sample before the analytical procedure. Such actions include the addition of preservatives, separation procedures, storage at low temperature, protection against light and irradiation, loading, etc.

SAMPLE INJECTOR: A device by which a liquid or gaseous sample is introduced into the apparatus. The sample can be introduced directly into the carrier-gas stream or into a chamber temporarily isolated from the system by values which can be changed so as to make an instantaneous switch of the gas stream through the chamber.

SAMPLE POINT: In statistics, a single observed value of a random variable that is a member of the sample space of an experiment.

SAMPLE SPACE (UNIVERSAL SAMPLE SPACE): A set of all possible outcomes often denoted by S, Ω or U of a statistical experiment, represented by points. For example, if the experiment is tossing a coin, the sample space is the set {head, tail}. For tossing a single six-sided die, the sample space is {1, 2, 3, 4, 5, 6}. For some kinds of experiments, there may be two or more plausible sample spaces available.

SAMPLE STATISTIC (SAMPLING STATISTIC): In statistics, any function of observed data, especially one used to estimate the corresponding parameter of the underlying distribution of the entire population. Examples are sample mean and sample variance.

SAMPLE SURVEY: A study that collects planned information from a sample of individuals about their history, habits, knowledge, attitudes or behaviour in order to estimate particular population characteristics.

SAMPLE UNIT: The discrete identifiable portion suitable for taking as a sample or as a portion of a sample. These units may be different at different stages of sampling.

SAMPLER: A device used to withdraw and deliver a volume or an amount of a sample.

SAMPLING: 1. The selection of an unbiased or random subset of individual observations within a population of individuals intended to yield some knowledge about the population of concern, especially for the purposes of making predictions based on statistical inference. In sampling with replacement, every individual chosen from the population is replaced before the next selection. In **Random Sampling** every individual has equal chance of being selected as part of the sample. In **stratified random sampling**, the population is divided into strata, each of which is randomly sampled and the data samples from the different strata are pooled. **Systematic sampling** involves choosing individuals at fixed intervals, for example, every 5th object in the population. 2. In converting an analogue signal to a digital one, sampling is used to convert signals into strings of numbers that can be processed mathematically by electronic circuits. In audio, a segment of sound is taken and played at a high rate of speed, causing a continuous sound. In video or moving image, multiple still images are taken, combined and played at a high rate of speed causing those images to be animated. Both digital video and digital audio files are created using samples. The quality of the sample is determined by the sampling rate or the bit rate at which the signal is sampled.

S

SAMPLING DESIGN: The procedure by which a sampling unit is selected from the population. In general, a particular design is determined by assigning to each possible sample S the probability P(S) of selecting that sample.

SAMPLING DISTRIBUTION: The probability distribution of a statistic calculated from a random sample of a particular size. For example, the sampling distribution of the arithmetic mean of the sample of size n taken from a normal distribution with mean μ and standard deviation σ, is a normal distribution also with mean μ but with standard deviation, $\frac{\sigma}{\sqrt{n}}$.

SAMPLING ERROR: That part of the total error (the estimate from a sample minus the population value) associated with using only a fraction of the population and extrapolating to the whole, as distinct from analytical or test error. It arises from a lack of homogeneity in the parent population. In chemical analysis, the final test result reflects the value only as it exists in the test portion. It is usually assumed that no sampling error is introduced in preparing the test sample from the laboratory sample. Therefore, the sampling error is usually associated exclusively with the variability of the laboratory sample. Sampling error is determined by replication of the laboratory samples and their multiple analyses. Since it is always associated with analytical error, it must be isolated by the statistical procedure of analysis of variance.

SAMPLING FRAME (FRAME): In statistics, an enumeration of a population for the purpose of sampling, especially as the basis of a stratified sample. It includes a numerical identifier for each individual, plus other identifying information about characteristics of the individuals, to aid in analysis and allow for division into further frames for more in-depth analysis.

SAMPLING PLAN: A predetermined procedure for the selection, withdrawal, preservation, transportation and preparation of the portions to be removed from a population as samples. Summarising the test values or observations from the selected portions yields an estimate for the concentration of an analyte or a value for a property determined with a calculable degree of uncertainty at a specified confidence level. A sampling plan includes the designation of the number, location and size of the portions and instructions for the extent of compositing and for the reduction (in amount and fineness) of the portions to a laboratory sample and to test portions. It may also contain acceptance criteria. Some sampling plans do not include more than instructions for the statistical selection of portions to be removed. Such plans should properly be designated as 'statistical sampling plans'.

SAMPLING THEOREM (NYQUIST THEOREM): A theorem, which states that a sound wave should be sampled no less than twice per period in order to properly represent that sound wave. It is used in the conversion of analogue signals to digital ones. This is due to the fact that if a sound wave at a certain frequency is sampled less than twice in its period, it will most likely be interpreted as a much lower frequency, referred to as aliasing. It is an undesirable condition, which, for example, in images, produces a jagged edge or stair-step effect; and for sound, produces a buzz.

SAMPLING VARIATION: The variation shown by different samples of the same size from the same population.

SAND: Tiny grains of of finely divided rock and mineral particles usually consisting mainly of quartz, with a diameter of between 0.05 and 2.0 mm. A sandy soil is defined as one containing at least 85% sand and not more than 10% clay. Sand is found mainly on seashores and in deserts.

SANDBOX: 1. A location on a computer, online service or network that allows testing or experimenting without causing problems. 2. In computer security, it is a location in memory designated by a program or user where a program can run without affecting the system. This enables a developer to test a program or script without affecting the system if anything goes wrong with the program or script.

SANDMEYER REACTION: A process of producing halogenated aromatic compounds through the substitution of the halide ion with a diazonium salt in the presence of copper(I) halide.

SANDSTONE: Sedimentary rock formed from sand-sized grains of between 0.06 -2 mm in diameter. The spaces between grains may be empty or filled with either a chemical cement of silica or calcium carbonate or a fine-grained matrix of silt and clay particles. The principal mineral constituents of the grain framework are quartz, feldspar and rock fragments. Sandstones are quarried for use as building stone. Because of their abundance, diversity and mineralogy, sandstones are also important to geologists as indicators of erosional and depositional processes.

SANDWICH COMPOUND: A chemical compound (organometallic) featuring a metal bounded to two arene ligands. The arenes have the formula C_nH_n, substituted derivatives, for example $C_n(CH_3)_n$ and heterocyclic derivatives, for example BC_nH_{n+1}. Because the metal is usually situated between the two rings, it said to be "sandwiched." A special class of sandwich complexes are the metallocenes.

SANDWICH RESULT (SQUEEZE RULE): In mathematics, it is one of a number of inequalities, used to confirm the limit of a function whose terms are bounded above and below by comparing it with two other functions whose limits are known or easily computed. For example, if $f(x) \leq g(x) \leq h(x)$ for all x; greater than some N and if f(x) tends to A and $h(x)$ tends to A as x tends to infinity, then $g(x)$ tends to A as x tends to infinity.

SANDY LOAM: A type of soil composed of about 10% clay, less than 50% silt, and about 50% sand used for gardening. This soil type is classified into four categories as coarse sandy loam, fine sandy loam, sandy loam and very fine sandy loam, based on its concentration of sand particles.

SANGER'S REAGENT (1-FLUORO-2,4-DINITROBENZENE; DNFB): A substance used in structural protein chemistry to arylate free amino groups with the chemical formula $C_6H_3F(NO_2)_2$, density 1.4718 gcm^{-3} (54 °C, 327 K), melting point 25.8 °C (299 K) and boiling point 296 °C (569 K).

SANITARY: 1. Free from dirt or substances that may cause disease or concerned with protecting health. 2. Relating to public health, especially general hygiene and the removal of human waste through the sewage system. **Sanitation** is the hygienic means of promoting health through prevention of human contact with the hazards of wastes. Hazards can be physical, microbiological, biological or chemical agents of disease. Wastes that can cause health problems are human and animal faeces, solid wastes, domestic waste water, industrial wastes and agricultural wastes. Hygienic means of prevention include using engineering solutions such as sewerage and wastewater treatment, septic tanks or even by personal hygiene practices such as simple hand washing with soap.

SANITARY LANDFILL: An engineered burial of refuse. The refuse is dumped into trenches and compacted by bulldozer, where, it is hoped, aerobic metabolism by microorganisms decomposes the organic matter to stable compounds (H_2O, CO_2, etc.). Moisture is essential for the biological degradation and groundwater assists the process except when it fills air voids and prevents the transport of oxygen to the refuse. Landfills of unsatisfactory design can be major sources of air, water and soil pollution.

SANITISATION/SANITIZATION: Making an object or environment free of pathogenic or harmful organisms, for example, by sterilisation.

SANKEY DIAGRAM: A specific type of flow diagram, in which the width of the arrows is shown proportionally to the flow quantity. It is used to visualise energy or material or cost transfers between processes.

SAP: A sugary fluid that is found in the phloem tissue of plants. It is the carbohydrate produced in photosynthesis and other organic molecules transported in xylem cells or phloem sieve tube elements of a plant. Fluid found in the vacuoles of other cells is sometimes referred to as **cell sap**.

SAPLING: A young tree, especially one with a diameter of less than 10 cm.

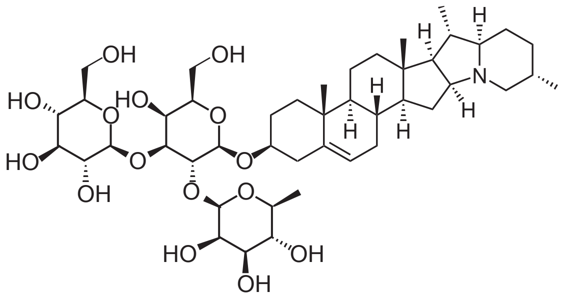
SAPONACEOUS: 1. Having the qualities of soap. 2. A substance or object that is soapy and slippery to touch.

SAPONIFICATION: Hydrolysis of an ester, using an alkali, to produce a free alcohol and a salt of the organic acid (carboxylic acid): $RCOOR'+OH^- \rightarrow RCOO^- + R'OH$. It is the chemical processes that produce soap from fatty acid derivatives. When triglycerides in fat/oil react with aqueous NaOH (sodium hydroxide, also known as caustic soda or lye) or potassium hydroxide, also known as caustic potash or potassium hydrate (KOH), they are converted into soap and glycerol: Fat/oil + alkali + heat → soap + glycerol. For example, glyceryl stearate-based fats react with sodium hydroxide to form sodium stearate (ordinary soap) and glycerine.

SAPONIFICATION EQUIVALENT: The quantity of fat in grams that can be saponified by 1 litre of normal alkalis.

SAPONIFICATION VALUE: Number of milligrams of potassium hydroxide required for the complete saponification of 1 g of the substance being tested. It is a measure of the average molecular weight or chain length of all the fatty acids present.

SAPONIN: Any of a class of glycosides, found widely in plants that have detergent properties and form lather when shaken with water. Chemically, saponins consist of a sugar group linked to a steroid or triterpene group. They are especially concentrated in the common soapwort (*Saponaria officinalis*), whose foliage is boiled and used as a soap substitute. An example of saponin is solanin whose structure is given below:



Chemical Structure of Solanin (An example of Saponin)

S

SAPOR: The flavour or taste found within a plant or a plant substance.

SAPPHIRE: Any of the gem varieties of corundum except ruby, especially the blue variety but other colours include yellow, brown, green, pink orange and purple. They are obtained from igneous and metamorphic rocks and from alluvial deposits. They are found chiefly in Thailand, India, Sri Lanka and Myanmar and also in Australia and in the United States. Sapphires can be synthesised by the fusion of aluminium oxide, with titanium oxide added as a colouring agent.

SAPRO-: Prefix meaning decaying or decomposition, for example, saprogenic or saprophyte.

SAPROBE: An organism that acts as a decomposer by absorbing nutrients from dead organic matter.

SAPROLEGNIALES (WATER MOULDS): An order of the Oomycetes containing some 32 genera and about 200 species of aquatic, mainly saprobic fungi, for example, Saprolegnia, often with diplanetic zoospores. Some species are parasitic on fish, for example, *Saprolegnia parasitica*, which often attack salmon and goldfish.

SAPROLITE: Naturally occurring deposit of soft, thoroughly decomposed and porous rock formed through chemical weathering of igneous, metamorphic or sedimentary rocks, for example, by the action of acids in rain water. It is especially common in humid and tropical climates, forming in the lower zones of soil. It is usually reddish brown or greyish white and contains those structures such as cross-stratification that were present in the original rock from which it formed.

SAPROMYOPHILY: Pollination of plants by carrion and dung flies. There are two types of fly pollination, the other one being **myophily**, which includes flies such as bee flies (Bombyliidae) and hoverflies (Syrphidae) that feed on nectar and pollen as adults. Carrion and dung flies are deceived into pollinating flowers that produce odours of decay and mimic the decaying flesh in which these flies normally oviposit. Sapromyophilic flowers are usually radial in shape, often with great depth or lantern shaped, frequently with window openings through which the flies crawl into the blossom or trap.

SAPROPHYTE: An organism, especially a fungus or bacterium that lives on and gets its nourishment from dead organisms or decaying organic material. Saprophytes recycle organic material in the soil, breaking it down into in simpler compounds that can be taken up by other organisms. **Saprophytism** is the process of obtaining nourishment from decaying organic matter. **Saprophagous** describes feeding on decaying organic matter.

SAPROTROPH (SAPROBE; SAPROVORE; SAPROBIONT): Any organism that absorbs soluble organic nutrients from dead organic matter, for example, dead plant or animal matter such as dung or faeces. Saprotrophs are important in food chain as they bring about decay and release of nutrients for plant growth. If the organism is fungi or bacterium it is called a **saprophyte**; if it is an animal or is animal-like it is called a **saprozoite**.

SAPWOOD: The relatively thin, youngest outer wood of a tree trunk or branch. It consists of living xylem cells which conducts water, dissolved materials and provide structural support. As the tree grows in diameter, the innermost layers of sapwood become heartwood and new sapwood is produced on the outside of the woody column.

SARCOCARP: The fleshy part of a fruit. The sarcocarp is situated in the epicarp and endocarp, as occurs in peach and plum.

SARCOCAULOUS: Having a fleshy stem.

SARCOLEMMA: The contractile membrane that surrounds a muscle fibre. It consists of a true cell membrane, called the **plasma membrane** and an outer coat made up of a thin layer of polysaccharide material that contains numerous thin collagen fibrils. The membrane is designed to receive and conduct stimuli.

SARCOMA: Malignant tumour of the soft tissues, which include muscle, fat, blood vessels, nerves, tendons and the joint linings.

SARCOMERE: Any of the functional units that make up the myofibrils of voluntary muscle. Each is bounded by two membranes: **z-lines**, which provide the point of attachment of actin filaments; and **M band**/**line**, which is the point of attachment for myosin filaments.

SARCOPLASM: The cytoplasm of a muscle fibre. It contains chemicals that are required for muscle contraction, including glycogen, adenosine triphosphate (ATP), phosphocreatine and myoglobin. Sacroplasm of active muscles tends to be rich in mitochondria. It has Golgi apparatus near the nucleus as well as a smooth endoplasmic reticulum.

SARCOPLASMIC RECTICULUM: Specialised endoplasmic reticulum found in the fibres of voluntary and cardiac muscles. It forms a network of membrane lined cavities surrounding the contractile myofibril that run through the fibres. It transmits the contractile impulse to all parts of a muscle fibre by releasing and then sequestering calcium ions that are essential for muscle contraction.

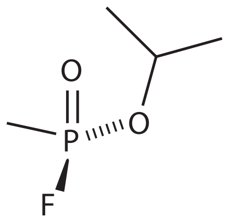
SARCOUS: Fleshy; of or relating to flesh or muscle tissue.

SARDINIAN: An important breed of Italian sheep which provides milk for the manufacture of cheese and coarse white wool. It is native to the island of Sardinia in Italy. It was developed from crosses of local lowland sheep, Merinos and North African sheep.

SARGASSUM: Any of the brown algae that make up the genus *Sargassum*. They are tropical marine seaweeds that drift in large, floating masses even though many species grow attached to rocks along the coast. They are used as fertiliser in New Zealand and as a component of soups and soy sauce in Japan.

SARIN: A highly toxic colourless liquid with the chemical formula $C_4H_{10}FO_2P$, density 1.0887 g/cm³ at 25 °C (297 K) and 1.102 g/cm³ at 20 °C (293 K), melting point -56 °C (217 K) and boiling point 158°C (431 K). It is miscible in water. Sarin is a chiral molecule (typically racemic), with four substituents attached to the tetrahedral phosphorus centre. It is prepared from methylphosphonyl difluoride and a mixture of isopropyl alcohol:

$CH_3P(O)F_2$ + $(CH_3)_2CHOH$ → $[(CH_3)_2CHO]CH_3P(O)F$ + HF. It inhibits the activity of cholinesterase and used as a nerve gas in chemical warfare. Production and stockpiling of sarin was outlawed by the Chemical Weapons Convention of 1993.



Chemical Structure of Sarin

SARMENT: A long, slender and prostrate filiform stem, as that of strawberry.

SATELLITE: 1. A relatively small object that moves in an orbit around a planet or a larger object, which in turn orbits a star. Satellites can be natural or artificial. Natural satellites that orbit planets are called **moons**. Artificial satellites are launched by man into space and used for scientific purposes, communication, forecasting and military application. In the solar system, all planets except Mercury and Venus have satellites. 2. In botany, it is also known as **trabant**, which is a spherical body seen attached to one end of a chromosome arm by a narrow filament at mitotic metaphase. The satellite stalk functions as a nucleolar organiser.

SATELLITE COMPUTER: A computer that is connected to a distant, more powerful computer.

SATELLITE DNA: The proportion of a DNA of eukaryotic cells that consists of very large numbers of copies of a short nucleotide sequence. It occurs mainly around the centromeres and telomeres of the chromosomes. It is so called because repetitions of a short DNA sequence tend to produce a different frequency of the nucleotides adenine, cytosine, guanine and thymine and thus have a different density from bulk DNA such that they form a second or 'satellite' band when genomic DNA is separated on a density gradient.

SATELLITE IMAGE: Digital data of the Earth or any other planet obtained from instruments carried in artificial satellites that provides a variety of information, such as vegetation patterns, sea surface temperature, weather and geology. It includes collecting data both in the visible and non-visible portions of the electromagnetic spectrum. One system is the multispectral scanner carried in Landsat satellites.

SATELLITE INFRARED SPECTROMETER: A spectrometer carried aboard satellites in the Nimbus series which measures the radiation of carbon dioxide in the atmosphere at several different wavelengths in the infra red region, giving the vertical temperature structure of the atmosphere over a large part of the Earth.

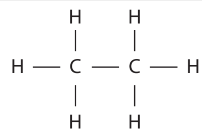
SATISFIABLE: In logic, an expression or formula is satisfiable if it is possible to find a model that makes it true. The **satisfiability problem** is the problem of determining whether there is an assignment of values to its variables that will satisfy any statement of a logical calculus. An example of satisfiability problem is the **Boolean satisfiability problem** or **propositional satisfiability problem** which is the problem of determining if there exists an interpretation that satisfies a given Boolean formula.

SATISFY: 1. In mathematics, to meet the conditions of a given theorem, assumption, etc. For example, $x = 3$ satisfies the equation $x^2 - 4x + 3 = 0$. 2. In logic, to yield a truth by the substitution of the given value or sequence of values in a predicate.

SATURATE: 1. To add as much of a gas, solid or liquid to a solution so that it can absorb or dissolve a maximum amount at a given temperature. 2. State of an organic compound in which all its carbon atoms are linked by single covalent bonds.

SATURATED AMMONIA: 1. Liquid ammonia in a state in which adding heat at constant pressure causes the liquid to vapourise at constant temperature. When the heat is removed at constant pressure, it causes immediate condensation at constant temperature. 2. Ammonia vapour in a state in which adding heat at constant pressure causes an immediate temperature rise (superheating). When the heat is removed at constant pressure, it starts immediate condensation at constant temperature.

SATURATED COMPOUND: An organic compound such as alkanes with all carbon bonds satisfied. It contains only single covalent bonds and does not contain double or triple bonds and thus cannot add elements or compounds. Those which contain double or triple bonds are said to be **unsaturated**. Saturated compounds can undergo substitution reaction but not addition reaction. An example of a saturated compound is ethene.



Chemical Structure of Ethane, an example of a Saturated Compound

SATURATED FAT: A fat with single covalent bonds between the carbons of its fatty acids. The molecules of a saturated fat have only single bonds between carbon atoms. If double bonds are present in the fatty acid portion of the molecule, the fat is said to be **unsaturated**. Unsaturated fats generally have lower melting points than saturated fats and are often liquids or oils at room temperature. Unsaturated fats can be converted to saturated fats by a process called **hydrogenation**.

SATURATED HYDROCARBON: A carbon-hydrogen compound such as olefins and acetylenics with all carbon being singly bonded. It contains the maximum number of hydrogen atoms bonded to each carbon atom. There are no double or triple bonds.

SATURATED VAPOUR: The state of any vapour, whose pressure exceeds that at which condensation usually occurs or vapour that can exist in equilibrium with its parent solid or liquid at a given temperature.

SATURATED VAPOUR PRESSURE: The vapour pressure of a thermodynamic system, at a constant temperature, in which the vapour of a substance is in equilibrium with a plane surface of that substance's pure liquid or solid phase. It can also be defined as pressure above a liquid at constant temperature which is confined so that the vapour from the liquid accumulates above it. Its value depends on the temperature and the properties of the liquid.

SATURATION: 1. The state of a solution or vapour in which it has the highest possible concentration of the dissolved or vapourised material at a given pressure and temperature. In other words, a saturated solution can dissolve no more solute at a given temperature and pressure or it contains the maximum amount of dissolved solid, liquid or gas so that no more will dissolve into the solution at a given temperature and pressure. A dynamic equilibrium exists between dissolved solutes and the undissolved particles. A solution which contains less than the maximum amount of solute is said to be **unsaturated**. A solution which contains more than the maximum amount of the solute at equilibrium is said to be **supersaturated**. Though it is sometimes possible to bring about supersaturation, such solutions or vapours are unstable and spontaneously revert to the saturated state, accompanied by the transformation of the excess material to the solid or liquid form (precipitation). 2. A state of an organic compound in which all of its carbon atoms are linked by single covalent bonds. 3. The condition in which a ferromagnetic material cannot be magnetised more strongly as all the domains are orientated in the direction of the field.

SATURATION FRACTION: Amount-of-substance of a component (solute) in a solution divided by the amount-of-substance of the component when it is saturating the system at constant temperature and pressure. It is also referred to simply as saturation, for example oxygen saturation.

SATURATION POINT: The point at which the greatest possible amount of a solute has been absorbed by a solution at a given temperature or the point at which no more can be added.

SATURATION SPECTROSCOPY: A branch of spectrometry in which the intense, monochromatic beam produced by a laser is used to alter the energy level range of particle velocities, giving rise to extremely narrow spectra lines that are free from Doppler broadening. It is used to study atomic, molecular and nuclear structure and to establish accurate values for fundamental physical constants.

SATURATION TRANSFER: A term used in nuclear magnetic resonance. When a nucleus is strongly irradiated, its spin population may partly be transferred to another nucleus by an exchange process.

SATURN: The second largest planet in the solar system. It has a thick atmosphere consisting of about 94% hydrogen and 6% helium, with traces of methane, ammonia, ethane, ethylene and phosphine. It has bright rings made up of orbiting fragments of rocks. The temperature is believed to be about -168 °C (105 K). It is the sixth in order of distance from the Sun and makes one revolution about the Sun in 29.42 years, at a mean distance of about 1,426,000,000 kilometres. The equatorial diameter of Saturn is about 120,540 km; a polar diameter about 108,700 km and a volume of 827,129,915,150,897 km^3, approximately 8.2713×10^{14} km^3 (769 times that of Earth); a mass of 5.6846×10^{26} kg (about 95 times that of Earth). Its mean density is 0.70 g/cm^3, the lowest mean density of all the planets. Saturn has 53 known moons with an additional 9 moons awaiting confirmation of their discovery. One day on Saturn takes 10.7 hours, the time it takes for Saturn to rotate or spin once. Saturn makes a complete orbit around the Sun (a year in Saturnian time) in 29 Earth years.

SAVANNAH (SAVANNA): Extensive open tropical or subtropical grassland with scattered trees and shrubs so that the canopy does not close. This allows sufficient light to reach the ground for the growth of herbaceous layer of vegetation. The climate of a savannah is characterised by a rainy period when the area is covered by grasses and dry period when the grasses wither. Savannah regions are subject to regular wildfires. These fires are usually confined to the herbaceous layer and do little long term damage to mature trees. However, these fires either kill or suppress tree seedlings, thus preventing the establishment of a continuous tree canopy which would prevent further grass growth.

SAVE: The process of writing data to a storage medium, such as a floppy disk, compact disc, flash drive or hard drive. The Save option is found in almost all programs commonly under the “File menu” or through an icon that resembles a floppy diskette. When you click the Save option, the file will be saved as its previous name. However, if the file is new, the program will ask the user to name the file and specify a location for it. Note: - This should precede “Save”

SAW-TOOTHED BEETLE: Saw-toothed grain beetle o*ryzaephilus surinamensis* and merchant grain beetle o*ryzaephilus Mercator*, that infest cereals, cornmeal, cornstarch, popcorn, rice, dried fruits, breakfast foods, flour, rolled oats, bran, macaroni, sugar, drugs, spices, herbs, candy, dried meats, chocolate, bread, nuts, crackers, raisins and other foodstuffs, making them unsalable and unpalatable. These beetles are capable of chewing into unopened paper or cardboard boxes, through cellophane, plastic and foil wrapped packages.

SAW-SCALING: An action of a snake curving its body in concentric curves and rasping its keeled scales together to make a sawing sound as a warning.

SAWFLY: Hymenopterous insect, chiefly of the family Tenthredinidae. The females have saw-like ovipositors used for cutting into plant tissue to deposit their eggs. Examples are the apple sawfly and gooseberry sawflies. The larvae or caterpillars cause serious damage to fruits and crops.

SAWHORSE PROJECTION: A perspective formula indicating the spatial arrangement of bonds on two adjacent carbon atoms. The bond between the two atoms is represented by a diagonal line, the left-hand bottom end of which locates the atom nearer the observer and the right-hand top end the atom that is further away. In other words, it views a carbon-carbon bond from an oblique angle, making it easy to visualise the molecule as a whole in 3-D.

SAWTOOTH WAVEFORM: A waveform, resembling a tooth of a saw in which the variable increases uniformly with time for a fixed period, drops sharply to its initial value and repeats the same sequence periodically. It is illustrated in practical electric circuits and its generators are frequently used to provide a time base for electronic circuits, as in the cathode-ray oscilloscope.

SCAB: A dry crust formed over a wound or sore as it heals or a disease of skin and plants causing scab-like roughness.

SCABERULOSE (SCABERULENT): Slightly rough to touch, owing to the presence of short, rigid hairs or the structure of the epidermal cells.

SCABIES: A contagious skin disease caused by the itch mite (*Sarcoptes scabiei*) which causes scabs and itching or causes a person’s skin to become rough and uncomfortable. It is acquired through close contact with an infested individual or contaminated clothing. It is most prevalent among those living in crowded and unhygienic conditions.

SCABRID: Having a roughened surface.

SCABRIDULOUS: Having a minutely roughened surface.

SCABROUS: Rough to touch, owing to the presence of short, rigid hairs or the structure of the epidermal cells.

SCALA: Any of three-fluid filled canals of the cochlea in the inner ear. They are the scala media, scala tympani and scala vestibule. They transduce the movement of air that causes the tympanic membrane and the ossicles to vibrate leading to the production of sound.

SCALAR: 1. In physics, a one-dimensional physical quantity, such as speed, mass, charge, volume, time and temperature, having magnitude (size) but not direction. This contrasts with velocity, which has magnitude and direction. 2. In mathematics, a real number that is defined in the context of vectors or matrices that a variable such as a is just a real number and not a vector or matrix. The real numbers relate to vectors in a vector space through the operation of scalar multiplication, in which a vector can be multiplied by a number to produce another vector. 3. An element of the ring over which a commutative group is module.

SCALAR FIELD: A function that maps a connected domain in Euclidean space into the real numbers. The scalar may either be a mathematical number or a physical quantity such as speed or mass. Examples of scalar fields in a three dimensional space include sin xyz, cos z, $x^2 + y^2 + z^2$. In physics examples include the temperature distribution throughout space, the pressure distribution in a fluid and spin-zero quantum fields, such as the Higgs field.

SCALAR MATRIX: A diagonal matrix whose diagonal entries are all equal scalars. Multiplication by a scalar matrix is equivalent to scalar multiplication by the constant scalar. A scalar operator is a multiple of the identity.

SCALAR MULTIPLICATION: The multiplication of a vector by a scalar quantity to obtain another vector. For example,
$3\langle 1, 2, 3\rangle = \langle 3, 6, 9\rangle$.

SCALAR OPERATOR: A linear operator that is a scalar multiple of the identity operator.

SCALAR POTENTIAL: A scalar function whose negative gradient is equal to some vector field, at least when this field is time-independent. An example is the potential energy of a particle in a conservative force field and the electrostatic potential. The potential functions of gravitation described by Newton’s law of gravitation and of electrostatics are scalar potentials.

SCALAR PRODUCT (DOT PRODUCT; INNER PRODUCT): The dot product of U with unit vector V, denoted $U \cdot V$, is defined as the projection of U in the direction of V or the amount that U is pointing in the same direction as unit vector V. This operation that can be defined algebraically as: $U \cdot V = U_1V_1 + U_2V_2 + U_3V_3$ or geometrically as: $U \cdot V = |U||V|\cos\theta$, where $|U|$ and $|V|$ are the magnitudes of the vectors **U** and **V** respectively and θ is the angle between them.

SCALAR QUANTITY: A measure that has quantity, amount, size or magnitude but does not show direction. Examples are length, area, mass, volume, density, speed, energy and temperature. Vector quantity has both magnitude and direction. For example, a scalar quantity can be exemplified by the distance 6 km; a vector quantity can be exemplified by the term 6 km north.

SCALAR TRIPLE PRODUCT: The scalar triple product of vectors $\mathbf{v}_1$, $\mathbf{v}_2$ and $\mathbf{v}_3$ from Euclidean three-dimensional space determines the volume of the parallelepiped with these vectors as edges. It is given by the determinant of the 3 × 3 matrix whose rows are the components of $\mathbf{v}_1$, $\mathbf{v}_2$ and $\mathbf{v}_3$.

SCALAR-VALUED: Of a mapping, taking values in a field of scalars as contrasted with a vector-valued mapping taking values in the corresponding vector space.

SCALARIFORM: Resembling the rungs of a ladder as found in certain secondary xylem elements possessing parallel bands of thickening or some perforation plates that have several pores separated by parallel bars of tissue.

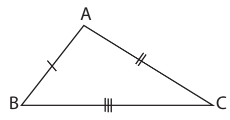
SCALARIFORM THICKENING (SCALARIFORM-RETICULATE THICKENING): A type of reticulate secondary wall patterning in tracheary elements in which the network is broken by elongate unthickened areas arranged in a more or less parallel fashion. Whereas annular thickening and spiral thickening permit further elongation, reticulate and scalariform thickenings allow little further extension and are therefore to be found in tissues, such as metaxylem and secondary xylem, that do not elongate after maturation.

SCALD: 1. Also known as **interdigital dermatitis**, it is a bacterial disease of sheep caused by *Fusobacterium necrophorum*. It causes lameness in lamb. 2. A defect in stored apple where brown patches appear on the skin and the tissue underneath becomes brown and soft due to accumulation of gases given off during ripening.

SCALE: 1. Any of the small bony or horny plates forming the body covering of fishes and reptiles. The wings of some insects especially the Lepidoptera are covered with tiny scales that are modified cuticular hairs. 2. Also called **bud scale**, it is a rudimentary body, usually a specialised leaf often covered with hair, wax or resin, enclosing an immature leaf bud. 3. A thin, scarious or membranous part of a plant such as a bract of a catkin. 4. The diseased condition of plants caused by scale insects. 5. A coating or incrustation such as on the inside of a boiler, formed by the precipitation of salts from the water. 6. A measuring instrument for weighing or showing amount of mass. 7. A sequence of collinear marks, usually at either regular intervals or representing equal steps that are used as a reference in making measurements. The one in which equal distances represent equal amounts is called a **linear scale**; and the one in which distances are proportional to the logarithms of the amounts represented is the **logarithmic scale**. The ratio between the size of a representation of something and its actual size is called a **scale factor**. 8. A ratio representing the size of an illustration or reproduction, especially a map or a model in relation to the object represented. For example, scale drawing is a drawing of an object in which all parts are drawn to the same scale as the object. For example, a map is a scale drawing of a geographical region. 9. A system of classification in which things or people are progressively ranked according to a specified criterion. 10. A series of musical notes, usually sequential, arranged in ascending or descending order of pitch. 11. A small structure that overlaps others to form the covering of the wings of butterflies and moths. 12. A thin flat piece or flake of something such as dead skin. 13. A flaky oxide that forms on the surface of some metals undergoing heat treatment, especially the black oxide that forms on iron or steel at high temperatures.

SCALE FACTOR: A number which scales or multiplies some quantity. For example, in the equation $y = cx$, c is the scale factor for x. For example, doubling distances corresponds to a scale factor of 2 for distance, while cutting a cake in half results in pieces with a scale factor of ½.

SCALENE TRIANGLE: A triangle with unequal sides and angles. In a scalene triangle, all interior angles are different; the shortest side is opposite to the smallest angle and the longest side is opposite to the largest angle. If two sides are equal, it is called an **isosceles triangle**. To find the area of a scalene triangle if all the three sides, a, b and c are known, Heron's formula: $\sqrt{s(s-a)(s-b)(s-c)}$ is applied, where $s = \left(\frac{1}{2}\right)(a+b+c)$. If two sides, a and b and an included angle C are known, then area, A is expressed as follows:



Scalene triangle

$$A = \left(\frac{1}{2}\right)ab\sin C.$$

With two angles, A and B and any side c known, angle C must be calculated using the property of triangle that the sum of three angles ie equal to 180°. The sine rule $\left(\frac{a}{\sin A}\right) = \frac{b}{\sin B} = \frac{c}{\sin C}$ is used to find one more side. The formula $\left(\frac{1}{2}\right)ab\sin C$ can then be applied. If the sine rule is used to find all the sides, then Heron's formula: $\sqrt{s(s-a)(s-b)(s-c)}$, can be used to find the area, where $s = \left(\frac{1}{2}\right)(a+b+c)$.

SCALE MODEL: A representation or copy of an object that is larger or smaller than the actual size of the object being represented; usually, it is a smaller replica. Examples are model buildings, architectural models, motor vehicle models, model railways, model ships, etc.

SCALER (SCALING CIRCUIT): An electronic counting circuit that provides an output when it has been activated by a defined number of input pulses. For example, a decade scaler produces an output when it has received ten or a multiple of ten input pulses. A binary scaler produces its output after two input pulses.

SCALING ANALYSIS: The analysis of a physical problem in which the terms in the equation describing the problem are expressed in powers of dimensionless units. This enables the orders of magnitude of terms to be estimated, allowing simplifications to be made by dropping very small terms. Scaling analysis is used extensively in theories describing phase transitions, disordered solids, polymers and turbulence.

SCALING CIRCUIT: An electronic circuit which produces an output pulse for each time a specified number of pulses has been received at its input.

SCALING FACTOR: A variational parameter used as a multiplier of each nuclear Cartesian and electronic coordinate chosen to minimise the variational integral and to make a trial variation function to satisfy the virial theorem. In practical calculations, the numeral factor to scale computed values, for example, vibrational frequencies, to those found in experiments.

SCALP: The skin of the head excluding the face or skin on top of a person's head, where hair usually grows.

SCAM: A term used to describe any fraudulent business or scheme that takes money or other goods from an unsuspecting person. With the world becoming more connected thanks to the Internet, online scams have increased, and it is often up to you to help stay cautious with people on the Internet.

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SCAN: 1. A term that describes the process of digitising an image, allowing it to be stored on or modified by a computer. This can be done by using a scanning device such as optical scanner. 2. The process where a software program reviews files for the purpose of detecting a virus attack and using an anti-virus program to heal the file by removing the virus.

SCAN RATE (REFRESH RATE): A CRT monitor measurement in Hz that indicates how many times per second a monitor screen image is renewed. It has a vertical and horizontal value. The higher the refresh rate, the less image flicker will be noticed on the screen. Typical refresh rates for CRT monitors include 60, 75 and 85 Hz. Some monitors support refresh rates of over 100 Hz.

SCANDENT: Having the ability to climb, used especially for plants.

SCANDIUM: A silvery-white metallic element belonging to the first row of transition metals in Group 3 (IIIB) of the Periodic Table with the symbol Sc, atomic number 21, relative density 2.99, relative atomic mass 44.956, melting point 1,540 °C (1813 K) and boiling point 2,850 °C (3123 K). At ordinary temperatures it crystallises in a hexagonal close-packed structure. It tarnishes slightly when exposed to air. It reacts with many acids. It is in various rare minerals and separated as a by-product in the processing of certain uranium ores. Its oxide (Sc_2O_3) is used as a catalyst and in making ceramics. An artificially produced radioactive isotope is used as a tracer in studies of oil wells and pipelines.

SCANNING ELECTRON MICROSCOPY (SEM): Any analytical technique which involves the generation and evaluation of secondary electrons (and to a lesser extent back scattered electrons) by a finely focused electron beam (typically or less) for high resolution and high depth of field imaging.

SCANNING PROBE MICROSCOPY (SPM): Any of several microscopic techniques that are based on measuring the interaction between a very sharp-tipped probe and the surface of the sample. The probe scans the sample surface in a systematic way and records the surface topography which can be processed by a computer to produce images. This is widely used in chemistry and now increasingly used in biology to study biomolecules and cell surfaces at the nanometre scale.

SCANNING TRANSMISSION ELECTRON MICROSCOPY (STEM): A special technique in which the electrons pass through a sufficiently thin specimen by focusing the electron beam into a narrow spot which is scanned over the sample in a raster. STEM essentially provides high resolution imaging of the inner microstructure and the surface of a thin sample (or small particles), as well as the possibility of chemical and structural characterisation of micrometre and sub-micrometre domains through evaluation of the X-ray spectra and the electron diffraction pattern.

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SCANNING TUNNELLING MICROSCOPE: A type of microscope for studying and imaging individual atoms on the surfaces of materials. A fine conducting probe is held close to the surface of a sample. Electrons travel between the sample and the probe, producing an electrical signal. The probe is slowly moved across the surface, raised and lowered so as to keep the signal constant. A profile of the surface is produced and a computer-generated contour map of the surface is produced. The technique is capable of resolving individual atoms, but works better with conductive materials. The instrument was invented in the early 1980s by Gerd Binnig and Heinrich Rohrer, who were awarded the 1986 Nobel Prize in physics for their work.

SCAPE: 1. The first segment in an insect's antenna, for example, as found in grain weevil. It has muscles and it pivots on a point called the antennifer, thus enabling the antenna to be moved in any direction. 2. The leafless stem of a solitary flower or inflorescence, such as that of the dandelion inflorescence; or the long leafless peduncle which grows directly from the ground, usually from a basal rosette.

SCAPHOID: Having the shape of a boat.

SCAPOSE (SCAPIFLOROUS): Having a solitary flower on a leafless peduncle or scape, usually arising from a basal rosette.

SCAPULA (SHOULDER BLADE): Either of two large, flat, triangular bones forming the back part of the shoulder of humans and many other vertebrates or a large bone forming part of the shoulder/pectoral girdle assisting in the articulation of the arm with the chest region.

SCAR: 1. A mark left on the skin after an injury such as a cut, a wound, sore, etc, after it has healed. 2. A mark left after a leaf falls from the stem or branch of a plant or the mark left on a seed after detachment from the placenta. 3. A rock submerged or partly submerged in the ocean.

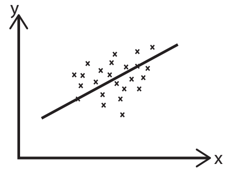
SCARECROW: A figure shaped like a human, wearing old cloths to scare away birds from crops.

SCARIFICATION: The abrasion or chemical treatment of the surface of a hard seed to make it permeable to water and so hasten germination.

SCARIFIER: A machine with tines, which stirs the soil surface without turning it over.

SCARIOUS: Thin, dry and membranous in texture and also not green, as certain bracts.

SCATTER DIAGRAM (SCATTER PLOT): A simple display graphically of the distribution of two random variables such as the observed distribution of salary and years of service in an organisation as a set of points, whose coordinates represent their observed paired values. One quantity is measured on the vertical axis, the other on the horizontal axis and each observation is represented by a dot.



A Scatter plot on a Cartesian plane

SCATTERING: In physics, it is a process in which a particle such as an electron, photon, or neutron collides with a material and changes energy and direction.

SCATTERING ANGLE (ANGLE OF OBSERVATION): The angle between the forward direction of the incident beam and a straight line connecting the scattering point and the detector.

SCATTERING OF ELECTROMAGNETIC RADIATION: The process in which energy is removed from a beam of electromagnetic radiation and reemitted with a change in direction, phase or wavelength. All electromagnetic radiation is subject to scattering by the medium such as gas, liquid or solid through which it passes.

SCATTERING PLANE: The plane containing the incident light beam and the line from the centre of the scattering system to the observer.

SCATTERING VECTOR: The vector difference between the wave propagation vectors of the incident and the scattered beam.

SCAVENGER: 1. Also known as **necrophagous**, it is any carnivorous predator that consumes corpses or carrions that are not killed to be eaten by the predator or others of its species. Scavengers play an important role in the ecosystem by contributing to the decomposition of dead animal remains. Examples are crows and vultures. Many large carnivores that hunt regularly, such as hyenas and lions, will scavenge if given the chance or use their size and ferocity to intimidate the original hunters. 2. A substance that reacts with (or otherwise removes) a trace component (as in the scavenging of trace metal ions) or traps a reactive reaction intermediate.

SCAVENGING: 1. Feeding on dead or decaying matter. 2. A binding radicals or free electrons with a receptive or reactive material. 3. The use of a precipitate to remove from solution by absorption or coprecipitation, a large fraction of one or more radionuclides. 4. The removal of pollutants from the atmosphere by natural processes, such as scavenging by cloud water, rainout and washout. This type of removal process is termed **precipitation scavenging**. Scavenging of airborne pollutants at the surfaces of plant, soil, etc, is termed **dry deposition**.

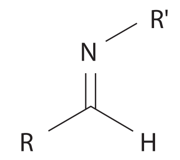
SCENT: Characteristic smells of something, especially a pleasant one or a smell produced by an animal which acts as a signal to other animals. Animals secrete them from sweat and sebaceous glands and special scent glands.

SCENT GLAND: A gland that opens into the outer surface of animals, producing odorous compounds that are used for communicating between members of the same species or for discouraging predators. Examples are sweat and sebaceous glands and special scent glands.

SCHEELITE: A mineral form of calcium tungstate, used as an ore of tungsten with the chemical formula $CaWO_4$ and Mohs scale hardness of 4.5-5. It occurs in contact metamorphosed deposits and vein deposits as colourless or white tetragonal crystals. It fluoresces bright bluish-white in ultraviolet radiation, a distinguishing feature utilised in prospecting and mining.

SCHEMA: In logic, an expression using meta variables that may be replaced by expressions of the object language to yield a well-formed formula; thus A = A is an axiom schema for identity, yielding the infinite set of axioms $x = x$, $y = y$, $z = z$, etc.

SCHIFF'S BASES: Organic compounds formed when an aldehyde or ketone condenses with a primary aromatic amine with the elimination of water. For example, $RNH_2 + R'CHO \rightarrow RN{:}CHR' + H_2O$. The compounds are often crystalline and are used in organic chemistry for characterising amines, chemical intermediates and perfume bases, in dyes and rubber accelerators and in liquid crystals for electronics.



Chemical Structure of Schiff's Base

SCHIFF'S REAGENT: A reagent used for testing for the presence of aldehydes and ketones. It consists of a solution of fuchsin dye that has been decolourised by sulfur dioxide. Aliphatic aldehydes restore the pink immediately, whereas aromatic ketones have no effect on the reagent. It is prepared by dissolving rosaniline hydrochloride in water and passing sulfur dioxide through it until the magenta colour is discharged.

SCHILLER LAYERS (IRIDESCENT LAYERS): In some systems, sedimenting particles form layers separated by approximately equal distances of the order of the wavelength of light. This gives rise to strong colours, when observed in reflected light and the system is said to form iridescent layers or Schiller layers.

SCHIMDT CAMERA (SCHMIDT TELESCOPE): A wide-field telescope that uses a thin aspheric front lens and a larger concave spherical mirror to focus the image. It was devised by Estonian optician, Bernhard Schmidt in 1930. Schmidt telescopes are very fast, some having focal ratios in the vicinity of *f/1*. Their distinguishing feature is the very wide angle over which good images are obtained.

SCHIST: A rock whose minerals have been aligned in one direction in response to deformation stresses, with the result that the rock can be split in parallel layers. They are a group of coarse-grained metamorphic rocks characterised by the presence of platy minerals such as micas, chlorite, talc, horn blende and graphite.

SCHISTOSOMIASIS (BILHARZIA): A parasitic disease caused by several species of parasitic flatworms of the genus *Schistosoma* (blood flukes), which require freshwater snails and mammals to complete their life cycle. While in the snail host it multiplies and produces larvae which are released into the water, where they penetrate human skin. The larvae then grow in the human host into mature forms and locate in veins around the intestines or bladder. The parasites can be released with urine into freshwater to begin the life cycle. The most common effect on humans is severe inflammatory reaction in the walls of the bladder and haemorrhage. An initial allergic reaction (inflammation, cough, late-afternoon fever, hives, liver tenderness) and blood in the stools and urine give way to a chronic stage, in which eggs impacted in the walls of organs cause fibrous thickening (fibrosis). This condition can lead to serious liver damage in the intestine and to bladder stones, fibrosis of other pelvic organs and urinary-tract bacterial infection. In most cases, early diagnosis and persistent treatment to kill the adult worms ensure recovery.

SCHIZOCARP: A dry indehiscent fruit formed from carpels that develop into separate one-seeded fragments called mericarps which may be dehiscent as in the regma or indehiscent as in the cremocarp and cercerulus.

SCHIZOGENY: The localised separation of plant cells to form a cavity in which secretions accumulate. Examples are the resin canals of some conifers and the oil ducts of caraway and aniseed fruits.

SCHIZOPETALOUS: Having cut petals, as seen in *Hibiscus schizopetalus*.

SCHIZOPHRENIA: A serious mental illness characterised by a disintegration of the process of thinking and of emotional responsiveness. Symptoms include hallucination, paranoia, disorganised speech, delusions, withdrawal from social relationships, etc.

SCHLICHT FUNCTION (SIMPLE FUNCTION; UNIVALENT FUNCTION): In mathematics, a complex function on some domain, that is analytic and taking no value more than once in the domain. A schlicht function mapping the finite complex plane onto itself is linear.

SCHLIEREN PHOTOGRAPHY: A technique that enables density differences in a moving fluid to be photographed. Short lived localised differences in density create differences of refractive index which show up on photographs taken by short flashes of light at streaks. It is used in wind-tunnel studies to show the density gradients created by turbulence and the shock waves around a stationery model.

SCHLOMILCH'S FORM OF THE REMAINDER: In mathematics, a form of the remainder in a Taylor series that includes the Cauchy and Lagrange form of te remainder as special cases.

SCHOENFLIES SYSTEM: A system for categorising symmetries of molecules, in which C_n groups contain only an *n*-fold rotation axis. C_{nv} groups, in addition to the *n*-fold rotation axis, have a mirror plane that contains the axis of rotation (and mirror planes associated with the existence of the *n*-fold axis). C_{nh} groups, in addition to the *n*-fold rotation axis, have a mirror plane perpendicular to the axis. S_n groups have an *n*-fold rotation–reflection axis.

SCHOOL (SHOAL): A large number of fish, whales, etc. swimming in a group.

SCHOTTKY DEFECT: A defect in an ionic crystal consisting of the smallest number of positive-ion vacancies and negative-ion vacancies which leave the crystal electrically neutral. It is named after Walter H. Schottky (1886-1976), a German physicist. The defect forms when oppositely charged ions leave their lattice sites, creating vacancies. These vacancies are formed in stoichiometric units, to maintain an overall neutral charge in the ionic solid. The vacancies are then free to move about as their own entities. Normally, these defects will lead to a decrease in the density of the crystal.

SCHRIER REFINEMENT THEOREM: In mathematics, the theorem that any two normal series of a group have isomorphic refinements.

SCHRODER-BERNSTEIN THEOREM (CANTOR-BERNSTEIN THEOREM): In mathematics, the theorem that establishes that two sets are equipollent if there is an injective mapping from each to the other.

SCHRÖDINGER EQUATION: A partial differential equation, formerly called the wave equation, developed in 1926 by Erwin Schrödinger (1887-1961), Austrian physicist that established the mathematics of quantum mechanics. The equation determines the behaviour of the wave function that describes the wavelike properties of a subatomic system. It relates kinetic energy and potential energy to the total energy and it is solved to find the different energy levels of the system. Schrödinger applied the equation to the hydrogen atom and predicted many of its properties with remarkable accuracy. The equation is used extensively in atomic, nuclear and solid-state physics.

SCHRÖDINGER, ERWIN (1887-1961): Austrian physicist, who became professor of physics at Berlin University in 1927. He made fundamental contributions to quantum mechanics and he shared the 1933 Nobel Prize with P.A.M. Dirac for his development in 1926 of the wave equation, now called the Schrödinger equation.

SCHRODINGER'S CAT: A thought experiment, devised by Austrian physicist, Erwin Schrödinger in 1935 to illustrate the paradox in quantum mechanics regarding the probability of finding, for example, a subatomic particle at a specific point in space.

SCHULTZES SOLUTION (CHLOR-ZINC IODIDE; CZI): A temporary stain containing a mixture of zinc dissolved in hydrochloric acid added to iodine, dissolved in potassium iodide. It is used to detect the presence of cellulose in plant tissue, which it turns blue. Lignified walls stain blue-green, lignin and suberin yellow and starch blue-black.

SCHUR COMPLEMENT: In mathematics, the quantity related to a given partitioned matrix by $D = B_4 - B_3 B_1^{-1} B_2$ when the original matrix is given as $\begin{bmatrix} B_1 & B_2 \\ B_3 & B_4 \end{bmatrix}$ with B_1 invertible and B_4 square.

SCHUR'S LEMMA (SCHUR'S THEOREM): In mathematics, the result that any square matrix is in unitary equivalence with an upper triangular matrix of which the diagonal entries are the eigenvalues of the original matrix. From this it follows easily that a normal matrix is unitarily equivalent to a diagonal matrix.

SCHWANN CELL (NEURILEMMA CELL): A cell that forms the myelin sheath of nerve fibres. Each cell is responsible for a given stretch along any particular axon and typically serves several neighbouring axon simultaneously. During its development the cell wraps itself around the axon so the sheath consists of concentric layers of Schwann cell membrane.

SCHWARZ INEQUALITY: In mathematics, the Cauchy-Schwarz inequality and in particular, its complex integral version.

SCHWARZ'S LEMMA: In mathematics, the consequence of the maximum principle that an analytic function that maps the set of complex numbers z such that $|z| < 1$ into itself and is zero at zero, either is a rotation or satisfies $|f(z)| < |(z)|$ in the punctured disk and has $|f'(0)| < 1$.

SCHWARZSCHILD RADIUS: A critical radius of an object of a given mass that must be exceeded if light is to be emitted from that body. It is equal to $2GM/c^2$, where G is the gravitational constant, c is the speed of light and M is the mass of the body.

S

SCHWEIZER'S REAGENT: A solution made by dissolving copper(II) hydroxide in concentrated ammonia solution. It has a deep blue colour and is used as a solvent for cellulose in the cuprammonium process for making rayon.

SCHWINGER EFFECT: An effect that is said to occur in quantum electrodynamics in which electron-positron pairs are sucked out of a vacuum by an electric field. The effect was predicted by Julian Schwinger in 1951. The Schwinger effect can be thought of to be similar to field emission in metals.

SCIARD FLY (DARK-WINGED FUNGUS GNATS): A family of flies, commonly found in moist environments. They are known to be pests of mushroom farms and are commonly found in household plant pots and greenhouse pot plants. The larvae feed on fine root causing plants to wilt.

SCIATIC NERVE: The largest nerve in the body, which is a sensory and motor nerve that originates at the lower end of the spinal cord and runs down the middle of the back of the thigh. It carries sensory information from the leg to the central nervous system and controls the action of many muscles.

SCIATICA: Pain that is felt in the back, hip and outer side of the leg. Its causes include inflammation of the sciatic nerve or pressure of a displaced disc on a nerve root leading out of the lower spine.

SCIENCE: Any systematic field of study or body of knowledge that aims, through experiment, observation and deduction to produce reliable explanation of phenomena with reference to the material and physical world or the systematic study of anything that can be examined, tested and verified. Today, different branches of science investigate almost everything that can be observed or detected and science as a whole shapes the way we understand the universe, our planet, ourselves and other living things.

SCIENTIFIC INVESTIGATION: An investigation that involves the systematic application of concepts and procedures such as observation, asking a question, forming a hypothesis (a possible answer to a scientific question) based on scientific knowledge, testing the hypothesis, experimentation and research, drawing conclusions, analysis and sharing of data.

SCIENTIFIC LAW (PHYSICAL LAW): Principles based on empirical observations that is universally applicable in science. Examples are Ohm's law, Kepler's three laws of planetary motion (describing how planets orbit the Sun) and Newton's laws of motion. A scientific law can often be expressed as a mathematical statement, such as $V = IR$ (Ohm's law) and $E = mc^2$, which is Albert Einstein's equivalent equation for the physical principle that a measured quantity of energy is equivalent to a measured quantity of mass.

SCIENTIFIC METHOD: Strategies or uniform rules of procedure used in advancing knowledge by formulating a question, collecting data about it through observation and experiment and testing a hypothetical answer or the procedures by which scientific investigation should be made. It involves problem statement, hypothesis, experiment, observation, data collection, data analysis, conclusion, theory, etc.

SCIENTIFIC NOTATION (STANDARD FORM; STANDARD INDEX FORM): A system of representing very large or very small numbers which expresses them as multiples of ten to a given power. It is of the form N × 10^n, where N is a decimal number between 0 and 10 that is rounded off to a few decimal places and n is known as the exponent and is an integer. For example, the scientific notation of 10677 is 1.0677 × 10^4. If $n > 0$ it represents how many times the decimal place in N should be moved to the right. If $n < 0$, then it represents how many times the decimal place in N should be moved to the left. For example 5.34 × 10^3 represents 5340, where the decimal moved three places to the right) and 3.24 × 10^{-3} represents 0.00324 where the decimal moved three places to the left.

SCIENTIFIC THEORIES: Unifying scientific explanations for a broad range of hypotheses and observations that have been supported by scientific experiments, and repeatedly tested and confirmed.

SCINTILLATION: 1. A burst of luminescence of short duration caused by an individual energetic particle or a flash of light produced in a phosphor by an ionising event. 2. Rapid variation in the light of a celestial body caused by turbulence in Earth's atmosphere; a twinkling.

SCINTILLATION CAMERA (GAMMA CAMERA; ANGER CAMERA): A camera that gives a complete image of radionuclide distribution in a particular area of the human body in one exposure, in contrast to line-scanning techniques known as scintigraphy. The applications of scintigraphy include early drug development and nuclear medical imaging to view and analyse images of the human body or the distribution of medically injected, inhaled or ingested radionuclides emitting gamma rays.

SCINTILLATION COUNTER: A device for detecting and measuring radiation by means of tiny visible flashes produced by the radiation when it strikes a sensitive substance such as zinc sulfide, sodium iodide, various liquids and organic phosphors. The scintillating medium is usually either solid or liquid and is used in connection with a photomultiplier, which produces a pulse of current for each scintillation.

SCINTILLATION DETECTOR: A radiation detector using a medium in which a burst of luminescence radiation, produced along the path of an ionising particle, is quantified.

SCINTILLATION SPECTROMETER: A measuring assembly incorporating a scintillation detector and a pulse amplitude analyser. It is used for determining the energy *spectrum* of certain types of radiation.

SCINTILLATORS: Materials used for the measurement of radioactivity, by recording the radioluminescence. They contain compounds (chromophores) which combine high fluorescence quantum efficiency, a short fluorescence lifetime and a high solubility. These compounds are employed as solutes in aromatic liquids and polymers to form organic liquid and plastic scintillators, respectively.

SCION: A piece of plant which is grafted onto a rootstock or a detached shoot or twig containing buds from a woody plant.

SCISSILE: Capable of being cut smoothly or split easily, for example, a scissile mineral or a scissile peptide bond.

SCIUROID: 1. Belonging to or resembling the Sciuridae, a family of rodents. **Sciuridae** is a family of true squirrels including animals such as ground squirrels, marmots, chipmunks, flying squirrels and spermophiles. 2. Having the shape of the tail of a squirrel, as in some inflorescences.

SCLERA: Commonly called 'the white of the eye', it is the tough white opaque, outer coat of the eyeball, covered by the conjunctiva, a clear mucus membrane that helps lubricate the eye. It forms the supporting wall of the eyeball, and is continuous with the clear cornea. The sclera covers approximately the posterior five sixths of its surface and continuous anteriorly with the cornea and posteriorly with the external sheath of the optic nerve. The sclera is made up of three divisions: the **episclera**, loose connective tissue, immediately beneath the conjunctiva; **sclera proper**, the dense white tissue that gives the area its colour; and the **lamina fusca**, the innermost zone made up of elastic fibres.

SCLERANTHIUM: An achene or a small dry thin-walled fruit enclosed within a hardened calyx tube.

SCLEREID: A type of sclerenchyma cells found in the ground tissue that is shorter than a fibre. Its lignified walls typically contain branched pits. Sclereids are found in abundance in seed coats, nut shells and in pear.

SCLERENCHYMA: A plant tissue with very long, narrow, thick and liquefied cells usually tapering at both ends which gives mechanical strength to the plant. It is found in the stem and also in the midrib of leaves. The cell walls contain pits enabling the exchange of substances between the adjacent cells. Mature sclerenchyma cells may not remain alive. They take the form of fibres or sclereids.

SCLERITE: Any of the hard chitinous plates forming the exoskeleton of an arthropod.

SCLEROMETER: A device for measuring the hardness of a material by determining the pressure on a standard point that is required to scratch it or by determining the height to which a standard ball will rebound from it when dropped from a fixed height. The rebound type is sometimes called a **scleroscope**.

SCLEROPHYLL: A type of vegetation that has stiff, firm leaves which retain the stiffness even when wilted. It is the dominant plant form in certain hot dry areas, for example, in the Mediterranean region.

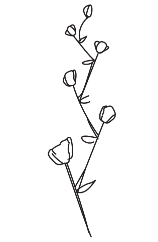
SCLEROPROTEIN: Any of a group of proteins found in the exoskeletons of some invertebrates, notably insects. They are formed by the conversion of relatively soft elastic larval proteins by natural tanning process involving orthoquinones. These are secreted and form cross linkages between polypeptides of the protein-producing, hard, rigid covering.

SCLEROSIS: 1. In medicine, it is the hardening and thickening of the body tissue as a result of unwarranted growth or from excessive formation of fibrous interstitial tissue, degeneration of nerve fibres or deposition of minerals, especially calcium. 2. In botany, it is the hardening and thickening of a plant cell wall that occurs as lignin is deposited, turning young green growth woody. 3. In zoology, a process which hardens animal exoskeletons by adding proteins such as resilin and sometimes mineral salts such as calcium carbonate.

SCLEROTIC: 1. The external protective layer of the vertebrate eyeball or relating to the dense outer coating of the eyeball that forms the white of the eye sclera. 2. Relating to the hardening and thickening of plant cell walls that turns young green growth woody. 3. Relating to or suffering from sclerosis of body tissue.

SCLEROTINA: Soil borne fungal disease affecting many crops, including potatoes, oilseed rape and pea.

SCLEROTIUM: Modified fungal hyphae that form a hard, dense, compact vegetative resting structure with a thick dark-coloured outer rind.



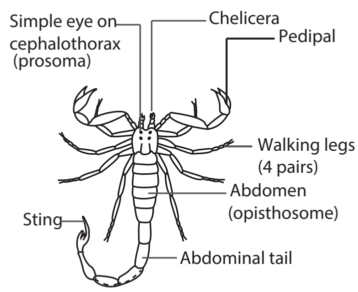
Scorpiod cyme

SCOBINA: The zigzag rachilla of some grass spikelets; the pedicel or immediate support of the spikelets of grasses.

SCOLEX: The head of a tapeworm, with hooks or suckers that enable the worm to attach itself to the digestive track of its host.

SCORPIOID CYME: A cyme with flowers or branches alternating in opposite ranks. An example is seen in the forget-me-not.

SCORPION: Any of some 1,300 arachnid species of the order Scorpionida, subphylum Chelicerata having a slender body, a segmented tail tipped with a venomous stinger and six pairs of appendages. They are found in warm and dry tropical regions. It is nocturnal and feeds mainly on spiders and insects.



Dorsal view of a Scorpion

SCOTOPIC VISION: The type of vision of the eye under low light conditions occurring when the rods in the eyes are the principal receptors and colour cannot be identified.

SCOTT'S TEST (COBALT THIOCYANATE TEST; RUYBAL TEST): A presumptive test for cocaine. It has an initial solution of 2% cobalt thiocyanate ($Co(SCN)_2$) in glycerine and water. This is followed by concentrated hydrochloric acid and then chloroform. A positive test is indicated by a blue colour in the chloroform layer.

SCOURS (DIARRHOEA): Excessive and frequent evacuation of watery faeces, usually indicating gastrointestinal distress or disorder caused by cholera, dysentery, etc.

SCRAPE NEST: A shallow depression without any soft lining made by some ground birds as a nest. Examples of birds that make such nests include ostriches, shorebirds, gulls, terns, falcons, pheasants, quails, partridges, bustards, nighthawks, vultures, etc. Usually the eggs laid in such nests are camouflaged to prevent detection by predators and the chicks are precocial, i.e., they leave the nests early.

SCRAPER: A steel-framed attachment for a tractor. It has a rubber scraping edge used for heavy duty work.

SCRAPIE: An infectious, fatal, chronic brain disease of sheep and goats. Affected animals suffer twitching and intense itching and thirst, grow lean and die.

SCRAPING (WEB SCRAPING): In computing, it is a process of extracting large amounts of information for a website. This may involve downloading several web pages or the entire site. The downloaded content may include just the text from the pages, the full HTML or both the HTML and images from each page.

SCREEN: 1. Upright, fixed, moveable or sometimes folding framework used for dividing a room, concealing something, protecting somebody from excessive heat, light, etc. 2. The broad flat end of a cathode-ray tube or liquid crystal display on which images are displayed such as in a television set or computer monitor.

SCREEN GRID: A grid placed between a control grid and an anode of an electron tube and usually maintained at a fixed positive potential, to reduce the electrostatic influence of the anode in the space between the screen grid and the cathode.

SCREEN SAVER (SCREENSAVER): On a computer, it is a program that is blanks the screen or fill it with moving images or patterns if the computer have been idle for a while. Screen savers were originally designed to help prevent images burning into the phosphor of older monitors. It is activated after the computer has been idle for a scheduled amount of time. You can voluntarily activate the screensaver by pressing the Windows key + L if the user is stepping away from the computer.

SCREEN SHOT (SCREEN DUMP; SNAPSHOT; SCREEN CAPTURE; SCREENSHOT): A picture taken of a computer's desktop. This may include the desktop background, icons of files and folders and open windows. Screenshot can be saved as an image file such as a GIF or JPEG, manipulated or printed. There are several ways to capture a screenshot, often each of these methods depend on what computer operating system is being used and what you intend to take a screenshot of. For example, when you use Microsoft Windows, the simplest way to take a screenshot is to press the print screen key on the keyboard. This takes a screenshot of the full screen and places it into the computer clipboard. Once in the clipboard, paste that screenshot into Microsoft Paint or other image editor.

SCREENING: 1. Investigation of a great number of something such as people, looking for those with particular problem or feature. Example is the screening of bags by x-ray at airports to detect those which contain weapons. 2. Preventing access of something by some sort of barriers. 3. In chemistry, screening effect or atomic shielding is the reduction of effective nuclear charge by intervening electron shells. 4. A process used in cleaning paper pulp, when manufacturing paper. 5. A mechanical method of separating a mixture of solid particles into fractions by size. The mixture to be separated, called the feed, is passed over a screen surface containing openings of definite size. Particles smaller than the openings fall through the screen and are collected as undersize. Particles larger than the openings slide off the screen and are caught as oversize. A single screen separates the feed into only two fractions. Two or more screens may be operated in series to give additional fractions. Screening occasionally is done wet, but most commonly it is done dry. 6. A screening test is a procedure that is performed to detect the presence of a specific disease. The individual or group of individuals does not present any symptoms of the disease.

SCREW: 1. A simple machine consisting of continuous groove or thread around the outside cylinder or cone; or a cylindrical or tapering piece of metal or plastic (or formerly wood) with a helical groove cut into it or a metal pin with a slot or cross cut into its head and a spiral groove around its shaft that can be turned or forced into wood, metal, etc., so as to fasten and hold things together. Each turn of a screw moves it forward or backwards by a distance equal to the pitch (spacing between machine threads). The mechanical advantage of a screw is $2\pi r/p$, where r is the radius of the thread and p is the pitch, i.e. the distance between adjacent threads of the screw measured parallel to the axis. The screw is by far the most useful form of inclined plane or wedge and finds application in the bolts and nuts used to fasten parts together; in lead and feed screws used to advance cutting tools or parts in machine tools; in screw jacks used to lift such objects as automobiles, houses and heavy machinery; in screw-type conveyors used to move bulk materials; and in propellers for airplanes and ships. 2. A symmetry element in a crystal lattice that consists of a combination of a rotation and a translation.

SCREW WORM: A fly similar to the blue bottle but dark green in colour. Screwworms are parasites that can cause great damage to domestic livestock and other warm blooded animals. The larvae of this pest enter open wounds of the host animal and feed on the raw flesh.

SCRIPT: A list of commands that are executed by a certain program or scripting engine. Scripts may be used to automate processes on a local computer or to generate Web pages on the Web. For example, DOS scripts and VB Scripts may be used to run processes on Windows machines, while AppleScript scripts can automate tasks on Macintosh computers. ASP, JSP and PHP scripts are often run on Web servers to generate dynamic Web page content.

SCROLL BAR: A vertical or horizontal bar commonly located on the far right or bottom of a window that allows a user to move the window viewing area up, down, left or right. Most users today are familiar with scroll bars because of the need to scroll up and down in almost every Internet web page.

SCROLL WHEEL (INTELLIMOUSE; SCROLL MOUSE): A mouse developed by Microsoft in 1996 that has a wheel between the left and right mouse buttons that enables the user to scroll up and down using the wheel instead of having to use the vertical scroll bar. The wheel is raised slightly, which allows the user to easily drag the wheel up or down using the index finger.

SCROTUM: The sac of skin and tissue that contains and supports the testes in most mammals. The scrotum lies outside the body and is therefore kept at a lower temperature needed for the formation of sperms. For human beings, the temperature should be one or two degrees lower than the body temperature of 37°C. Higher temperatures may damage sperm count.

SCRUB: 1. Small trees and bushes. 2. An area of land covered with small trees and bushes.

SCRUBBER: In chemistry, it is an apparatus used in sampling and in flue gas cleaning. The gas is passed through a space containing wetted 'packing' or spray. In general, particles are collected in scrubbers by one or a combination of the following: (i) impingement of particles on a liquid medium; (ii) diffusion of the particles onto a liquid medium; (iii) condensation of liquid medium vapours on the particles; and (iv) partitioning of the gas into extremely small elements to allow collection of the particles by Brownian diffusion and gravitation settling on the gas-liquid interface. The devices include spray towers, jet scrubbers, Venturi scrubbers, cyclonic scrubbers, inertial scrubbers, mechanical scrubbers and packed scrubbers. Normally the gas flow in the scrubber is counter to the liquid flow. Efficient scrubbers will collect particles as small as 1 to 2 μm in diameter.

SCRUBBING: 1. The process of selectively removing contaminating solutes (impurities) from an extract that contains these as well as the main extractable solute by treatment with a new immiscible liquid phase. The term stripping has a different meaning and should not be used in this sense although this usage has been customary in certain industries. 2. A process used in gas sampling or gas cleaning in which components in the gas stream are removed by contact with a liquid surface or a wetted packing, on spray drops, droplets or in a bubbler, etc.

SCSI (SMALL COMPUTER SYSTEM INTERFACE): In computer science, it is one of the methods developed by the American National Standards Institute (ANSI) for connecting peripheral devices such as printers drives, scanners drives, tape drives, CD-ROM drives to a computer. A group of peripherals linked in series to a single SCSI port is called a **daisy-chain**.

SCUBA (SELF-CONTAINED UNDERWATER BREATHING APPARATUS; AQUALUNG SCUBA): A device that allows divers to breathe and move about freely underwater for extended periods. It consists of a mouthpiece, a connecting tube, a valve and at least one compressed-air cylinder. The key component is the demand valve which allows the diver to breathe air at the pressure prevailing in the surrounding water however great the pressure in the supply cylinder. Used air is vented into the water.

SCURS: Small horns that are not part of an animal's skull but are attached to the skin.

SCURVY: 1. A nutritional disorder caused by deficiency of vitamin C (ascorbic acid), which is contained in fresh vegetables and fruit. The deficiency interferes with tissue synthesis, causing spongy gums, loosening of the teeth, bleeding into the skin and mucous membranes, aching joints and muscles, progressing to bleeding of the gums and then other organs and drying up of the skin and hair. 2. Covered with tiny, scale-like particles.

SCUTE: A large, dermal bony or horny plate such as turtle's shell or the shell on the back of a crocodile, found on many reptiles or some fishes.

SCUTELLUM: 1. Scale or hard plate, such as on a toe of a bird or the thorax of an insect. 2. The shield-shaped embryonic leaf or cotyledon of a grass seed or the tissue in a grass seed that lies between the embryo and the endosperm. It is the modified cotyledon of grasses being specialised for the digestion and absorption of stored food material on the endosperm.

SCYTHE: A hand implement with a long slightly curved blade with a sharp edge on the inner side of the curve. It is fitted with a long wooden handle and used for mowing grass or reaping crops.

SDR-RAM (SYNCHRONOUS DYNAMIC RANDOM ACCESS MEMORY, SDRAM): A type of dual in-line memory module (DIMM) that synchronises itself with the computer's system clock to provide synchronisation between the memory and the computer processor. DIMM is a small circuit board that holds semiconductor memory chips with two independent rows of input/output contacts. Synchronisation allows the memory to run at higher speeds than previous memory types.

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SDSL (SYMMETRIC DIGITAL SUBSCRIBER LINE): A technology, which is used for transferring data over existing copper telephone lines. It has a transfer speed of 1.544 Mbps for both the upstream and downstream data transmissions. SDSL is called symmetric because it supports the same data rates for upstream and downstream traffic. A similar technology that supports different data rates for upstream and downstream data is called **asymmetric digital subscriber line** (**ADSL**).

SEA: The great body of salt water that covers a large portion of Earth or a body of salt water that is surrounded by land on all or most sides or forming part of an ocean. An example is the Mediterranean Sea.

SEA ANEMONE: A tube-shaped sea animal with petal-like tentacles round its mouth or a solitary and often colourful sea animal with a squat cylindrical body topped by a ring of tentacles that attaches itself to rock or other nonliving material. Anemones may be very small or may be more than 90 cm in length or diameter.

SEA BREEZE: A gentle coastal wind blowing off the sea towards the land that occurs during the night. It is most noticeable in summer when the warm land surface heats the air above it and causes it to rise to be replaced by the colder air from the sea.

SEA FLOOR: The floor of the oceans. The major features of the seafloor are the continental shelf, the continental rise, the abyssal floor, seamounts, oceanic trenches and mid-ocean ridges. The abyssal or deep ocean floor is about 3 km deep and is mostly made of basaltic rock covered with fine-grained (pelagic) sediment consisting of dust and the shells of marine organisms.

SEA-FLOOR SPREADING: The process by which new oceanic crust (sea floor) forms as magma rises to Earth's surface and solidifies at a mid-ocean ridge.

SEA-LEVEL RISE: An increase in the mean level of the ocean. Eustatic sea-level rise is a change in global average sea level brought about by an alteration to the volume of the world ocean. Relative sea-level rise occurs where there is a net increase in the level of the ocean relative to local land movements. Climate modellers largely concentrate on estimating eustatic sea-level change. Impact researchers focus on relative sea-level change. Sea-level rise is a major effect of climate change. Higher sea levels may cause serious impacts in various parts of the world, especially in countries which are located in low-lying areas as well as small islands.

SEA-LEVEL: The level of the surface of the ocean relative to the land, halfway between low and high tide, used as a standard in calculating elevation. The sea level at any location changes constantly with changes in tides, atmospheric pressure and wind conditions. Longer-term changes are influenced by changes in the Earth's climates. Consequently, the level is better defined as mean sea level, the height of the sea surface averaged over all stages of the tide over a long period of time.

SEABORGIUM (UNNILHEXIUM): An artificially produced radioactive element with atomic number 106 that is produced by bombarding Californium-249 nuclei with oxygen-18 nuclei, chemical symbol Sg. It is named after the scientist Glenn Seaborg.

SEALED RADIOACTIVE SOURCES: Small metal containers in which a specific amount of a radioactive material is sealed in order to prevent the escape of the radiation.

SEAMOUNT: An isolated steep-sided hill up to 1 000 metres tall rising from the ocean seafloor that does not reach to the water's surface. Most are conical in shape and volcanic in origin. In recent years, several active seamounts have been observed, for example Loihi in the Hawaiian Islands.

SEARCH: In computer science, it is a term used to describe the process of finding information and web pages. A number of operating systems, software programs, and websites contain a search, seek or find feature to locate data. For example, when on the Internet, you use a search engine (e.g., Google) to search through web pages.

SEARCH COIL: A small coil in which current can be induced to detect and measure a magnetic field. A fluxmeter is used in conjunction with the coil.

SEARCH ENGINE: 1. In computer science, it refers to a software program or script available through the Internet that searches documents and files for keywords and returns the results of any files containing those keywords. 2. A website whose primary function is to provide a means for gathering and reporting information available on the Internet or a portion of the Internet. Examples of Search Engines are Google, Bing and Yahoo.

SEARLE'S BAR (SEARLE'S BAR METHOD): Equipment for determining the thermal conductivity of a bar material by lagging the bar while one end is heated and the other cooled by cold water and steam respectively. The temperature is measured along the bar, from which the conductivity is measured.

SEASON: 1. The oestrus period of female animals or the period when they are ready to accept a male for mating. 2. Period of the year having a characteristic climate. The change in season is mainly due to the change in position of the earth's axis in relation to the sun. Tropical regions have two seasons – the wet and the dry whilst the temperate regions have four seasons which are recognised as spring, summer, autumn (fall) and winter.

SEAWALL (GROYNE; GROIN): A massive human-made wall or embankment along a shore to prevent wave erosion and other damage due to wave action and storm surge, such as flooding. Seawalls are normally constructed with concrete, steel and stone, or by placing large boulders in front of a cliff or sea.

SEAWATER: The water of the seas and ocean covering about 70% of the Earth's surface and comprising about 97% of the world's water. It contains large amounts of dissolved solids, the most abundant of which is sodium chloride. On average, seawater in the world's oceans has a salinity of about 3.5%. This means that every kilogramme or every litre, of seawater has approximately 35 grams of dissolved salts. The average density of seawater at the ocean surface is 1.025 g/ml. Seawater is denser than freshwater because of the salts' added mass. The freezing point of sea water decreases with increasing salinity and is about -2 °C (271 K) at 35 g/L (equivalent to 599 mM).

SEAWEEDS: Large multicellular algae living in the sea or in the intertidal zone. They are common species of Chlorophyta, Phaeophyta and Rhodophyta. Seaweeds are used as food and brown algae are used in fertilisers. The red alga *Gelidium* is used to make the gelatin-like product called agar.

SEBACEOUS GLAND: A tiny oil-producing gland that is commonly found in human skin, except in the soles of the feet and in the palms of the hands. Its ducts open into a hair follicle through which it discharges sebum onto the skin surface.

SEBUM: Oily secretion from the sebaceous glands consisting chiefly of fat, keratin and cellular material, that protects, lubricates and water-proofs skin and hair and helps prevent drying up or desiccation.

SEC: Symbol for the hyperbolic function hyperbolic secant, the reciprocal of the hyperbolic cosine cosh. Its derivative is -sech$x$tanh$x$ and an antiderivative or indefinite integral is $\tan^{-1}|\sinh x|$.

SECANT: 1. Abbreviated sec, in geometry, it is a straight line cutting a curve or surface. If it intersects the curve in two different points, as in the secant of a circle, the segment of the secant between the points is called a chord. The limiting position of a secant is called the tangent to the curve or surface at that point. 2. The **secant** sec A is the reciprocal of cos A or the ratio of the length of the hypotenuse to the length of the adjacent side: $\sec A = \frac{1}{\cos A} = \frac{hypotenuse}{adjacent}$.

SECANT METHOD (METHOD OF LINEAR INTERPOLATION): A variation of Newton's method for finding a root of the nonlinear equation $f(x) = 0$. It is given by

$$x_{n+1} = x_n - f(x_n)\left[\frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})}\right],\ n \geq 1.$$

SECATEURS: A tool which has two sharp blades used for cutting plant stems.

SECO-: Cleavage of a ring with addition of one or more hydrogen atoms at each terminal group thus created is indicated by the prefix 'seco'.

SECOND: 1. The basic SI unit of time with the symbol s. There are 60 seconds in one minute and 3,600 seconds in one hour and 86,400 seconds in one day. 2. Also known as **second of arc**, it is the unit of angular measurement in mathematics, which is equal to one-sixtieth of a minute and which in turn, is equal to one-sixtieth of a degree.

SECOND CLASS LEVER: A type of a lever in which the load is situated between the effort and the pivot (fulcrum). Examples are nutcrackers and wheelbarrows. The force arm is longer than the work-producing arm; thus the work produced is always greater than the energy used, with a resultant high efficiency.

SECOND DERIVATIVE: In mathematics, it is the derivative from a given function by differentiating its first derivative; written as $\frac{d^2t}{dx^2}$. There are also higher derivatives such as third derivatives.

SECOND DERIVATIVE TEST: A test for the optimality of a critical point of a function to determine if a critical point of a function is a local minimum, maximum or saddle point. For a function of one variable, the test of the second derivative at the point is a **local minimum** if it is positive; if it is negative it is a **local maximum**; and if it is zero, it is known as **indeterminate**. For a function of several variables the test is to determine if the Hessian at the point is positive definite or a local minimum; negative definite or a local maximum; indefinite, called a saddle point; or singular, called indeterminate. If the determinant of the Hessian at the point is negative the point is a saddle point.

SECOND DIAGONAL (SUPERDIAGONAL): In a matrix, the line of elements lying immediately above the diagonal, of the form a_{i+1}. A **superdiagonal matrix** is a matrix in which all the entries are zero except in the superdiagonal.

SECOND FILIAL GENERATION (F2): The offspring resulting from crosses between individuals of the first filial generation (F1).

SECOND GENERATION BIOFUEL (ADVANCED BIOFUEL): Fuel that can be derived from plant materials that do not have an alternative use as food. Such materials include wastes, agricultural and forestry residues, and dedicated energy crops, using a different technology. Second generation biofuel has been developed as first generation biofuel is derived from the main crop and cannot produce enough biofuel without threatening food supplies and biodiversity. The end product of second generation biofuel such as advanced bioethanol or advanced biodiesel is the same as that produced by first generation technology.

SECOND GENERATION CELLULAR TECHNOLOGY (2G): Telecommunication network technologies that succeeded first Generation (1G) which was analogue. Its notable features include digital encryption of telephone conversations, and considerably higher efficiency on the spectrum, which allows for greater penetration level for mobile phones. It also introduced mobile data services, beginning with SMS text messaging.

SECOND IONISATION ENERGY: The amount of energy required to remove one mole of electron from one mole of singly charged positive ions to form doubly charged positive ions. The energy for the removal of an electron from a neutral atom other than hydrogen is more designated as the first ionisation energy.

SECOND LAW OF THERMODYNAMICS: It states that the universe tends towards a state of greater disorder in spontaneous process or all processes occur spontaneously in the direction that increases the total entropy of the universe. There are many versions of the second law, but they all have the same effect, which is to explain the phenomenon of irreversibility in nature.

SECOND LIFE: An online 3D virtual world platform where users can socialise, connect and create using free voice and text chat. It was launched on June 23, 2003 by the San Francisco-based firm Linden Lab.

SECOND MESSENGER: A chemical within a cell that is responsible for initiating response to a signal from a chemical messenger that cannot enter the target cell itself, for example, because it is not lipid-soluble and is therefore unable to cross the plasma membrane. A common second messenger is cyclic AMP. These molecules trigger a cascade of events by activating other cellular components. The pathways involve cascades of sequential molecular activation steps that are organised into three major components: (i) a receptor that recognises and binds agonists; (ii) second messengers or signal transducing molecules that couple receptors to intracellular pathways; and (iii) effectors or molecules responsible for the ultimate response. A central feature of all signalling cascades is that they discriminate

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among a variety of signals and provide a mechanism for signal amplification.

SECOND-COUNTABLE SPACE: In mathematics, topological space that has a countable base. A metric space is second-countable if and only if it is separable. For example, the usual topology of Euclidean space is second-countable.

SECOND-KIND INDUCTION (COMPLETE INDUCTION; GENERAL INDUCTION): A mathematical induction in which the step is from all integers less than k to all less than $k + 1$, rather than from the single integer n to $k + 1$.

SECOND-ORDER: 1. In chemistry, **second-order reactions** are chemical reactions in which the rate of reaction is proportional to the concentration of the square of a single reactant rate: $r = k[A]^2$, or to the product of the concentration of two reactants: $r = k[A][B]$. Second-order reactions are of two types: Second-order reactions of Classes I and II. Second-order reactions of Classes I reactions are reactions in which the rate varies with concentration of a single species, but the stoichiometric coefficient is 2. The rate varies with the reciprocal of the concentration, so that a plot of 1/(concentration) against time is linear. Second-order reactions of Class II are reactions in which the rate varies with the concentration of two substrates, each of which has a stoichiometric coefficient of 1. A plot of ln(concentration A/concentration B) against time is linear. Examples of second-order reaction are found in many important biological reactions, such as the formation of double-stranded DNA from two complementary strands. 2. The second derivatives; for example, a quadratic term in a polynomial is second-order if one considers the polynomial as a Taylor series. 3. The first and second derivatives of an ordinary differential equation of the dependent variable with respect to the independent variable. 4. The partial differential equation with no partial derivative of an order greater than 2.

SECOND-ORDER TRANSITION (CONTINUOUS TRANSITION): A transition in which a crystal structure undergoes a continuous change and in which the first derivatives of the Gibbs energies (or chemical potentials) are continuous but the second derivatives with respect to temperature and pressure (i.e. heat capacity, thermal expansion, compressibility) are discontinuous. Example is the order-disorder transition in metal alloys, for example, CuZn.

SECONDARY: 1. Happening as a result of another or primary condition or disorder, for example secondary infection. 2. In chemistry, a compound characterised or formed by replacement of two atoms or radicals within a molecule. 3. In geology, produced from another mineral by decay or alteration. 4. In botany, of relating to or derived from a lateral meristem, especially a cambium. 5. Of second ranking; not primary or main, inferior or minor.

SECONDARY ATMOSPHERE: An atmosphere produced by the outgassing of volatile compounds from the rocky material of a terrestrial-type planet. Venus, Earth and Mars, for example, all possess secondary atmospheres.

SECONDARY CELL: An electrochemical cell in which the chemical reaction producing the e.m.f. is reversible and the cell can therefore be recharged by passing a current through it. An example is lead-acid cell.

SECONDARY COLOUR: Any colour that can be obtained by mixing two primary colours. Secondary colours of light are sometimes referred to as the pigmentary primary colours. For example, if beams of red light and green light are made to overlap, the secondary colour, yellow, will be formed.

SECONDARY COMPOUNDS (SECONDARY METABOLITIES): Plant products that are not directly involved in the normal growth, development or reproduction of organisms. Absence of secondary metabolities does not result in immediate death, but rather in long-term impairment of the organism's survivability, fecundity or aesthetics or perhaps no significant change at all. They often play an important role in plant defence mechanisms against herbivores and other interspecies defences.

SECONDARY CONSUMER: In a food chain, an organism that feeds on the primary consumers. The primary consumers, which themselves feed on green plants (producers), are predominantly herbivores. Secondary consumers are, therefore, mainly carnivores that feed on the herbivores. If a secondary consumer has no predators (that is, there are no tertiary consumers), it is known as the top carnivore.

SECONDARY CORTEX (PHELLODERM): Parenchyma tissue of secondary origin, derived centripetally from the phellogen. Cells of the phelloderm may be distinguished from those of the cortex by their arrangement in radial columns, reflecting their origin from the phellogen.

SECONDARY CRYSTALLISATION: Crystallisation occurring after primary crystallisation, usually proceeding at a lower rate.

SECONDARY DIAGONAL (OFF DIAGONAL): A set of entries in a square matrix that runs from the upper right to lower left elements. Alternatively, any other related sequence of elements of a matrix, such as the elements above the main diagonal.

SECONDARY ELECTRON MULTIPLIER: A device that multiplies current in an electron beam (or in a photon or particle beam by first conversion to electrons) by incidence of accelerated electrons upon the surface of an electrode which yields a number of secondary electrons greater than the number of incident electrons. These electrons are then accelerated to another electrode (or another part of the same electrode), which in turn emits further secondary electrons so that the process can be repeated. It is recommended that one should refer to the abundance of an ion, to the intensity of an ion beam and to the height or area of a peak.

SECONDARY ELECTRON YIELD: The number of secondary electrons generated per primary electron for a given specimen and experimental conditions. It depends on the (mean) atomic number of the excited area of the sample, the angle between electron beam and sample surface, the primary electron energy, thickness of the sample and sample potentials.

SECONDARY ELECTRONS (SE): All electrons emitted from the surface of a solid except back scattered primary electrons. In practice, electrons emitted from the surface of a solid under particle bombardment which have a kinetic energy of less than 50 eV.

SECONDARY EMISSION: The emission of electrons or ions from the surface of a substance bombarded with ions or electrons. Most commonly used secondary emissive materials include alkali antimonide, beryllium oxide (BeO), magnesium oxide (MgO), gallium phosphide (GaP), gallium arsenide phosphide (GaAsP) and lead oxide (PbO).

SECONDARY ENERGY: Energy that has undergone some conversion by man before it can be used. For example, petrol is derived from the refinement of crude oil and electric energy is obtained from the conversion of mechanical energy in the form of hydroelectric and wind plants. In contrast, **primary energy** is a natural resource such as coal, crude oil, sunlight, and uranium that has not undergone any conversion or transformation by man and can be used directly, as it appears in the natural state.

SECONDARY EXTINCTION: 1. In ecology, it is the death of one population due to the extinction of another, often a food species. Species are bound together in ecological communities to form a food web of species interactions. Once a species is lost, those species that fed on it, were fed on by it or were otherwise benefited or harmed by that species will all be affected, for better or worse. These species, in turn, will affect yet other species. 2. In physics, it is the increased absorption or decreased diffraction of x-rays by a crystal lattice, due to previous reflection of the x-rays from suitably placed crystal planes.

SECONDARY FLIGHT FEATHERS: Short, innermost stiff, long flight feathers close to the body of a bird, positioned just behind the primary flight feathers on the rear edge of the wings. Many birds have six secondary feathers. Primary, secondary and tertiary flight feathers are collectively known as **remiges**.

SECONDARY FLUORESCENCE: Ionisation of the analyte element in a sample by characteristic radiation from other elements in the sample resulting in an enhancement of the signal measured.

SECONDARY GROWTH (SECONDARY THICKENING): Increase in girth or diameter of plant shoots and roots of plants as a result of the production of new cells by the vascular cambium. It occurs in most dicotyledons and gymnosperms but not in monocotyledons. The tissues produced by secondary growth are called **secondary tissues** and the resultant plant is the secondary plant body.

SECONDARY HOST: A plant or animal which carries a disease or parasite for a short transition period during part of the disease or parasites' life cycle, where it develops and eventually gets transmitted into the primary host or victim. For example, the pig is the secondary host for tapeworm, which eventually gets transferred into the gut of man and many vertebrate animals. For trypanosomes, the cause of sleeping sickness, humans are the primary host, while the tsetse fly is the secondary host.

SECONDARY IMMUNE RESPONSE (SECONDARY IMMUNITY): Resistance to an antigen the second time it appears because of the presence of long lasting antibodies and very long-lived lymphocytes that arise during proliferation and differentiation of antigen-stimulated B cells and T cells. Any new encounter with the same antigen results in a rapid proliferation of memory cells. The second response is faster and more massive and lasts longer than the primary immune response. For example, if someone had recovered from flu or chicken pox and later encounters the same strain of that virus, antibodies that were made specifically for that antigen will rise dramatically without delay.

SECONDARY INFECTION: An infection that is acquired during the course of a separate initial infection. Examples are vaginal yeast infection that occurs after antibiotic treatment of a bacterial infection; and bacterial pneumonia after a viral upper respiratory infection.

SECONDARY IONISATION: The process in which ions are ejected from a sample surface (which may be a solid or substrate dissolved in a solvent matrix) as a result of bombardment by a primary beam of atoms or ions.

SECONDARY KEY: In computer science, it is a key that holds the physical location of a record or a portion of a record in a file or database, and provides an alternative means of accessing data.

SECONDARY LEAFLET: The leaflet that occurs below the terminal leaflet.

SECONDARY MACRONUTRIENTS: Elements such as calcium, magnesium and sulfur that plants require in relatively small quantities.

SECONDARY MERISTEMS (LATERAL MERISTEMS): Plant meristems that produce secondary growth from a cambium. They are also called the lateral meristems because they surround the established stem of a plant and cause it to grow laterally or in diameter. There are two types of secondary meristems, namely, **vascular cambium**, which gives rise to wood in plant; and **cork cambium**, which gives rise to the periderm which replaces the epidermis.

SECONDARY METABOLITES: Metabolites which are produced by routes other than the normal metabolic pathways, mostly after the phase of active growth and under conditions of deficiency. The biological significance of many secondary metabolites is not exactly known.

SECONDARY MINERAL: A mineral formed at a later time through processes such as weathering and hydrothermal alteration, from the decomposition of a primary mineral or from the precipitation of the products of decomposition of a primary mineral. In contrast, a primary mineral is formed during the original solidification (crystallisation) of the rock by igneous, hydrothermal, or pneumatolytic processes, and retains its original composition and form and has not been altered chemically.

SECONDARY PEDUNCLE: An inflorescence branch.

SECONDARY PHLOEM: Phloem produced by the vascular cambium in a woody plant stem or root. It consists of two interpenetrating systems, the vertical or axial and the horizontal or ray.

SECONDARY POLLUTION: The products of the primary pollutants which form through photochemical and thermal reactions in the atmosphere (O_3, peroxyacetyl nitrate, etc.).

SECONDARY PRODUCTIVITY (SECONDARY PRODUCTION): The rate of biomass formation or energy fixation by heterotrophic organisms such as grazers and decomposers; driven by the transfer of organic material between trophic levels and represents the quantity of new tissue created through the use of assimilated food. Organisms responsible for secondary production include animals, protists, fungi and many bacteria.

SECONDARY RADIATION: Radiation emitted by any matter irradiated with electromagnetic or ionising radiation.

SECONDARY SEXUAL CHARACTERISTICS: External features of organisms that are characteristic of its gender but not the reproductive organs themselves; controlled by sex hormones (androgens or oestrogens). Examples include facial hair in human males, large breasts in human females, long feathers of male peacocks, bright colours in some male birds or mane in lions. They are developed at puberty in humans. In many cases they are involved in attraction of mates.

SECONDARY SOURCE: A data used for investigation after it has been used by another person or document or recording that relates or discusses information originally presented elsewhere. A secondary source contrasts with a primary source, which is an original source of the information being discussed; a primary source can be a person with direct knowledge of a situation or a document created by such a person. Secondary sources involve generalisation, analysis, synthesis, interpretation or evaluation of the original information.

SECONDARY STANDARD: 1. A solution that has been titrated against a primary standard such as a standard solution. 2. A unit such as length, capacitance or weight, used as a standard of comparison in individual countries or localities, but checked against the one primary standard in existence somewhere. 3. A unit defined as a specified multiple or submultiple of a primary standard, such as the centimetre.

SECONDARY STORAGE DEVICE (SECONDARY STORAGE; EXTERNAL MEMORY; AUXILIARY STORAGE): A storage medium that is not always directly accessible by a computer. This differs from primary storage technology, such as an internal hard drive, which is constantly available. Examples of secondary storage devices include external hard drives, USB flash drives and tape drives. These devices must be connected to a computer's external I/O ports in order to be accessed by the system. They may or may not require their own power supply. Examples of secondary storage media include recordable CDs and DVDs, floppy disks and removable disks, such as Zip disks and

Jaz disks. Each one of these types of media must be inserted into the appropriate drive in order to be read by the computer.

SECONDARY STRUCTURE: Any repetitive folded pattern that results from the interaction of the corresponding polymeric chains in a protein or a nucleic acid. It is characterised by folding of the peptide chain into an alpha helix, beta sheet or random coil.

SECONDARY SUCCESSION: The ecological succession that occurs on a pre-existing soil after the primary succession has been disrupted or destroyed due to a disturbance that reduced the population of the initial inhabitants. An example is the development of new inhabitants to replace the previous community of plants and animals that has been disrupted or disturbed by an event such as forest fire, flood, harvesting, epidemic disease and pest attack.

SECONDARY WAVE (S-WAVE; SHEAR WAVE): A seismic body wave that travels through the interior of the earth and is usually the second conspicuous wave to reach a seismograph. It shakes the ground back and forth perpendicular to the direction the wave is moving. S-waves move slower than the corresponding P-wave and cannot move through liquids at all. P-wave is a longitudinal earthquake wave that travels through the interior of the earth and is usually the first conspicuous wave to be recorded by a seismograph.

SECONDARY WINDING: The winding on the output side of a transformer or an induction coil. The secondary winding of a transformer may step up or reduce the voltage on the primary side and supplies the load.

SECONDARY XYLEM: Xylem produced by the vascular cambium (wood) in a woody plant stem or root. It is composed of two interpenetrating systems, the horizontal (ray) and vertical (axial).

SECONDARY-ION MASS SPECTROMETRY (SIMS): A technique for analysing the chemical structure of a solid by bombarding it with ions. A sample of the solid to be analysed is bombarded with ions, referred to as primary ions, collecting and analysing ejected secondary ions, having energies between 5 and 20keV. These secondary ions are measured with a mass spectrometer to determine the elemental, isotopic or molecular composition of the surface.

SECONDCUT: Grass which has been cut a second time in the season for hay or silage.

SECRETIN: A hormone produced by the anterior part of the small intestine (duodenum) in response to the presence of hydrochloric acid from the stomach which causes the pancreas to secrete alkaline pancreatic juices and stimulate bile production in the liver. Secretin was the first substance to be described as hormone in 1902.

SECRETION: The production or discharge of specific substances, usually fluid by the cell or specialised glands such as sweat, saliva, enzymes and hormones into the external medium. Plant secretions include nectar and various enzymes concerned with the digestion of nutrients within the plant cells. Some secretions perform special functions in the body (true secretions); others such as urine from the kidneys and perspiration from the sweat glands are eliminated as waste products or excretions. Digestive secretions include saliva, gastric juice, intestinal juice, pancreatic juice and bile. Certain secretions serve as lubricants, for example, the synovial fluid in joints or the secretions from mucous membranes and from the lachrymal or tear glands. The mammary glands secrete milk. The endocrine or ductless glands secrete hormones that enter directly into the bloodstream. **Secrete** is to produce a substance such as a hormone, oil or enzyme or bodily fluids from cells.

SECTION: 1. Any of the parts into which something may be or has been divided into. 2. In botany, a section is a taxonomic hierarchy above series but subordinate to genus. 3. In mathematics, it is the intersection between a plane and a surface or solid, a plane figure formed by cutting through a solid. 4. A factor group of a subgroup of a given group.

SECTION FORMULA (RATIO THEOREM): The theorem that if a point P divides a directed line segment, AB, in the ratio $m : n$, then the position vector, p, of P, can be expressed in terms of those of A and B as $\mathrm{p} = \frac{m\mathrm{a} + n\mathrm{b}}{m+n}$.

SECTOR: 1. A part of a circle bounded by two radii and the part of the circumference that lies between them. It divides a circle into two sectors, the minor sector and the major sector. The area of a sector is given by $1/2r^2\theta$, where r is the radius and θ is the central angle (in radians) subtended by the arc. 2. A measuring instrument in mathematics consisting of two arms marked with graduations and hinged together at one end. 3. The smallest addressable unit of a magnetic storage device.

SECTORIAL CHIMAERA: A chimaera that results from a cell mutation in a meristem that subsequently gives rise to a group or sector of mutant cells. Such chimaeras are usually unstable.

SECULAR MAGNETIC VARIATION: A perturbation of the Moon's motion caused by variations in the effect of the Sun's gravitational attraction on the Earth and Moon as their relative distances from the Sun vary during the synodic month.

SECULATE: Shaped like a sickle.

SECUND: Arranged or turned only to one side of the axis; or the arrangement of all organs or flowers to one side such as the flowers of lily of the valley.

SECUNDINE: The inner integument surrounding the ovule of a plant.

SECURE DIGITAL CARD (SD CARD): A type of memory card used for storing data in devices such as digital cameras, PDAs, mobile phones, portable music players and digital voice recorders. It is easily recognised by a slanted top-right corner. The card is one of the smaller memory card formats, measuring 24 mm wide by 32 mm long and is just 2.1 mm thick; usually with a storage capacity of 2 GB.

SECURE SOCKET LAYER (SSL): In computer science, it is a standard computer networking protocol, developed by Netscape Communications that allows secure communications over the Internet using two keys to encrypt data: a public key known to everyone and a private or secret key known only to the recipient of the message. SSL technology is commonly used in electronic commerce transactions where users send personal information such as credit card numbers and passwords over the Internet that need to be encrypted.

SECURITY ADMINISTRATOR TOOL FOR ANALYSING NETWORKS (SATAN): A set of testing and reporting tools used to deliver information about networked hosts. It has a web interface, tabulated results and if a security hole is found it provides context-sensitive tutorials about how to proceed.

SEDATIVE: Any drug that has a calming effect, reducing anxiety and tension. It is capable of inducing sleep if taken in larger doses. Sedatives can be addictive and must be used under supervision. Examples are babiturates, narcotics and benzodiazepines.

SEDENTARY: 1. Fixed and not moving or involving little exercise or physical activity. In humans, sedentary lifestyle leads to obesity and associated conditions such as hypertension and diabetes mellitus. 2. Describes marine organisms that remain in one place, usually attached to a rock, for most of their lives. 3. Describes birds that remain in the same area throughout the year and do not migrate.

SEDENTARY AGRICULTURE (SEDENTARY FARMING:) A type of farming in which the farmer is based in the same location all the time. The farmer may keep livestock in which animals are kept in stock or guarded by herdsmen to graze on pastures and returned to their base, grow crops or engage in mixed farming.

SEDENTARY ORGANISMS: Organisms that do not possess a means of self-locomotion and are normally fixed to one spot, permanently attached to something, usually a substrum such a rock or other surface. Examples of such organisms are barnacles, some bivalve molluscs, all bryozoans, and brachiopods.

SEDGE: One of a number of grass or rush-like herbs of the family cyperaceae, common in marshlands and poorly drained areas. They have minimal nutritional value. Sedges differ from true grasses in having solid, angular (usually triangular) stems. Most are perennial, reproducing by rhizomes.

SEDIMENT: Any loose material that has 'settled' or deposited from suspension in water, air or generally as the water or wind speed decreases or solid particles originally in a suspension, that settles downwards and forms a layer. It is a naturally-occurring material that is broken down by processes of weathering and erosion and is subsequently transported by the action of fluids such as wind, water or ice and/or by the force of gravity acting on the particle itself. Beach sands and river channel deposits are examples of fluvial transport and deposition, though sediment also often settles out of slow-moving or standing water in lakes and oceans. Desert sand dunes and loess are examples of aeolian transport and deposition. Glacial moraine deposits and till are ice transported sediments.

SEDIMENT LOAD: The amount of sediments carried by a stream or running water in a specified period of time, expressed in millions of tons (mt). The majority of a stream's sediment load is carried in solution (dissolved load) or in suspension. The remainder is called the bed load.

SEDIMENTARY ROCK: A group of rocks formed as a result of the accumulation and consolidation of sediments. Sedimentary rocks are one of the three major rock groups forming the Earth's crust. The others are metamorphic and igneous rocks. The rocks are deposited in layers as strata, forming a structure called bedding. The study of sedimentary rocks and rock strata provides information about the sub surface that is useful for civil engineering, for example, in the construction of roads, houses, tunnels, canals or other constructions. Sedimentary rocks are also important sources of natural resources such as coal, fossil fuels, drinking water or ores.

SEDIMENTATION: 1. In chemistry, it refers to separation of a dispersed system under the action of a gravitational or centrifugal field according to the different densities of the components.2. In the atmospheric sciences, the process of removal of an air borne particle from the atmosphere due to the effect of gravity.

SEDIMENTATION COEFFICIENT: The rate of sedimentation in centimetres per second that occurs when a solution is subjected to a centrifugal force of one dyne (10^{-5} newton). Its symbol is s. These rates are extremely low and are usually multiplied by 10^{13} to give a reasonable figure. In this modified form the unit is termed the Svedberg unit, with the symbol S. Different molecules and cellular inclusions tend to have characteristic sedimentation rates depending on the weight and shape of the molecule. For example, the ribosomes of prokaryotes have a value of 70S and those of eukaryotes 80S.

S

SEDIMENTATION EQUILIBRIUM: The equilibrium established in a centrifugal field when there is no net flux of any component across any plane perpendicular to the centrifugal force.

SEDIMENTATION POTENTIAL DIFFERENCE (SEDIMENTATION POTENTIAL; DORN EFFECT): The potential difference E or E_{sed} at zero current caused by the sedimentation of particles in the field of gravity or in a centrifuge, between two identical electrodes at different levels (or at different distances from the centre of rotation). E is positive if the lower (peripheral) electrode is negative.

SEDIMENTATION RATE (ERYTHROCYTE SEDIMENTATION RATE; SED RATE; WESTERGREN SEDIMENTATION RATE): In medicine, it is a non-specific blood test that is a measure of the rate at which red blood cells (erythrocytes) settle out of suspension in blood plasma. ESR is useful in detecting inflammation associated with conditions such as serious infections, cancers and autoimmune diseases.

SEDIMENTATION VELOCITY METHOD: A method by which the velocity of motion of solute component(s) or dispersed particles is measured and the result is expressed in terms of its (their) sedimentation coefficient(s).

SEE: In mathematics, of a pair of points of a set in a Euclidean vector space, having the line segment between the two points lying entirely within the given set. The convex set of points that sees the entire set is the star of the set.

SEE-BECK EFFECT (THERMOELECTRIC EFFECT): The generation of electromotive force (emf) in a circuit containing two dissimilar metals or semiconductors when the junctions between the two are kept at different temperatures.

SEED: 1. The reproductive structure of a higher plant or the structure that develops from an ovule after fertilisation in flowering plants and which grows into a new plant or the reproductive structure of a plant (angiosperms and gymnosperms) that develops from the ovules after fertilisation. 2. A crystal used to induce other crystals to form from a gas, liquid or solution. 3. A form of a radioactive isotope that is used to localise and concentrate the amount of radiation administered to a body site, such as a tumour.

SEED BED (SEEDLING BED): An area of land in which seeds and seedlings are cultivated in specially prepared cold frame, hotbed or raised bed in a controlled environment into larger young plants before transplanting them into a garden or field. A seedling bed is used to increase the number of seeds that germinate.

SEED BED WHEELS: A set of wheels bolted onto the front of a tractor which will give even compaction and a uniform sowing depth.

SEED BOX: 1. A box in which seeds can be sown. 2. The part of the plant head which contains the seed.

SEED CERTIFICATION: The testing, sealing and labelling of seed sold to farmers. This ensures that the seed is free from disease and from weeds.

SEED COAT (TESTA): The lignified or fibrous protective covering of a seed that develops from the integuments or outer covering of the ovules after fertilisation.

SEED CULM: Stem which supports the inflorescence of the plant.

SEED DISPERSAL: The movement or transport of mature seeds away from the parent plant. This decreases competition with the parent plant. The various means of seed dispersal include by gravity, wind, ballistic (seeds being forcefully ejected from the pod), water, animals and humans.

SEED DORMANCY: A condition of seed plants that prevents germination when the seeds are under optimal environmental conditions for germination. Seed dormancy is divided into two major categories based on what part of the seed produces dormancy: exogenous and endogenous dormancies. Exogenous dormancy is caused by conditions outside the embryo and is often broken down into three subgroups: physical, mechanical and chemical dormancies. Endogenous dormancy is caused by conditions within the embryo itself and it is also often broken down into three subgroups: physiological dormancy, morphological dormancy and combined dormancy; each of these groups may also have subgroups.

SEED DRESSING: The treatment of seeds with a fungicide and/or an insecticide to prevent certain soil and seed-borne diseases.

SEED DRILL (PLANTER): An implement or tool used by farmers in ancient Mesopotamia and China to deposit seeds evenly into the ploughed furrow, greatly enhancing agricultural production.

SEED HEAD: The top of a stalk with seeds, either in a seedcase or separately attached to the stem.

SEED LEAF (COTYLEDON): The first leaf or a pair of leaves produced by the embryo in seed plants that either remains in the seed or emerges upon seed germination to perform photosynthesis, a function similar to a true leaf. The difference between the seed leaf and the true leaf is that the seed leaf develops at the embryonic stage while the true leaf develops after the embryonic stage. Seed leaves are always opposite, simple and entire, regardless of how the subsequent or true leaves will look. In many dicots such as marigolds, the emerging shoot lifts the two cotyledons above ground, where they expand, turn green and function as the seedling's first leaves.

SEED LEGUME: A crop such as peanuts, beans, peas, alfalfa, clover and lentil that is capable of fixing nitrogen in the soil. These plants have nitrogen-fixing bacteria in their roots. This enables them to grow in nitrogen-poor soil.

SEED MIXTURE: Seeds of different plants supplied by seed merchants to farmers to produce a new ley. It includes grasses and legumes.

SEED PIECES: A vegetative means of propagating sugarcane.

SEED PLANT (SPERMATOPHYTA): Any plant that produces seeds. Most of these plants belong to the phyla Anthophyta (flowering plants) or Coniferophyta (conifers).

SEED RATE: The amount of seed sown per hectare shown as kilos per hectare.

SEED RIPENESS: The stage at which a seed can be successfully harvested.

SEED ROYALTIES: Money paid by seed growers to breeders of seed.

SEED STALK: The funiculus or the stalk of an ovule or seed.

SEED TREE: A tree left standing when others are cut down, to allow it to drop seeds on the cleared land around it.

SEED TRIALS: Test of new seeds to check the level of viability.

SEED WEEVIL: An insect pest affecting brassica seed crops. Seeds are destroyed in the pod by the larvae.

SEED-BORNE: Carried by seeds. For example, **seed-borne disease** is any disease carried by seeds.

SEEDER: A machine for sowing seeds.

SEEDER UNIT: A seed drill which sows seeds separately at set intervals.

SEEDING YEAR: The calendar year in which a seed is sown.

SEEDLESS FRUIT: A fruit such as pineapple, grape and banana that develops without mature seeds. Seedless fruits can develop because the fruit develops without any fertilisation, called **parthenocarpy** or pollination triggers fruit development, but the ovules or embryos abort without producing mature seeds.

SEEDLESS HAY: Hay obtained from a grass crop after threshing out the seed heads.

SEEDLING: A young plant newly grown from a seed and still supported by stored food in the seed.

SEEDLING BLIGHT: A fungal infection of germinating seedlings most commonly caused by Pythium, Fusarium, Diplodia, Rhizoctonia and Penicillium. It causes growth to slow down and foliage may become yellow or wilt. Roots of affected plants are found to be rotten and there may be lesions at soil level. Correct planting conditions, seed dressings and soil sterilisation can prevent the disease.

SEEK TIME: In computing, the average time it takes a hard disk controller to locate a specific piece of stored data.

SEEPAGE: Slow oozing out of water from the ground surface. **Seep** is to flow slowly through a substance.

SEGER CONES: A series of cones made from different mixtures of clay, limestone, etc. It is used to indicate the temperature inside a furnace or kiln.

SEGMENT: 1. Any of the individual units that make up part of the body or the body of some animals such as annelids and insects. 2. The portion of a solid cut by a plane or the portion of a curve or line between any two of its points. 3. Part of a circle bounded by an arc and a chord. 4. In chemistry, for bulk materials, it refers to each of the single, large portions of material pre-existing either in space such as bags, bales, drums or accumulated during a fixed time, for example, discharge from a conveyor belt or formable as increments by a sampling device. Segments may be actual or conceptual. 5. In clinical chemistry, it is of particular importance and refers to the set of samples which can be analysed between two successive calibrations. A segment includes samples, control materials and blank samples.

SEGRE CHART: A diagram in which the number of protons in a nucleus is plotted against the number of neutrons or a chart of the nuclides which is laid off in squares, each square displaying data about a nuclide; each column contains nuclides with a given neutron number and each row contains nuclides with a given atomic number; successive columns and rows represent successively higher numbers of neutrons and protons.

SEGRE PLOT: A graph of neutron number plotted against atomic number for all stable nuclides. It is used to ascertain the stability of nuclei by understanding the nature of strong interaction and competition between this attractive force and the repulsive electric force. The strong interaction is independent of electric charge (at any given separation the strong force between two neutrons is the same as that between two protons or between a proton or a neutron). Therefore, in the absence of the electrical repulsion between protons, the most stable nuclei would be those having equal numbers of neutrons and protons. The electrical repulsion shifts the balance to favour a greater number of neutrons, but a nucleus with too many neutrons is unstable, because not enough of them are paired with protons.

SEGREGATION: 1. Separation of the two versions of alleles of each gene and their distribution to separate sex cells during meiosis of these cells in organisms with paired chromosomes. Alternatively, segregation may be defined as the separation of pairs of alleles during the formation of reproductive cells so that they contain one allele only of each pair. 2. In taxonomy, a segregate or a segregate taxon is created when a taxon is split off, from another one. This main taxon will be better known, usually bigger and will continue to exist. 3. In chemistry, it refers to the process that differentiates the composition at an interface or surface from the average or bulk composition.

SEGREGATION RATIO: The proportion of one type of offspring to another that occurs as a result of separation of alleles at meiosis.

SEINE NET: Long flat net like a fence that is used to encircle a school of fish, with the boat driving around the fish in a circle. It is used to harvest anchovies, salmon and tuna. There are two main types of seine net: purse seines and Danish seines also called **anchor seine**. Purse seines are pulled by two boats, with ends that are pulled together around a shoal of fish so that the net forms a pouch. Danish seine consists of a conical net with two long wings with a bag where the fish collect. Drag lines extend from the wings and are long so they can surround an area. Boats equipped for seine fishing are called seiners.

SEINING (SEINE FISHING): Catching fish by using a large commercial fishing net that is weighted so that it hangs vertically in the water; used to harvest anchovies, salmon and tuna.

SEISMIC WAVE: Vibrational waves of energy that travel through the core of the Earth or other elastic bodies, for example as a result of an earthquake, explosion or some other process that imparts low-frequency acoustic energy. Seismic waves are studied by seismologists and geophysicists. They are measured by a seismograph, geophone, hydrophone (in water) or accelerometer.

SEISMOGRAM: Paper record graphing earthquake motions, created by a seismograph.

SEISMOGRAPH: An instrument that records ground oscillations such as those caused by earthquakes, volcanic activity or explosions. A seismograph consists of a pendulum mounted on a support base. The pendulum in turn is connected to a recorder, such as an ink pen. When the ground vibrates, the pendulum remains still while the recorder moves, thus creating a record of the earth’s movement. A typical seismograph contains three pendulums: one to record vertical movement and the others to record horizontal movement.

SEISMOLOGY: The scientific study of earthquakes and the propagation of elastic waves through the Earth. The field also includes studies of earthquake effects, as well as diverse seismic sources such as volcanic, tectonic, oceanic, atmospheric and artificial processes such as explosions. A related field that uses to infer information regarding past earthquakes is paleoseismology. A recording of Earth motion as a function of time is called a **seismogram**.

SEISMONASTY: A nastic movement of a plant or fungus in response to touch or vibration, especially the rapid folding of the leaflets of a sensitive plant.

SEJUGOUS: A pinnate leaf having six pairs of leaflets.

SELACHII: The major subclass of the Chondrichthyes containing the sharks, rays, skates and similar but extinct forms. Their sharp teeth develop from tooth-like, placoid scales and are rapidly replaced as they wear out.

SELECT: 1. To identify plants or animals with desirable characteristics such as high yield or disease resistance as part of the activity of breeding with new varieties. 2. In databases, it is a command used in query languages such as SQL to pick one more of physical options or to query data.

SELECTION: 1. The process by which one or more factors acting on a population produce differential mortality and favour the transmission of specific characteristics of subsequent generations. The process results in adaptation of an organism to its environment by means of selectively reproducing changes in its genotype. Variations that increase an organism's chances of survival and procreation are preserved and multiplied from generation to generation at the expense of less advantageous variations. Natural selection is the mechanism by which evolution occurs. It may arise from differences in survival, fertility, rate of development, mating success or any other aspect of the life cycle. 2. In biotechnology, it is a laboratory method applying a mixture of microorganisms to particular growth conditions under which only the cells with particular characteristics can survive and may be isolated. 3. In mathematics, the selection from a correspondence is a single-valued mapping whose value lies inside the image under a set-valued function at each argument.

SELECTION COEFFICIENT: A measure of the sum of forces acting to oppose the reproductive success of a particular organism, phenotype or genotype. Its symbol is s or t. It varies between 0 and 1 and is related to the fitness (W) by the equation $W = 1 - s$. When the selection coefficient increases fitness increases. It compares the fitness of a phenotype to another favoured phenotype and is the proportional amount that the considered phenotype is less fit as measured by fertile progeny. When $s = 0$ then the considered phenotype is selectively neutral compared to the favoured phenotype; $s = 1$ indicates complete lethality. For example, if the favoured phenotype produces 100 fertile progeny and only 90 are produced by the phenotype selected against then $s = 0.1$. An alternative way of expressing this is to describe the fitness of the favoured phenotype as 1.0 and that of the phenotype selected against as 0.9.

S

SELECTION PRESSURE: The extent to which an organism possessing a particular characteristic are either eliminated or favoured by environmental demands. It indicates the degree of intensity of natural selection or factors that influence the direction of natural selection.

SELECTION RULES: 1. Rules that determine which transitions between different energy levels are possible in a system, such as an elementary particle, nucleus, atom, molecule if crystal, described by quantum mechanics. 2. Rules that summarise the changes that must take place in the quantum numbers of a quantum-mechanical system for a transition between two states to take place with appreciable probability. Transitions that do not agree with the selection rules are called forbidden and have considerably lower probability. Selection rules have been derived for electronic, vibrational and rotational transitions and may differ according to the technique used to observe the transition. Example of selection rule is that in a centro-symmetric environment electronic transition between like atomic orbitals (i.e. s-s or p-p or d-d) are forbidden.

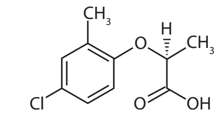
SELECTIVE: A term expressing qualitatively the extent to which other substances interfere with the determination of a substance according to a given procedure.

SELECTIVE BREEDING (ARTIFICIAL SELECTION): The act of mating selected parents of animal species to get desirable characteristics in the offspring. Typically, strains which are selectively bred are domesticated and the breeding is sometimes done by a professional breeder. Bred animals are known as breeds, while bred plants are known as varieties, cultigens or cultivars. The product of the cross of animals is called a **crossbreed** and crossbred plants are called **hybrids**.

SELECTIVE GENE EXPRESSION: The process in which only certain genes are expressed in one particular tissue, which can produce specialised cells from a single cell.

SELECTIVE FREQUENCY LOSS: A conductive loss of hearing for a particular range of frequencies. This may be due to unintentionally blocking out certain sounds that would otherwise cause a distraction.

SELECTIVE HERBICIDE: A weed killer designed to kill only plants with specific characteristics and not others. It is often used on lawns to kill broad-leaved weeds without damaging the grasses. Examples are 2,4-D (weedone), mecoprop (MCPP) and 2,4-D plus 2,4-DP (weedone DPC).

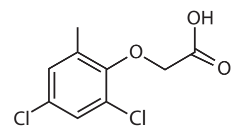


Chemical Structure of Mecoprop, a Selective Herbicide

SELECTIVE MEDIUM: A growth medium that allows only certain types of microorganisms to grow and inhibits the growth of others.

SELECTIVE MICRO-SAMPLE: A sample which results where a small portion has been separated from the lot or laboratory sample by selective means such as magnetic-, density- or manual separation, by micro-drilling or by centrifugation, for example, the separation of magnetic materials from a geological material or the separation of metal particles from a lubricating oil. If individual particles are analysed the term individual particle analysis is applied.

SELECTIVE PESTICIDE: A pesticide whose toxic effects is against specific pests but does not affect growing crops. An example is 2,4-dichlorophenoxyacetic acid (2,4-D), a synthetic auxin.



Chemical Structure of 2,4-D, a Selective Pesticide

SELECTIVE POISONING: A process in which a poison retards the rate of one catalysed reaction more than that of another or it may retard only one of the reactions.

SELECTIVE PRECONCENTRATION: An operation (process) as a result of which microcomponents are selectively isolated from a sample. It is used when the simultaneous presence of several components in the concentrate may distort the results of analysis. Selective preconcentration is usually achieved by isolation of the microcomponent to be determined.

SELECTIVE PRESSURE: Any cause that limits the survival and reproduction of individuals within a population. If it persists for generations, it can lead to genetic change. Selective pressures are influenced by many factors, for example, environmental conditions, availability of food and energy sources, predators, diseases, and human effects.

SELECTIVE REABSORPTION: The absorption of some of the components of the glomerular filtrate such as salt, glucose and amino acids, vitamins and water back into the blood as the filtrate flows through the nephrons of the kidney.

SELECTIVE SAMPLE: A sample that is deliberately chosen by using a sampling plan that screens out materials with certain characteristics and/or selects only material with other relevant characteristics.

SELECTIVE SOLVENT: A medium that is a solvent for at least one component of a mixture of polymers or for at least one block of a block or graft polymer, but a non-solvent for the other component(s) or block(s).

SELECTIVELY PERMEABLE (SELECTIVE PERMEABILITY): Describing a barrier such as cell membrane and the thin film on the inside of an egg that allows some chemicals to pass through but not others. The rate of passage depends on the pressure, concentration and temperature of the molecules or solutes on either side, as well as the permeability of the membrane to each solute.

SELECTIVITY: 1. It sometimes refers to the discrimination shown by a given reactant A when it reacts with two alternative reactants B and C or in two different ways, for example, at two different sites with a reactant B. 2. The term also sometimes refers to the ratio of products obtained from given reactants. This meaning is of importance for catalysts, which can have a wide range of selectivities. Selectivity is quantitatively expressed by ratios of rate constants for the alternative reactions or by the decadic logarithms of such ratios.

SELECTRON: A boson with which the fermion electron is paired in supersymmetric theories.

SELENIDES: Binary compounds of selenium with other more electropositive elements. Selenides of non-metals are covalent. Most metal selenides can be prepared by direct combination of the elements. Selenides are unstable to oxidation by air. They are rare and include ferroselite and umangite.

SELENIUM: A non-metallic element, red in powder form, black in vitreous form and metallic grey in crystalline form in Group 16 (VIA) of the Periodic Table, resembling sulfur. Its chemical symbol is Se, atomic number 34, valence 2, 4 or 6 and relative atomic mass 78.96. It is obtained from flue dust in refineries that use sulfide ores. One of its allotropes conducts electricity in the presence of light and is used in photocells and rectifiers.

SELENIUM CELL: Either of one or two photoelectric cells consisting of insulated selenium strip between two suitable electrodes. One type relies on the photoconductive effect and the other on the photoelectric effect. In the photoconductive selenium cell, an external emf is applied while in the photoelectric selenium cell, the emf is generated.

SELENODONT: A term for describing molar teeth in which the grinding surfaces possess crescent-shaped ridges.

SELENOLOGY: The branch of astronomy which deals with scientific study of the Moon.

SELENOPHYTE: A plant which grows on seleniferous soils and takes up selenium from the soil. It is frequently used as an indicator of the presence of selenium in the soil.

SELF-ABSORPTION: It occurs in emission sources of finite thickness when radiant energy quanta emitted by atoms (or molecules) are absorbed by atoms of the same kind present in the same source. The absorbed energy is usually dissipated by collisional transfer of energy or through emission of radiant energy of the same or other frequencies. In consequence, the observed radiant intensity of a spectral line (or band component) emitted by a source may be less than the radiant intensity would be from an optically thin source having the same number of emitting atoms. Self-absorption may occur in all emitting sources to some degree, whether they are homogeneous or not.

SELF-ABSORPTION BROADENING: Photons emitted in one region of a source are partly absorbed in their passage through the plasma. Because of the fact that the absorption profile is of the same shape as the emission profile, energy is selectively absorbed from the emission line, i.e. the absorption coefficient is a maximum at the centre of the line or central wavelength. The actual line profile is changed as a result of the lowering of the maximum intensity accompanied by a corresponding increase in apparent halfwidth.

SELF-ABSORPTION EFFECT: The reabsorption of luminescence by the analyte and interfering impurities within the excitation volume.

SELF-ADJOINT: 1. Of a matrix or linear operator on Hilbert space, Hermitian, equal to its own Hermitian conjugate or adjoint; thus $\langle Ax, y\rangle = \langle x, Ay\rangle$, for all x and y in the Hilbert space. The concept makes sense for a mapping from a reflexive normed space to its dual. 2. Of an algebra, having the complex conjugate of any member of the algebra in the algebra.

SELF-CONJUGATE: Of a line or point, lying on the polar or dually, running through its pole. The only self-conjugate point on a self-conjugate line is its pole.

SELF-CONSISTENT FIELD (SCF; SELF-CONSISTENT FIELD METHOD): A method of finding an approximate wave function for a system of many electrons, in which one attempts to find a product of single-particle wave functions, each one of which is a solution of the Schrödinger equation with the field deduced from the charge density distribution due to all the other electrons.

SELF-CONTAINED HERD: A dairy herd which breeds its own replacements with the calves being kept and reared.

SELF-CONTAINED UNDERWATER BREATHING APPARATUS, SCUBA (AQUALUNG SCUBA): A device that allows divers to breathe and move about freely underwater for extended periods. It consists of a mouthpiece, a connecting tube, a valve and at least one compressed-air cylinder. The key component is the demand valve which allows the diver to breathe air at the pressure prevailing in the surrounding water however great the pressure in the supply cylinder. Used air is vented into the water.

SELF-ENERGY: The energy that is contributed by the interactions between different parts of a system.

SELF-EXCITING GENERATOR: A type of electrical generator in which the field magnets are excited by currents from the generator output.

SELF-FEED: To take a controlled amount of feed from a large container as required.

SELF-FEED SILAGE: A feeding system where stock feed from silage is centrally controlled.

SELF-FERTILISATION (AUTOGAMY; ENDOGAMY): The fertilisation of a plant or invertebrate animal with its own pollen or sperm or the fusion of male and female sex cells (gametes) produced by the same individual. Many organisms capable of self-fertilisation can also reproduce by means of cross-fertilisation. As an evolutionary mechanism, self-fertilisation allows an isolated individual to create a local population and stabilises desirable genetic strains, but it fails to provide a significant degree of variability within a population and thereby limits the possibilities for adaptation to environmental change. This type of fertilisation occurs in pioneer populations (many weeds are autogamic), where insect vectors may be rare (as in the tundra ecosystems) and in bisexual organisms, including most flowering plants, numerous protozoans and many invertebrates.

SELF-INCOMPATIBILITY (SELF-STERILITY): The failure of gametes from the same plant to form a viable embryo. In angiosperms this is usually due to complex interactions between the pollen and stigmatic tissues. It may be achieved by the prevention of pollen germination, the retardation of growth or the disorientation of the pollen tube or by failure of nuclear fusion. Self incompatibility caused by the stigmatic tissues is described as **sporophytic**, while that due to the pollen tube is termed **gametaphytic**.

SELF-INDUCTION (ELECTROMAGNETIC INDUCTION): The production of an electromotive force (emf) in a conductor or coil of wire, when there is a change of magnetic flux linkage with the coil or by the motion of a conductor through a magnetic field so as to cut across the magnetic flux. The magnitude of the emf (E) is directly proportional to the rate of change of the flux linkage: $E = d\varphi/dt$.

SELF-INTERSECTING: A self-intersecting polygon is a polygon with edges that cross other edges. A self-intersecting polyhedron is a polyhedron with faces that cross other faces.

SELF-INVERSE: In mathematics, of an element of a group, ring, etc., being its own inverse or combining with itself to produce the identity element so that xx = I or -1 × -1 = 1. An element of a group of order 2; for example, is the function $f(x)$ .. $1/x$, defined on the interval]0, ∞[. A reflection when repeated gives the identity transformation and the function $x \to 1 - x$, when combined with itself gives the identity function and are said to be self-inverse.

SELF-MONITORING ANALYSIS AND REPORTING TECHNOLOGY (S.M.A.R.T.): A technology that is used to protect and prevent errors in hard drives. It monitors and analyses hard drives, then checks its health for problems. The main purpose of SMART is to keep the hard drive running smoothly and prevents it from crashing.

SELF-MULCHING SOIL: A soil in which the surface layer becomes so well aggregated that it does not crust and seal under the impact of rain, but instead serves as a surface mulch when it dries.

SELF-ORGANISATION (SELF-ORGANIZATION): Spontaneous events occurring in a system when certain parameters of the system reach critical values. It is the process where a structure or pattern appears in a system without a central authority or external element imposing it. This globally coherent pattern appears from the local interaction of the elements that make up the system, thus the organisation is achieved in a way that is parallel (the entire elements act at the same time) and distributed (no element is a coordinator). In physics, chemistry and biology, self-organisation occurs in open systems driven away from thermal equilibrium. The process of self-organisation can be found in many other fields such as economics, sociology, medicine, technology, religion, robotics, cybernetics, human society, ecology, etc. A few examples of self organisations are: (i) in physics, it is a structural (order-disorder, first-order) phase transitions and spontaneous symmetry breaking such as spontaneous magnetisation, crystallisation in the classical domain and the laser, superconductivity and Bose-Einstein condensation, in the quantum domain (but with macroscopic manifestations); (ii) in chemistry, it is a molecular self-assembly, reaction-diffusion systems and oscillating chemical reactions, autocatalytic networks and liquid crystals; (iii) in biology, it is a spontaneous folding of proteins and other biomacromolecules, formation of lipid bilayer membranes, homeostasis (the self-maintaining nature of systems from the cell to the whole organism), pattern formation and morphogenesis or how the living organism develops and grows; and the creation of structures by social animals, such as social insects (bees, ants, termites) and many mammals; (iv) in mathematics and computer science, it is a phenomena such as cellular automata, random graphs and some instances of evolutionary computation and artificial life; and (v) in economics, it is a market economy, growth, competition, extinction of companies, etc.

SELF-ORGANISED CRITICALITY: A mathematical theory that describes how systems composed of many interacting parts can tune themselves toward dynamical behaviour that is critical in the sense that it is neither stable nor unstable but at a region near a phase transition.

SELF-POISONING (AUTOPOISONING): A phenomenon in which a product of a reaction may cause poisoning or inhibition.

SELF-POLLINATION: The transfer of pollen from the anther to a stigma of the same flower or to another flower on the same plant or pollination within the same flower or between flowers on the same plant. The mechanism is seen most often in some legumes such as peanuts. In another legume, soybeans, the flowers open and remain receptive to insect cross pollination during the day; if this is not accomplished, the flowers self pollinate as they are closing. Other plants that can self pollinate are many kinds of orchids, peas, sunflowers and *tridax*.

SELF-PURIFICATION: The ability of water to cleanse itself of polluting substances.

SELF-QUENCHING: Quenching of an excited atom or molecular entity by interaction with another atom or molecular entity of the same species in the ground state.

SELF-REFERENCE: In logic, the property of an expression of referring to itself, which gives rise to such semantic paradoxes as that of determining the truth-value of the sentence "this sentence is false", which is true if it is false and false if it is true.

SELF-REGULATION: An ecosystem controlling itself without external intervention. The term may also refer to homeotasis, which is the property of a system that regulates its internal environment and tends to maintain a stable, constant condition of properties like temperature or pH. It can be either an open or closed system. All animals also regulate their blood glucose, as well as the concentration of their blood. Mammals regulate their blood glucose with insulin and glucagon. The human body maintains glucose levels constant most of the day, even after a 24-hour fast.

SELF-REVERSAL: A case of self-absorption, when a line is self-absorbed to such an extent that the peak or central wavelength intensity is less than at the wings or non-central wavelengths.

SELF-SEEDED: A plant that geminates from seed that has fallen to the ground.

SELF-SHIELDING: The lowering of the flux density in the inner part of an object due to absorption in its outer layers.

SELF-SIMILARITY: The property an object has when a part of itself looks the same or similar to the whole. Many objects in the real world, such as coastlines, are statistically self-similar; parts of them show the same statistical properties at many scales. Self-similarity is a defining characteristic of fractals.

SELF-STERILITY (SELF-INCOMPATIBILITY, SI): The condition found in many hermaphrodite organisms in which male and female reproductive cells produced by the same individual will not fuse to form a zygote or if they do the zygote is unable to develop into an embryo. In plants with self-sterility, when a pollen grain produced in a plant reaches a stigma of the same plant or another plant with a similar genotype, the process of pollen germination, pollen tube growth, ovule fertilisation and embryo development is halted at one of its stages and consequently no seeds are produced. This may be caused by pollen compatibility, the timing of pollen shed, dioeciousness or other problems. A plant that will fertilise a self-sterile plant is called a **polleniser**. Self-sterile plants include apples, sweet cherries, nuts (almonds, walnuts, chestnuts, pecans, filberts, etc.), Asian pears, peaches, apricots, plums and blueberries.

SELFISH DNA: Regions of DNA that apparently have no function and exits between those regions of DNA that represent the genes. An example is transposons or DNA segment that moves. Selfish DNA is so called because it seemingly exists only to pass copies of itself from one generation to another and does so by acting like a molecular parasite using the organism in which it is contained as the survival machine. This is known as the selfish DNA theory. The greatest amount of selfish DNA is found in vertebrates and higher plants.

SELIWANOFF'S TEST: A biochemical test to identify the presence of ketonic sugar such as fructose in solution. It is a colour test which develops a red colour with resorcinol in hydrochloric acid.

SELLOTAPE: A British brand of transparent, cellulose-based, pressure sensitive adhesive tape. It is generally used for joining, sealing, attaching and mending.

SEMANTIC TABLEAU: In logic, a tree diagram constructed in order to demonstrate the consistency or otherwise of a set of statements by successively breaking down the given statements into simpler components. As soon as a contradiction is obtained, that branch of the tableau need be considered no further and is closed.

SEMANTICS (MODEL THEORY): The study of mathematical structures that satisfy a particular set of axioms, especially in the field of logic. Its fields of study include the concepts of truth, satisfaction and validity, which are defined extrinsically for a formal system, as contrasted with proof theory, which is concerned with the study of only the intrinsic property of syntactic deducibility.

SEMANTIDE: An information-carrying molecule that carry the information of the genes or a transcript or a translation. The genes themselves are primary semantides, messenger RNA molecules are secondary semantides and most polypeptides are tertiary semantides.

SEMAPHORE: 1. A visual device or technique that communicates messages using flags, lights or mechanically moving arms. An example is a railway semaphore signal, which consists of a vertical post on which a single projecting arm is mounted, whose angle indicates "all clear" or "danger". 2. In operating systems, it is an abstract data type used to control access by multiple processes to the processor by the operating system.

SEMAPHORIN: One of a class of proteins that act as guidance molecules during the development of nerve cells, immune cells, blood vessels, bone and other tissues. There are 8 major classes of semaphorins. The first 7 are ordered by number, from class 1 to class 7. The eighth group is class V, where V stands for virus. Classes 1 and 2 are found in invertebrates only, whilst classes 3, 4, 6 and 7 are found in vertebrates only. Class 5 is found in both vertebrates and invertebrates and class V is specific to viruses. Some are secreted by cells, whereas others remain bound to the plasma membrane.

SEMELPARITY: The system of reproducing only once in a lifetime, after which death is inevitable even if the reproductive event is fruitful. Examples of semelparity organisms include annual and biennial plants and most invertebrate animal taxa. The most well known one is Pacific salmon that perishes after spawning. Other examples are squid, mayflies, Atlantic eel, some species of butterflies, cicadas, some molluscs such as squid and octopus and many arachnids. Annual plants, including all grain crops and most domestic vegetables, are semelparous; and also long lived plants such as century plant (agave) and some species of bamboo.

SEMEN (SEMINAL FLUID): A whitish fluid containing spermatozoa and consisting of secretions of the testes, seminal vesicles, prostate and bulbourethral glands. It is produced by the male animal and introduced into the vagina of the female during ejaculation.

SEMI-: A prefix that denotes half (1/2) of something, for example, semicircle means half of a circle.

SEMI-ARID: An area which is partially deficient in rainfall or is very dry. They are climates that are intermediate between desert climates and humid climates in ecological characteristics and agricultural potential. Semi-arid climates tend to support short or scrubby vegetation and usually dominated by either grasses or shrubs.

SEMI-AXIS: Half the length of a semi-major or semi-minor axis of a conic section (ellipses and hyperbolas). The semi-major axis is the longest radius and the semi-minor axis the shortest. If they are equal in length then the ellipse is a circle.

SEMI-DIGGER: A type of mould board on a plough.

SEMI-ELLIPTICAL: Shaped like half of an ellipse, being divided into two by its major axis.

SEMI-EMPIRICAL CALCULATIONS: A technique used to calculate parameters, which define atomic and molecular properties using integrals obtained from experiments such as ionisation energy by using ab-initio method.

SEMI-GROUP: An algebraic structure endowed with an associative binary operation, usually called addition, under which it is closed. Semi-groups are called monoids if they also have, an identity element. Semi-groups include various sets of numbers which are closed under addition or multiplication; semi-groups of matrices with respect to multiplication; semi-groups of functions with respect to "pointwise" multiplication, $*$, defined by $(f * g)(x) = f(x)g(x)$; semi-groups of sets with respect to intersection or union; etc.

SEMI-INTERQUARTILE RANGE (QUARTILE DEVIATION): In statistics, it is a measure of the spread of a distribution measure of spread or dispersion and equal to half the difference between third quartiles (75th percentile) and the first (25th percentile), i.e., $\frac{Q_3 - Q_1}{2}$, where Q_3 and Q_1 are respectively the third and first quartiles.

SEMI-MAGIC SQUARE: A square that fails to be a magic square only because one or both of the main diagonal sums do not equal the magic constant. **Magic square** is an array of n numbers such that the sum of the n numbers in any row or column is a constant.

SEMI-MOUNTED: An implement which is supported by tractor but also has its own wheels.

SEMI-NORM: In mathematics, it is the generalisation of the concept of norm that does not demand that only the origin has value zero. Thus a norm is a semi-norm with trivial kernel.

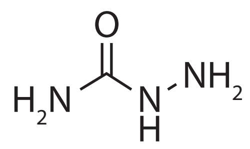
SEMI-PRECOCIAL: Young birds (hatchlings) that are capable of leaving the nest but are dependent on their parents for feeding.

SEMI-RING OF SETS: In mathematics, it is family of subsets containing the empty set, closed under finite intersection and union and having the property that if E is a subset of F with E and F in the family, then F \ E is a countable union of disjoint members of the family.

SEMI-SIMPLE: 1. A semi-simple module is a module genarated by or being the direct sum of simple submodules. A module over a (not necessarily commutative) ring with unity is said to be semi-simple (or completely reducible) if it is the direct sum of simple (irreducible) submodules. A ring that is a semi-simple module over itself is known as an Artinian semi-simple ring. Some important rings, such as group rings of finite groups over fields of characteristic zero, are semisimple rings. A ring is semi-simple if and only if it is the direct sum of (finitely many) minimal left ideals. Alternatively, of a ring such that a given radical is zero. 2. Of a commutative Banach algebra, such that the intersection of maximal two-sided ideals is zero.

SEMI-TRANSCENDENTAL FUNCTION: The general solution of a non-linear second order differential equation for which the general solution is not an algebraic function of two constants of integration but the equation admits of a first integral that is an algebraic function of one constant of integration. For example, the first integral of $w'' + 2ww' = q(z)$ is $w' + w^2 = \int q(z)\,dz + A$; the general solution is therefore at worst a semi-transcendental function of A and the second constant of integration.

SEMICARBAZIDE: A derivative of urea in which one NH_2 has been replaced by hydrazine $H_2N\text{-}CO\text{-}NH\text{-}NH_2$; they are therefore organic bases. They react with aldehydes and ketones to form semicarbazones, which are useful in analysis and characterisation of the carbonyl compounds. Semicarbazide is used as a detection reagent on thin layer chromatography. It stains α-keto acids on the TLC plate, which must then be viewed under a UV light to see the results.



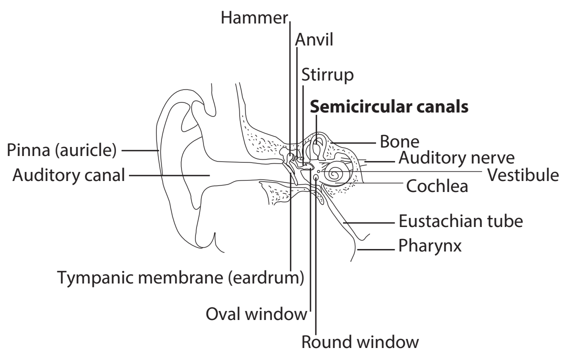
Chemical Structure of Semicarbazide

SEMICARBAZONE: Crystalline organic compound containing the group $R_2C{=}N_2HCONH_2$, formed when an aldehyde or a ketone reacts with semicarbazide ($NH_2NHCONH_2$) with the elimination of water. They are used to identify the original aldehyde or ketone. It is also used in separating ketones from reaction mixtures, the derivative is crystallised out and hydrolysed to give the ketone.

SEMICARPOUS: With ovaries of adjacent carpels partly fused, the styles and stigmas separate.

SEMICIRCLE: 1. A half circle formed by the diameter cutting through the circle. 2. An arc of a circle that is half of its circumference.

SEMICIRCULAR CANALS: One of the three looped tubes that form part of the labyrinth in the inner ear. It consists of three looped canals set at right angles to each other and attached to the utriculus. The three canals are the horizontal semicircular canal (also known as the lateral semicircular canal), superior semicircular canal (also known as the anterior semicircular canal) and the posterior semicircular canal. They are filled with fluid and detect changes in the position of the head, contributing to the sense of balance, sensitivity to movement and gravity and the maintenance of physical equilibrium.



Semicircular canals of the human ear

SEMICLASSICAL APPROXIMATION: An approximation technique used in calculating quantities in quantum mechanics. It is called semiclassical because the wave function is written as an asymptotic series with ascending powers of the Planck's constant h, with the first term being purely classical. The semiclassical approximation is particularly successful for calculations involving radioactive decay producing alpha particles, tunnel effect, etc.

SEMICOKE: A carbonaceous material intermediate between a fusible mesophase pitch and a non-deformable green coke produced by incomplete carbonisation at temperatures between the onset of fusion of coal and complete devolatilisation. Semicoke still contains volatile matter, therefore. Semicoke may be conceived as covering a continuous range from coal that has not yet been fused to coke breeze. Semicoke can also be used as a filler in carbon mixtures.

SEMICONDUCTOR: A pure crystalline material, whose electrical conductivity lies between that of an insulator and a metal conductor. It is an insulator at absolute zero temperature. Its valence electrons are loosely bound enough for some to be released by thermal agitation at room temperature and at high temperatures its conductivity is nearly as great as that of metal conductors. Semiconductors' conductivity can be increased and controlled by doping with impurities and by the application of heat or light. When a doped semiconductor contains mostly free holes it is called "p-type" and when it contains mostly free electrons it is known as "n-type". The electrical conductivity of semiconductors ranges from about 10^3 to 10^{-9} ohm^{-1} cm^{-1}, as compared with a maximum conductivity of 10^7 for good conductors and a minimum conductivity of 10^{-17} ohm^{-1} cm^{-1} for good insulators. Examples of semiconductor materials are silicon and germanium They are used in the manufacture of electronic devices such as diodes, transistors and integrated circuits. Semiconductors exhibit conduction properties that may be temperature-dependent, permitting their use as thermistors or temperature-dependent resistors or voltage-dependent, as in varistors.

SEMICONDUCTOR DETECTOR: A radiation detector using a semiconductor, in which free electric charges are produced along the path of an ionising particle, in combination with a high voltage and electrodes to collect the induced electric charges.

SEMICONDUCTOR-METAL TRANSITION: Any transition from a semiconductor to a metallic state under the influence of a temperature or pressure change or both.

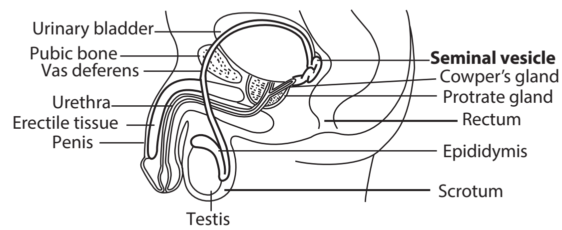
SEMICONSERVATIVE REPLICATION: The process by which DNA makes exact copies of itself. During this process, double-stranded DNA uncoils and the separated polynucleotides act as templates for the formation of new complementary polynucleotide chains. Hence any given DNA molecule will consist of one old strand and one new one and the original molecule is semi conserved during the manufacture of an identical copy (replica).

SEMILUNAR VALVE: Either of two valves in the heart, found in the pulmonary artery and in the aorta. It prevents the back flow of blood into the right and left ventricles from the pulmonary artery and aorta respectively, thus maintaining blood flow in a single direction.

SEMINAL PROPAGATION: The process of growing new plants from seeds or from tuber.

SEMINAL ROOTS: Roots formed at the time of seed germination to anchor the seed in the soil.

SEMINAL VESICLE: 1. A pouch or sac in many male invertebrates and lower vertebrates that is used for storing sperm. 2. Either of a pair of glands that secrete the fluid component of semen into the ejaculatory duct in males. This secretion is alkaline and neutralises the acidic conditions in the female genital tract. It contains fructose, which gives energy to the sperm.



The male reproductive system showing the seminal vesicle and other parts

SEMINIFEROUS: Producing or bearing seed or semen.

SEMINIFEROUS TUBULES: Tubules located in the testes which are the specific location of meiosis and the subsequent creation of spermatozoa.

SEMIOCHEMICAL: A chemical that affects the behaviour of an organism. Such chemicals include pheromones which are used for communication between members of the same species and allelochemicals which act as chemical signals between members of different species.

SEMIPERMEABLE MEMBRANE: A membrane that contains microscopic pores and is permeable to molecules or ions of the solvent but not the solute in osmosis. During osmosis, small solvent molecules pass freely through the pores but larger solute molecules or ions cannot pass through it. An example is the lipid bilayer, on which is based the plasma membrane that surrounds all biological cells. Many natural and synthetic materials thicker than a membrane are also semipermeable. One example of this is the thin film on the inside of an egg.

SEMIPLUME: Any of the several feathers having a long shaft found under the contour feather on a bird's body. Semiplumes are responsible for providing insulation as well as some flexibility to the bird.

SEMIPRIME: In mathematics, it is a natural number that is the product of two (not necessarily distinct) prime numbers. The square of any prime number is a semi prime; and the largest known semiprime is therefore the square of the largest known prime. The semiprimes less than 100 are 4, 6, 9, 10, 14, 15, 21, 22, 25, 26, 33, 34, 35, 38, 39, 46, 49, 51, 55, 57, 58, 62, 65, 69, 74, 77, 82, 85, 86, 87, 91, 93, 94 and 95. Semiprimes that are not perfect squares are called discrete or distinct, semiprimes.

SEMISYSTEMATIC NAME (SEMITRIVIAL NAME): A name in which at least one part is used in a systematic sense; or a name of a chemical compound that contains both systematic and trivial components. For example, calciferol includes the -ol suffix denoting an —OH radical, whereas calcifer-, which has no systematic meaning, is used only in this word.

SEMOLINA: Coarse flour made from rice or wheat after the fine flour has been grounded and used for pudding.

SEMPERFLOROUS (AIANTHOUS): Continuously flowering throughout the year.

SEMTEX: A general purpose nitrogen-based, stable, odourless plastic explosive containing RDX (cyclonite) and PETN (pentaerythritol tetranitrate). It is used in commercial blasting, demolition and in certain military applications.

SENESCENCE (BIOLOGICAL AGING): The change that occurs in an organism between maturity and death. Characteristically, there is deterioration in functioning as the cells become less efficient in maintaining and replacing the vital cell components. In humans, there is a reduction in mental ability and also a decline in physical activity.

SENNA: A genus of shrubs and trees and herbs of about 250 species in the Fabaceae/Leguminosae family with pinnately compound leaves and showy, usually yellow flowers many of which are often classified as members of the genus Cassia. They grow in various parts of the world. The leaves are pinnate with opposite paired leaflets. The inflorescences are racemes, occurring at the ends of branches or emerging from the leaf axils. The flower has five sepals and usually yellow petals. The fruit is a legume pod containing several seeds. Some Senna species are used as ornamental plants and also have medicinal. One species, *Senna occidentalis* with the synonym *Cassia occidentalis* and the common names negro coffee, coffee weed or stinking weed is a small many-branched annual or perennial shrub or herb growing up to 2 m high. It has compound leaves with ovate to ovate-elliptic or lanceolate leaflets occurring in 4-5 pairs. The herb has been traditionally used to normalise bowel movements. The seeds are also beneficial in treating whooping cough, convulsions and hypertension and heart diseases. The leaves of *Senna fistula*, *Senna marilandica* and *Senna alexandrina* are used to treat constipation; that of *Senna reticulata* are used to treat bacterial and fungal infections; *Senna surattensis* and *Senna sieberiana* leaves are used to treat venereal disease; *Senna obtusifolia* leaves are used to treat hypertension, high cholesterol, constipation, skin diseases and eye disorders; *Senna alata*, *Senna obtusifolia* and *Senna sophera* leaves are used to treat ringworm.

SENSATION: 1. The raw data detected by the senses. For example, red is colour sensation. 2. A physical feeling such as cold, caused by having one or more of the sense organs stimulated. 3. A general feeling, especially one not attributable to any obvious cause, for example, discomfort, anxiety or doubt.

SENSE: 1. In mathematics, it is one of the two opposite directions measured on a directed line describing whether it is positive or negative; or one of the two opposite directions of rotation that is either clockwise or anticlockwise. 2. Any of the faculties such as sight or taste by which a person or animal obtains information about the physical world or about the state of their body in relation to this environment. Specific receptors are sensitive to pain, temperature, chemicals, etc. 3. A perception or feeling produced by a stimulus.

SENSE ORGAN: A part of the body of an animal such as eye, ear, nose and taste bud that contains or consists of a concentration of receptors that are sensitive to stimuli or any organ that an animal uses to gain information about its surrounding. The stimulators of these receptors initiate the transmission of nervous impulses to the brain, where sensory information is analysed and interpreted.

SENSILLUM (SENSILLA): Any of the various hair-like or peg-like sense organs found in insects and other arthropods, comprising a cluster of basal receptor cells whose dendrites extend inside a sheath and hair shaft. They respond to a wide range of chemical and mechanical stimuli, acting as chemoreceptors of both smell and taste.

SENSITISATION (SENSITIZATION): 1. Alteration of the integrity of a plasma membrane resulting from the reaction of specific antibodies with antigens on the surface of the cell wall. In the presence of a complement the cell raptures. 2. Initial exposure to a specific antigen such that re-exposure to the same antigen causes severe immune response. 3. A simple form of non-associative learning in which, there is an increase in an organism's responsiveness to stimuli following an especially intense or irritating stimulus. For example, a sea snail that receives a strong electric shock will afterward withdraw its gill more strongly than usual in response to a simple touch. Depending on the intensity and duration of the original stimulus, the period of increased responsiveness can last from several seconds to several days. 4. Addition of small amounts of a hydrophilic colloid to a hydrophobic sol to possibly make the latter more sensitive to flocculation by electrolyte.

SENSITIVITY: 1. The ability of organisms to detect changes such as heat, light and sound in their environment and thus make appropriate responses. 2. In climate change, sensitivity is the degree to which a system is affected, either adversely or beneficially, by climate-related stimuli. There are two types of effect: (i) direct, for example, a change in crop yield in response to a change in the mean, range, or variability of temperature; and (ii) indirect, for example, damages caused by an increase in the frequency of coastal flooding due to sea-level rise. 3. In electronics, it is the degree of response of a receiver or instrument to an incoming signal or to a change in the incoming signal. For example, the signal strength required by an FM tuner to reduce noise and distortion or the degree of response of a plate or film to light, especially to light of a specified wavelength.

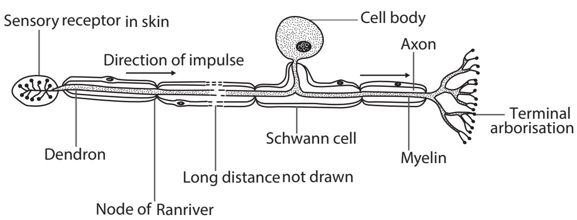
SENSITOMETER: An instrument for measuring a degree of sensitivity. An example is the device used for measuring the sensitivity of photographic film to light; or a device for measuring the sensitivity of eyes to light.

SENSOR: 1. A device capable of detecting and responding to physical stimuli such as movement, light or heat or a device that takes in and reacts to some kinds of input energy and outputs a related electrical signal. 2. A device designed to detect a physical state or measure a physical quantity to produce an input signal for a computer. For example, a sensor may detect the fact that a printer has run out of paper.

SENSORY (AFFERENT): Specialised tissues connected with the physical senses of touch, smell, taste, hearing and seeing. They are the routes by which the peripheral nervous system carries information from the organs and tissues of the body to the central nervous system.

SENSORY HEARING LOSS: A type of hearing loss due to dysfunction of the organ of Corti. The problem is most commonly located in the external hair cells.

SENSORY NEURON: A nerve cell that transmits information about changes in the internal and external environment to the central nervous system. Sensory neurons are of two types: somatic sensory neurons and visceral sensory neurons. Somatic sensory neurons occur in peripheral nerves in the skin, skeletal joints and bones. Visceral sensory neurons occur in sympathetic and parasympathetic nerves in the heart, lungs and other organs. They are activated by sensory input such as vision, touch and hearing and send projections into the central nervous system that convey sensory information to the brain or spinal cord.



Sensory neuron

SENTENTIAL CALCULUS (PROPOSITIONAL CALCULUS; PROPOSITIONAL LOGIC; SENTENTIAL LOGIC): In logic, the formal theory of which the intended interpretation concerns the logical relations between sentences treated only as a whole and without regard to their internal structure. Some examples are "and" (conjunction), "or" (disjunction), "not" (negation) and "if" (but only when used to denote material conditional).

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SENTENTIAL FUNCTION (OPEN SENTENCE; PROPOSITIONAL FUNCTION): In logic, a sentence containing one or more variables that can be either true or false depending on the value of the variable(s). Examples are $3x = 5$ and $3x + y = 4$.

SEPAL: Parts of a flower which are usually green but brightly coloured in some plants like the lily and the orchid or a modified leaf in the outermost whorl calyx of a flower that encloses the petals and other parts. They surround and protect the flower and bud. They have a simple structure and collectively are known as the calyx.

SEPALOID: Resembling a sepal in colour and texture.

SEPARABLE: 1. Of a topological space, containing a countable dense subset; every compact metric space or second countable space is separable, as is Euclidean space since it contains the rational n-tuples, which are countable and dense. 2. Of a function, written so that the variables are separated, additively or multiplicatively, as, for example, $f(x,y,z)=f_1(x)+f_2(y)+f(z)$. 3. Of a polynomial, having irreducible factors with no repeated roots. For example, $2x^2y+4y=2y(x^2+2)$, $x^2+y^2-4x+6y+13=(x-2)^2+(y+3)^2$. Optimisation procedures can be applied to each of the variables individually, where a function is separable. 4. Of an extension field, having the property that every element in the extension has a minimum polynomial that is separable. Every extension of a field of characteristic zero is separable. 5. Of a first order ordinary differential equation, having the property that it can be written so that the coefficients of the differentials of the independent and dependent variables are, respectively, functions of these variables alone. It is of the form $\int \frac{1}{g(y)}\mathrm{d}y = \int h(t)\,\mathrm{d}t + \mathrm{A}$.

SEPARATES (SOIL SEPARATES): Any of a group of rock or mineral particles (sand, silt, and clay), separated from a soil sample, having diameters less than 2 mm and ranging within the limits of one of the standard classifications of soil particle size.

SEPARATE CONCENTRATE FEEDING: A winter feeding system for livestock in which the animals are allowed free feeding of roughage and concentrates are fed separately in restricted quantity.

SEPARATE POINTS: Of an algebra, satisfying the condition that given any two distinct points of the set there is a member of the algebra for which the value at the points differs.

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SEPARATED: 1. Of two sets in a topological space, having the property that neither intersects the closure of the other. A space is connected if and only if it cannot be written as a union of two non-empty separated sets. 2. Constituting a Hausdorff space.

SEPARATING: In mathematics, of a set of real-valued functions, having the property that for any x and y in the domain, there is a function, f in the set for which $j(x) \neq f(y)$.

SEPARATING FUNNEL (SEPARATION FUNNEL; SEP FUNNEL): A type of laboratory conical glassware with a hemispherical end and a stopper at the top and stopcock, at the bottom that is used to separate a mixture into two immiscible solvent phases of different densities. If a mixture of two liquids are placed inside the separating funnel, it shows two layers with the denser liquid (known as aqueous solution, for example, iodine in aqueous solution) sinking to the bottom and the lower density one (known as organic solvent, for example, chloroform or ethyl acetate) floating on top. When the stopcock on the separating funnel is opened, the liquid with the higher density starts to flow into a container placed underneath the funnel. Before the liquid with the lower density starts to flow into the container below, the stopcock is closed. The liquid with the lower density remaining in the separating funnel can then be drained into a different container to separate the two liquids.

SEPARATION AXIOM: Any of a number of possible additional axioms for a topological space that assert at least the existence of open sets that contain one but not the other of every pair of points.

SEPARATION OF VARIABLES (FOURIER METHOD): Any of several methods for solving ordinary and partial differential equations, using algebra to rewrite the equation with each of the two variables occurring on a different side of the equation.

SEPARATION THEOREM OF MAZUR (GEOMETRIC FORM OF THE HAHN-BANACH THEOREM): The theorem that in n-dimensional Euclidean geometry, disjoint convex sets lie entirely on opposite sides of some closed hyperplane. This requires one of the sets to have a non-empty topological interior.

SEPSIS: Infection of the soft tissues or blood by pathogenic microorganisms arising usually after they have entered the body through a skin wound. It results in the destruction of the tissue by the pathogen or their toxin.

SEPT-: Prefix for seven; for example, a septangle is a geometric figure that has seven sides and seven connecting edges.

SEPTATE: Divided by one or more partitions into locules or cells.

SEPTIC TANK: A small-scale sewage treatment system comprising of a watertight receptacle, usually made of concrete or fibreglass in which solid organic sewage is decomposed and purified by anaerobic bacteria. It is commonly found in areas that lack connection to main sewage.

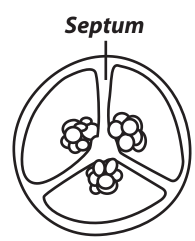
SEPTICAEMIA (BACTEREMIA): Bacterial infection of the bloodstream, often characterised by elevated body temperature, chills and weakness.

SEPTICIDAL: A term used to describe fruit dehiscence in which the slits in the pericarp arise along the junctions of the carpels.

SEPTIVALENT (HEPTAVALENT): An element having a valency of seven.

SEPTORIA LEAF BLOTCH: A fungal disease which causes numerous leaf spot diseases on field crops, forages and many vegetables and is responsible for yield losses. It is caused by *Mycosphaerella graminicola* (*Septoria tritici*), a species of filamentous fungus, which affects wheat and occasionally other grasses including barley. It is the major disease of wheat in the United Kingdom. *Septoria apiicola* causes late blight of celery. The symptoms include chlorotic spots that turn brown and necrotic.

SEPTUM: 1. A partition dividing something into two or more cavities. Examples include each of the muscular membranes separating the chambers of the heart, the tissue separating the nostrils and the internal dividing walls in the seed heads of poppies. 2. A thin partition that separates components in a machine. 3. Thin walls or partitions between the internal chambers of the shell of a cephalopod, namely nautiloids or ammonoids. As the creature grows, its body moves forward in the shell to a new living chamber, secreting septa behind it. This adds new chambers to the shell, which can be clearly seen in cross-sections of the shell of the living nautilus or in ammonoid and nautiloid fossils.



SEQUENCE: 1. An ordered set of elements whose domain of definition is the set of positive integers that can be put into a one-to-one correspondence with the set of positive integers. 2. A collection of numbers in a prescribed order such as a_1, a_2, a_3, ... It may be used to generate subsequent numbers in the list. 3. The order in which subunits appear in a chain, for example, amino acids in a polypeptide or nucleotide bases in a dNA or rNA molecule.

SEQUENCE ANALYSIS: The process of characterising sequences of biomolecules, particularly the nucleotides of nucleic acids or the amino acids of proteins. Computer software can be used to analyse the sequence data once the order of nucleotides has been established. This helps to identify features like open reading frames, promoters, enhancers and repetitive DNA.

SEQUENCE DATABASE: A database containing the sequences of biomolecules, which may be the sequences of nucleotides for nucleic acids or of amino acids for proteins. The data also include relevant annotations. A database can include sequences from only one organism, as in databases including all the proteins in *Saccharomyces cerevisiae* or it can include sequences from all organisms whose DNAs have been sequenced.

SEQUENCE-TAGGED SITE (STS): A unique short sequence of DNA, typically less than 400 nucleotides long that can be used as a tag or marker in physical mapping of cloned DNA segments. STS is derived by sequencing a short length of a particular clone, then using the sequence data to design primers for the polymerase chain reaction.

SEQUENCING (PROTEINS, NUCLEIC ACIDS): Analytical procedures for the determination of the order of amino acids in a polypeptide chain or of nucleotides in a DNA or RNA molecule.

SEQUENT: In logic, it is the formal representation of an argument in a logical calculus, as a set of premises and a conclusion, usually written using the tumstile or gatepost symbol '$\vdash$'. For example, the inference of A from A & B is written A & B $\vdash$ A. The sequent $\vdash$ A represents the derivation of A from no assumptions and thus indicates that A is a theorem. A sequent may have any number of condition formulae, called antecedents and any number of asserted formulae, called succedents or consequents. If all of the antecedent conditions are true, then at least one of the consequent formulae is true. **Sequent calculus** is any logical calculus presented in terms of line-by-line logical arguments.

SEQUENTIAL CONVERGENCE: In mathematics, it is the convergence of a sequence, as contrasted with net convergence.

SEQUENTIAL DECISION MAKING: In statistics, a stepwise decision making aiming to identify short-term strategies in the face of long-term uncertainties, by incorporating additional information over time and making mid-course corrections.

SEQUENTIAL MEASURING CELL: A measuring cell which operates according to a succession of operations on the sample or on the sensitive elements (or on both), these operations being carried out according to one or more repetitive programs.

SEQUENTIAL SAMPLE: Units, increments or samples taken one at a time or in successive predetermined groups, until the cumulative result of their measurements (typically applied to attributes), as assessed against predetermined limits, permits a decision to accept or reject the population or to continue sampling. The number of observations required is not determined in advance, but the decision to terminate the operation depends, at each stage, on the results of the previous observations. The plan may have a practical, automatic termination after a certain number of units have been examined.

SEQUENTIAL SPECTROMETER: A spectrometer which enables the intensity of several spectral bands of radiation to be measured one after the other in time.

SEQUENTIALLY COMPACT SET: Of a set in a topological space, having the property that every sequence contains a convergent subsequence with limit in the set. If the limit is not necessarily in the set, it is called **relative sequential compactness**. In a metric space or in the weak topology of a Banach space, sequential compactness and compactness coincide for closed sets; so that in a metric space such as the reals, a set is sequentially compact if and only if it is compact.

SEQUESTRATION: 1. In chemistry, the process of forming coordination complexes of an ion in solution. Sequestration often involves the formation of chelate complexes and is used to prevent the chemical effect of an ion without removing it from the solution. 2. Also known as **carbon sequestration**, it is the process of removing the atmospheric carbon dioxide (CO_2) content and storing it in a reservoir or carbon sinks (such as oceans, forests or soils) through physical or biological processes. Biological approaches to sequestration include direct removal of carbon dioxide from the atmosphere through land-use change, afforestation, reforestation, and practices that enhance soil carbon in agriculture. Physical approaches include separation and disposal of carbon dioxide from flue gases or from processing fossil fuels to produce hydrogen- and carbon dioxide-rich fractions and long-term storage in underground in depleted oil and gas reservoirs, coal seams, and saline aquifers.

SERE: A complete succession of plant communities which results in the climax community. It is composed of a series of different plant communities that change with time, known as seral stages or seral communities. A **seral community** is the name given to each group of plants within the succession. A primary succession describes those plant communities that occupy a site that has not previously been vegetated. An example of seral communities in secondary succession is a recently logged coniferous forest. During the first two years, grasses, heaths and herbaceous plants such as fireweed will be abundant. A seral community depends on the substratum and climate and includes the following: **hydrosere** is a community developed in water; **lithosere** is a community developed on rock; **psammosere** is a community developed on sand; **xerosere** describes a community developed in dry area; **halosere** describes a community developed in saline body such as a marsh.

SERIAL: In mathematics, of a relation, connected, transitive and asymmetric, thereby imposing an order on all the members of the domain, as less than on the natural numbers.

SERIAL CORRELATION (AUTOCORRELATION): A measure of the correlation of a particular time series delayed by k lags (the distance between the observations that are so correlated). It is obtained by dividing the covariance between two observations, separated by k lags, of a time series (auto-covariance) by the standard deviation y_t and y_{k-1}. If the autocorrelation is calculated for all values of k the autocorrelation function is obtained. For a time series that does not change over time, the autocorrelation function decreases exponentially to zero.

SERIAL DILUTION: Successive dilution of a specimen solution. Usually the dilution factor at each step is constant, resulting in a geometric progression of the concentration in a logarithmic fashion. For example, 1:10 dilution equals 1 ml of a specimen plus 9 ml of diluent (e.g., sterile water); 1:100 dilution equals 1 ml of a 1:10 dilution plus 9 ml of diluent.

SERIAL LINE INTERNET PROTOCOL (SLIP): In computer science, it is a method of connecting a computer to the Internet through a dial-up connection, using a serial line, such as telephone circuits and RS-232 cables.

SERIAL MOUSE: A mouse that connects to a computer via a serial port. A serial port is a general-purpose interface that can be used for almost any type of device, including modems, mice and printers, although most printers are connected to a parallel port. Mice can also use the PS/2 and USB ports.

SERIAL PORT (COM PORT; RS-232 PORT): A type of connection on PCs that is used for peripherals such as mouse, gaming controllers, modems and older printers. A serial port can only transmit one bit of data at a time, whereas a parallel port can transmit many bits at once.

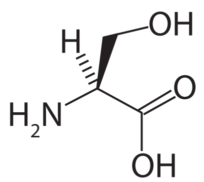
SERIATE: Arranged in rows or series.

SERICEOUS: Covered with long, soft, slender, silky hairs.

SERICULTURE: Raising silkworms for the production of silk. It involves the incubation of the tiny eggs of the silkworm moth until they hatch and become worms. After hatching, the worms are placed under a layer of gauze, on which is spread a layer of finely chopped mulberry leaves. For six weeks, the worms eat almost continuously. At the end of this period, they are ready to spin their cocoons.

SERIES: 1. The sum of the terms of a sequence, each of which can be written in a form that is an algebraic function of its position or a number of things, events, etc., each of which is related in some way to the others, especially to the one before it or the indicated sum of a finite or infinite sequence of terms, with each term being added to those that precede it. An **arithmetic series** is the sum of the terms of an arithmetic sequence. For example, 3 + 7 + 11 + 15 + ⋯, which has a constant difference of 4 between terms. A **geometric series** is the sum of the terms of geometric or exponential sequence, for example 2 + 4 + 8 + 16 +... 2. A group of related chemicals that is similar in structure or properties. 3. Also known as **series circuit**, it is a set of two or more electrical or electronic components connected end to end that the entire current (I) passes through each element without division or branching into parallel circuits. For resistors in series, the total resistance R, is the sum of all the resistances: $R = r_1 + r_2 + r_3$ ... . For capacitance in series, total capacitance $C = c_1 + c_2 + c_3$ ... . 4. A succession of rock strata deposited during a particular period of geologic time. 5. A rank in the taxonomic hierarchy above species but subordinate to section.

SERINE(2-AMINO-3-HYDROXYPROPANOIC ACID): White crystalline nonessential amino acid, present in many proteins and also an important component of phospholipids with the chemical formula $C_3H_7NO_3$, density 1.603 g/cm^3 (22°C, 295 K), melting point 246°C (519 K) and decomposes. It is used in biochemical and microbiological research and as a dietary supplement and feed additive.



Chemical Structure of Serine

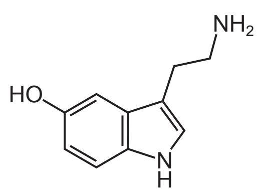
SEROCONVERSION: The stage in an immune response when antibodies to the infecting agent are first detected in the bloodstream. For example, people infected with HIV typically seroconvert about 4-6 weeks following the initial infection, when antibodies against vital proteins are first produced. Serology, which is the testing for antibodies, is used to determine antibody positivity. Prior to seroconversion, the blood test is seronegative for the antibody; after seroconversion, the blood test is seropositive for the antibody.

SEROLOGY: The scientific study of body fluids, for the purpose of detecting and treating illness or the laboratory study of blood serum and its constituents, particularly antibodies and complements which play a part in the immune response.

SEROTINOUS: Late in flowering, with flowers developing or blooming after the leaves are fully developed.

S

SEROTONIN (5-HYDROXYTRYPTAMINE; 5-HT): A compound with the chemical formula $C_{10}H_{12}N_2O$, melting point 121 °C (394 K) with a boiling point of 416 °C ±30.0 °C (at 760 Torr). It is derived from amino acid tryptophan, which is derived from foods such as meat and dairy products high in protein. Serotonin transmits messages across the synapses or gaps between adjacent cells. It is released from blood cells called platelets to activate blood vessel constriction and blood clotting. In the gastrointestinal tract, serotonin inhibits gastric acid production and stimulates muscle contraction in the intestinal wall. It is also believed to function in the central nervous system and controls mood, memory and appetite.



Chemical Structure of Serotonin

SEROUS MEMBRANE (SEROSAL): A tissue consisting of a layer of mesothelium (a thin layer of cells) attached to a surface by a thin layer of connective tissue. Examples are pleura, peritoneum and pericardium serous. Serous membranes line body cavities that do not open to the exterior where they secrete a lubricating fluid which reduces friction from muscle movement.

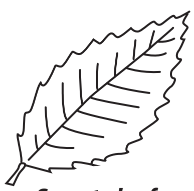
SERPENTINE: 1. Any of a group of hydrous magnesium silicate minerals with the general composition $Mg_3Si_2O_5(OH)_4$. They occur in two forms: chrysotile, which is fibrous and the chief source of asbestos; and antigorite, which occurs as platy masses. Serpentine is usually greyish, white or green but may be yellow or green-blue. It has a high polish and is sometimes used as an ornamental stone. 2. In mathematics, it is a curve that is symmetric about the origin and asymptotic to the x-axis. Its canonical equation is $x^2y + b^2y - a^2x = 0$.

SERPENTINE LOCOMOTION (LATERAL UNDULATION): A type of movement in snakes characterised by waves of lateral bending along the body from head to tail.

SERPIN: Any of a group of proteins found in all groups of organisms except fungi. They are characterised by irreversibly inhibiting serine proteases involved in a wide array of metabolic processes including regulation of complement activation, blood coagulation, tumour suppression, inflammation and blood vessel formation. The most abundant serpin in human plasma is α_1-antitrypsin, which inhibits trypsin and most significantly elastase released by leucocytes.

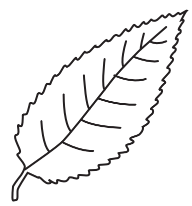
SERRET-FRENET FORMULAE (FRENET FORMULAE): The fundamental formulae that describe the kinematic properties of a particle moving along a continuous, differentiable curve in three-dimensional Euclidean space or the geometric properties of the curve itself irrespective of any motion. In other words, the formulae are for a space curve that recaptures the unit vector tangent (to the curve, pointing in the direction of motion) denoted T, the normal unit vector (the derivative of T with respect to the arclength parameter of the curve, divided by its length), denoted N and the binormal unit vector B (the cross product of T and N), from the curvature κ and torsion τ of the curve. The formulae are: (i) $\mathrm{N}' = -\kappa\mathrm{T} + \tau\mathrm{B}$, (ii) $\mathrm{B}' = -\tau\mathrm{N}$ and (iii) $\mathrm{T}' = \kappa\mathrm{N}$.

SERRATE: A term used to describe a leaf margin that is toothed, with forward-pointing notches. Leaf margins finely toothed in this manner are termed **serrulate**. **Serra** is the tooth of a serrate leaf. **Serration** is a serrated condition or one of the teeth along a serrated margin.



Serrate leaf

SERRULATE: Toothed along the margin with minute, forward pointing teeth, as in a leaflet of the rose.



Serrulate leaf

SERTOLI CELLS: Cells that line the seminiferous tubules in the testis. They protect the spermatids and convey nutrients to both the developing and matured spermatozoa. They produce the hormone, inhibin, which can inhibit follicle-stimulating hormones and thereby regulate production of spermatozoa.

SERUM: A clear fluid that remains after blood clots or blood plasma from which substances that cause blood to clot has been removed or the fluid that separates from clotted blood, similar to plasma, but without clotting agents. Serum includes all proteins not used in blood clotting and all the electrolytes, antibodies, antigens, hormones and any exogenous substances such as drugs and microorganisms.

SERVER: A computer that provides services to other computers, called **clients**. Servers include web servers, mail servers and file servers. Servers can be attached to local area networks and/or be hooked up to the internet. With the proper software and connections, servers can control the distribution of email, store World Wide Web documents and provide access to files that are shared by many users. For example, when you click on a link in a Web page your Browser sends a request to a remote computer, known as a Web Server, which serves the requested page to your browser, which then displays it on your computer screen.

SERVER MESSAGE BLOCK (SMB): A network protocol used by Windows-based computers that allows systems within the same network to share files. It allows computers connected to the same network or domain to access files from other local computers as easily as if they were on the computer's local hard drive.

SERVICE PACK (SP): A large update containing several fixes and updates for Microsoft Windows operating systems. Service packs are a collection of fixes that help resolve software and hardware related issues with Microsoft Windows.

SERVICE PROVIDER (INTERNET SERVICE PROVIDER; ISP): A company or organisation that provides internet access. Service providers often provide e-mail capability, Web hosting and other services in addition to connectivity. The ISP connects to its customers using a data transmission technology appropriate for delivering Internet Protocol datagrams, such as dial-up, DSL, cable modem, wireless or dedicated high-speed interconnects.

SERVOMECHANISM: In computer terms, it is an automatic control device which controls the position, velocity or acceleration of a high-power output device by means of a command signal from a low-power reference device. By feedback, the error between the actual output state and the state commanded is measured, amplified and made to drive a servomotor which corrects the output. The drive may be electrical, hydraulic or pneumatic.

SESAMOID BONE: Any of several small round bones such as kneecap (patella) embedded in a tendon or joint capsule. They are also found in the hand, for example, at the base of the first metacarpal and also in the wrist; and also in the foot, where the first metatarsal connects to the big toe. Sesamoid bones help to protect and increase the mechanical effect of tendons and also to prevent the tendons from pressing into the joint.

SESQUI-: A prefix denoting a ratio of 2:3 in a chemical compound. For example, a sesquioxide has the formula M_2O_3.

SESQUILINEAR: In mathematics, of a function of two variables on a complex vector space, having the property that it is linear in the first variable and conjugate linear in the second variable. This applies for the inner product.

SESSILE: 1. Animals called sedentary animals that live permanently attached to a surface. These include many marine animals such as sea anemones and limpets. 2. Attached directly to a branch or stem without a stalk, for example an oak leave. An example is an oak leave.

SESSION: In computing, it refers to a limited time of communication between two systems. Some sessions involve a client and a server, while other sessions involve two personal computers. A session begins when a user logs in to or accesses a particular computer, program or web page and ends when the user logs out of or shuts down the computer, closes the program or web page.

SESSION INITIATION PROTOCOL (SIP): A protocol defined by the Internet Engineering Task Force (IETF). It is used for establishing sessions between two or more telecommunications devices over the Internet. SIP has many applications, such as initiating video conferences, file transfers, instant messaging sessions and multiplayer games.

SET: 1. Also known as **class**, it is a well-defined collection of elements in mathematics or logic, for example, numbers or terms or a collection of objects. Members of a set usually have something in common and may be defined by a membership rule or formula or by listing its members within braces, {}. For example, the set of common domestic animals may include {cat, dog, cow, sheep, etc.}. An object x that is a member of a set A is written as $x \in$ A, if x is not a member of A, it is written $x \notin$ A. The operator, $\subset$ is used to denote a union between two sets, for example, the union between two sets, A and B is written as A $\subset$ B. The intersection symbol, $\cap$ is used when two sets, A and B intersect, which is defined as the set composed of all elements that belong to both A and B and read as "A intersection B" or "the intersection of A and B". 2. To adjust a mechanical or electronic device to a desired level. 3. In agriculture, a set means the number of eggs a hen lays at a time; or seedlings that are ready to be transplanted. 4. To become bent from strain.

SET STOCKING: A grazing system associated with extensive grazing.

SET THEORY: 1. In mathematics, it is the branch of mathematics that studies the properties of well-defined collections of objects (finite sets or classes) and their relations; it also extends to the study of the properties of infinite sets. 2. In logic, set theory is a theory constructed within first-order predicate calculus that yields the mathematical theory of classes, especially one that distinguishes sets from proper classes as a means of avoiding certain paradoxes. In axiomatic set theory the consequences of various sets of axioms are studied in the abstract, while naive set theory models the intuitive properties of sets as the consequences of a set of interpreted axioms.

SET-ASIDE: To use a piece of formerly arable land for something other than growing crops.

SET-VALUED FUNCTION (MULTIVALUED FUNCTION; MULTIFUNCTION; CARRIER; POINT-TO-POINT MAPPING): A mapping that associates at least one input element with multiple output elements. It is a mapping from a set into the power set of another set. In mathematics or a mapping that associates a number of different elements of the output with the same element of the input; a mapping from a set into the power set of another set. The inverses of trigonometric, hyperbolic, exponential and integer power functions are multivalued. For example, the function z^2 maps each complex number z to a well-defined number z^2, while its inverse function $\sqrt{z}$ maps, for example, the value $z = 1$ to $\sqrt{1} = \pm 1$

SETA: 1. The stalk of a bryophyte sporophyte that supports the capsule. At maturity, the seta usually elongates rapidly, bearing the capsule to a height suitable for spore dispersal. A seta often contains vascular tissue, enabling the sporophyte capsule to obtain nutrients from the parent gametophyte. 2. A nerve-like extension to a leaf, as seen in certain species of Selaginella.

SETON: To foster an orphaned animal to another female as a lamb onto an ewe.

SETOSE: Covered with bristles. **Setulose** means covered with minute bristles.

SETSQUARE: An instrument in the form of a right-angled triangle used for drawing parallel lines. It is used in engineering and technical drawing, with the aim of providing a straight edge at a right angle or other particular planar angle to a baseline. A simple set square consits of a triangular piece of transparent plastic with the centre removed. It usually comes in two forms, both right triangles: one with 90-45-45 degree angles, the other with 30-60-90 degree angles. Combining the two forms by placing the hypotenuses together will also yield 15° and 75° angles.

SETTLING CHAMBER: A chamber designed to reduce the velocity of gases in order to permit the settling out of fly ash. It may be part of, adjacent to or external to an incinerator.

SETTLING ERROR: An error caused by allowing insufficient time for the reading to settle or if an absorption peak is scanned too rapidly.

SETTLING VELOCITY: The terminal rate of fall of a particle through a fluid as induced by gravity or other external force.

SETTO: An orphaned animal given to a foster mother.

SEVERE COMBINED IMMUNODEFICIENCY (SCID): A genetic disorder in which afflicted individuals have no functional immune system and are prone to infections. Both the cell-mediated immune response and the antibody-mediated response are absent. Children with SCID are vulnerable to recurrent severe infections, retarded growth and early death.

SEWAGE (SEWERAGE): Waste matter from industrial or domestic sources that is dissolved or suspended in water. Raw sewage is a pollutant. It contains a high content of organic matter and therefore provides a rich source of food for many decomposers and detritivores, some of which are pathogenic to humans.

SEWAGE DISPOSAL: The disposal of human excreta and other waste products from houses, streets and factories, conveyed through sewers to sewerage works.

SEWAGE SLUDGE (SLUDGE): A semiliquid waste with a solid concentration in excess of 2500 parts per million, obtained from the purification of municipal sewage.

SEWALL WRIGHT EFFECT (GENETIC DRIFT): A change in gene frequency that is due purely to chance, as opposed to natural selection. Genetic drift is more likely to occur in very small populations, since in such populations mating is likely to be nonrandom compared to matings in larger populations.

SEX- (SEXI-; SEXTI-): A prefix that denotes six; for example, a sexivalent is an element having a valency of six.

SEX: 1. Either of the two reproductive categories, male or female, of animals and plants. 2. The set of characteristics that determine whether the reproductive role of an animal or plant is male or female.

SEX CELL: A reproductive cell in an organism that reproduces sexually. Examples are sperm cells and ova in animals and pollen and ovules in flowers.

SEX CHROMOSOMES: Chromosomes that determine the sex of an organism. Many animals have two different types of sex chromosomes: XY in males and XX in females. In other animal groups such as butterflies and moths, birds and reptiles, the situation is reversed; males have two matching W chromosomes while the female has one W and Z chromosomes. The X chromosome is larger and carries more genetic information than the Y. Traits such as haemophilia, red-green colour blindness, which are controlled only by genes found on the X chromosome are said to be sex-linked.

SEX DETERMINATION: 1. Also known as **cell determination**, it is the commitment of an undifferentiated embryonic cell to a particular path of differentiation, even though there may be no morphological features that reveal this determination. It is generally irreversible, but in the case of imaginal discs of drosophila that are maintained by serial passage, transdetermination may occur. 2. The enetic mechanisms of establishing the sex of an organism, usually by the inheritance at the time of fertilisation of certain genes commonly locasised on a particular chromosome. In animals, it is determined by the presence or absence of the Y chromosome. There are two aspects of sexuality: the primary form involves the gametes and the secondary aspect is gender. The sex of the foetus before birth can be performed by examination of foetal fluids obtained by a process called amniocentesis.

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SEX FACTOR: A plasmid or an episomal bacterial sex factor found in the cytoplasm of certain bacteria that is responsible for initiating conjugation and gene transfer.

SEX HORMONES (SEX STEROID; GONADAL STEROIDS): Steroid hormones that control sexual development. The most important ones are the androgens and oestrogens. Natural sex steroids are made by the gonads (ovaries or testes), by adrenal glands or by conversion from other sex steroids in other tissue such as liver or fat. There are also many synthetic sex steroids. Synthetic androgens are often referred to as anabolic steroids.

SEX LINKAGE: The tendency for certain inherited characteristics occurring far more frequently in one sex than the other. For example, red-green colour blindness and haemophilia affect men more often than women. This is because the genes governing normal colour vision and blood clotting occur on the X sex chromosome. Women have two X chromosomes. If one carries an abnormal allele it is likely that its effects will be masked by a normal allele on the other X chromosome. However, men have only one X chromosome and any abnormal alleles, therefore, will not be masked.

SEX RATIO: The ratio of the number of females to males in a population. Because the mortality rates in both sexes may be different, the sex ratios in different age classes may differ. The **primary sex ratio** is the ratio at the time of conception, **secondary sex ratio** is the ratio at time of birth and **tertiary sex ratio** is the ratio of mature organisms.

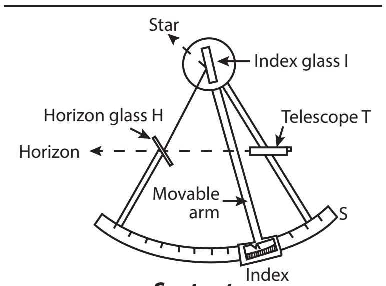
SEX-LIMITED INHERITANCE: The restriction of a trait to one sex as a result of incomplete penetrance rather than allelic differences.

SEXAFS (SURFACE-EXTENDED X-RAY ABSORPTION FINE-STRUCTURE SPECTROSCOPY): A technique that makes use of the oscillations in x-ray absorbance, found in the high-frequency side of the absorption edge to investigate the structure of surfaces. This technique involves the illumination of the sample by high intensity x-ray beams from a synchrotron and monitoring their photo absorption by detecting in the intensity of Auger electrons as a function of the incident photon energy.

SEXAGESIMAL: A number system with the base 60 used in measuring time, angles and geographic coordinates. The number 60, has twelve factors, which includes 1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60 of which 2, 3 and 5 are prime numbers. With so many factors, many fractions involving sexagesimal numbers are simplified. For example, one hour can be divided evenly into sections of 30 minutes, 20 minutes, 15 minutes, 12 minutes, 10 minutes, six minutes, five minutes, etc. It originated with the ancient Sumerians in the 3rd millennium BC.

SEXIVALENT (HEXAVALENT): Elements having a valency of six. Examples include sulfur in SF_6 and SO_3, tungsten in WCl_6, manganese in manganese trioxide, MnO_3 and manganates, K_2MnO_4. All elements are classified in nine groups, according to their valencies. The same element may fall into several groups, since the valency is not a constant for an element but may vary according to the nature of the other element with which it is combined, for example, with hydrogen (HCl, H_2S) or with oxygen (Cl_2O_7, SO_3).

SEXTANT: An instrument used in navigation to measure the angle between the horizon and a celestial body such as the Sun, the Moon or a star. It uses two mirrors called the horizon glass and the index glass and an arm attached to the index mirror to take angle readings. The horizon is observed through the clear glass of a fixed horizon glass H by applying the eye to the eye piece of the telescope T. The index glass I is rotated until the image of a star is reflected from I into the lower silvered half of H through to T. The angle is read from the graduated scale S at the point indicated by an arm attached to I.



Sextant

SEXTILE: 1. In statistics, one of five values of a variable that divide its distribution into six equiprobable intervals; for example, the fifth sextile is the value of the variable below which lie 5/6 of the population. 2. An aspect formed when two planets or celestial bodies are 60° distant from each other.

SEXUAL DIMORPHISM: A special case of polymorphism based on the distinction between the secondary sex characteristics of males and females. For example, one obvious of the secondary sexual characteristics is that males, when adequately nourished, tend to be physically larger in size than females.

SEXUAL INTERCOURSE (COITUS; COPULATION; MATING): The act in which the male reproductive organ enters the female reproductive tract. In mammals, erected penis is inserted into the vagina of the female. Thrusting movements of the penis result in ejaculation in which semen containing spermatozoa is deposited into the vagina.

SEXUAL REPRODUCTION: Reproduction involving male and female sex cells or reproductive processes in organisms that require the union of two sex cells (gametes) such as eggs and sperm.

SEXUAL SELECTION: The means by which it is assumed that certain secondary sexual characteristics particularly of male animals have evolved. These features would be inherited by its male offspring and would thus tend to become exaggerated down the generations. For example, females tend to mate with the male that give the best courtship display and, therefore has the brightest colouration. These features would be inherited by its male offspring and would thus tend to become exaggerated down the generations.

SEXUALLY TRANSMITTED INFECTION (STI; SEXUALLY TRANSMITTED DISEASE, STD; VENEREAL DISEASE, VD): Any disease that is passed on from one individual to another through sexual contact including vaginal intercourse oral sex and anal sex. Some STIs can also be transmitted by sharing drug needles or syringes, as well as through childbirth or breastfeeding. Examples of STIS are gonorrhoea, syphilis and HIV/AIDS.

SEYFERT GALAXY: A type of galaxy, typically a spiral or barred spiral with an active galactic nucleus. The nucleus is brighter than the spiral arms, too bright to derive its radiation only from stars. It is thought that there is probably a low-power quasar-like object at the centre.

SH_2 DOMAIN (SRC HOMOLOGY 2 DOMAIN): A functional unit of protein that forms a binding site for another protein.

SHADE: Comparative darkness caused by the cutting off of direct rays of light. A **shade plant** is a plant that is able to flourish in conditions of low light intensity. Examples of shade plants are mosses, enchanter's nightshade (*Circaea lutetiana*), certain violets (Viola) and dog's mercury (*Mercurialis perennis*). Shade plants usually have thinner epidermal and palisade layers and fewer stomata than normal. They also tend to have a short compensation period, i.e. the food reserves used in respiration at night are quickly replaced by photosynthesis in the day.

SHADOOF: A water-raising device consisting of a suspended pivoting pole with a bucket on one end and a counterweight on the other used in ancient Egypt to pump water for irrigation purposes.

SHADOW: An area of darkness formed on a surface when the path of light is blocked by an opaque object. Shadows usually take the shape of the objects on which the light is cast.

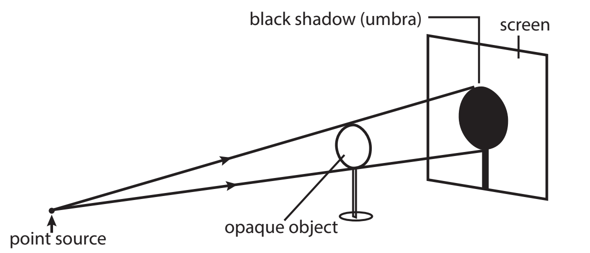


Fig. I: Shadow from a point source of light

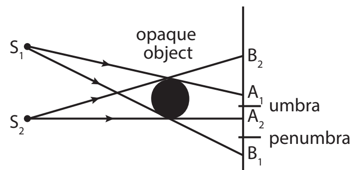


Fig. II: Shadow from an extended source of light

SHADOWING (HEAVY-METAL SHADOWING): A technique used in electron microscopy to determine the scale of surface features on a specimen. The specimen is partially coated with a thin layer of a heavy metal, for example, platinum, palladium or gold, which is evaporated in a vacuum chamber.

SHADOW ZONE: Area of the Earth from angular distances of 104 to 140 degrees shielded from earthquake waves by the outer core, where stationary waves (S-waves) are absorbed and primary waves (P-waves) are refracted.

SHALE: A fine-grained dark sedimentary rock composed of layers of compressed silt, clay or mud. Shale constitutes roughly 60% of the sedimentary rock in the Earth's crust. They are valuable raw material for tile, brick and pottery and constitute a major source of alumina for Portland cement.

SHALLOT: A mildly aromatic herbaceous plant (*Allium cepa L.*, var. *aggregatum* and *A. oschaninii*) of the family Alliaceae, used to flavour foods. Closely related to the onion and garlic, the shallot is a hardy perennial with short, small, cylindrical and hollow leaves. It has lavender to red flowers in a compact umbel; and small, elongated, angular bulbs. Much like garlic, the bulbs develop in clusters on a common base. The leaves are sometimes eaten when green.

SHANK: The lower part of a horse's leg between the knee and the foot.

SHANNON ENTROPY (ENTROPY): A random variable X with values in a finite set X, defined as $H(X) \equiv -\sum_{x} P(x)\log_2[P(x)]$, where $P(X)$ is the probability that X is in the state x and $P\log_2 P$ is defined as 0 if $P = 0$. The joint entropy ofvariables (X_1, ..., X_n) is defined by $H(X_1,...,X_n) \equiv -\sum_{x_1}...\sum_{x_n} P(x_1,...,x_n)\log_2[P(x_1,...,x_n)]$,

SHAPE: A geometric form such as a triangle, square, cube or cone which may be 2-dimensional or 3-dimensional.

SHAPE-MEMORY POLYMER: Polymer that resumes its original shape when heated above its glass-transition or melting temperature and being subjected to a plastic deformation. An example is crystalline *trans*-polyisoprene.

SHAPIRO TIME DELAY EXPERIMENTS (GRAVITATIONAL TIME DELAY EFFECT): Experiments in which the time taken for a radar pulse to travel to a nearby planet and return takes longer than they would if the mass of the object were not present. The time delay is caused by spacetime dilation, which increases the path length. The gravitational field of the Sun affects the time taken. These measurements confirm the predictions made by general relativity as to the amount of effect on time.

SHARE CROPPING: A system of land tenure, whereby tenants called share croppers pay an agreed portion of their crop to the landlord as rent.

SHAREFARMING: A joint enterprise between a party with an interest in the land and another party involved in the farming operations. Usually, one provides the capital and the other the farm management inputs such as labour and equipments.

SHAREWARE: A software that allows a user to try it free before buying it. Unlike freeware, shareware often has limited functionality or may only be used for a limited time before requiring payment and registration. Once you pay for a shareware program, the program is fully functional and the time limit is removed.

SHARP EYESPOT: A soil-borne fungus affecting cereals which can cause lodging and shrivelled grain.

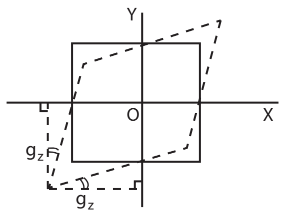
SHAVINGS: Thin curled pieces of wood removed and used as litter for animals.

SHEA BUTTER: A slightly yellowish or ivory coloured natural fat obtained from the seeds of the African shea butter tree, *Vietellaria paradoxa*. It is the only species in genus Vitellaria. The shea nut tree grows wild in the savannahs of west and central Africa. A tree can take from 18-25 years to produce the nuts from which shea butter is made. The shea fruit consists of a thin, tart, nutritious pulp that surrounds a relatively large, oil-rich seed from which shea butter is extracted. Shea butter is composed of fatty acids of palmitic, stearic, oleic, linoleic and arachidic. The high concentration of non–saponifiable fatty acids promote cellular growth, helping to restore damaged skin and the butter has been used for its ability to protect and regenerate the skin. The traditional use of the butter is as oil for cooking; and to reduce the appearance of fine lines, scars and stretch marks and to ease a variety of skin irritations, such as psoriasis, eczema and sunburn. It is an ingredient in skin and hair care products. It is also used in soap, candle and food manufacture and also in the chocolate industry as a substitute for cocoa butter.

SHEAF: 1. In mathematics, a family of planes that pass through a single point. 2. In agriculture science, it is a bundle of reaped and not yet threshed cereal plants such as wheat that are tied with one or two bonds.

SHEAR: 1. To cut hair, fleece or foliage from the surface of something using a sharp tool, for example, the removal of wool from sheep. A person who clips the fleece off a sheep is called a **shearer**. 2. To cause something to deform or break by applying a twisting force or to deform or break in this way.

SHEAR MODULUS: Shear stress divided by shear strain. Its symbol is *G*.



Shear strain

SHEAR RATE: The velocity gradient in a flowing fluid.

SHEAR STRAIN: Displacement of one surface with respect to another divided by the distance between them. Its symbol is γ.

SHEAR STRESS: A force acting tangentially to a surface divided by the area of the surface.

SHEAR THINNING: If viscosity is a univalued function of the rate of shear, a decrease of the viscosity with increasing rate of shear is called shear thinning and an increase of the viscosity shear thickening.

SHEAR TRANSITION: A diffusionless transition that involves a change of the shape of the unit cell by a process that can be described as shear.

SHEAR VISCOSITY: For a Newtonian fluid, the shear viscosity is often termed simply viscosity since in most situations it is the only one considered. It relates the shear components of stress and those of rate of strain at a point in the fluid.

SHEAR WAVE (S-WAVE): A seismic body wave that travels through the interior of the earth and is usually the second conspicuous wave to reach a seismograph. It shakes the ground back and forth perpendicular to the direction the wave is moving. S-waves move slower than the corresponding P-wave and cannot move through liquids at all. P-wave is a longitudinal earthquake wave that travels through the interior of the earth and is usually the first conspicuous wave to be recorded by a seismograph.

S

SHEARING FORCE: 1. A force directed parallel or at a tangent to a surface. A shear force tends to cause one portion of an object to slide, displace or shear with respect to another portion of the object. 2. A force identified on the basis of the formula; force = mass x acceleration, in which the mass component or resistance to be overcome, is more important than acceleration. Shear force is particularly important in weight-lifting.

SHEARING TRANSFORMATION: Transformation that maps parallel lines onto parallel lines. In the transformation, all points along a given line *L* remain fixed while other points are shifted parallel to *L* by a distance proportional to their perpendicular distance from *L*.

SHEARLING: A young sheep which has been sheared for the first time.

SHEARS: A large cutting instrument shaped like scissors.

SHEATH: 1. Cover of tissue, skin, such as the wing of some insects fitting over part of an animal or plant. 2. A contraceptive device used on the penis during sexual intercourse. It is also called **condom**.

SHED: 1. To separate one or more animals from a flock or herd. 2. To let leaves of grains fall.

SHEEP: Common name for a collection of grazing mammals of the genus Ovis in the family Bovidae, in the order Artiodactyla that may be either wild or domesticated (*Ovis aries*). Sheep are even-toed, hoofed animals. They are cud-chewing with the upper incisor teeth missing and with a four-compartmented stomach. You can tell the approximate age of a sheep by the teeth. A lamb is born with eight milk teeth at the front; a pair of milk teeth is replaced each year starting with the pair in the middle, so by the time a sheep is four years old, it is said to be full-mouthed. After this stage their teeth may deteriorate and they become broken mouth. Broken mouth is the term used for a sheep that has lost some of her teeth. They have paired, hollow, unbranched horns that are not shed. An adult may weigh 75 to 200 kg. It is bred for its flesh called **mutton** used as food, its wool for clothing and carpets and milk for drinking and cheese making. A female sheep is called an **ewe**, a male one is a **ram** and young sheep under one year old is known as a **lamb**.

SHEEP MAGGOTFLY: A type of maggot fly (*Lucilia sericata*) of the genus Lucilia that lays its eggs on the wool of sheep. The eggs hatch into maggots that burrow into the flesh causing a condition called strike.

SHEEP POX: A highly contagious viral disease caused by a poxvirus in the genus Capripoxvirus. Symptoms include fever, loss of appetite, difficulty in breathing etc.

SHEEP RUN: An extensive area used for sheep grazing.

SHEEP TICK (SHEEP KED): A small wingless dipterous insect, *Melophagus ovinus*, which is parasitic on sheep.

SHEEP WALK: An area of land on which sheep are pastured.

SHEET: 1. In mathematics (Euclidean geometry), any of the maximal continuous parts of a surface on which it is possible to draw a curve from any point to any other point without leaving the surface. A hyperboloid of two sheets is a surface of two sheets separated by a finite distance between the vertices. 2. In complex junctions, it is a portion of a Riemann surface.

SHEET EROSION: Erosion that takes place evenly over the whole area of a slope, caused by the runoff from saturated soil after heavy rainfall. A more or less uniform layer of fine particles is removed from the entire surface of an area, sometimes resulting in an extensive loss of rich topsoil. Sheet erosion commonly occurs on recently ploughed fields or on other sites having poorly consolidated soil material with scant vegetative cover.

SHEFFER'S STROKE: In logic, a truth function of two sentences, equivalent to the negation of their conjunction and denoted with the vertical bar and written P | Q, where P and Q are the arguments; in English, it denotes "not both". P | Q is false only when P and Q are both true and therefore also called nand ("not and") or the alternative denial, since at least one of its operands is false. In Boolean algebra and digital electronics it is known as the NAND operation.

SHELF-LIFE: The number of days or weeks for which a product can stay on the shelf of a shop and still be useful to consumers or buyers.

SHELL: 1. The space that contains a group of electrons orbiting around the nucleus of an atom or the orbit or energy level of electrons. 2. The hard outer covering of birds' eggs, nuts such as walnut or coconuts; some seeds such as peas; fruits; and of some animals such as oysters, lobsters and snails. 3. A metal case filled with explosives to fire from a large gun. 4. Also known as **command shell**, it is a program that works with the operating system as a command processor and used to enter commands and initiate their execution. It is a text-only user interface for Linux and other Unix-like operating systems and the most fundamental way that a user can interact with the system and the shell hides the details of the underlying operating system from the user. Examples of shells are MS-DOS Shell, Korn Shell, Bourne shell and the C Shell.

SHELL MODEL: A model of atomic nucleus in which nucleons are thought to move under the influence of a central field in shells that are similar to atomic electron shells. The nucleons act as independent particles filling a pre-assigned set of energy levels as permitted by the quantum numbers and Pauli principle.

SHELLAC: A hard resin produced as a secretion by a plant parasite, for example, the female lac bug. It is used in sealing wax, varnish and electrical insulators.

SHELTERBELT: A row of trees planted to give protection from wind.

SHELTERWOOD: A large area of trees left standing when others are cut, to act as shelter for seedling trees.

SHEPHERD: A person who looks after sheep.

SHERARDISING (SHERARDIZING; VAPOUR GALVANISING): The process of coating iron or steel with a zinc corrosion resistant layer by heating the iron or steel in contact with zinc dust to about 370 °C (643 K). The two metals amalgamate to form internal layer of zinc-iron alloy and an external layer of zinc.

SHETLAND: A rare breed of cattle, native to the Shetland Isles in northern Scotland. It is medium-sized, black and white with short legs, short horn and bulky body.

SHIELD CONE: A broad cone of a volcano resulting from successive smooth lava flows.

SHIELDING: 1. A barrier surrounding a region to prevent it from the influence of an energy field or to protect a region from an electric field by earthing. 2. The barrier provided by inner electron shells to the influence of nuclear charge on outer electrons. 3. The reduced attraction by the protons for the outermost electrons, caused by the core electrons. 4. A barrier used to surround a source of harmful or unwanted radiations as in a nuclear reactor where cement or lead shield is used to absorb neutrons. 5. A broad piece of armour made of rigid material and strapped to the arm or carried in the hand for protection against hurled or thrusted weapons. 6. Also called **continental shield**, it is the ancient, stable, interior layer of continents composed primarily of Precambrian igneous or metamorphic rocks.

SHIELDING CONSTANT: The difference between the external magnetic flux density and the local magnetic flux density at a resonating nucleus affected by the neighbouring electrons divided by the external flux density. its symbol is σ.

SHIFT KEY: A modifier key on a personal computer keyboard that allows a user to type a single capital letter when used in combination with an alphabet. For example, pressing and holding the Shift key and then pressing the letter 'm' would generate the capital letter M. The Shift key is also found on Apple computer keyboards and performs the same function as the Personal Computer. With Apple keyboard shortcuts, the Shift is represented as an up arrow (⇑). On U.S. keyboard and other similar keyboards, you can also hold down the Shift key and press the number across the top of the keyboard to get the symbols (e.g., !, @, #, and $) on those keys. The Shift key is also used with shortcut keys. For example, 'Control + Shift + S is the shortcut for 'Save As'.

SHIFTING CULTIVATION: A farming system in which plots of land are cultivated temporarily and then abandoned. This system often involves clearing of a piece of land followed by several years of wood harvesting or farming, until the soil loses fertility. Once the land becomes inadequate for crop production, it is left to be reclaimed by natural vegetation or sometimes converted to a different long-term cyclical farming practice.

SHIGELLA: A rod-shaped bacterium that lives in the intestinal tracts of humans and animals and can cause dysentery or shigellosis.

SHIGELLOSIS (SHIGELLA INFECTION): An intestinal disease caused by a family of bacteria known as shigella. The main sign of shigella infection is diarrhoea, which often is bloody. Shigella can be passed through direct contact with the bacteria in the stool. Shigella bacteria also can be passed in contaminated food or by drinking or swimming in contaminated water. Children between the ages of 2 and 4 are most likely to get shigella infection.

SHIKIMIC ACID: An aromatic acid, with the chemical formula $C_7H_{10}O_5$, molar mass 174.15 g mol^{-1} and a melting point of 185-187 °C. It is an important biochemical metabolite in plants and microorganisms, found in abundance in certain higher plants. It is an intermediate in the synthesis of the aromatic amino acids phenylalanine, tyrosine and tryptophan; an intermediate in the synthesis of lignin and of the electron carriers ubiquinone and plastoquinone. The shikimic acid pathway is the normal route for the synthesis of phenolics and is one of the few pathways in which aromatic compounds are formed from aliphatic precursors.

SHIKIMIC ACID PATHWAY: The key metabolic pathway in the biosynthesis of the aromatic amino acids tyrosine, phenylalanine and tryptophan which occurs in plants, bacteria and fungi but not in animals.

SHIN: 1. The lower part of the foreleg of cattle. 2. The upper part of a ploughshare. 3. The front part of the leg below the knee and above the ankle; or the front edge of the tibia.

SHINE-DALGARNO SEQUENCE: A sequence of five to nine nucleotide preceding the start codon in prokaryotic messenger RNA that is recognised by the ribosome as the correct site for binding the mRNA molecule prior to the translation.

SHINGLES (HERPES ZOSTER): A viral infectious disease that affects the sensory nerves, characterised by pain and eruption of blisters along the course of the affected nerves. It is caused by the varicella-zoster virus that also causes chickenpox.

SHIREHORSE: A tall heavy breed of draught horse, with coats which may be of various colours and long hair growing from its fetlocks.

SHIVERING: A common affliction of the nervous system with involuntary muscular contraction. It is a stimulus to rapid muscular contractions, set off from the temperature-regulating centre in the hypothalamus, in response to cooling of the skin and the blood. The contractions generate heat, helping to maintain deep body temperature despite increased heat loss during cold exposure. Shivering occurs in fever, when there is effectively a re-setting of the hypothalamic 'thermostat'.

SHOCK WAVE: A type of propagating disturbance. It carries energy and can propagate through a medium such as solid, liquid, gas or plasma or in some cases in the absence of a material medium, through a field such as the electromagnetic field. Shock waves are characterised by an abrupt, nearly discontinuous change in the characteristics of the medium. In a shockwave, there is always an extremely rapid rise in pressure, temperature and density of the flow.

SHOCKLEY, WILLIAM BRADFORD (1910-1989): American physicist and inventor, who received a Ph.D. from Harvard University and joined Bell Labs in 1936, where he began experiments that led to the development of the transistor. Along with John Bardeen and Walter Houser Brattain, Shockley co-invented the transistor, for which all three were awarded the 1956 Nobel Prize in Physics.

SHOCKWAVE PLAYER: A software developed by Adobe that enables Web pages containing interactive multimedia materials to be played on the Web. Such materials may contain games, product demonstrations and online learning applications.

SHODDY: A waste product of the wool industry containing 15% nitrogen and used as fertiliser.

SHOOT: 1. Parts of a vascular plant growing above the ground, comprising a stem bearing leaves, buds and flowers. It develops from the plumule in a seed. 2. A new growth from the stem of a plant.

SHOOT SYSTEM: The aerial portion of a plant body, consisting of stems, leaves and flowers.

SHOOT-TIP CULTURE (MERISTEM CULTURE): The culture of excised meristems on suitable nutrient media under aseptic conditions. The stem apex is usually used though axillary meristems may also be taken. Often, gibberellic acid must be added to the medium to promote normal growth into a plantlet. Once a suitable medium has been found for the regeneration of plantlets, the technique can be used as a means of rapid propagation since the plantlets can be divided at intervals into segments that can be grown on individually. Meristem culture may be used to propagate infertile plants, nursery stock or F1 hybrids that do not breed true. As few virus diseases affect the plant apex, the technique can be used to produce virus-free stock.

SHORT CHAIN: A chain of low relative molecular mass.

SHORT CIRCUIT: A failure in an electrical circuit caused by an accidental flow of excessive current or a low resistance connection between two nodes of an electrical circuit that are meant to be at different potentials or voltages such as a live wire and a neutral wire or an earthwire. In a three phase system, it occurs when any of the phases links either to themselves or to a neutral wire or an earthwire. Its relatively low resistance permits large current to flow through it, bypassing the rest of the circuit and this may cause the circuit to overheat dangerously. To protect against damage, devices such as a fuse or a circuit breaker are used to interrupt the circuit.

SHORT CUT KEY (KEYBOARD SHORTCUT): A keyboard key or a combination of keys used to perform a menu function or other common functions in an application. Depending on the operating system, shortcut keys usually combine the Ctrl or Alt keys with some other keys. For example, in Windows operating systems, the shortcut keys for "copy" is "Ctrl + C" and "Ctrl + V" for "paste". Keyboard shortcuts usually are not as intuitive as point-and-click mouse actions; however, they can be utilised by the novice users in frequently used programs to get to locations much faster than using the mouse.

SHORT-DAY PLANT (SDP): A plant in which flowering can be induced or enhanced by short days, usually of less than 12 hours of daylight. Examples are strawberry and chrysanthemum.

SHORT EXACT SEQUENCE: A five-term exact sequence of which the initial and terminal objects are trivial. For example, a short exact sequence of groups A, B and C is given by two maps α : A → B and β: B → C and is written $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0,$ when the image of f is the kernel of g, f is monic and g is epi.

SHORT INTERFERING DNA (SMALL INTERFERING RNA, siRNA; SILENCING RNA): Any of several types of small RNA molecules that are produced in cells from double-stranded (ds) precursors by the action of the protein Dicer. This mechanism of RNA interference probably evolved as a means of combating infection by RNA viruses, where it interferes with the expression of a specific gene. In addition to their role in the RNA interference pathway, siRNAs also acts in shaping the chromatin structure of a genome.

SHORT INTERSPERSED ELEMENT (SINE): Any of a class of dispersed moderately repetitive DNA found in eukaryotes consisting of numerous copies of relatively short sequence scattered throughout the genome. SINEs are not translated into proteins and occur mostly in interons. They are thought to be degenerate copies of retrotransposons with unknown function. The most notable example in human and other primates are the Alu family.

SHORT MESSAGE SERVICE (SMS): A wireless service that allows individuals to send short text messages with devices such as mobile phones, pagers and personal computers. The messages can typically be up to 160 characters in length, though some services use 5-bit mode, which supports 224 characters.

SHORT RADIUS: In mathematics, the perpendicular distance between the centre of a regular polygon and any of its sides.

SHORT ROTATION COPPICE: Varieties of willow or poplar which yield a large amount of fuel and are grown as an energy crop. This woody solid biomass can be used in applications such as district heating, electric power generating stations, alone or in combination with other fuels.

SHORTSIGHTED (MYOPIA; NEARSIGHTED): The inability of the eye to focus on distant objects or a common condition in which light entering the eye is focused in front of the retina and distant objects cannot be seen sharply. In high myopia, the eyeball is unusually long, whereas in physiological myopia the eyeball length is normal but the power of the cornea is too great for the axial length. Short sight occurs when the lens loses its ability to become flatter or when the eyeball is too long. As a result the images of distant objects are focused on a point in front of the retina.

SHORT TANDEM REPEATS (STRs): Short sequences of nucleotides in DNA, 2-7 bp long, that are repeated in tandem at particular loci throughout the genome. The number of repeats at one site varies typically between five and thirty. STRs are widely used as the basis for DNA profiling because they are highly variable and their size makes them readily amenable to detection and amplification by the polymerase chain reaction.

SHORT-TERM ENVIRONMENTAL CHANGE: Environmental change that occurs quickly that do not give organisms enough time to adapt and affecting them immediately. Examples of such changes are oil spills, droughts, floods, and fires.

SHOTGUN CLONING: A method for cloning the entire genome of an organism in which the DNA is broken into fragments at random using the restriction enzymes. Each fragment is then incorporated with a cloning vector and introduced into a host organism. The fragments are maintained as DNA library.

SHOTGUN SEQUENCING: A strategy for sequencing large DNA fragments or entire genomes based on sequencing and ordering smaller fragments obtained by shotgun cloning. It relies primarily on computational power to assemble the numerous sequence reads to cloned fragments into a whole-genome sequence.

SHREDDER: A machine for shredding waste vegetable matter before composting.

SHRINKAGE: Decrease in volume of a network, gel or solid associated with the exudation of a fluid.

SHRINKING PATTERN: A pattern that diminishes in value or size from term to term, for example, 10, 8, 6, 4.

SHRUB: Perennial woody plant that typically produces several separate branches, none of which is dominant. It is usually less than 3 m tall, at or near ground level rather than the single trunk of most trees. A shrub is usually smaller than a tree but there is no clear distinction between large shrubs and small trees. A **shrub layer** is the undergrowth of a forest consisting usually of shrubby vegetation.

SHUNT (BYPASS): 1. An electrical resistor or other material connected in parallel with some other circuit or device to take part of the current passing through it. 2. A passage between two natural body channels, such as blood vessels, especially one created surgically to divert or permit flow from one pathway or region to another.

SHUNT VESSEL: A blood vessel that links an artery directly to a vein, allowing the blood to bypass the capillaries in certain areas. Shunt vessel can control blood flow by constriction and dilation. In endotherm, the shunt vessels dilate in response to cold, thereby cutting off blood flow to the extremities and prevent heat loss.

SHUT-DOWN STATE: The condition of an instrument when it is switched off to conserve energy or reagents or to protect working parts. This term is of particular importance in clinical chemistry.

SHUT-DOWN TIME: The time interval between production of the last result of an instrument and shut-down state.

SHUTTLE: 1. A bobbin-like device used in weaving for passing the weft thread between the warp threads. 2. A small bobbin-like device used to hold the thread in a sewing machine or in tatting, knitting, etc. 3. A space shuttle is a reusable spacecraft designed to transport people and cargo between Earth and space. 4. A form of transport that travels regularly between two places.

SHUTTLE VECTOR: A DNA molecule, for example, plasmid that is able to replicate in two different host organisms and can, therefore, be used to convey genes from one to the other.

SIAL: The rocks that form the Earth's upper or continental crust. These are granite rock types rich in silica (SiO_2) and aluminium (Al).

SIAMESE TWINS (CONJOINED TWINS): Identical twins born with some part of their bodies joined together as a result of incomplete division of one fertilised ovum.

SIBLING: 1. Individuals of the same parents. 2. A member of a group of people, who trace their descent from a single real or presumed ancestor. 3. Fraternal twins of opposite sexes.

SIBLING SPECIES (CRYPTIC SPECIES): Two or more related groups of organisms that cannot be readily distinguished by their appearance or other traditional taxonomic criteria but whose members cannot interbreed successfully.

SIBS: Plants derived either by selfing or by crossing between genetically similar parents. In the production of F1 hybrid varieties occasional crossing may occur within rather than between the parent lines giving a proportion of nonhybrid seed. The term sibling species is sometimes used to describe closely related populations having a common ancestry.

SICKLE: A short-handled tool with a curved blade for cutting grass, grain, etc.

SICKLE CELL: A red blood cell that is crescent-shaped as a result of a single change in the amino acid sequence of the cell's haemoglobin, which causes the cell to contort, especially under low-oxygen conditions.

SICKLE CELL DISEASE: A hereditary chronic blood disorder common among people of black African descent. It is characterised by distortion of and fragility of the red blood cells which assume an abnormal, rigid, sickle shape and are lost rapidly from circulation. It often results in anaemia. People who are homozygous for sickle cell genes are said to have sickle cell disease.

SICKLE CELL TRAIT: A hereditary condition of the blood in which some red cells become sickle-shaped, but they are not enough affected cells to cause anaemia. This trait, which usually gives some resistance to malaria, occurs when the responsible gene is inherited from only one parent. Such individuals are heterozygous for sickle cell gene.

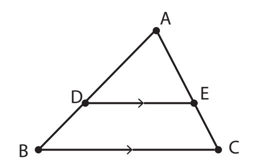
SIDA: A large genus of shrubby dicotyledonous flowering plants in the mallow family, Malvaceae, distributed in tropical and subtropical regions worldwide. The leaf blades are usually unlobed with serrated edges, but may be divided into lobes. They are borne on petioles and have stipules. Flowers are solitary or arranged in inflorescences of various forms. An example is *Sida Acuta*, commonly known as common wireweed, spiny headed sida or broom weed has a fibrous to woody stem with a tough and stringy bark and a strong branching taproot. The leaves have serrated margins, petiolated with stipules and lanceolate. The bisexual flowers have petals which are fused at the base and notched slightly at the top. The flowers are yellow in colour and are solitary. The fruit is a capsule, that splits into 8-10 mericarps. *Sida acuta* is not usually eaten by herbivores, giving it an advantage in pastures. It is therefore, classified as a weed. *Sida cordifolia*, is a weed that grows in wastelands and along roadsides. Its various parts are used medicinally.

SIDE CHAIN ALIPHATIC: A radical or group that is attached to one of the atoms in the ring of a cyclic compound or to one of the atoms of the longer straight chain of atoms in an acyclic compound.

SIDE RAKE: A machine, which picks up two swaths and combine them into one before baling.

SIDE REACTION: A chemical reaction that takes place at the same time as the main reaction but to a lesser extent, leading to other products mixed with the main products.

SIDE SPLITTING THEOREM (SIDE SPLITTER THEOREM): A theorem used to find the sides of triangles, which states that if a line is parallel to one side of a triangle and intersects the other two sides, then it divides those sides proportionally. For example, in the diagram, if ABC is a diagram and a line DE is drawn parallel to BC, then $\frac{AD}{AB} = \frac{AE}{AC}$.



Side Splitting Theorem

SIDE-BY-SIDE BAR GRAPH (SIDE-BY-SIDE COLUMN GRAPH): A bar graph that uses pairs of bars to compare two related data sets.

SIDEBAND: Any frequency added to the carrier as a result of modulation. Sidebands carry the actual information while the carrier contributes none at all. Those frequency components that are higher than the carrier frequency are known as upper sidebands; those lower are called lower sidebands. The upper and lower sidebands contain equivalent information; thus only one needs to be transmitted.

SIDEREAL PERIOD: The time taken by a satellite or a planet to complete one revolution of its orbit, measured in reference to the background of the fixed stars. A planet's sidereal period can be calculated from its synodic period.

SIDEREAL TIME: In astronomy, it is time measured by the rotation of the Earth with respect to the fixed stars, using an astronomical clock. It is divided into sidereal hours, minutes and seconds, each of which is proportionally shorter than the corresponding SI unit. A sidereal day is 23 hours 56 minutes 4 seconds and is defined as the time taken by the Earth to turn once with respect to the stars. Solar time loses about 4 minutes a day compared with sidereal time because of the Earth's orbital movement.

SIDERITE (CHALYBITE): 1. A yellow-brown mineral consisting of iron carbonate obtained from sedimentary rocks and used as a source of iron. 2. A dense metallic meteorite.

SIDEROCHROMES: Compounds that are synthesised by microorganisms, which are responsible for iron uptake.

SIDEROPHORE: A metabolite that is formed by some bacteria and fungi growing under low iron stress that forms a strong coordination compound with iron in the environment, which is secreted and transported across microbial cell membranes.

SIEMENS: The SI unit of electrical conductance, named after the German inventor and industrialist, Ernst Werner von Siemens (1816-92). It is equal to the conductance of a circuit that has a resistance of 1 ohm or to the reciprocal of one ohm (1 S ≡ $10^{-1}\Omega$ or Ω^{-1}) and is sometimes referred to as the mho. Its symbol is the mho, an inverted Greek letter omega ⍰ and sometimes Ω^{-1}.

SIEVE: 1. A device with wire network or gauze, usually with a circular frame, used for separating finer substances or particles such as grains, flour or sand from coarse ones or for straining liquid. 2. A process of finding prime numbers which was discovered by Eratosthenes, a Greek mathematician.

SIEVE AREA: A specialised part of the primary wall of a sieve element, perforated by a number of pores, derived from a primary pit field. These pores are lined with callose and penetrated by cytoplasmic strands. The sieve areas of gymnosperm sieve cells are relatively unspecialised, whereas those of angiosperm sieve tubes are usually differentiated as sieve plates.

SIEVE CELL: A relatively primitive vascular cell in the phloem of the lower vascular plants and gymnosperms. Its main function is the translocation of sugars and other nutrients. Sieve cells have relatively unspecialised sieve areas compared to those of sieve tube elements and are longer and narrower in shape.

SIEVE OF ERATOSTHENES: A simple method for finding all prime numbers up to a specified integer. It was supposed to be created by Eratosthenes, a Greek mathematician. It works efficiently for the smaller primes, below 10 million. For example, for the first 100 integers as shown in the diagram, the integers that are not circled are the prime numbers. The prime numbers are obtained by circling all the numbers that are divisible by 2,3,5,7,...

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

Sieve of Erastosthenes

SIEVE TUBE: A continuous longitudinal tube composed of numerous sieve tube elements.

SIEVE TUBE ELEMENT (SIEVE TUBE MEMBER): A type of plant cell occurring in the phloem. They are long wide cells with end plates and wide pores joined end-to-end to form the track for transporting food materials. Mature elements contain little cytoplasm and no nucleus.

S

SIEVEPLATE: 1. A highly specialised sieve area or collection of sieve areas, perforated by relatively large pores. It is usually located on the end walls of sieve tube elements. A sieve plate composed of a single sieve area is called a **simple sieve plate**. A number of sieve areas may be arranged collectively to form a compound sieve plate, as in Nicotiana. 2. A distillation-tower tray that is perforated so that the vapour emerges vertically through the tray, passing through the liquid holdup on top of the tray. It is used as a replacement for bubble-cap trays in distillation.

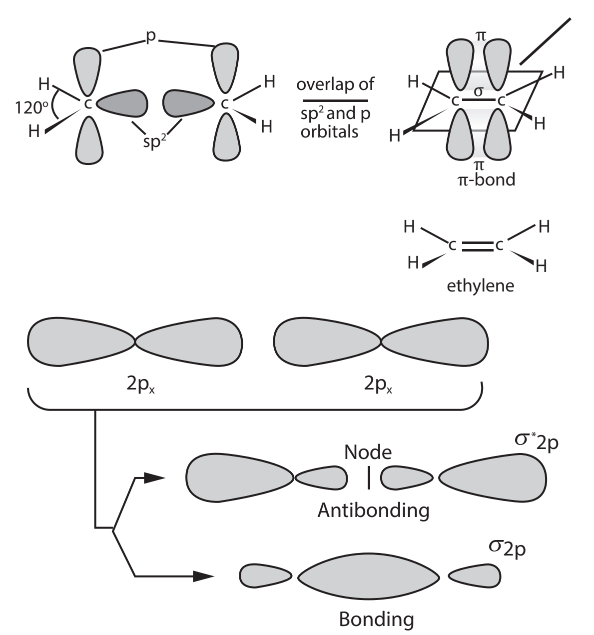
SIEVERT: The SI unit of dose equivalent. It is the measurement of the probability that a specific dose of a particular radiation type will cause a biological effect. Its symbol is Sv and it was named after Rolf Sievert, a Swedish physicist. One Sievert is equal to one joule per kilogram. Humans can absorb up to 0.25 Sv without immediate ill effects, but 1 Sv may produce radiation sickness and more than 8 Sv causes death.

SIGHT: The sense of sight, which is typically rays of light entering the eyes and forming images on the retina at the back of the eyeball, which perceives the form, colour, size, movement and distance of objects. Of all the senses, vision provides the most detailed and extensive information about the environment. In the higher animals, especially the birds and primates, the eyes and the visual areas of the central nervous system have developed a size and complexity far beyond the other sensory systems.

SIGMA: The eighteenth letter of the Greek alphabet, which carries the 'S' sound written as Σ (upper case) or σ (lower case). It is used to indicate summation or addition of the numbers or quantities indicated. When the last letter in a word of lower case letters is sigma the alternative (ς) form is used. Also in Greek numerals, Σ′ indicates the number 200.

SIGMA-ALGEBRA (σ-ALGEBRA): In mathematics, a collection of subsets of a set that contains the set itself, the empty set, the complements in the set of all members of the collection and the union of countably many sets and intersection of countably many sets.

SIGMA BOND (Σ (σ)-BOND): A bond formed by a head-on-overlap between two atomic orbitals. They are the strongest type of covalent chemical bond. Sigma bonding is most clearly defined for diatomic molecules using the language and tools of symmetry groups. In this formal approach, a σ-bond is symmetrical with respect to rotation about the bond axis. By this definition, common forms of sigma bonds are s + s, $p_z + p_z$, s + p_z and $d_z{}^2 + d_z{}^2$, where z is defined as the axis of the bond.



Sigma Bonding

SIGMA HYPERON: An unstable elementary particle of the baryon group with positive, negative or neutral electric charge. It is a collective name for three semistable baryons with charges of +1, 0 and -1 times the proton charge, designated Σ^+, Σ^0 and Σ^-, all having masses of approximately 1193 mega electron volts, a spin of ½ and positive parity. They form an isotopic spin multiplet with a total isotopic spin of 1 and a hypercharge of 0. It can also be defined as any baryon belonging to an isotopic spin multiplet having a total isotopic spin of 1 and a hypercharge of 0; designated $\Sigma_J^P(m)$, where m is the mass of the baryon in mega electron volts and J and P are its spin and parity; the $\Sigma_{3/2+}(1385)$ is sometimes designated Σ*.

SIGMA MOLECULAR ORBITAL: A molecular orbital in which the electron density is concentrated around a line between the two nuclei of bonding atoms. **Sigma electron** is an electron in a sigma orbital.

SIGMA PARTICLE: A type of spin ½ baryon. There are three types of semi stable baryons (sigma particles) denoted by Σ^+, Σ^0 and Σ^- for positively charged, electrically neutral and negatively charged respectively.

SIGMA-COMPACT (σ-COMPACT): Of a subset of a topological space, having a union of countably many compact subspaces. If a space is both σ-compact and locally compact, it is said to be **σ-locally compact**.

SIGMA-FINITE (σ-FINITE): Of a measure, such that every measurable set is the countable union of sets of finite measure; for example, Lebesgue measure on the real numbers is not finite, but it is σ-finite.

SIGMA-FUNCTION (σ-FUNCTION): 1. In mathematics, the sum-of-divisors function $\sigma_a(n)$, that sums the distinct divisors of n, including 1 and n. The sum of the proper factors of n is therefore, $\sigma(n) - n$. When p is prime, $\sigma(a) = \frac{p^{a+1}-1}{p-1}$, and since a is multiplicative, the value for any other argument can be easily computed from its prime factorisation. In terms of this function, a perfect number is one with $\sigma(n) = 2n$ and a pair of amicable numbers have $\sigma(a) = \sigma(b) = a + b$. 2. The function $\sigma_k(n)$ that sums the k^{th} powers of the divisors of n. In this notation, $\sigma_1(n)$ is $\sigma(n)$ and $\sigma_0(n)$ is the divisor function, $d(n)$.

SIGMA-RING (σ-RING): A collection of subsets of a set that is closed under symmetric difference and countable union.

SIGMATROPIC REACTION: A type of rearrangement reaction involving sigmatropic shifts. In sigmatropic reactions, the energy released in breaking one sigma bond is used in forming another sigma bond between two atoms which were not previously linked. This type of reaction is an example of a pericyclic reaction. Sigmatropic reactions often have transition-metal catalysts that form intermediates in analogous reactions. The most well-known of the sigmatropic rearrangements are the [3,3] Cope rearrangement, Claisen rearrangement, Carroll rearrangement and the Fischer indole synthesis. .

SIGMOID: Shaped in the form of the letter S.

SIGMOID COLON (PELVIC COLON): The S-shaped part of the colon in the lower abdomen which extends from the brim of the pelvis, usually down to the third segment of the sacrum (the triangular bone in the lower spine). The sigmoid colon is connected to the descending colon above and the rectum.

SIGN: 1. In mathematics, any symbol, usually used with a name, for example, "plus sign" to indicate a mathematical operation of addition. If used with a number, quantity or expression, it denotes positive or negative. For example, 3 subtracted from zero changes the sign to negative, -3. 2. **Signed** is the ability to take either sign; for example, signed numbers or signed measures, as are treated by Jordan decomposition. **Signed measure** is a countably additive set function that may take either sign. 3. In statistics, a **sign test** is a statistical test used to test the consistent differences of scores between the same or matched pairs of subjects under two experimental conditions. 4. An indication of the presence of a disease or disorder especially if observed by a medical officer but not obvious to the victim. 5. A trace of a wild animal, which includes scent, footprints, droppings, etc.

SIGN CONVENTION: A set of rules that is determined by convention for assigning plus or minus signs to distances in the formulae involving lenses and mirrors. Two conventions are used: the real-is-positive convention and the Cartesian convention. The real-is-positive is the convention now widely used. The Cartesian convention is preferred by some for more complicated calculations because it has the advantage of conforming to the sign convention used with Cartesian coordination in mathematics.

SIGN STIMULUS (RELEASER): The essential features of a stimulus, which is necessary to elicit response or a specific external stimulus that initiates certain behavioural sequences that typically occur in a fixed stereotyped fashion. For example, a red belly is the sign stimulus necessary to provoke an attack from a rival male. Even a very crude model fish is attacked if it has a red under surface. Similarly territorial fighting in male robin is triggered by a sign stimulus of a red breast. A bunch of red feathers is enough to induce territorial behaviour.

SIGNAL: 1. Anything that serves to indicate, warn, direct or that serves to communicate information or that acts as an incitement to action. 2. The parameter that contains information and by which information is transmitted in an electronic circuit. The signal is often the source of voltage in which the amplitude, frequency and wavelength can be varied. 3. Any kind of coded message sent from one organism to another or from one place in an organism to another place. Examples are vocal calls, some social behaviours and chemical signals. 4. Also known as **biopotential**, it is an electric quantity such as voltage, current or field strength that is caused by chemical reactions of charged ions.

SIGNAL HYPOTHESIS: A hypothesis to explain how ribosomes become attached to membranes within cells in order to deliver the appropriate proteins to the cell organelles, such as mitochondria and chloroplast or transport protein outside the cell membrane. It proposes that the leading end of the nascent polypeptide chain consists of a signal peptide which is recognised by a signal recognition particle that draws the ribosome to the membrane surface by interaction with a docking protein.

SIGNAL MOLECULE: A molecule of a hormone, neurotransmitter, growth factor or other chemical that binds specifically with a cell-surface receptor and thereby initiates a sequence of activities that triggers a response inside the cell or transmitting information between cells.

SIGNAL PEPTIDASE: An integral membrane protein located on the luminal surface of the endoplasmic reticulum that cleaves the signal peptide and frees the protein for folding and export.

SIGNAL PROCESSING (DIGITAL SIGNAL PROCESSING, DSP): In computer science, it is the process of manipulating analogue and digital signals using various mathematical and computational algorithms to filter, enhance image, compress data, etc., in order to improve the accuracy and reliability of communications. Examples include audio signal processing for electrical signals, such as speech or music; speech signal processing for processing and interpreting spoken words; image processing used in digital cameras, computers and various imaging systems; and video processing for interpreting moving pictures.

SIGNAL RECOGNITION PARTICLE (SRP): A protein-RNA complex that is composed of six different proteins and one small RNA that binds to the signal peptide as it emerges from the ribosome, temporarily halting the synthesis of the protein while the complex is transported to the endoplasmic reticulum where the SRP recognises and docks to a docking protein located on the cytoplasmic surface followed by the transfer of the ribosome to a ribosome receptor on the membrane in eukaryotes and plasma membrane in prokaryotes. The SRP and its docking protein are released from the ribosome and translation resumes.

SIGNAL RECOGNITION PARTICLE RECEPTOR (DOCKING PROTEIN): A receptor on the rough endoplasmic reticulum that specifically binds to the signal recognition particle.

SIGNAL SEQUENCE: A sequence in a protein that serves as a signal to guide the protein to a specific location.

SIGNAL TOPOLOGY (LOGICAL TOPOLOGY): The way in which data flows within a network from node to node, irrespective of its physical design.

SIGNAL TRANSDUCTION: Any mechanism by which binding of an extracellular signal molecule to a cell-surface receptor triggers a response inside the cell (the transmission of molecular signals from a cell's exterior to its interior). The mechanism depends on the type of signal molecule but it often involves changes in concentration of a second messenger such as calcium or cyclic AMP within the cell which can in turn affect numerous cell activities. The ability of an organism to function normally is dependent on all the cells of its different organs communicating effectively with their surroundings. Once a cell picks up a hormonal or sensory signal, it must transmit this information from the surface to the interior parts of the cell, for example, to the nucleus. This occurs via signal transduction pathways that are very specific, both in their activation and in their downstream actions.

SIGNAL TRANSDUCTION PROTEINS: Proteins which may be a protein kinase, an ion channel forming protein (as innerve cells) or a protein that undergoes some energy dependent change in which the energy is supplied by the hydrolysis of a higher energy compound such as guanosine triphosphate.

SIGNAL-TO-NOISE RATIO (SNR; S/N): The ratio of the amplitude of a desired signal at any point to the amplitude of noise signals at that same point; often expressed in decibels. Peak value is usually used for pulse noise, while the root-mean-square (rms) value is used for random noise.

SIGNATURE: 1. In mathematics, it is a number, $\varepsilon i_1, i_2, i_k$, denoting whether a permutation (i_1, i_2, i_k) differs from the natural order by an odd or even number of steps $\varepsilon = +1$ if the permutation is even; $\varepsilon = -1$ if it is odd. 2. Of an Hermitian matrix or quadratic form, the surplus of positive over negative coefficients in any real diagonal matrix or form similar to the given one; this equals the excess of positive over negative eigenvalues.

SIGNATURE OF SPACE-TIME: The division of space-time into space dimensions and time dimensions.

SIGNED MINOR (COFACTOR): A determinant obtained by deleting the row and column of a given element of a matrix or determinant. Consider the square matrix $A[a_{ij}]$. The cofactor, A_{ij} of the entry a_{ij} which is given by $(-1)^{i+j}|A|$, where $|A|$ is the determinant of the matrix A, obtained by deleting the i^{-th} row and j^{-th} column of the matrix A.

SIGNIFICANCE LEVEL: In statistics, estimation or calculation of the property of randomness or deviation from the normal, in a statistical sample.

SIGNIFICANCE TEST: In statistics, a hypothesis test to determine if the alternative hypothesis, H_a achieves the predetermined significance level required for it to be accepted in order for a null hypothesis, H_0 to be rejected.

SIGNIFICANT FIGURES: 1. The required number of digits used to express the degree of accuracy of a number, starting from the first non-zero digit. For example, in 2,350,086 all the digits are significant; in 47,6000, only the digits 4,7 and 6 are significant; and in 0.005769,

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only the digits 5, 7, 6 and 9 are significant. To express a number *n* of significant figures if the $(n + 1)^{th}$ digit in the number is greater than or equal to five, it is approximated to one and added to the n^{th} digit to give the number of significant digits required. 2. A digit which makes a contribution to the value of a number.

SIGNUM FUNCTION: A real function represented by sgn(x) or sg(x). The value of the function depends on the sign of x, which can be +1, 0 or -1.

SILAGE: Green fodder packed tightly to exclude air and stored in a silo or pit without drying to feed cattle.

SILAGE ADDITIVE: A substance containing bacteria and/or chemical, used to speed up or improve the fermentation process in silage or to increase the amount of nutrients in it.

SILAGE EFFLUENT: An acidic liquid produced by the silage process which can be a serious pollutant, especially if it drains into a water body.

SILAGE LIQUOR: Liquid which forms in silage and drains away from the silo.

SILAGE TOWER: A container used for making and storing silage.

SILANE: Any of a class of inorganic compounds of silicon and hydrogen or a hydride of silicon with the general formula Si_nH_{2n+2}. The first three in the series are silane (SiH_4), which is a colourless gas insoluble in water. The others are disilane (Si_2H_6) and trisiline (Si_3H_8). The compounds are analogous to the alkanes but are much less stable and only the lower members of the series can be prepared in any quantity. It is produced by the reduction of silicon with lithium tetrahydroaluminate(III). It is also formed by the reaction of magnesium silicate (Mg_2Si) with acids. It is stable in the absence of air, but spontaneously flammable even at low temperatures. It is a reducing agent, used for the removal of corrosion in inaccessible plants.

SILASESQUIAZANES: Compounds in which every silicon atom is linked to three nitrogen atoms and every nitrogen atom is linked to two silicon atoms, thus consisting of SiH and NH units and having the general formula $(SiH)_{2n}(NH)_{3n}$. By extension hydrocarbyl derivatives are commonly included.

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SILASESQUIOXANES: Compounds in which every silicon atom is linked to three oxygen atoms and every oxygen atom is linked to two silicon atoms and having the general formula $(SiH)_{2n}O_{3n}$. By extension hydrocarbyl derivatives are commonly included.

SILASESQUITHIANES: Compounds in which every silicon atom is linked to three sulfur atoms and every sulfur atom is linked to two silicon atoms and having the general formula $(SiH)_{2n}S_{3n}$. By extension hydrocarbyl derivatives are commonly included.

SILATHIANES: Compounds having the structure $H_3Si[SSiH_2]_nSSiH_3$ and branched-chain analogues. They are analogous in structure to siloxanes with –S– replacing –O–. By extension hydrocarbyl derivatives are commonly included.

SILAZANES: Saturated silicon-nitrogen hydrides, having straight or branched chains. They are analogous in structure to siloxanes with –NH– replacing –O–, for example, $H_3SiNHSiH_2NHSiH_3$ trisilazane. By extension hydrocarbyl derivatives are commonly included.

SILENCING RNA, siRNA (SMALL INTERFERING RNA; SHORT INTERFERING DNA): Any of several types of small RNA molecules that are produced in cells from double-stranded (ds) precursors by the action of the protein Dicer. This mechanism of RNA interference probably evolved as a means of combating infection by RNA viruses, where it interferes with the expression of a specific gene. In addition to their role in the RNA interference pathway, siRNAs also acts in shaping the chromatin structure of a genome.

SILENT MUTATION: An alteration in the genetic code that has no apparent effect on the phenotype of an organism.

SILICA (SILICON DIOXIDE; SILICON(IV) OXIDE): A colourless or white vitreous solid, which is insoluble in water and soluble in hydrofluoric acid and in strong alkali. It is a widely found mineral form of silicon II dioxide (SiO_2), which naturally occurs in various crystalline and amorphous forms, for example, quartz, opal, sand, flint and agate. It is used for the manufacture of glass, abrasives, concrete, etc.

SILICA GEL: A rigid gel made by coagulating a sol of sodium silicate and heating it to drive off water. It is a porous amorphous variety of silica (SiO_2) and is capable of absorbing large quantities of water and other solvents. It is therefore used as a support for catalysts, as a drying agent, desiccant or adsorbent.

SILICA-ALUMINA RATIO: The molecules of silicon dioxide ($Si0_2$) per molecule of aluminum oxide (AI_20_3) in clay minerals or soils.

SILICA-SESQUIOXIDE RATIO: The molecules of silicon dioxide ($Si0_2$) per molecule of aluminum oxide (AI_20_3) plus ferric oxide (Fe_20_3) in clay minerals or soils.

SILICATE: Any of a group of substances containing negative ions composed of silicon and oxygen. They are derivatives of silica or a salt of a silicic acid. They are a very extensive group. Natural silicates, found in many rocks and minerals, contain varying proportions of silica and of a wide range of metal oxides. The basic structural unit is the tetrahedral SiO_4, group.

SILICATE MINERALS: A group of rock-forming minerals that makes up the bulk (about 90%) of the Earth's outer crust and constitutes one-third of all minerals. All silicate minerals are based on fundamental structural of SiO_4 tetrahedron.

SILICIC ACID (MONOSILICIC ACID ORTHOSILICIC ACID): A family of chemical compounds of the element silicon, hydrogen and oxygen, with the general formula $[SiO_x(OH)_{4-2x}]_n$. It is a hydrated form of silica (SiO_2) made by reacting a soluble silicate with an acid. It forms a colloidal or gel-like mass. Silicic acids may be formed by acidification of silicate salts such as sodium silicate in aqueous solution.

SILICIDE: A compound of silicon with a more electropositive element. The silicides are structurally similar to the interstitial carbides but the range encountered is more diverse. They react with mineral acids to form a range of silanes.

SILICLE: A dry, dehiscent fruit of the Cruciferae (Brassicaceae), usually having a length less than three times its width or broader than long with two valves separating from the persistent placentae and septum (replum).

SILICON: A brittle non-metallic or metalloid element, with the Si, belonging to Group 14 (IVA) of the Periodic Table with symbol Si, atomic number 14, atomic weight 28.086, relative density 2.33, melting point 1,410 °C (1683 K); boiling point 2,355 °C (2628 K) and have a valence of 4. It is obtained from sand, granite, clay and many minerals. It is the second most abundant element (after oxygen) in the Earth's crust and occurs in amorphous and crystalline forms. It is used doped or in combination with other materials in glass, semiconducting devices, concrete, brick, refractories, pottery and silicones.

SILICON CARBIDE (CARBORUNDUM MOISSANITE): A very hard black material, which is insoluble in water but soluble in molten alkali with the chemical formula SiC. It is made by heating silicon (IV) oxide with carbon in an electric furnace. It is heat resistant, decomposing when heated to 2730 °C (3003 K). It is used in refractory materials, such as rods, tubes, firebrick and in special parts for nuclear reactors. It is also used as an abrasive and in cutting, grinding and polishing instruments.

SILICON CHIP: A single crystal of a semiconducting silicon material, typically having millimetre dimensions, fabricated in such a way that it can perform a large number of independent electronic functions. It contains hundreds of thousands of micro miniature electronic circuit components, mainly transistors, packed and interconnected in layers beneath the surface. These components can perform control, logic and memory functions.

SILICON TETRACHLORIDE (SILICON(IV) CHLORIDE; TETRACHLOROSILANE): Colourless corrosive fuming liquid, with suffocating aroma and the chemical formula $SiCl_4$, density 1.483 g/cm^3, melting point -68.74 °C (204 K) and boiling point 57.65 °C (330.65 K). Silicon tetrachloride is prepared by the chlorination of various silicon compounds such as ferrosilicon, silicon carbide or mixtures of silicon dioxide and carbon. In the laboratory, it can be prepared by treating silicon with chlorine: $Si + 2\ Cl_2 \rightarrow SiCl_4$. It is a source of pure silica. It is used in the production of silica glass and in warfare smoke screen.

SILICON-STRIP DETECTORS (SSD): Detectors made of tiny strips of silicon, which create voltage pulses when traversed by high-energy charged particles, such as protons, electrons, or positrons.

SILICONES: Largely inert artificial polymers that include silicon, carbon, oxygen, hydrogen and sometimes other elements. Chains of the silicon atoms alternate with oxygen atoms with the silicon atoms being linked to the organic groups. Silicones are synthesised from chlorosilanes, tetraethoxysilane and related compounds. Some of the useful properties include thermal stability, good electrical insulation, non-sticky, low chemical reactivity, resistance to oxygen, ozone and UV radiation (sunlight). A variety of silicone materials exist including oils, waxes and rubbers.

SILIQUA: A type of capsule such as mustard (Brassca species) formed from two fused carpels. It is an elongated dry, dehiscent fruit, characteristic of the family Brassicaceae (formerly Cruciferae).

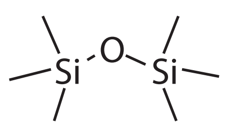
SILK: A material produced by the silk glands of spiders, some insects and certain other invertebrates. It exudes as liquid by a protruding appendage, the spinneret, but quickly hardens after leaving the gland. Silk is composed of keratin crystals embedded in a rubbery matrix of amino acid chains giving the material its flexibility and strength. The best-known type of silk is obtained from the cocoons of the larvae of the mulberry silkworm (*Bombyx mori*) reared in sericulture. Only this type of silk has been used for textile manufacture.

SILK COTTON TREE (KAPOK TREE; *Ceiba pentandra*): A drought deciduous tropical emergent tree of the family Malvaceae. Native to South America, it is now found in the tropical rainforests of the world. The tree is buttressed and can grow up to 70 metres with a trunk diameter of 3 metres. Its straight trunks are cylindrical, smooth and grey in colour with large spines protruding from the trunk. The branches grow in horizontal tiers and spread widely. The wood is a pinkish white to ashy brown in colour, with a straight grain. The leaves are palmate and compound. Flowers usually open before the leaves appear and are clustered on small, new branches. The 5 petals of a flower are about 2.5 centimetres long and are a creamy white or pale pink in colour. The tree produces lots of pods containing fluffy, yellowish fibre which is a mixture of lignin and cellulose. The fibre is very flammable, resistant to water and light in weight. It is used to stuff mattresses, pillows, upholstery and toys. The seeds in the pods produce oil for soap making and fertilisers. In certain traditions, its parts such as the seeds, leaves, bark and resin have been used to treat dysentery, fever, asthma and kidney disease. The wood is also used to make canoes in some areas.

SILO: 1. A tall, air-tight cylindrical tower used for storing grain, animal feed or other material or for making silage. 2. A reinforced protective underground chamber where a missile or missiles can be stored and from which they can be launched.

SILOPRESS: A polythene sack into which silage is forced.

SILOXANE: A member of a group of compounds that contain the linkage Si-O-Si with organic groups bound to the silicon atoms. They are used for making silicones.



Chemical Structure of Siloxane

SILT: Fine grained sediment which is intermediate in coarseness between clay and sand. Silt, like clay and sand, is a product of the weathering and decomposition of pre-existing rock. Its grains have a diameter of 0.002 – 0.02 mm. Silt is usually carried by moving or running water and deposited as a sediment. **Silting** is the deposition of water-borne sediments in stream channels, lakes, reservoirs, or on floodplains, usually resulting from a decrease in the velocity of the water.

SILURIAN: The third period of geological time in the Palaeozoic era from about 440 to 395 million years ago, during which there is the first evidence of the invasion of the land by plants and animals.

SILVER: A white lustrous extremely malleable and ductile metallic element with the symbol Ag, atomic number 47, liquid density at melting point 9.320 g/cm^{-3}, melting point 961.78 °C (1234.93 K) and boiling point 2162 °C (2435 K). It occurs in nature in ores and as a free metal. It has the highest thermal and electric conductivity of any substance. It is used as coins ornaments, jewellery, dental materials, solders, photographic chemicals, conductors, etc.

SILVER BROMIDE (SILVER(I) BROMIDE; BROMYRITE): A pale yellow insoluble crystalline salt with the chemical formula AgBr, density 6.473 g/cm^3 (solid), melting point 432 °C (705 K) and boiling point 1502 °C (1775 K). It is used for making light-sensitive photographic emulsions. It can be precipitated from silver(I) nitrate solution by adding a solution containing bromide ions. It dissolves in concentrated ammonia solutions but not in dilute ammonia.

SILVER CHLORIDE: A white insoluble crystalline salt with the chemical formula AgCl, density 5.56 g/cm^{-3}, melting point 455 °C (728 K) and boiling point 1547 °C (1820 K). It is nearly insoluble in water but is soluble in a water solution of ammonia, potassium cyanide or sodium thiosulfate. On exposure to light it becomes a deep greyish blue due to its decomposition into metallic silver and molecular chlorine. It is used in the manufacture of pure silver and in photographic emulsions.

SILVER IODIDE (SILVER(I) IODIDE): Pale yellow insoluble crystalline salt used in photographic emulsions with the chemical formula AgI, density 5.675 g/cm^3 (solid), melting point 558 °C (831 K) and boiling point 1506 °C (1779 K). Silver iodide is prepared by the reaction of an iodide solution such as potassium iodide with a solution of silver ions such as silver nitrate. It darkens on exposure to light and is used in photographic emulsions, rainmaking and medicine, especially as an antiseptic.

SILVER NITRATE (NITRIC ACID SILVER(1+) SALT): A poisonous, colourless crystalline compound with the chemical formula $AgNO_3$, density 4.35 g/cm^3, melting point 212 °C (485 K) and boiling point 444 °C (717 K). It becomes greyish black when exposed to light in the presence of organic matter. Silver nitrate can be prepared by reacting silver, such as a silver bullion or silver foil, with nitric acid, resulting in silver nitrate, water and oxides of nitrogen:

$3\ Ag + 4\ HNO_3 \rightarrow 3\ AgNO_3 + 2\ H_2O + NO$ and $3\ Ag + 6\ HNO_3 \rightarrow 3\ AgNO_3 + 3\ H_2O + 3\ NO_2$. It is used in manufacturing photographic film, silvering mirrors, dyeing hair, plating silver and in medicine as a cautery and antiseptic.

SILVER OXIDE (SILVER RUST; ARGENTOUS OXIDE): An odourless, dark-brown powder with a metallic taste with the chemical formula Ag_2O and decomposes above 300 °C (573 K). It can be prepared by combining aqueous solutions of silver nitrate and an alkali hydroxide. It is soluble in nitric acid and ammonium hydroxide and insoluble in alcohol. It is used in medicine and in glass polishing and colourings, as a catalyst and to purify drinking water.

SILVER PLATING: Electrolytic deposition of a layer of silver on the surface of another metal, usually from a hot alkaline solution of complex silver cyanides. Silver plating is used in electronics such as plating copper, as its electrical resistance is lower and for plating variable capacitors. Silver plating has also been used for items such as cutlery, vessels of various kinds and candlesticks.

SILVER-LACED WYANDOTTE: A dual purpose breed of poultry. The feathers are silvery, with black edges, especially on the tail.

SILVICIDE: A substance that can kill trees, used to control undesirable brush and trees.

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SILVICULTURE: The cultivation, management and study of forests. It involves the relationship of forests to their environments as well as their products including timber, pulp, energy, fruits, fodder and benefits such as water, wildlife habitat, recreation, microclimate amelioration and carbon sequestration.

SIM CARD (SUBSCRIBER IDENTITY MODULE; SUBSCRIBER IDENTIFICATION MODULE): A small card that is used in cellular phones and sometimes other devices that contains subscriber information. One advantage of a SIM card is that it can be moved from phone to phone, making it easier to change phones if needed.

SIMA: The rocks that form the Earth's oceanic crust and underlie the upper crust. These are basaltic rock types rich in silica and magnesium, hence the name. It is denser and more plastic than the sial that forms the continental crust.

SIMIAN VIRUS 40 (SIMIAN VACUOLATING VIRUS 40; SV 40): An infectious DNA-containing virus belonging to the agent of the genus Lentivirus in the family Retroviridae of the papovavirus group that is much used in cancer research for its ability to induce transformation in cells. The virus was originally isolated from monkeys and infects primates of the infraorder Simiiformes (hence its name), which includes the so-called anthropoids: apes, monkeys and humans. The virus attacks and integrates itself into our DNA which can lead to cancerous cell activity. **Simian immunodeficiency viruses** (**SIVs**) are retroviruses that naturally infect a wide range of African nonhuman primates and are the source of the human immunodeficiency viruses (HIV-1 and HIV-2).

SIMILAR: 1. Of two geometric figures, having the property that one is the image of the other under a transformation from the plane into itself and multiplies all distances by the same positive scale factor, k. 2. In Euclidean geometry, **similar figures** are have the same shape, with corresponding sides in proportion to each other. For example, two triangles are similar if 3 angles of 1 triangle are the same as 3 angles of the other or 3 pairs of corresponding sides are in the same ratio or the angle of 1 triangle is the same as the angle of the other triangle and the sides containing these angles are in the same ratio. 3. Of two sets of points, related by homothety, without translation. 4. Of two classes, equipollent. 5. Of two matrices or operators A and B, having the property that there exists an invertible transformation, C, with $A = C^{-1}BC$; A and B then represent the same linear transformation with respect to bases related by C. 6. Of polynomial terms in several variables, containing the same powers of each indeterminate.

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SIMILAR TRIANGLES: Triangles that have equal corresponding angles and corresponding sides in proportion to each other. Two triangles are similar if 3 angles of 1 triangle are the same as 3 angles of the other or 3 pairs of corresponding sides are in the same ratio or the angle of 1 triangle is the same as the angle of the other triangle and the sides containing these angles are in the same ratio.

SIMILARITIES (DIAGONAL RELATIONSHIP): A relationship within the Periodic Table by which certain elements in the second period have a close chemical similarity to their diagonal neighbours in the next group of the third period. For example, lithium (Li) and magnesium (Mg) show a diagonal relationship.

SIMILARITY TRANSFORMATION (SIMILARITY): 1. In Euclidean geometry, a transformation that preserves similarity, composed of some or all of translation, rotation and homothety. 2. A mapping of a set by which each element in the set is mapped into a positive constant multiple of itself, the same constant being used for all elements. 3. An operation performed upon a square matrix that leaves invariant its characteristic polynomial, trace and determinant. The transformation is obtained by multiplying the given matrix on one side by any nonsingular matrix and on the other by the inverse of that nonsingular matrix.

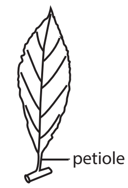
SIMILITUDE (RATIO OF SIMILITUDE): In mathematics, it is the ratio of proportionality between similar figures. In other words, it is the ratio of the sides of one figure to the sides of the other figure.

SIMM (SINGLE IN-LINE MEMORY MODULE): An older type of computer memory, which is a small circuit board with six to nine memory chips per board. SIMMs use a 32-bit bus, which is not as wide as the 64-bit bus dual in-line memory modules (DIMMs) use. SIMMs have been widely replaced by DIMMs.

SIMMONS-SMITH REACTION: A reaction in which a cyclopropane ring is produced from an alkene. It uses the Simmons-Smith Regent which was originally diiodomethane (CH_2I_2) with a Zn/Cu couple. Because the methylene unit is delivered to both carbons of the alkene simultaneously, the configuration of the double bond is preserved in the product and the reaction is stereospecific.

SIMON TEST: A presumptive test for methamphetamines. The Simon reagent consists of 1 gram of sodium nitroprusside ($Na_2Fe(CN)_5NO$) and 2 ml of acetaldehyde (ethanol) in 50 ml of water. A 2% solution of sodium carbonate is added. It reacts with secondary amines such as methamphetamine to give a blue solution.

SIMPLE: 1. Not composed of more than one anatomically or morphologically equivalent unit. 2. In botany, it describes a leaf blade, which is not divided into leaflets, although the blade may be deeply lobed or cleft. A **simple ovary** is an ovary composed of only one ovary. 3. In mathematics, the term is used for any of the following: (i) A **simple equation** is one that is linear (has variables only in the first degree). (ii) A **simple graph**, also known as a **strict graph** is a graph that does not have loops or multiple paths between the same pair of vertices. (iii) For a root of an equation, occurring only once; not multiple. (iv) A **simple group** contains no proper non-trivial normal subgroup. Simple groups can be finite or infinite and include the infinite families of alternating groups of degree ≥ 5, cyclic groups of prime order, Lie-type groups and the 26 sporadic groups. (v) A **simple module** is one that has only itself and the zero module as submodules. (vi) A non-zero ring with the property that its only two-sided ideals are itself and zero. Every simple ring is prime, every Artinian simple ring is Noetherian and every commutative simple ring is a field.



simple leaf

SIMPLE CLOSED CHAIN: A graph whose initial and final nodes are identical and in which no other node occurs more than once.

SIMPLE CLOSED CURVE (JORDAN CONTOUR; JORDAN CURVE): A curve, such as a circle, whose shape is closed by a curved line or line-segments, does not intersect itself and has no end points. Every Jordan curve separates the plane into exactly two components.

SIMPLE CONTINUED FRACTION: A continued fraction with unit numerators and integral denominators; a continued fraction in which the value of $b_n = 1$ for all n. A finite simple continued fraction is a simple continued fraction with only a finite number of terms. An infinite simple continued fraction is a simple continued fraction with an infinite number of terms.

SIMPLE FIELD EXTENSION: In mathematics, it is a subfield of a given extension field generated from a given base field by a single element. It is referred to as algebraic or transcendental over the base field according as the element is algebraic or transcendental.

SIMPLE FLUID: In physics, a Newtonian fluid for which the variation of the deviatoric part of the stress tensor as a function of the velocity gradient is independent of direction.

SIMPLE FRACTION (COMMON FRACTION; VULGAR FRACTION): A fraction having an integer both as a numerator and a denominator, for example, $\frac{4}{5}$ and $\frac{7}{4}$.

SIMPLE FUNCTION (SCHLICHT FUNCTION; UNIVALENT FUNCTION): In mathematics, a complex function on some domain, that is analytic and taking no value more than once in the domain. A schlicht function mapping the finite complex plane onto itself is linear.

SIMPLE HARMONIC APPROXIMATION: The approximation of the motion of a particle by simple harmonic motion.

SIMPLE HARMONIC MOTION (SHM; HARMONIC MOTION): Repetitive back-and-forth movement through a central or equilibrium position in which the maximum displacement on one side is equal to the maximum displacement on the other. A body in simple harmonic motion experiences a single force which is given by Hooke's law; that is, the force is directly proportional to the displacement x and points in the opposite direction. Each complete vibration takes the same time, the period; the reciprocal of the period is the frequency of vibration. Each oscillation is identical and thus the period, frequency and amplitude of the motion are constant. If the equilibrium position is taken to be zero, the displacement x of the body at any time t is given by $x(t) = A\cos(2\pi ft + \varphi)$, where A is the amplitude, f is the frequency and φ is the phase. The force that causes the motion is always directed toward the equilibrium position and is directly proportional to the distance from it. Other examples include the electrons in a wire carrying alternating current and the vibrating particles of a medium carrying sound wave.

SIMPLE INTEREST: An interest that is calculated on a loan or an investment once a period, usually annually, on the amount of the capital (principal) alone and not on any interest earned. Simple interest is calculated based on the following formula: Interest = (Principal × Rate × Time)/100, where **Interest** is the total amount of interest paid, **Principal** is the amount lent or borrowed or invested and **Rate** is the percentage of the interest charged on the principal or capital for a period of time. **Time** is the period over which the loan is to be repaid. The total amount payable, therefore, after a period of Time (T) is the Principal (P) plus Interest (I).

SIMPLE MACHINE: The simplest mechanisms that use mechanical advantage or leverage to multiply force. A simple machine changes the direction and magnitude of a force. It uses a single applied force to do work against a single load force. Examples are the six devices formerly considered to be the basic components from which all machines were composed. These are the inclined plane, lever, pulley, screw, wedge and wheel and axle. Simple machines fall into two categories: those dependent on the vector resolution of forces (inclined plane, wedge, screw) and those in which there is equilibrium of torques (lever, pulley, wheel).

SIMPLE MAIL TRANSFER PROTOCOL (SMTP): The protocol used to send e-mail over the Internet over port 25. E-mail clients such as Outlook, Eudora or Mac OS X Mail) uses SMTP to send a message to the mail server and the mail server uses SMTP to relay that message to the correct receiving mail server. The messages can then be retrieved from a server with either POP or IMAP. **Post Office Protocol** (**POP** or **POP mail**) is a protocol used to receive e-mail on many e-mail clients. There are two versions of POP. The first, called POP2, requires SMTP to send messages. The newer version, POP3, can be used with or without SMTP. **Internet Message Access Protocol** (**IMAP**) is a protocol that allows users to retrieve e-mail, but still uses SMTP for sending e-mail messages. IMAP4 is the latest version of IMAP.

SIMPLE NETWORK MANAGEMENT PROTOCOL (SNMP): A technique used for exchanging management information between network devices. For example, SNMP may be used to configure a router or simply check its status. There various types of SNMP commands used to control and monitor managed devices read, write, trap and traversal operations.

SIMPLE OBJECT ACCESS PROTOCOL (SOAP): A method of transferring messages or small amounts of information, over the Internet. SOAP messages are formatted in XML and are typically sent using HTTP (hypertext transfer protocol). Both are widely supported data transmission standards.

SIMPLE EVENT: An event with a single outcome, for example, drawing the ace of spades from a standard deck of cards. The probability of simple events is denoted by P(E) where E is the event. The probability will lie between 0 and 1.

SIMPLE SHEAR: An idealised treatment of a fluid between two large parallel plates (to permit ignoring edge effects) of area, separated by a distance.

SIMPLEST FORM FRACTIONS: A fraction whose numerator and denominator share no factors other than the number one.

SIMPLEX: 1. An Euclidean geometric figure having the minimum number of boundary points. An n-dimensional simplex is a polytope with n + 1 affinely independent vertices, thus in 1-dimension, a simplex is a line, in 2-dimensions, it is a triangle and in 3-dimensions, a tetrahedron, etc. 2. Also known as **simplex communication** or **simplex transmission**, it is a type of communication in which data can only be transmitted in one direction. Broadcast information or data, can only travel in one direction, in contrast to duplex which allows for two-way broadcasting. Examples of simplex include radio broadcasting, television broadcasting, computer to printer communication and keyboard to computer connections.

SIMPLEX METHOD: In mathematics, it is a numerical method for solving problems in linear programming by changing constraints to equations and solving the problem by matrix manipulation.

SIMPLICIAL COMPLEX: In mathematics, a set consisting of a finite number of simplices, K with the property that any two that intersect is a face of each of them; and every face of a simplex of K is in K.

SIMPLICIAL MAPPING: In mathematics, a mapping between two simplificial complexes with the property that the image of any component simplex is a simplex.

SIMPLIFIED MOLECULAR INPUT LINE ENTRY SPECIFICATION (SMILES): A line notation for the chemical structure of molecules. It uses a short series ASCII characters to represent structures. Most molecule editor computer programs can draw a two-dimensional diagram or a three-dimensional model of a molecule based on its SMILES code. It uses standard atom symbols but hydrogen atoms are omitted. A double bond is represented by an equal (=) sign and a triple bond by a hash (#) sign.

SIMPLIFY: To convert a mathematical expression, such as an equation or fraction, to a simpler form by regrouping terms or removing common factors. For example, $2x + 3y = y$ simplifies to $x = -y$.

SIMPLY-CONNECTED: The condition of a geometrical object if it consists of one piece and does not have any holes or "handles." For example, a line, a disk and a sphere are simply-connected but a torus (doughnut) and teapot are not.

SIMPSON'S RULE (PARABOLIC RULE): 1. A rule for finding the approximate area under a curve. It is a basic approximation formula for definite integrals which states that the integral of a real-valued function f on an interval $[a,b]$ is approximated by

$$\frac{h\left[f(a)+4f(g+h)+f(b)\right]}{3},$$

where $h = \frac{(b-a)}{2}$. This is the area under a parabola which coincides with the graph of f at the abscissas a, $a + h$ and b. 2. In petrochemical engineering, it is a mathematical relationship for calculating the oil- or gas-bearing net-pay volume of a reservoir. It uses the contour lines from a subsurface geological map of the reservoir, including gas-oil and gas-water contacts.

SIMSON LINE (SIMSON): In mathematics, in a triangle ABC and an arbitrary point P on its circumcircle, it is the line containing the feet of the perpendiculars from P to the sides of ABC or their extensions. This forms a triangle which is known as the pedal triangle of P.

SIMULATION TECHNIQUE: A technique for reducing or eliminating analytical errors resulting from interferences, using a reference solution sufficiently similar in quantitative composition to the sample solutions to be analysed that the interferences in the reference and sample solution are equivalent.

SIMULTANEITY: The condition in which two or more events occur at the same instant. In Newtonian physics, any two events found to be simultaneous by one observer will also appear to be simultaneous to any other observer in uniform relative motion. The two events

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have simultaneity, relative to an observer, if they take place at the same time according to a clock which is fixed relative to the observer.

SIMULTANEOUS DIFFERENTIAL EQUATION: A set of differential equations for an indefinite function of one or more variables that relate the values of the function. It is of the form $\frac{dx}{P}=\frac{dy}{Q}=\frac{dz}{R}$, where P, Q and R are functions of the three variables x, y, z. The equations are solved by transforming them to a total differential equation. The method is to find constants a, b, c such that $\frac{a\,dx+b\,dy+c\,dz}{zP+bQ+cR}$ has either zero denominator and a zero numerator or a numerator that is the differential of the denominator.

SIMULTANEOUS EQUATION: One of the two or more algebraic equations that contain two or more unknown quantities that may have a unique solution. For example, in the case of two linear equations with two unknown variables such as (i) $x + 3y = 6$ and (ii) $3y - 2x = 4$, the solution will be those unique values of x and y that are valid for both equations.

SINE: 1. A trigonometric function, which is periodic. Its symbol and abbreviation is sin. The symbol for its inverse function is $\sin^{-1}$. The period of the sine function is 360°: Sin(360 + θ) = Sinθ. 2. A mathematical function which is equal to the vertical coordinates of a circumference point divided by the radius of a circle with its centre at the origin of a Cartesian coordinate system. 3. Also called **short interspersed element**, it is any of a class of dispersed moderately repetitive DNA found in eukaryotes consisting of numerous copies of relatively short sequence scattered throughout the genome. SINEs are not translated into proteins and occur mostly in interons. They are thought to be degenerate copies of retrotransposons with unknown function. The most notable example in human and other primates are the Alu family.

SINE CURVE: 1. The ratio of the size of an angle to its sine, a sine function and is a curve with the equation $y = \sin x$. It is a smooth curve that varies from $y = -1$ to $y = 1$, is continuous, has maxima at $\frac{\pi}{2}+2n\pi$, minima at $\frac{3\pi}{2}+2n\pi$ and is zero at $n\pi$ for all integers n. 2. Also known as **sinusoid**, it is any curve derivable from the sine curve by multiplication or addition of a constant.

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SINE RULE (SINE LAW): An equation relating the lengths of the sides of an arbitrary triangle to the sines of its angles. According to the law, $\frac{a}{\sin A}=\frac{b}{\sin B}=\frac{c}{\sin C}$, where a, b and c are the lengths of the sides of a triangle and A, B and C are the opposite angles. Sometimes the law is stated using the reciprocal of this equation: $\frac{\sin A}{s}=\frac{\sin B}{b}=\frac{\sin C}{c}$.

SINE WAVE: Any oscillation whose waveform is that of a sine curve. Its mathematical equation is represented by the sine function and the simplest one is $y = \sin x$. Its is generated by oscillations such as sound, alternating current and earthquakes.

SINGE: 1. To burn something slightly so that only the surface, edge or tip is affected or be burned in this way. 2. To expose the carcass of a bird or animal to a flame in order to remove unwanted feathers, bristles or hair. 3. To burn the short fuzzy ends of fibres from cloth in the manufacturing process.

SINGLE-BLIND TRIALS: An experiment in which information that could introduce bias that could affect the result is withheld from the participants, but the experimenter will be in full possession of the facts. For example, in a clinical trial, the researchers, but not the subjects know subjects that are receiving the active medication or treatment and those who are not.

SINGLE BOND: Covalent bond formed by the sharing of one pair of electrons between two atoms. Covalent bonding includes many kinds of interaction, including σ-bonding, π-bonding, metal-to-metal bonding, agostic interactions and three-centre two-electron bonds. Covalent bonds are affected by the electronegativity of the connected atoms. Two atoms with equal electronegativity will make nonpolar covalent bonds such as H–H. An unequal relationship creates a polar covalent bond such as with H–Cl. There are three types of covalent substances: individual molecules, molecular structures and macromolecular structures. Individual molecules have strong bonds that hold the atoms together, but there are negligible forces of attraction between molecules. Such covalent substances are gases. Examples are hydrogen chloride (HCl), sulfur dioxide (SO_2), carbon dioxide (CO_2) and methane (CH_4). In molecular structures, there are weak forces of attraction. Such covalent substances are low-boiling-temperature liquids such as ethanol and low-melting-temperature solids such as iodine and solid CO_2. Macromolecular structures have large numbers of atoms linked in chains or sheets such as graphite or in 3-dimensional structures such as diamond and quartz. These substances have high melting and boiling points, are frequently brittle and tend to have high electrical resistivity. Elements that have high electronegativity and the ability to form three or four electron pair bonds, often form such large macromolecular structures.

SINGLE CELL PROTEIN (SINGLE-CELL PROTEIN; SCP): Protein produced by microorganisms, such as bacteria, yeast and unicellular algae that are extracted for use as a component of human and animal food. Microorganisms are considerably more efficient in producing protein than crops and farm animals. Yeast and bacteria can double their mass in a minimum of 20 minutes, while cereals would take 7-14 days and young cattle 4-8 weeks to achieve this.

SINGLE CIRCULATION: A type of circulatory system that occurs in fishes in which the blood passes only once through the heart in each complete circuit of the body.

SINGLE IN-LINE PACKAGE (SIP): A computer chip packaging that contains only one row of connection pins, instead of two rows, like those found on a dual in-line packages (DIPs).

SINGLE NEUCLEOTIDE POLYMORPHISM (SNP): A DNA sequence variation occurring when a single nucleotide A, T, C or G in the genome or other shared sequence differs between members of a species or paired chromosomes in an individual. For example, two sequenced DNA fragments from different individuals, AAGC**C**TA to AAGC**T**TA, contain a difference in a single nucleotide. In such case there are two alleles: C and T. Almost all common SNPs have only two alleles. Some are known to be associated with disease-causing alleles.

SINGLE-COPY DNA (scDNA): A region of the genome, the sequence of which is present only once per haploid complement.

SINGLE-ELECTRON TRANSFER MECHANISM (SET): A reaction mechanism characterised by the transfer of a single electron between the species occurring on the reaction coordinate of one of the elementary steps.

SINGLE-FACTOR INHERITANCE: The determination of a character by one major gene, although the gene may exist in various allelic forms. Mendel's genes are examples of single-factor inheritance.

SINGLE-GRAIN STRUCTURE: A soil structure composed completely of individual or primary soil particles which are totally unattached to each other. It is usually found only in extremely coarse textured soils.

SINGLE-SIDEBAND MODULATION (SSB): A refinement of amplitude modulation that more efficiently uses electrical power and bandwidth, in which the transmitter suppresses one sideband and therefore transmits only a single sideband. Amplitude modulation produces a modulated output signal that has twice the bandwidth of the original baseband signal. Single-sideband modulation avoids this bandwidth doubling and the power wasted on a carrier, at the cost of somewhat increased device complexity.

SINGLE-STRAND CHAIN: A polymer chain that comprises constitutional units connected in such a way that adjacent constitutional units are joined to each other through two atoms, one on each constitutional unit.

SINGLE-STRAND MACROMOLECULE: A macromolecule that comprises constitutional units connected in such a way that adjacent constitutional units are joined to each other through two atoms, one on each constitutional unit.

SINGLE-SUCKING: A natural method of rearing beef cattle, whereby calves are permitted to suckle their own mother.

SINGLE-VALUED FUNCTION (ONE-VALUED FUNCTION): A mathematical relation such that each element in the domain of the function or of a given set is associated with an element in the range of the function or of another set. For example, $f(x) = x^2$ is a single-valued function, while $f(x) = \pm\sqrt{x}$ is not because to each value of x other than zero there correspond two values of $f(x)$, which differ in sign. Single-valued functions include trigonometric, hyperbolic, exponential and integer power functions, but their inverses are not.

SINGLET STATE: An atomic state in which two spin angular momenta of electrons cancel each other, resulting in zero net spin. It is one of the two ways in which the spin of two electrons can be combined; the other being a triplet. A singlet state usually has a higher energy than a triplet state because of the effect of correlations of spin on the Coulomb interactions between electrons.

SINGLETON: 1. In mathematics, it is a set having only one member, for example, the set S of all integer s that are neither positive nor negative. In this case, S = {0}. 2. A class that allows only a single instance of itself to be created and gives access to that created instance. It is used in programming languages such as Java and .NET to define a global variable.

SINGLING: The process of reducing the number of plants in a row or the process of reducing the number of plants from a multigerm seed to a single plant.

SINGULAR: 1. In mathematics, of a continuous linear operator, not invertible or having a discontinuous inverse. 2. A singular solution to a differential equation is a solution that is singular or one for which the initial value problem fails to have a unique solution at some point on the solution. The set on which a solution is singular may be as small as a single point or as large as the full real line. Solutions which are singular in the sense that the initial value problem fails to have a unique solution need not be singular functions. 3. A singular measure v with respect to a measure μ, is the property that there exists a measurable set E with μ(E) = 0 and $\mathrm{V(F)} = \mathrm{v(F \cap E)}$, for all measurable sets F; this relationship is symmetric. If two measures v and μ are finite, then there is a Lebesgue decomposition $v = v_1 + v_2$, where v_1 is singular with respect to $\mu(V_1$ [illegible] $\mu)$ and v_1 is absolutely continuous with respect to μ.

SINGULAR CYANOBACTERIA (BLUE-GREEN ALGAE; BLUE-GREEN BACTERIA; CYANOPHYTA): A phylum of single-celled organisms that resemble bacteria in their internal cell structure and lack an enclosed nucleus and other specialised cell structures. They are prokaryotic and are the earliest known form of life on the earth. They are widely distributed in aquatic habitats, on the damp surfaces of rocks and trees and in the soil; where they are joined together in colonies or filaments; and obtain energy through photosynthesis and reproduce by simple cell division or by fragmentation of the filaments. Some fix nitrogen and thus are necessary to the nitrogen cycle, while others live a symbiotic existence; for example, living in association with fungi to form lichens. Sometimes they form algae blooms on fresh water bodies, where the water is polluted by nitrates and phosphates from fertilisers and detergents and cover the water's surface. Overtime, they give off toxins as they decay, which kill fish and other wildlife and can be harmful to domestic animals, cattle and people.

SINGULAR MATRIX: A matrix whose determinant is zero. It does not possess an inverse.

SINGULAR POINT: 1. In mathematics, it is also called **singularity** and refers to any point at which an algebraic curve does not have a unique smooth tangent either, for example, because it is isolated, has a cusp or a point of self-intersection. At this point a function of a complex variable has no derivative but every neighbourhood contains points at which the function has derivatives. 2. A point on the boundary of an open disk that is not a regular point. It is a point at which the derivative of a given function of a complex variable does not exist but every neighbourhood of which contains points for which the derivative does exist.

SINGULAR SOLUTION: A solution of an ordinary differential equation for which the initial value problem fails to have a unique solution at some point on the solution and cannot be obtained from the general solution by choosing suitable values of the arbitrary constants. The set on which a solution is singular may be as small as a single point or as large as the full real line.

SINGULAR VALUES: In mathematics (linear algebra), the singular values of a real matrix A, is any of the square roots of the eigenvalues that are non-negative real numbers, usually listed in decreasing order usually listed in decreasing order, of the product of a real malrix, A with its transpose. These values can be obtained from **singular values decomposition** of A, which is a factorisation of a real or complex matrix A into the product of three matrices $A = UDV^T$ where the columns of U and V are orthonormal and the matrix D is diagonal with positive real entries. function returns the hyperbolic sine of a number, that can be expressed using the constant e.

SINGULARITY: 1. In physics, it is an infinite dense point in space–time when matter is infinitely dense, for example at the centre of the black hole where the laws of physics no longer apply and a point of a function takes an infinite value. 2. In mathematics, it is a point at which a function of a complex variable does not exist and cannot be differentiated although every neighbourhood of that point contains points, where function is differentiable. Also, it is a discontinuity that is not removable.

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SINGULARY (MONADIC; UNARY): Of an operator, predicate, etc., having only a single argument place. Negation, inversion and set complementation are monadic operators. A monadic relation is a (one place) predicate.

SINH: The symbol for the hyperbolic function hyperbolic sine. It is related to the sine function, for any complex number z, by the identity $\sinh z = -i \sin iz$, where $i = \sqrt{-1}$. It is an exponential function that returns the hyperbolic sine of a number and can be defined in terms of the exponential function using the constant e as $\sinh z = \frac{e^z - e^{-z}}{2}$. It is an odd function of which both the derivative and antiderivative or indefinite integral are cosh, the hyperbolic cosine function. Together with cosh, they satisfy $\cosh^2 z - \sinh^2 z = 1$ and $\sinh(2z) = 2\sinh z \cosh z$.

SINISTRORSE: In botany, it describes a growing leaf that is turned spirally anticlockwise or from the right towards the left, as occurs in the leaves of some stems. On the other hand, a **dextrorse** grows upward spirally from left to right.

SINK: 1. The receptor for materials such as greenhouse gas, an aerosol, or a precursor of a greenhouse gas which is removed from the atmosphere. For example, **carbon sink** is anything that collects and stores carbon dioxide (CO_2) from the atmosphere. They include trees and other vegetation, forests and oceans. Plants collect carbon dioxide from the atmosphere to use in photosynthesis; some of this carbon is transferred to soil as plants die and decompose. The oceans are a major carbon storage system for carbon dioxide. The process by which carbon sinks remove carbon dioxide (CO_2) from the atmosphere is known as **carbon sequestration**. 2. A site within a plant or cell where a demand exists for particular substrates or catalysts. Thus the mitochondria are sinks for oxygen and respiratory substrates while the chloroplasts require carbon dioxide. At a higher level fruits, roots and shoot apices are sinks for photosynthates. 3. In mathematics, it is a vertex in a network, whose flows are directed towards one direction.

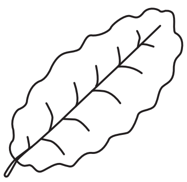
SINK HABITAT: A habitat where mortality exceeds reproduction and immigration exceeds emigration. When the environmental conditions in a habitat are not constant but fluctuate, allocating offspring to sink habitats can increase the long term growth rate of a population.

SINKHOLE (POT-HOLE): A saucer-shaped or funnel-shaped depression in a limestone landscape formed when the rock is dissolved away by standing water. If a stream of river flows into the depression and disappears underground, it is called **swallow hole**.

SINOATRIAL NODE (SA NODE; SINOAURICULAR NODE; SINUS NODE): A region of modified muscle cells in the right atrium that sends timed electrical impulses to the heart's other muscle cells, causing them to contract and determines the heart rate. It is the heart's natural pacemaker.

SINTERED GLASS: Porous glass made by sintering powdered glass. It is used for filtration of precipitates in gravimetric analysis.

SINTERING: The process of heating and compacting a powered material at a temperature below its melting point in order to weld the particles together into a single rigid shape. The process is used to form complex shapes, to produce alloys and to allow work on metals with very high melting points. Sintering is also used in the preliminary moulding of ceramic or glass powders into forms that can then be permanently fixed by firing.



Sinuate leaf

SINUATE: In botany, having the margin strongly or distinctly wavy, as a in leaf.

SINUOUS: In botany, a wavy or serpentine form; or having many curves and turns.

SINUS: 1. A sac-like cavity or organ such as the sinus venosus in an animal or a wide channel containing blood, especially venous blood. 2. A cavity filled with air in the bones of the face and skull, especially one opening into the nasal passage. 3. An elongated tract leading from a pus-filled region of the body to the exterior or to the cavity of a hollow organ. 4. An indentation or cleft between the lobes of a leaf or the fused petals of a corolla.

SINUS VENOSUS: A thin-walled chamber of the heart of vertebrate embryos that conveys deoxygenated blood to the single atrium. It is retained in adult fishes but in higher vertebrates it becomes incorporated into the left atrium.

SINUSOID: 1. In mathematics and physics, it is any curve described by the equation $y = \sin x$ and has the same shape as the sine curve and derived from the sine curve by multiplication or addition of a constant, but with possibly different amplitude, period or intercepts with the axis. **Sinusoidal** is anything that relates to a sine curve or wave. 2. In biology, it is a tiny blood vessel or blood filled space in an organ. Sinusoids replace capillaries in certain organs notably the liver. They allow more direct contact between the blood and the tissue it is supplying.

SINUSOIDAL ANGULAR MODULATION (ANGLE MODULATION): The variation in the angle or phase of a sine-wave carrier to transmit data, instead of varying the amplitude. These variations are controlled by both the frequency and the amplitude of the modulating wave. When used in an analogue manner, it is referred to as either frequency modulation (FM) or phase modulation (PM), depending upon whether the instantaneous frequency or instantaneous phase varies with the modulation. When used in a digital modulation format, it is referred to as either frequency-shift keying (FSK) or phase-shift keying (PSK). Both FSK and PSK can be used with either a binary or an *M*-ary alphabet, where *M* is an integer greater than 2.

SINUSOIDAL OSCILLATOR (SINE-WAVE OSCILLATOR): One of the two general categories of oscillators; the other being relaxation. Sinusoidal oscillators consist of amplifiers with resistance/capacitance (RC) or inductor/capacitance (LC) circuits that have adjustable oscillation frequencies or crystals that have a fixed oscillation frequency. Its output voltage is a sine-wave function of time. Relaxation oscillators generate triangular, sawtooth, square, pulse or exponential waveforms. Oscillators are used as a reference in such applications as audio, function generators, digital systems and communication systems.

SINUSOIDAL PROJECTION: A map projection in which areas are equal to corresponding areas on a globe, the parallels and the prime meridian being straight lines and the other meridians being increasingly curved outward from the prime meridian.

SINUSOIDAL WAVE (SINE WAVE): Any waveform that has an equation in which one variable is proportional to the sine of the other or a waveform that represents periodic oscillations in which the amplitude of displacement at each point is proportional to the sine of the phase angle of the displacement and that is visualised as a sine curve. Such a waveform can be generated by an oscillator that executes simple harmonic motion. It occurs often in pure mathematics, as well as physics, signal processing, electrical engineering and many other fields. An example of where it occurs is in alternating current supply, which reverses 50 or 60 times per second depending on which country you live in. Also, an ideal, unmodulated wireless signal has a sine waveform, with a frequency usually measured in megahertz (MHz) or gigahertz (GHz). Varying electrical voltages can be visualised on an instrument called an **oscilloscope**. The **amplitude** of a sine wave is the maximum vertical distance it reaches in either direction from zero or from the centre line of the wave. As a sine wave is symmetrical about its centre line, the amplitude of the wave is half the peak to peak value. In general, a sine wave is given by the formula $A \sin(\omega t)$, where A is the amplitude and ω is the frequency. In electrical voltage measurements, amplitude is sometimes used to mean the peak-to-peak voltage (Vpp). This number will be twice the mathematical amplitude. The **frequency** of a sine wave is the number of complete cycles that occurs every second. Its unit is the Hertz (abbreviated Hz). One Hertz (1Hz) is equal to one cycle per second. In the power supply system in Ghana and most countries, the frequency is 50 cycles in one second or 50 Hz. The **period** of a wave is the time it takes to perform one complete cycle. The wave is a series of identical cycles happening repeatedly.

SIPHON: 1. A pipe or tube system consisting of two flexible or rigid legs in the form of an inverted U-tube with one side longer than the other. It is used to convey a liquid from one vessel to another vessel at a lower level, over an intermediate point that is higher than either, with the flow maintained by gravity and atmospheric pressure as long as the tube remains filled. 2. A tubular organ, especially of aquatic invertebrates such as squids or clams, by which water is taken in or expelled.

SIPHONACEOUS (SIPHONOUS): Algae and other protoctists in which the thallus is coenocytic or contains many nuclei not separated by cell walls and often take the form of a tubular structure.

SIPHONAPTERA: An order of secondarily wingless insects comprising the fleas. They are bloodsucking pests of humans and animals and transmit serious diseases from animals to humans.

SIPHUNCULATA (ANOPLURA): An order of secondarily wingless insects comprising the sucking lice. These pests cause irritation to humans and domestic animals and can transmit diseases, including typhoid.

SIRE: The male parent of a four-legged animal, especially a domesticated animal such as a bull or stallion.

SIROCCO: A warm, humid wind occurring in southern Europe, from the south or southeast originating over the Sahara Desert and gaining humidity from the Mediterranean.

SISAL: *Agave sisalana*, a plant with fleshy elongated leaves which provides a strong fibre used for making ropes and twines for marine, agricultural, shipping and general industrial use, as well as for matting, rugs, hats and brushes. It grows to a height of about 1 metre and a diameter of about 38 centimetres. The stalk bears fleshy, rigid, grey to dark green, lance-shaped leaves in a dense rosette. Tanzania and Brazil are the main producers of sisal.

SISTER CHROMATIDS: Chromatids joined by a common centromere and carrying identical genetic information, unless crossing-over has occurred. Each member of sister chromatids contains a full copy of the DNA sequence.

SISTER SPECIES: Either of the two descendant species formed when one species splits during evolution. The sister species is the one closely related to any given species, since both share an ancestral species not shared by any other. In classification system based on the principles of cladistics, a system of biological classification based on a common ancestry, sister species are always grouped together. Moreover the sister group is the prime choice for use in outgroup comparison.

SISTROID: In mathematics, of an angle, contained between the convex sides of two intersecting curves as opposed to cissoid.

SITE INDEX (SI): A measurement commonly used in forestry to describe the productivity of a particular location, called site, for forest growth under the existing or specified environment, given as the average height of dominant and co-dominant trees in a stand at a specified age such as 50 and 100 years. For example, a red oak with an age of 50 years and a height of 21 metres will have a site index of 70. Methods used to determine site index are based on tree height, plant composition and the use of soil maps.

SITE MAP (NAVIGATION MAP): In computing, it is a specialised tool to help users find their way around a website; it is an overview of the pages within a website. Site maps of smaller sites may include every page of the website, while site maps of larger sites often only include pages for major categories and subcategories of the website. Site maps help visitors or search engines get to any popular section of a web site or see what the web site has to offer.

SITE OF SPECIAL SCIENTIFIC INTEREST (SSSI): Any area or location earmarked for scientific research purposes and as a heritage site due to its uniqueness in plant and animal diversity and contribution to the environment. Sites for biological interest are known as Biological SSSIs, and sites for geological or physiographic interest are Geological SSSIs.

SITE-DIRECTED MUTAGENESIS: A technique used in genetic engineering for introducing particular changes to the base sequence of a gene at a specific site. It allows specific and precise mutation to be made and is done, for example, to modify the amino-acid sequence of the protein expressed by the gene to investigate how such a change affects the protein structure and function.

SITKA SPRUCE: *Picea sitchensis*, is a fast-growing temperate softwood coniferous tree of Pinales Order, Pinaceae family and genus Picea. It grows to 50-70 metres tall and sometimes reaches 100 metres tall, with a trunk diameter of up to 5 m, exceptionally 6-7 metres diameter. It is a long-lived tree, with individuals over 700 years old known. It is used for making paper and as timber.

SIVAPITHECUS (RAMAPITHECUS): A genus of extinct primates that lived about 8-12 million years ago. It was about 1.5 metres in body length, similar in size to a modern orang-utan. In most respects, it would have resembled a chimpanzee, but its face was closer to that of an orang-utan. The shape of its wrists and general body proportions suggest that it might have spent a significant amount of its time on the ground, as well as in trees. It had large canine teeth and heavy molars, suggesting a diet of relatively tough food, such as seeds and savannah grasses. Fossil remains have been found in India, Pakistan, the Near East and East Africa.

SIX'S THERMOMETER (MAXIMUM AND MINIMUM THERMOMETER): A self-registering thermometer used to measure and record maximum and minimum temperatures over a period of time. It was named after British born scientist and retired business man, James Six (1731 1793), who invented it in 1782. It consists of a U-shaped glass tube whose bottom half is filled with mercury and graduated right-hand and left-hand tubes for recording the minimum and maximum temperatures registered. The left-hand tube is filled with alcohol, which expands (when the temperature rises) or contracts (when the temperature drops). The right-hand tube is partially filled with alcohol and a portion is filled with gas. When the temperature increases, the alcohol in the left side expands and forces the mercury and the alcohol column on the right side into the gas chamber. When the temperature decreases, the alcohol in the left side contracts and the gas expands, forcing the liquid back into the left side and pushing the index along with it. In all cases, the maximum and minimum temperatures recorded remain at their positions until they are reset.

SIZE CHANGE: A transformation in which the image of a figure is an enlargement (stretch) or reduction (shrink) of the original figure by a given scale factor.

SKELETAL ATOM: An atom of a skeletal structure.

SKELETAL BOND: A bond connecting two skeletal atoms.

SKELETAL DIVISION (SKELETON DIVISION): A long division in which most or all of the digits are replaced by symbols (usually asterisks) to form a cryptarithm.

SKELETAL MUSCLE (STRIATED MUSCLE; STRIPED MUSCLE): A type of muscle attached to bones, that is responsible for skeletal movements. The peripheral portion of the central nervous system (CNS) controls the skeletal muscles. These muscles are under conscious or voluntary, control. The basic unit is the muscle fibre with many nuclei. Skeletal muscle makes up the bulk of muscles in the body.

SKELETAL STRUCTURE: The sequence of atoms in the constitutional unit(s) of a macromolecule, an oligomer molecule, a block or a chain which defines the essential topological representation.

SKELETAL SYSTEM: The bodily system that consists of the bones, their associated cartilages and the joints and supports and protects the body, produces blood cells and stores minerals. **Skeletal** relates to a skeleton.

SKELETON: The hard framework of an animal that provides mechanical support for the body as well as protection for the internal organs such as brain, heart and lungs and for attachment of muscles. The skeleton is known as **endoskeleton** when it is internal as in humans and other vertebrates or **exoskeleton** when it is external such as the shell of an insect, a mollusc or a crustacean. The skeleton makes up about 30-40% of an adult's body mass. The skeleton is composed of bones. Each bone is a complex living organ that is made up of many cells, protein fibres and minerals. At birth, babies have about 300 bones. As we grow older, various bones fuse together and the typical adult human skeleton has 206 bones. Some individuals may have more or fewer bones because of anatomical variations. The breakdown is as follows: the axial skeleton has 80 bones composed of the skull with 28 bones and the torso with 52; the appendicular skeleton has 126 bones composed of the upper extremity with 64 (32 x 2) and the lower extremity with 62 (31 x 2). The axial skeleton runs along the body's midline axis and includes the skull, hyoid, auditory ossicles, ribs, sternum and the vertebral column. The appendicular skeleton includes the folowing regions: upper limbs, lower limbs, pelvic girdle and pectoral (shoulder) girdle. There are different types of bones, categorised by shape: flat, long, short and irregular. Flat bones are like those found in the skull, ribs, sternum and scapula. Long bones are the type found in the extremities, like the femur or thighbone in the leg or ulna in the arm. Short bones are found in the wrists and ankles and some of the irregular bones are found in the vertebrae and at the sutures in the skull. The longest bone in our bodies is the femur (thigh bone). The smallest bone is the stirrup bone inside the ear. Bones are connected to other bones at joints. The different types of joints include: fixed joints such as in the skull, which consists of many bones; hinged joints such as in the fingers and toes; and ball-and-socket joints such as the shoulders and hips. **Skeletal muscles**, also known as **striated muscles** or **striped muscles** are muscles attached to bones and are responsible for skeletal movements. The centre of the bone is filled with marrow and this is surrounded by the hardened bone tissue itself. In bone, the extracellular matrix is composed of collagen with calcium phosphate deposited in it. Vitamin C is needed for collagen synthesis. Bone tissue is composed of repeating, circular units called **Haversian systems**. Within each Haversian system, there is a central canal where blood vessels and nerves can be found. This is surrounded by concentric layers of the matrix material called **lamellae**. In lamellae are spaces called lacunae which contain the osteocytes. Osteoblast is a large cell responsible for the synthesis and mineralisation of bone during both initial bone formation and later bone remodelling. Osteoblasts form a closely packed sheet on the surface of the bone, from which cellular processes extend through the developing bone. They produce many cell products, including the enzymes alkaline phosphatase and collagenase, growth factors, hormones such as osteocalcin and collagen (the structural protein of bones, tendons, ligaments and skin). A large part of the bone formed during the growth period is destroyed by specialised cells called **osteoclasts** in the growth process. Some other functions of the skeleton include: (i) providing support for the soft tissues of the body, enabling us to stand upright; (ii) providing anchorage for muscles so they can pull effectively, with bones and also acting as levers for pulling; (iii) the rib cage protects the heart and lungs, while the brain is protected by the skull.

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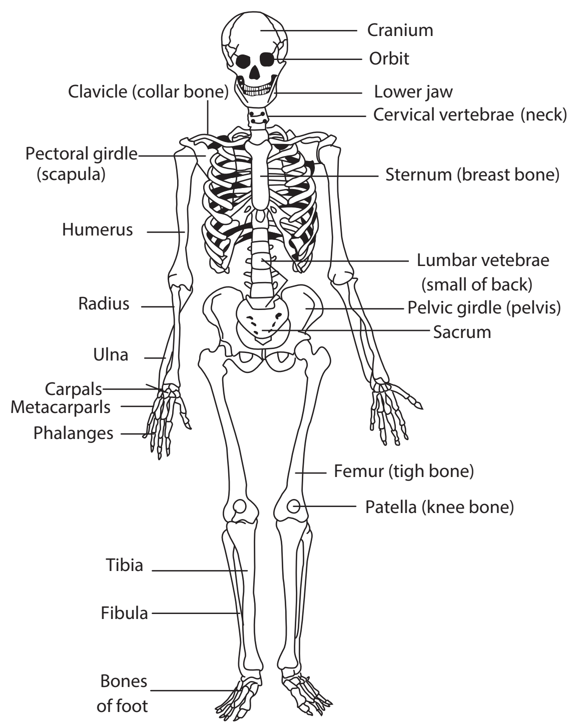
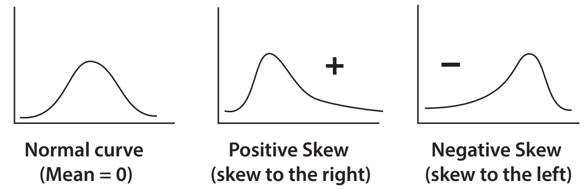


Anterior view of of the human skeleton

SKETCHPAD: An interactive computer-graphics program developed by an American, Ivan Sutherland (1938 -) in 1963 for his doctoral thesis project at Massachusetts Institute of Technology (MIT), United States of America. It allowed a user to interact with objects displayed on a screen using light-pen, a type of mouse, allowing users to visualise and control program functions. From it modern programs such as computer-aided design (CAD), computer graphics, the graphical user interface (GUI), etc., have been developed.

SKEW: 1. In mathematics, also called **agonic**, it is neither parallel nor intersecting. The term is used for the non-intersecting diagonals of adjacent faces of a cuboid or lines that lie in different planes and do not intersect one another and therefore, not forming an angle. 2. The skew of a curve, also called **twisted** is when it does not lie in a plane. 3. In statistics, it refers to a distribution that is not symmetric.

SKEW DISTRIBUTION: Non-symmetrical or asymmetrical unimodal distribution of data on a graph, in which values tail off more slowly in one direction than the other. Often it is a log-normal distribution. A positive skew distribution has a longer tail of high values than low ones, signifying low performance; a negative skew distribution signifies high performance. Mean, median and mode do not coincide as they do with a normal distribution.



Skew Distribution

SKEW FIELD (SKEW-FIELD): In mathematics, it is a division ring, in which every nonzero element has a multiplicative inverse, such that the equations $ax = b$ and $ya = b$ with $a \neq 0$ are uniquely solvable. It satisfies all the field axioms except commutativity of multiplication. An example of a non-commutative associative skew-field is the system of quaternions.

SKEW LINES (CROSSING LINES): Lines that lie in different planes and do not intersect one another.

SKEW-HERMITIAN MATRIX: In mathematics, it is a square matrix, A with complex entries whose conjugate transpose is equal to the negative of its adjoint. Its relation is written $A^{(H)} = -A$, where $A^{(H)}$ is the adjoint. An example is the 2 × 2 matrix $A = \begin{bmatrix} 1 & -i \\ i & 1 \end{bmatrix}$.

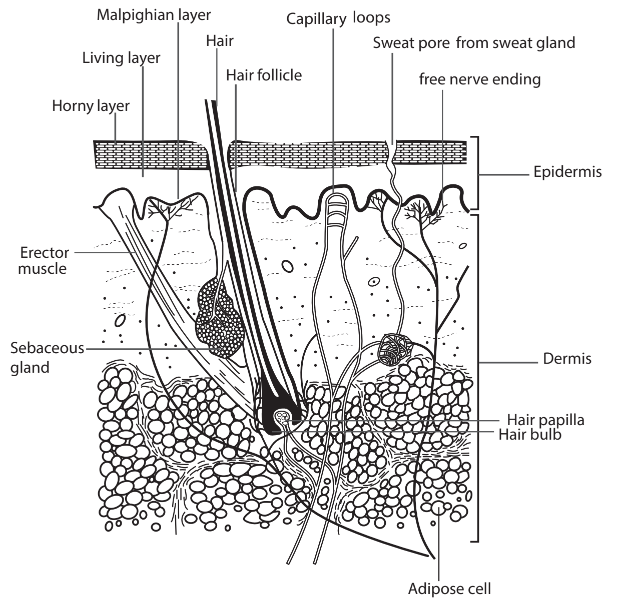
SKEW-SYMMETRIC MATRIX: In mathematics, it is a square matrix, which is equal to the negative of its transnspose. Its principal diagonal elements are equal to zero. Its relation is written $A = -A^{(T)}$, where $A^{(T)}$ is the matrix transpose. An example is the 2 x 2 matrix $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$.

SKEWNESS: In statistics, a measure of the asymmetry of a distribution of a real-valued random variable about its mean. The skewness can be positive or negative or even undefined. A **positive skewness** is when the data points are skewed to the right of the data average; when skewed to left it is **negative skewness**.

SKIM COULTER: The part of a plough, which turns a small slice off the corner of the furrow about to be turned and throws it into the bottom of the one before.

SKIMMED MILK: Milk that has had both fat and fat solubles removed, used as milk substitutes for calves and for human consumption. Skimmed milk contains less fat than whole milk and as such many nutritionists and doctors recommend it for people who are trying to lose weight or maintain a healthy weight.

SKIN: The outer covering of the body of a vertebrate that protects it from injury, water loss, entry of germs and also receives external sensory stimuli such as temperature and pressure. It consists of three layers: an outer one called epidermis, an inner one called dermis with a complex nervous and blood supply and a fat layer, also called the subcutaneous layer. The skin may have a variety of structures such as hair, scales and feathers. The human skin is continually being renewed, in contrast with that of reptiles, which moult. The **epidermis** is the relatively thin, tough, outer layer of the skin. Most of the cells in the epidermis are keratinocytes, which originate from cells in the deepest layer of the epidermis called the **basal layer**. New keratinocytes slowly migrate up toward the surface of the epidermis. Once the keratinocytes reach the skin surface, they are gradually shed and are replaced by younger cells pushed up from below. The outermost portion of the epidermis, known as the **stratum corneum**, is relatively waterproof and, when undamaged, prevents most bacteria, viruses and other foreign substances from entering the body. The epidermis (along with other layers of the skin) also protects the internal organs, muscles, nerves and blood vessels against trauma. In certain areas of the body such as the palms of the hands and the soles of the feet that require greater protection , the outer keratin layer of the epidermis, called **stratum corneum** is much thicker. The **dermis**, the skin's next layer, is a thick layer of fibrous and elastic tissue, made up mostly of collagen, elastin and fibrillin that gives the skin its flexibility and strength. The dermis contains nerve endings, sweat glands and oil (sebaceous) glands, hair follicles and blood vessels. The nerve endings sense pain, touch, pressure and temperature. The sweat glands produce sweat in response to heat and stress. The sebaceous glands secrete sebum into hair follicles. The hair follicles produce the various types of hair found throughout the body. Below the dermis lies the **fat layer**, also called the **subcutaneous layer** that helps insulate the body from heat and cold, provides protective padding and serves as an energy storage area. The fat is contained in living cells, called fat cells, held together by fibrous tissue. In botany, epidermis is the outermost cellular layer which covers the whole plant structure (roots, stem, leaves, flowers and fruit).



Vertical section through the human skin

SKIN DEPTH: The depth beneath the surface of a conductor, which is carrying current at a given frequency due to electromagnetic waves incident on its surface, at which the current density drops to one neper below the current density at the surface.

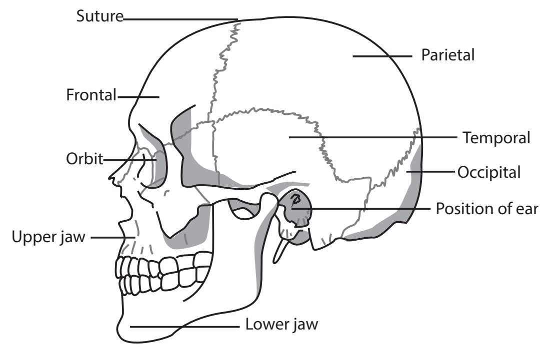
SKIN SPOT: A potato disease, causing pimple-like dark brown spots which can harm the buds in the eyes of seed tubers.

SKINK (SNAKE-LIZARD): Terrestrial and burrowing lizard belonging to the family Scincidae, with small or absent limbs and covered with shiny scales.

SKIP DISTANCE: The smallest separation between a transmitter and a receiver that permits radio signals of a specific frequency to travel from one to the other by reflection from the ionosphere.

SKOLEM FORM: In logic, a prenex normal form of a formula in which all universal quantifiers precede any existential quantifier and that contains no function symbols. In a Skolem normal form, a formula of first-order logic has a prenex normal form with the existential quantifiers removed, leaving only the universal first-order quantifiers.

SKULL: The bony framework of the head of a vertebrate animal. In mammals, the skull consists of the cranium and the bones of the face and jaws. With the exception of the lower jaw, its bones meet in immovable joints (sutures) to form a unit that encloses and protects the brain and sense organs and gives shape to the face.



The Human Skull

SKY WAVE: The propagation of electromagnetic waves refracted back to the Earth's surface by the ionosphere. As a result of skywave

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propagation, a broadcast signal from a distant AM broadcasting station at night or from a shortwave radio station or during sporadic E season, a low band TV station can sometimes be heard as clearly as local stations. Most long-distance HF radio communication, between 3 and 30 MHz is a result of sky wave propagation.

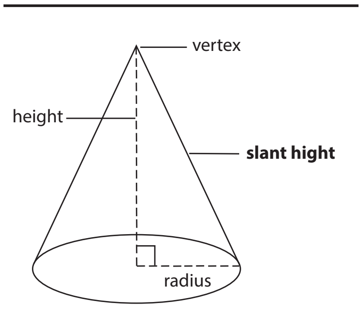
SKYRMION: A mathematical representation of baryon in which it is regarded as a soliton for a meson field theory. It was formulated by British physicist Tony Hilton Royle Skyrme (1922–1987). This representation can be justified using quantum chromodynamics and gives very useful insight into both baryon and nuclear structure.

SLACK VARIABLE: A set of nonnegative variables temporarily added to a linear program to replace an inequality of the form $g(x) \leq 0$ by the equality $g(x) + y = 0$ and the inequality $y \geq 0$.

SLAG: A substance containing impurities in metal and made during the smelting or refining of metals. Slag consists mostly of mixed oxides of elements such as silicon, sulfur, phosphorus and aluminium, ash and products formed in their reactions with furnace linings and fluxing substances such as limestone; for example, calcium oxide (CaO) reacts with silicon(IV) oxide (silicon dioxide, SiO_2) to form calcium silicate (slag). During smelting or refining, slag floats on the surface of the molten metal, protecting it from oxidation by the atmosphere and keeping it clean. It cools into a coarse aggregate used in certain concretes, used as a road-building material, as ballast and as a source of available phosphate fertiliser.

SLAKED LIME (CALCIUM HYDROXIDE): A white alkaline powder with the chemical formula $Ca(OH)_2$. It is obtained from the action of water on calcium oxide. It is used in the manufacture of cement, plaster and glass. A solution of slaked lime is known as **limewater**.

SLANT HEIGHT: The height of a shape such as a cone, measured along the slope or the distance from any point on the circle to the apex of the cone.



Slant height of a cone

SLAPMARK: The herd mark allocated by the Department for Environment, Food and Rural Affairs (Defra) in the United Kingdom. It is put on both shoulders of a pig.

SLASH AND BURN AGRICULTURE: A form of shifting agriculture in which forest or vegetation is cut down and burnt to create open space for growing crops. The ash from the burnt plants provides nutrients to fertilise crops. It is used for growing crops for just a short period of time (about 2 - 3 years), because the land begins to lose its fertility as the nutrients are used up. The land is therefore left fallow for a longer period of time so that the vegetation can regenerate to restore nutrients. Its disadvantages include deforestation, erosion, loss of nutrients and loss of biodiversity. For example, it is largely responsible for the deforestation of the virgin Amazon forest in Brazil.

SLATE: A blue to grey fine-grained metamorphic rock characterised by the ease with which it cleaves into large thin sheets. It is formed mainly by the metamorphosis of mudstone or shale, in which platy minerals become aligned in parallel planes. Slate is used for roofing, flooring, panels (both decorative and electrical) and chalkboard.

SLATER'S CONDITION: In mathematics, the constraint qualification placed on a set of inequalities $g_1(x) \leq 0, ..., g_n(x) \leq 0$ (usually with convex functions) that the inequalities should have a simultaneous strict solution: a vector z with $g_1(x) \leq 0, ..., g_n(x) \leq 0$.

SLATTED MOULDBOARD: A type of mouldboard which breaks up the soil as it is being ploughed. **Mouldboard** is the curved blade of a plough, which turns over the furrow

SLEEP: A readily reversible state of reduced awareness and metabolic activity that occurs periodically in many animals or a state of partial or full unconsciousness in people and animals, during which voluntary functions are suspended and the body relaxes, rests and restores itself. The onset of sleep in humans and other mammals is marked by a change in the electrical activity of the brain.

SLEEP APNOEA (APNOEA; APNEA): A disorder in which breathing stops for periods longer than 10 seconds during sleep. It can be caused by failure of the automatic respiratory system to respond to elevated blood levels of carbon dioxide.

SLEEP MOVEMENT: The movement of the leaves or petals in response to daily rhythms of dark and light. The leaves are horizontal in daylight and folded vertically at night. For example, crocuses, gazanias, moss rose and many other flowers close at night and reopen the following day. Mimosa trees, clover, garden beans and many other plants have leaves that fold up or down at night.

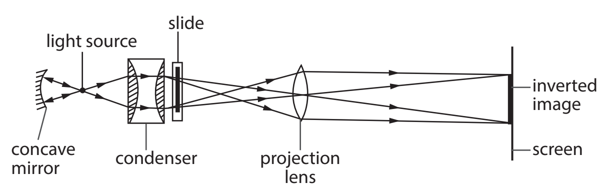
SLEEPING SICKNESS (HUMAN AFRICAN TRYPANOSOMIASIS; TRYPANOSOMIASIS): A vector-borne parasitic disease caused by protozoan parasites belonging to the genus Trypanosoma, usually found in tropical Africa caused by two protozoa, *Trypanosoma brucei rhodesiense* and *Trypanosomoa brucei gambiense* that are carried by tsetseflies (Glossina genus), which have acquired their infection from human beings or from animals harbouring the human pathogenic parasites. Only certain species of tsetse-flies transmit the disease. there are many regions in Africa where tsetse flies prevail but where there are no cases of sleeping sickness. Sleeping sickness spreads more easily in poor settings, generally occurring in remote rural areas where health systems are weak or non-existent. The rural populations that depend on agriculture, fishing, animal husbandry or hunting, are the most exposed to the tsetse bite and therefore to the disease. In cattle the disease is called **Nagana**. The disease that results in the more severe form of the illness is caused by Trypanosoma brucei rhodesiense. Some of the symptoms are sleepiness, anxiety, weakness, fever, headache and swollen lymph nodes all over the body. Another form of trypanosomiasis, known as American trypanosomiasis or Chagas disease occurs mainly in Latin America.

SLEEP WALKING (SOMNAMBULISM): The act of walking around and performing other activities such as eating while sleeping with no recollection of it on waking up. Common in childhood, it may continue into the adulthood. It may be caused by stress, seizure disorders such as epilepsy, bipolar disorders or other neurological conditions; however, in most cases the causes are not known.

SLEET: Rain or melted snow that freezes into ice pellets with diameters of about 5 mm or less before hitting the ground. Sleet may occur when a warm layer of air lies above a below-freezing layer of air at the Earth's surface.

SLEPTON: A symmetry that relates elementary particles of one spin to other particles that differ by half a unit of spin and are known as superpartners. In a theory with unbroken supersymmetry, for every type of boson, there exists a corresponding type of fermion with the same mass and internal quantum numbers and vice-versa.

SLIDE PROJECTOR: A device for displaying images of transparent slides on a screen.



Slide projector

SLIDE RULE: A mathematical instrument with pairs of logarithmic sliding scales, used for rapid calculations, including multiplication, division and the extraction of square roots. It consists of two rulers marked with graduated logarithmic scales, one sliding inside the other.

SLIDING FILAMENT THEORY: A theory of muscle contraction in which the actin and myosin filaments of striated muscles slide over one another, shortening the entire length of the sarcomere. In sliding over one another, the myosin heads will interact with the actin filaments and, using adenosine triphosphate (ATP), bend to pull past the actin. Myosin-binding sites on the actin filaments are exposed when calcium ions bind to troponin molecules in these filaments.

SLIDING GROWTH: A type of cell growth in which an area of the cell wall expands and slides over that part of the wall of the adjacent cell with which it was originally in contact. This results in the severance of plasmodesmata between adjacent cells but does not lead to cell disruption.

SLICK SPOT: Small areas in a field, usually of a B horizon soil which becomes slick (smooth or slippery) when wet, because of a high content of alkali or exchangeable sodium that interferes with the growth of most crops.

SLIME BACTERIA (MYXOBACTERIALES): An order containing flexible creeping bacteria that form colonies in self-produced slime. They lack a rigid cell wall and flagella and move by gliding across the substrate; and inhabit moist soils or decaying plant matter or animal waste. Most of its members are terrestrial. They differ from other bacteria in producing resting cells, termed microcysts that are often borne in fruiting bodies.

SLIME LAYER: A diffuse layer found immediately outside the cell wall in certain bacteria consisting mostly of exopolysaccharides, glycoproteins, and glycolipids that protects it from dessication, loss of nutrients, and chemicals such as antibiotics. Exopolysaccharides are high molecular weight polymers that are composed of sugar residues and are secreted by a microorganism into the surrounding environment.

SLIME MOULD: 1. Small simple organisms of the phylum Acrasiomycota, especially of the genus *Dictyostelium* that grow on dung or decaying vegetation. They exist either as uninucleate cells or as multinucleate mass of cytoplasm without cell walls at the vegetative stage. Slime moulds show amoeboid movement and feed by ingesting small particles of food. They reproduce by means of spores with protective walls. 2. Various organisms of the phylum Myxomycota that grow on decaying vegetation and in moist soil and have a similar but more advanced life cycle.

SLING: A type of harness that is used to support the weight of an animal that is suffering from some kind of disability.

SLING PSYCHROMETER: A handheld psychrometer that spins around, used to measure dewpoint and relative humidity.

SLINK CALF: A calf born early, before the normal gestation period is completed.

SLIP: 1. A small piece of plant stem, used as a cutting or in budding or a piece taken from a stem, leaf or root used to grow a new plant. 2. To dislocate or displace a bone, especially in the spine. 3. The deformation of a metallic crystal by shearing along a plane. 4. In computer science, it is the acronym for **Serial Line Internet Protocol**, a method of connecting a computer to the Internet through a dial-up connection, using a serial line, such as telephone circuits and RS-232 cables.

SLIP RING: Part of the rotor in an electric alternator or a metal ring in a generator or motor to which current is delivered or from which it is removed by brushes. A slip ring is attached to each end of the coil of wire that spins with the rotor to provide a continuous electrical connection between rotating and stationary conductors. Typical applications of slip rings are in electric rotating machinery, synchros and gyroscopes. Slip rings are also employed in large assemblies, where a number of circuits must be established between a rotating device, such as a radar antenna and stationary equipment.

SLIPPERY BURR (SLIPPERY DICK; *Corchorus siliquosus*): A short-lived subshrub in the Malvaceae family having a thin, erect stem with lateral branches and small leaves, distributed in Caribbean, Central America and North America. It is an inconspicuous weedy plant (about 1 metre tall) with a taproot and blends well in the surrounding vegetation. It produces small brightly coloured flowers arranged in umbels arising from nodes. The calyx has 5 unfused greenish sepals. The corolla has 5 unfused bright yellow petals. There are numerous stamens. The ovary is superior with 5 locules and numerous seeds. They are short-lived or ephemerals and usually open after noon and close the following morning. The flowers develop into fruits which are capsules. The plant is important for its leaves, which are used to make a tea and sometimes cooked and eaten like spinach.

SLOPE (GRADIENT): 1. A position or direction at an angle less than 90° to the Earth's surface. 2. An inclined line, surface, plane, position or direction. 3. The rate at which an ordinate of a point of a line on a coordinate plane changes with respect to a change in the abscissa. 4. The tangent of the angle of inclination of a line or the slope of the tangent line for a curve or surface.

SLOPE FORM: The slope form of the equation of a straight line is $y = mx$, where m is the gradient of the line.

SLOPE INTERCEPT FORM: The slope-intercept form of equation of a straight line is $y = mx + c$. The slope is m and the y-intercept is c.

SLOPE OF THE TANGENT: The slope of the tangent to the graph of $y = f(x)$ at the point (x, y) is $f'(x)$, defined as follows:

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}.$$

SLOT (EXPANSION SLOT): In computers, it is an opening in the motherboard of a computer that allows additional units to be added.

SLOTH: Medium-sized arboreal mammal of the order Pilosa, inhabiting the jungles of Central and South America characterised by very slow movement.

SLOW NEUTRON (THERMAL NEUTRON): A low energy neutron, which is free and not bound within an atomic nucleus and is in thermal equilibrium with the surrounding medium, especially one ejected from atomic nuclei during nuclear reactions such as fission and slowed by a moderator. Its thermal energy per particle is about 0.025 electron volt with neutron velocity of 2200 m/s at 293.6 K.

SLOW VIRUS: Any of a group of quasi-viral or subviral agents associated with a disease having a long incubation period of months to years with a gradual onset frequently terminating in severe illness and/or death. They are now known to be caused by several different conventional viruses and also unconventional infectious agents. Examples of slow virus diseases are scrapie of sheep, bovine spongiform encephalopathy in cattle and Creutzfeldt-Jakob in humans.

SLUDGE: 1. A thick wet substance especially wet mud or snow. 2. The solid or semi-solid left from industrial wastewater or sewage treatment processes. It can also refer to the settled suspension obtained from conventional drinking water treatment and numerous other industrial processes.

SLUDGE COMPOSTING: The decomposition of sewage for use as a fertiliser or mulch. Aerobic composting involves the activity of aerobic microbes and hence the provision of oxygen during the composting process. The four most common methods for composting sewage sludge are aerated static pile, windrow, aerated windrow and in vessel. Aerobic composting generally is characterised by high temperatures, no foul odours and is more rapid than anaerobic composting. Anaerobic composting is characterised by low temperatures, the production of odorous intermediate products and generally proceeds at a slower rate than does aerobic composting. In mesophillic composting the temperatures are kept at intermediate temperatures, 15 to 40 °C (288-313 K, which in most cases is the ambient temperature. Thermophillic

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composting is conducted at temperatures from 45 to 65 °C (318-338 K). The general climate for sludge composting is healthy. As it gets increasingly difficult to site combustion facilities and where land application is not feasible, composting is becoming the preferred method for handling sewage sludge.

SLUG: 1. Any gastropod mollusc that lacks a shell, has a much reduced shell or has a small internal shell. All slugs are descended from snails that gradually lost or reduced their shells over time. Slugs have a soft, slimy body and live in moist habitats on land except for one freshwater species. Certain species, such as *Limax maximus*, have become serious pests in gardens. Note that slug is not a snail. Snails are gastropods that have coiled shells that are big enough to retract into. 2. A gravitational unit of mass in the foot-pound-second system, which is equal to the mass that will acquire an acceleration of 1 ft sec^{-2} when acted on by a 1 pound-force. One slug is equivalent to 14.5939~ kg.

SLUG PELLET: A small hard piece of mixture, usually blue-green coloured, containing a substance such as metaldehyde which kills slugs.

SLUICE: 1. A channel for water especially through a dam or other barrier; controlled by a valve or gate. It is often used for carrying excess water. 2. A long inclined trough used to separate gold ore from sand or gravel. 3. An artificial stream or channel for floating log.

SLURRY: A suspension of solids in a liquid. An example is a mixture of water, cement and sand to form concrete. A thin mixture of a liquid, especially water and any of several finely divided substances, such as cement, plaster of Paris or clay particles.

SLURRY GUN: A powerful spraying device that spread slurry. **Slurry spreader** is a machine which spreads slurry; and **slurry injector** is a tractor-hauled machine which injects slurry into the soil.

SMALL CIRCLE: In mathematics, a circular section of a sphere whose plane does not contain the sphere's centre.

SMALL INTERFERING RNA (siRNA; SILENCING RNA; SHORT INTERFERING DNA): Any of several types of small RNA molecules that are produced in cells from double-stranded (ds) precursors by the action of the protein Dicer. This mechanism of RNA interference probably evolved as a means of combating infection by RNA viruses, where it interferes with the expression of a specific gene. In addition to their role in the RNA interference pathway, siRNAs also acts in shaping the chromatin structure of a genome.

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SMALL INTESTINE: The long, narrow, convoluted tube portion of the alimentary canal, about 6.7-7.6 metres long between the stomach and the large intestine, consisting of the duodenum, jejunum and ileum, where digestion of food and most absorption of nutrients take place. It plays an essential role in the final digestion and absorption of food.

SMALL NETTLE (ANNUAL NETTLE, DWARF NETTLE; *Urtica urens*): An annual stinging temperate weed, *Urtica urens* of the genus Urtica. It has medicinal and therapeutic uses and important in the diet of many farmland birds.

SMALL UBIQUITIN-RELATED MODIFIER (SUMO): Any of a family of proteins with structural similarity to ubiquitin that are attached to other proteins within cells that modify their function or change their location.

SMALLTALK: In computer science, it is the first high-level programming language used in object-oriented applications.

SMALLPOX: An acute infectious viral disease caused by the variola virus, a brick-shaped, deoxyribonucleic acid-containing member of the Poxviridae family. The disease was spread by transfer of the virus in respiratory droplets during face-to-face contact. It was characterised by severe systemic involvement and a single crop of skin lesions that proceeds through macular, papular, vesicular and pustular stages. The incubation period is an average of 10-12 days, with a range of 7-17 days. In 1980, smallpox was declared eradicated by the World Health Assembly.

SMART BATTERY (INTELLIGENT BATTERY): A portable computer battery that has the ability to keep the computer up-to-date with its power status and other specific battery characteristics.

SMALL COMPUTER SYSTEM INTERFACE (SCSI): In computer science, it is one of the methods developed by the American National Standards Institute (ANSI) for connecting peripheral devices such as printers drives, scanners drives, tape drives, CD-ROM drives to a computer. A group of peripherals linked in series to a single SCSI port is called a **daisy-chain**.

SMART DRIVE (USB SMART DRIVE): A USB flash drive that has installed software to interact with a computer or other host device.

SMART DRUG (COGNITIVE ENHANCERS; NOOTROPICS): Any drug or combination of nutrients such as vitamins, amino acids, minerals and sometimes and herbs that is said to enhance the functioning of the brain such as improvement of memory, learning, attention, concentration, problem solving, reasoning, social skills, decision making and planning, etc. The drugs have been used to treat people with neurological or mental disorders, but there is no scientific evidence to suggest that these drugs have significant effect on healthy people.

SMART TERMINAL: In computing, it is a terminal (monitor and keyboard) that has some processing capabilities, but not as much as an intelligent terminal.

SMARTPHONE: An advanced mobile phone such as Apple's iPhone that includes advanced functionality beyond making phone calls and sending text messages. In addition to functioning as a mobile phone, smartphones can be used as a media player, a camera, a GPS navigation device and a Web browser - and in many other ways. Apple's iPhone uses a touchscreen for typing and to run applications.

SMELL (OLFACTION): A sense that enables an organism to perceive and distinguish the odours of various substances. It works by having receptors or olfactory receptors in the nasal cavity for particular chemical groups, into which airborne chemicals must fit to trigger a message to the brain. The sense of smell is not as strongly developed in humans as in many other vertebrates, particularly carnivores which employ olfactory organs to locate food and detect dangerous predators, in finding mating partners and in recognising other animals.

SMELT: 1. To melt ore in order to obtain metal from it or to produce metal in this way. The separation of the metal usually requires a chemical change. Smelting requires heating the ore in a furnace to a high temperature in the presence of a reducing agent such as carbon and a fluxing agent like a limestone. 2. To undergo fusing or melting in the process of smelting. 3. A small silvery northern sea or freshwater fish of Family Osmeridae.

SMETIC: A distinct mesophase of liquid crystal substances. Molecules in this phase show a degree of translational order not present in the nematic. In the smectic state, the molecules maintain the general orientational order of nematics, but also tend to align themselves in layers or planes. Motion is restricted to within these planes and separate planes are observed to flow past each other. The increased order means that the smectic state is more solid-like than the nematic.

SMOG: Fog made heavier and darker by smoke and chemical fumes. It also refers to photochemical haze caused by the action of solar ultraviolet radiation on atmosphere polluted with hydrocarbons and oxides of nitrogen, especially from automobile exhaust.

SMOG CHAMBER: A large confined volume in which sunlight or simulated sunlight is allowed to irradiate air mixtures of atmospheric trace gases (hydrocarbons, nitrogen oxides, sulfur dioxide, etc.), which undergo oxidation. In theory these chambers allow the controlled study of complex reactions which occur in the atmosphere.

SMOG INDEX: A mathematical correlation between smog and meteorological and/or pollutant concentrations associated with it. These are qualitative indices which are sometimes used in some urban communities to predict the degree of the air pollution which is expected for the coming day.

SMOKE: A colloid consisting of a collection of airborne solid, liquid particulates and gases emitted when a material undergoes combustion or pyrolysis, together with the quantity of air that is entrained or otherwise mixed into the mass.

SMOKELESS FUEL: Fuel that does not give off any visible smoke when burned, because all the carbon is fully oxidised to carbon dioxide (CO_2). Examples include natural gas, oil and solid fuels such as coke, anthracite and charcoal.

SMOKER (BLACK SMOKERS): An active hydrothermal vent, usually at a mid-ocean ridge on the sea floor, from which superheated water at about 350 °C, with very high concentrations of dissolved minerals such as copper, zinc, gold and manganese and other sulfur-bearing minerals or Sulfides erupt at high pressure. When the saturated mineral solutions exit the vent, cooling by contact with the ocean causes the metals to precipitate, mainly as sulfide or sulfate salts forming chimney-like structures.

SMOOTH ENDOPLASMIC RETICULUM (AGRANULAR ENDOPLASMIC RETICULUM): A part of endoplasmic reticulum that is tubular in form and lacks ribosomes. Its functions include lipid synthesis, carbohydrate metabolism, calcium concentration, drug detoxification and attachment of receptors on cell membrane proteins.

SMOOTH FUNCTION: A function that has continuous derivatives up to some desired order over some domain. A function can therefore be said to be smooth over a restricted interval such as (a, b) or $[a, b]$. The number of continuous derivatives necessary for a function to be considered smooth depends on the problem at hand and may vary from two to infinity.

SMOOTH MUSCLE (INVOLUNTARY MUSCLE): A muscle that acts independently of the will, especially in reflex functions or a non-striated muscle associated with internal tissues and organs in mammals, which is not normally under voluntary, conscious control. Unlike the voluntary muscle, it has no cross stripes; consists of narrow spindle-shaped cells with a single, centrally located nucleus; and contracts slowly and automatically. Its presence in the wall of the alimentary canal, for example, allows rhythmic movements known as peristalsis, which causes food to be mixed and forced along the gut.

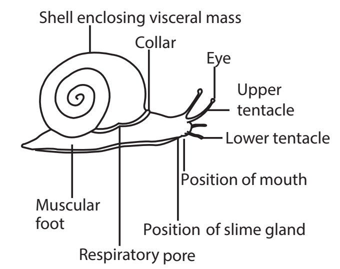
SMOOTH-STALKED MEADOW GRASS: A species of grass which can withstand quite dry conditions. It is a perennial grass with smooth greyish-green leaves and green purplish flowers.

SMS: 1. Abbreviation for **Short Message Service**, a wireless service that allows individuals to send short text messages with devices such as mobile phones and pagers. The messages can typically be up to 160 characters in length, though some services use 5-bit mode, which supports 224 characters. 2. Abbreviation for **Systems Management Server**, which is a tool developed by Microsoft that helps administrators perform diagnostics, maintain and create an inventory of computers that are connected to the Network and are running SMS. 3. Abbreviation for **Storage Management Subsystem**, which was developed by IBM to help manage storage resources. With SMS a user is capable of managing backup, data security, data migration, data recovery and data deletion.

SMUDGING: The process of burning oil to produce smoke to prevent loss of heat from the ground and so minimise or prevent frost damage to crops and orchards.

SMUT: A group of fungal disease of cereals and other grasses belonging to the phylum Basidiomycota that attack the ears of cereal crops. It is characterised by sooty black masses of spores forming on the grains, leaves and other parts.

SNAIL: Small soft-bodied invertebrate belonging to the molluscs, mostly with a spiral shell and a retractable muscular foot on which it crawls. Snails belong to the class Gastropoda. The African land snail is classified as *Achatina fulica* and the cultivated land snail as *Helix pomatia*. Abalones are classified in the genus *Haliotis* and periwinkles in the genus *Littorina*. The queen conch is classified as *Strombus gigas*. Snails are found in the ocean, in fresh waters and on land. Some species are used as food and the shells of some are used as ornaments. Snails feed mainly on algae and decaying matter and are important members of the food web, being a source of food to fish and waterfowl. Many species of snails are hermaphroditic and capable of self-fertilisation.



Lateral view of a Garden Snail

SNAIL MAIL: A term used to refer to mail sent via the conventional postal service, in contrast to electronic-mail (e-mail). E-mail can deliver messages within minutes while a mail sent through conventional postal services take several hours, days or weeks, depending on its destination.

SNAKE: Any of numerous limbless, crawling reptiles with a scaly elongated body tapering toward the tail, with lidless eyes and often venomous fangs of the suborder Serpentes or Ophidia (order Squamata). Most snakes live on the ground, but some are burrowers, arboreal or aquatic. One group is exclusively marine and ranges in length from about 10 cm to 9 m. Snakes are cold-blooded or cannot produce their own body heat. They rely on the Sun to heat their bodies and then regulate their temperature with behaviour. Because they do not rely on energy from food to generate body heat, snakes can survive on an extremely meagre diet. Snake bodies are covered in overlapping scales composed of a horny material called **keratin**. They regularly shed the outer layer of their skin as they grow or the scales become drab and worn over time and must be periodically replaced. Snakes have three-chambered hearts that can move sideways to accommodate large prey animals travelling from the mouth to the stomach. A snake obtains information about its environment by continuously flicking out its forked tongue to collect scent particles from the air, which an organ called Jacobson's organ located in the roof of its mouth detects. Snakes have small bones in their heads that conduct sound. They are able to hear low-frequency sounds and to sense vibrations that travel through the ground or water. The majority of snakes have good eyesight, especially for detecting moving objects, although most burrowing snakes can only distinguish between light and dark. Several groups of snakes inject their prey with poisonous secretions called venom.

SNAKE-LIZARD (SKINK): Terrestrial and burrowing lizard belonging to the family Scincidae, with small or absent limbs and covered with shiny scales.

SNAPCHAT: A mobile social media application and service for sharing photos, videos, and messages. The application can be downloaded and installed to create an account and add friends in order to communicate with them. Once a message is received, it lasts for only set a time limit and is automatically deleted from.

SNAPSHOT (SCREENSHOT, PRINT SCREEN; SCREEN CAPTURE; SCREEN DUMP): A term used to describe an image that is a copy of what is currently being displayed on a screen. This image is created by using an operating system or a software program and saved as a GIF or JPEG file.

SNEEZE: Sudden, forceful and involuntary expulsion of air through the nose and mouth because of irritation of the nasal passages.

SNELL'S LAW (DESCARTES' LAW; THE SNELL–DESCARTES LAW; LAW OF REFRACTION): A law of refraction discovered by Dutch astronomer and mathematician Willebrord Snellius in 1620. It states that for light rays passing from one transparent medium to

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another, the ratio of the sine of the angles of incidence to the sine of the angle of refraction is constant. In other words, for a given pair of media, the ratio of the sine of the angle of incidence to the sine of angle of refraction is a constant. It is defined mathematically as $\mu = \frac{\sin i}{\sin r}$, where i is the angle of incidence and r is the angle of refraction and μ is the refractive index. In optics, the law is used in ray tracing to compute the angles of incidence or refraction and in experimental optics and gemology (the science dealing with natural and artificial gems and gemstones) to find the refractive index of a material.

SNORKEL: A breathing tube used by skin divers swimming near the surface, connecting the diver's mouth with the atmosphere. A similar tube was used by submarines in World War II to supply air to their engines without completely surfacing.

SNOUT: 1. The nose and mouth of some animals, including the pig. 2. The projecting part of the head of an insect or other invertebrate such as weevil. 3. Something that projects or sticks out such as the nozzle of a gun, for example.

SNOW: 1. A precipitation in the form of flakes of crystalline water ice that fall from the clouds. It is water vapour in the atmosphere that has frozen into ice crystals and falls to the ground in the form of flakes. Snowfall tends to form within regions of upward movement of air around a type of low-pressure system known as an **extratropical cyclone**. 2. Random patterns of small white specks on a television or radar screen caused by electrical interference.

SNOW BLINDNESS (PHOTOKERATITIS): A type of keratitis that occurs in the cornea of the eye due to exposure to ultraviolet light (UVL) reflected off snow or ice. UVL can also be reflected from sources such as welder's arc, photographic flood lamps, lightning, carbon arcs, photographic flood lamps, lightning, electric sparks and halogen desk lamps. Its signs and symptoms include red and painful eye, excess tears or other discharge from the eye, blurred and decreased vision, sensitivity to light and difficulty in opening the eye. It is prevented by sun-glasses or protective goggles.

SNOW MOULD: A fungal pre-emergent blight and root rot of cereals, especially winter barley, although other winter cereals are also occasionally affected. It is caused by the pathogen *Monographella nivalis (Microdochium nivale)*. Infected plants often have an extensive covering of white mycelium which spreads on overlapping leaves, causing a matting of leaf tissue. The fungus often infects the oldest leaves directly from the soil but eventually the whole plant can be affected.

SNOW ROT: A fungus disease caused by *Typhula incarnata* affecting only winter cereals, especially barley and wheat, causing leaves to shrivel or brown. Plants can be killed, but often good growing conditions in the spring allow crop recovery.

SOAKAWAY: A channel in the ground filled with gravel, which takes rainwater from a down pipe or liquid sewage from a septic tank and allows it to be absorbed into the surrounding soil.

SOAP: 1. A cleaning or washing agent which is made by the action of an alkali of sodium hydroxide solution (caustic soda) or potassium hydroxide (caustic potash) on fats of animals or fats of vegetable origin. Soaps of metals heavier than sodium are not very soluble. The curdy precipitate made by soap in hard water is the calcium or magnesium salt of the fatty acid in the soap. **Metallic soap** is salt of stearic, oleic, palmitic, lauric or erucic acid with a heavy metal such as cobalt or copper. Heavy-metal soaps are used in lubricating greases, as gel thickeners, in paints and inks, in fungicides, decolourising varnish and waterproofing. **Napalm** is an aluminium soap of various fatty acids. When mixed with gasoline it makes a firm jelly, which is used in some bombs and in flamethrowers. 2. An abbreviation for **Simple Object Access Protocol**, which is a method of transferring messages or small amounts of information, over the Internet. SOAP messages are formatted in XML and are typically sent using HTTP (hypertext transfer protocol). Both are widely supported data transmission standards.

SOAY: A rare horned breed of sheep which sheds its fleece naturally. It is thought to be the link between wild and domesticated breeds. The short hairy fleece is tan and black in colour.

SOBOLE: A creeping underground stem that produces roots and buds; a shoot arising from the base of a stem or from the rhizome. **Soboliferous** means of or pertaining to sobols.

SOCIAL BEHAVIOUR: Any behaviour exhibited by interactions among individuals in a group of animals, normally of the same species that is usually beneficial to one or more of the individuals. It ranges from moving as a herd in order to minimise the effects of predators to performing designated roles in highly organised societies. Social behaviour serves many purposes and is exhibited by an extraordinary wide variety of animals, including invertebrates, fish, birds and mammals.

SOCIAL DOMINANCE: A hierarchical pattern of social organisation involving domination of some members of a group by other members in a relatively orderly and long-lasting pattern.

SOCIAL INSECTS: Insects that form permanent family groups called colonies and share the work and other resources involved, for survival. Examples are termites, bees, wasps and ants.

SOCIAL MEDIA: A collection of online communication channels that allows individuals to interact through sharing or exchanging information, career interests, ideas and pictures/videos in virtual communities and networks. Some common examples are Facebook, Twitter, Google+ and LinkedIn.

SOCIAL NETWORKING SITE (SNS; SOCIAL NETWORKING WEBSITE): An online platform that allows users to create a public profile with personal information, photos, etc., and interact with other users who share similar personal or career interests, activities, backgrounds or real-life connections. Examples of SNS are LinkedIn and Facebook.

SOCIETY: A colony or community of organisms, usually of the same species in which there are divisions of resources, divisions of labour and mutual dependence, often held together by stimuli exchanged among members of the group.

SOCIOBIOLOGY: The study of social behaviour based on evolutionary theory that such behaviour is often genetically transmitted and subject to evolutionary processes.

SODA: An acid salt, formed by combining an acid (carbonic) and a base (sodium hydroxide), which reacts with other chemicals as a mild alkali. It is a weak base and at temperatures above 149 °C (422 K), dissociates into sodium carbonate (a more stable substance), water and carbon dioxide. Examples of soda include: (i) sodium hydroxide or caustic soda (NaOH); (ii) baking soda or sodium bicarbonate ($NaHCO_3$); (iii) sodium carbonate decahydrate or washing soda ($Na_2CO_3 \cdot 10H_2O$); (iv) sodium monoxide (Na_2O); and (v) caustic soda or sodium hydroxide (NaOH). It is used as an antacid, cleaner and deodoriser. It is also used in manufacturing effervescent salts and beverages and baking powder, as well as in the production of other sodium salts, treatment of wool and silk, used in pharmaceuticals, sponge rubber, fire extinguishers, cleaners, laboratory reagents, mouthwash, gold and platinum.

SODA ASH (SODIUM CARBONATE; WASHING SODA; SODA CRYSTALS): A white, crystalline, slightly alkaline salt, sodium salt of carbonic acid with the chemical formula Na_2CO_3, density 2.20 g cm^{-3}, melting point 50 °C (323 K) with a boiling point of 851 °C (1124 K) and decomposes. It most commonly occurs as a crystalline heptahydrate, which readily effloresces to form a white powder, the monohydrate. It is domestically well known for its everyday use as a water softener. It has a cooling alkaline taste and can be extracted from the ashes of many plants. It is synthetically produced in large quantities from table salt in a process known as the Solvay process. It is used in photography, in cleaning, in pH control of water, in textile treatment, glasses and glazes, as a food additive and a volumetric reagent. For aquatic organisms it is a very important or only source of carbon.

SODA LIME: A mixed hydroxide of sodium and calcium made by slaking lime with caustic soda solution and recovering greyish white granules by evaporation. It is used in granular form in closed breathing environments, such as general anaesthesia, submarines, rebreathers and recompression chambers to remove carbon dioxide from breathing gases to prevent CO_2 retention and carbon dioxide poisoning.

SODA WATER (CLUB SODA; SELTZER WATER; CARBONATED WATER): Carbon dioxide dissolved in water under pressure and used as a fizzy drink. Soda water is combined with sweeteners and flavourings to produce a variety of soft drinks. Many cocktails also use soda water as an ingredient.

SODDING: Removal of a strip of soil with grass and roots for vegetative establishment purpose.

SODDY, FREDERICK (1877-1956): British chemist, who worked with Ernest Rutherford in Canada and William Ramsay in London before finally settling in Oxford in 1919. With Ernest Rutherford, they explained that radioactivity is due to the transmutation (nuclear reactions) of elements. In 1913, he proved the existence of isotopes, which won him the 1921 Nobel Prize for Physics. Soddyite, a radioactive uranium mineral and a small crater on the far side of the Moon were named after him.

SODIC SOIL: A soil in which much of the chlorine has been washed away, leaving sodium ions attached to clay particles in the soil. Such a soil contains sufficient exchangeable sodium (Na) to adversely affect crop production and soil structure under most conditions of soil and plant type. In sodic soils the exchangeable sodium percentage (ESP) is more than 6%.

SODIUM: A soft, light, extremely malleable, wax-like, silver-white or silvery or grey metallic element, which is a member of Group 1 (IA) of the Periodic Table. It is obtained from common salt, sodium chloride. Its chemical symbol is Na, atomic number 11, atomic weight 22.99, melting point 97.8 °C (370.8 K), boiling point 882.940 °C (1156.090 K), relative density 0.971 and a valence of 1. It reacts readily with other substances, especially water and non-metals and is essential to the body's fluid balance. Sodium is an essential element for all animal life including human and for some plant species. In animals, sodium ions are used in opposition to potassium ions, to allow the organism to build up an electrostatic charge on cell membranes and thus allow transmission of nerve impulses when the charge is allowed to dissipate by a moving wave of voltage change. Sodium's relative rarity on land is due to its solubility in water, thus causing it to be leached into bodies of long-standing water by rainfall. Such is its relatively large requirement in animals, in contrast to its relative scarcity in many inland soils that herbivorous land animals have developed a special taste receptor for the sodium ion.

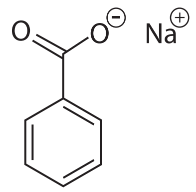
SODIUM ACETATE (SODIUM ETHANOATE): A sodium salt of acetic acid, a colourless crystalline compound with the chemical formula CH_3COONa. As an anhydrous salt, it has a relative density of 1.52 and melting point of 324 °C (594 K) and as a trihydrate, it has a relative density of 1.45 and loses water at 58 °C (328 K). The compound may be prepared by the reaction of ethanoic acid with sodium carbonate or with sodium hydroxide. Due to its property of being a strong base and a weak acid, it is used in buffers for pH control, in numerous laboratory applications, in electroplating in pharmaceuticals, dyeing, soaps and in photography. It is added to food as seasoning and to alcoholic beverages to reduce risk of hang over etc.

SODIUM ALUMINATE: A white solid with the chemical formula $NaAlO_2$, density 1.5 g/cm^3 and melting point 1650 °C. It is insoluble in ethanol but soluble in water, giving a strongly alkaline solution. It is made by heating bauxite with sodium carbonate and extracting the residue with water. It may be prepared in the laboratory adding excess aluminium to hot concentrated sodium hydroxide. It is used as a mordant, in glass manufacturing, in cleansing compounds and in effluent treatment, as a coagulant in water purification, etc.

SODIUM AMIDE (SODAMIDE): A white crystalline powder with the chemical formula $NaNH_2$, melting point 210 °C (480 K) and boiling point of 400 °C (670 K). It has an ammonia-like odour. It is produced by passing dry ammonia over metallic sodium at 350 °C (620 K). It decomposes in water and in warm ethanol. Sodium amide can be prepared by the reaction of sodium with ammonia gas, but it is usually prepared by the reaction in liquid ammonia using iron(III) nitrate as a catalyst. It reacts with red-hot carbon to give sodium cyanide and with nitrogen(I) oxide to give sodium azide. Sodium amide is used in the industrial production of indigo, hydrazine and sodium cyanide

SODIUM AZIDE: A white poisonous crystalline solid with the chemical formula NaN_3, density 1.846 g/cm^3 (20 °C), melting point 275 °C (548 K) and decomposes. It is made by the action of nitrogen(I) oxide on hot sodium amide. It is soluble in water, slightly soluble in alcohol. It is used in the manufacture of detonators.

SODIUM BENZENECARBOXYLATE (SODIUM BENZOATE; SODIUM PHENYL METHANOATE): A colourless crystalline or white amorphous powder with the chemical formula C_6H_5COONa, density 1.497 g/cm^3 and melting point 300 °C (573 K). It is soluble in water and slightly soluble in ethanol. It is made by the reaction of sodium hydroxide with benzoic acid. It is used in the dyestuff industry and as a food preservative. It was formerly used as an antiseptic.



Chemical Structure of Sodium Benzenecarboxylate

SODIUM BICARBONATE INDICATOR: A mixture of sodium bicarbonate solution and the dyes cresol red and thymol blue. It is used as an indicator to detect small changes in pH. It changes from red to orange and yellow with a slight increase in acidity.

SODIUM BROMIDE: A white, crystalline solid with the chemical formula NaBr. It occurs in two states: the anhydrous with a density of 3.21 g/cm^3 and melting point of 747 °C (1020 K); and the dihydrate with a density of 2.18 g/cm^3 and melting point of 36 °C (309 K). Its boiling point is 1396 °C (1669 K). It is prepared by the action of bromine on hot sodium hydroxide solution or of hydrogen bromide on sodium carbonate solution. It is used in photographic processing and in analytical chemistry.

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SODIUM CARBONATE [SODA ASH; WASHING SODA; SODA CRYSTALS; SODIUM TRIOXOCARBONATE(IV)]: A white, crystalline, slightly alkaline salt or sodium salt of carbonic acid with the chemical formula Na_2CO_3, density 2.20 g cm^{-3}, melting point 50°C (323 K) and decomposes at a boiling point of 851 °C (1124 K). It most commonly occurs as a crystalline heptahydrate, which readily effloresces to form a white powder, the monohydrate. It has a cooling alkaline taste and can be extracted from the ashes of many plants. It is synthetically produced in large quantities from table salt in a process known as the Solvay process. It is used in photography, in cleaning, in pH control of water or used domestically as a water softener, in textile treatment, glasses and glazes, as a food additive and a volumetric reagent. For aquatic organisms it is a very important or only source of carbon.

SODIUM CHLORATE [SODIUM CHLORATE(V); SODIUM TRIOXOCHLORATE(V)]: A white soluble crystalline solid with the chemical formula $NaClO_3$, density 2.5 g/cm^3, melting point 248 °C (521 K), boiling point 300 °C (573 K) and decomposes. It is a powerful oxidising agent, soluble in water and ethanol. It is prepared by the reaction of chlorine on hot concentrated sodium hydroxide. It is used as a weed killer and in the textile industry and also used in the manufacture of matches and soft explosives.

SODIUM CHLORIDE (COMMON SALT; TABLE SALT): A colourless crystalline compound with the chemical formula NaCl, density 2.165 g cm^{-3}, melting point 801 °C (1074 K) and boiling point 1413 °C (1686 K). It is widely found in nature, mainly from sea water, halide deposits, etc. It is used as a preservative, for food seasoning, etc. Sodium chloride has a key role in biological systems in maintaining electrolyte balance. It is used industrially as the starting point for a range of sodium based products; in Solvay process for Na_2CO_3 and Custer-Keller process for NaOH.

SODIUM CYANIDE: A white or colourless crystalline solid which is deliquescent, soluble in water and in liquid ammonia and slightly soluble in ethanol with the chemical formula NaCN, density 1.595 g/cm^3, melting point 563.7 °C (837 K) and boiling point 1496 °C (1769 K). It is made by absorbing hydrogen cyanide in sodium hydroxide or sodium carbonate solution. Its compound is very poisonous because it reacts with the iron in haemoglobin and so prevents oxygen from reaching the tissues of the body. It is used in electroplating and precious metals extraction.

SODIUM DICHROMATE: A red crystalline hygroscopic solid which is insoluble in ethanol but soluble in water with the chemical formula $Na_2Cr_3O_7.2H_2O$. It is made by melting chrome iron ore with lime and soda ash and acidification of the chromate thus formed. It is used as an oxidising agent in organic chemistry and in analytical chemistry. It is also useful in dyeing.

SODIUM DIHYDROGEN PHOSPHATE(V) (SODIUM DIHYDROGEN ORTHOPHOSPHATE): A white crystalline solid with the chemical formula NaH_2PO_4, density 2.52 g/cm^3, melting point 356.7 °C (630 K), dehydrates at 100 °C (373 K), boiling point 400 °C (673 K) and decomposes. It is insoluble in alcohol but soluble in water. It is prepared by treating sodium carbonate with an equimolar quantity of phosphoric acid with sodium hydroxide. It is used in detergents but this use is being discouraged because it causes eutrophication. It is also used in the preparation of sodium phosphate, in baking powders, as food additives and to study phosphate participation in the metabolic process.

SODIUM FLUORIDE: A crystalline compound, with the chemical formula NaF, density 2.558 g/cm^3, melting point 993 °C (1266 K) and boiling point 1704 °C (1977 K). It is soluble in water and slightly soluble in ethanol. It occurs naturally as villiaumite and may be prepared by the reaction of sodium hydroxide or of sodium carbonate with hydrogen fluoride. It is very toxic but in dilute solution it is used in fluoridation of water for the prevention of tooth decay because it is able to replace OH group with F group in the material for dental enamel.

SODIUM HEXAFLUORALUMINATE (SODIUM FLUOROALUMINATE; SODIUM ALUMINOFLUOROALUMINATE; CRYOLITE; KRYOLITE): A colourless monoclinic solid, with the chemical formula Na_3AlF_6, relative density 2.9, melting point 1000 °C (1273 K). It changes to a cubic form at 580 °C (853 K). It is very slightly soluble in water and occurs naturally as the mineral cryolite but also manufactured by the reaction of aluminium fluoride with alumina and sodium hydroxide or directly with sodium aluminate. Its most important use is in the manufacture of aluminium in the Hall-Heroult cell. It is also used in the manufacture of enamels, opaque glasses and ceramic glazes. Other uses include the manufacture of enamels, ceramic glazes and opaque glasses.

SODIUM HYDRIDE: A white crystalline solid, with the chemical formula NaH and density 1.396 g/cm^3 which decomposes slowly above 300 °C (573 K). It reacts violently with water to give sodium hydroxide and hydrogen and it ignites spontaneously with oxygen at 230 °C (503 K). It is prepared by the reaction of pure dry hydrogen with sodium at 350 °C (623 K). It is a powerful reducing agent with several laboratory applications such as organic synthesis, reducing agent, hydrogen storage and drying agent.

SODIUM HYDROGEN CARBONATE (BAKING SODA; SODIUM BICARBONATE): A white, crystalline, slightly alkaline salt or sodium salt of carbonic acid with the chemical formula $NaHCO_3$, density 2.20 g cm^{-3}, melting point 50°C (323 K) and decomposes with a boiling point of 851 °C (1124 K). It most commonly occurs as a crystalline heptahydrate, which readily effloresces to form a white powder, the monohydrate. It is domestically well known for its everyday use as a water softener. It has a cooling alkaline taste and can be extracted from the ashes of many plants. It is synthetically produced in large quantities from table salt in a process known as the Solvay process. It is used in photography, in cleaning, in pH control of water, in textile treatment, glasses and glazes, as a food additive and a volumetric reagent. For aquatic organisms it is a very important or only source of carbon.

SODIUM HYDROGEN SULFATE (SODIUM ACID SULFATE BISULFATE OF SODA): A white solid with the chemical formula $NaHSO_4.H_2O$. It is soluble in water and slightly soluble in ethanol. It occurs in two states: the anhydrous with a density of 2.742 g/cm^3 and melting point of 315°C (588 K); and the monohydrate with a density of 1.8 g/cm^3 and melting point of 58.5°C (331.6 K). Its boiling point is 315°C (588 K) and decomposes to $Na_2S_2O_7$ (+ H_2O). It may be manufactured by the reaction of sodium hydroxide with sulfuric acid or by heating equimolar proportions of sodium chloride and sulfuric acid. It is used in making paper, glass and for textile finishing.

SODIUM HYDROGEN SULFITE [SODIUM BISULFITE; SODIUM HYDROGEN TRIOXOSULFATE(IV)]: A white solid with the chemical formula $NaHSO_3$, which is very soluble in water (yellow in solution) and slightly soluble in ethanol. It decomposes on heating to give sodium sulfate, sulfur dioxide and sulfur. It is formed by saturating a solution of sodium carbonate with sulfur dioxide. The compound is used in the brewing industry and in the sterilisation of wine casks. It is a general antiseptic and bleaching agent.

SODIUM HYDROXIDE (CAUSTIC SODA): A brittle white or translucent deliquescent solid, which is the commonest alkali. Its chemical formula is NaOH, density 2.13 g/cm^3, melting point 318 °C (591 K) and boiling point 1388 °C (1661 K). It is made by the electrolysis of brine using mercury cells on any of a variety of diaphragm cells. In this process, the principal product is chlorine and sodium hydroxide is almost reduced to by-products. The solution is extremely corrosive to body tissue and very dangerous to the eyes. It is used particularly in soap and paper production. It is also used for the treatment of effluent for the removal of heavy metals and of acidity and used to absorb acidic gases such as carbon dioxide.

SODIUM IODIDE: A white crystalline solid with the chemical formula NaI, density 3.67 g/cm^3, melting point 661 °C (934 K) and boiling point 1304 °C (1577 K). It is soluble in water, ethanol and ethanoic acid and prepared by the reaction of hydrogen iodide with sodium carbonate or sodium hydroxide in solution. It is used in photography, in medicine as an expectorant and in the administration of radioactive iodine for studies of thyroid function and treatment of thyroid diseases.

SODIUM METHANOATE (SODIUM FORMATE): A colourless deliquescent solid with the chemical formula HCOONa, density 1.92 g/cm^3 at 20 °C (293 K), melting point 253 °C (526 K) and decomposes. It is soluble in water and slightly soluble in ethanol. It may be produced by the reaction of carbon monoxide with solid sodium hydroxide at 200 °C (473 K) and 10 atmospheres of pressure. It can be prepared in the laboratory by the reaction of methanoic acid and sodium hydroxide. It is used in the production of ethanedioic acid and methanoic acid. In the laboratory it is a convenient source of carbon monoxide.

SODIUM MONOXIDE (SODIUM OXIDE): A white solid that reacts violently with water to give sodium hydroxide, a strong base. Its chemical formula is Na_2O, density 2.27 g/cm^3, melting point 1132 °C (1405 K), boiling point 1950 °C (2223 K) and decomposes. It sublimes at 1275 °C (1548 K). It is manufactured by the oxidation of the metal in a limited supply of oxygen and purified by sublimation. It reacts with water to produce sodium hydroxide. Its commercial applications are in the chemical industry. Sodium oxide is produced by the reaction of sodium with sodium hydroxide as $2NaOH + 2Na \rightarrow 2Na_2O + H_2$, with sodium peroxide as $Na_2O_2 + 2Na \rightarrow 2Na_2O$ or sodium nitrite as $2NaNO_2 + 6Na \rightarrow 4Na_2O + N_2$. Most of these reactions rely on the reduction of hydroxide, peroxide or nitrite by sodium.

SODIUM NITRATE [CHILE SALTPETRE; CALICHE; CHILE SALTPETER; NITRATE OF SODA; SODIUM HYDROGEN TRIOXONITRATE(V)]: A white crystalline salt with the chemical formula $NaNO_3$, density 2.257 g/cm^3 (solid), melting point 308 °C (581 K) and a boiling point of 380 °C (653 K). It is soluble in water and ethanol. It is obtained from deposits of caliche. It is prepared by the reaction of nitric acid with sodium hydroxide or sodium carbonate. It is used mainly in the production of nitrate fertilisers, as a food preservative and in the manufacture of explosives and fireworks.

SODIUM NITRITE [NITROUS ACID SODIUM SALT; DIAZOTIZING SALT; SODIUM DIOXONITRATE(V)]: A hygroscopic, colourless, odourless crystalline compound powder, which is very soluble in water, with the chemical formula $NaNO_2$, melting point 281 °C (551 K) and a boiling point of 320 °C (591 K). It decomposes above 320 °C. It is a strong oxidising agent, which is slowly oxidised by oxygen in the air to sodium nitrate ($NaNO_3$). Sodium nitrite prevents cured meat from becoming rancid and stabilises taste and prevents the formation of the botulism toxin. It is used for pickling and colouring meat and fish products, as an intermediate for dyestuffs, in dyeing and printing textile fabrics and bleaching fibres, rust-proofing and in medicine as a vasodilator, a bronchodilator and an antidote for cyanide poisoning. It is also used in photography, as a laboratory reagent, etc. Sodium nitrate reacts with the amine groups in protein to form nitrosoamines. Nitrosoamines in food cause cancer and sodium nitrite is implicated in the formation of suspected carcinogens in meat.

SODIUM PEROXIDE (DISODIUM PEROXIDE; SODIUM DIOXIDE; FLOCOOL): A whitish or pale yellow powdery solid with the chemical formula Na_2O_2, density 2.805 g/cm^3, melting point 460 °C (733 K), boiling point 657 °C (930 K) and density 2.8, which decomposes above its boiling point. It is a strong oxidising agent and reacts violently with water. It is soluble in ice-water and decomposes in warm water or alcohol. The compound is formed by the combustion of sodium metal in excess oxygen. Contact with combustible materials may cause fire or explosion. It is used as a bleach for cloth and wood pulp; for the extraction of minerals from various ores; as an oxidising agent; and medically as a germicide, antiseptic and disinfectant. It is also used as an oxygen source by reacting it with carbon dioxide to produce oxygen and sodium carbonate.

SODIUM PUMP: A mechanism by which sodium ions are transported out of eukaryotic cells across the plasma membrane against an opposing concentration gradient.

SODIUM SESQUICARBONATE (TRISODIUM HYDROGENDICARBONATE): A white crystalline hydrated double salt of sodium bicarbonate and sodium carbonate, which has a needle-like crystal structure with the chemical formula $Na_3H(CO_3)_2$, molecular formula, $C_2HNa_3O_6$, molar mass 190.00 g mol^{-1}, density 2.112 g/cm^3 (dihydrate). The dihydrate is soluble in water as follows: 13 g/100 mL at 0 °C and 42 g/100 mL at 100 °C. The dihydrate, $Na_3H(CO_3)_2 \cdot 2H_2O$, occurs in nature as the evaporite mineral trona. It is soluble in water but less alkaline than sodium carbonate and it decomposes on heating. It may be prepared by crystallising equimolar quantities of the constituent materials. Sodium sesquicarbonate is used in bath salts, swimming pools and as an alkalinity source for water treatment. It is also used in the archaeological conservation of objects made of copper and copper alloys that have been corroded by salt water.

SODIUM SILICATE SOLUTION: Concentrated solution of sodium silicate in water, used to prepare silica gel and precipitated silica.

SODIUM SILICATES: Any of several compounds containing sodium oxide (Na_2O) and silica (Si_2O) or a mixture of sodium silicates. Sodium *ortho*silicate is Na_4SiO_4 or $2Na_2O \cdot SiO_2$, sodium *meta*silicate is Na_2SiO_3 or $Na_2O \cdot SiO_2$, sodium *di*silicate is $Na_2Si_2O_5$ or $Na_2O \cdot 2SiO_2$ and sodium *tetra*silicate is $Na_2Si_4O_9$ or $Na_2O \cdot 4SiO_2$. All these compounds are transparent, glassy or crystalline solids that have high melting points above 800°C (1073 K) and are water soluble. They are produced chiefly by fusing sand and sodium carbonate in various proportions. The product is commonly known as water glass. The greatest single use of sodium silicates is as a raw material for making silica gel.

SODIUM SULFATE [SODIUM TETRAOXOSULFATE(VI)]: A white orthorhombic crystalline compound at ordinary temperatures with the chemical formula Na_2SO_4. It forms two hydrates: the decahydrate is Glauber's salt; and the anhydrous, which is found in nature as the mineral thenardite. The density for the anhydrous is 2.664 g/cm^3 and 1.464 g/cm^3 for the decahydrate. The melting point for the anhydrous is 884 °C (1157 K) and that for the decahydrate is 32.4 °C (305.4 K). The boiling point for the anhydrous is 1429 °C (1702 K). Sodium sulfate is soluble in cold water and very soluble in hot water. The major commercial source of sodium sulfate is salt cake, a by-product of the production of hydrochloric acid from sodium chloride (common salt) by treatment with sulfuric acid. It is obtained with other chemicals by evaporation of natural brines. It is also obtained as a by-product of viscose rayon manufacture and in several other ways. The principal use of sodium sulfate is in processing wood pulp for making kraft paper. It is also used in glass manufacture, textile dyeing and synthetic detergents.

SODIUM SULFATE DECAHYDRATE: A colourless crystalline solid, which is a decahydrate of sodium sulfate with the chemical formula $NaSO_4.10H_2O$. Its density is 1.46 g/cm^3 and melting point is 32.4 °C (305.4 K). It is used in the manufacture of paper, glass, dyes and detergents.

SODIUM SULFIDE (DISODIUM SULFIDE): A yellow-red solid with the chemical formula Na_2S. It is formed by the reduction of sodium sulfate with carbon (coke) at high temperatures. It is known in an anhydrous form with relative density 1.85, melting point 1180 °C (1453 K); and as a nonahydrate, a chemical compound with nine molecules of water ($Na_2S.9H_2O$), with relative density of 1.43 and decomposes at 920 °C (1193 K). When exposed to moist air, Na_2S and its hydrates emit hydrogen sulfide, which smells much like rotten eggs. It is a corrosive and readily oxidised. It usually contains polysulfides of the type Na_2S_2, Na_2S_3 and Na_2S_4, which cause the variety of colours. The compound is deliquescent, soluble in water with extensive hydrolysis and slightly soluble in alcohol. It is used in wood pulping, dyestuffs manufacture and metallurgy on account of its reducing properties. It has also been used for the production of sodium thiosulfate (for the photographic industry) and as a depilatory agent in leather preparation. It is a strong skin irritant.

SODIUM SULFITE: A white soluble crystalline solid with the chemical formula Na_2SO_3 which reacts with an acid to produce sulfur dioxide (SO_2). It exists in an anhydrous form with a relative density of 2.63 and as a heptahydrate with a relative density of 1.59. It is prepared by reacting sulfur dioxide with either sodium carbonate or sodium hydroxide. Dilute mineral acids reverse this process. Sodium sulfite is soluble in water and because it is readily oxidised it is widely used as a convenient reducing agent. Sodium sulfite is used as a bleaching agent in textiles and in paper manufacture. Its use as an antioxidant in some canned foodstuffs gives rise to a slightly sulfurous smell immediately on opening, but its use is prohibited in meats or foods that contain vitamin B_1. Sodium sulfite solutions are occasionally used as biological preservatives.

SODIUM SUPEROXIDE (SODIUM DIOXIDE): A whitish yellow solid with the chemical formula NaO_2, density 2.805 g/cm^3, melting point 675 °C (948 K) and decomposes violently above 250 °C (523 K). It reacts with carbon dioxide to give sodium carbonate and oxygen. It reacts with water to form hydrogen peroxide and oxygen. Mixtures with combustible materials and most organic compounds are readily ignited by friction, heat or contact with moisture. It is formed by the reaction of sodium peroxide with excess oxygen at high temperature and pressures.

SODIUM THIOSULFATE: A white soluble crystalline solid with the chemical formula $Na_2S_2O_3$. It is commonly found as the pentahydrate, a monoclinic with relative density of 1.73 and melting point of 42 °C (315 K). This loses water at 100 °C (273 K) to give the anhydrous form with relative density of 1.66. It is soluble in water but insoluble in ethanol. It is prepared by the reaction of sulfur dioxide with a suspension of sulfur in boiling sodium hydroxide solution. It is produced chiefly from liquid waste products of sodium sulfide or sulfur dye manufacture. It is also produced from sodium carbonate, sulfur dioxide and sulfur by a process that involves several steps. It is a reducing agent, used as photographic fixing agent (when it reacts with unexposed silver halides) and in dyeing and volumetric analysis. It is also used in chrome-tanning leather and in chemical manufacture.

SODIUM-SULFUR BATTERY: A type of secondary battery that has molten electrodes of sulfur and sodium separated by a solid electrolyte consisting of beta alumina (about 11 parts aluminium oxide and 1 part sodium oxide). When the battery is producing current, sodium ions flow through the alumina to the sulfur where they form sodium polysulfide. The electrolyte does not operate effectively until the temperature reaches around 350°C (623 K), which is the operating temperature of the battery. The cells have high peak power levels and relatively low weight so they are considered for use in electric vehicles.

SODIUM-VAPOUR LAMP: A type of gas discharge lamp that uses sodium in an excited state to produce light. There are two types: low pressure sodium-vapour lamp and high pressure sodium-vapour lamp. The low pressure sodium-vapour lamp consists of an inner discharge tube made of borosilicate glass that is fitted with metal electrodes and filled with neon and argon gas and a little metallic sodium. The glass tube has a protective layer of special borate glass on the inner surface of the glass tube to reduce the rate of attack by the chemically corrosive sodium vapour. When current passes between the electrodes, it ionises the neon and argon, giving a red glow until the hot gas vaporises the sodium, which ionises, producing a monochromatic yellow light. Low pressure sodium vapour lamps are widely used as street lights because of their high luminous efficiency and and the ability of the yellow light to penetrate fog. High-pressure sodium-vapour lamps have an inner discharge tube made of translucent alumina that can withstand the corrosive effects of a mixture of mercury and sodium under greater pressure and higher temperature. The tube contains a mixture of xenon, sodium and mercury. The lamp requires a ballast and ignitor to start a small arc between the starting electrode and the main electrode, which are made of tungsten and carry a high-voltage, high-frequency pulse to strike the arc. The xenon gas is easily ionised and facilitates striking the arc when voltage is applied across the electrodes. The heat generated by the arc then vaporises the mercury and sodium. The mercury vapour raises the gas pressure and operating voltage and the sodium vapour produces light when the pressure within the arc tube is sufficient enough. High pressure sodium-vapour lamps produce a whiter light and are used for illuminating public thoroughfares, storage yards, tunnels, sports stadiums, process plants, interiors with high ceiling heights, etc.

SOFT ACIDS: Metal ions with the highest affinity for soft bases. They tend to be cations of less electropositive metals and have lower charge-to-radius ratios and are more polarisable. Examples are M0 (metal atoms), Cu^+, Ag^+, Hg^+, Pd_2^+, Pt_2^+, $Co(CN)_5^{2-}$, $InCl_3$, BH_3, RS^+, Br_2. Metal ions with the highest affinities for hard bases are called hard acids. They are usually cations of electropositive metals and are relatively nonpolarisable and have higher charge-to-radius ratios. Examples are H^+, Na^+, Ca_2^+, Mn_2^+, Al_3^+, N_3^+, Cl_3^+, Gd_3^+, Cr_3^+, Co_3^+, Fe_3^+, Be_2^+ and Mg_2^+.

SOFT BASES: Lewis bases that contain larger, relatively polarisable donor atoms. Examples are R_2S, RSH, I^-, SCN^-, $S_2O_3^{2-}$, R_3P, $(RO)_3P$, CN^-, RNC, CO, C_2H_4, C_6H_6, H^- and R^-. Hard bases are Lewis bases that contain small, relatively nonpolarisable donor atoms. Examples are NH_3, RNH_2, N_2H_4, H_2O, H_2O, OH^-, F^-, $CH_3CO_2^-$, SO_4^{2-}, CO_3^{2-}, NO_3^-, PO_4^{3-},ClO_4^- and ROH.

SOFT COPY: A document saved in a computer and only accessible through the computer. It is the electronic version of a document, which can be opened and edited using a software program. **Hard copy** is anything that has been printed on paper. Hard copies allow data to be read without the need of a computer.

SOFT FRUIT: A general term for all fruits and berries such as strawberries, raspberries and currants, which are stoneless and have relatively softer skins and a short shelf life. They are borne mainly on low-growing plants or bushes.

SOFT IRON: A form of iron that contains little carbon with a high relative permeability. An example is silicon steel. It is easily magnetised and demagnetised. It is used in making parts exposed to rapid changes of magnetic flux such as cores of electromagnets, motors, generators and transformers.

SOFT MATTER (SOFT CONDENSED MATTER): A general name given to noncrystalline condensed matter that is easily deformed by thermal stresses or thermal fluctuations. It includes liquids, colloids, polymers, foams, gels, granular materials and a number of biological materials.

SOFT PALATE: The movable fold, consisting of muscular fibres enclosed in mucous membrane, that is suspended from the rear of the hard palate and closes off the nasal cavity from the oral cavity during swallowing or sucking. During swallowing (deglutition) the soft palate is raised and helps to shut off the nasal part of the pharynx from the portion below.

SOFT RADIATION: Ionising radiation of low penetrating power, usually used with reference to high-energy electromagnetic radiation, typically high energy x-rays of long wavelength or gamma rays. Soft radiation has the ability to penetrate a given thickness of material, typically a lead shield. Soft radiation is able to cross 5 gcm^{-2} of brass but not 167 gcm^{-2} of lead.

SOFT ROT: A disease of fruits and vegetables, caused by fungi or bacteria, in which the tissues of the affected parts become soft. Soft rots are most notable as postharvest diseases. Bacteria of the genus *Erwinia* and fungi of the genus *Rhizopus* cause soft rots of stored vegetables and fruits, particularly in humid badly ventilated conditions. The tissues become soft as the enzymes from the pathogen break down the cell walls.

SOFT WATER: Water that does not contain salts of calcium (Ca^{2+}) or magnesium (Mg^{2+}), especially water containing less than 85.5 parts per million of calcium carbonate and does not form scum with soap. It is simply water that lathers easily with soap.

SOFT WHEAT: Wheat containing grains which, when milled, break down in a random manner. Soft wheat has less protein than hard wheat and has poor milling qualities. Flour from soft wheat is most suitable for making biscuits and pastries.

SOFT WOOD: The open grained wood produced by pine trees and other conifers.

SOFTWARE: The programs and data used by a computer. The two major software categories are system software and application software. System software is made up of the operating system and other control programs for managing the hardware and running the applications. Application software is any program including inventory, payroll, spreadsheet, word processor, games and DVD playback software that processes data for the user.

SOFTWARE AS A SERVICE (SAAS): Software such as Microsoft Office 365 that is provided over the Internet as a service instead of a product. Users access the software program by visiting a website and pay monthly service fees or pay by the total time the software is used. SaaS is considered part of cloud computing since the software is hosted on the Internet.

SOFTWARE PIRACY (PIRACY; PIRATING): In computer science, it is the act of illegally using, copying or distributing software without purchasing the software or having the legal rights.

SOHCAHTOA: An acronym coined for easy remembrance of the trigonometrical ratios: (**S**ine **O**pposite **H**ypotenuse - **SOH**), (**C**osine **A**djacent **H**ypotenuse - **CAH**) and (**T**angent **O**pposite **A**djacent - **TOA**). For example,

SOH expresses the Sine ratio: $Sine\,\theta = \dfrac{opposite}{hypotenuse}$

CAH expresses the Cosine ratio: $Cos\,\theta = \dfrac{Adjacent}{hypotenuse}$;

TOA expresses the Tangent ratio: $Tan\,\theta = \dfrac{Opposite}{Adjacent}$.

SOIL: A natural body consisting of superficial layers of the earth's crust. It consists of a mixture of rock particles, humus, air, micro organisms, mineral salts and water that provides a suitable environment for growing plants. Its formation depends on the original material from which it is derived, climate topography of the area, the organism present and time over which it has been developing. Soil particles pack loosely, forming a soil structure filled with pore spaces. These pores contain soil solution (liquid) and air (gas). Most soils have a density between 1 and 2 g/cm³.

SOIL AIR (SOIL ATMOSPHERE): The air content of the soil or the air and other gases in spaces in the soil, specifically, those found within the zone of aeration.

SOIL AMENDMENT: 1. An alteration of a soil to improve its physical or chemical properties such as its permeability, water retention, the fertility as well as alter the acidity of the soil characteristics by the addition of substances such as lime, gypsum, and sawdust to make the soil more suitable for the growth of plants. 2. Any substance such as manure, lime, gypsum, sawdust and peat used for this purpose.

SOIL BALL: The rooting system of a plant, complete with the ball of soil surrounding it, as when a plant is lifted from a pot or seedbed.

SOIL BULK DENSITY (BULK DENSITY, SOIL): The ratio of dry mass of soil to a known bulk soil volume, including both the solids and the pore space. It is expressed in g/cm³ using undisturbed, oven-dried soil obtained from a core sample with a known volume. The bulk volume is determined before the soil is dried to constant weight at 105°C. Soils with a bulk density higher than 1.6 g/cm³ tend to restrict root growth. Bulk density typically increases with depth since subsurface layers have reduced organic matter, aggregation, and root penetration compared to surface layers and therefore, contain less pore space. Bulk density influences water infiltration and plant root health. Sandy soils are more prone to high bulk density.

SOIL CAPPING: A hard crust on the surface of the soil which can be caused by heavy rain drop or the passage of heavy farm machinery.

SOIL CLASSIFICATION: The systematic arrangement of soils into categories on the basis of similar characteristics. Broad groupings are made on the basis of general characteristics, and subdivisions on the basis of more detailed differences in specific properties. The simplest arrangement distinguishes between pedocals, rich in calcium carbonate and pedalfers, low in calcium but high in compounds of aluminium and iron. The Great Soil Groups are zonal, azonal and intrazonal. Of the zonal soils, podzols are found beneath coniferous forest and latosols develop in warm, moist conditions. Chernozems, prairie soils and chestnut soils are formed beneath grassland. Grey and red desert soils occur in hot, arid areas and tundra soils form in periglacial environments.

SOIL COMPACTION: The process in which the soil is pressed down as found in foot paths or by trampling of large heads of hoofed animals or by heavy weight of machinery. The soil becomes very firm with little space between its particles. Heavily compacted soils contain few large pores and have a reduced rate of both water infiltration and drainage from the compacted layer, thus increasing runoff and erosion. Plants have difficulty in compacted soil because the mineral grains are pressed together, leaving little space for air and water, which are essential for root growth.

SOIL CONSERVATION: The use of a range of methods to protect the soil against physical loss by erosion, overcultivation or against chemical deterioration such as excessive loss of fertility by either natural or artificial means in order to maintain soil fertility and productivity.

SOIL CONTAMINATION (SOIL POLLUTION): The addition or presence of chemical or biological elements in the soil which causes deterioration of soil and renders it unsuitable for plant growth. The contamination may arise from the rupture of underground storage tanks, application of pesticides, percolation of contaminated surface water to subsurface strata, oil and fuel dumping, leaching of wastes from landfills or direct discharge of industrial wastes to the soil. The most common chemicals involved are petroleum hydrocarbons, solvents, pesticides, lead and other heavy metals.

SOIL CREEP (CREEP): A gradual steady movement of soil and loose rock down a slope caused usually by the action of rain and gravity.

SOIL DEPLETION: The loss of mineral salts or nutrients from soil. This is caused by regularly growing crops without the use of crop rotation or fertilisers and by leaching.

SOIL DRAINAGE: The flow or removal of excess water from soil, either naturally or through pipes and drainage channels inserted into the ground. Proper drainage improves soil structure, increases efficiency of phosphorus fertiliser, conserves soil nitrogen and controls waterlogging, leaching and salinisation of soils caused by irrigation.

SOIL EROSION: The removal of the soil layer usually by wind, water, ice and by the mass movement of soil down slope. Wind erosion is by deflation; water erosion takes place in gullies, rills or by sheet wash. Down slope mass movement ranges from soil creep to landslides. The complementary actions of erosion and deposition or sedimentation operate through wind, moving water and ice to alter existing landforms and create new landforms. Erosion will often occur after rock has been disintegrated or altered through weathering. Moving water is the most important natural agent of erosion. The removal of forests or other vegetation or by repeated ploughing and harvesting of crops often leads to serious soil erosion.

SOIL FACTORS (EDAPHIC FACTORS): The physical, chemical and biological properties of soil that influence the life of organisms. The main edaphic factors include water content organic content, texture and pH.

SOIL GENESIS: 1. Also called **pedogenesis**, it is the processes that lead to the formation of soil from parent material by by factors such as humification, weathering, leaching, and calcification. 2. The division of soil science that deals with soil genesis with special reference to the soil-forming factors responsible for development of the solum, or true soil: time, parent material, topography, climate and organisms.

SOIL HORIZON (SOIL PROFILE): Layers of soil profiles lying one above the other, parallel to the soil surface whose physical characteristics differ from one another. They are formed through a number of geological, chemical, and biological processes over thousands of years. They are identified mainly by colour, structure, texture, particle size, as well as biological and chemical composition. Most soils have three main soil horizons or layers which are Horizon A, Horizon B, Horizon C, and sometimes Horizon O. **Horizon A** refers to the upper layer of soil, nearest the surface and forms the mineral horizon providing plants with nutrients they need for growth. It is commonly known as **topsoil**. In the woods or other areas that have not been ploughed or tilled, this layer would probably include organic litter, such as fallen leaves and twigs. The litter helps prevent erosion, holds moisture and decays to form a very rich soil known as **humus**. **Horizon B** is the layer immediately below horizon A, which has no accumulation of litter and therefore much less humus. Horizon B does contain some elements from horizon A because of the process of leaching. **Horizon C** is below horizon B and consists mostly of weatherised big rocks. This solid rock, gave rise to the horizons above it. Sand, silt and clay are the basic types of soil. Leaching may also bring some minerals from horizon B down to horizon C. **Horizon O** is the soil layer on the surface of the soil above Horizon A with a high percentage of organic matter. The "O" stands for organic. O horizons may be divided into three distinct categories: the one with undecomposed leaves and other organic matter (Oi); the one underlain by a partially decomposed layer (Oe); and then a very dark layer of well decomposed humus (Oa). There is no set order for these horizons within a soil. Some soil profiles have a combination or, for example, just an O. Some profiles may have all the horizons.

SOIL IMPROVEMENT: The practice of making the soil more fertile by methods such as drainage and applying manure.

SOIL LOOSENER: A trailed implement which loosens the surface of the soil.

SOIL MANAGEMENT: The operations, practices and treatments used to maintain a healthy and functional soil system. Soil management concerns all operations, practices and treatments used to protect soil and enhance its performance. Soil management practices that affect soil quality include: (i) **Controlling traffic** on the soil surface to reduce soil compaction, which can reduce aeration and water infiltration; (ii) **Cover crops** keep the soil anchored and covered in off-seasons so that the soil is not eroded by wind and rain; (iii) **Crop rotations** for row crops alternate high-residue crops with lower-residue crops to increase the amount of plant material left on the surface of the soil during the year to protect the soil from erosion; (iv) **Nutrient management** can help to improve the fertility of the soil and the amount of organic matter content, which improves soil structure and function; (v) **Tillage**, especially reduced-tillage or no-till operations limit the amount of soil disturbance while cultivating a new crop and help to maintain plant residues on the surface of the soil for erosion protection and water retention.

SOIL MAPPING (SOIL SURVEY): The process of classifying soil types and other soil properties in a given area and geo-encoding such information. It applies the principles of soil science and encompasses geomorphology, theories of soil formation, physical geography and analysis of vegetation and land use patterns.

SOIL MECHANICS: A branch of engineering that studies the nature and properties of the soil. It is the application of the laws of solid and fluid mechanics to soils and similar granular materials, as a basis for design, construction and maintenance of stable foundations and earth structures.

SOIL MICROBIOLOGY: The study of organisms in soil, their functions and activities in the soil. Microorganisms in soil are important because they do very useful things, for example, they fix nitrogen to make plants grow and release oxygen into the atmosphere, maintain fertility, produce products that might stimulate plant growth, detoxify harmful chemicals, and aerate the soil. Soil microorganisms can be classified as bacteria, actinomycetes, fungi, algae and protozoa. Each of these groups has characteristics that define them and their functions in soil.

SOIL MOISTURE: Water stored in the spaces between soil particles. Soil moisture is a key variable in controlling the exchange of water and heat energy between the land surface and the atmosphere through evaporation and plant transpiration. As a result, soil moisture plays an important role in the development of weather patterns and the production of precipitation.

SOIL MOISTURE DEFICIT: The difference between the amount of water that is in the soil and the amount needed for crops to grow successfully.

SOIL MONOLITH: A vertical section of the soil profile that is not disturbed with the natural appearance of the soil, including its horizonation, colour, and structurethat is preserved and mounted in a cabinet for display and teaching purposes.

SOIL NUTRITION: Substances that provide nourishment to plants or the minerals that plants take from the soil to help them grow in a healthy manner. **Soil fertility** is the potential capacity of soil to provide nutrients to support plant growth. In all, the soil requires sixteen chemical elements to maintain fertility and the growth of all plants. They include carbon, oxygen and hydrogen obtained from carbon dioxide and water, nitrogen, phosphorus, potassium, calcium, magnesium, sulfur, iron, manganese, zinc, copper, boron, molybdenum and chlorine. Some plant species also require one or more of the elements cobalt, sodium, vanadium and silicon. The nutrients most necessary for proper plant growth are nitrogen, potassium, phosphorous, iron, calcium, sulfur and magnesium all of which usually exist in most soils in varying quantities. In addition most plants require minute amounts of substances known as trace elements, which are present in the soil in very small quantities. These include manganese, zinc, copper and boron. The supply of usable minerals in the soil is often increased by enrichment with artificial fertilisers and by treatments, hastening the breakdown of complex compounds.

SOIL ORGANIC MATTER: Decay or decaying vegetation and animal residue that form part of soil.

SOIL ORGANISM: Any organism that lives in the soil. There are two categories, micro fauna/microorganisms and macrofauna. The main soil microorganisms include bacteria, fungi, and protozoa. The macro fauna include oligochaeta, arthropods, molluscs, and nematodes. Soil organisms play an essential role in soil formation and the soil environment. Bacteria are the most important microorganisms in the soil as among other things the various types do the following: (i) provide nutrients through decomposition of plant and organisms' waste which is converted into energy in forms useful to the rest of the organisms in the soil; (ii) nitrogen-fixing bacteria convert atmospheric nitrogen (N_2) into ammonia (NH_3); (iii) nitrifying bacteria mix ammonia (NH_3) with water (H_2O) to create ammonium ion (NH_{4+}) and nitrate (NO_{3-}), which can be used by other organisms; (iv) denitrifying bacteria convert nitrate to nitrogen (N_2) which is returned into the atmosphere.

SOIL PAN (HARD PAN): A soil or a layer of compacted subsoil that often prevents the penetration of water or of shrub or tree roots. It can occur naturally or be caused by repeated cultivation with a tiller or plough.

SOIL PARENT MATERIAL: Material, rock or deposit, from which soil is formed. The nature of the parent material will largely determine the nature of the regolith and hence the soil texture; thus, basalt tends to produce clay soils while granite generally gives rise to sandy soils.

SOIL PERSISTENCE: The length of time that a herbicide remains active in soil after application. Factors such as soil (soil composition, soil chemistry, ph, and microbial activity), climatic conditions (moisture, temperature, and sunlight), and herbicide properties (water solubility, vapour pressure) determine the length of time herbicides persist in the soil.

SOIL PHASE: A subdivision of a soil type or other unit of classification outside the system of soil taxonomy having characteristics that affect the use and management of the soil, but that do not vary sufficiently to differentiate it as a separate type.

SOIL POROSITY: The amount of pore or open space between soil particles of clay, silt and sand. This spaces, or pores, between the soil particles varies depending upon the type of soil and its composition. The spaces between the particles that make up the structure of soil hold air and water. Fine-textured soils have more total pore space than coarse-textured soils; however, the pores are much smaller than those in soils with a high sand content. Thus, clay soils with more fine particles hold more water than coarse-textured sandy soils, but because of the large surface area associated with the smaller clay particles, much of this water is adsorbed and difficult for plants to extract.

SOIL PRODUCTIVITY: The capacity of a soil, in its normal environment, to produce a specified plant or sequence of plants under a specified system of management. The limitations are needed because no soil can produce all crops with equal success and a single system of management cannot produce the same effect on all soils.

SOIL PROFILE (VERTICAL SECTION Of SOIL): The vertical section of a soil, showing the nature and sequence of the various layers, as developed by deposition or weathering or both.

SOIL REACTION: The degree of soil acidity or alkalinity measured in units of pH. Soil reaction is important because it affects nutrient availability, microbial activity and plant growth. It is one of the most important physiological characteristics of soil solution. The presence and development of microorganisms and higher plants depend upon the chemical environment of soil. There are three types of soil reactions as follows: **(i) acidic,** in which high precipitation leaches appreciable amounts of exchangeable bases from the surface layers of the soils so that the exchange complex is dominated by H ions. (ii) **Alkaline**, in which salts of strong base such as sodium carbonate go into soil solution and hydrolyse, consequently giving rise to alkalinity. (iii) **Neutral**, which has a pH of between 6.6 and 7.3.

SOIL RESOURCES: The physical and chemical materials in the soil which support defective plant growth.

SOIL RETENTIVITY PROFILE: A graph showing the retaining capacity of a soil as a function of depth. The retaining capacity may be for water at any given tension, for cations, or for any other substances held by soils.

SOIL SALINITY: The quantity of salt content in a soil. Salt-affected soils are caused by excess accumulation of salts, typically most pronounced at the soil surface. Salts can be transported to the soil surface by capillary transport from a salt laden water table and then accumulate due to evaporation. They can also be concentrated in soils due to human activity. As soil salinity increases, salt effects can result in degradation of soils and vegetation.

SOIL SCIENCE: The study of the properties and management of soil. It involves: (i) the study of the soil as a natural resource on the surface of the Earth including soil formation, classification and mapping; (ii) physical, chemical, biological and fertility properties of soils in relation to the use and management of soils.

SOIL SERIES: The classification of soils based on their similarities used in soil mapping or a group of soils formed from the same parent rock and having similar horizons and soil profiles, but with varying characteristics according to their location.

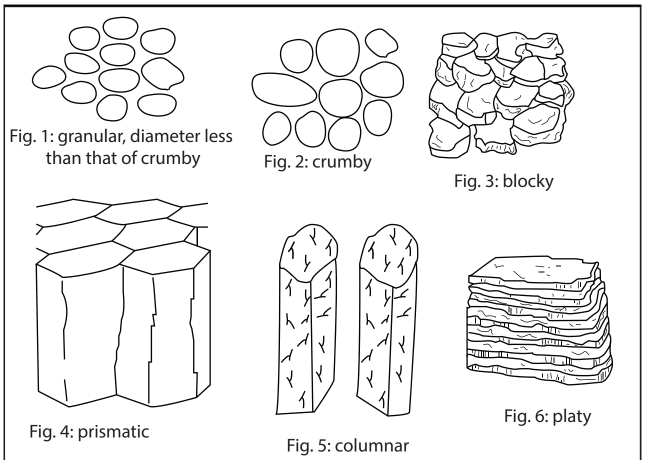
SOIL SIEVE: A device used to separate the four types of inorganic particles in soil, which are gravels, sand, silt and clay.

SOIL STABILISATION/STABILIZATION: Permanent physical or chemical treatment of soils and aggregates to improve their engineering properties. This can enhance its shear strength and/or control the shrink-swell properties, thus improving the load bearing capacity of a sub-grade to support pavements and foundations.

SOIL STERILANT: A chemical that temporarily or permanently prevents the growth of all plants and animals, depending on the chemical. Soil sterilants must be registered as pesticides.

SOIL STERILISATION (SOIL STERILIZATION): The physical or chemical treatment of glasshouse, greenhouse and other horticultural soils in order to kill weed seeds, plant disease organisms and pests. This control method affects many organisms, even though the elimination of only specific weeds, fungi, bacteria, viruses, nematodes or pests is desirable. Physical control measures include steam and solar energy. Chemical control methods include herbicides and fumigants.

SOIL STRUCTURE (SOIL FABRIC): The arrangement of soil separates into units called soil aggregates differing in shape, size, stability and degree of adhesion to one another. Aggregates may exist individually, such as sand particles or they may be grouped together in larger units, called peds. An aggregate possesses solids and pore space. Aggregates are separated by planes of weakness and are dominated by clay particles. Silt and fine sand particles may also be part of an aggregate. The aggregates can be grouped in various forms described as, for example, crumby, granular, blocky, platy, columnar and prismatic. The various forms are descrbed as follows: (i) **crumby** and **granular** structures are individual particles of sand, silt and clay grouped together in small, nearly spherical grains, commonly found in the A-horizon of the soil profile and has very good water circulation . The crumby form is larger than the granular one. (ii) **Prismatic** and **columnar** structures are formed into rectangular vertical columns resembling pillars separated tiny vertical cracks and commonly found in the B-horizon of the soil profile. Water circulates with greater difficulty and drainage is poor. (iii) **Blocky structures** form in nearly square or angular blocks with flattened surfaces and sharp edges, commonly found in the B-horizon. The soil resists penetration and movement of water. (iv) **Platy structure** is formed in a rectangular shape with thin overlapping plates or sheets piled horizontally on one another and has very poor water circulation. It is commonly found in forest soils, in part of the A- horizon, and in claypan soils.



Examples of aggregate of soil structure

SOIL SURVEY (SOIL MAPPING): The process of classifying soil types and other soil properties in a given area and geo-encoding such information. It applies the principles of soil science and draws heavily from geomorphology, theories of soil formation, physical geography and analysis of vegetation and land use patterns.

SOIL TEXTURE: Classification of soil based on type and proportions of inorganic particles in it. It affects soil properties such as aeration and water retention. Particles are grouped according to their size into

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what are called **soil separates**. Soil texture classification is based on the fractions of soil separates present in a soil. These separates are typically named clay (less than 0.002 mm in diameter), silt (between 0.002 mm and 0.05 mm in diameter) and sand (more than 0.05 mm in diameter). The **soil texture triangle** is a diagram often used to figure out soil textures. The sides of the soil texture triangle are scaled for the percentages of sand, silt and clay. Clay percentages are dashed lines read from left to right across the triangle . Silt is indicated by light, dotted lines and read from the upper right to lower left. Sand is indicated by bold, solid lines and read from lower right towards the upper left portion of the triangle . The boundaries of the soil texture classes are highlighted in blue. The intersection of the three sizes on the triangle give the texture class. For example, a soil with 20% clay, 60% silt and 20% sand falls in the "silt loam" class.

SOIL THERMOMETER (EARTH THERMOMETER): A Celsius thermometer, usually the mercury-in-glass type adapted for measuring soil temperature.

SOIL TYPE: A finer subdivision of a soil series. It includes all soils of a series which are similar in all characteristics, including texture of the surface layer.

SOIL WATER HOLDING CAPACITY: The amount of water that a given soil can hold after gravitational water loss has ceased. This determines the **available soil water holding capacity**, which is the amount of water volume between field capacity (the maximum amount of water the soil can hold) and wilting point (where the plant can no longer extract water from the soil). Beyond the wilting point there is still water in the soil profile, however it is contained in pores that are too small for the plant roots to access. Soil texture, soil structure, soil organic matter (SOM) and plant rooting depth are the crucial factors in determining the amount of water available for plants to access. Soil that is made up of smaller particle sizes, such as silt and clay, has larger surface area and therefore easier for the soil to hold onto water so it has a higher water holding capacity. The water holding capacity for sandy soil is low because it has large particle sizes which results in smaller surface area. Soil organic matter (SOM) is another factor that can help increase water holding capacity.

SOIL WATER TENSION: A measure of suction, defined as the force per unit area that is necessary for plant roots to extract water from the soil. When more pressure (energy) is needed for the plant to extract water from the soil, the value is represented by an increasing number. Saturated soil has a soil tension of about 0.001 bars, while soils around plants in the desert are roughly around 60 bars.

SOIL-BORNE FUNGUS: Fungus which is carried in the soil. Soils contain microscopic organisms including fungi, bacteria, viruses, nematodes and protozoa that are capable of damaging plants.

SOIL-FORMATION FACTORS: The interaction of five major factors that is responsible for the formation of soil. These include the following: parent rock, climate, organisms, relief, and time. Climate and organisms are labelled as "active" factors of soil formation as their influence over soil development can be directly observed. For example, rain, heat, cold, wind, microorganisms (algae, fungi), earthworms, and burrowing animals can be directly observed influencing soil development. Time, topography, and parent material are noted as "passive" factors because their effects are not immediately observed. The relative influence of each factor varies from place to place, but the combination of all five factors normally determines the kind of soil developing in any given place.

SOILAGE: A system of cutting and carrying green forage crops to feed animals grazing on unproductive pastures in order to supplement their diets. Crops commonly used for soilage are clovers and lucerne.

SOILING: Visible damage to materials by deposition of air pollutants.

SOL: A type of colloid, in which two or more solid particles are dispersed in a liquid. Examples of sol are emulsion, protoplasm, gel and starch in water.

SOL FRACTION: Mass fraction of the dissolved or dispersed material, resulting from a network-forming polymerisation or crosslinking process, that is constituted of molecules of finite (statistically definable) relative molecular masses.

SOL-GEL CRITICAL CONCENTRATION: Concentration of an added electrolyte above which a particulate sol undergoes coagulation instead of gelation.

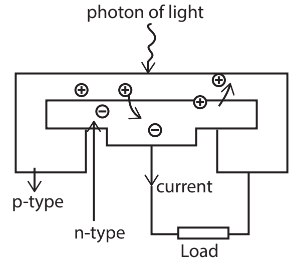
SOL-GEL PROCESS: Process through which a network is formed from solution by a progressive change of liquid precursor(s) into a sol, to a gel and in most cases finally to a dry network.

SOLANACEAE: A large family of dicotyledonous plants commonly called the potato or nightshade family. It contains about 90 genera and some 3000 species have been described. Many solanaceous plants are poisonous due to their possession of alkaloids, for example, atropine found in deadly nightshade (*Atropa belladonna*) and nicotine found in tobacco (*Nicotiana tabacum*). Others are important as food plants, especially the Irish potato (*Solanum tuberosum*) and tomato (*Lycopersicon esculentum*). The aubergines and the various capsicums also belong to this family. The genus Nicotiana includes the important cash crop tobacco and various ornamental including species of Petunia, Schizanthus (butterfly flowers) and Salpiglosis (velvet flowers).

SOLAR: Of the Sun or relating to or originating from the Sun or using the Sun's radiation as a source of energy or measured with reference to the Earth's movement in relation to the Sun. For example, solar energy is a type of energy derived from the Sun's radiation or energy transmitted from the Sun in the form of heat and light.

SOLAR ACTIVITY: The variations in the output of the Sun in all forms, which are not constant. It varies with both time (from seconds to millions of years) and position on the Sun. Examples of solar activity is solar flares, coronal mass ejections, high-speed solar wind, and solar energetic particles. All solar activities are driven by the solar magnetic field.

SOLAR CELL (PHOTOVOLTAIC CELL): A solid state device, usually a semiconductor such as silicon, which absorbs photons with energies higher than or equal to the bandgap energy and simultaneously produces electric power. In other words, it obtains energy from solar radiation and coverts it into electrical energy. It consists of a thin semiconductor such as silicon, which is specially treated to form an electric field, positive on one side and negative on the other. Any electron generated by sunlight in the depletion region may pass from the p-side to the n-side of the junction and constitutes current flow. If a wire or any load is connected between the ends of the n-type and p-type region with metal contacts, electron will flow to the p-type through an external load. The electricity generated by photovoltaic panels is direct current DC and an inverter is used to convert it to alternating current (AC). A group of cells are connected electrically and packaged into a frame called a module or solar panel.



Schematic diagram of a solar cell showing single crystal silicon p-n junction.

SOLAR CONJUNCTION: The period of time during which the Sun is in or near the communications path between a spacecraft and Earth, thus disrupting the communications signals.

SOLAR CONSTANT: The rate at which the energy of the Sun is received per unit area at the outer limit of the Earth's atmosphere at the mean distance between the Sun and the Earth. Its value is 1.353 kWm^{-2}.

SOLAR CYCLE: A period in which several important kinds of solar activities repeat including the two 11-year cycles of macula (sunspot), whose magnetic polarities alternate between the Sun's northern and southern hemispheres, with two peaks. It declines in a phenomena such as solar prominences and auroras that vary in the same period.

SOLAR DRYER: A device for drying crops using the heat of the Sun. There are mainly three types of solar dryers: (i) the absorption or hot box type dryers in which the product is directly heated by the Sun; (ii) the indirect or convection dryers in which the product is exposed to warm air which is heated by means of a solar absorber or heat exchanger; (iii) dryers combining the principles of the above two, where the product is simultaneously exposed to the Sun and a stream of pre-heated air.

SOLAR ECLIPSE: An eclipse in which the Earth passes through the shadow cast by the moon. In other words, it occurs when the moon comes between the sun and the Earth and the shadow of the moon is cast on Earth. Solar eclipses only happen when the moon is new and when it lies close to the node of its orbit (when it is roughly in the same plane as the Earth's orbit).

SOLAR ENERGY: Energy from the Sun's radiation or energy transmitted from the Sun in the form of heat and light, used by green plants for photosynthesis and harnessed as solar power. It is the original source of all forms of energy on Earth except nuclear energy. The upper atmosphere of Earth receives about 1.5×10^{21} watt-hours (thermal) of solar radiation annually. This amount of energy is more than 23,000 times that used by the human population of this planet and only about one two-billionth of the Sun's energy, about 3.9×10^{20} MW. Solar energy is considered a renewable source of energy. It does not produce by-products or pollutants that harm the environment.

SOLAR FLARE: A sudden explosive release of particles and energy from the Sun's chromospheres with intense brightness. Flares develop in a few minutes and may last several hours, releasing intense x-rays and streams of energetic particles. They appear to be connected with changes in the Sun's magnetic fields during the solar cycle. The ejected particles take a day or two to reach the vicinity of Earth, where they can disrupt radio communications and cause auroras and may pose a radiation hazard to astronauts.

SOLAR HEATING: A form of domestic and industrial heating that makes use of direct solar energy. There are two types: passive and active. **Passive heating** relies on architectural design. The building's siting orientation, layout, materials and construction are utilised to maximise the heating effect of sunlight falling on it. A well-insulated building with a large south-facing window, for instance, can trap heat on sunny days and reduce reliance on gas, oil or electricity. Brick, stone or tile capacity walls are often incorporated to absorb the Sun's energy and radiate it into the interior, usually after a time lag of several hours. In **active solar heating**, mechanical means are used to collect, store and distribute solar energy. In liquid-based systems, a blackened metal plate on the exterior absorbs sunlight and traps heat, which is transferred to a carrier fluid. Alternatively, fluid may be pumped through a glass tube or volume of space onto which sunlight has been focused by mirrors. After picking up heat from the collector, the warm fluid is pumped to an insulated storage tank. The system can supply a home with hot water from the tank or provide space heating with the warmed water flowing through tubes in floors and ceilings.

SOLAR LONGITUDE: The angular distance along Earth's orbit measured from the intersection of the ecliptic and the celestial equator where the Sun moves from south to north; it gives the position of Earth in its orbit.

SOLAR MASS: The mass of the Sun, which is approximately 2,000 trillion trillion tons (2×10^{30} kg).

SOLAR MAXIMUM: The month(s) during the solar cycle when the 12-month mean of monthly average sunspot numbers reaches a maximum.

SOLAR NEBULA: The rotating cloud of gas and dust that began to collapse about 5 billion years ago to form the solar system.

SOLAR NEBULAR HYPOTHESIS (NEBULAR HYPOTHESIS): A hypothesis that the Solar System evolved from a nebula cloud made from a collection of dust and gas. It was first developed in the 18th century by Emanuel Swedenborg, Immanuel Kant and Pierre-Simon Laplace.

SOLAR NEUTRINO PROBLEM: The problem that the number of neutrinos detected on Earth as being given off by the Sun is only one-third of the expected solar neutrino flux. It is now thought that this is because some of the neutrinos given off by the Sun are converted from electron neutrinos to the two other types of neutrinos on their way to Earth. Neutrinos are produced in nuclear reactions inside the Sun; and can also be produced in laboratory nuclear reactions.

SOLAR OXYGENATING PUMP: An air pump, powered by direct sunlight, which adds oxygen to ponds providing a healthier environment for fish. The advantages of such an oxygenator are that the pump has no operating costs, requires no power outlet or wires to operate it and works hardest when it is needed most (on hot summer days when the sun is beating down and the oxygen content of the water in the pond would otherwise be at its lowest).

SOLAR PARALLAX: The angle subtended by the Earth's equatorial radius at the centre of the Sun at the mean distance between the Sun and the Earth. Its adopted value is 8.794148 arc seconds and constitutes the fundamental datum for the dimensions of the solar system.

SOLAR PLEXUS: The ganglion of nerve cells and fibres situated at the back of the abdomen which supply the abdominal viscera.

SOLAR POND: Natural or artificial pool of very salty water, in which salt becomes more soluble in the Sun's heat. Water at the bottom becomes saltier and hotter and is insulated by the less salty water layer at the top. Temperatures at the bottom reach about 100 °C can be used as a direct source of heat and the resulting solar radiation may be used in generating electricity. An example of a natural solar pond is the Dead Sea.

SOLAR PROMINENCE: A cloud composed of luminous hydrogen, calcium, sodium and other gases forming temporarily in the upper chromosphere or inner corona of the Sun. It is specially numerous in regions above sunspots and is observed as a bright projection. It has a lower temperature but higher density than its surroundings.

SOLAR RADIATION: The electromagnetic radiation emitted by the Sun. The total range of wavelengths of light emitted by the Sun (99.9% in the range from 150 to 4000 nm) is filtered on entering the Earth's atmosphere, largely through the absorption by oxygen, ozone, water vapour and carbon dioxide. Near sea level, only light of wavelengths longer than about 290 nm is present. The light from 290 – 400 nm is effective in inducing important photochemical processes since absorption by the important trace gases, ozone, nitrogen dioxide, aldehydes, ketones, etc., is significant in this region.

SOLAR RADIO EMISSION: Emissions of the Sun at radio wavelengths from centimetres to decametres, under both quiet and disturbed conditions, of which several types are recognised. Type I is a noise storm composed of many short, narrowband bursts in the metric range (50 to 300 MHz). Type II is narrowband emission that begins in the meter range (300 MHz) and sweeps slowly (over tens of minutes) toward decametre wavelengths (10 MHz). Type II emissions occur in loose association with major flares and are indicative of a shock wave moving through the solar atmosphere. Type III are narrowband bursts that sweep rapidly (in a matter of seconds) from decimetre to decametre wavelengths (0.5 to 500 MHz). This type often occurs in groups and is an occasional feature of complex solar active region. Type IV is a smooth continuum of broadband bursts primarily in the metre range (30–300 MHz). These bursts are associated with some major flare events beginning 10 to 20 minutes after the flare maximum and can last for hours.

SOLAR SYSTEM: The Sun and everything that moves around it or the Sun, the nine major planets and their natural satellites or moons, the asteroids, the comets and the meteoroids.

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SOLAR TELESCOPE: A telescope designed specifically for observing the Sun.

SOLAR TIME: Time measured with respect to the Sun. Apparent solar time is based on the daily motion of the true sun, in other words, the actual position of the Sun in the sky. This is the version of solar time that a sundial shows, but it runs unevenly, for the two main reasons that Earth's axis is tilted with respect to its orbital plane and Earth's orbit is elliptical rather than circular. A better version of solar time is mean solar time, which is used for accurate timekeeping, including all civil timekeeping, is based on how the Sun would move in the sky if Earth had no axial tilt and had a perfectly circular orbit and there were no other irregularities in Earth's motion. Solar time loses about 4 minutes a day compared with sidereal time because of Earth's orbital movement.

SOLAR TOWER: A solar telescope that is mounted on a tall tower in order to place the optics above the image-distorting layer of Sun-heated air near the ground.

SOLAR UNITS: Units based on the physical properties of the Sun such as mass, diameter, density and luminosity used to describe other celestial objects. Examples include solar luminosity, solar mass and solar radius.

SOLAR WIND: A stream of particles, mostly protons and electrons that continuously escape from the Sun or the flow of high speed ionised particles from the Sun's surface into interplanetary space.

SOLAR YEAR (TROPICAL YEAR): The time taken by the Earth to complete one revolution round the Sun, which is about 365 days 5 hours, 48 minutes and 46 seconds or the length of time that the Sun takes to return to the same position in the cycle of seasons, as seen from Earth. An example is the time from vernal equinox to vernal equinox or from summer solstice to summer solstice.

SOLARI PIGGERY: A type of housing for pigs, with fattening pens on each side of a central feeding passage, housed in an open-sided Dutch barn.

SOLARISATION (SOLARIZATION): 1. Exposure to the rays of the Sun, especially for the purpose of killing pests in the soil, by covering the soil with plastic sheets and letting it warm up in the sunshine. 2. A phenomenon in photography in which the image recorded on a negative or on a photographic print is wholly or partially reversed in tone. Dark areas appear light or light areas appear dark.

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SOLDER: An alloy with a low melting point, which is usually a mixture of tin and lead, used in the molten state to join electrical components to a circuit board or to join metal objects together. The type of solder used is determined by the metals to be joined. Soft solders are commonly composed of lead and tin and have low melting points of between 200 °C and 300 °C. Hard solders such as silver solders have high melting points of above 450 °C and are suited for use with ferrous and high-melting-point nonferrous alloys.

SOLE: 1. The bottom surface of the human foot, especially the part stood on. 2. A group of several flatfish belonging to several families, especially family Soleidae The most common of them is the common sole, *Solea solea*. Sole is commonly used for food. 3. In botany, it is the part of the carpel, which is furthermost from the apex.

SOLE CROPPING (MONOCROPPING): The practice of growing a single crop on a piece of land. This is common with cash crop farming. One of its advantages is that crops that are best suited for the land can be planted, leading to higher yields. Its disadvantages include depleting soil nutrients but is compensated for by using fertilisers; and factors such as crop failure and decreasing market price of the product, put the farmer at risk.

SOLENOGLYPHOUS: Describing a snake with long movable fangs that fold against the roof of the mouth when the jaws are closed, as in vipers and adders.

SOLENOID: A coil of wire, usually cylindrical in which a magnetic field is created by passing an electric current through it or a device consisting of a cylindrical coil of wire surrounding a movable iron core that moves along the length of the coil when an electric current is passed through it. Solenoids are used in control circuits to operate an electrical switch; for example, an automobile starter solenoid or a linear solenoid, which is an electromechanical solenoid. Certain mechanical devices such as a plunger can be attached to a solenoid to operate certain devices such as a valve, called solenoid valves to open/close valves to the passage of gas and fluid such as oil, air and water. Solenoid valves are used in automatic irrigation sprinkler systems, domestic washing machines and dishwashers to control water entry to the machine, in controlling pneumatic and hydraulic systems, etc. Solenoids are also used in locking applications such as vault doors, cash registers, disk drives and missile systems; and as an interlock device for integration into automatic gearbox drive selectors.

SOLENOIDAL VECTOR FIELD: In mathematics, a vector field that has a divergence of zero at all points in the field.

SOLID: 1. One of the three states in which matter occurs (the others are liquid and gas). Solids resemble liquids in having a definite volume, but differ from both liquids and gases in having a definite shape. A solid's atoms or molecules are fixed in positions and do not have the freedom of movement found in a liquid or gas. All solids have the ability to support loads applied either perpendicular (normal) or parallel (shear) to a surface. Solids can be crystalline, for example, metals; amorphous such as in a glass; or quasicrystalline as in certain metal alloys, depending on the degree of order in the arrangement of the atoms. 2. A three-dimensional geometric figure consisting of points continuously connected and separated from the rest of space by a surface called the boundary of the solid. Points on the boundary are usually considered part of the solid. A solid is bounded if there exists a sphere having finite radius that could enclose the solid. A solid is convex if all points of any line segment having end points on the boundary also are points of the solid.

SOLID ANGLE: The three dimensional angle formed by the vertex of a cone or intersection of three planes. When this vertex is the centre of a sphere of radius r and the base of the cone cuts out an area s on the surface of sphere, the solid angle in steradians (according to SI) is defined as $\frac{s}{r^2}$. Its symbol is Ω.

SOLID FIGURE: An object with a three dimensional (3D) shape.

SOLID GEOMETRY: The branch of geometry that is concerned with the properties of three-dimensional figures and surfaces.

SOLID OF REVOLUTION: A solid formed when a plane surface is revolved about a straight line (the axis) that lies on the same plane. Assuming that the curve does not cross the axis, the solid's volume is equal to the length of the circle described by the figure's centroid multiplied by the figure's area (Pappus's second centroid Theorem).

SOLID SOLUTION: A crystalline solid that is a mixture of two or more components, with ions, atoms or molecules of one component replacing some of the ions, atoms or molecules of other component in a crystalline lattice. Solid solutions can be found in alloys. Also, in nature, some solid solutions appear in the form of minerals made under conditions of heat and pressure. For example, gold and copper form solid solution in which some of the copper atoms in the lattice are replaced by gold atoms.

SOLID STATE: Physical state of matter that has a definite shape and resists having it changed; the volume of a solid changes only slightly with temperature and pressure. A true solid state is associated with a definite crystalline form such as metals and amorphous solids such as glass.

SOLID STATE DRIVE (SSD): A type of mass storage device similar to a hard disk drive (HDD). It supports reading and writing data and maintains stored data in a permanent state even without power. Internal SSDs connect to a computer like a hard drive, using standard IDE or SATA connections.

SOLID STATE PHYSICS: The study of the physical properties of solids. It deals with the properties of crystal-lattice arrangements of atoms and dislocations and defects in the arrangements with special emphasis on the electrical properties of semiconducting materials in relation to their electronic structure.

SOLID-NOT-FAT PERCENTAGE: A measure of milk quality, showing the percentages of all substances other than fat in the milk.

SOLID-STATE DETECTOR (SEMICONDUCTOR RADIATION DETECTOR): A radiation detector in which a semiconductor material such as a silicon or germanium crystal constitutes the detecting medium.

SOLIDIFICATION: A change of state from liquid or vapour to solid that occurs at the freezing point of the substance.

SOLIDUS: 1. The slanted line (/) in a fraction dividing the numerator from the denominator. For example, in the fraction ¾ the solidus is the line between 3 and 4. 2. The line or curve on a phase diagram below which a given substance is completely solid (crystallised). The solidus is applied to metal alloys, ceramics and natural rocks and minerals.

SOLIFLUCTION: Slow, downhill mass movement of sediment saturated with meltwater which may contain regolith. This occurs in regions which experience regular periods of freezing and thawing.

Solitary flower

SOLITARY: 1. In botany, it refers to flowers occurring singly and not borne in a cluster. 2. In zoology, it describes an animal that spends a majority of its life without others of its species except during time of mating and raising young.

SOLITON: A stable particle-like single wave that maintains its shape while it travels at constant speed. Solitons are caused by a cancellation of nonlinear and dispersive effects in the medium. Solitons occur in many areas of physics and applied mathematics such as fluid mechanics, lasers, optics, solid state physics, etc. It is used in the solution of certain equations for propagation.

SOLONCHAK (WHITE ALKALI SOIL): A type of intrazonal infertile soil with a high salt content (halomorphic). It is found in arid and semiarid inland regions and in areas where sea spray has blown inland resulting in the deposition of salt. The evaporation of the surface water results in salts from the underlying layers being drawn upwards and deposited on the surface.

SOLSTICE: Either of the points (about 23½° or 23.45°) at which the Sun is furthest north or south of each year. It also refers to the time at which the Sun reaches either of these points. The summer solstice in the Northern Hemisphere occurs on June 21, when the North Pole is tilted toward the Sun. The winter solstice occurs on December 21 in the Northern Hemisphere, when the South Pole is tilted toward the Sun.

SOLUBILITY: The amount of a solute that will dissolve in a fixed volume of a solvent at a given temperature to form a saturated solution or a measure of one substance's ability to dissolve in a specific amount of another substance at standard temperature and pressure. It may be expressed as moles per dm^3 (moles/dm^3) or grams of solute per dm^3 (grams/dm^3) of solvent.

SOLUBILITY CURVE: A plot of solubility against temperature or a graph showing the concentration of a substance in its saturated solution in a solvent as a function of temperature.

SOLUBILITY PRODUCT: The product of the concentrations of ions in a saturated solution raised to the power of the coefficient of the substance in the dissociation equation. For example, if a compound A_xB_y is in equilibrium with its solution $A_xB_y(s) \rightleftharpoons xA^+(aq) + yB^-(aq)$ the equilibrium constant is $K_c = [A^+]^x[B^-]^y/[A_xB_y]$. Since the concentration of the undissolved solid can be put equal to 1, the solubility product is given by $K_s = [A^+]^x[B^-]^y$. The expression is only true for sparingly soluble salts. If the product of ionic concentrations in a solution exceeds the solubility product, then precipitation occurs.

SOLUBLE: 1. Having the ability to dissolve in another substance such as water. 2. In mathematics, also known as solvable, it is able to be solved, for example, a solvable equation.

SOLUBLE GROUP (SOLVABLE GROUP): In mathematics, a group with normal series in which every normal factor is Abelian. Equivalently, for some positive integer, the derived series terminates in a trivial subgroup or, for a finite group, the composition factors have a prime order. For $n \geq 5$, the symmetric group S_n is insoluble, while for $1 \leq n \leq 4$, S_n is insoluble.

SOLUBLE NSF ATTACHMENT RECEPTOR (SNARE): Any of a family of proteins that facilitate fusion between intracellular transport vesicles and the plasma membrane and essential for secretion of cellular materials by exocytosis.

SOLUM: Soil, including both top and sub soil or the layers above the parent material, which are strongly influenced by climate and vegetation.

SOLUTE: The minor component of a solution which is regarded as having been dissolved by the solvent. It is the dissolved particles in a solution. For example, salt is the main solute in seawater.

SOLUTE POTENTIAL (OSMOTIC POTENTIAL): The component of water potential that is due to the presence of solute molecules. Pure water has a solute potential (Ψπ) of zero as solutes lower the water potential of the system. The phenomenon of osmotic pressure arises from the tendency of a pure solvent to move through a semi-permeable membrane and into a solution containing a solute to which the membrane is impermeable. This process is of vital importance in biology as the cell's membrane is selective towards many of the solutes found in living organisms. Solutions which have the same solute potential are isotonic and will have the same concentration so that there is no concentration gradient between them and therefore, no net movement of water between the two.

SOLUTION: 1. In mathematics, it is a set of values that satisfies the requirements of an equation or set of equations. 2. In chemistry, it is a homogenous mixture of two or more substances whose properties vary continuously with varying proportions of the components. Solutions can exist in all three physical states (solid, liquid or gas). In a solution, the particles of the solute and solvents (the atoms or molecules) are uniformly distributed. A liquid solution can be distinguished experimentally from a pure liquid by the fact that during transfers into other single phases at equilibrium (freezing and vapourising at constant pressure) the temperature and other properties vary continuously, whereas those of a pure liquid remain constant.

SOLUTION BY RADICALS: In mathematics, the possibility of obtaining an expression for the roots of polynomial equations involving only radicals or, more precisely, as the end product of a tower of radicals: a finite sequence of numbers each of which is the radical of a polynomial in the previous members of the sequence. This is possible for all equations of degree less than 5, but not in general for quintic or higher-degree polynomials as a consequence of Abel and Galois.

SOLUTION CURVE: The curve $\{(t, y(t)) : t \in I\}$, where y is the solution of a system of ordinary differential equations and I is the interval of existence of y.

SOLUTION SET: The set of value(s) for a variable that satisfy an equation or inequality or a set of given equations or inequalities. For example, for a set $\{f_i\}$ of polynomials over a ring R, the solution set is the subset of R on which the polynomials all vanish or evaluate to 0, defined formally as $\{x \in R: \forall i \in I, f_i(x) = 0\}$. Examples are: (i) the solution set of the single equation $f(x) = x$ is the set {0}; (ii) for any non-zero polynomial, f over the complex numbers in one variable, the solution set is made up of finitely many points; (iii) for a complex polynomial in more than one variable the solution set has no isolated points.

SOLVABLE PROBLEM: In computing and logic, a decision problem that possesses an algorithm that, given a suitably presented instance of the problem, returns the answer for that instance. If no such algorithm exists, the problem is unsolvable.

SOLVATION: The association or combination of a solute unit (ionic, molecular or particulate) with solvent molecules or the interaction of ions of solute with molecules of solvent. For example, when sodium chloride is dissolved in water, the sodium ions attract polar molecules with the negative oxygen atoms pointing towards the positive sodium ions. The greater the polarity of the solvent, the greater is the attraction between the solute and solvent molecules. This is what causes ionic compounds to dissolve. Water is a good polar solvent. Solvation by water is called **hydration**.

SOLVATION ENERGY: The change in Gibbs energy when an ion or molecule is transferred from a vacuum (or the gas phase) to a solvent.

SOLVAY PROCESS (AMMONIA-SODA PROCESS): An industrial process for producing washing soda (sodium carbonate) from common salt. A solution of salt is saturated with ammonia and carbon dioxide is passed through it. This causes sodium hydrogen carbonate to precipitate. It is then heated to obtain the sodium carbonate.

SOLVE: To work through a problem, an equation or a system of equations to arrive at a suitable answer or to reach a satisfactory solution.

SOLVENT: A substance, usually a liquid that will dissolve another substance or the dissolving medium in a solution. Water is the most common solvent. In solutions of solids or gases in a liquid, the liquid is the solvent. In all other solutions such as liquids in liquids or solids in solids the constituent that is present in larger quantity is considered the solvent. Other compounds valuable as solvents because they dissolve materials that are insoluble or nearly insoluble in water are acetone, alcohol, benzene or benzol, carbon disulfide, carbon tetrachloride, chloroform, ether, ethyl acetate, furfural, gasoline, toluene, turpentine and xylene or xylol.

SOLVENT EXTRACTION: 1. The separation of materials of different chemical types and solubilities by selective solvent action. Some materials are more soluble in one solvent than in another, hence there is a preferential extractive action. Solvent extraction is used to refine petroleum products, chemicals, vegetable oils and vitamins. 2. Also known as **liquid extraction**, it is a process for removing uranium fuel residue from used fuel elements of a reactor. It generally involves decay cooling under water for up to 6 months, removal of cladding, dissolution, separation of reusable fuel, decontamination and disposal of radioactive wastes.

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SOLVENT PARAMETER: Quantitative measures of the capability of solvents for interaction with solutes. Such parameters have been based on numerous different physicochemical quantities, for example, rate constants, solvatochromic shifts in ultraviolet/visible spectra, solvent-induced shifts in infrared frequencies, etc. Some solvent parameters are purely empirical in nature (they are based directly on some experimental measurement). It may be possible to interpret such a parameter as measuring some particular aspect of solvent–solute interaction or it may be regarded simply as a measure of solvent *polarity.* Other solvent parameters are based on analysing experimental results. Such a parameter is considered to quantify some particular aspect of solvent capability for interaction with solutes.

SOLVENT SHIFT: A shift in the frequency of a spectral band of a chemical species arising from interaction with its solvent environment.

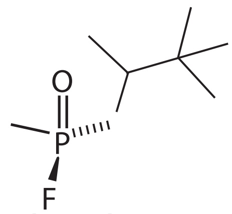
SOLVENT-INDUCED SYMMETRY BREAKING: Breaking of symmetry of a molecular species by interactions with the solvent that can modify the molecular charge distribution, to favour asymmetrical configurations.

SOLVOLYSIS: Chemical reaction between solvent and solute molecules in which a dissolved solute and its solvent combine to form a new compound.

SOLVUS: A line on a binary phase diagram (or a surface on a ternary phase diagram) that defines the limit of solid solubility under equilibrium conditions.

SOMNAMBULISM (SLEEP WALKING): The act of walking around and performing other activities such as eating while sleeping with no recollection of it on waking up. Common in childhood, it may continue into the adulthood. It may be caused by stress, seizure disorders such as epilepsy, bipolar disorders or other neurological conditions; however, in most cases the causes are not known.

SOMAN (*O*-PINACOLYL METHYLPHOSPHONOFLUORIDATE): A highly toxic, colourless, volatile liquid used as a nerve gas with chemical the formula $C_7H_{16}FO_2P$, density 1.022 g/cm³, melting point -42 °C (231 K) and a boiling point of 198 °C (471 K). It is an organophosphorous compound and a nerve agent, interfering with normal functioning of the mammalian nervous system by inhibiting the cholinesterase enzyme. As a chemical weapon, it is classified as a weapon of mass destruction by the United Nations according to UN Resolution 687. Its production is strictly controlled and stockpiling is outlawed by the Chemical Weapons Convention of 1993, where it is classified as a Schedule 1 substance. Soman was the third of the so-called *G-series* nerve agents to be discovered along with GA (tabun), GB (sarin) and GF (cyclosarin).



Chemical Structure of Soman

SOMATIC: 1. Relating to all the cells as opposed to the germ cells or reproductive cells. 2. Of, relating to or affecting the body, especially as distinguished from the viscera, head or limbs.

SOMATIC CELL: Cells forming the body of an organism except the reproductive cells. Somatic cells are not or will not become a gamete. Internal organs, skin, bones, blood and connective tissue are all made up of somatic cells.

SOMATIC HYBRIDISATION (SOMATIC HYBRIDIZATION): The production of cells, tissues or organisms by fusion of nongametic nuclei. The phenomenon may be induced under laboratory conditions in cells that never normally fuse together and used as a plant breeding or genetic tool. It may also occur naturally, especially in fungi.

SOMATIC HYPERMUTATION (SHM): A mechanism by which activated B cells can generate antibodies of increased specificity and with improved antigen-binding over the cause of an adaptive immune response. SHM diversifies the receptors used by the immune system to recognise foreign elements (antigens) and allows the immune system to adapt its response to new threats during the lifetime of an organism

SOMATIC NERVOUS SYSTEM: The part of the peripheral nervous system that serves the sense organs and muscles of the limbs and body wall and brings about voluntary muscle activity. It consists of efferent nerves responsible for sending brain signals for muscle contraction. The system includes all the neurons connected with skeletal muscles, skin and sense organs.

SOMATIC RECOMBINATION: A mechanism of genetic recombination that occurs in vertebrates, which randomly selects and assembles segments of genes encoding specific proteins with important roles in the immune system. This site-specific recombination reaction generates a diversity of T cell receptor (TCR) and immunoglobulin (Ig) molecules that are necessary for the recognition of diverse antigens from bacterial, viral and parasitic invaders and from dysfunctional cells such as tumour cells.

SOMATIC SENSES: All senses that enable pain, temperature change, touch, pressure and the body's position in space. It does not include vision, hearing, taste and smell.

SOMATOSTATIN (GROWTH HORMONE INHIBITING HORMONE, GHIH): A hormone secreted by the hypothalamus that inhibits the growth hormone of the anterior pituitary gland. Somatostatin acts primarily as a negative regulator of a variety of different cell types, blocking processes such as cell secretion, cell growth and smooth muscle contraction. It is secreted from the hypothalamus into the portal circulation and travels to the anterior pituitary gland, where it inhibits the production and release of both growth hormone and thyroid-stimulating hormone. Many tissues other than the hypothalamus contain somatostatin.

SOMATOTROPIN (GROWTH HORMONE): A hormone, produced by the pituitary gland, that stimulates protein synthesis and promotes the growth of bone.

SOMITE (PRIMITIVE SEGMENT): Any of a series of segmented blocks of tissue in animal embryos that develop from the mesoderm. In vertebrate embryos they lie on the dorsal side of the notochord and give rise to the vertebral column, ribs, dermis and striated muscle. In invertebrates showing metameric segmentation, the somites give rise to the body segments.

SONAR (SOUND NAVIGATION AND RANGING): A remote sensing technique or device that uses sound waves to detect, locate and sometimes identify objects in water. There are many applications, using a wide variety of equipment. Naval uses include detection of submarines, sea mines, torpedoes, swimmers, torpedo guidance, acoustic mines and for navigation. Civilian uses include determining water depth, finding fish, mapping the ocean floor, locating various objects in the ocean such as pipelines, wellheads, wrecks and obstacles to navigation; measuring water current profiles; and determining characteristics of ocean bottom sediments.

SONIC BOOM: A strong shock wave generated by an object moving through the air at supersonic speeds (faster than the speed of sound). The loudness depends on the speed and the altitude of the aircraft. An object such as an airplane, moving through the air, generates sound. When the speed of the object reaches or exceeds the speed of sound, the object catches up with its own noise. At higher speeds, it forces the sound ahead of itself faster than the noise would ordinarily travel. The piled-up sound takes the form of a violent shock wave (sonic boom) that propagates behind the object. Sonic booms occasionally have mechanically destructive effects in addition to their role as noise pollutants.

SONIC HEDGEHOG HOMOLOGUE (SHH): One of three proteins in the mammalian signalling pathway family. The others are desert hedgehog (DHH) and Indian hedgehog (IHH). SHH plays a crucial role in patterning and development of tissues in vertebrates, particularly of the nervous and skeletal systems. It is homologous to Hedgehog protein, which performs a similar function in the fruit fly *Drosophilia*.

SONIC SPEED: The speed of sound at the current conditions of air density, humidity and temperature; or the speed of a body travelling at Mach 1, which is about 1,220 kilometres per hour at sea level.

SONICATION: 1. A process of applying ultrasound energy to agitate particles in a sample for various purposes, including dissolution, breaking up intermolecular forces, removing dissolved gases, deactivating a biological material, etc. In biological applications, sonication may be sufficient to disrupt or deactivate a biological material, a process called **sonoporation**. It is also used in ultrasonic cleaning to free particles adhering to surfaces such as spectacles and jewelry. Sonication is also used to extract microfossils from rock. 2. The process of shaking pollen from flowers in buzz pollination using sound waves by vibrating bee wing muscles.

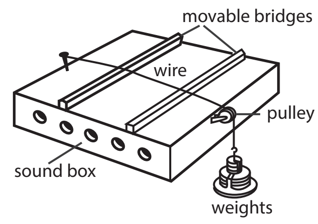
SONICATOR: An instrument for producing high-intensity ultrasound waves by converting an electrical signal into a mechanical vibration that can be directed toward a substance to agitate particles in a sample for various purposes such as breaking apart compounds or cells for further examination. It consists of ultrasonic electric converter, which creates a signal at about 20 KHz that powers a transducer to convert the electric signal into mechanical vibration by using piezoelectric crystals or crystals. The vibration is amplified by the sonicator and transmitted to the solution being sonicated through a probe. This causes the molecules in the solution to break apart and cells to rupture. The probe tips are varied to suit many applications.

SONOCHEMISTRY: The study of chemical reactions in liquids subjected to high-intensity sound or ultrasound. This causes the formation, growth and collapse of tiny bubbles within the liquid, generating localised centres of high temperature with extremely rapid cooling rates.

SONOGEL: Colloidal gel produced by the action of ultrasonically induced cavitation.

SONOLUMINESCENCE: The light that is emitted by passing high-frequency sound waves through a liquid medium. It is caused by rapid changes of pressure induced by the sound resulting in minute bubbles to form in the liquid, which then collapse, squeezing and heating gas inside the bubble.

SONOMETER (MONOCHORD): A device used to verify the relationship between the vibration frequency of the sound produced by a plucked string, the tension, length and mass per unit length of the string. It consists essentially of a hollow sounding box with two bridges attached to its top. The string, fixed to the box at one end is stretched between the two bridges so that the free end can run over a pulley and support a measured load. The relationships are usually called Mersenne's laws, named after Marin Mersenne (1588-1648), who investigated and codified them. The Mersenne's laws state that for small amplitude vibration, the frequency is proportional to the following: (i) the square root of the tension of the string;(ii) the reciprocal of the square root of the linear density of the string; and (iii) the reciprocal of the length of the string.



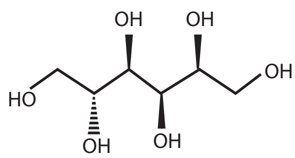
A monochord (sonometer)

SONOSOL: Sol produced by the action of ultrasonically induced cavitation.

SOOT: A randomly formed particulate carbon material, which may be coarse, fine and/or colloidal in proportions depending on its origin. Soot consists of variable quantities of carbonaceous and inorganic solids together with absorbed and occluded tars and resins. An unwanted by-product of incomplete combustion or pyrolysis, soot is generated within flames consisting essentially of aggregates of spheres of carbon. Soot found in domestic fireplace chimneys contains few aggregates but may contain substantial amounts of particulate fragments of coke or char. Soot from diesel engines consists essentially of aggregates together with tars and resins.

SOOTY MOULD: A fungal disease caused by a genus of fungus *Scorias* in the Capnodiaceae family. It is characterised by dark patches of a black sticky substance on the plant leaves, stems and twigs. It is transmitted by insects such as aphids, scale, mealybugs and white flies that secrete honeydew.

SORBITOL (HEXAHYDRIC ALCOHOL): A white, sweetish, crystalline alcohol with the chemical formula $C_6H_8(OH)_6$. It is an isomer of mannitol, formed by the reduction of glucose. It is also found in various berries and fruits. It is used as a flavouring agent, a sugar substitute for people with diabetes and a moisturiser in cosmetics, in the manufacture of vitamin C and other products.


Chemical Structure of Sorbitol

SOREDIUM: A type of asexual reproductive structure found on lichen that functions as a means of vegetative propagatio. Soredia, which range from 25 to 100µm in diameter, form as a result of the rupture of the cortex exposing the algal layers, which break into spore-like structures composed of a few algal cells and hyphae are released like spores and dispersed by air currents.

SORET BAND: A very strong absorption band in the blue region of the optical absorption spectrum of a haem protein.

SORGHUM: A drought-resistant cereal plant, widely cultivated in tropical and warm areas. It can grow up to three metres tall and bear seeds on terminal heads or panicles. The two major types of sorghum are the grain or nonsaccharine type, cultivated for grain production and to a lesser extent for forage; and the sweet or saccharine type, used for forage production and for making syrup and sugar.

SOROSIS: A type of composite fruit such as pineapple fruit formed from an entire inflorescence spike or arising from many flowers.

SORPTION: A physical and chemical process by which one substance becomes attached to another. It is the action of absorption or adsorption. Absorption is the incorporation of a substance in one state into another of a different state, for example, liquids being absorbed by a solid or gases being absorbed by a liquid. Adsorption is the physical adherence or bonding of ions and molecules onto the surface of another phase, for example, reagents adsorbed to solid catalyst surface.

SORPTION PUMP: A type of vacuum pump in which gas is removed from a system by absorption on a solid. It creates a vacuum by adsorbing molecules on a very porous material like molecular sieve which is cooled by a cryogen, typically liquid nitrogen. It is mainly used as a roughing pump for a sputter-ion pump in ultra-high vacuum experiments, for example, in surface physics. The main advantages are the absence of oil or other contaminants, low cost and vibration free operation because there are no moving parts. The main disadvantages are that it cannot operate continuously and cannot effectively pump hydrogen, helium and neon or all gases with lower condensation temperature than liquid nitrogen.

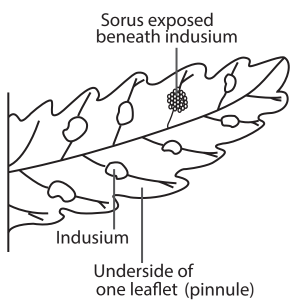
SORPTION TECHNIQUES: Techniques based on the distribution of components being separated between two phases, one of which is stationary and the other mobile. The advantage of some chromatographic methods is the possibility of combining the preconcentration and determination steps, as well as improving the speed of determination and the possibility of separation of components with similar properties and of achieving high values of the preconcentration coefficient. The methods enable analyses of micro amounts of substances. Ion-exchange chromatography is not widely used owing to the great volumes of solutions being treated and, consequently, to a great degree to the variation in the blank and to some diffusional limitations.

SORREL: 1. Any of several plants of the genus *Rumex*, having acid-flavoured leaves, sometimes used as salad greens, especially *R. acetosella*, a widely naturalised Eurasian species. 2. Any of various plants of the genus *Oxalis*, having usually compounded leaves with three leaflets.

SORT KEY: A field in a record that is used as the basis to determine the sequence of the file. For example, addresses could be sorted using the city as primary or major sort key and the street as secondary or minor sort key.

SORTING: In computer science, it is a specific function used in a program to arrange selected items into ascending or descending numerical or alphabetical order.

SORUS: Any of a spore-producing structure in certain fungi. In fungi and lichens, the **sorus** is surrounded by an external layer. In some red algae it may take the form of a depression into the thallus. The plural of sorus is **sori**.



Sorus/Sori

SOS RESPONSE: A post replication DNA repair system used by bacteria, such as *Escherichia coli* to respond to treatments that damage DNA or prevent DNA replication. It is a metabolic regulatory system, which is activated in a bacterial cell in response to damage by agents such as ultraviolet light, certain chemicals and other mutagens, which interfere with DNA replication, causing large postreplicative gaps in newly synthesised strands. During this response, about twenty genes are expressed at increased rates; controlled by two regulatory proteins: the LexA repressor and the RecA protein. LexA repressor, inhibits expression of the SOS genes during normal cell growth and RecA protein, is activated by treatments that turn on the SOS response causing the LexA protein to cleave itself into two pieces, which inactivates it as a repressor. These genes are individually termed SOS genes and collectively the SOS regulon. In the absence of active LexA, the recA and SOS genes are expressed in large amounts.

SOUND: 1. Energy that moves as longitudinal waves through a medium, solid, liquid or gas and can be detected by the ears of an animal or vibrations travelling through air, water or some other medium, especially those within the range of frequencies that can be perceived by the human ear. A sound wave is generated by a vibrating object. The vibrations cause alternating compressions (regions of crowding) and rarefactions (regions of scarcity) in the particles of the medium. The particles move back and forth in the direction of propagation of the wave. The speed of sound through a medium depends on the medium's elasticity, density and temperature. At sea level and freezing point, the speed of sound through the air is 1,220 km per hour. Ordinarily, however, sound moves through air at a velocity of 344 m/s and through water at 1,460 m/s. The frequency of a sound wave, perceived as pitch, is the number of compressions or rarefactions that pass a fixed point per unit time. The frequencies audible to the human ear range from approximately 20 hertz to 20 kilohertz. 2. The music, speech or other sounds heard through an electronic device such as a television, radio or loudspeaker, especially with regard to volume or quality. 3. To observe the sound made by an organ of the body for testing or diagnostic purposes.

SOUND CARD: A printed circuit board that, coupled with a set of speakers, enables a computer to reproduce music and sound effects and also perform functions such as to record sound input from a microphone connected to the computer and manipulate sound stored on a disk.

SOUND INTENSITY (ACOUSTIC INTENSITY): Amount of energy that a sound wave brings to a unit area every second in a direction perpendicular to that area. Its unit is ergs per second per square centimetre or in units of power, microwatts (10^{-6} watt) per square centimetre.

SOUND INTENSITY LEVEL (IL): One tenth of the common logarithm of the ratio of the intensity of a sound to the sound intensity at the threshold of hearing. Its unit of measurement is decibels (dB). The human ear responds to the intensities ranging from 10^{-12} W/m^2 to more than 1 W/m^2 (which is loud enough to hurt the ears). Because the range is so wide, intensities are scaled by factors of ten. The barely audible and the faintest intensity of sound is 10^{-12} W/m^2 is taken as reference intensity, called **zero bel**. The sound intensity level, β in decibels of a sound having an intensity I in watts per metre squared is defined to be $\beta = 10\log_{10}\left(\frac{I}{I_0}\right)$, where I_0 = 10^{-12} W/m^2 is a reference intensity.

SOUND PRESSURE LEVEL (SPL): A ratio of the absolute, sound pressure and a reference level, usually the threshold of hearing, or the lowest intensity sound that can be heard by most people. SPL is measured in decibels (dB). It can be calculated using the following formula: $L_p = 20\log_{10}\left|\frac{p}{p_0}\right| dB,$ where p is the sound pressure of the sound wave and p_0 is the reference sound pressure. SPL can be measured with a sound pressure level meter.

SOUND WAVES: Vibrations made in the air or other medium by which sound is carried or an audible pressure wave caused by a disturbance in water or air and carried forward in a ripple effect. In air, it is a succession of outwardly travelling layers of compression and rarefaction, capable of being detected by the ear.

SOUR: Having a sharp taste such as that of vinegar, a lemon, unripe mangoes and oranges.

SOUR SOIL (SOIL ACIDITY): A soil which is excessively acidic and hence needs treatment with lime. Acidity in soils comes from hydrogen ions (H^+) and aluminium (Al^{3+}) ions in the soil solution and sorbed to soil surfaces. Potential of hydrogen (pH) is the measure of H^+ in solution. Al^{3+} is important in acid soils because between pH4 and 6, Al^{3+} reacts with water (H_2O) forming $AlOH_2^+$ and $Al(OH)_2^+$, releasing extra H^+ ions. Every Al^{3+} ion can create $3H^+$ ions. Many other processes including rainfall, fertiliser, plant root activity and the weathering of primary and secondary soil minerals contribute to the formation of acid soils. Acid soils can also be caused by pollutants such as acid rain and mine spoilings. Soil pH controls many chemical processes that take place. It specifically affects plant nutrient availability by controlling the chemical forms of the nutrient. The optimum pH range for most plants is between 6 and 7.5, however many plants have adapted to thrive at pH values outside this range.

SOURCE: 1. The electrode in a field effect transistor from which electrons or holes enter the electrode space. 2. The starting point of a river. 3. In computing, it is the location where information is obtained from. For example, if you are copying information from a CD to a computer, the CD is considered the source and the computer is the destination. In a website, the source is the HTML code or other code used to generate a web page on a browser.

SOURCE CODE: A software program, which is used to create a program. It consists of the programming statements that are created by a programmer with a text editor or a visual programming tool and then saved in a file. For example, a programmer using the C language types in a desired sequence of C language statements using a text editor and then saves them as a named file. This file is said to contain the source code. It is now ready to be compiled with a C compiler and the resulting output, the compiled file, is often referred to as object code. Commonly, the source code is not released to others outside of the company that developed the software.

SOURCE HABITAT: A habitat where reproduction exceeds mortality and from which excess individuals disperse.

SOURCE LANGUAGE: In computer science, it is the language in which a program is written, which must be translated. It is usually classified as either high-level language or low-level language, according to whether each notation in the source language stands for many or only one instruction in machine code.

SOURCE OF ERROR: A critical part of part of the conclusion in an experiment that tells what went wrong and how it can be changed to make the data more accurate next time.

SOURCE OF LIGHT: The various sources that provide energy to the charged particles, such as electrons, whose motion creates light. If the energy comes from heat, then the source is called **incandescent**. If the energy comes from another source, such as electric or chemical energy, the source is called **luminescent**.

SOUSLIN SET (ANALYTIC SET): In mathematics, the continuous image of a Polish space. Analytic subsets of Polish spaces are closed under countable unions and intersections, continuous images and inverse images.

SOUTH: One of the four cardinal points of the compass on the right of a person facing the sunrise, opposite to the north or part of any place, country, etc, lying further in this direction than other parts. By convention, the bottom side of a map is south. The southern direction has azimuth or a bearing of 180°. **True south** is the direction towards the southern end of the axis about which the Earth rotates, called the South Pole and located in Antarctica. **Magnetic south** is the direction towards the west magnetic pole, some distance away from the south geographic pole or the South Pole.

SOUTH POLE: The south end of the axis of rotation of a planet or other astronomical object or the point where the southern end of the Earth's axis intersects the celestial sphere. It is located at latitude 90° south. It is the southern point from which all meridians of longitude start. The area around it is a lofty plateau in west-central Antarctica, with ice as much as 2,700 metres thick. It has six months of complete daylight and six months of total darkness each year. It was first reached by the Norwegian explorer, Roald Amundsen (1872-1928) in 1911, one month before the expedition led by British explorer Robert Falcon Scott.

SOUTHBRIDGE: In computer science, it is a chip on the motherboard that connects the northbridge to other components inside the computer, including hard drives, network connections, USB and Firewire devices, the system clock and standard PCI cards. The southbridge sends and receives data from the CPU through the northbridge chip, which is connected directly to the computer's processor.

SOUTHERN BLOTTING: A chromatographic method for isolating and identifying a specific DNA restriction fragment of DNA such as the fragment formed as a result of DNA cleavage by restriction enzymes. It was developed by Edward Southern (1938 -), a British biochemist. DNA from a gel electrophoresis pattern is blotted into nitrocellulose paper, then the DNA is denatured and fixed on the paper. Subsequently, the pattern of specific sequences in the southern blot can be determined by hybridisation to a suitable probe and autoradiography. A **northern blot** is similar, except that RNA is blotted instead onto the nitrocellulose paper.

SOUTHERN LIGHTS (AURORA AUSTRALIS): Aurora that appears in the Southern Hemisphere skies characterised by streamers of reddish or greenish light. It is caused by collisions between gaseous particles in the Earth's atmosphere with charged particles released from the Sun's atmosphere.

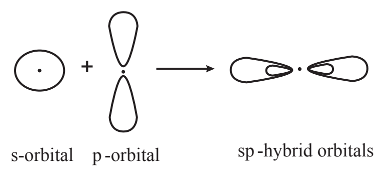
SOW: 1. Put seed on or in the ground or in soil or in pots, seed boxes, etc. 2. The female of the swine family that has given birth.

SOXHLET APPARATUS: A piece of laboratory apparatus for extracting components from a solid. The material used is placed in a thimble made of thick filter paper and this is held in a specially designed reflex condenser with a suitable solvent. It was invented in 1879 by Franz von Soxhlet (1848-1926), a German agricultural chemist.

SOYA: An annual legume plant belonging to the subfamily Papilionoideae, family Fabaceae (formerly Leguminosae) and classified as *Glycine max*. It is an erect, hairy plant from 0.6 to 1.5 metres in height, with large trifoliate leaves, small white or purple flowers and short pods with one to four seeds. Its seeds, called soya beans or soybeans are high in protein and edible fat. A product from the beans is soya milk (soymilk), a highly nutritious milk substitute often with vitamins and sugar added. Soya milk is also used to make curd or a type of cheese. Soya is also a pasture plant planted to improve agricultural soil. Soya beans contain no starch and are a good source of protein for diabetics.

SOYOIL (SOYABEAN OIL): A pale yellow, fixed drying oil produced by solvent extraction from soya beans. It is soluble in alcohol, chloroform and ether. It is used for soap manufacture, cattle feeds and printing inks and in margarine, salad dressing and high-protein foods.

sp-HYBRIDISATION (sp-HYBRIDIZATION): A process in which one s- and one p- orbitals of the same atom combine to form two similar hybrid sp orbital. The angle between the sp orbitals is 180°. The chemical bonding in compounds such as alkynes with triple bonds is explained by *sp*-hybridisation. The



Formation of sp-hybrid orbitals

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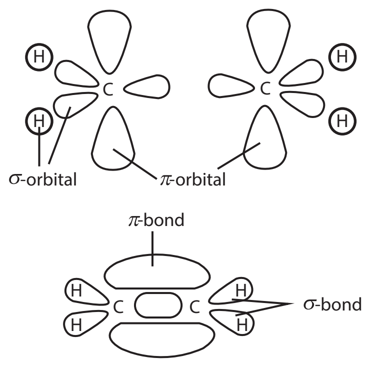
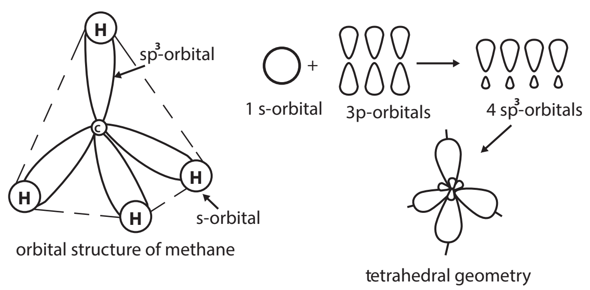
chemical bonding in acetylene (ethyne) (C_2H_2) consists of *sp-sp* overlap between the two carbon atoms forming a sigma (σ) bond and two additional pi (π) bonds formed by *p-p* overlap. Each carbon also bonds to hydrogen in a sigma *s-sp* overlap at 180° angles.

sp^2-HYBRIDISATION (SP^2 HYBRIDISATION): A process in which one s- and two p- orbitals of the same atom combine to form three similar hybrid sp^2-orbitals. In ethylene (ethene) the two carbon atoms form a sigma (σ) bond by overlapping two sp^2-orbitals and each carbon atom forms two covalent bonds with hydrogen by s-sp^2 overlap all with 120° angles. The pi (π) bond between the carbon atoms perpendicular to the molecular plane is formed by $2p$-$2p$ overlap. The hydrogen-carbon bonds are all of equal strength and length, which agrees with experimental data.



sp^2-HYBRIDISATION

sp^3-HYBRIDISATION (sp^3-HYBRIDIZATION): A process in which one s- and three p- orbitals of the same atom combine to form four similar hybrid sp^3 orbitals. The angle between any two of the four sp^3 hybrid orbitals is the tetrahedral angle 109.5°. For instance, the carbon in methane is sp^3 hybridised and is represented as follows:



sp^3-hybridisation

SPACE: 1. The boundless, three-dimensional extent in which objects and events occur having relative position and direction. 2. A collection of points that have geometric properties in that they obey set rules, axioms, etc. An example is a Euclidean space that is governed by Euclidean geometry. 3. Also known as **Outer space**, it is the region, usually of negligible density, existing between all astronomical objects in the universe including the Earth.

SPACE ADAPTATION SYNDROME (SPACE SICKNESS): Symptoms that occur among most astronauts caused by changes in gravitational forces, such as the transition to weightlessness. The symptoms include pallor, increased body warmth, cold sweating, malaise, loss of appetite, nausea, fatigue, vomiting and anorexia.

SPACE ALLOWANCE: The amount of space a farm animal should have in which to move around.

SPACE BAR (SPACE; SPACEBAR; SPACE KEY): The largest and longest key that is situated on the last row of the computer keyboard. When the spacebar key is pressed, it creates an empty space, also known as a space character or whitespace that helps separate words and other characters in a sentence. For example, if there were no spaces in a sentence all of the text would be together and would be very difficult to read.

SPACE CHARGE: Any net electrical charge that exists in a given region of space. In electronics, this usually refers to the electrons in the space between the filament and plate of an electron tube. In atmospheric electricity, space charge refers to a preponderance of either negative or positive ions within any given portion of the atmosphere. A net positive space charge is found in fair weather at all altitudes in the atmosphere and is largest near the Earth's surface.

SPACE CURVE: A curve that may pass through a three-dimensional space. It is the boundary of a bounded surface. This contrasts with a plane curve ,which must lie in a single plane.

SPACE EQUIVALENCE: The upper reaches of Earth's atmosphere where conditions for survival are almost the same as those required for survival in space.

SPACE GROUP: A group formed by the set of all symmetry operations of a crystal lattice. This set consists of translations, rotations and reflections and their combinations such as glide and screw. Space groups are used in the quantum theory of solids and in structure analysis in crystallography.

SPACE PROBE: An automated unmanned spacecraft that gathers information from planets or moons or other bodies in the solar system. Radio signals carry the information back to the Earth.

SPACE SHUTTLE: Any of a series of reusable US space vehicles that can be launched into earth orbit transporting astronauts and equipment for a period of observation, research, etc, before re-entry. It has an unpowered landing on a runway. They include Columbia, which exploded in 2003, Challenger, also exploded in 1986, Discovery, Atlantis and Endeavour. The first operational flight was in 1982.

SPACE STATION: Any large structure designed to contain astronauts for extended periods of time. Space stations are used for carrying out astronomical observations and surveys of Earth, as well as for biological studies and the processing of materials in weightlessness. A space station consists of many parts called **modules** that were assembled together in space by astronauts, connected by **nodes**. On the sides of the space station are solar panels, which collect energy from the Sun to generate electricity. The first space station was the Soviet Salyut 1 (1971). In 1973, NASA launched Sky. The core of the Soviet space station Mir was launched in 1986. The **International Space Station** (ISS) is a large spacecraft located at an altitude of about 400 km high, constructed by the USA, Russia and 14 other nations. Its core module was launched in 1998. It contains an international crew of 6 people at a time that carry out research. It travels around the Earth at an average speed of 27,700 km/h orbiting Earth every 90 minutes.

SPACE SUIT (SPACESUIT): Protective clothing worn by astronauts inside spacecraft as a safety precaution in case of loss of cabin pressure. It is also necessary for extra-vehicular activity (EVA) and work done outside the spacecraft such as on the surface of the Moon. Space suits are designed to have the following: (i) a pressurised atmosphere; (ii) oxygen to remove carbon dioxide; (iii) maintain a comfortable temperature despite strenuous work and movement into and out of sunlit areas; (iv) protection from micrometeoroids and from radiation to some degree; (v) clear visibility to move easily inside and outside of the spacecraft and communicate with ground controllers and fellow astronauts.

SPACE-FILLING CURVE: A curve that completely fills up every point in two or more dimensional spaces, such as squares or cubes. An example is a Peano curve.

SPACE-REFLECTION SYMMETRY: The theory that the behaviour of a group of subatomic particles and its mirror-image counterpart cannot be separated in strong interactions.

SPACE-TIME (SPACE TIME; SPACE-TIME CONTINUUM): A geometry of the physical universe as suggested by the theory of relativity that relates space and time in a four-dimensional structure consisting of length, width, depth and time. It was postulated by A. Einstein and H. Minkowski. In Newtonian physics, space was considered as a separate entity which could be expressed by Cartesian coordinates; time was viewed as an independent one-dimensional concept. Based on a system of geometry devised by H. Minkowski, space and time are regarded as entwined or twisted together.

SPACE-TIME DIAGRAM: The representation of the motion of an object in space-time. A space-time diagram is a graph showing the position of objects as a function of time. The usual convention is that time runs up the diagram, so the bottom is the past or early times and the top is the future or late times. A point on this graph describes both a position (the horizontal or x coordinate) and a time (the vertical or t coordinate). A "point" in space-time is called an **event**.

SPACECRAFT: A vehicle of a kind used to travel beyond the Earth's atmosphere. It is designed to operate, with or without a crew, in a controlled flight pattern above Earth's lower atmosphere. Most spacecrafts are not self-propelled; they are accelerated to the necessary high velocity by staged rockets, which are jettisoned when their fuel is used up. For internal power, Earth-orbiting spacecraft use solar cells and storage batteries, fuel cells or a combination, whereas craft designed for deep-space missions usually carry thermoelectric generators heated by a radioactive element.

SPACER: A flexible segment used to link successive mesogenic units in the molecules of MCPLCs (main-chain polymer liquid crystals) or to attach mesogenic units as side-groups onto the polymer backbone of SGPLCs (side-group polymer liquid crystals).

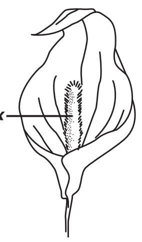
SPACER DNA: A non-repetitive DNA that lies between transcribed sequences and has no apparent function except to separate other genetic elements. In bacteria, spacer DNA sequences are only a few nucleotides long. In eukaryotes, they can be extensive and include repetitive DNA, comprising the majority of the DNA of the genome.

SPACEWALK: An excursion outside a spacecraft or space station by an astronaut wearing only a spacesuit and, possibly, some sort of manoeuvring device. A **standup spacewalk** is when the astronaut remains partially within the spacecraft, for example, in the case of a Gemini astronaut, who stands up in his seat with one of the capsule hatches open while in orbit.

SPACING: The space between several things such as between lines or words in type or the way this space is arranged.

SPADE: A hand held tool for digging and scooping sand, mortar, etc, with a wide shallow blade flattened where it meets the shaft so that it can be pushed into the ground with the foot.

SPADING MACHINE: A machine which uses rotation digger blades to cultivate compacted soil and dig out pans created by other cultivators.



SPADIX: A flowering shoot with a large fleshy floral axis bearing small usually unisexual flower typically surrounded by a leaf-like curved bract called a **spathe**. It is characteristic of plants of the families Araceae and Musaceae.

SPALLATION: 1. In general, a process in which fragments of material (spall) are ejected from a body due to impact or stress. 2. In impact mechanics it describes ejection or vapourisation of material from a target during impact by a projectile. 3. In planetary physics, spallation describes meteoritic impacts on a planetary surface and the effects of a stellar wind on a planetary atmosphere. 4. In mining or geology, spallation can refer to pieces of rock breaking off a rock face due to the internal stresses in the rock. It commonly occurs on mine shaft walls. 5. In anthropology, spallation is a process used to make stone tools such as arrowheads by knapping. 6. In nuclear physics, spallation is the process in which a heavy nucleus emits a large number of nucleons as a result of being hit by a high-energy particle, thus greatly reducing its atomic weight.

SPAM: 1. Also known as **unsolicited commercial email**, UCE or **bulk e-mail**, it is a term used to describe junk e-mail or unsolicited e-mail messages received on the Internet. It is sent to people without prior approval, promoting a particular product, service or a scam to get other people's money. 2. When talking in chat, forum or a newsgroup, spam, also known as **flooding**, is the process of posting multiple lines of the same text two or more times. In a newsgroup, if a message is posted two or more times, this is also considered spam or a flood of messages. 3. A general term used in any form of online communication to describe someone who is advertising a product or service.

SPAN: 1. In mathematics, also called **hull**, it is the closure of a set under some operation; the smallest set containing a given set and having a specified property. The linear span of a set in a vector space is the smallest linear subspace containing the set. 2. To have (a given set) as its span; to contain all elements of (the given set) in the set of linear combinations of its elements. For example, the vectors (0, 1) and (1, 0) span the real plane.

SPANISH CEDAR (BARBADOS CEDAR; BRAZILIAN CEDAR; BRITISH GUIANA CEDAR; BRITISH HONDURAS CEDAR; *Cedrela odorata*): A monecious, deciduos, medium-sized to large tree in the family Meliaceae that is commonly found in tropical and subtropical forests and is the most commercially important and widely distributed species in the genus Cedrela. It can grow up to 40 metres tall, sometime reaching up to 60 metres in South America. The bole is straight, cylindrical and branchless for up to 25 metres and about 120-300 centimetres in diameter. The butresess is absent or very small. The surface of the bark is rough and fisured, redish-brown especialy near the base of the bole and greyish higher up. The leaves alternate, paripinate with pairs of leaflets that are oposite to alternate, entire, ovate oblong-lanceolate. It has large and much-branched inflorescences, which bear numerous small, five-part, symmetrical greenish-white flowers. Trees are monoecious; male and female flowers are borne on the same inflorescence but the species is proterogynous (female flowers open first). The fruit is a large woody capsule, borne near branch tips. Fruits ripen, split and shed seeds while still attached to the parent tree. The seed is a sharply angled or winged columela. Its aromatic wood is rot resistant and also resistant to termites. It is, therefore, used for making panelling and veneer wood including cigar boxes, light construction, mouldings, cabinets, furniture, for building boats and canoes and for making musical instruments. It is also used as firewood. Medicinally, the root and bark are used to tree several ailments such as fever and pain; and the seeds are believed to have vermifugal properties. It is also planted as an ornamental tree alongside roads and in parks.

SPANNER (WRENCH): A hand or power tool with fixed or movable jaws, used to seize, turn or twist objects such as nuts and bolts.

SPANNING TREE: A subgraph linking a given set of nodes of an undirected graph. The tree is then said to span those nodes.

SPARSE: In mathematics, the sparse of a matrix or an equation system is one in which most of the elements are zero, as frequently occurs in application. This is in contrast with a dense matrix that has a high proportion of non-zero entries.

SPARSE MATRIX TECHNIQUE: In mathematics, any technique that exploits the properties of sparse matrices, such as the structure and the desired operations to perform on it, in order to reduce dramatically the work involved in solving, storing or manipulating systems of equations.

SPARK: A high energy discharge through a gas that lasts for a very short time. It is accompanied by a flash of light and a sharp crackling noise. Electric sparks play an important part in many physical effects. Uncontrolled sparks have harmful and undesirable effects, such as the gradual destruction of contacts in a conventional electrical switch and large-scale havoc resulting from lightning discharges. Controlled spark can be used in the ignition system of an automobile, as an intense short-duration illumination source in high-speed photography and as a source of excitation in spectroscopy.

SPARK CHAMBER: A device for detecting the passage of charged particles produced by reaction in a particle accelerator. It consists of a chamber, filled with helium and neon at atmospheric pressure, in which a stack of 20 to 100 plates are placed. The plates are connected alternately to the positive and negative terminals of a source of high potential, 10,000 Volts or more.

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SPARK COUNTER: A particle detector in which high-speed charged particles ionise a gas. It consists of argon mixed with an organic gas, triggering a spark between two plane parallel metal electrodes. It is a useful step towards understanding the Geiger-Müller tube.

SPARK PLUG: A device that ignites the fuel mixture in the cylinder in an internal-combustion engine by emitting a spark. It fits into the cylinder head of an internal-combustion engine and carries two electrodes separated by an air gap, across which current from a high-tension ignition system discharges, creating a spark and igniting the fuel.

SPAT: Young shellfish just after they attach to underwater structures or an immature bivalve mollusc. An example is the oyster.

SPATHE: A large, conspicuous bract or pair of bracts subtending and often enclosing an inflorescence or its peduncle.

SPATIAL AND TEMPORAL SCALES: Climate may vary on a large range of spatial and temporal scales. Spatial scales indicate a geographic region of climate change which may range from local (less than 100,000 km^2), through regional (100,000 to 10 million km^2) to continental (10 to 100 million km^2). Temporal scales indicate the time period of climate change which may range from seasonal to geological (up to hundreds of millions of years).

SPATIAL EQUATION OF CONTINUITY (CONTINUITY EQUATION): In physics, a mathematical statement about the conservation of mass in a continuum material body defined as $\rho + \rho\, div\, v = 0$, where v is the velocity of the particles of a body with density ρ and the divergence is taken with respect to its current configuration.

SPATIAL DESCRIPTION (EULERIAN DESCRIPTION): The description of physical phenomena associated with the deformation of a body in terms of fields defined over the current rather than the reference.

SPATHODEA: A genus of trees in the family Bignoniaceae, containing only 1 species, *Spathodea campanulata*, commonly known as the Fountain Tree or African Tulip Tree. It grows between 15 and 30 m in height with a trunk of 80 cm in diameter. It is found throughout tropical Africa, growing naturally in secondary forests in the high forest zone and in deciduous, transition and savannah forests. It has short branches and a compact rounded crown. Older trees have rounded buttresses towards the base of the trunk. It is easily recognised by its conspicuous flaming inflorescence comprising large and brilliant orange-red or fire-red bell-shaped flowers with orange-yellow margin. The leaves are elliptic or oblong, asymmetrical at the base, imparipinnate, with 4 - 8 pairs of opposite leaflets plus a terminal leaflet. The flowers have unpleasant smell and are hermaphrodite and large, occurring in terminal compact with erect racemes standing up right above the foliage. The fruit splits along one side to release very numerous small flat seeds with papery transparent wings all round, which are edible. The soft, light brownish-white wood is used for carving and making drums. The bark has laxative and antiseptic properties and the seeds, flowers and roots are used as medicine. The bark is used for cleansing fresh wounds and may also be boiled in water to use for bathing newly born babies to heal body rashes.

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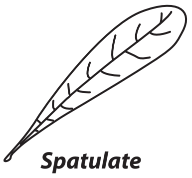
SPATIAL-DISTRIBUTION INTERFERENCE: Interference which may occur when changes in concentration of concomitants affect the mass flow rates or mass flow patterns of the analyte species in the flame. If they are caused by changes in volume and rise velocity of the gases formed by combustion, in extreme cases manifesting themselves by changes in the size and/or shape of the flame, they are non-specific and are called **flame-geometry interferences**.

SPATTER: Very fluid fragments of molten lava ejected from a vent that flatten and congeal on the ground. Typically, spatter will build walls of solidified lava around a single vent to form a circular-shaped spatter cone or along both sides of a fissure to build a spatter rampart.

SPATULA (TONGUE DEPRESSOR): 1. A flat wide wooden or plastic stick that a doctor uses to hold down the tongue in order to examine the throat and mouth. 2. A thin hand tool often made of nickel for handling chemicals or other materials when weighing. 3. A kitchen tool that has a handle which is bent upward and a wide, thin blade used for lifting and turning foods on a hot surface; or a kitchen tool that has a long handle and short, soft blade and that is used especially for mixing, spreading, etc.



Spatulate

SPATULATE (SPATHULATE): Describing structures that have a broad apex and a long narrow base, such as the leaves of the daisy (*Bellis perennis*).

SPAWN: 1. The egg-laying process of fish or a mass of eggs of a fish, amphibian or other water animal. **Spawning** is a type of external fertilisation practised by organisms such as sea urchins, starfish, clams, mussels, frogs, corals and many fish such as trout, salmon and lamprey in which gametes are released or spawned, by the adults into the ocean or a pond. Fertilisation takes place in the water, where embryos start to develop. 2. A mass of microscopic fungal threads mycelium, especially when prepared on a growth medium for starting a new culture of the fungus.

SPAY: The surgical removal of the ovaries of a female animal. In female animals, spaying involves abdominal surgery to remove the ovaries and uterus called **hystero-oophorectomy**. It is also possible to remove only the ovaries in a process called **oophorectomy**, which is mainly done in cats and young dogs. Spaying is performed commonly on household pets such as cats and dogs, as a method of birth control. It is performed less commonly on livestock, as a method of birth control or for other reasons. The surgery can be performed using a traditional open surgery approach or by laparoscopic keyhole surgery.

SPEAKER: 1. In computer science, it is a term used to describe the user who is giving vocal commands to a software program. 2. A hardware device that connects to a computer to generate sound. The signal used to produce the sound that comes from a computer speaker is created by the computer's sound card.

SPEAR: In botany, it is shoot stalk or blade of a green plant such as a grass, asparagus or broccoli.

SPEAR GRASS (FEATHER GRASS; NEEDLE GRASS): Any of various perennial grasses of the genus *Stipa* having feathery clusters of spikelets.

SPEARMAN'S RANK CORRELATION: A statistical method of determining the correlation between two variables or bivariate data. It is the approximation of the Pearson's product moment correlation.

Mathematically, it is given by $\rho = 1 - \frac{6\sum_{i=1}^{n} d_i^2}{n(n^2-1)}$, where n is the number of observations and d is the difference between the ranks of the observations.

SPECIAL CREATION: The theological doctrine which asserts that the origin of the universe and all life in it suddenly sprang into being or that every species was individually created by God in the form in which it exists today. This is based on the historical account of creation in the Book of Genesis that creation of the universe in its present form occurred over the course of six 24-hour days. This is in contrast to Darwin's theory of evolution through natural selection or the complex processes by which the characteristics of living organisms change over many generations as traits are passed from one generation to the next.

SPECIAL FUNCTION: A function, often named after the person who introduced it, that has a particular use in physics or some branch of mathematics. Examples include Bessel functions, Lagrange polynomials, beta functions, gamma functions and hypergeometric functions.

SPECIAL INTEGRAL: In mathematics, a solution of a partial differential equation that cannot be obtained from the general solution by substituting suitable functions for the arbitrary functions.

SPECIAL LINEAR GROUP: In mathematics, the normal subgroup of the general linear group consisting of the set of $n \times n$ matrices whose determinant is equal to 1. It is denoted SL(n, F). In some cases the base field is replaced by the integers.

SPECIAL ORTHOGONAL GROUP: In mathematics, the orthogonal group consisting of all orthogonal matrices having a determinant that is equal to 1. It is denoted SO(n).

SPECIAL REPORT ON EMISSIONS SCENARIOS (SRES): A report by the Intergovernmental Panel on Climate Change (IPCC) that was published in 2000. The greenhouse gas emissions scenarios described in the Report have been used to make projections of possible future climate change.

SPECIAL THEORY OF RELATIVITY, STR (SPECIAL RELATIVITY, SR): The physical theory of space and time proposed in 1905 by Albert Einstein, based on the postulates that all the laws of physics are equally valid in all frames of reference moving at a uniform velocity and that the speed of light from a uniformly moving source is always the same, regardless of how fast or slow the source or its observer is moving. The theory has as consequences the relativistic mass increase of rapidly moving objects, the Lorentz-Fitzgerald contraction, time dilatation and the principle of mass-energy equivalence.

SPECIALISATION (SPECIALIZATION): In biology, it is the adaptation of an organism to a particular environment or activity or functions.

SPECIATION: 1. Development of one or more distinct species from an existing species or the evolutionary formation of new biological species, usually by one species that divides into two or more species that are genetically unique. The four main types of speciation are allopatric, peripatric, parapatric and sympatric. 2. The analytical activity of identifying chemical species and measuring their distribution. For example, separation of an ion or element into the various valence state of the ion or element.

SPECIATION ANALYSIS: Analytical activities of identifying and/or measuring the quantities of one or more individual chemical species in a sample.

SPECIES: 1. A subdivision of biological organisms which share the same general physical characteristics and which can mate and produce fertile offspring. Species ranks below a genus. Differing measures are often used, such as similarity of DNA, morphology or ecological niche. Presence of specific locally adapted traits may further subdivide species into "infraspecific taxa" such as subspecies (and in botany other taxa are used, such as varieties, subvarieties and formae). A species is given a two-part name: the generic name and the specific name or specific epithet or in other words a binomial that consists of the name of a genus followed by a Latin or latinised uncapitalised noun. For example, onion is known as *Allium cepa*. 2. Chemical species is any set of chemically identical molecular entities, the members of which have the same composition and can explore the same set of molecular energy levels within the time scale of a particular experiment; or the various valence state of a given ion or element, for example, As^{+3} and As^{+5} are species of As. 3. In mathematics, it is a classification of a set in a topological space in terms of whether the process of computing the sequence each member of which is the derived set of the preceding, starting with the given set, terminates finitely with the empty set. If this does happen the set is of the **first species**; if It does not happen the set is of the **second species**.

SPECIES DIVERSITY: The number of species in an area and their relative abundance on Earth. The most common index of species diversity is a family of equations called Simpson's Diversity Index. For example, D = $(n/N)^2$, where n is the total number of organisms of a particular species and N is the total number of organisms of all species. D is the value of diversity. It can range between 0 and 1, where 0 is infinite diversity and 1 is the least diverse an ecosystem can possibly be with only one species present. Humans have a huge effect on species diversity through the following activities: (i) destruction, modification and fragmentation of habitat; (ii) introduction of exotic species; and (iii) overharvesting.

SPECIES PACKING: The phenomenon in which present-day communities generally contain more species than earlier communities because organisms have evolved more adaptations over time.

SPECIES RICHNESS: The number of species present in a community.

SPECIES SELECTION: A theory maintaining that species living the longest and generating the greatest number of species determine the direction of major evolutionary trends.

SPECIES-SPECIFIC: Characteristic of and limited to a particular species.

SPECIFIC: 1. Detailed and precise or relating to one particular thing, not general. 2. Relating to a biological species. 3. A term which describes a disease caused by a particular infectious agent. 4. A term used to indicate that a physical property is being expressed with reference to a particular quantity such as mass, volume or length.

SPECIFIC ACID-BASE CATALYSIS: Catalysis by acids or bases in solution is said to be specific when the only observable catalytic effects are those due to the ions formed from the solvent itself. For example, when water is the solvent the only observable catalysis is that due to the H^+ and OH^- ions.

SPECIFIC ACTIVITY: 1. Number of disintegrations of a radio-isotope per unit time per unit mass. Specific activity can be expressed in such units as millicuries per gram, disintegrations per second per milligram or counts per minute per milligram. Its symbol is α. 2. In biology, also known as **specific enzyme activity**, it is the amount of enzyme reaction with a substrate. It is expressed as micromoles formed per unit time per milligram of protein. It is a measure of enzyme purity and the larger the value, the purer the enzyme preparation becomes since the amount of protein (mg) is typically less.

SPECIFIC ADSORPTION: Ions become specifically adsorbed when short-range interactions between them and the interphase become important. They are believed then to penetrate into the inner layer and may (but not necessarily) come into contact with the surface. They are usually assumed to form a partial or complete monolayer.

SPECIFIC BURN-UP: The total energy released through induced nuclear transformations divided by the mass of a nuclear fuel.

SPECIFIC CATALYSIS: The acceleration of a reaction by a unique catalyst, rather than by a family of related substances. The term is most commonly used in connection with specific hydrogen-ion or hydroxide-ion (lyonium ion or lyate ion) catalysis.

SPECIFIC CHARGE: The ratio of the charge of an elementary particle or other charged body to its mass expressed as e/m. Specific charge can be determined by measuring the velocity which the particle acquires in falling through an electric potential by measuring the frequency of revolution in a magnetic field or by observing the orbit of the particles in combined electric and magnetic fields.

SPECIFIC DETECTOR: A detector which responds to a single sample component or to a limited number of components having similar chemical characteristics.

SPECIFIC GRAVITY (RELATIVE DENSITY): The ratio of the density of a substance to the density of a standard substance at the same temperature and pressure. Its symbol is ρ. For liquids and solids the standard substance is usually water; for gases it is air. Also, for liquids and solids it is the ratio of the density, usually at 20 degrees Celsius to the density of water at its maximum density. Gold has a density of 19.3 g/cm^3, so its specific gravity is 19.3 g/cm^3 divided by 1 g/cm^3 (the density of water at 4°C) or 19.3.

SPECIFIC HEAT CAPACITY: The quantity of heat required to raise a unit mass (one kilogram) of a substance by one Kelvin. Its unit is joule per kilogram per Kelvin (Jkg^{-1}k^{-1}).

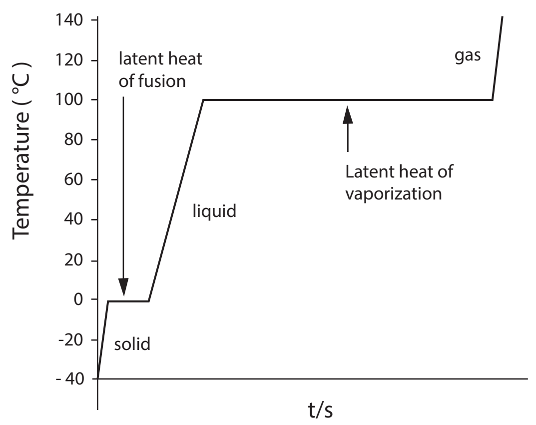
SPECIFIC HUMIDITY: In a system of moist air, the ratio of the mass of water vapour to the total mass of the system. Specific humidity ratio is expressed as a ratio of kilograms of water vapour, m_w per kilogram of total moist air m_t.

SPECIFIC IMPULSE: A measure of the thrust available from a rocket propellant. It is the ratio of the thrust produced to the fuel consumption. The higher the specific impulse, the less propellant is needed to gain a given amount of momentum. Its symbol is I_{sp}. It is a widely used measure of performance for chemical, nuclear and electric rockets. It is usually given in seconds in International System (SI) units. The impulse produced by a rocket is the thrust force F times its duration, t in seconds. The specific impulse is given by the equation: $I_{sp} = \frac{Ft}{m_p}$, where m_p is the propellant mass flow rate. Its equivalent, specific thrust F_{sp} that is sometimes used alternatively, is the rocket thrust divided by the propellant mass flow rate F/m_p.

SPECIFIC INTENSITY: 1. In photometry, a measure of the amount of radiant energy received per unit solid angle, per unit time, per unit area of a surface element orthogonal to the direction of propagation of the radiation. 2. In visual perception, one of the three basic parameters along with hue and saturation that may be used to describe the physical perception of colour. 3. In radio astronomy, it is a term often used much more loosely to mean either the flux density of an unresolved radio source or the surface brightness of an extended source.

SPECIFIC IONISATION: The number of ion pairs formed per unit distance along the track of an ionising particle passing through matter.

SPECIFIC LATENT HEAT: The amount of heat needed to change a unit mass of a substance from a solid to a liquid or from a liquid to a gas, without changing its temperature. The equation relating energy to specific latent heat is given as: Energy (J) = mass (kg) × specific latent heat (J/kg).



Temperature-time graph showing a heating curve of water heated from -40°c to 140°c

SPECIFIC LATENT HEAT OF FUSION: The heat required to change one kilogram of a solid substance from solid to liquid without any temperature change. Heat of fusion is the latent heat associated with melting a solid or freezing a liquid.

SPECIFIC LATENT HEAT OF VAPORISATION/ VAPORIZATION: The heat required to change one kilogram of a liquid from liquid to vapour or vice versa without any temperature change. For example, when water reaches its boiling point and is kept boiling, it remains at that temperature until it has all evaporated; all the heat added to the water is absorbed as latent heat of vaporisation and is carried away by the escaping vapour molecules.

SPECIFIC LEAF AREA (SLA): The ratio of the leaf area of a plant to the dry mass of the leaf, typically measured in square metres per kilogram. It can be used to estimate the reproductive strategy of a particular plant based upon light and moisture (humidity) levels, among other factors. A high value of SLA is typical of fast-growing plants with high photosynthetic capability, whereas slow-growing or drought-resistant plant would typically have a low SLA.

SPECIFIC RATE CONSTANT: A constant which is determined by experiment and varies for different reactions and which changes only with temperature, k in the rate-law expression: Rate = $k[A]$ x $[B]v$.

SPECIFIC RESISTANCE (RESISTIVITY): The electrical resistance offered by a homogeneous unit cube of material to the flow of a direct current of uniform density between opposite faces of the cube. It is usually determined by calculation from the measurement of electrical resistance of samples having a known length and uniform cross section according to the following equation: $\rho l/A$, where ρ is the resistivity, R the measured resistance, A the cross-sectional area and l the length. In SI, the unit of resistivity is the ohm-metre. The room-temperature resistivity of pure metals extends from approximately 1.5 × 10^{-8} ohm-metre for silver, the best conductor, to 135 × 10^{-8} ohm-metre for manganese, the poorest pure metallic conductor. Most metallic alloys also fall within the same range. Insulators have resistivities within the approximate range of 10^{8} to 10^{16} ohm-metres.

SPECIFIC SURFACE: The surface area of a particular substance per unit mass expressed in metre squares per kilogram (m^2/kg). It provides a measure of the surface area available for a process, such as adsorption, for a given mass of a powder or porous substance. It can also be defined by surface area divided by the volume (units of m^2/m^3 or m^{-1}).

SPECIFIC VOLUME: The volume of a substance per unit mass of a material or the volume occupied by a unit of mass of a material. It is equal to the reciprocal of mass density. Its unit is m^3kg^{-1}.

SPECIFIC WEIGHT: The weight per unit volume or the ratio between the mass of an object and its volume. The higher the density, the tighter the particles are packed inside the substance.

SPECIFICALLY ABSORBING ION: Ions which possess a chemical affinity for the surface in addition to the Coulomb interaction, where chemical is a collective adjective, embracing all interactions other than purely Coulombic. Examples are van der Waals or hydrophobic bonding, pi-electron exchange and complex formation. Specifically adsorbing ions can adsorb on an initially uncharged surface and hence provide it with a charge. The term **specifically adsorbed** applies to the sorption of all other ions having an affinity to the surface in addition to the purely Coulombic contribution.

SPECIFICALLY LABELLED: An isotopically labelled compound is designated as specifically labelled when a unique isotopically substituted compound is formally added to the analogous isotopically unmodified compound. In such a case, both position(s) and number of each labelling nuclide are defined.

SPECIFICALLY LABELLED TRACER: A tracer in which the label is present in a specified position.

SPECIFICITY: 1. The characteristic of having a specific range or use. 2. The selective reactivity that occurs between substances, such as between an antigen and its corresponding antibody.

SPECIMEN: A specifically selected portion of a material taken from a dynamic system and assumed to be representative of the parent material at the time it is taken. It may be considered as a special type of sample, taken primarily in time rather than in space. Although the specimen may not be reproducible in time, for example, it may be taken from a flowing stream or a portion of blood, no separable sampling error exists since this error is unavoidably included with the corresponding error of the estimate of the property, function or analyte being studied.

SPECKLE INTERFEROMETRY: A method of improving the angular resolution of telescopes that are impaired by atmospheric turbulences so that spatial information on astronomical objects at the diffraction-limited resolution of a telescope can be obtained. The instrument used to do this is called **speckle interferometer**.

SPECKLE PATTERN: A random granular pattern that can be observed, when light from a highly coherent source, such as a laser is scattered by a rough surface (for example, a piece of paper, white paint, a display screen and a metallic surface) or an inhomogeneous medium.

SPECKLED YELLOWING: A disease of sugar beet, caused by the deficiency of manganese.

SPECTROSCOPY: 1. The use of absorption, emission or scattering of electromagnetic radiation by matter to perform an analysis to obtain spectrum, which can be used in chemical analysis and in the studies of the properties of atoms, molecules, ions, etc. Examples include mass spectroscopy, which uses spectroscopy to determine the masses of small electrically charged particles and microwave spectroscopy, which uses spectroscopy to study atomic or molecular resonances in the microwave spectrum. Some of the instruments used include a laser or an ion or radiation source and a device such as a spectrophotometer or interferometer to measure the spectrum of light emitted or absorbed by a given material. 2. The scientific field of study that deals with the use of the spectroscope and with spectrum analysis.

SPECTACLES: A pair of lenses fitted in a frame and worn in front of the eyes to correct or assist defective vision. Most lenses are made of glass or plastic. Sunglass lenses are tinted to reduce glare and often treated to reduce ultraviolet light exposure.

SPECTATOR IONS: Ions which exist in a chemical reaction process but do not take part in the reaction. An example is $2Na^+_{(aq)} + 2Cl^-_{(aq)} + Cu^{2+}_{(aq)} + SO_4{}^{2-}_{(aq)} \rightarrow 2Na^+_{(aq)} + SO_4{}^{2-}_{(aq)} + CuCl_{2\,(s)}$. The $Na^+_{(aq)}$ and $SO_4{}^{2-}_{(aq)}$ ions are spectator, since they remain unchanged on both sides of the equation. They simply "observe" the other ions react, hence the name.

SPECTATOR-STRIPPING REACTION: An extreme type of stripping reaction in which one reaction product has almost the same direction and momentum as one of the reactant molecules had before the reactive collision occurred. **Stripping reaction** is a chemical process, studied in a molecular beam, in which the reaction products are scattered forward with respect to the moving centre of mass of the system.

SPECTRAL CLASS: A form of classification used for stars based on their spectra and luminosity. The Harvard classification is based on the seven star types known as O, B, A, F, G, K, M. The implications of these are as follows: (i) O stars are very hot and extremely luminous, blue in colour and are ionised helium lines dominant; (ii) Class B stars are very luminous and blue and are neutral helium lines dominant, no ionised helium; (iii) Class A stars are blue-white and are hydrogen lines dominant; (iv) Class F are white stars and are metallic lines strengthen, hydrogen lines weaken; (v) Class G are yellow stars and are ionised calcium lines dominant; (vi) Class K are orange red stars and are neutral metallic lines dominant; (vii) Class M are coolest red stars and are molecular bands dominant. The hottest stars (O and B) are blue-white in colour, while the coolest (M) is red. Each of the letter classes has subdivisions indicated by numerals 0 through 9. Thus, a B0 is the hottest B-type star, B5 is halfway between types B and A and B9 is only slightly hotter than type.

SPECTRAL DECOMPOSITION: In mathematics, it is the expression of a normal matrix A as UDU*, where U is unitary and D is diagonal; U may be taken to be real if A is real and symmetric.

SPECTRAL DISTRIBUTION: The variation of the spectral radiance with wavelength.

SPECTRAL FORM: In mathematics, the representation $s = \sum_{i=1}^{n} \lambda_i u_i \otimes u_i$ of a symmetric second order Cartesian tensor a over an n-dimensional space, where λ_i, are the eigenvalues and U_i the eigenvectors of s.

SPECTRAL INTERFERENCE: Interference due to the incomplete isolation of the radiation emitted or absorbed by the analyte from other radiation detected by the instrument. Its occurrence may be established by comparing the measures of the analyte-free blank solution and the solvent blank.

SPECTRAL LINE: Any of a number of lines corresponding to definite wavelengths of an atomic emission or absorption spectrum; representing the energy difference between two energy levels. It is a dark or bright line in an otherwise uniform and continuous spectrum, resulting from an excess or deficiency of photons in a narrow frequency range, compared with the nearby frequencies.

SPECTRAL QUANTITIES: Quantities characterising electromagnetic radiation (radiant power, energy, energy density, intensity, exitance, radiance, irradiance, etc.) derived by differentiation with respect to wavelength, frequency or wavenumber. For example, spectral (concentration of) irradiance is the derivative of the irradiance with respect to wavelength, frequency or wavenumber.

SPECTRAL RADIANCE: The radiance, L, at wavelength λ per unit wavelength interval. The SI unit is $Wm^{-3}\,sr^{-1}$, but a commonly used unit is $Wm^{-3}\,sr^{-1}\,nm^{-1}$.

SPECTRAL RADIANT ENERGY: Derivative of radiant energy, Q, with respect to wavelength λ. SI unit is $J\,m^{-1}$; a common unit is $J\,nm^{-1}$.

SPECTRAL RADIANT EXITANCE: The radiant exitance, M, at wavelength per unit wavelength interval. The SI unit is Wm^{-3}, but a commonly used unit is $Wm^{-3}\,nm^{-1}$.

SPECTRAL RADIANT INTENSITY: The radiation intensity, I at wavelength λ per unit wavelength interval. The SI unit is $Wm^{-1}\,sr^{-1}$, but a commonly used unit is $Wnm^{-1}\,sr^{-1}$.

SPECTRAL RADIANT POWER: The radiant power at wavelength λ per unit wavelength interval. The SI unit is $W\,m^{-1}$, but a commonly used unit is $W\,nm^{-1}$.

SPECTRAL RADIUS: In mathematics, the supremum among the absolute values of members of the spectrum of a given square matrix or a bounded linear operator.

SPECTRAL SENSITISATION/SENSITIZATION: The process of increasing the spectral responsivity of a (photoimaging) system in a given wavelength region.

SPECTRAL SERIES: The set of spectral lines of a given atom produced by electron jumps from different initial energy levels to the same final one; for example, the Balmer series of hydrogen.

SPECTRAL THEOREM: In mathematics, the theorem that provides conditions under which an operator or a matrix can be represented as a diagonal matrix in some basis. It states that a bounded linear operator can be reconstructed in a canonical fashion as a spectral integral with respect to a family of commuting self-adjoint projections defined on the spectrum of the given operator. For normal matrices or for compact normal operators, this becomes $T = \sum \lambda_i P_i$, where P_i is the orthogonal projection on the null space of $\lambda_i I - T$.

SPECTRIN: A cytoskeletal protein that forms part of the fibrous network underlying the intracellular side of the plasma membrane in eukaryotic cells. Together with actin and other cytoskeleton proteins, they form a network that gives the red blood cell membrane its shape and flexibility. In certain types of brain injury such as diffuse axonal injury, spectrin is irreversibly cleaved by the proteolytic enzyme calpain, destroying the cytosketelon.

SPECTROCHEMICAL BUFFER: Buffer added to samples and reference samples with the intention of making the measure of the analytical element less sensitive to changes in concentration of an interferent.

SPECTROCHEMICAL CARRIER: An additive which gives rise to a gas which can help to transport the vapour of the sample material

S

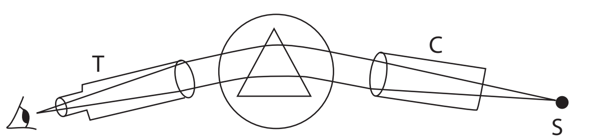
into the excitation region of the source, for example, carbon in an air atmosphere when carbon dioxide is formed.

SPECTROCHEMICAL SERIES: A list of ligands ordered on ligand strength and how they cause splitting of the energy levels of d-orbitals in metal complexes.

SPECTROCHEMISTRY: A branch of chemistry concerned with the study of the spectra of substances using spectroscopy. It includes the analysis of spectra in chemical terms and the use of spectra to derive the structure of chemical compounds and to qualitatively and quantitatively analyse their presence in samples.

SPECTROHELIOGRAM: A photograph of the Sun's chromospheres taken at a particular wavelength of light with the aid of a high dispersion spectroscope.

SPECTROMETER (SPECTROPHOTOMETER; SPECTROGRAPH; SPECTROSCOPE): An instrument used to study the composition of light emitted by a source over a specific portion of the electromagnetic spectrum. The variable measured is most often the light's intensity but could also, for instance, be the polarisation state. The independent variable is usually the wavelength of the light or a unit directly proportional to the photon energy, such as wave number or electron volts, which has a reciprocal relationship to wavelength. A spectrometer is used in spectroscopy for producing spectral lines and measuring their wavelengths and intensities. They operate over a very wide range of wavelengths, from gamma rays and x-rays into the far infrared. If the region of interest is restricted to the near visible spectrum, the study is called **spectrophotometry**. In the diagram, light from the source S is collimated by C and dispersed by the prism. The telescope T is used to observe this light. The light can be rotated around the prism table and the angular deviation measured.



Simple spectrometer

S

SPECTROMETRY: Measurement of the intensity of spectral lines or spectral series as a function of wavelength or energy, using a spectrometer or spectroscope to derive the physical constants of materials.

SPECTROPHOTOMETER: A combination of spectrograph and photometer used to analyse a spectrum that has been recorded on a photographic plate. By passing a narrow beam of light through the recorded image the original intensity of light from the source at various wavelengths can be recovered. This information can then be displayed as a graph or stored as a digitised spectrum for further processing.

SPECTROPHOTOMETRY: The measurement of the intensity of different colours (wavelengths) present in a beam of light, normally using a prism or diffraction grating and a photoelectric cell. It is used for colour comparison and in chemical analysis.

SPECTROSCOPE: An optical instrument for splitting various wavelengths of electromagnetic radiation into a spectrum, using a prism or diffraction grating. The instrument produces a spectrum for visual observation and analysis.

SPECTROSCOPIC BINARY: A binary star system in which the two components are so close together or so far from the Sun that they cannot be resolved simply by looking at them, even through a powerful telescope. Their binary nature can, however, be established because of the Doppler shift of their spectral lines.

SPECTROSCOPIC ELECTRON CONFIGURATION NOTATION: A type of notation used to show an atom's electron configuration that describes the energy levels, orbitals and the number of electrons in each. For example, the electron configuration of lithium is $1s^2 2s^1$. The numbers and letters describe the energy level and orbital and the superscript numbers show how many electrons are in that orbital.

SPECTROSCOPY: An analytic technique concerned with the measurement of the interaction, usually the absorption or the emission of radiant energy with matter, with the instruments necessary to make such measurements and with the interpretation of the interaction both at the fundamental level and for practical analysis. Spectroscopy is employed in a wide range of research areas such as chemistry, biochemistry and physics for the identification of metabolites and other compounds of biological significance.

SPECTRUM: 1. A range of related things listed in a set of order. 2. In mathematics, the set of complex numbers, denoted σ(T), for which the resolvent of a matrix or (bounded) linear operator on a normed space fails to exist as a bounded linear operator by virtue either of lying in the point spectrum where $\lambda_i I - T$ is not one-to-one or of lying in the continuous spectrum where $\lambda_i I - T$ is one-to-one with proper dense range or of lying in the residual spectrum that is, the remainder of the spectrum, which is the set of numbers for which the range of $\lambda_i I - T$ is not dense. For a matrix, only the point spectrum occurs and is the set of eigenvalues. For a bounded linear operator in Banach space, the spectrum is bounded and non-empty, while the residual spectrum is empty if T is normal. 3. In physics, it is an arrangement of frequencies or wavelengths when electromagnetic radiations are separated into their constituent parts. It is a plot of the intensity of light as a function of frequencies; the distribution of wavelengths and frequencies. The band of colours, as seen in a rainbow is the portion of the electromagnetic spectrum that is visible as light to the human eye, produced by separation of the components of light by their different degrees of refraction according to wavelength. Some sources emit only certain wavelengths and produce an emission spectrum of bright lines with dark spaces between. Such line spectra are characteristic of the elements that emit the radiation.

SPECTRUM ANALYSIS: The interpretation of the information present in an energy spectrum in terms of radiation energy and intensity.

SPECULUM: 1. An alloy of copper and tin formerly used in reflecting telescopes to make the main mirror cast, ground and polished to make a highly reflective surface. It is now replaced by silvered glass for the same purpose. 2. A metal or plastic instrument used to dilate an orifice or canal in the body to allow for medical examination. 3. A bright patch of plumage on the wings of certain birds such as ducks, especially a strip of metallic sheen on the secondary flight feathers; a transparent spot in the wings of some butterflies or moths. 4. The iridescent blue patch occurring on some Ophyrs (a genus of orchids). 5. A mirror-like surface on an organ.

SPEECH RECOGNITION: In computer science, any computer system that enables data to be input by voice to be converted into a machine-readable format. It includes a microphone sound card that plugs into the computer and converts analogue voice to digital signals. It is used in applications such as call routing, speech-to-text, voice dialling and voice search. In contrast, **voice recognition** is the ability of a computer system to recognise the unique voice characteristics of a particular person, just as a fingerprint as used in security systems.

SPEED: The ratio of the distance covered by a body to the time taken. Its unit is the metre per second, m/s. Speed is a scalar quantity as its direction does not matter. **Average speed** is the total distance travelled divided by the time in which it has taken to do it. It is a scalar quantity. **Instantaneous speed** is the speed of an object at a particular moment in time. It is a scalar quantity.

SPEED OF LIGHT: The speed at which electromagnetic radiation travels. The speed of light in a vacuum is 299,792,458 metres per second (299 792 458 ms^{-1}). This is the highest speed attainable in the universe. It is thought to be a universal constant and is independent of the speed of the observer. It has the symbol c. In the theory of relativity, c connects space and time and appears in the famous equation of mass-energy equivalence, $E = mc^2$.

SPEED OF SOUND: The speed at which sound waves are propagated through an elastic medium. Its symbol is c or c_s. In air at 20 °C, sound travels at 343 metres per second (344 ms^{-1}) or 1,236 kilometres per hour, referred to as Mach 1 by aerospace engineers. In water at 20 °C it travels at 1,484 m/s (1, 461 ms^{-1}) or 5,342.4 kilometres per hour. In steel at 20 °C it travels at 5,120 m/s (5, 120 ms^{-1}) or 1,800 kilometres per hour. The speed of sound in a medium depends on the medium's modulus of elasticity (E) and its density (ρ) according to the relationship $\sqrt{(\frac{E}{\rho})}$.

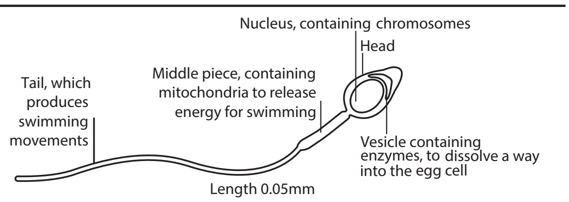
SPEEDOMETER: An instrument showing the speed of a motor vehicle consisting of cable attached to the rear of the transmission of an automobile; the cable turns at a rate proportional to the speed of the car.

SPEEDWELL: A widespread weed of the genus *Veronica* found in cereal crops and oilseed rape having opposite leaves and clusters of small, usually blue flowers. It is hazardous to row crops because of its rapid spread.

SPEMANN'S ORGANISER (SPEMANN'S ORGANISER): A cluster of cells in the dorsal mesoderm that determines the development of the spinal cord and brain in the early amphibian embryo. They form near the lip of the blastopore following gastrulation and releases spinal proteins that diffuse to overlying ectoderm.

SPERM (SPERMATOZOON): The mature mobile reproductive cell (gamete) of male animals which is produced by the testes. Most sperms are shaped like a tadpole and move with the help of a long tail. Sperms fuse with ova during fertilisation. Sperm cells contribute half of the genetic information to the diploid offspring. In mammals, the sex of the offspring is determined by the sperm cell: a spermatozoon bearing a Y chromosome will lead to a male (XY) offspring; while one bearing an X chromosome will lead to a female (XX) offspring. The ovum always provides an X chromosome.

SPERM CELL: One of the two male gametes formed in the pollen tube of gymnosperms after mitotic division of the body cell. In some species, for example, Cycas and Ginkgo, they are flagellate and swim from the pollen tube to the archegonia. Such antherozoids, in contrast to the nonmotile generative nuclei of angiosperms, are associated with a large amount of cytoplasm and in Cycas reach a diameter of 300 μη. In other gymnosperms, for example, the Gnetales and Coniferales, the sperm cells are not motile.



Sperm cell

SPERM COMPETITION: An intrasexual or male to male competition that occurs after the initiation of copulation, when females indulge in multiple mating over a relatively short period of time in which sperms from two or more males compete to reach and fertilise the egg cell of a single female. This may take the form of anatomical, physiological and psychological adaptations. If females mate in a way that concurrently places sperm from two or more males in her reproductive tract, this generates several selection pressures on males. For example, in rodents a male may mate a number of times with the same female with a rest period between successive mating during which the sperm journeys toward the egg. If a second male mates with the female during the rest period his sperm may disrupt the movement of the sperm of the first male, succeeding in fertilising the egg. In human females, prostitution, communal sex such as wife-swapping, short-term matings, rape and female infidelity can place the sperm of different men into competition. Many male animals guard their mates after copulation in order to prevent other males from mating with her.

SPERM DUCT (VAS DEFERENS): A partially coiled tube or duct in the male reproductive system that transports matured sperm from the epididymis (a tube attached to the testis where sperm is stored) to the urethra that carries semen and urine outside the body. There are two ducts, connecting the left and right epididymis. Each vas ends by joining the duct of a seminal vesicle, to enter the urethra (the urinary passage), where that traverses the prostate gland. It is a thick-walled tube, about 0.61 metres long that begins at the tail of the epididymis and eventually joins the duct of the seminal vesicle to form the ejaculatory duct. When each of the two vas deferentia is sealed to prevent sperm from mixing with the semen produced by the male during ejaculation to prevent conception, it is called **vasectomy**.

SPERMATHECA (SEMINAL RECEPTOR): A sac or receptacle in some female or hermaphrodite animals such as insects, some molluscs, oligochaeta worms and certain other invertebrates and vertebrates in which sperm from the mate is stored until the eggs are ready to be fertilised.

SPERMATICIDE (SPERMATOCIDE; SPERMICIDE): A contraceptive chemical substance that prevents pregnancy by deactivating and killing sperm and blocking the cervix to prevent sperm from encountering an egg. It is often in the form of a cream or a gel, used in conjunction with a birth control device such as diaphragms, condoms, cervical caps and sponges.

SPERMATID: A young or unripe spermatozoon formed after the division of the secondary spermatocyte and contains haploid number of chromosomes and cannot cause fertilisation unless it matures into a spermatozoon. Four spermatids are formed after two meiotic divisions of primary spermatocytes and, therefore contain the haploid number of chromosomes.

SPERMATIUM (SPERMATIA): A type of non motile male sex cell that is produced by the red algae and by certain fungi.

SPERMATOCYTE: A diploid cell in the testis that divides by meiosis to give rise to four spermatids. A primary spermatocyte develops from a spermatogonium. Two secondary spermatocytes are formed from the first meiotic division of a primary spermatocyte and each of these produces two spermatids after the second meiotic division.

SPERMATOGENESIS (SPERM PRODUCTION): The series of cell divisions that result in the development, formation or production of spermatozoa in the somniferous tubules of testes. A single spermatogonium divides by mitosis to form primary spermatocytes, each of which undergoes the initial division of meiosis to form two secondary spermatocytes. Each of these then undergoes the second meiotic division to form two spermatids, which mature into spermatozoa without further cell division.

SPERMATOGONIUM (SPERMATOGONIA): Any of the diploid cells in the walls of the seminiferous tubules of the testis that gives rise to the primary spermatocytes. There are three subtypes: (i) Type A(d) cells, with dark nuclei, which replicate to ensure a constant supply of spermatogonia to fuel spermatogenesis; (ii) Type A(p) cells, with pale nuclei, which divide by mitosis to produce Type B cells; (iii) Type B cells, which divide to give rise to primary spermatocytes. Each primary spermatocyte duplicates its DNA and subsequently undergoes meiosis I to produce two haploid secondary spermatocytes. Each of the two secondary spermatocytes further undergoes meiosis II to produce two spermatids (haploid).

SPERMATOPHORE: A mass that contains spermatozoa and is created by males in various species of animal, for example, newts and cephalopods. During copulation the male transfers the spermatophore to the female's ovipore, which the female uses to fertilise her eggs.

SPERMATOPHYTA (PHANEROGAMS; SEED PLANTS): A division of the plant kingdom containing plants that reproduce by means of seed in traditional classification. They are a subset of the embryophytes or land plants. The living spermatophytes form five groups: (i) **cycads**, a subtropical and tropical group of plants with a large crown of compound leaves and a stout trunk; (ii) *Ginkgo*, a single living species of tree; (iii) **conifers**, cone-bearing trees and shrubs; (iv) **gnetophytes**, woody plants in the genera *Gnetum*, *Welwitschia* and *Ephedra*; and (v) **angiosperms**, the flowering plants, a large group including many familiar plants in a wide variety of habitats.

SPERMATOZOID (ANTHEROZOID): A motile male gamete produced by lower plants and some gymnosperms, which moves by means of flagella. In lower plants antherozoids are released from an antheridium but in the gymnosperms they are formed in the pollen tube prior to fertilisation. Most antherozoids consist of an elongated nucleus contained within a ribbon-like cell. This form enables them to penetrate the narrow neck of an archegonium. The number of flagella borne by an antherozoid may be used as a diagnostic character.

SPERMICIDE (SPERMATOCIDE; SPERMATICIDE): A contraceptive chemical substance that prevents pregnancy by deactivating and killing sperm and blocking the cervix to prevent sperm from encountering an egg. It is often in the form of a cream or a gel, used in conjunction with a birth control device such as diaphragms, condoms, cervical caps and sponges.

SPERMOGONIUM (PYCNIUM; PYCNIDIUM): 1. A flask-like structure produced by certain fungi, for example, Puccinia and Leptosphaeria in which asexual pycnospores (spermatid) are formed. 2. A spore-producing body formed by the fungal component of certain lichens, for example, *Parmelia physodes*. It is a flask-shaped structure sunken in the thallus and opens to the surface by a pore. Inside the pycnidium are a number of conidiophores, which produce pycnoconidia. The functions of the pycnoconidia are uncertain though it appears that in some species they may, by behaving as male gametes, initiate ascocarp formation.

SPERMOSPHERE: 1. The area around a germinating seed, where there is increased microbiological activity. 2. A mass or ball of cells formed by the repeated division of a male germinal cell (spermospore), each constituent cell (spermoblast) of which is converted into a spermatozoid; a spermatogemma.

SPHAERIALES: An order of the Pyrenomycetes mainly containing saprobic fungi, with some parasitic on plants. It contains over 5000 species in some 500 genera. They typically have hard dark perithecia, as seen, for example, in Daldinia concentrica.

SPHAEROPSIDALES: An order of the Coelomycetes containing many fungi that are plant parasites, for example, *Septoria apiicola*, which causes leaf spot of celery. It includes about 6000 species in some 750 genera. The conidiophores are formed within pycnidia.

SPHAERORAPHIDE (DRUSE): A globular mass of needle-like crystals found either attached to the cell wall or free in the cytoplasm.

SPHAGNUM: A genus of between 151 and 350 species of mosses commonly called **peat moss**, due to its prevalence in peat bogs and mires. Bogs are dependent on precipitation as their main source of nutrients, thus making them a favourable habitat for *Sphagnum* as it can retain water and air quite well. Members of this genus can hold large quantities of water inside their cells. Some species can hold up to 20 times their dry weight in water, which is why peat moss is commonly sold as a soil conditioner. The empty cells help retain water in drier conditions. In wetter conditions, the spaces contain air and help the moss float for photosynthetic purposes. *Sphagnum* and the peat formed from it do not decay readily because of the phenolic compounds embedded in the moss's cell walls. An additional reason is that the bogs in which *Sphagnum* grows are submerged, deoxygenated and favour slower anaerobic decay rather than aerobic microbial action. Peat moss can also acidify its surroundings by taking up cations such as calcium and magnesium and releasing hydrogen ions.

SPHALERITE (ZINC BLENDE): A yellow-brown to brownish-black form of zinc sulfide (ZnS), which is the principal ore of zinc. It crystallises in the cubic system; and often occurs with galena in metasomatic deposits and also in hydrothermal veins and replacement deposits. It is usually yellow-brown to brownish-black in colour

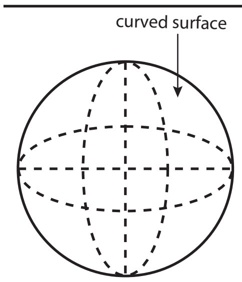
SPHALERITE STRUCTURE (ZINC-BLENDE STRUCTURE): A type of ionic crystal structure with each metal ion surrounded by six oppositely charged ions arranged tetrahedrally. Examples of compounds with the sphallerite structure are zinc sulfide (ZnS), copper(I)chloride (CuCl), cadmium sulfide (CdS) and indium arsenide (InAs).

SPHENOCENTRUM JOLLYANUM (SPHENOCENTRUM JOLLYANUM PIERRE): A small erect evergreen, dioecious shrub of the family Menispermaceae, which is sparingly branched and grows up to 1.5 m tall. It is mostly found in the tropical forest zones of West Africa from Côte d'Ivoire to Cameroon as an undergrowth plant. It has glabrous branchlets and bright yellow roots which have a sour taste. The leaves are very variable in shape on the same plant. The blade is obviate-elliptic and entire or shaped like an oak leaf, pinnately lobed, acuminate at the apex, cuneate at the base, with six pairs of looped lateral nerves, the basal pair along the leaf margin. The flowers are yellowish-white and are solitary on older branches or formed in axillary unstalked bunches on the stem. The fruits are ovoid-ellipsoid and are formed in small clusters on the stem just below the leaves. *Sphenocentrum jollyanum* is traditionally used widely for medicinal purposes for treating various human diseases such as swellings, oedema, gout, food poisoning, paralysis, epilepsy, convulsions and spasm.

SPHENOID BONE: An irregular, wedge-shaped bone at the base of the skull. The sphenoid bone is notable in that it makes contact with all of the other cranial bones. It spans the skull laterally and helps form the base of the cranium, the sides of the skull and the floors and sides of the orbits (eye sockets).

SPHENOPHYTA (ARTHROPHYTA): A phylum of tracheophyte or vascular plants with only one living genus (Equisetum), the horsetails. Its members reproduce by spores rather than seeds.

SPHERE: 1. A perfectly round geometrical object in three-dimensional space such as the shape of a round ball. It is completely symmetrical around its centre with all points on the surface lying the same distance r, from the centre point. The distance r is the radius. The surface area of a sphere is given by the formula $A = 4\pi r^2$ and the volume inside the sphere is given by $V = \frac{4}{3\pi r^2}$, where A is the surface area, V the volume and r the radius. In mathematics, a careful distinction is made between the sphere such as a two-dimensional spherical surface embedded in three-dimensional Euclidean space and the ball, which is the three-dimensional shape consisting of a sphere and its interior. 2. In a metric space, it is the set of points equidistant in metric from a given point. 3. In algebraic topology, the bicontinuous image of the unit sphere in Euclidean n-space, written s^{-1}. 4. In physics, a sphere is an object capable of colliding or stacking with other objects which occupy space. 5. An astronomical object such as a planet, moon or star. 6. **Spherical** is an object shaped like a sphere or relating to a sphere or astronomical objects.



Curved surface of a sphere

SPHERE-PACKING PROBLEM: In mathematics, any of the class of problems, involving the most efficient way to pack equal-sized spheres together, in a region of n-space of in such a way as to optimise the volume of the spheres. In three dimensions, there are three periodic packings for identical spheres, which are cubic lattice, cubic close packing (face-centred cubic) lattice and hexagonal lattice. In 1611, Johannes Kepler (1571-1630) hypothesised that the cubic close packing (face-centred cubic) and hexagonal close packing arrangements with density of around 74.04% have greater average density. In 2014, Thomas Callister Hales (1958-), an American mathematician proved Kepler's conjecture.

SPHERICAL ABERRATION: A blurring of an image occurring in an optical device such as a lens and a mirror that is due to the increased refraction of light rays when they strike a lens or a reflection of light rays when they strike a mirror near its edge, in comparison with those that strike nearer the centre. It signifies a deviation of the device from the norm, which results in an imperfection of the produced image.

SPHERICAL ANGLE: The angle formed at the intersection of the arcs of two great circles of a sphere. It is equal to the angle between their tangents at that point.

SPHERICAL CAP: A portion of a sphere cut off by a plane and lying on either side of the. If the height of the cap is equal to the radius of the sphere, the spherical cap is called a **hemisphere**.

SPHERICAL COORDINATES: A system for specifying positions in terms of angles on a sphere, such as the celestial sphere or the surface of a planet or large moon. In spherical coordinates, a point P in space is located by its position relative to three mutually perpendicular axes (the x-, y- and z-axes) in terms of (i) its distance r from the origin, O, (ii) the angle (θ) between the x-axis and the projection of OP onto the x-y plane and (iii) the angle (φ) between OP and the z-axis. The coordinates of P are thus expressed in the form (r, θ, φ).

SPHERICAL EXCESS: In mathematics, it is the amount by which the angles of a spherical triangle exceeds two right angles or the amount by which the sum of the angles of a spherical polygon exceeds $\pi(n - 2)$ radians, where n is the number of sides of the polygon. This difference is related to the area of the figure by the formula $A = \pi r^2 E$, where E is the ratio of spherical excess to two right angles and r is the radius of the sphere.

SPHERICAL GEOMETRY: 1. The branch of geometry that deals with the properties of figures formed on the surface of a sphere, especially by the intersection of great circles. 2. A non-Euclidean geometry that is a Riemannian geometry and has a model on the surface of a sphere.

SPHERICAL HARMONICS: The natural vibrations of an elastic sphere, which are used to describe the normal modes of oscillation of spherically shaped objects. They are the angular portion of the solution to Laplace's equation in spherical coordinates where azimuthal symmetry is not present. They have been used mathematically for models such as the surface of the Sun and the atomic nucleus. Each harmonic has two unique identifying indices. On the surface of a vibrating sphere, certain nodal circles appear, where the surface is at rest. The number of these nodal circles for a given spherical harmonic is called its order n, where n is one of the indices used to describe the normal mode. The second identification index corresponds to the number m of nodal circles which pass through the poles of the vibrating sphere. Consequently all nodal circles are lines of definite latitude or longitude.

SPHERICAL MIRROR: A curved mirror formed by a part of a hollow glass sphere with a reflecting surface. There are two types of spherical mirrors: concave and convex. Spherical mirrors form an image without chromatic aberration. The main parts of a spherical mirror as shown in the diagram include the following: (i) the **principal axis**, which is a line drawn though the centre of the mirror; (ii) the **centre of curvature** (C) which lies on the principal axis; (iii) the **radius of curvature** of the mirror is the distance from the centre of curvature (C) to any point along the mirror; (iv) The focal point (F) is the point of intersection of all rays at the mirror from an infinite distance and parallel to the principal axis. The distance from the focal point to the mirror along the principal axis is called the **focal length** (f). The focal length is one half of the radius of curvature, which is expressed mathematically as $f = \frac{C}{2}$.

SPHERICAL POLAR COORDINATES: A coordinate system for three-dimensional space, where the position of a point is specified by three numbers: (i) the **radial distance** of that point from a fixed origin; (ii) its **inclination angle** measured from a fixed zenith direction; (iii) the **azimuth angle** of its orthogonal projection on a reference plane that passes through the origin and is orthogonal to the zenith, measured from a fixed reference direction on that plane. The inclination angle is often replaced by the elevation angle measured from the reference plane.

SPHERICAL POLYGON: A closed geometric figure formed on the surface of a sphere that is bounded by three or more arcs of great circles.The area of the polygon is proportional to its spherical excess.

SPHERICAL RADIANCE: It is the same as **radiant exitance**, *M*. It is the integration of the radiant power, *P*, leaving a source over the solid angle and over the whole wavelength range.

SPHERICAL SURFACE: A surface of constant and positive total curvature.

SPHERICAL TRIANGLE: A closed geometric figure formed on the surface of a sphere and bounded by intersecting minor arcs of three great circles. The spherical angles between these arcs may be all acute, all right or all obtuse and their sum must be strictly between π and 3π radians. The area of the triangle is proportional to its spherical excess. There are eight triangles formed by three distinct great circles, but only one is formed from minor arcs and is referred to as a spherical triangle. The properties of the others can be deduced from it. The study of these properties is spherical trigonometry.

SPHERICAL TRIGONOMETRY: A branch of spherical geometry which deals with polygons especially triangles on the sphere and the relationships between the sides and the angles. This is of great importance for calculations in astronomy and earth-surface orbital and space navigation.

SPHERICITY: The state or fact of being spherical.

SPHERICS: The geometry and trigonometry of figures on the surface of a sphere.

SPHEROID: A surface in three dimensions obtained by rotating an ellipse about one of its principal axes. If the ellipse is rotated about its major axis, the surface is called a **prolate spheroid** (similar to the shape of a rugby ball). If the minor axis is chosen, the surface is called an oblate spheroid (similar to the shape of the Earth). The sphere is a special case of the spheroid in which the generating ellipse is a circle. A spheroid is a special case of an ellipsoid where two of the three major axes are equal.

SPHEROIDICITY: The state or fact of being spheroidal.

SPHEROMETER: An instrument for measuring the curvature of a surface and the exact radius of a sphere. The usual instrument for this purpose consists of a tripod, the pointed legs of which rest on the spherical surface at the corners of an equilateral triangle. At the centre of this triangle is a fourth point, the height of which is adjusted by means of a micrometer screw gauge.

SPHEROSOME (OLEOSOME): Small spherical organelles, generally spherical or oblate and 0.4-3 μm in diameter that occur in lipid-storage tissues in plants such as endosperm of castor-bean seeds. Spherosomes are enclosed within a phospholipid monolayer (a half-unit membrane) with embedded oleosins and other proteins and contain triacylglycerols formed by enzymes in the endoplasmic reticulum (ER) and partitioned into the interior of the ER membrane at sites defined by the presence of oleosins.

SPHERULITE: A polycrystalline, roughly spherical morphology consisting of lath, fibrous or lamellar crystals emanating from a common centre. Space filling is achieved by branching, bending or both, of the constituent fibres or lamellae.

SPHINCTER: A circular ring of muscle that constricts a passage or closes a natural orifice. When relaxed, a sphincter allows materials to pass through the opening. When contracted, it closes the opening. Examples are anal, ileal, pharyngoesophageal, pupillary, pyloric, reticulo-omasal, teat, urethral, vaginal and vesical sphincter muscles.

SPHINGOLIPID: A type of complex lipid containing a long-chain amino alcohol, such as sphingosine, joined by an amide linkage to a long-chain fatty acid. Sphingolipids are important as membrane components in both animal and plant cells. In yeasts and in higher plants the amino alcohol is commonly phytosphingosine.

SPHYGMOMANOMETER: An instrument used to measure blood pressure in an artery. It consists of an inflator bulb or pressure pump, a pressure gauge and an inflatable cuff placed around the upper arm. The cuff is wrapped around the arm and pumped up sufficiently to stop the pulse as felt at the wrist or heard with a stethoscope placed on the artery at the bend of the elbow. As the applied pressure is reduced, blood starts to flow again in the artery and the pressure reading on the gauge at this point represents systolic pressure. The pressure at which there is a full flow of blood, indicated by a marked change in the sound heard through the stethoscope, represents the diastolic pressure.

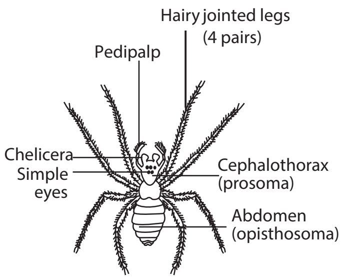
SPICATE: Arranged in or resembling a spike.

SPICE: Aromatic plant substances such as nutmeg, pepper, mustard, ginger, cinnamon and cloves used as flavouring or condiment.

SPICIFORM: An inflorescence with the general appearance, but not necessarily the structure of a true spike.

SPICULE: A short, pointed epidermal projection.

SPIDER: An eight-legged predatory silk producing arachnid of the order Araneae, class Arachnida. Spiders have an unsegmented body divided into a cephalothorax bearing eight legs, two poison fangs and two feelers. The unsegmented body has two to eight spinnerets, depending on the species, on the hind part of the abdomen, into which numerous silk glands open and used to produce silk to make nests, cocoons or webs for trapping insects. Spiders come in a variety of colours and with a great diversity of markings. Some rely on camouflage in order to blend in with their background, while others are bright coloured as a warning that they are poisonous to predators. For locomotion, spiders use hydraulic pressure.

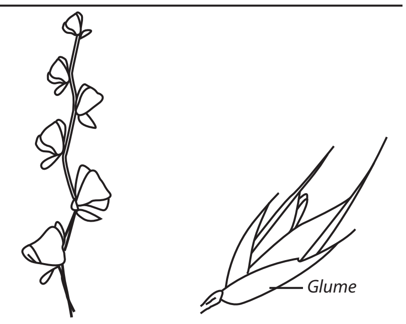


Dorsal view of a spider

SPIDER PLANT (*Chlorophytum comosum*): A herbaceous perennial flowering plant which grows to about 60 centemetres and native to tropical Africa and southern regions of Africa, but has become naturalised in some parts of the world. It has fleshy, tuberous roots and elongated variegated green leaves with a creamy white stripe growing from a central crown and can reach up to 30 centimetres long. Tiny white flowers are produced in a long branched inflorescence, each flower having a pedicel and a six three-veined tepals. At the tip of a branch of an inflorescence is a plantlet which droops, eventually touching the soil and developing adventitious roots. There are evidence to show that spider plants reduce indoor pollution. The plants are grown as house plants for their beautiful trailing foliage.

SPIEGEL: A form of pig iron containing 15-30% manganese and 4-5% carbon. It is added to steel in a Bessemer converter as a deoxidising agent and to increase the manganese content of steel.

SPIKE: 1. A thin, sharp-pointed piece of metal, wood or another rigid material. 2. A type of racemose inflorescence in which stalkless flowers arise from an unbranched axis with the oldest flowers at the base, as found, for example, in plantain and orchid. In the family of Gramineae the flowers are grouped in clusters called **spikelets**, which may be arranged to form a compound spike. In other words, **spikelets** are parts of the flower head characteristic of grasses and sedges, attached to the main stem without a stalk, having a varying number of reduced flowers each subtended by one or two scale-like bracts. 3. A pointed element in a graph or tracing. 4. A young mackerel of about 15 centimetres or less in length.



Spike ***Spikelet***

SPIKE TOOTH HARROW: A tractor trailed implement consisting of a simple frame with tines attached where the frame members cross.

SPILITE: A fine-grained, extrusive igenous rock found near volcanoes. It is dark-coloured and occurs as pillow lava, which is formed where lava from a volcanic eruption flows into the sea. Spilite often contains cavities formed by gas bubbles. It is a basic rock containing up to 40% silica and more sodium than basalt, the common volcanic rock.

SPILL-OVER EFFECT: The economic effects of domestic or sectoral mitigation measures on other countries or sectors. Spill-over effects can be positive or negative and include effects on trade, carbon leakage, transfer, and diffusion of environmentally sound technology and other issues.

SPIM: An unsolicited commercial communications received on a computer through an instant-messaging system.

SPIN: 1. Rotating motion of an electron in an atom. 2. The intrinsic angular momentum of an elementary particle or system of such particles independent of its motion or the quantum number of an elementary particle that is a measure of its intrinsic angular momentum and magnetic moment. It is measured in multiples of {Planck constant} (h-bar), which is equal to Planck's constant divided by 2. Its symbol is S. Electrons, neutrons and protons have a spin of 1/2, for example, while pions and helium nuclei have zero spin. The spin of a complex nucleus is the vector sum of the orbital angular momentum and intrinsic spins of the constituent nucleons. For nuclei of even mass number, the multiple is an integral; for those of odd mass number, it is a half-integer. 3. To create yarn from raw materials such as cotton, flax, sisal, wool and jute to make textiles.

SPIN CONSERVATION RULE (WIGNER RULE): For both radiative and radiationless transitions, the principle that transitions between terms of the same multiplicity are spin-allowed, while transitions between terms of different multiplicity are spin forbidden.

SPIN CROSSOVER: A type of molecular magnetism that is the result of electronic instability caused by external constraints (temperature, pressure or electromagnetic radiation), which induce structural changes at molecular and lattice levels. The phenomenon is most characteristic of first-row transition metal complexes, for example, those of Fe^{II}.

SPIN DENSITY: The unpaired electron density at a position of interest, usually at carbon, in a radical. It is often measured experimentally by electron paramagnetic resonance (EPR) and ESR (electron spin resonance) spectroscopy through hyperfine coupling constants of the atom or an attached hydrogen.

SPIN DRAG: A resistance to the transport of spin in a quantum-mechanical system.

SPIN GLASS: An alloy of a small amount of a magnetic metal with a nonmagnetic metal, in which the atoms of the magnetic element are randomly distributed through the crystal lattice on the magnetic element.

SPIN HALL EFFECT (SHE): A transport phenomenon predicted by the Russian physicists M.I. Dyakonov and V.I. Perel in 1971. It consists of an appearance of spin accumulation on the lateral surfaces of a current-carrying sample, the signs of the spin directions being opposite on the opposing boundaries. In a cylindrical wire, the current-induced surface spins will wind around the wire. When the current direction is reversed, the direction of spin orientation is also reversed.

SPIN LABEL: A molecule or group that contains an unpaired electron and can be attached to another molecule. The spin of the unpaired electron can be detected by electron paramagnetic resonance spectroscopy. The technique of spin labelling is used to investigate proteins and biological systems. Groups containing unpaired electrons include organic free radicals and a variety of types of transition-metal complexes such as vanadium, copper, iron and manganese.

SPIN NETWORK: A type of diagram which can be used to represent states and interactions between particles and fields in quantum mechanics. From a mathematical perspective, the diagrams are a concise way to represent multilinear functions and functions between representations of matrix groups. The diagrammatic notation often simplifies calculation because simple diagrams may be used to represent complicated functions.

SPIN QUANTUM NUMBER: One of the four quantum numbers that characterises the electron configuration of an atom. The spin quantum number (s) can take the values +/-½.

SPIN STATISTICS THEOREM: A fundamental theorem of relativistic quantum field theory that relates the spin of a particle to the particle statistics it obeys. It states that: (i) the wave function of a system of identical integer-spin particles has the same value when the positions of any two particles are swapped. Particles with wavefunctions symmetric under exchange are called **bosons**. (ii) The wave function of a system of identical half-integer spin particles changes sign when two particles are swapped. Particles with wavefunctions anti-symmetric under exchange are called **fermions**. This theorem enables one to understand the results of quantum statistics that wave functions for bosons are symmetric and wave functions for fermions are antisymmetric. It also provides the foundation for the Pauli's exclusion principle.

SPIN TENSOR (BODY SPIN; VORTICITY TENSOR): The skew, symmetric part of the velocity gradient; if Q is the body spin and L is the velocity gradient then $\Omega = 1/2(L - L^T)$. This is the local angular velocity expressed in tensor form.

SPIN TRANSPORT: The transport of spin in a quantum mechanical system, it is of a key importance in spintronics. Spintronics (magnetoelectronics) is the study of magnetic and electric fields produced by electron spin.

SPIN WAVE: A collective excitation associated with magnetic systems. Spin waves occur in both ferromagnetic and antiferromagnetic systems. As temperature is increased, the thermal excitation of spin waves reduces a ferromagnet's spontaneous magnetisation. The energies of spin waves are typically only μeV in keeping with typical Curie points at room temperature and below.

SPIN-ALLOWED ELECTRONIC TRANSITION: An electronic transition which does not involve a change in the spin part of the wavefunction.

SPIN-FLIP METHOD: A quantum mechanical method for the calculation of open-shell excited states. The method accurately describes low-lying multi-configurational electronic states of diradicals and triradicals in an efficient and robust single-reference scheme.

SPIN-FLIP TRANSITION: A rotation of electron spins, above a critical magnetic field, in an antiferromagnet from parallel to largely perpendicular alignment, relative to an applied magnetic field.

SPIN-GLASS TRANSITION: A second-order transition from a paramagnetic or ferromagnetic state to a spin-glass state in which spins from moment-bearing solute atoms become ordered randomly in a non-magnetic host such that the net magnetisation of any region is zero.

SPIN-LATTICE RELAXATION: A process in which electrons in a crystal return to the distribution for equilibrium statistical mechanics after some perturbation such as magnetic resonance has caused more electron spins to be in high-energy states.

SPIN-ORBIT COUPLING (SPIN-ORBIT INTERACTION; SPIN-ORBIT EFFECT): An interaction or relationship between the orbital angular momentum and the spin angular momentum of an individual particle such as an electron. An example is the spin-orbit interaction that causes shifts in an electron's atomic energy levels detectable as a splitting of spectral lines, due to electromagnetic interaction between the electron's spin and the nucleus's electric field, through which it moves. A similar effect, due to the relationship between angular momentum and the strong nuclear force, occurs for protons and neutrons moving inside the nucleus, leading to a shift in their energy levels in the nucleus shell model.

SPIN-ORBIT SPLITTING: Removal of state degeneracy by spin-orbit coupling.

SPIN-SPIN COUPLING: The interaction between the spin magnetic moments of different electrons and/or nuclei. It causes, for example, the multiplet pattern in nuclear magnetic resonance spectra.

SPIN-SPIN COUPLING CONSTANT: Coefficient of the indirect spin–spin coupling between two nuclei (A and B) in a magnetic resonance Hamiltonian.

SPIN-STATE TRANSITION: An electronic transition from a high-spin state to a low-spin state or vice versa.

SPINACH: An annual plant, *Spinacia oleracea* of the goosefoot family (Chenopodiaceae), which is a common garden plant, widely cultivated for its succulent leaves and eaten raw in salads or cooked as a vegetable. Spinach is rich in iron and vitamins A and C.

SPINACH BEET (SWISS CHARDS; CHARD): *Beta vulgaris* of the family *Chenopodiaceae* is a variety of beet grown as a vegetable for its edible leaves and stalks. It does not have a swollen root.

SPINAL CORD: The portion of the central nervous system within the spinal canal of the vertebral column comprising of the entire central nervous system except the brain. It carries nerve impulses between various parts of the body and the brain.

SPINAL NERVE: Pairs of nerves that arise from the spinal cord. There are 31 pairs in humans, each nerve arising from a dorsal root and a ventral root and contains both motor and sensory fibres. The spinal nerve forms an important part of the peripheral nervous system carrying motor, sensory and autonomic signals between the spinal cord and the body.

SPINDLE: 1. A rotating rod on a device such as a door handle, turntable or lathe. 2. A rod or pin tapered at one end and usually weighted at the other, on which fibres are spun by hand into thread and then wound. A similar rod or pin is used for spinning on a spinning wheel. It is also a pin or rod holding a bobbin or spool on which thread is wound on an automated spinning machine. 3. Any of various long thin stationary rods, such as a spike on which papers may be impaled. 4. A **spindle fibre** is a structure formed by microtubules in the cytoplasm during cell division that moves chromatids or chromosomes apart and gathers them into two clusters at opposite ends of the cell. 5. Any of various mechanical parts that revolve or serve as axes for larger revolving parts, such as in a lock, axle or lathe.

SPINE: 1. The axis of the skeleton of a vertebrate animal, extending from the head and consisting of a series of interconnected vertebrae that enclose and protect the spinal cord. 2. A sharp stiff projection on the body of an animal, such as the quill of a porcupine or the ray of a fish's fin. 3. A modified leaf or part of a leaf forming a stiff sharp pointed outgrowth on a plant such as those on a rose or cactus. If some or all of the leaves are modified into spines, then leaf area and hence transpiration, is reduced. Spine points sometimes act as nuclei on which water droplets can condense, run down to the soil and provide moisture for the plant. In such plants the functions of the leaf may be taken over by the stem. Spines also protect against herbivores and excessive sunlight. 4. A continuous ridge in a range of mountains or hills. 5. The back of a book cover to which the pages are attached.

SPINEL: A group of oxide minerals, of general formula $A^{2+}B_2^{3+}C_4^{2-}$ that crystallise in a cubic (isometric) system with the oxide anions arranged in a cubic close-packed lattice and the cations A and B occupying some or all of the octahedral and tetrahedral sites in the lattice. A and B can be divalent, trivalent or quadrivalent cations, including magnesium, zinc, iron, manganese, aluminium, chromium, titanium and silicon. Although the anion is normally oxide, structures are also known for the rest of the chalcogenides. A and B can also be the same metal under different charges, such as the case in Fe_3O_4 as $Fe^{2+}Fe_2^{3+}O_4^{2-}$.

SPINEL STRUCTURE: An ionic crystal structure shown by compounds of the type AB_2O_4. There is a face-centred cubic arrangement of O^{2-} ions.

SPINESCENT: Bearing a spine or a spinelike point at the tip.

SPINNER: 1. A device for harvesting potatoes. 2. In mathematics, a top with numbers marked on it.

S

SPINNERET: 1. A small tubular organ of spiders and some insects larvae such as silkworms, used to produce silk. In spiders, four to six spinnerets are located on the hind part of the abdomen, into which numerous silk glands open. 2. A finely perforated metal plate or cup with tiny holes through which a viscous liquid is extruded, used in the production of rayon, nylon and other synthetic fibres.

SPINNING: 1. Twisting wool, cotton, silk or other material to convert them into yarn, either by hand or with machinery. 2. Rotation on an axis.

SPINOCEREBELLAR TRACTS: A set of axonal fibres originating in the spinal cord and terminating in the ipsilateral cerebellum, which is the pathway of neurons that transfer sensory information from the spinal cord to the cerebellum of the brain.

SPINODAL CURVE: A curve that separates a metastable region from an unstable region in the coexistence region of a binary fluid. Above the spinodal curve the process of moving towards equilibrium occurs by droplet nucleation, while below the spinodal curve there are periodic modulations of the order parameter, which begins with small amplitude.

SPINODAL DECOMPOSITION: A mechanism by which a solution of two or more components can separate into distinct regions (or phases) with distinctly different chemical compositions and physical properties. It is observed in the quenching of binary mixtures.

SPINOR: A mathematical entity similar to a vector but having the property that it changes sign on each rotation through 360° or a vector with two complex components, which undergoes a unitary unimodular transformation when the three-dimensional coordinate system is rotated. It can represent the spin state of a particle of spin ½. Spinors have been used extensively in general relativity theory.

SPINTRONICS: A branch of technology that specifically makes use of the quantum-mechanical spin of electrons in electronic devices. It is likely to lead to fast components that both store and process information and to devices used in quantum computing.

SPIRACLE: 1. A small paired aperture along the side of the thorax or abdomen of a spider or an insect used for gaseous exchange. 2. A small gill slit or opening behind the eye area of some fishes such as rays and skates. 3. A blowhole or nostril of a dolphin, whale or similar ocean mammal. 4. A small vent in a lava flow that allows the escape of built-up gases.

SPIRAL: 1. A plain curve formed by a point winding round a fixed point from which it distances itself at regular intervals. Examples are the spiral traced by a flat coil of robe, a three-dimensional curve that turns around an axis at a constant or continuously varying distance while moving parallel to the axis; a helix, something having the form of such a curve, for example, a spiral of black smoke and spiral binding. 2. The course or flight path of an object rotating on its longitudinal axis. 3. A continuously accelerating increase or decrease, for example, the wage-price spiral.

SPIRAL CLEAVAGE: A type of holoblastic cleavage characteristic of protostomes that has an arrangement of the blastomeres of each upper tier over the cell junctions of the next lower tier so that the blastomeres spiral around the pole to pole axis of the embryo.

SPIRAL GALAXY: A galaxy that consists of a nucleus of older stars and a disk with spiral arms made mainly of dust, gas, and young stars. The bulge of a spiral galaxy is composed primarily of old, red stars. Very little star formation goes on in the bulge.

SPIRAL THICKENING (HELICAL THICKENING): A type of secondary wall patterning in tracheary elements in which the secondary cell wall is laid down in a helical pattern. Spiral thickening, like annular thickening, permits further extension of the cell. It is therefore found in tissues such as protoxylem, which develop in young organs that are still elongating.

SPIRAL SIMILARITY: In mathematics, it is the combination of central dilatation and rotation about the same centre. For a rotation, the coefficient of homothety is 1. For a homothety, the angle of rotation is 0.

SPIRILLUM: Any spiral-shaped bacteria making up the genus *Spirillum*, which are aquatic except for one species that causes a type of rat-bite fever in humans. Spirilla are gram-negative and move by means of tufts of flagella at each end.

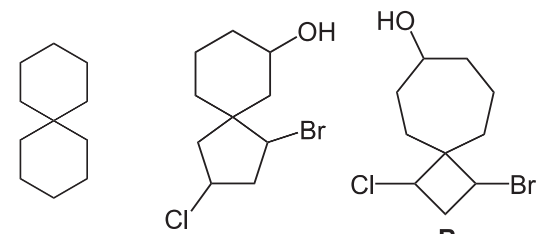
SPIRIT: 1. A volatile liquid obtained by distillation or a volatile distillate such as aviation spirit. 2. A solution that consists of a volatile substance dissolved in ethanol (ethyl alcohol). They include beverages of high alcohol content made by distillation of fermented liquors such as brandy, gin, rum, vodka and whisky usually containing 40% alcohol by volume, about 31.7 g per 100 mL. 3. A supernatural being such as a ghost and an angel without a physical body.

SPIRITS OF SALT: A former name for **hydrochloric acid** because this compound can be made by adding common sulfuric acid to common salt (sodium chloride). Hydrochloric acid is a solution of hydrogen chloride (HCl) in water. It is found naturally in gastric acid. It is a highly corrosive, strong mineral acid with many industrial uses. It is used in the chemical industry as a chemical reagent in the large-scale production of vinyl chloride for PVC plastic and methylene diphenyl diisocyanate(MDI) or toluene diisocyanate (TDI) for polyurethane.

It has numerous smaller-scale applications, including household cleaning, production of gelatin and other food additives, descaling and leather processing.

SPIRO CHAIN: A double-strand chain consisting of an uninterrupted sequence of rings, with adjacent rings having only one atom in common. Alternatively, a spiro chain is a double-strand chain with adjacent constitutional units joined to each other through three atoms, two on one side and one on the other side of each constitutional unit.

SPIRO COMPOUND: A type of organic compound in which there are two rings linked through a single atom, the spiro atom. In most cases, the spiro atom is a carbon atom and the rings are not in the same plane.



A B

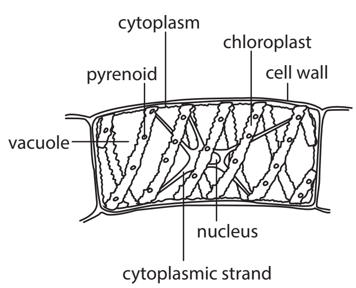
Structure of Spiro [5.5]undecane ***Two spiro compounds to illustrate nomenclature***

SPIRO MACROMOLECULE: A double-strand macromolecule consisting of an uninterrupted sequence of rings, with adjacent rings having only one atom in common. Alternatively, a spiro macromolecule is a double-strand macromolecule with adjacent constitutional units joined to each other through three atoms, two on one side and one on the other side of each constitutional unit.

SPIRO UNION: A union formed by a single atom which is the only common member of two rings. The common atom is designated as the spiro atom. A free spiro union is one which constitutes the only union direct or indirect between the two rings.

SPIROCHAETE: Any of a group of spirally coiled bacteria that moves by means of the flexions of the cell. Most spirochaetes are gram-negative, anaerobic and feed on dead organic matter and are parasitic or free-living and are particularly common in sewage-polluted waters. Some are pathogenic to humans and other mammals.

SPIROGYRA: A genus of filamentous freshwater green alga, belonging to the order Zygnematales, with more than 400 species. It is commonly found in freshwater habitats and reproduces sexually.



Spirogyra

SPIROMETER. An instrument used for measuring lung capacity or very deep inspiration rates.

SPLASH: To make substance such as a liquid or chemical fall or scatter in irregular drops or particles so that it goes in all directions.

SPLAY LEG: A disorder in piglets which are born with inability to stand properly.

SPLEEN: An organ in the abdomen near the stomach in most vertebrates or a ductless vascular organ in the left upper abdomen of humans and other vertebrates that helps to destroy old red blood cells, form lymphocytes, store blood and filter foreign bodies from the blood.

SPLENIC ARTERY: A very wide vessel, the widest branch of the celiac artery that originates from the celiac trunk and supplies blood to the spleen.

SPLENIC VEIN: A vein formed by several small veins on the surface of the spleen. It joins the superior mesenteric to form the portal vein that returns blood from the stomach, pancreas and spleen to the liver.

SPLICEOSOME: A complex assembly that interacts with the ends of an RNA intron in splicing RNA; releases an intron and joins two adjacent exons**.**

SPLICING: 1. To join together or unite, for example, two ropes or parts of a rope by the interweaving of strands.2. To join, for example, timbers and spars, by overlapping and binding their ends. 3. To join, for example, film and magnetic tape, by butting and cementing. 4. Splicing of RNA is the procedure by which introns are removed from eukaryotic precursor mRNA molecules and adjacent exon sequences are joined together (spliced). 5. Splicing of DNA is the manipulation for joining together double-stranded DNA fragments with protruding single-stranded 'sticky ends' by means of ligases.

SPLINE: 1. In mathematics, a piecewise polynomial function that can have a very simple form and a high degree of smoothness at the places where the polynomial pieces connect. The points of connection are known as **knots**. It can be used for approximation and interpolation of functions. Alternatively, it can be defined as a curve that connects two or more specific points or that is defined by two or more points. 2. A key fitting that fits into grooves in the hub and shaft of a wheel; and used to prevent movement between two parts, especially in transmitting torque.

SPLINE-FITTING: In mathematics, a popular type of piecewise approximation by polynomials of degree n (or more general functions) on an interval in which they are fitted to the function at specified points (nodes or knots) where the polynomial used can change but the derivatives of the polynomials are required to match up to degree n - 1 at each side of the nodes or meet related interpolatory conditions. In addition, boundary conditions are imposed at the ends of the interval. For the natural cubic spline, one uses cubic polynomials and requires that the second derivative of the spline vanishes at the endpoints (a natural boundary condition). The clamped boundary condition demands that the first derivatives of the function and the spline agree at the end points.

SPLINT: 1. A strip of rigid material such as wood bound to an arm, leg, etc., to keep a broken bone or any other injured part from moving. 2. A condition that occurs in young horses, consisting of painful bony outgrowths in or near the splint bones on the inner sides of the legs.

SPLIT EXACT SEQUENCE: In mathematics, a short exact sequence in which the second nontrivial mapping, g, has a right inverse, g', with $g \circ g' = 1$ (or equivalently, the first non-trivial mapping has a left inverse).

SPLIT VEIN CUTTING: Cutting made during plant propagation by cutting transversely the large veins before placing it flat on a rooting medium.

SPLIT-RING: A small steel ring with two spiral turns, often used as a means of fastening two parts together or as a key ring or a formed wire fastener that is shaped like a circle, hence the name. The open end of the wire is in the middle of the cotter so when it is installed the inner tab is first installed in the hole. Because of this feature it is often used in applications where sharp edges cannot be tolerated, such as fabric applications.

SPLITTING FIELD: In mathematics, the smallest extension field in which a given polynomial over a given field splits or decomposes into linear factors. Splitting fields correspond to finite normal extension fields. Every polynomial has a splitting field that is unique up to automorphisms.

SPODUMENE: Variously coloured, monoclinic mineral, which is a source of lithium with the chemical formula $LiAlSi_2O_6$, hardness 6.5-7 and relative density of 3. Its transparent lilac and green varieties are used as a gemstone

SPONDYLOSIS: A general term for degenerative processes in the spine due to ageing, injury or other diseases. If it affects the neck, it is called **cervical spondylosis**; if it affects the lower back, it is called **lumbar spondylosis**; that of the middle back is **thoracic spondylosis**. General symptoms include low back pain, neck pain, leg pain or arm pain.

SPONGE: 1. Any sac-like simple invertebrate of the phylum Porifera, with a porous fibrous skeleton composed of calcium carbonate, silica and spongin. They often live in colonies and attach themselves to underwater objects in oceans. Sponge is also the light, fibrous, flexible, absorbent skeleton of certain of these organisms, used for bathing, cleaning and other purposes. 2. A folded gauze pad used in surgery or medicine to absorb discharges, dress wounds or apply medications. 3. An absorbent contraceptive device that contains a spermicide, inserted into the vagina to cover the cervix. 4. An absorbent porous plastic, rubber, cellulose or other material, used for absorption. 5. A metal in a porous, brittle form, such as after the removal of other metals in processing. It is used as a raw material in manufacturing.

SPONGY BONE (SPONGIOSA; SUPPORTING BONE; TRABECULAR BONE): The inner layer of bone, which is found at the ends of long bones. It is less dense than compact bone. Some spongy bones contain red marrow.

SPONGY MESOPHYLL (SPONGY PARENCHYMA): The lower, layer of a bilateral leaf. It contains irregular-shaped chlorophyll-bearing parenchymatous cells interspersed with lots of intercellular air spaces that facilitate air circulation in the leaf.

SPONTANEOUS COMBUSTION: Combustion in which a substance unexpectedly bursts into flame by producing sufficient heat within itself, usually by a slow oxidation process, for ignition to take place without the need for an external high-temperature energy source. Spontaneous combustion often occurs in things such as piles of oily rags, green hay, leaves or coal and can constitute a serious fire hazard.

SPONTANEOUS EMISSION: The emission of a photon by an atom molecule, nanocrystal or nucleus as it reverts from the excited state to the ground state. It occurs independently of any external electromagnetic radiation. Interactions between atoms and vacuum fluctuations of the quantised electromagnetic field cause the transition.

SPONTANEOUS FISSION: Nuclear fission which occurs without the addition of particles or energy to the nucleus.

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SPONTANEOUS GENERATION (EQUIVOCAL GENERATION): An absolute theory regarding the origin of life that living organisms can somehow be produced by nonliving matter. For example, it was believed that microorganisms arose from decay and that vermin spontaneously developed from household rubbish.

SPOOFING: A term used in hacking or deception that imitates another person, software program, hardware device or computer, with the intentions of bypassing security measures. One of the most commonly known is IP spoofing.

SPOOLING: The process of transferring computer data for printing into a computer's memory store so that it can be printed later without slowing down the computer's operations.

SPORADIC: 1. Describing or referring to a disease that appears in isolated or scattered instances or locations. 2. Describing or referring to a situation which occurs at intervals that have no apparent pattern or order in time.

SPORANGIOPHORE: A structure bearing one or more sporangia. It may be a simple stalk, such as that formed on germination of the zygospore of such fungi as *Rhizopus* and Mucor. Alternatively it may be multicellular or branched as in certain pteridophytes.

SPORANGIUM (SPORANGIA): A spore-producing organ in angiosperms, gymnosperms, ferns, fern allies, bryophytes, algae and fungi or a reproductive structure in plants that produce asexual spores. Their spores are sometimes called **sporangiospores**. A stalk bearing sporangia is called **sporangiosphore.**

SPORE: A reproductive cell that can develop into an individual without fusing with another reproductive cell. Spores thus differ from gametes, which must fuse in pairs in order to create a new individual. Spores are produced by plants, fungi, bacteria and some protozoa. Bacterial spores serve largely as a resting or dormant stage in the life cycle, preserving the bacterium through periods of unfavourable conditions. Many bacterial spores are highly durable and can germinate even after years of dormancy. Fungal spores serve a function similar to that of seeds in plants; they germinate and grow into new individuals under suitable conditions of moisture, temperature and food availability

SPORE MOTHER CELL (SPOROCYTE): A diploid cell that gives rise to four haploid spores by meiosis or a cell from which spores are produced.

SPORICIDE: A substance used to kill spores. **Sporicidal** is the ability to kill spores.

SPOROCARP: A hard, nut-like specialised structure containing sporangia.

SPOROCYST: 1. A resting cell that produces asexual plant spores. 2. A protective case or cyst in which sporozoites develop and from which they are transferred to different hosts. 3. A structure formed by a parasite within its host that produces spores.

SPOROGENESIS: The production of spores or the process of reproduction via spores. Reproductive spores are formed in many eukaryotic organisms, such as plants, algae and fungi, during their normal reproductive life cycle.

SPOROGONIUM: The sporophyte generation in mosses and liverworts. It is made up of an absorptive foot, a stalk and a spore-producing capsule that never separates from the mother plant. It may be completely or partially dependent on the gametophyte.

SPOROPHORE (FRUCTIFICATION): The aerial spore-producing organ of certain fungi. Examples are the stalk and cap of mushroom.

SPOROPHYLL: A leaf that bears sporangia. In ferns the sporophyll are the normal foliage leaves but in other plants they are modified and arise in specialised structures such as the strobili of club mosses, horsetail and gymnosperms and the flower of angiosperm.

SPOROPHYTE: The diploid spore producing generation in the life cycle of a plant that undergoes alternation of generation or a plant in its asexual spore-producing phase.

SPOROPOLLENIN: The highly resistant material making up the exine of spores and pollen grains. It is a polymerised carotenoid, capable of withstanding temperatures approaching 300°C and strong acids.

SPOROZOA: A parasitic single-celled protozoan which has alternating sexual and asexual generations and reproduces by means of spores. It includes plasmodium, the cause of malaria and belongs to the class sporozoa.

SPORT: In biology, it refers to an individual organism that shows the effect of mutation. The term is usually applied to a typical form of a plant in horticulture.

SPORULATION: The act and process of spore formation.

SPOT TEST (PRESUMPTIVE TEST): A simple test for a given substance using a reagent that changes colour when mixed with the substance; or the addition of a drop of reagent to a drop or two of sample solution to obtain distinctive colours or precipitates. It is used in qualitative analysis.

SPRAING: A virus disease, tobacco rattle virus (TRV) of potatoes spread by nematodes from the genera *Trichodorus* and *Paratrichodorus* referred to as trichodorid nematodes in the soil. It causes brown streaks in the flesh of the potato.

SPRAWLING: Bending or curving downward; lying upon something and supported by it.

SPRAY: 1. A mass of tiny drops of liquid. 2. Special liquid for spraying onto a plant to prevent insect infestation or disease. 3. A slender shoot or branch with its leaves, flowers or fruits.

SPRAY BAR: An attachment consisting of a horizontal tube with nozzle or jets used for spraying over a wide area.

SPRAY IRRIGATION: A system of irrigation using sprinklers which are located along a boom.

SPRAY LINES: A method of distributing irrigation water using flexible hose. It is used mainly in horticulture.

SPRAYER: A machine which forces a liquid through a nozzle under pressure. It is used for discharging liquids, pesticides, fungicides, insecticides, etc.

SPREAD: 1. To put something such as manure, fertiliser or mulch on an area of ground. 2. Also known as **dispersion**, it is the distribution of frequency about the mean. 3. In mathematics, it is a tree whose paths are infinite and whose nodes are sequences of natural numbers corresponding to the initial segments of the infinite sequences that are generated in accordance with some spread law.

SPREAD PLATE TECHNIQUE: A method of isolation and enumeration of microorganisms in a mixed culture and distributing it evenly by using a sterilised spreader with a smooth surface made of metal or glass to apply a small amount of bacteria suspended in a solution over a plate. The plate needs to be dry and at room temperature so that the agar can absorb the bacteria more readily. A successful spread plate will have a countable number of isolated bacterial colonies evenly distributed on the plate.

SPREAD SPECTRUM: A term used in telecommunications, to refer to a signal generated in one bandwidth, which is spread across several frequencies to create a signal with a wider bandwidth. This can result in more secure communications, a resistance to interference, increased difficulty for someone or something to jam that signal and a lower chance of detection by another source. It can also benefit satellite downlinks by limiting power flux density.

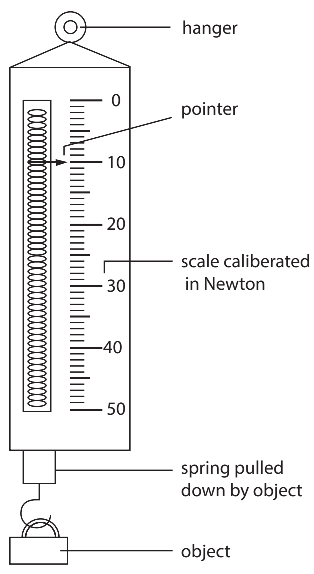
SPREADING HOGWEED (PIG WEED; RED HOGWEED; RED SPIDERLING TARVINE; *Boerhavia diffusa*): A perennial plant in the Nyctaginaceae, the Four O'Clock family, that grows as common weed in wet areas, especially in the rainy seasons with sunshine. It occurs throughout India, Africa, the Pacific and southern United States and has become naturalised in many places. It has a fusiform root with a stem, which is prostrate, decumbent or ascending. The leaves are broadly ovate and obtuse at the base and arranged oppositely or sub-oppositely. The small red or white flowers are in pendunculate, glomerulate clusters arranged in slender, long stalked, axillary or terminal corymbs. The fruits are rounded above, obovoid, slightly cuneate below and glandular throughout. Medicinally, the various parts such as the root, leaves and seeds have been used to treat various ailments such as leucorrhoea, anaemia, inflammations, heart diseases, asthma, diabetes, dyspepsia, tumours, spleen enlargement and abdominal pains. It also has antibacterial and antioxidants properties.

SPREADSHEET: A program in computing that imitates a sheet of ruled paper, divided into columns and rows, commonly used for budgets, forecasting and other finance-related tasks. Spreadsheet programs use rows and columns of cells; each cell can hold text or numeric data or a formula that uses values in other cells to calculate a desired result. To ease computation, these programs include built-in functions that perform standard calculations. Depending on the program, a single spreadsheet can contain anywhere from thousands to millions of cells.

SPRIGGING: A method of plant propagation in which cuttings of stolons or rhizomes are planted instead of seed in the soil surface or into furrows or small holes. This method is common for establishing Cynodon (Bermuda Grass) and Zoysia (Zoysiagrass). The stolons/rhizomes can also be broadcast in slurry as in hydroseeding.

SPRING: 1. An elastic material used to store mechanical energy. Springs are usually made out of spring steel. Small springs can be wound from pre-hardened stock, while larger ones are made from annealed steel and hardened after fabrication. Some non-ferrous metals are also used including phosphor bronze and titanium for parts requiring corrosion resistance. Beryllium copper is used for springs carrying electrical current (because of its low electrical resistance). As long as they are not stretched or compressed beyond their elastic limit, most springs obey Hooke's law, which states that the force with which the spring pushes back is linearly proportional to the distance from its equilibrium length: $F = -kx$, where x is the displacement vector (the distance and direction the spring is deformed from its equilibrium length), F is the resulting force vector (the magnitude and direction of the restoring force the spring exerts) and k is the rate, spring constant or force constant of the spring, a constant that depends on the spring's material and construction. Springs can be classified depending on how the load force is applied to them: (i) **tension/extension spring**, is a spring designed to operate with a tension load, so that the spring stretches as the load is applied to it; (ii) **compression spring**, is designed to operate with a compression load, so that the spring gets shorter as the load is applied to it; (iii) **torsion spring**, in which the load is an axial force, the load applied to a torsion spring is a torque or twisting force and the end of the spring rotates through an angle as the load is applied; (iv) **constant spring**, in which the supported load will remain the same throughout deflection cycle; (v) **variable spring** in which the resistance of the coil to load varies during compression. The classification based on shape includes:(i) **coil spring**, which is made of a coil or helix of wire; (ii) **flat spring**, which is made of a flat or conical shaped piece of metal; (iii) **machined spring**, which is made by machining bar stock with a lathe and/or milling operation rather than coiling wire. Since it is machined, the spring may incorporate features in addition to the elastic element. Machined springs can be made in the typical load cases of compression/extension, torsion, etc. Coil springs and other common springs typically obey Hooke's law. Springs based on beam bending can, for example, produce forces that vary nonlinearly with displacement and do not obey Hooke's law. 2. Also known as **spring water**; it is a natural flow of water from the ground formed where the water table or the aquifer surface meets the ground surface, which is a component of the hydrosphere. A spring may be the result of karst topography, where surface water has infiltrated the Earth's surface (recharge area), becoming part of the area groundwater. The groundwater then travels through a network of cracks and fissures (openings ranging from intergranular spaces to large caves). The water eventually emerges from below the surface, in the form of a karst spring. A confined aquifer in which the recharge area of the spring water table rests at a higher elevation than that of the outlet spring water can force water to the surface. Such water is called **artesian well**. Non-artesian springs may simply flow from a higher elevation through the earth to a lower elevation and exit in the form of a spring. Other springs are the result of pressure from an underground source in the earth, in the form of volcanic activity. The result can be water at elevated temperature such as a hot spring. 3. The season of the year, occurring between winter and summer, during which the weather becomes warmer and plants revive, extending in the Northern Hemisphere from the vernal equinox to the summer solstice and popularly considered to comprise March, April and May, at which time it is autumn in the Southern Hemisphere. At the spring equinox, days are close to 12 hours long with day length increasing as the season progresses. Spring is considered to occur in the Southern Hemisphere from September to November.

SPRING BALANCE: A simple form of balance with which a force is measured by a helical spring or a device to determine the weight of objects by measuring the tension it creates on a spring. The tension read off a scale is directly proportional to the force, provided that the elastic limit of the spring is not exceeded. The device is often used to approximately measure the weight of a body.



Spring Balance

SPRING CONSTANT: The force per unit extension of spring, or the ratio of the force affecting the spring to the displacement caused by it. Its SI unit is Newton per meter (N/m).

SPRING LAMB: Lamb between the ages of three to seven months old.

SPRING WHEAT: Wheat sown in spring and harvested toward the end of the summer.

SPRING TIDE: An unusually high tide that occurs two times a month, at the new and full moons, when the Sun, the Moon, and the Earth are aligned.

SPRINGER: A cow almost ready to calve.

SPRINGTAIL: A small, six-legged primitive wingless insect of the order Collembola, which is very common in soils, where they cause damage to fine roots.

SPRINKLER: A device that sends out a moving spray of water through a plastic or metal nozzle perforated with many small holes for watering gardens or for suppressing fires. **Sprinkler irrigation** is a system of watering fields using pipelines which carry water under pressure from a pump or elevated source with sprinklers spaced along a pipe, which sprays droplets of water in a continuous circle until the moisture reaches the root level of the crop.

SPROCKET: A projecting tooth on a wheel or cylinder that connects with the links of a chain to make the chain or film move forward or a wheel that drives or is driven by a chain. A sprocket is different from a gear in that sprockets are never meshed together directly and differs from a pulley in that sprockets have teeth and pulleys are smooth. Sprockets are used in bicycles, motorcycles, cars, tanks and other machinery to transmit rotary motion between two shafts, where gears are unsuitable or to impart linear motion to a track, tape, etc.

SPROUT: 1. A little shoot growing out from a seed, with a stem and small leaves. 2. To begin to grow; to send out a new shoot from a seed or bud.

SPRUCE: An evergreen coniferous tree of the pine family, belonging to the genus Picea of the temperate regions. It has short needles, drooping cones and is pyramid shaped. Tough, finely grained, resonant and pliable, spruce wood is used for sounding boards in pianos and bodies of violins, as well as in construction, for boats and barrels and as pulpwood. Some common species include black spruce (*Picea mariana*), a source of spruce gum; and white spruce (*Picea glauca*), a source of good timber. Blue or Colorado spruce (*Picea pungens*) is used as an ornament because of its bluish leaves and symmetrical growth habit.

SPRUCE-LARCH ADELGID: A relative of the aphid which may cause serious damage on spruce grown for Christmas tree.

SPUR: 1. A ridge of rock jutting out from a mountain into a valley or plain. 2. A tubular projection from a part of a flower, such as found in a columbine or larkspur or a short branch of literal shoot from a stem or branch of a plant. 3. A pointed projecting part on some animals such as the stiff outgrowth on the legs of some birds and insects or the sharp honey projection on the legs of some male birds and domestic roosters above the claws. 4. A bony outgrowth which is usually a normal part of the body but sometimes often an injury. 5. A small spike attached to a rider's boot used to make a horse run faster.

Spurred flower

SPUTNIK: Any of a series of Earth-orbiting spacecraft launched by the former Soviet Union from 1957 to 1961, marking the start of the space age. Sputnik 1, the world's first artificial satellite (October 1957), remained in orbit until early 1958, when it re-entered the Earth's atmosphere and burned up. Sputnik 2 carried a dog, Laika, the first living creature to orbit Earth. Since Sputnik 2 was not designed to sustain life, Laika did not survive the flight. Eight more missions with similar satellites carried out experiments on various animals to test life-support systems and re-entry procedures and to furnish data on space temperatures, pressures, particles, radiation and magnetic fields.

SPUTTERING (CATHODE SPUTTERING): A process in which some of the atoms of an electrode, usually the cathode is ejected as a result of bombardment by heavy positive ions. The process can be used to obtain a clean surface or to deposit a uniform film of a metal on an object in an evacuated enclosure.

SPY HOPPING: A vertical rise out of water or tall grasses performed by certain cetaceans or land mammals, respectively.

SPYWARE: A software program that has been designed to secretly gather information about a user's activity to track users' habits in order to better target them with advertisements. Spyware is usually installed onto a user's machine without their knowledge when downloading free music sharing programs, visiting adult oriented web pages and through other downloads and plug-ins on the Internet.

SQL (STRUCTURED QUERY LANGUAGE): In computer science, it is a high-level computer language used to access and modify information in a database. Some common SQL commands include "insert," "update," and "delete."

SQUALENE: An intermediate unsaturated aliphatic hydrocarbon with the chemical formula $C_{30}H_{50}$. It is found especially in human sebum and in the liver oil of sharks, formed in the synthesis of cholesterol. The immediate oxidation of squalene to squalene 2,3-epoxide is the last common step in the synthesis of sterols in mammals, plants and fungi. It is also used in biochemical research.

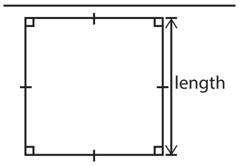
SQUALL: A sudden increase in wind speed to at least 11 m/s and lasting for one minute or longer. Commonly associated with thunderstorms and heavy rain, squalls may do great damage. A **squall line** is a line of thunderstorms often hundreds of kilometres long with squalls along its advancing edge.

SQUAMA: Shaped like the scale of a fish, as occurs in some types of pappus in the Compositae (Asteraceae). **Squamate** or **squamaceous** means covered with scales (squamae). **Squamella** or squamellae means small or minute scale or scales.

SQUAMATA: An order of reptiles comprising the lizards and snakes. They are distinguished by a highly modified skull that has only a single temporal opening or none, by the lack of shells or secondary palates and by possession of paired hemipenes in the males.

SQUAMOUS EPITHELIUM: A type of epithelium that is made up of its most superficial layer consisting of flat scale-like cells called squamous that allows materials to pass through relatively quickly. This type of epithelium is found lining the alveoli in the lungs and Bowman's capsules in the kidney. Epithelium may be composed of one layer of these squamous cells, in which case it is referred to as **simple squamous epithelium**. If it possesses multiple layers, it is referred to as **stratified squamous epithelium**. Both types perform differing functions, ranging from nutrient exchange to protection.

SQUARE: 1. A two-dimensional four sided plane geometric figure with all sides equal and whose four interior angles are right angles (90°). If the sides of a square have a length x, the perimeter of the square is simply four times the length of a side, represented algebraically by the formula $P = 4x$. The area of a square is found by multiplying the length of a side by itself: $A = x^2$. 2. Also known as **square number**, it is the product resulting from multiplying a number or algebraic term by itself. For example, 49 is the square of 7 because 7 × 7 = 49. 3. An "L"- or "T"-shaped instrument made of plastic, wood or metal, used for drawing or measuring right angles.



Square

SQUARE BRACKET: Any of the pair of brackets, symbolised [ ], used to indicate that the expression between them takes precedence over those surrounding it and is to be evaluated before the remainder of the formula. Sometimes they are only used in expressions that already contain parentheses and have lower priority than them, but higher priority than braces. Square brackets are also used to represent numbers in an interval and a square bracket at one end of an interval indicates that the interval is closed at that end, meaning that it includes the number at that end.

SQUARE FREE: An integer that is not divisible by a perfect square, n^2, for $n > 1$.

SQUARE MATRIX: A mathematical matrix that has equal number of rows and columns or a matrix of the order $m \times n$, where m is the number of rows, n the number of columns and $m = n$. An item in a matrix is called an entry or an element. A square matrix has an inverse if and only if its determinant is non-zero.

SQUARE NUMBER: An integer such as 1, 4, 9, 16 and 25 that is a perfect square of another integer.

SQUARE PLANAR: A term used to describe molecules and polyatomic ions that have one atom in the centre and four atoms at the corners of a square.

SQUARE PLANAR COMPLEX: A complex having a metal in the centre of a square plane with ligand donor atoms at each of the four corners.

SQUARE PLOUGHING: A method of ploughing, in which a piece of land is ploughed in a clockwise direction starting from the centre. This method is suitable for ploughing large fields.

SQUARE PYRAMID: A solid figure with a base that is a square and four faces that are triangles meeting at a common point. The volume of a pyramid and for any cone is given by: $V = \frac{1}{3}Bh$, where B is the area of the base and h the height from the base to the apex. The volume can also be calculated without knowing B, the area of the base. The volume of a pyramid whose base is an n-sided regular polygon with side length s and whose height is h is therefore: $V = \frac{n}{12} hs^2 \cot\frac{\pi}{n}$. The surface area of a pyramid is $A = B + \frac{PL}{2}$, where B is the base area, P is the base perimeter and L is the slant height, $L = \sqrt{h^2 + r^2}$, where h is the pyramid altitude and r is the inradius of the base.

SQUARE ROOT: A number or a value that when multiplied by itself is equal to a given number or quantity. In arithmetic expressions, it is usually written as $\sqrt{x}$; and $x^{1/2}$ in algebraic expressions. Every positive definite operator, A has a unique positive definite square root, B, for which A = BB and every non-singular matrix over the complex field possesses square root.

SQUARE ROOT FUNCTION: A function where the outputs are the square roots of the inputs. For example, square root function can be represented as $f(x) = \sqrt{x}$. It can also be written in exponent form, $f(x) = x^{1/2}$.

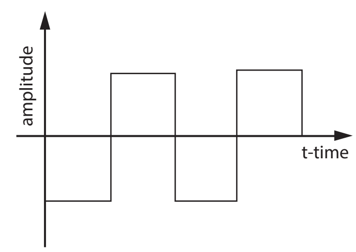
SQUARE ROOT THEOREM: The theorem that if H is a positive definite Hermitian matrix then there exists a unique positive definite Hermitian matrix G such that H = G^2.

SQUARE SUMMABLE: A sequence whose series of the squares of its terms converge to a finite sum.

SQUARE THE CIRCLE: To construct a square of the same area as a given circle using a straight-edge and compasses only and only a finite number of steps. This is impossible since it demands the construction of the number $\sqrt{\pi}$ and the fact that π is a transcendental number that cannot be constructed.

SQUARE UNIT: The area of a square each of whose sides measures 1 unit. It is used to measure area. The area of a 1 cm by 1 cm object is 1 cm^2 (1 square centimetre or 1 centimetre squared).

SQUARE WAVE: A waveform that rises quickly to particular amplitude, remains constant for a time period and drops fast at the end. In digital systems, square waves are the norm, because they represent a binary digit (0 or 1). Square waves can also be generated in musical synthesisers and have a raspy sound.



Square Wave

SQUARE-FREE (QUADRATFREI): In mathematics, of an integer, containing no repeated prime factors. Examples include 1, 2, 3, 5, 6, 7, 10, 11, 13, 14 and 15.

SQUARE-INTEGRABLE: Of a real- or complex-valued measurable function on a set, having the property that the square of the modulus of the function has a finite integral. The set of all Lebesgue square-integrable functions defined on an interval comprises the Hilbert space L_2 when functions that differ on sets of zero measure are identified and their common square-integral value is used as the norm.

SQUARED: In mathematics, it is an operation in which a number is multiplied by itself or raised to the second power or having an exponent of 2. For example, 6 squared is written as 6^2.

SQUARK: The hypothetical spin-zero superpartner of the quark.

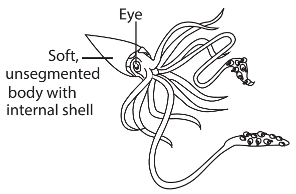
SQUARROSE: Abruptly recurved or spreading above the base; rough or scurvy due to the presence of recurved or spreading processes.

SQUASH: 1. To crush or squeeze (something) with force so that it becomes flat, soft or out of shape. 2. A microscopical preparation in which the material is flattened before examination. It is commonly used in the study of chromosomes. Some tissues, such as anthers, may be squashed directly while others such as root tips, must first be softened. This is commonly achieved by adding a cellulase enzyme. 3. Any of various tendril-bearing plants of the genus Cucurbita, having fleshy edible fruit with a leathery rind and unisexual flowers. 4. A concentrated liquid made from fruit juice and sugar, which is diluted to make a drink.

SQUEEZE RULE (SANDWICH RESULT): In mathematics, it is one of a number of inequalities, used to confirm the limit of a function whose terms are bounded above and below by comparing it with two other functions whose limits are known or easily computed. For example, if $f(x) \leq g(x) \leq h(x)$ for all x; greater than some N and if f(x) tends to A and $h(x)$ tends to A as x tends to infinity, then $g(x)$ tends to A as x tends to infinity.

SQUEEZED STATE: A quantum state in a system in which one of a pair of conjugate variables is specified more accurately than in the vacuum state and the product of the uncertainties of the conjugate variables is the minimum allowed by the Heisenberg uncertainty principle.

SQUID: 1. Diverse marine cephalopod molluscs of the genus *Loligo* and related genera with ten arms (eight short arms and two usually longer tentacles), a long tapered body, a caudal fin on each side and usually a slender internal chitinous support. Squids body sizes range from about 10 centimetres to 16.5 metres long. They are highly adaptable to changing environments. 2. A device that measures minute changes in magnetic flux by means of a pair of Josephson junctions, often used to detect extremely small changes in magnetic fields, electric currents and voltages.



Squid (marine cephalopod)

SQUIRREL: A small rodent of the family Sciuridae, found throughout the world except in Australia, Madagascar and the Polar Regions. It lives in trees, but some species are ground-living and has a long bushy tail and eats nuts and seeds.

SRAM (STATIC RANDOM ACCESS MEMORY): In computer science, it is a type of RAM that stores data using a static method, in which the data remains constant as long as electric power is supplied to the memory chip. This is different to DRAM (dynamic RAM), which stores data dynamically and constantly needs to refresh the data stored in the memory. Because SRAM stores data statically, it is faster and requires less power than DRAM.

SRC TYROSINE KINASE (C-SRC TYROSINE KINASE; CSK): Any of a family of proteins that can add a phosphate group to specific tyrosine residues of other proteins. They are homologous to the Src protein encoded by the viral oncogene *v-src* which can produce tumours in chickens. Commonly associated with cell receptors, these kinases are involved in signal transduction pathways controlling cell division and proliferation and antigen recognition by lymphocytes.

SSL (SECURE SOCKET LAYER): In computer science, it is a standard computer networking protocol, developed by Netscape Communications that allows secure communications over the Internet using two keys to encrypt data: a public key known to everyone and a private or secret key known only to the recipient of the message. SSL technology is commonly used in electronic commerce transactions where users send personal information such as credit card numbers and passwords over the Internet that need to be encrypted.

STABILISATION/STABILIZATION: In climate change, the achievement of stabilisation of atmospheric concentrations of one or more greenhouse gases such as carbon dioxide or a CO^2-equivalent basket of greenhouse gases.

STABILISATION/STABILIZATION ANALYSIS: In climate change, analyses or scenarios that address the stabilisation of the concentration of greenhouse gases.

STABILISATION ENERGY (STABILIZATION ENERGY): The amount by which the energy of a delocalised chemical structure is less than the theoretical energy of a structure with localised electrons. It is obtained by subtracting the experimental heat of formation of the compound (in $kjmol^{-1}$) from that calculated on the basis of a classical structure with localised bonds.

STABILISATION LAGOON (STABILIZATION LAGOON): A pond used for purifying waste by allowing sunlight to fall on the sewage and water. It consists of shallow man-made basins comprising a single or several series of anaerobic, facultative or maturation ponds. The primary treatment takes place in the anaerobic pond, which is mainly designed for removing suspended solids and some of the soluble element of organic matter. During the secondary stage in the facultative pond most of the remaining soluble element of organic matter is removed through the coordinated activity of algae and heterotrophic bacteria. The main function of the tertiary treatment in the maturation pond is the removal of pathogens and nutrients (especially nitrogen). Stabilisation ponds are particularly well suited for tropical and subtropical countries because the intensity of the sunlight and temperature are key factors for the efficiency of the removal processes.

STABILISE (STABILIZE): 1. To make stable or steadfast. 2. Take measures to prevent soil from being eroded especially from a hillside.

STABILISER (STABILIZER): 1. An artificial substance added to processed food such as sauces containing fat to stop the mixture from changing. 2. A chemical compound added to another substance to make it resistant to change. 3. In mathematics, it is the subgroup of elements of a group of permutations of a non-empty set, under which the image of a given subset is itself. 4. A device used to maintain a constant voltage from a source. 5. A substance added to a fast-drying paint to improve the dispersion of pigment. 6. Pairs of submerged "fins" controlled gyroscopically and used to minimise the rolling of a ship in rough waters. 7. An airfoil or combination of airfoils that keeps an aircraft or missile aligned with the direction of flight.

STABILISING SELECTION (STABILIZING SELECTION; NORMALISING/NORMALIZING SELECTION): A type of natural selection in which genetic diversity decreases as the population stabilises on a particular trait value. This is probably the most common mechanism of action for natural selection. Stabilising selection commonly uses negative selection or purifying selection to select against extreme values of the character.

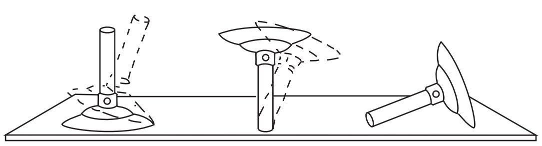
STABILITY: 1. A measure of how an animate or inanimate object can resist attempts to change its position. 2. A measure of how a substance can resist attempts to make it take part in a chemical change, for example, calcium carbonate has greater stability to heat than copper carbonate. 3. In mathematics, it is a condition in which a slight disturbance in a system does not produce a significant disrupting effect on that system. A solution to a differential equation is said to be stable if a slightly different solution that is close to it when $x = 0$ remains close for nearby values of x. Stability of solutions is important in physical applications because deviations in mathematical models inevitably result from errors in measurement. A stable solution will be usable despite such deviations. 4. In physics, the ability of atomic particles to have no known mode of decay and therefore indefinitely long-lived or incapable of becoming a different isotope or element by radioactive decay. 5. In chemistry, not easily decomposed or otherwise modified chemically; the ability of a substance to stay in the same state.

STABILITY CONSTANT: An equilibrium constant that expresses the propensity of a substance to form from its component parts. The larger the stability constant, the more stable is the species. The stability constant (formation constant) is the reciprocal of the instability constant (dissociation constant).

STABILITY OF MATTER: The conclusion that matter consisting of a very large number of protons and electrons described by non-relativistic quantum mechanics is stable.

STABLE: 1. In chemistry, not easily decomposed or otherwise modified chemically. A stable substance tends to stay in the same chemical or atomic state. 2. In mathematics, of a problem or computational method, not overly sensitive to marginal perturbations in the underlying data. In other words, the output should be continuous in some sense as a function of the perturbation. The term is used both numerically and theoretically. 3. For a system of differential equations, having the property that any solution that starts close enough to a stationary point returns over time. 4. For an equilibrium point, y^{ε} of linear ordinary differential equations, having the property that for all positive ε, there is a $\delta > 0$ such that if $|| y(0) - y^{\varepsilon} || < \delta$, then $|| y(t) - y^{\varepsilon} || < \varepsilon$ for all non-negative t. If in addition, there is a positive R such that given positive ε, there exists a positive T, such that if $|| y(0) - y^{\varepsilon} || < R$, then $|| y(t) - y^{\varepsilon} || < \varepsilon$ for all $t \geq T$, then y^{ε} is asymptotically stable; if y^{ε} is not stable, it is is unstable. 5. **Lyapunov stable** defines an initial value of a differential equation in dynamical systems having the property that any solution that starts close enough to the initial value (equilibrium point) stay near forever and is asymptotically stable, if in addition, as time tends to infinity the solution also converges to the initial value. 6. In physics, it refers to atomic particles having no known mode of decay and therefore indefinitely long-lived or incapable of becoming a different isotope or element by radioactive decay. 7. A building in which horses are lodged and fed.

STABLE EQUILIBRIUM: The state of a system that will return to its original position after experiencing a slight disturbance.For example, a body resting on a table has a reaction from the table; this is an upward force, which equals the weight of the body. The reaction and the weight are equal and opposite forces, so the body does not move. When the body is slightly tilted, the reaction acts at the corner or point where there the body is in contact with the table. The weight still acts through the centre of gravity, G. The resultant effect is two opposite parallel forces, which tend to rotate the body to its original position. In stable equilibrium, the centre of gravity, G is at its lowest position.



(a) Stable (b) Unstable (c) Neutral

A diagram of Stable, Unstable and Neutral Equilibrium

STABLE FLY (STABLE FLY; BARN FLY; BITING HOUSE FLY; DOG FLY; POWER MOWER FLY): A fly belonging to the family Muscidae, classified as *Stomoxys calcitrans*. It resembles a housefly but with a distinct proboscis which can pierce the skin and suck blood. It breeds in stable manure and is a serious pest to animals as their bites cause irritation.

STABLE ION: An ion which is not sufficiently excited to dissociate spontaneously into a daughter ion and associated neutral fragment(s) or to react further before reaching the detector.

STACHYOSE: A nonreducing tetrasaccharide made up of two galactose units and one unit each of glucose and fructose. It is found in many of the Labiatae, especially the tubers of *Stachys tuberifera* and in some of the Leguminosae.

STACK: 1. A pile of sheaves of grain, hay or straw. 2. A pillar of rock that forms an islet at the end of a headland.

STACKED BAR GRAPH (STACKED BAR CHART): A graph in which the bars are sub-divided to show additional information. Each bar in the chart represents a unit, and segments in the bar represent different parts or categories of that unit.

STACKING ENERGY: The energy of interaction that favours the face to face packing of purine and pyrimidine base pairs.

STAG: Male hog or cattle castrated after reaching sexual maturity.

STAGGER: A condition of animals in which they stagger about, as in looping-ill and swayback disease. Grass staggers in cattle is caused by hypomagnesaemia.

STAIN: A discolouration or a dye that can be clearly distinguished from the surface, material or medium it is found upon, caused by the chemical or physical interaction of two dissimilar materials. Stains are used in a variety of fields, including in research such as biochemical staining; in technology such as metal staining; in art, for example, wood staining and stained glass. In biology and medicine stains are used to colour cells or thin sections of biological tissues that are normally transparent to make them more clearly visible through a microscope. **Staining** heightens the contrast between the various cell or tissue components. Biological staining is also used to mark cells in flow cytometry and to flag proteins or nucleic acids in gel electrophoresis.

STAINLESS STEEL (INOX STEEL; CORROSION-RESISTANT STEEL, CRES): A widely used alloy of iron, chromium and nickel that resists rusting. It contains a minimum of 10.5 or 11% chromium content by mass. Stainless steel does not stain, corrode or rust as easily as ordinary steel, but it is not stain-proof. There are different grades and surface finishes of stainless steel to suit the environment to which the material will be subjected in its lifetime. It is used where both the properties of steel and resistance to corrosion are required. Stainless steels are used in petroleum refineries and chemical plants for the pipes and tanks, for jet planes and for space capsules, for surgical instruments and equipment, for replacing broken bones because the steels can withstand the action of body fluids. It is also commonly used for cutlery and kitchen fittings because it does not taint food and can be easily cleaned.

STAIRCASE STRUCTURE: A linear-programming problem in which the non-zero entries are arranged in blocks around the main diagonal such that they resemble a staircase.

STALACTITE: Long ice-like mass of accumulated calcium carbonate hanging from the roof or sides of a limestone cavern. It is formed by the evaporation of waters bearing calcium carbonate that has seeped very slowly through the roof of the cavern.

STALAGMITE: Formation similar to a stalactite, but which grows upwards from the floor of a cave that rises from the floor of a limestone cave due to the dripping of mineralised solutions and the deposition of calcium carbonate. The corresponding formation on the ceiling of a cave is known as a **stalactite**. If these formations grow together, the result is known as a **column**.

STALE SEEDBED PLANTING SYSTEM: A useful weed control technique in which a seedbed has received no seedbed preparation tillage just prior to planting. A crop is planted in this unprepared seedbed, and weeds present before or at planting are killed with herbicides.

STALK: 1. The main stem or axis of a plant that is fleshy rather than woody. 2. A supporting part of a plant, for example, a leaf stem petiole or flower stalk pedicel. 3. A slender supporting structure for an organ or body of an animal.

STALK CELL: One of two cells, the other being the body cell, formed by division of the generative cell in the male gametophyte of certain gymnosperms.

STALK ROT: Stalk rot fungus disease of cereal crops such as maize or corn caused by a species of *Fusarium mycotoxins* that can affect human and animal health if they enter the food chain.

STALL: A single compartment in a stable or cow shed where an animal can stand or lie.

STALLION: Adult male horse that is more than four years old which has not been castrated, especially one that is kept for breeding.

STAMEN: The male reproductive organ of a flower in which pollen grains are produced. It consists of an upper fertile part called the anther on a thin sterile stalk known as the filament.

STAMINATE FLOWER: A flower possessing male parts (stamens) but no female parts. An example is the male flowers of holly (Ilex aquifolium).

STAMINODE: A sterile stamen, which may remain rudimentary or partially developed, as in figworts (Scrophularia) or become highly specialised and modified, as in Iris, where it forms a conspicuous part of the flower. **Staminody** describes a condition in which other organs such as petals or sepals become stamens.

STANDARD: 1. An accepted reference sample which is used for establishing a unit for the measurement of physical quantities. A physical quantity is specified by a numerical factor and a unit. For example, a mass could be expressed as 8 grams, a length as 6 centimetres and a time interval as 2 minutes. The **gram** is a mass unit defined in terms of the international kilogram, which serves as the primary standard of mass. The centimetre is defined in terms of the international metre, which is the primary standard of length and is defined as the length of path travelled by light in a vacuum during a time interval of 1/299,792,458 of a second. The **minute** is a time interval defined as 60 seconds, where the second is the international standard of time and is defined as the duration of 9,192,631,770 periods of the radiation corresponding to the transition between the two hyperfine energy levels of the ground state of the caesium-133 atom. 2. In botany, the term is used to refer the upper and largest petal of a papilionaceous flower such as occurring in peas and sweet peas. 3. In botany, a tree of natural size supported by its own stem and not dwarfed by grafting on the stock of a smaller species nor trained upon a wall or trellis.

STANDARD ATMOSPHERE: Pressure defined as 101 325 Pa and used as unit of pressure with the symbol atm.

STANDARD ATOMIC WEIGHTS: Recommended values of relative atomic masses of the elements, revised biennially by the IUPAC Commission on Atomic Weights and Isotopic Abundances and applicable to elements in any normal sample with a high level of confidence. A normal sample is any reasonably possible source of the element or its compounds in commerce for industry and science that has not been subject to significant modification of isotopic composition within a geologically brief period.

STANDARD BOND DISSOCIATION ENTHALPY (BDE; D_0): A measure of the bond strength in a chemical bond. It is defined as the standard enthalpy change when a bond is cleaved by homolysis, with reactants and products of the homolysis reaction at 0 K (absolute zero). It is the enthalpy change that follows the breaking of one mole of covalent bonds, with the molecules and resulting fragments being in their standard states and at standard conditions.

STANDARD CANDLE: An object, usually a type of star or a galaxy, of known intrinsic brightness that can be used in setting up a distance scale. Measurement of the apparent brightness of a standard candle, such as a Cepheid variable, yields its distance.

STANDARD CELL: Electrolytic cell characterised by having a known constant electromotive force (e.m.f). It is a primary cell whose voltage is accurately known and remains sufficiently constant for instrument calibration purposes. For example, the Weston standard cell has a voltage of 1.018636 volts at 20 °C.

STANDARD CONCENTRATION: Chosen value, amount or concentration, denoted by and usually equal to 1 mol dm^{-3}.

STANDARD DEVIATION: A measure of the spread of a set of data. For a normal distribution, the standard deviation gives a measure of the width of the tails of the distribution function. The **sample standard deviation** is the positive square root of the sample variance s^2, where variance is defined as the arithmetic mean of the squares of the deviations of the members of a sample from the arithmetic mean of the sample. The **population standard deviation** σ is a given measure of the variability or dispersion of a population, a data set or a probability distribution. Its formula is: $\sigma = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(X_i - \mu)^2}$, where σ = standard deviation, ⍰ = mean of the set of data points, n = number of data points, X_i = random variables X_1, X_2, X_3, ...X_n. A low standard deviation indicates that the data points tend to be very close to the mean, while a high standard deviation indicates that the points are spread over a large range of values.

STANDARD ELECTRODE POTENTIAL (E^0; E^θ): An electrode potential measured under standard conditions of a temperature of 25°C (298 K), 1 atmosphere pressure and at 1 mole of the activity of redox participants of the half-reaction. In other words, it is the electrode potential specified by a comparison with a standard electrode or the measure of individual potential of a reversible electrode at standard state, which is with solutes at an effective concentration of 1 mol dm^{-3} and gases at a pressure of 1 bar. Electrode potential is the potential difference difference between the electrode in question acting as a cathode and the standard hydrogen electrode acting as an anode. The larger the value of the standard electrode potential, the further the reaction is from equilibrium.

$Li^+ + e^- \longrightarrow Li$	-3.05
$K^+ + e^- \longrightarrow K$	-2.93
$Ca^{2+} + 2e^- \longrightarrow Ca$	-2.87
$Na^+ + e^- \longrightarrow Na$	-2.71
$Mg^{2+} + 2e^- \longrightarrow Mg$	-2.36
$Al^{3+} + 3e^- \longrightarrow Al$	-1.66
$Mn^{2+} + 2e^- \longrightarrow Mn$	-1.18
$Zn^{2+} + 2e^- \longrightarrow Zn$	-0.76
$Fe^{2+} + 2e^- \longrightarrow Fe$	-0.44
$Cd^{2+} + 2e^- \longrightarrow Cd$	-0.40
$Sn^{2+} + 2e^- \longrightarrow Sn$	-0.14
$Pb^{2+} + 2e^- \longrightarrow Pb$	-0.13
$2H^+ + 2e^- \longrightarrow H_2$	0.00
$Cu^{2+} + 2e^- \longrightarrow Cu$	+0.34
$I_2 + 2e^- \longrightarrow 2I^-$	+0.54
$Fe^{3+} + e^- \longrightarrow Fe^{2+}$	+0.77
$Hg_2^{2+} + 2e^- \longrightarrow 2Hg$	+0.79
$Ag^+ + e^- \longrightarrow Ag$	+0.80
$NO_3^- + 4H^+ + 3e^- \longrightarrow NO + 2H_2O$	+0.96
$Br_2 + 2e^- \longrightarrow 2Br^-$	+1.07
$MnO_2 + 4H^+ + 2e^- \longrightarrow Mn^{2+} + 2H_2O$	+1.23

$Cr_2O_7^{2-} + 14H^+ + 6e^- \longrightarrow 2Cr^{3+} + 7H_2O$	+1.33
$Cl_2 + 2e^- \longrightarrow 2Cl^-$	+1.36
$ClO_3^- + 6H^+ + 6e^- \longrightarrow Cl^- + 3H_2O$	+1.45
$MnO^-_4 + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4H_2O$	+1.51
$F_2 + 2e^- \longrightarrow 2F^-$	+2.87

Standard electrode potential (standard redox potential)

STANDARD ENTHALPY OF COMBUSTION: The enthalpy or heat given off, when one mole of a substance is completely burnt in excess oxygen under standard conditions. It is denoted by the symbol ΔH_c^θ.

STANDARD ENTHALPY OF FORMATION (MOLAR ENTHALPY OF FORMATION): The enthalpy or energy change that takes place, when one mole of the substance (compound) is formed from its constituent elements under standard conditions. It is assigned the symbol ΔH_f^θ, where the subscript $_f$ indicates the formation reaction and the superscript $^\theta$ means standard conditions.

STANDARD ENTHALPY OF NEUTRALISATION/ NEUTRALIZATION: The enthalpy or heat given off, when one mole of H^+ ions from an acid reacts with one mole of OH^- ions from an alkali under standard conditions (298 K and 1 atmosphere) to form one mole of water. It is assigned the symbol ΔH_n^θ. The standard enthalpy change of neutralisation for a strong acid and base is -57.3 kJ/mol. Standard enthalpy of neutralisation is calculated by: Q = mcΔT, where Q is the heat energy liberated, m is the mass of the solution, c is the specific heat capacity of the solution and ΔT is the temperature change of the reaction between an acid and a base.

STANDARD ENTHALPY OF SOLUTION: The enthalpy or energy change that occurs when 1 mole of a substance dissolves under standard conditions in 1 dm^3 of water. It is assigned the symbol ΔH^θ_{soln}.

STANDARD ENTROPY: The entropy content of one mole of substance under its standard temperature and pressure (298 K and 1 atmosphere).

STANDARD EQUATION: The canonical form of the equation of a curve derived by a suitable change of variables.

STANDARD ERROR OF THE MEAN, SEM (STANDARD DEVIATION OF THE MEAN): A method used to estimate the standard deviation of a sampling distribution, which is defined as the standard deviation divided by the square root of the number of samples. The standard deviation of a population provides a measure of the scatter of the data and SEM measures the accuracy of the calculated population mean or it is used to give a level of confidence about the sample mean, given as $\sigma_y = \frac{\sigma}{\sqrt{n}}$, where σ is the standard deviation and n is the number of observations. It is statistical index of the probability that a sample mean is representative of the mean of the population from which the sample was drawn.

STANDARD FORM (SCIENTIFIC NOTATION; STANDARD INDEX FORM): A way of writing or expressing very large or very small numbers in a compact form that is easy to use in computations. It is often used in physics, where this extrema values are used regularly. Only one integer appears before the decimal point, the value being adjusted by multiplying by the appropriate power of 10. Any number is expressed as a number between 1 and 10 multiplied by a power of 10 that indicates the correct position of the decimal point in the original number. Numbers greater than 10 are expressed by positive powers of 10 and numbers less than 1 are expressed by negative powers of 10. For example, 43,700 is written as 4.37×10^4 and 0.00526 as 5.26×10^{-3}. The larger the converted number, the more compact it is. For example, 400,000,000,000,000 can be written as 4×10^{14}. In general this is given by $k \times 10^n$, where $1 \leq k < 10$ and n can be either positive or negative integer.

STANDARD GRAVITY (STANDARD ACCELERATION DUE TO FREE FALL): The nominal acceleration of a body in a vacuum near the surface of the Earth. It is defined to be precisely 9.80665 m/s^2. Its symbol is g_0 or g_n.

STANDARD HYDROGEN ELECTRODE (NORMAL HYDROGEN ELECTRODE): A redox electrode which forms the basis of the scale of oxidation-reduction potential. Its absolute electrode potential is estimated to be 4.44 ± 0.02 volts at 25°C, which forms a standard for comparison with other electrodes reactions. Hydrogen's standard electrode potential (E^0) has been set to zero at all temperatures. The potential of any other electrodes are compared with that of the standard hydrogen electrode at the same temperature. The standard hydrogen electrode consists of a platinised platinum electrode dipped in an acidic solution of 1M H^+ and in contact with H_2 at 1 atm pressure. Hydrogen electrode is based on the redox half cell: $2H^+_{(aq)} + 2e^- \rightarrow H_{2(g)}$.

STANDARD MODEL (STANDARD MODEL OF PARTICLE PHYSICS): The theory that explains the three major interactions of elementary particle physics: the strong interaction responsible for nuclear forces; the weak interaction radioactive decay; and the electromagnetic interaction. The standard model extends quantum electrodynamics to explain all three interactions of subnuclear physics in terms of similar basic constituents. It was developed throughout the early and middle 20th century. The current formulation was finalised in the mid 1970s upon experimental confirmation of the existence of quarks. Since then, particles such as bottom quark in 1977, the top quark in 1995 and the tau neutrino in 2000 have been discovered. It has been successful in explaining a wide variety of experimental results and sometimes regarded as a theory of almost everything.

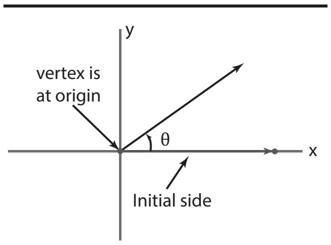
STANDARD MOLALITY: Chosen value of molality, which is usually equal to 1 mol kg^{-1}.

STANDARD NORMAL DISTRIBUTION (Z-DISTRIBUTION): In statistics, a special case of normal distribution when a normal random variable has a mean of 0 and a standard deviation of 1. The normal random variable of a standard normal distribution is called a **standard score** or a **z score**, which is the result of converting from normal distribution to standard normal distribution by the formula $z = (X - \mu)/\sigma$, where X is a normal random variable, μ is the mean and σ is the standard deviation.

STANDARD NOTATION (DECIMAL NOTATION): The normal way in which whole numbers, integers, and decimals are written. It is base-ten place-value numeration. The decimal notation system has ten as its base and it is the numerical base most widely used in the world.

STANDARD PART: The function applicable only to finite non-standard reals, returning the unique 'nearest' standard real number. As every finite non-standard real number is a standard real number plus an infinitesimal, this function is readily defined. A standard infinitesimal is one of the lowest possible order.

STANDARD POSITION OF AN ANGLE: In trigonometry an angle is usually drawn in the standard position as shown in the diagram. An angle is in standard position if its vertex is located at the origin and one ray is on the positive x-axis. The ray on the x-axis is called the initial side and the other ray is called the terminal side. In the diagram, the vertex of the angle B is on the origin of the x and y axis.



Standard Position Of An Angle

STANDARD REACTION QUANTITIES: Infinitesimal changes in thermodynamic functions with extent of reaction divided by the infinitesimal increase in the extent when all the reactants and products are in their standard states.

STANDARD REAL NUMBER: Any non-standard real number that corresponds to an element of the ordinary real numbers.

STANDARD REDUCTION POTENTIAL (STANDARD REDOX POTENTIAL): The potential in volts generated by a reduction half-reaction compared to the standard hydrogen electrode at 25 °C, 1 atm and a concentration of 1 M. The more positive the potential is the more likely it will be reduced. Standard reduction potentials are denoted by the variable E^0 and written in the form of a reduction half reaction. For example, the reduction of water is
$2H_2O + 2e^- \rightarrow H_2 + 2OH^-$ has a $E^0 = 1.776$ V .

STANDARD SCORE (Z SCORE): A statistical measurement, that expresses the degree of standard deviations from the mean of the distribution scores in a group; it is the normal random variable of a standard normal distribution. A z-score of 0 means the score is the same as the mean; if less than 0, the score is less than the mean; if greater than 0, the score is greater than the mean; if equal to 1 the score is 1 standard deviation greater than the mean; if equal to 2, the score is 2 standard deviations greater than the mean; etc.

STANDARD SOLUTION: A solution with an accurately known concentration or a solution of a substance whose concentration is accurately known, expressed in units of moles per Litre (mol/L, often abbreviated to M for molarity). Standard solutions are normally used in titrations to determine the concentration of a substance in solution.

STANDARD STATE: A reference point of a substance used to calculate its properties under different conditions. For a given material or substance, the standard state is the reference state for the material's thermodynamic state properties such as enthalpy, entropy, Gibbs free energy and for many other material standards. In principle, the choice of standard state is arbitrary, although the International Union of Pure and Applied Chemistry (IUPAC) recommends a conventional set of standard states for general use. IUPAC recommends using a standard pressure, P = 105 Pa. Standard state involves a reference value of pressure (usually 1 atmosphere or 100 kPa) or concentration (usually 1M or mol/L) and a temperature of 298.15 K or 273.15 K. Standard state is not standard temperature and pressure (STP) for gases, neither is it standard solutions used in analytical chemistry.

STANDARD SUBTRACTION METHOD: A variation of the standard addition method. In this procedure, changes in the potential resulting from the addition of a known amount of a species which reacts stoichiometrically with the ion of interest, for example, a complexing agent is employed to determine the original activity or concentration of the ion.

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STANDARD TEMPERATURE AND PRESSURE (STP): Formerly known as normal temperature and pressure (N.T.P.), it is a set of standard conditions used as a basis for calculations involving quantities that vary with temperature and pressure. It is used when comparing the properties of gases. Standard temperature is the freezing point of pure water, 0°C (273.15°K). The standard pressure is the pressure exerted by a column of mercury (symbol Hg) 760 mm high, often designated 760 mm Hg. This pressure is also called one atmosphere and is equal to 1.01325×10^6 dynes per sq cm.

STANDARD THERMODYNAMIC QUANTITIES: Values of thermodynamic functions in the standard state characterised by a standard pressure, molality or amount concentration, but not by temperature. Standard quantities are denoted by adding the superscript $^\theta$ or ^{o} to the symbol of the quantity.

STANDARD TIME: Time in any of the 24 time zones into which the Earth is divided, each an hour or 15^o apart, which corresponds to the distance between the longitudes. Longitude 0^o, known as the Greenwich Meridian is the reference point, with each time zone to the east gaining an hour from each other, whilst those due to the east lose an hour.

STANDARD UNIT OF MEASUREMENT: A unit of measure that has been defined by a recognised authority, such as Système Internationale or International (S.I.) System and adopted by convention or by law. Examples of standard units in the metric system are metre, kilogram, second, ampere, kelvin, candela, and mole.

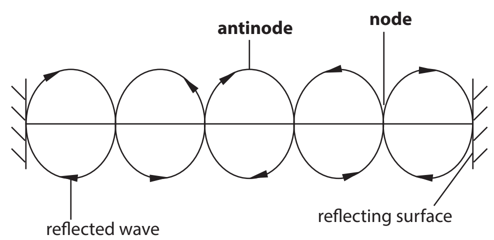
STANDARD VOLUME (MOLAR VOLUME): The volume occupied by one kilogram of molecule (the molecular mass in kilograms) or one mole of any substance at standard temperature of 0 °C (273.15 K) and standard pressure of 1 atmosphere (760 mm mercury, 760mmHg) or 101.325KPa.

STANDARDISED DATA (STANDARDIZED DATA): Data from which the mean of the observations related to the random variable (associated to the data) is subtracted and then divided by the standard deviation of the observations. Thus standardised data have a mean of 0 and a variance of 1. It is expressed as $\frac{x-\mu}{\sigma}$, where x is the observed data, u is the mean of the data set and σ is the standard deviation of the data set.

STANDARDS: Set of rules or codes mandating or defining product performance, for example, grades, dimensions, characteristics, test methods, and rules for use. International product and/or technology or performance standards establish minimum requirements for affected products and/or technologies in countries where they are adopted. On environmental issues, standards reduce greenhouse gas emissions associated with the manufacture or use of the products and/or application of the technology.

STANDING CROP: 1. The total number or dry weight or energy content of individuals of organisms alive in a particular area at any moment. It is measured as the weight or biomass of a given species in a sample selection. 2. A crop such as wheat which is still growing in the field.

STANDING WAVE (STATIONARY WAVE): A form of wave in which the profile of the wave does not move through the medium but remains stationary. This wave is in contrast to a travelling (progressive) wave, in which the profile moves through the medium at the speed of the wave. A standing wave is formed when a travelling wave is reflected back along its own path or when two waves of equal wavelength and amplitude travel in opposite directions at the same velocity through a medium. In a standing wave there are points at which the displacement is zero; these points are known as **nodes**. When the displacement is however maximum, the points are called **antinodes**. The distance between a node and its adjacent antinode is one quarter of a wavelength. Standing waves are present in the vibrating strings of musical instruments. For example, when a violin string is bowed or plucked, it vibrates as a whole, with nodes at the ends and also vibrates in halves, with a node at the centre and in various other fractions, all simultaneously. The vibration as a whole produces the fundamental tone and the other vibrations produce the various harmonics.



Antinodes and nodes

STANDSTILL: The keeping of animals in the same place for 6 days to prevent the spread of disease.

STANNATE: A compound formed by reaction of tin oxides or hydroxides with alkali containing SnO_4^{4-}. Tin oxides are amphoteric or weakly acidic and react to give stannate ions.

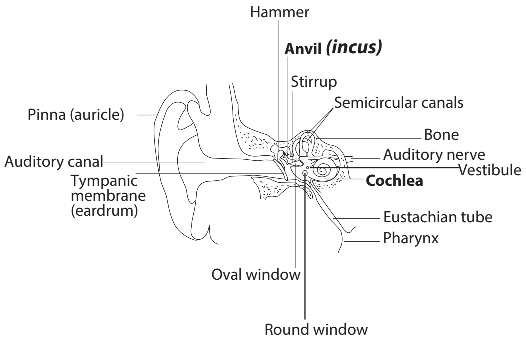
STANNIC: Tin in tetravalent form or relating to or containing tin, especially with a valency of four.

STANNIC ACID (STANNIC ANHYDRIDE; TIN DIOXIDE; TIN OXIDE; TIN PEROXIDE): A white amorphous solid with the chemical formula H_2SnO_3 and melting point 1127 °C (1400 K). It is insoluble in water but soluble in concentrated sulfuric acid. It is used in ceramic glazes and colours, special glasses, putty and cosmetics and as a catalyst.

STANNIC COMPOUNDS: Compounds of tin in its higher (+4) oxidation state with the chemical formula $SnCl_4$. An example is tin(IV) chloride.

STANNOUS COMPOUNDS: Compounds of tin in its lower (+2) oxidation state, an example is tin(II) chloride.

STAPES (STIRRUP): The third of the three ear ossicles of the mammalian middle ear. It is the innermost of the three small bones that transmit vibration to the inner ear.



The anvil (incus) of the human ear

STAPHYLOCOCCAL MASTITIS: A condition of cows caused by several types of staphylococci, especially when accompanied by stress resulting from liver fluke or cold conditions.

STAPHYLOCOCCUS (STAPHYLOCOCCI): A genus of spherical, non-motile, gram-positive bacteria that occur widely as saprotrophs or parasites. The cells occur in grapelike cluster. Many species infect the eyes, skin, mucous membranes and urinary tract and some produce toxins responsible for septicaemia and food poisoning in humans and animals.

STAR: 1. A self-illuminating celestial body that generates nuclear energy within its core or a gaseous mass in space that generates energy by thermonuclear reactions. For example, the Sun is a small star that produces its energy from the fusion of hydrogen into helium. 2. A star-shaped symbol (*) put beside a word, etc. to show importance, merit, rank or quality. 3. The collection of sets containing a given member of a family of sets as a subset. The star of a simplex in a simplicial complex is the set of all simplices in the complex containing the given simplex as a face. 4. The set of points of a given set in a Euclidean vector space such that any segment from such a point to another point of the set lies in the set. The members of the star, see the set which is star-shaped if the star is non-empty. A set is convex precisely if it is its own star. The *Krasnoselskii theorem* is that, in an n-dimensional Euclidean space, a compact set is such that every $n + 1$ points of the set can be seen from within the set is star-shaped.

STAR CLUSTER: A group of stars that are close enough to each other for them to be physically associated and held together by mutual gravitational attraction. Stars belonging to the cluster are formed together from the same cloud of interstellar gas and have approximately the same age and initial chemical composition. Observations of star clusters are of great importance in the studies of stellar evolutions since the stars in a given cluster are approximately at the same distance from Earth. There are two types of star clusters: Open or galactic clusters and Globular clusters. Open or galactic clusters are fairly loose systems of between a few hundred and a few thousand members. Globular clusters are approximately spherical collections of between ten thousand and a million stars.

STAR CURVE (ASTROID): A hypocycloid (the path of a point on a circle rolling inside another circle) with the diameter of the fixed circle four times the diameter of the rolling circle. It has four cusps.

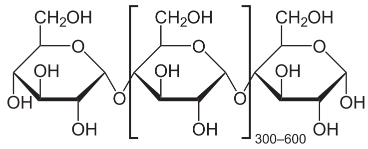
STAR MACROMOLECULE: A macromolecule (a molecule containing a very large number of atoms) containing a single branch point from which linear chains (arms) emanate. A star macromolecule with linear chains (arms) attached to the branch point is termed an ***n*-star macromolecule**, for example, a five-star macromolecule. If the arms of a star macromolecule are identical with respect to constitution and degree of polymerisation, the macromolecule is termed a **regular star macromolecule**. If different arms of a star macromolecule are composed of different monomeric units, the macromolecule is termed a **variegated star macromolecule**.

STAR TOPOLOGY: A type of physical topology with a single access point or a switch at the centre of the topology; all the other nodes are connected directly to this point.

STAR-LIKE REGION: In mathematics, it is a region, R, that contains a point z_0, having the property that if z_1 is any other point in R, then the segment $[z_0, z_1]$ connecting a given point and any other point in a region lies completely within the region R.

STARBURST GALAXY: A galaxy undergoing an unusually fast rate of star formation as compared to the long-term average rate of star formation in the galaxy; or the star formation rate observed in most other galaxies.

STARCH (AMYLUM): The white, granular, polysaccharide with the general chemical formula $(C_6H_{10}O_5)_n$, where n may range from 100 to several thousands. It is manufactured in the green leaves of all green plants from excess glucose produced during photosynthesis and serves the plant as a reserve food supply. In the dark, the starch thus formed is rapidly broken down to sucrose and transported to other organs. It is contained in large amounts in plants such as the roots of cassava (*Manihot esculenta*), which yield tapioca, the rhizomes of arrowroot (*Maranta arundinacea*), the stem pith of some sago palms such as species of Metroxylon and Arenga, the tubers of potato and the grains of various cereals, especially maize, wheat, rice and sorghum. It is used by humans and animals to produce energy, thus forming the most important component in their diet. It is also processed to produce many of the sugars in commercial foods. It can be used as a thickening, stiffening or gluing agent. It is also used as adhesive in the papermaking industries.



Chemical Structure of Starch (amylose)

STARCH-STATOLITH HYPOTHESIS: A possible explanation for the way plants perceive gravity. Most gravity-sensitive organs contain specialised cells called statoliths, in their cytoplasm. The statoliths gradually fall to the lower side of the cell under the influence of gravity. The settling out of these grains in reoriented cells is said to provide an internal stimulus, which is transmitted to the growing region of the organ so that appropriate amendments to the growth pattern can be made. These growth movements restore the original position of the cells and hence of the whole plant part.

STARK EFFECT (ELECTROCHROMIC EFFECT; ELECTRIC FIELD EFFECT): The shifting and splitting of lines in the spectra of atoms due to the presence of a strong electric field. In terms of quantum mechanics the stark effect is described by regarding the electric field as a perturbation on the quantum states and energy levels of an atom in the absence of an electric field. This application of perturbation theory was first used in quantum mechanics. Stark effect can also be understood in terms of the classical electron theory of Lorentz. The stark effect for hydrogen atoms was also described by the Bohr's theory of the atom.

STARK-EINSTEIN LAW (EINSTEIN PHOTOCHEMICAL EQUIVALENCE LAW): The second law of photochemistry, which states that in a photochemical process such as a photochemical reaction, for each photon of light absorbed by a chemical system, only one molecule is activated to initiate a process involving several molecules.

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START CODON (INITIATION CODON): The triplet of nucleotide on a messenger RNA molecule at which the process of translation is initiated. In eukaryotes, the start codon is AUG, which codes for the amino acid methionine. In bacteria the start codon can be either AUG coding for N-formlymethionine or GUG, coding for valine.

START MENU: A feature of the Windows operating system that provides quick access to programs, folders and system settings. By default, the Start menu is located in the lower-left corner of the Windows desktop. In Windows 95 through Windows XP, the Start menu can be opened by clicking the "Start" button. In newer versions of Windows, such as Windows Vista and Windows 7, the Start menu can be opened by clicking the Windows logo. Some keyboards also have a Windows key that opens the Start menu when pressed.

START/STOP TRANSMISSION (ASYNCHRONOUS TRANSMISSION): A data transmission where the communication can start and stop at any time. Data sent through an asynchronous transmission contains a start bit and stop bit, helping the receiving end know when it has received all of its data.

STARTER: 1. A culture of bacteria used to inoculate or start growth in a fermentation process. 2. A device used with a ballast to start an electric-discharge lamp such as a fluorescent lamp. 3. An electric controller for starting an electric motor, for bringing it up to normal speed and for stopping it. It consists of a circuit of devices such as relays, magnetic contactors and safety devices such as cut-off switches. Examples of methods used in motor starters to achieve special effects are star-delta and direct-on-line.

STARTER SOLUTIONS: Dilute mixtures of soluble fertilisers used when plants are transplanted. The fertiliser material easily dissolves in water and the nutrients are readily available for plant uptake. Starter solutions are used primarily for transplanted vegetables such as tomato, eggplant, pepper, muskmelon, watermelon, cabbage, cauliflower and broccoli.

STARTLE DISPLAY: A response by an animal to discovery by a predator in which the potential victim exposes some previously hidden markings, for example, eyespot in order to surprise its attacker. This may enable the victim to escape or divert the attention of its attacker elsewhere. Some camouflaged insects use this as a second form of defence.

STARY: In zoology, the term refers to an individual animal that has been left alone or has parted ways with others of its flock during movement or migration.

STASIS: 1. The cessation of blood flow in the blood vessels or other body fluid, such as lymph resulting from a static balance between opposing forces. 2. The phenomenon in which types of living creatures show little or no change over long periods of time. An example is the similarity between living species of shrimps and their ancestors, the shrimp-like crustacean dated at 425 million years old.

-STAT: A suffix used to denote instruments such as thermostats or substances that maintain a controlled state.

STAT-: A prefix attached to the name of an electrical unit to obtain a name for a unit in the electrostatic system of units. Examples are statvolt and statcoulomb.

STATE: 1. In mathematics, any of the indexed random variables of a Stochastic process. Also, any of the possible outcomes of a Markov chain. 2. Denoting or pertaining to a state variable; for example, a state equation or state constraint is a restriction that involves only the state variables. 3. In thermodynamics, it is a description of a system characterised by macroscopic independent parameters or variables, such as pressure, volume, absolute temperature, as well as the number of particles in the system and any other constraints to which the system may subject be subjected to. This is often referred to as the **macrostate**. In statistical physics the states of a system will be the solutions of the time-independent Schrodinger equation for that system, referred to as **microstates**. The state is often represented by a point in a $2s$ dimensional space, the phase space, where s is the number of degrees of freedom of the system.

STATE CHANGE (PHASE TRANSITION): A change in the physical state (solid, liquid, gas) or the phase transition of a material. For example, melting, boiling, evaporation, solidification and condensation are changes of state or phase transition. Phase transition is brought about by a change in the temperature of the system. The temperature at which the change of state occurs is called **transition temperature**, denoted by T_c. Examples of phase transitions are the **gas-liquid transition** called **condensation**, the **liquid-solid transition** called **freezing**, the **normal-to-superconducting transition** in electrical conductors, the **paramagnet-to-ferromagnet transition** in magnetic materials and the **superfluid transition** in liquid helium. It also includes transitions involving amorphous or glassy structures, spin glasses, charge-density waves and spin-density waves.

STATE FUNCTION (FUNCTION OF STATE; STATE QUANTITY; STATE VARIABLE): A property that is determined by the state of the system, not on the way in which the system acquired that state. A state function describes the equilibrium state of a system. For example, internal energy, enthalpy and entropy are state functions because they describe quantitatively an equilibrium state of thermodynamic systems. In contrast, mechanical work and heat are process quantities because they describe quantitatively the transition between equilibrium states of thermodynamic systems.

STATE MACHINE (FINITE-STATE MACHINE): A mathematical abstraction of computation used in the design of programs of sequential logic circuits. In diagrammatic representation, it is depicted as states (nodes) and the transitions between these states.

STATE OF A SYSTEM: The values of all pertinent macroscopic variables such as composition, volume, pressure and temperature of a system.

STATE SYMBOL: A symbol usually in brackets, used in chemical equation to indicate the physical state of the substances present. The symbols include (s) for solid, (l) for liquid, (g) for gas and (aq) for aqueous solution.

STATE-TO-STATE KINETICS: A branch of chemical kinetics concerned with the dynamics of reactions in which the reactant species are in known quantum states and in which the quantum states of the products are determined.

STATEMENT: 1. Information given in the form of words or mathematical symbols. 2. In computer science, one of the sequences of commands written in a high-level language that directs the computer to perform a specified action.

STATES OF MATTER (STATE): The form or condition in which a substance exists, either as a solid, liquid or a gas. Plasma is sometimes regarded as the fourth state of matter. Solid is the state in which intermolecular attractions keep the molecules in fixed spatial relationships. Liquid is the state in which intermolecular attractions keep molecules in proximity, but do not keep the molecules in fixed relationships. Gas is that state in which the molecules are comparatively separated and intermolecular attractions have relatively little effect on their respective motions.

STATIC ELECTRICITY: The build-up of electric charge on the surface of objects. The basic principles behind static electricity is that objects are made of atoms that are normally electrically neutral because they contain equal numbers of positive charges (protons in their nuclei) and negative charges (electrons in shells surrounding the nucleus). The phenomenon of static electricity requires a separation of positive and negative charges. When two objects are in contact, electrons may move from one object to the other, which leaves an excess of positive charge on one material and an equal negative charge on the other. When the objects are separated they retain this charge imbalance. The static charges remain on an object until they either are grounded or are quickly neutralised by a discharge. Static electricity can be with current (or dynamic) electricity, which can be delivered through wires as a power source. Although charge exchange can happen whenever any two surfaces come into contact and separate, a static charge only remains when at least one of the surfaces has a high resistance to electrical flow such as an electrical insulator.

STATIC FIELDS MASS SPECTROMETER: An instrument which can separate selected ion beams with fields which do not vary with time. The fields are generally both electric and magnetic.

STATIC PRESSURE: The pressure of a fluid at rest or in motion exerted perpendicularly to the direction of flow.

STATIC RANDOM ACCESS MEMORY (SRAM): A type of RAM that stores data using a static method, in which the data remains constant as long as electric power is supplied to the memory chip. This is different to DRAM (dynamic RAM), which stores data dynamically and constantly needs to refresh the data stored in the memory. Because SRAM stores data statically, it is faster and requires less power than DRAM.

STATIC STABILITY: In meteorology, the stability of the atmosphere in the vertical direction to vertical displacements.

STATIC WEBSITE: A website, which contains Web pages with fixed content. Each page is coded in HTML and displays the same information to every visitor. Static sites are the most basic type of website and are the easiest to create. Unlike dynamic websites, they do not require any Web programming or database design. A static site can be built by simply creating a few HTML pages and publishing them to a Web server.

STATICS: The study of objects and particles at rest or in equilibrium under the action of outside forces; or the branch of mechanics concerned with bodies that are acted upon by balanced forces and couples so that they remain at rest or in uniform motion in a straight line.

STATION: 1. A very large farm, especially in New Zealand and Australia specialising in raising sheep or cattle. 2. A fixed point used by surveyors as a reference.

STATION MODEL: A diagram on a weather map showing weather data for a specific area made by a specific weather station at a given time. Data shown include temperature, dew point, wind, cloud cover, air pressure, pressure tendency, and precipitation.

STATIONARY ORBIT: An orbit lying in or approximately in the plane of the equator for which the orbital period is equal to the spin period of the central body. From the ground, the satellite would appear to stand still, hovering above the surface in the same spot, day after day.

STATIONARY FRONT: A region where a warm air mass and a cold air mass meet but neither has enough force to move the other. On a weather map, it is shown as alternating red semicircles and blue triangles. The blue triangles point in one direction and the red semicircles point in the opposite direction.

STATIONARY PHASE: 1. The phase of a culture of microorganisms or animal and plant cells cultured *in vitro* that follows the exponential growth phase and in which there is little or no growth. In some cases it is a phase of product formation, for example, formation of secondary metabolites. 2. One of the two phases forming a chromatographic system. It may be a solid, a gel or a liquid. If a liquid, it may be distributed on a solid. This solid may or may not contribute to the separation process. The liquid may also be chemically bonded to the solid (bonded phase) or immobilised onto it (immobilised phase).

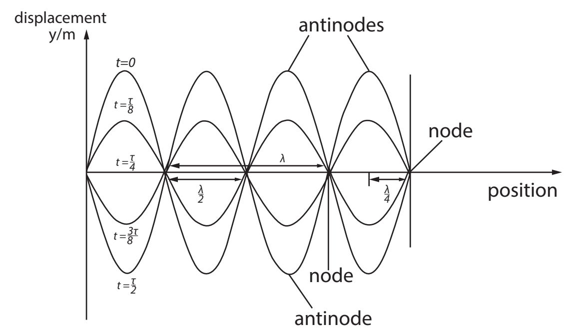
STATIONARY PHASE VOLUME: The volume of the liquid stationary phase or the active solid in the column. The volume of any solid support is not included. In the case of partition chromatography with a liquid stationary phase, it is identical to the liquid phase volume.

STATIONARY POINT: 1. A point on a graph where the tangent to the curve is parallel to the x axis. 2. A point at which a planet's apparent motion changes from direct to retrograde motion or vice versa. 3. For a function of several variables, a point at which all partial derivatives are zero.

STATIONARY PROCESS: In probability theory, a stochastic process is said to be stationary if the joint distribution of any subset of the sequence of random variables is invariant with respect to shifts in the time index.

STATIONARY STATE: The state of a system when it has an energy level permitted by quantum mechanics. Transitions from one stationary state to another can only occur by the emission or absorption of appropriate quanta of energy which can be in the form of photons.

STATIONARY WAVE (STANDING WAVE): A form of wave in which the profile of the wave does not move through the medium but remains stationary. This wave is in contrast to a travelling (progressive) wave, in which the profile moves through the medium at the speed of the wave. A stationary wave is formed when a travelling wave is reflected back along its own path or when two waves of equal wavelength and amplitude travel in opposite directions at the same velocity through a medium. In a stationary wave there are points at which the displacement is zero; these points are known as **nodes**. When the displacement is however maximum, the points are called **antinodes**. The distance between a node and its adjacent antinode is one quarter of a wavelength. Stationary waves are present in the vibrating strings of musical instruments. For example, when a violin string is bowed or plucked, it vibrates as a whole, with nodes at the ends and also vibrates in halves, with a node at the centre and in various other fractions, all simultaneously. The vibration as a whole produces the fundamental tone and the other vibrations produce the various harmonics.



Stationary wave

STATISTIC: In statistics, any function of a number of random variables usually identically distributed and used to estimate the true value of a corresponding parameter such as population parameter. It is a characteristic of a sample.

STATISTICAL ANALYSIS: A collection of methods used to process large amounts of data and report overall trends. It is particularly useful when dealing with noisy data. It provides ways to objectively report on how unusual an event is based on historical data.

STATISTICAL DEPENDENCE: In statistics, a type of relation in which two random variables are not independent; the result of one is affected by that of another. If X and Y are dependent events, the probability of both events occurring is the product of the probability of the first event and the probability of the second event once the first event has occurred. Two events, X and Y are said to be positively dependent if the conditional probability, P(X/Y), of X given Y is greater than the probability, P(X) of X alone, written as P(X & Y) > P(X) · P(Y). They are negatively dependent if the inequalities are reversed and these relations may depend upon the specific values of Y. In the case of equality for all values, the variables are independent.

STATISTICAL EQUILIBRIUM: The state of a system in which the coefficient of probability, P, is constant with respect to time, expressed as $\frac{\partial P}{\partial t}=0$.

STATISTICAL INFERENCE: The process of drawing conclusions about a population by conducting hypothesis testing and obtaining estimates on data collected from a sample.

STATISTICAL MECHANICS: A branch of physics in which statistical methods are applied to the microscopic components of a system in order to predict its macroscopic properties. In classical statistical mechanics each particle is regarded as occupying a point in phase space.

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STATISTICAL TABLES: Tables that show values of the cumulative distribution functions, probability density functions or probability functions of certain common distributions for different values of their parameters. Statistical tables conserve space and reduce explanatory and descriptive statements to a minimum and used to determine whether or not a particular statistical result exceeds some required significance level.

STATISTICS: 1. A branch of mathematics concerned with the collection, analysis and making inferences from data or concerned with the inferences that can be drawn from numerical data on the basis of probability in terms of samples and populations. Statistical tools summarise past data through such indicators as the mean and the standard deviation and also predict future events using frequency distribution functions. 2. Quantitative data on any subject, especially data comparing the distribution of some quantity for different subclasses of the population, such as government publications of birth and death rates, often called vital statistics. Statistics as computed values is two or more computed values from a set of data or distribution. One value is a "statistic" (singular). For example, the range of a data is statistic, but the range, and the mean are statistics (plural).

STATOBLAST: A bud that is produced asexually by bryozoans that survives when the bryozoan dies and stores food. It is encased in a chitinous covering, which enables it to remain dormant and withstand long periods of unfavourable conditions such as drought and extreme temperatures until it germinates to produce a new colony.

STATOCYST (OTOCYST): A sense organ found in many invertebrates such as crustaceans, bivalves and cnidarians. It consists of a fluid filled sac lined with sensory hairs and particles of sand, lime, etc., that stimulates different hairs to maintain balance and to indicate position in space.

STATOLITH: A membrane bound group of starch grains in plant cells that is believed to act as a sensor to gravity. Starch statoliths are found in cells at the root tips and in the tissues close to the vascular bundles in shoots. Under the influence of gravity they drift to the lower side of the cell. Their mechanism of action in triggering the transport of growth substances across the plasma membrane is not understood. One theory is that statolith exerts pressure on membrane systems inside the cell.

STATOR: The stationary part of a motor, dynamo, turbine or other working machine about which a rotor turns. It may be either a permanent magnet or an electromagnet. Where the stator is an electromagnet, the coil which energises it is known as the **field coil** or **field winding**.

STATUS BAR: The bottom portion of Internet browsers and other program windows that displays the current status of the web page or window currently being viewed. For example, an open folder window on the desktop may display the number of items in the folder and how many items are selected. Photoshop uses the status bar to display the size of the current image, the zoom percentage and other information. Web browsers use the status bar to display the Web address of a link when the user moves the cursor over it. It also shows the status of loading pages and displays error messages.

STEADY MOTION: A motion in which the partial derivative with respect to time of the velocity of the body at fixed positions in the current configuration is zero.

STEADY STATE: 1. In enzyme kinetic analysis, the time interval when the rate of reaction is approximately constant with time. 2. The state of a living cell in which the concentrations of many molecules are approximately constant because of a balancing between their rates of synthesis and breakdown.

STEADY-STATE THEORY: The theory of an expanding universe whose average density remains constant, matter being continuously created throughout it to form new stars and galaxies at the same rate that old ones recede from sight. A steady-state universe has no beginning or end and its average density and arrangement of galaxies are the same as seen from every point. Galaxies of all ages are intermingled. Much evidence obtained since the 1950s contradicts the steady-state theory and supports the big-bang model.

STEALTH TECHNOLOGY (LOW OBSERVABLE TECHNOLOGY; LO TECHNOLOGY): Technology used to reduce radar, heat, visual and infra-red detection of aircraft, missile, naval vessels and also other military vehicles.

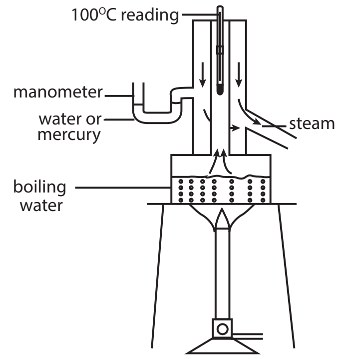
STEAM: Water vapour above its boiling point or the vapour that is formed when water is boiled. It is widely used in chemical and other industrial processes and for the generation of power. It is also much used for space heating, for propelling vessels, ships and also locomotive train engines.

STEAM DISTILLATION: A method of distilling, separating or purifying by bubbling steam through a liquid such as oil that is immiscible with water.

STEAM ENGINE: A heat engine in which the thermal energy of steam, usually supplied by a boiler is converted into mechanical energy. It consists of a cylinder fitted with a piston and valve gear to enable high-pressure steam to be admitted to the cylinder when the piston is near the top of its stroke. As the steam expands, it pushes the piston, which is usually connected to a crank on a flywheel to produce rotary motion.

STEAM JACKET: A covering or casing surrounding the cylinders and heads of a steam engine to keep the surfaces hot and dry.

STEAM POINT (UPPER FIXED POINT): The temperature at which the maximum vapour pressure of water is equal to the standard atmospheric pressure on the Celsius scale. It is the boiling point of pure water whose isotopic composition is the same as that of sea water at standard atmospheric pressure. It is assigned a value of 100°C on the International Practical Temperature Scale of 1968. To calibrate the upper fixed point of a thermometer, a double-walled copper vessel known as hypsometer has some water poured into it to cover the lower part of the vessel. The vessel has an outlet attached to its side to allow excess steam that may increase the inside pressure to flow out. In order to monitor the pressure, a manometer is attached to the side as shown in the diagram. The thermometer is pushed through a hole in a cork at the top part of the vessel. The thermometer is adjusted in such a way that the top part of the mercury thread is just outside the cork but with its bulb above the water. The water in the vessel is heated until it boils. To ensure that boiling takes place at 100°C, the surrounding jacket and the flame are adjusted so that the mercury levels in the two arms of the manometer are the same. When the top part of the mercury thread becomes stationary, its position is labelled as 100°C (steam point or upper fixed point).



Calibration of steam point/ upper fixed point

STEAM UP: To feed a cow before it calves to prepare it for the next lactation.

STEARATE (OCTADECANOATE): A salt or ester of stearic acid or the ananoic form of stearic acid with chemical formula $C_{17}H_{35}COO^-$. Examples are sodium stearate, $Na(C_{17}H_{35}COO)$, calcium stearate $Ca(C_{17}H_{35}COO)_2$ and magnesium stearate, $Mg(C_{17}H_{35}COO)_2$. Scum formed in hard water is mainly calcium and magnesium stearate.

STEARIC ACID (OCTADECANOIC ACID): A solid saturated fatty acid with the chemical formula $C_{18}H_{36}O_2$. It is soluble in alcohol and ether but not in water. It is prepared by treating animal fat with water at a high pressure and temperature, leading to the hydrolysis of triglycerides. It can also be obtained from the hydrogenation of some unsaturated vegetable oils. It is found in many fats and oils and it occurs widely in animal and vegetable fat. It is used to make soap and candles and also as a lubricant.

STECKLING: Young sugar beet plants grown in seedbeds in summer, to be transplanted in the autumn or following spring.

STEEL: Any of a number of alloys consisting predominantly of iron with varying proportions of carbon of up to 1.7% as the major alloying element and other elements such as manganese, chromium and nickel. An alloy of steel and 11-12% of chromium is called **stainless steel**.

STEER: A male of the cattle family, especially a young bull that has been castrated before reaching sexual maturity and reared for beef.

STEERAGE HOE: A hoe mounted behind a tractor and steered by the driver to avoid damage.

STEEPEST DESCENT (GRADIENT METHOD): In mathematics, an iterative method, due to Fermat, of finding the minimum of a real valued differentiable function of n variables by moving to the minimum of the function (an exact line search method) along the line in the direction of the steepest negative gradient (a steepest descent direction) at successive points on the curve or surface. This is the prototype of non-linear optimisation methods. The corresponding maximisation technique is called **steepest ascent**.

STEGANOGRAPHY: In information technology, it is an encryption technique used to conceal data within a seemingly innocuous one.

STEINER POINT: The point of a compact convex set C in Euclidean n-space as $s(C) = \int_S x\delta_C^*(x)\sigma(dx)$, where S is the $n-1$ sphere, δ_C^* is the support function of C and σ is normalised Lebesgue measure. This produces an element of C and s(.) is Lipschitz in the metric defined by Hausdorff distance.

STEINER TRIPLE SYSTEM: In mathematics, a type of block design consisting of a collection of n-element base set in such away that each l-element subset of the base set lies in exactly one of the m-element subsets, denoted S(l, m, n). The seven-point finite projective plane is an example of an S(2, 3, 7), that is, this set of 7 points has lines of 3 points determined by any pair of points. These systems have uses in group theory and sphere-packing problems.

STEINITZ' EXCHANGE THEOREM: The theorem that if, for $k < m$, u_i $(1 \le i \le m)$ and v_j $(1 \le j \le m)$ are linearly independent subsets of a vector space then there exists a permutation $\{1, ..., m\}$ such that $u_1, ..., u_k, v_{\pi(1)}, ..., v_{\pi(m-k)}$ are linearly independent. Consequently, any finite-dimensional vector space has a unique dimension and any subspace of the same dimension is the space itself.

STELE (VASCULAR CYLINDER): The cylindrical core of the stem and roots of a plant that contains the sap-conducting vascular tissues and varying amounts of packing tissue pith

STELL: A stone shelter for sheep and cattle in an upland area.

STELLAR EVOLUTION: The sequence of changes that occur in a star during its lifetime, from birth to extinction.

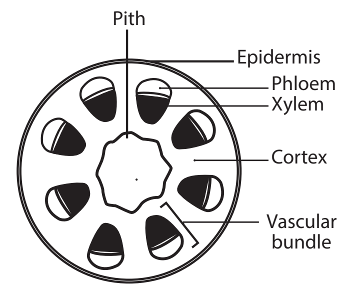
STELLAR NURSERY: An area within a nebula where active new star formation occurs or a large and relatively dense cloud of cold gas and dust in interstellar space from which new stars are born.

STELLAR POPULATION: In astronomy, it is the classification of stars according to their chemical composition as determined by spectroscopy. The two classes include population I stars that are relatively young, found in spiral galaxies, especially the blue stars of high luminosity; and population II, which are much older stars, more evolved stars of lower metallic content that includes many high luminosity red giants and many variable stars.

STELLAR WIND: The radial outflow of matter, chiefly protons, electrons and helium nuclei, from the atmosphere of a very hot star or a stream of ionised particles ejected from the surface of a star. In the case of the Sun, the stellar wind is called **solar wind**. The rate at which the stellar matter is emitted varies with the lifetime of the star.

STELLATE: Star-shaped, as in hairs with several branches radiating from the base. **Stelpilous** describes such things with stellate hairs.

STEM: That part of a flowering plant that grows vertically upwards, bearing the buds, leaves and flowers. Its functions include transporting water, mineral salts and carbohydrates, raising the leaves above the soil for maximum light, raising the flowers and thus enhancing pollination and in green stems, carrying out photosynthesis.



Transverse section of a stem

STEM CANKER: A fungal disease affecting many types of trees causing death of the affected tissues of the stem.

STEM CELL: A cell that does not differentiate itself but undergoes unlimited division to form specialised cells, which either remain as stem cells or differentiate to form specialised cells. Examples are cells in the bone marrow which divide to produce daughter cells that differentiate into various types of immune cell. Also stem cells in the intestine continually divide to replace cells sloughed off the gut lining.

STEM EELWORM: Plant nematodes that affect cereals, in particular oats. The stem swells and this prevents it from growing any ears.

STEM TUBER: A tuber that forms from thickened rhizomes or stolons. The top sides of the tuber produce shoots that grow into typical stems and leaves and the under sides produce roots. They tend to form at the sides of the parent plant and are most often located near the soil surface. Examples of plants that produce stem tubers are potato and yam.

STEM-AND-LEAF DIAGRAM: In statistics, it is a type of histogram used in displaying quantitative data in graphical format, with the tens digit of each piece of data forming the stem and the units digit forming the leaves. The data is then grouped in class intervals of ten.

STENO: A prefix meaning narrow. For example, stenohaline aquatic organisms can tolerate only small variation in the salinity of water; or stenothermal organisms live only within a narrow temperature range.

STENOPETALLOUS: Of or relating to petals which are narrow.

STENOPHYLLOUS: Of or relating to leaves which are narrow.

STEP FUNCTION: The function that takes only a finite number of different constant values on each of a succession of disjoint intervals whose union is its domain, such as, for example, $[x]$, the integral part of any real x. Such functions are important in the definition of some forms of integration. Its graph is a series of line segments that look like a series of small steps.

STEP-LENGTH METHOD (STEP-SIZE METHOD): An approximate descent method that is based on finding the appropriate length of step to take in the descent direction, as opposed to a trust region method.

STEFAN-BOLTZMANN LAW: The law, which states that the total energy radiated per unit surface of a black body in unit time is proportional to the fourth power of its thermodynamic temperature. It is the constant of proportionality known as Stefan constant (or Stefan Boltzmann constant) and has the value 5.6697 × 10^{-8} $Js^{-1}m^{-2}K^{-2}$. The law was postulated by an Austrian physicist, Joseph Stefan (1835-1893) and theoretically derived by Ludwig Boltzmann (1844-1906). It is represented by the equation, $Q = e\sigma AT^4t$, where Q is the thermal energy emitted, e is the emissivity of the body, which varies from zero to one. σ is a constant called the Stefan – Boltzmann constant and it has a value of σ = 5.67 × 10^{-8} $Wm^{-2}k^{-4}$, A is the area of the emitting body, T is absolute temperature and t is the time.

STEPPE: A wide grassy plain with no trees apart from those near rivers and lakes found in the temperate regions of Eurasia. It is like the prairie in North America. It may be semi-desert or covered with grass or shrubs or both, depending on the season and latitude with summer temperatures of up to 40°C (313 K) and in winter -40°C (233 K).

STEPPING STONE METHOD: A method of solving transportation problems that adapts the simplex method by exploiting the special structure of the problem.

STEPWISE REACTION: A chemical reaction with at least one reaction intermediate and involving at least two consecutive elementary reactions.

STERADIAN: The dimensionless supplementary SI unit of solid angle, defined as the solid angle subtended at the centre of a sphere of radius r by a portion of the surface of the sphere whose area, A, equals r^2. Its symbol is sr.

STEREO MICROSCOPE: A microscope with two optically separate eyepieces to make viewed objects look three dimensional. It is often used to study the surfaces of solid specimens or to carry out close work such as dissection, microsurgery, watch-making, circuit board manufacture or inspection and fracture surfaces as in fractography and forensic engineering.

STEREOBLOCK MACROMOLECULE: A block macromolecule composed of stereoregular and possibly nonstereoregular, blocks.

STEREOCHEMICAL FORMULA (STEREOFORMULA): A three-dimensional view of a molecule either as such or in a projection.

STEREOCHEMICAL NON-RIGIDITY: The capability of a molecule to undergo fast and reversible intramolecular isomerisation, the energy barrier to which is lower than that allowing for the preparative isolation of the individual isomers at room temperature. It is conventional to assign to the sterochemically non-rigid systems those compounds, whose molecules rearrange rapidly enough to influence NMR line shapes at temperatures within the practical range (from -100 ^oC to 200^oC) of experimentation.

STEREOCHEMISTRY: A branch of chemistry concerned with the study of the spatial arrangement or the three-dimensional configuration of the atoms within a molecule and the way that these affect the properties of that molecule.

STEREOCILIUM (STEREOCILIA): 1. Nonmotile tufts of secretory microvilli on the free surface of cells of the male reproductive tract. 2. A relatively short non-motile cilium, some 20 to 300 of which form an array on the apical surface of a sensory hair cell. The stereocilia are usually associated with a single, much longer characteristic of undulipodia. Instead, their shaft contains numerous fine longitudinal actin filaments. Bending of the kinocilium causes the array of stereocilia to bend en masse, thereby changing the pattern of impulses in the adjacent sensory neuron.

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STEREOCONVERGENCE: The predominant formation of the same stereoisomer or stereoisomer mixture of a reaction product when two different stereoisomers of the reactant are used in the same reaction. When that product involved in the reaction is one enantiomer the result has been called **enantioconvergence**.

STEREOELECTRONIC: Pertaining to the dependence of the properties (especially the energy) of a molecular entity in a particular electronic state (or of a transition state) on relative nuclear geometry. The electronic groundstate is usually considered, but the term can apply to excited states as well. Stereoelectronic effects arise from the different alignment of electronic orbitals in different arrangements of nuclear geometry.

STEREOELECTRONIC CONTROL: Control of the nature of the products of a chemical reaction (or of its rate) by stereoelectronic factors. The term is usually applied in the framework of an orbital approximation. The variations of molecular orbital energies with relative nuclear geometry (along a reaction coordinate) are then seen as consequences of variations in basis-orbital overlaps.

STEREOGRAPHIC PROJECTION: 1. Also known as **azimuthal orthomorphic** in mapping, it is a perspective conformal, azimuthal map projection in which points on the surface of a sphere or spheroid, such as the Earth, are conceived as projected by radial lines from any point on the surface to a plane tangent to the antipode of the point of projection. Circles project as circles through the point of tangency, except for great circles which project as straight lines. The principal navigational use of the projection is for charts of the Polar Regions. 2. In crystallography, it is a method of displaying the positions of the poles of a crystal in which poles are projected through the equatorial plane of the reference sphere by lines joining them with the South Pole for poles in the upper hemisphere and with the North Pole for poles in the lower hemisphere. 3. In mathematics, it is the projection of the Riemann sphere onto the Euclidean plane performed by emanating rays from the north pole of the sphere through a point on the sphere.

STEREOISOMER: One of two or more isomers with the same molecular formula and functional groups, but different configurations. They differ in the spatial arrangement of the atoms. For example, 1-bromopropane (C_3H_7Br) and 2-bromopropane ($CH_3CH(Br)CH_3$) are stereoisomers. **Stereoisomerism** is the existence of compounds that have the same molecular formula and functional groups, but differ in the arrangement of groups in space.

STEREOLOGY: The study of the properties of three dimensional structures and objects based on two-dimensional views of them. It provides practical techniques for extracting quantitative information about a three-dimensional material from measurements made on two-dimensional planar sections of the material. It is an important and efficient tool in many applications of microscopy such as petrography, materials science and biosciences including histology, bone and neuroanatomy.

STEREOMETRY: A branch of geometry dealing with solid figures or the art or science of measuring solid figures. Stereometrically pure forms include the cone, cube, pyramid and sphere and were important elements in Neo-Classicism.

STEREOMORPHIC: A radially symmetrical object such as cotton, hibiscus, crocus, lily, okra flowers and starfish, so that a line drawn through the middle of the structure along any plane will produce a mirror image on either side.

STEREOMUTATION: A change of configuration at a stereogenic unit brought about by physical or chemical means.

STEREOREGULAR: Describing a compound that has a regular spatial arrangement of atoms within its molecules or a polymer that has a regular pattern of side groups along a chain.

STEREOREGULAR MACROMOLECULE: A regular macromolecule essentially comprising only one species of stereorepeating unit.

STEREOREGULAR RUBBER: A member of a group of stereoregular synthetic rubbers which are produced by a solution polymerisation process using special catalysts.

STEREOREPEATING UNIT: A configurational repeating unit having defined configuration at all sites of stereoisomerism in the main chain of a regular macromolecule, a regular oligomer molecule, a regular block or a regular chain.

STEREOSCOPE: An optical instrument that simulates binocular vision by presenting slightly different pictures to the two eyes so that an apparent three-dimensional image is produced. The stereoscope is useful in making relief maps by aerial photographic survey.

STEREOSCOPIC VISION: Ability to perceive a single, three-dimensional image from the simultaneous but separate images delivered to the brain by each eye. In other words, it is the sensations from a common object received by the two eyes are superimposed and as a result of the slight differences in the fields and the superimposition of the fields, the effects of depth and shape of the object are attained.

STEREOSELECTIVE POLYMERISATION: Polymerisation in which a polymer molecule is formed from a mixture of stereoisomeric monomer molecules by incorporation of only one stereoisomeric species.

STEREOSELECTIVE SYNTHESIS: A chemical reaction (or reaction sequence) in which one or more new elements of chirality are formed in a substrate molecule and which produces the stereoisomeric (enantiomeric or diastereoisomeric) products in unequal amounts.

STEREOSELECTIVITY: The preferential formation in a chemical reaction of one stereoisomer over another. When the stereoisomers are enantiomers, the phenomenon is called **enantioselectivity**

and is quantitatively expressed by the enantiomer excess; when they are diastereoisomers, it is called **diastereoselectivity** and is quantitatively expressed by the diastereoisomer excess.

STEREOSPECIFIC POLYMERISATION: Polymerisation in which a tactic polymer is formed. However, polymerisation in which stereoisomerism present in the monomer is merely retained in the polymer is not regarded as stereospecific. For example, the polymerisation of a chiral monomer, for example, D-propylene oxide (Dmethyloxirane), with retention of configuration is not considered as a stereospecific reaction. However, selective polymerisation, with retention, of one of the enantiomers present in a mixture of D- and L-propylene oxide molecules is classified as stereospecific.

STEREOSPECIFICITY (STEREOSPECIFIC): A reaction is termed stereospecific if starting materials differing only in their configuration are converted into stereoisomeric products. According to this definition, a stereospecific process is necessarily stereoselective but not all stereoselective processes are stereospecific. Stereospecificity may be total (100%) or partial. The term is also applied to situations, where a reaction can be performed with only one stereoisomer. For example, the exclusive formation of *trans*-1,2-dibromocyclohexane upon bromination of cyclohexene is a stereospecific process, although the analogous reaction with (*E*)-cyclohexene has not been performed.

STERIC EFFECT: The influence of the spatial configuration of reacting substances upon the rate, nature and extent of reaction. The sizes and shapes of atoms and molecules, the electrical charge distribution and the geometry of bond angles influence the courses of chemical reactions.

STERIC FACTOR: A factor introduced into simple versions of the collision theory of reactions to take care of the fact that the reaction probability depends on the certain mutual orientations of the reactant molecules.

STERIC HINDRANCE: Repulsion of an approaching potential reactant by a sterically congested compound. The chemical reaction is slowed down or prevented because large groups on a reactant molecule hinder the approach of another reactant molecule.

STERIC ISOTOPE EFFECT: A secondary isotope effect attributed to the different vibrational amplitudes of isotopologues. For example, both the mean and mean-square amplitudes of vibrations associated with C–H bonds are greater than those of C–D bonds. The greater effective bulk of molecules containing the former may be manifested by a steric effect on a rate or equilibrium constant.

STERIGMA (STERIGMATA): A spore-bearing stalk found in certain fungi. In the Basidiomycota, it develops as an outgrowth of the basidium.

STERILISATION (STERILIZATION): 1. Any surgical operation to terminate the possibility of reproduction. 2. The killing or elimination of living organisms such as bacteria and fungi. Drying or freezing kills many species of bacteria and causes others to become inactive. Heat or moist heat above a certain temperature kills all bacteria. Sterilisation of many different objects, such as spacecraft and surgical instruments, are important facets of bacteriological work.

STERILISED/STERILIZED MILK: Milk prepared for human consumption by the elimination of nearly all microorganisms. Sterilised milk goes through a severe form of heat treatment, which destroys nearly all the bacteria in it. The milk is pre-heated to around 50ºC (323 K), then homogenised, after which it is poured into glass bottles which are closed with an airtight seal. The bottles are carried on a conveyor belt through a steam chamber, where they are heated to a temperature of between 110-130ºC (383-403 K) for approximately 10-30 minutes. Then they are cooled using a cold water tank, sprays or in some cases, atmospheric air and then crated. The sterilisation process results in a change of taste and colour and also slightly reduces the nutritional value of the milk, particularly the B group vitamins and vitamin C. Sterilised milk can be stored for approximately six months without the need for refrigeration. Once opened it must be treated as fresh milk and used within 5 days.

STERILITY: 1. The state of being free from microorganisms. Drying or freezing kills many species of bacteria and causes others to become inactive. Heat or moist heat above a certain temperature kills all bacteria. Sterilisation of many different objects, such as spacecraft and surgical instruments, are important facets of bacteriological work. 2. The permanent inability to reproduce offspring. Sterility can be caused by improper functioning of the sex organs such as the testes in the male and the ovaries in the female of humans. It can also be caused by the abnormal release of hormones that control the activities of the sex organs. Sometimes it results from surgical removal of the womb called **hysterectomy**, the ovaries or testes in order to treat a serious disorder such as cancer. In many cases, failure to reproduce is caused not by sterility but by disorders that cause infertility. A **sterile** is a living organism that cannot reproduce; or an object which is free from contamination by living micro-organisms or free of germs.

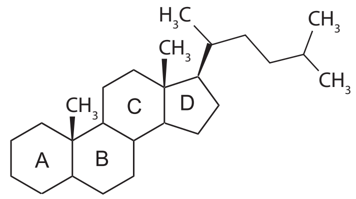
STERN-GERLACH EXPERIMENT: A classic experiment which demonstrated the quantum mechanical nature of atomic magnetic moments. A beam of silver atoms entered a strong non-uniform magnetic field. Classical physics leads to the expectation that the beam should be broadened into a continuous band, however, in agreement with quantum mechanics, the beam was split into two separate beams.

STERNOCLAVICULAR JOINT: A synovial joint (specifically, a plane joint) at which the medial end of the clavicle fits into the shallow socket provided by the clavicular notch of the manubrium sterni and the upper surface of the first costal cartilage. It is the only point of articulation of the upper limb with the axial skeleton and it determines the range of its movements.

STERNOMASTOID MUSCLE (STERNOCLEIDOMASTOID MUSCLE): A long muscle in the neck, extending from the mastoid process of the mastoid bone to the sternum and clavicle. It serves to rotate the neck and head.

STERNUM (BREASTBONE): 1. An elongated or rod shaped bone in terrestrial vertebrates, located on the ventral side of the thorax that articulates with and provides support for the clavicles (collarbones) of the pectoral girdle and with most of the ribs. It connects to the rib bones via cartilage, forming the anterior section of the rib cage, helping to protect the lungs, heart and major blood vessels from physical trauma. In mammals, from anterior to posterior the sternum has three sections, which include the following: (i) the manubrium that articulates with the clavicles and first ribs; (ii) the mesosternum, often divided into a series of segments called the sternebrae, to which the remaining true ribs are attached; and (iii) the xiphisternum, the posterior segment. 2. The ventral portion of each segment of the exoskeleton of arthropods .

STEROID: A group of natural or synthetic organic compounds, which are chemically related with a molecular core or nucleus of 17 carbon atoms in a characteristic three-dimensional arrangement of four rings. Examples are cholesterol, dexamethasone (a synthetic glucocorticoid), the corticosteroids, anabolic steroids and androgens (male sex hormones), oestrogens and progesterone (female sex hormones) and the bile acids (steroid carboxylic acids).



Chemical structure of a steroid

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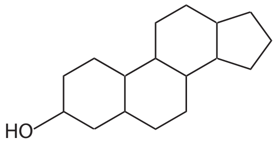
STEROL: Any of a group of steroid-based alcohol having a hydrocarbon side-chain of 8-10 carbon atoms at the 17-beta position and a hydroxyl group at the 3-beta position. Sterol exits as either free sterol or as esters of fatty acids. Examples of animal sterols known as **zoosterol** include cholesterol and lanosterol. Plant sterol is known as **phystosterol** and includes beta-sitosterol. Fungal sterol is known as **mycosterol** of which ergosterol is an example.



Chemical Structure of Sterol

STETHOSCOPE: Any of various instruments used for listening to sounds produced within the body, especially the heart and lungs, to help evaluate their condition.

STIBANES: The saturated hydrides of tervalent antimony, having the general formula Sb_nH_{n+2}. Hydrocarbyl derivatives of SbH_3 belong to the class stibines.

STIBINE: A colourless slightly soluble poisonous gas with an offensive odour and the chemical formula SbH_3. It is made by the action of hydrochloric acid on an alloy of antimony and zinc. **Stibines** are stibines and compounds derived from it by substituting one, two or three hydrogen atoms by hydrocarbyl groups R_3Sb. $RSbH_2$, R_2SbH and R_3Sb ($R \neq H$) are called primary, secondary and tertiary stibines, respectively.

STIBNITE: A mineral form of antimony(III)sulfide with the chemical formula Sb_2S_3. It is a steel-grey crystalline solid, often containing extractable amounts of lead, mercury and silver in addition to the antimony. It is the chief ore of antimony.

STICKER: A substance added to a fungicide or bactericide preparation to help it to stick to the sprayed surface.

STICKING COEFFICIENT: The ratio of the rate of adsorption to the rate at which the adsorptive strikes the total surface, covered and uncovered. It is usually a function of surface coverage, of temperature and of the details of the surface structure of the adsorbent.

STICKY END (COHESIVE END): A single unpaired strand of nucleotides protruding at the end of double stranded DNA molecules. It is able to join with a complementary single strand, such as the sticky end of another DNA molecule, forming a large single stranded molecule.

STIFF LAMB DISEASE: Muscle degeneration in lambs caused by a deficiency of vitamin E.

STIGMA: 1. The glandular sticky surface at the tip of the pistil of a flower involved in reproduction which receives the pollen. 2. A mark on the skin indicating a medical condition. 3. A coloured mark or spot, often resembling an eye, found on some protozoans and invertebrates, especially butterflies and other lepidopterans. 4. The area of the ovarian surface where the Graafian follicle bursts through during ovulation to release the ovum.

STILL: Apparatus for the distillation of liquids, especially alcohol.

STILLBIRTH: The death of the product of conception before complete expulsion or extraction from its mother. Death is indicated by the absence of all signs of life, such as beating of the heart or pulsation of the umbilical cord. Death of the foetus before this duration of pregnancy is referred to as **miscarriage**.

STIMULANT: A drug that stimulates an organ. Nervous system stimulants range from alcohol (an apparent stimulant only), caffeine and hallucinogenic drugs, to drugs liable to induce convulsions. Cardiac stimulants include digitalis and adrenaline and are used in cardiac failure and resuscitation respectively. Bowel stimulants have a laxative effect. Uterus (womb) stimulants (oxytocin and ergometrine) are used in obstetrics to induce labour and prevent postpartum haemorrhage.

STIMULATED EMISSION: The emission of electromagnetic radiation by an atomic electron or an excited molecular state, when it interacts with incident radiation of the same frequency. A photon created in this manner has the same phase, frequency, polarisation and direction of travel as the photons of the incident wave.

STIMULUS: Any detectable change in the environment of an organism that provokes a behavioural or physiological response in the organism. When a stimulus is applied to a sensory receptor, it elicits or influences a reflex via stimulus transduction. The ability of an organism or organ to respond to external stimuli is called **sensitivity**. A stimulus is often the first component of a homeostatic control system. When a sensory neuron and a motor neuron communicate with each other, it is called a **nerve stimulus**. Any of the five senses will accommodate to a particular stimulus.

STIMULUS FILTERING: The process of selecting only a few useful sensory information of the great number of stimuli that an animal receives from its environment, so that only relevant stimuli reach the brain. Many sensory organs are adapted so that they receive stimulus only within a certain range. For example, the human eye only detects colours in the visible region of the spectrum, but not in the ultraviolet or infrared regions.

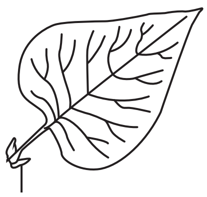
STIMULUS GENERALISATION: The act or process by which an organism responds to a new stimulus similar to but distinct from the conditioned stimulus.

STIPE: A supporting stalk or stem-like structure, especially the stalk of a pistil, the petiole of a fern frond or the stalk that supports the cap of a mushroom. **Stipel** is the small structure at the base of a leaflet, giving it support.

STIPES: The second or bottom mouth part of some insects and crustaceans or the eyestalk of a crab or crayfish.

STIPULA: A newly sprouted feather.

STIPULE: Any of a pair of small growths resembling leaves found at the base of a petiole in some plants such as roses and beans.


Stipule

STIRLING ENGINE: A heat engine consisting of a hot cylinder and a cold cylinder separated by a regenerator acting as a heat exchanger. It consists of oscillating pistons enclosed in cylinders. In its operation, heat is applied to the hot cylinder, causing the working fluid within it to expand and drive the piston. The fluid is cooled in the regenerator before entering the cold cylinder, where it is compressed by the piston and driven back to be heated in the regenerator before entering the hot cylinder again. It was invented by Reverend Dr Robert Stirling (1790-1878), Scottish clergyman. Although it is costly to produce, it is very efficient.

STIRLING NUMBERS: The coefficients in an expansion of positive integral powers of a variable in terms of factorial powers or vice-versa. **Stirling number of the first kind**, denoted $s(n, k)$ is generated by $s(0, 0) = 1$; $s(n, 0) = 0 (n > 0)$, and for $0 < k < n$, $s(n + 1, k) = s(n, k - 1) + ns(n, k)$. 2. **Stirling number of the second kind**, denoted $S(n, k)$ is the number of partitions of $[n]$ into exactly k non-empty subsets. It is generated by $S(n, n) = 1$ $(n > 0)$, $S(n, 0) = 0$ $(n \geq 0)$ and for $0 < k < n$, $S(n + 1, k) = S(n, k - 1) + kS(n, k)$.

STIRLING'S FORMULA: The asymptotic formula that approximates the value of the factorial function or the gamma function. Its simplest formula is of the form $\frac{\Gamma(s+1)}{(s/e)^s\sqrt{2\pi s}} = 1 + o(1)$ as $s \to \infty$. Elements of the matrix must be real numbers in the closed interval [0, 1].

STIRRING: Moving an implement in a circular motion repeatedly in a liquid or other substance such as paint in order to, for example, mix, agitate, dissolve its contents. A **stirring rod** is a device that is used in science laboratories for agitating or stirring a solution in order to cause a quicker reaction. It is also used to spread liquid substances on a solid surface and to pour liquids slowly, especially dangerous chemicals. Its length is between 10 and 40 centimetres with a thickness of about half a centimetre and usually made of borosilicate. It is mostly used in chemistry and biology experiments.

STIRRUP (STAPES): The third of the three ear ossicles of the mammalian middle ear. It is the innermost of the three small bones that transmit vibration to the inner ear.

STOCHASTIC: 1. Of, or involving a random element or a sequence of random variables. 2. In mathematics, a stochastic matrix is a square matrix in which the elements are non-negative and the sum of the elements in all rows equals 1, called **row stochastic** or in all columns called **column stochastic**. It is doubly stochastic if it satisfies both conditions. 3. A stochastic process is any process in which there is a random element or a sequence of random variables known as a time series, which depends on some parameter, which may be discrete or continuous, but is often taken to represent time. Another basic type of a stochastic process is a random field, whose domain is a region of space, in other words, a random function whose arguments are drawn from a range of continuously changing values. Stochastic processes are very important in non-equilibrium statistical mechanics and disordered solids. These processes can be time dependent, in which case a variable that changes with time does so in such a way that there is no correlation between different time intervals. The Fokker-Planck equation and the Langevin equation that describe stochastic processes are called **stochastic equations**. Statistical methods and theory of probability are necessary in analysing stochastic processes and their equations. Brownian movement is an example of a stochastic process. Other examples of processes modelled as stochastic time series include stock market and exchange rate fluctuations, signals such as speech, audio and video, medical data such as a patient's electrocardiograph (EKG), echoencephalograph (EEG), blood pressure or temperature and random movement such as Brownian motion or random walks.

STOCHASTIC PROCESS: Any process in which there is a random element or a sequence of random variables known as a time series. Another basic type of a stochastic process is a random field, whose domain is a region of space, in other words, a random function whose arguments are drawn from a range of continuously changing values. Stochastic processes are very important in non-equilibrium statistical mechanics and disordered solids. These processes can be time dependent, in which case a variable that changes with time does so in such a way that there is no correlation between different time intervals. The Fokker-Planck equation and the Langevin equation that describe stochastic processes are called **stochastic equations**. Statistical methods and theory of probability are necessary in analysing stochastic processes and their equations. Brownian movement is an example of a stochastic process. Other examples of processes modelled as stochastic time series include stock market and exchange rate fluctuations, signals such as speech, audio and video, medical data such as a patient's electrocardiograph (EKG), echoencephalograph (EEG), blood pressure or temperature and random movement such as Brownian motion or random walks.

STOCHASTIC THEORIES: Theories that treat reaction rates in terms of the probabilities of transitions between the various energy levels in the reactant molecules.

STOCHASTIC VARIABLE (CHANCE VARIABLE; RANDOM VARIABLE): A variable whose value is determined by the result of a random experiment and may take any of a range of values, which are either continuous or discrete that cannot be predicted with certainty but only described probabilistically. It may also be defined as a measurable function defined on a probability space having range in the interval [0, 1].

STOCHASTICALLY CLOSED: In statistics, of a set of states of a Markov chain, a subset of states such that there is a zero probability of leaving the subset once it is entered.

STOCK: 1. Animals or plants derived from a common ancestor. 2. A plant with root onto which another plant is grafted. 3. A software for fixed assets management and stock control developed in 2004. Stocktaking process is carried using a hand-held mobile terminal equipped with barcode reader or RFID technology. Every object is identified by a barcode with the unique Asset Register Number linked with the exact record in the Fixed Assets Register (FAR). 4. A product or material kept in storage until needed for use or transferred to some ultimate point for use, for example, crude oil tankage or paper-pulp feed. 5. Designation of a particular material, such as bright stock or naphtha stock. 6. An usually discordant, batholith-like body of intrusive igneous rock not exceeding 103.6 square kilometres in surface exposure and usually discordant. 7. Common name for any species of the genus *Matthiola*, *Malcomia maritima* (Virginia stock) and for the wallflower, all belonging to the family Cruciferae (mustard family) and for a carnation of the family Caryophyllaceae (pink family). Stock is classified in the division Magnoliophyta, class Magnoliopsida order Capparales, family Cruciferae; and order Caryophyllales, family Caryophyllaceae. Most are herbs indigenous to the Mediterranean region and to South Africa. A few are widely cultivated, both in greenhouses and in gardens, for the fragrant blossoms, usually purplish in the wild but of various colours in horticultural types.

STOCK BREEDER: A farmer who specialises in breeding livestock.

STOCK BULL: A bull kept for breeding purpose in a pedigree herd.

STOCK FARMING: The rearing of livestock for sale.

STOCKHOLM CONVENTION: IUPAC convention on signs of electromotive forces and electrode potentials.

STOCKING DENSITY: The number of animals kept on a specific area of land or the number of animals on a particular piece of land at a particular point in time. Stocking density is calculated by dividing the total number of livestock units by the area (in hectares). The amount of forage required by one animal unit (AU) for one month is called an Animal Unit Month (AUM).

STOCKING RATE: The number of animals on a pasture during a month or grazing season, usually expressed in animal unit months (AUM) per unit area. One AUM is the amount of forage required by an animal unit (AU) for one month or the tenure of one AU for a one-month period. If one AU grazes on an area of rangeland for six months, that tenure is equal to six AUs for one month or six AUMs. In general, the number of animal units, multiplied by the number of months they are on the range equals the number of AUMs used. Using a stocking rate too high for the land to support animals over a period of time can result in overgrazing. Conversely, a stocking rate which does not use the available forage to its optimum is an inefficient use of the land. Stocking rates must be reduced when drought is likely to extend into the rapid-growth windows of dominant forage and browse species.

STOCKMAN: A farm worker who looks after animals, especially cattle.

STOCKPROOF: A fence of pen which prevents livestock from escaping.

STOCKY: An animal with short strong legs.

STOICHIOMETRIC AMOUNTS (STOICHIOMETRIC RATIO): The exact molar amounts (number of moles) or ratio of reactants and products that appear in a balanced chemical equation, assuming that the reaction proceeds to completion and all reagent is consumed, there is no shortfall of reagent and no residues remain.

STOICHIOMETRIC COMPOUND: A compound whose molecules have the component elements present in exact proportions as demanded by a simple molecular formula or combined in exact whole-number ratios.

STOICHIOMETRIC MIXTURE: A mixture of substances that are in the exact proportions required for a given reaction. For example, for hydrocarbon fuels in air, that mixture which combusts completely to produce carbon dioxide and water.

STOICHIOMETRIC REACTION: A reaction in which the reactants combine in simple whole-number-ratios to form products or a reaction which goes to completion, rather than stopping partway at an equilibrium point.

STOICHIOMETRY: The quantitative study of reactants and products in a chemical reaction or the relative proportions in which elements form compounds or in which substances react. In other words, it is a branch of chemistry that deals with numerical relationships in reactions.

S

STOKES: A centimetre–gram–second (cgs) unit of kinematic viscosity, which is equal to the ratio of the viscosity of a fluid in poises to its density in grams per cubic centimetre. One stokes = $10^{-4}m^2s^{-1}$. It was named after Sir George Gabriel Stokes (1819-1903), British physicist and mathematician. The International System of Units (SI) is the currently used system of units, and the square metress per second ($m^2 s^{-1}$) has replaced stokes.

STOKES PARAMETERS: Four parameters, denoted s_0, s_1, s_2, s_3 that determine the state of polarisation of a wave of monochromatic radiation from observations of the beam. In the field strength, the Stokes' parameters are quadratic and can be determined by intensity measurements or by a polariser. The parameters are not independent but related by $s_0^2 = s_1^2 + s_2^2 + s_3^2$. The determination of Stokes' parameters has astrophysical applications, which include gaining information concerning the mechanism of electromagnetic radiation from such objects as pulsars.

STOKES RADIATION: The electromagnetic radiation that occurs in the Raman Effect, when the frequency of the scattered radiation is lower than the frequency of the unscattered radiation (energy in transition from the photon to the scattering molecule).

STOKES SHIFT: The difference, usually in frequency units, between the spectral positions of the band maxima or the band origin of the absorption and luminescence arising from the same electronic transition. Generally, the luminescence occurring at a longer wavelength than the absorption is stronger than the opposite. The latter may be called an anti-Stokes shift.

STOKES, SIR GEORGE GABRIEL (1819-1903): British physicist and mathematician, born in Ireland and studied at Cambridge University, where he became a professor of mathematics all his life. He is best known for Stokes' law, concerning the movement of objects in a fluid. A centimetre–gram–second (cgs) unit of kinematic viscosity, stokes, was named after him.

STOKES' LAW: A law that predicts the frictional force, F on a spherical ball moving through a viscous medium. It was named after Sir George Gabriel Stokes (1819-1903), British physicist and mathematician. It states that at low velocities, the frictional force on a spherical body moving through a fluid at constant velocity is equal to 6π times the product of the velocity (v), the fluid viscosity (η) and the radius of the sphere $(a): F = 6\pi\eta av$. The sphere accelerates until it reaches a steady terminal speed. At the terminal speed, the viscous force (F) acting upwards and the upthrust (U) due to the fluid on the sphere are together equal to the weight (W) of the sphere acting downward: $W = F + U$. It implies that $\frac{4}{3}\pi r^3\rho g = 6\pi\eta r v + \frac{4}{3}\pi r^3 \sigma g$, where ρ and σ are the densities of the spherical ball and the fluid, respectively.

S

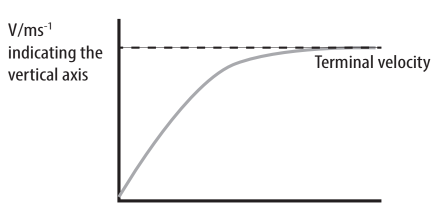


Fig I: Velocity/Time graph showing the motion of a body through a viscous fluid.

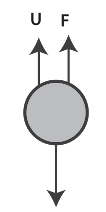


Fig. II: Forces acting on a body moving in a viscous fluid

STOKES' THEOREM (THE GENERALISED STOKES' THEOREM): A theorem that relates a surface integral of the curl of the vector field to a line integral of the vector field around the boundary of the surface. The theorem states that if a surface S, which is smooth and simply connected (any closed curve on the surface can be contracted continuously into a point without leaving the surface) is bounded by a line L the vector F defined in S satisfies $\int_S curlF.ds = \int_L F.dl$ where l is the distance.

STOLON: 1. Also known as a runner, it is a long stem or shoot that arises from the central rosette of a plant that runs along the ground and may form new plants where it touches the ground. 2. A budding of the body wall in simple organisms like the extension of some colonial organism such as hydroids that anchors to a rock or a substrate.

STOLONISING (STOLONIZING): A process of establishing vegetation in which sprigs (a shoot, stem or twig cut from a plant) are broadcast onto the soil surface.

STOMA (STOMATA): 1. Tiny pore(s), usually on the lower side of a leaf or on the stem of vascular plants which is the site where water vapour and gaseous exchange between the plant tissues and the atmosphere takes place. 2. Any small opening or pore on the surface of a living organism. 3. An opening either natural such as the mouth or surgically created, which connects a portion of the body cavity to the outside environment. Any hollow organ can be manipulated into an artificial stoma as necessary. This includes the oesophagus, stomach, duodenum, ileum, colon, pleural cavity, urethras, bladder and kidney pelvis.

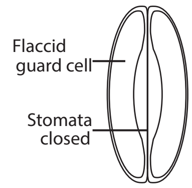


Fig. I: Stoma (closed)

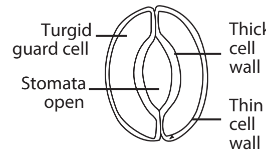


Fig. II: Stoma (open)

STOMACH: The portion of the vertebrate alimentary canal between the small intestine and the oesophagus, which expands or contracts with the amount of food in it. It is a muscular sac just below the diaphragm, where ingested food is stored and undergoes preliminary digestion. Peristalsis mixes food with enzymes and hydrochloric acid from glands in its lining and moves the resulting chyme toward the small intestine. It has four regions: the cardia leads down from the oesophagus; the fundus curves above it; the body is the largest part; and the antrum narrows to join the duodenum at the pyloric valve.

STOMACH INSECTICIDE: Any insecticide that must be taken into the alimentary canal to be effective. Stomach poisons are often inorganic compounds containing such elements as arsenic, mercury or fluorine.

STOMACHICS: Agents or substances containing properties that assist digestion and increase appetite. Certain herbs can be effective to achieve this; such herbs include cannabis, celery, cloves, dandelion, ginseng, and papaya.

STOMATAL APPARATUS: The stomata and guard cells that control the size of the stoma.

STOMATAL CONDUCTANCE: The rate at which water vapour passes through the stomata of a plant per unit area, typically measured in millimoles per square metre per second. It varies between plants, depending on the distribution density, size and pore thickness of the stomata and in the same plant over time according to the difference in vapour pressure between the inside of the plant and the external environment and the degree of opening of the stomatal pores.

STOMIUM: A region of thin-walled cells in certain spore-producing structures that rupture to release the spores. For example, in the sporangium of the fern *Dryopteris*, the stomium ruptures when the annulus dries out.

STONE: 1. A single piece of rock. 2. A piece of rock shaped or finished for a particular purpose, especially a piece of rock that is used in construction. Examples are a coping stone, a paving stone, a gravestone or tombstone, grindstone, millstone or whetstone, milestone or boundary stone. 3. A gem or precious stone. 4. Something, such as a hailstone, resembling a stone in shape or hardness. 5. The hard covering enclosing the seed in certain fruits, such as the cherry, plum or peach. 6. A calculus or mineral concretion in an organ, such as the kidney or gallbladder or other body part.

STONE CELL BRACHYSCLEREID (BRACHYSCLEREID): A relatively short, more or less isodiametric, sclerenchyma cell (sclereid). Brachysclereids are usually present singly or in small groups, although they occur in large numbers in some tissues, such as the fleshy part of the pear fruit.

STONE TRAP: A trough with a trap door in a combine harvester, which prevents stone from passing into the concave.

STONE-GROUNDFLOUR: A type of flour made by grinding with millstone.

STONE-CECH COMPACTIFICATION: A method for constructing a universal map from a topological space X to a completely regular Hausdorff space S, such that any continuous function from the original space to a compact space has a unique continuous extension to the Hausdorff space.

STONE-WEIERSTRASS THEOREM: The extension of the Weierstrass approximation theorem that gives conditions for an algebra of continuous complex-valued functions that every continuous function defined on a closed interval [a, b] can be uniformly approximated as closely as desired by a polynomial function. Because polynomials are among the simplest functions and because computers can directly evaluate polynomials, this theorem has both practical and theoretical relevance, especially in polynomial interpolation.

STOOK: Several corn sheaves gathered together in a field to form a small pyramid.

STOOL: 1. A piece of excrement. 2. The base of a plant from which shoots or shockers sprout. **Stooling** is the cutting back of a tree or shrub to ground level to encourage new growth from the base.

STOP: 1. In botany, to remove the growing tip of a shoot to encourage lateral growth. 2. In physics, it is a circular aperture that limits the effective size of a lens in an optical system. It may be adjustable, as the iris diaphragm in a camera or have a fixed diameter, as the disc used in some telescopes. 3. In computing, it is a command to bring a program or command to a halt. At the command prompt, the keyword KILL and the process ID or name is typed to halt the program or command.

STOP COCK: A valve inserted in a pipe by which the flow of a liquid or gas through the pipe can be regulated.

STOP CODON (TERMINATION CODON): The triplet of nucleotides on a messenger RNA molecule which signals a termination of the process of translation. It is recognised by proteins called release factors, which bind to the A side of the ribosome. It effectively stops the formation of polypeptide chain at that point.

STOPPED-FLOW TECHNIQUE: A technique for investigating fast reactions in solution. The reactant solutions are rapidly mixed and flow through a tube. The composition or properties of the mixture are monitored at some point in the tube. The flow tube is suddenly stopped and the change of signal with time is used to elucidate the kinetics of the process. This technique is especially useful for reactions occurring in the millisecond range.

STOPPER: A plug or cork used for closing a bottle, tube, drain, etc. Laboratory rubber stopper or rubber bung is a tapered plug used mainly in chemical laboratory to seal the openings of test tubes, flasks and other laboratory glassware to prevent a gas or liquid from escaping its container during a scientific experiment or contamination of samples. Some specific experiment requires a specific material. For example, the M35 Green Neoprene is used for chemical resistance. For food fermentation, M18 white natural gum is preferred. For high temperature application, red or white silicone rubber stopper should be used.

STOPPING POTENTIAL: 1. The potential needed to stop the ejection of an electron from a metal surface when an incident beam of energy greater than the work potential of a metal is directed on it. 2. The negative potential on a collector at which a photoelectric current becomes zero. The stopping potential is independent of its intensity for a given frequency of incident radiation.

STOPPING POWER (LINEAR ENERGY TRANSFER, LET; LINEAR STOPPING POWER): The energy transferred per unit path length by a moving high-energy charged particle such as an electron or proton to the atoms and molecules along its path. It is a measure of the ability of matter to reduce the kinetic energy of a particle passing through it, which is the energy loss of a particle per unit distance or the energy lost by a charged particle passing through a substance per unit length of path. It is proportional to the square of the charge on the particle and increases as the velocity of the particle decreases. It is of particular importance when the particles pass through living tissues as it modifies the effect of a specific dose of radiation.

STOPPING RULE (TERMINATION CRITERION): A criterion that determines when to terminate an approximative procedure, either because enough accuracy has been obtained (measured variously) or because a solution can be ruled out or too much effort has been expended.

STOPWATCH: A watch with a hand that can be started and stopped when desired or a hand held timepiece designed to measure the amount of time elapsed from a particular time when activated to when the piece is deactivated.

STORAGE: 1. The act of keeping commodities, possessions and property and the accommodations for those goods and products. 2. In computer science, any physical device in or on which computer information can be kept. A microcomputer has two main types of storage. Its random access memory (RAM) represents temporary storage that the microprocessor uses for programs, work in progress and various types of internal work-control information. The computer's disk drives and other external storage media represent facilities for holding information on a more permanent basis. It also includes read-only memory (ROM), which is a permanent, non erasable medium for holding necessary information, including startup instructions and input/output procedures, etc.

STORAGE AREA NETWORK (SAN): A network of storage devices that can be accessed by multiple computers. Each computer on the network can access hard drives in the SAN as if they were local disks connected directly to the computer. This allows individual hard drives to be used by multiple computers, making it easy to share information between different machines.

STORAGE DRYING: A method of drying bales of hay by blowing air through them.

STORAGE MANAGEMENT SUBSYSTEM (SMS): A system developed by IBM to help manage storage resources. With SMS a user is capable of managing backup, data security, data migration, data recovery and data deletion.

STORAGE POLYSACCHARIDE: The energy reserves which are stored in plants and animal cells when there is excess of carbon available. The types of energy stored are carbohydrates or starch in plants and glycogen in animal cells

STORAGE RING: A container consisting of a set of magnets set in a doughnut-shaped ring around which charged particles from an accelerator can be kept circulating until they are used. The rings are designed like synchrotrons, except that they do not accelerate the particles circling within them but supply just sufficient energy to make up for losses (mainly synchrotron radiation) or annular vacuum chambers in which charged particles can be stored, without acceleration, by a magnetic field of suitable focusing properties. Two interlaced storage rings are used in Geneva at CERN, containing protons rotating in opposite directions. At the intersections very high collision energies (up to 1700G eV) can be achieved.They are used to effectively stretch the duty cycle of a particle accelerator or to produce colliding beams of particles, resulting in a greater possible centre of mass energy.

STORE: 1. Cattle or lambs bred or bought for fattening. The animals are usually reared on farm and then sold on to dealers or other farmers. 2. To copy the data in the computer's memory to an external storage device such as a disk, tape or flash memory card.

STORIED (STOREYED): A term used to describe the axial cells and rays of wood that are arranged in horizontal series as viewed in tangential section.

STORM: Occasion of violent weather conditions or disturbance of the ordinary conditions of the atmosphere attended by wind, rain, snow, sleet, hail or thunder and lightning. Types of storms include: (i) the **extratropicalcyclone**, the common, large-scale storm of temperate latitudes; (ii) the **tropical cyclone** or **hurricane**, which is somewhat smaller in area than extratropical cyclone and accompanied by high winds and heavy rains; (iii) the **tornado** is a small but intense storm with very high winds, usually of limited duration; (iv) **thunderstorm** is local in nature and accompanied by brief but heavy rain showers and often by hail. The term is also applied to blizzards, sandstorms and dust storms, in which high wind is the dominant meteorological element.

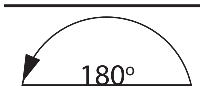
STORM SURGE: The temporary increase, at a particular locality, in the height of the ocean due to extreme meteorological conditions such as heavy winds and low pressure.

STOVER: An inferior type of fodder or leaves and stalks of corn that are left in a field after harvesting and are dried for use as fodder.

STP: Abbreviation for standard temperature (273.15 K or 0°C) and pressure (10^5 Pa). It was formerly known as (Normal Temperature and Pressure NTP). It is the standard conditions used as a basis for calculating quantities that vary with temperature and pressure and usually employed in reporting gas volumes. Flow meters calibrated in standard gas volumes per unit time often refer to volumes at 25°C, not 0 °C .

STRAIGHT: 1. Animal foodstuff composed of one type of food. 2. Extending or proceeding in one single direction, without bends, curves, irregularities or deviations and having the property that the lines passing through any pair of points on the given line are all identical.

STRAIGHT ANGLE: A 180° angle or one-half of a revolution.



Straight Angle

STRAIGHT CHAIN COMPOUND (STRAIGHT-CHAIN COMPOUND): A molecule in which the atoms are linked in a long chain with no side chains attached.

STRAIGHT FERTILISER: Fertiliser that supplies only one nutrient such as nitrogen.

S

STRAIGHT LINE: A line segment joining two points that has no curves.

STRAIGHT-LINE GRAPH: A graph with a general equation $y = mx + c$, where m is slope of a line and c is a constant that determines where the line crosses the y-axis.

STRAIGHTEDGE: An idealised mathematical instrument having a straight edge, which can be used to draw a line segment but not for measurement as it has no equally marked spaces along its edges. This is in contrast to a ruler that has straight edges but no equally marked spaces along its edges.

STRAIN: 1. A group of organisms such as virus, bacterium or fungus of the same species but with different characteristics. For example, a flu strain is a certain biological form of the influenza or flu virus. 2. In chemistry, a molecule experiences strain when its chemical structure undergoes some stress which raises its internal energy in comparison to a strain-free reference compound. 3. In physics, it is also known as **deformation**, it is the change in the metric properties of a continuous body B in the displacement from an initial placement $\kappa_0(B)$ to a final placement $\kappa(B)$. A change in the metric properties means that a curve drawn in the initial body placement changes its length when displaced to a curve in the final placement. If all the curves do not change length, it is said that a rigid body displacement occurred.

STRAIN ENERGY: The excess energy due to steric strain of a molecular entity or transition state structure. In other words, it is the potential energy stored in a body by virtue of an elastic deformation, equal to the work that must be done to produce this deformation. The strain energy components involve the following destabilising terms: non-bonded repulsions, bond-angle distortions, bond stretch or compression, rotation around or twisting of double bonds and electrostatic strain. In general, the contributions of these components are inseparable and interdependent.

STRAIN GAUGE: A device used to measure strain or small mechanical deformation in a body. The most widely used devices are metal wires or foils or semiconductor materials, such as a single silicon crystal, which are attached to structural members. When the resistance is a component in a Wheatstone-bridge circuit an estimate of the strain can be made by noting the change in resistance. Other types of strain gauge rely on changes of capacitance or the magnetic induction between two coils, one of which is attached to the stressed member.

STRAIN HARDENING (WORK HARDENING): An increase in the resistance to the further plastic deformation of a body as a result of a rearrangement of its internal structure when it is strained, particularly by repeated stress as in elasticity. In strain hardening, ductility is markedly reduced.

STRAKE: 1. An attachment bolted onto the rear wheels grisp of a farm machinery. The strake has spikes which can be extended beyond the tyre and which dig into the soil. 2. The slender forward extension of the inboard region of the wing of a combat aircraft used to provide increased lift in the high angle-of-attack manoeuvring condition. 3. A part of the shell of the hull of a boat or ship, which in conjunction with the other strakes, keeps the sea out and the vessel afloat. It is a strip of planking in a wooden vessel or of plating in a metal one, running longitudinally along the vessel's side, bottom or the turn of the bilge, usually from one end of the vessel to the other. 4. A straight edge used for levelling a bed of sand or striking poured concrete or plaster level with the edges of the formwork or mould into which it has been poured. A strake used for flooring or paving work is often called a **screed**.

STRAMENOPILES (HETEROKONTS): An assemblage of eukaryotic organisms, containing more than 100,000 known species, most of them diatoms regarded as constituting the phylum Heterokontophyta, proposed on the basis of recent evidence from molecular systematic. The stramenopiles include a diversity of forms, ranging from unicellular or colonial forms to large multicellular forms such as the brown algae.

STRAND: 1. A length of animal, plant or mineral fibre or tissue that resembles a thread. 2. An element that with others make up a larger complex whole. 3. The sandy shore of a lake, sea or river. 4. Any of a single filament, threads, hairs, wires, etc. twisted together into a rope, cable or a textile material.

STRANGE ATTRACTOR: The set of points in phase space to which the representative point of a dissipative system with internal friction tends as the system evolves. The various attractors are single point; a closed curve (a limit cycle), which describes a system with periodic behaviour; or a strange attractor, in which case the system exhibits chaos.

STRANGE MATTER: A matter composed of up, down and strange quarks rather than the up and down quarks found in normal nucleons. Suggestions are made that these strange matters may have been formed in the early universe and that pieces of this matter which are called the S-drop may still exist.

STRANGE PARTICLE: An elementary particle that has non-zero strangeness. Strange particles have anomalously long lifetimes (10^{-10} to 10^{-7} second) compared to those of most other hadrons (10^{-23} s).

STRANGENESS: The property of certain elementary particles called hadrons (K-mesons and hyperons) that decay more slowly than would have been expected from the large amount of energy released in the process. To account for this behaviour, these particles were assigned the quantum number s with the symbol s. For nucleons and other non-strange particles s = 0 and for strange particles s does not equal zero but has an integral value. In quark theory, hadrons with the property of strangeness contain a strange quark or its antiquark.

STRANGLES: 1. A disease of marigold and sugar beet which occurs in fairly large seedling after singling. 2. An infectious disease of horses and related animals, caused by the bacterium *Streptococcus equi* characterised by inflammation of the nasal mucous membrane and abscesses under the jaw and around the throat that causes strangling or choking sensation.

STRAP: In botany, it is the ligule of a ray flower in the Compositae (Astetraceae).

STRATA: 1. A horizontal layer of material, especially one of several parallel layers arranged on top of each other. 2. A bed or layer of sedimentary rock having approximately the same composition throughout. 3. A layer of the atmosphere or the sea, regarded as lying between horizontal planes. 4. A layer of living cells.

STRATEGY: 1. In biology, it is a group of related traits, such as anatomical, physiological and behavioural characteristic evolved under the influence of natural selection, that solves particular problems encountered by living organisms. 2. In mathematics, a sequence of actions executed to make sense of a mathematical situation and/or obtain mathematical results.

STRATIFICATION: 1. The arrangement of the components of an entity in layers. It is a feature of sedimentary rocks and soils. It is also seen in stratified epithelium. Thermal stratification can occur in some lakes. 2. In horticulture, the practice of pre-treating seeds to simulate natural winter conditions by placing them between layers of peat and sand and then exposing them to low temperatures for a period which is required before they germinate. 3. Layering that occurs in most sedimentary rocks and in igneous rocks that are formed at the Earth's surface, such as from lava flows and volcanic deposits. The layers called **strata** may range from thin sheets that cover several square kilometres to thick lens-like bodies that are only a few metres wide. 4. In mathematical logic, stratification is any consistent assignment of numbers to predicate symbols guaranteeing that a unique formal interpretation of a logical theory exists. 5. In hydrology, water stratification occurs when water of high and low salinity (halocline), oxygenation (chemocline), density (pycnocline) or temperature (thermocline), forms layers that act as barriers to water mixing.

STRATIFIED DRIFT (SORTED DRIFT): Materials that are distinctly sorted according to size and weight of their component fragments, indicating a medium of transport (water or wind) more fluid than glacier ice.

STRATIFIED EPITHELIUM (LAMINATED EPITHELIUM): Epithelium that is made up of a number of layers of cells, each varying in size and shape, present in areas of the body that are subject to wear and tear. Stratified epithelium containing keratin forms the outer layer of the skin. Those that do not contain keratin are found in moist areas such as the mouth and vagina.

STRATIFIED SAMPLE: A sample consisting of portions obtained from identified subparts (strata) of the parent population. Within each stratum, the samples are taken randomly. The objective of taking stratified samples is to obtain a more representative sample than that which might otherwise be obtained by random sampling.

STRATIGRAPHY: The branch of geology that studies the origin, composition, sequence and correlation of rock strata. It forms the basis of historical geology and has also found practical application in mineral exploration, especially that of petroleum.

STRATOCUMULUS CLOUD: A layer of clouds that is sometimes accompanied by weak intensity precipitation. Stratocumulus vary in colour from dark grey to light grey and may appear as rounded masses, rolls, etc., with breaks of clear sky in between. Very often the rolls are so close together that their edges join and give the under surface a wavy character. The process of formation, called **cumulogenesis** involves the spreading out of the tops of cumulus clouds, at less than 2000 m.

STRATOPAUSE: That region of the atmosphere which lies between the stratosphere and the mesosphere and in which a maximum in the temperature occurs.

STRATOSPHERE: That part of the atmosphere 10-40 km from the Earth's surface, where most of the Earth's ozone is concentrated and the temperature slowly rises from as low as -55 °C (273 K) to around 0°C (373 K) because of absorption of solar ultraviolet light by ozone.

STRATUM: A layer characterised by certain unifying characteristics, properties, or attributes distinguishing it from adjacent layers. Its plural is strata.

STRATUM CORNEUM: The layers of dead keratinised cells that form the outermost layers of mammalian epidermis. It provides a water resistant barrier between the external environment and the living cells of the skin.

STRATUS CLOUD: A uniform, featureless layer of cloud resembling fog but not resting on the ground. When this very low layer is broken up into irregular shreds, it is designated as fractostratus.

STRAW: The dry stems and leaves of crops such as wheat and oilseed rape. Its uses include as fuel, livestock bedding and fodder, thatching and basket-making.

STRAW BURNING: A cheap method of disposal of straw, which helps to control diseases.

STRAW CHOPPER: A device fitted to the back of a combine harvester which cuts straw into short lengths and drops them on the stubble to make it easier to plough.

STRAW SPREADER: A device attached to the back of a combine harvester which spreads unwanted straw over the ground and then ploughs them in.

STRAW WALKER: The part of a combine harvester where straw is carried away from the threshed grain after it has been separated from the stalk.

STRAWBERRY: Common name for low, perennial herbs with small sweet red fruit containing many achenes (dry single-sealed fruit that does not open to release its fruits) resembling seeds. They are native to temperate regions. Strawberries make up the genus *Fragaria* of the family Rosaceae. They have white flowers, which are borne in cymes, have a five-cleft calyx, five rounded petals, many stamens and numerous pistils. They reproduce sexually through the flowers and also asexually by means of rooting new stems, called runners or stolons, which emerge from the base of the strawberry plant and grow horizontally along the ground. It is used as a dessert fruit and also preserved as jam.

STRAWBERRY FOOTROT (DERMATOPHILUS INFECTION; CUTANEOUS STREPTOTHRICHOSIS; LUMPY WOOL): A bacterial disease, *Dermatophilus congolensis*, which affects sheep, cattle and horses causing ulcers. It also affects people, who attend to infected animals if elementary hygiene measures are not observed. It causes severe skin infections.

STRAY CAPACITANCE: An undesirable capacitance that arises in a circuit because of interactions between components of the circuit. It can be minimised by placing these components as far away from each other as possible in a circuit.

STREAK: 1. A viral disease of plants such as potatoes or tomatoes that produces discoloured markings on stems and leaves. 2. A bacterial culture inoculated by drawing a bacteria-laden needle across the surface of a solid culture medium. 3. A long, thin line or mark of a different substance or colour from its surroundings. 4. The colour of the fine powder produced when a mineral is rubbed against a hard surface, used as a distinguishing characteristic.

STREAK TUBE: An image converter adapted to provide scanning or time-resolved images. If the image is recorded the whole device is an example of a streak camera.

STREAMFLOW: Water that flows in streams, rivers, and other channels, which is a major element of the water cycle, defined as the volumetric rate of flow of water (volume per unit time) in an open channel. It is usually expressed in cubic metre per second (m^3 sec^{-1}). Streamflow is one important route of water from the land to lakes and oceans.

STREAM FUNCTION: A function describing the streamlines of a body and is defined for incompressible flows in two dimensions and also in three dimensions with axisymmetry. It is defined completely by the geometry of the streamlines and must be the same in any coordinate system. It can be used to plot streamlines, which represent the trajectories of particles in a steady flow.

STREAMING: The method of constantly sending and receiving content over the Internet, commonly seen in the forms of audio and video streaming, especially when a multimedia file can be played back without being completely downloaded. The content is viewed live instead of being recorded for later playback. Certain audio and video files like Real Audio and QuickTime documents can be streaming files.

STREAMING BIREFRINGENCE (FLOW BIREFRINGENCE): The birefringence induced by flow in liquids, solutions and dispersions of optically anisotropic, anisometric or deformable flow molecules or particles due to a non-random orientation of the molecules or particles.

STREAMING CURRENT: The electric current flowing in a streaming cell if the electrodes, which are supposed to be ideally depolarised, are short-circuited. It is positive if the current in the membrane, plug, etc., is from high to low pressure side (and in the outside lead from low to high pressure side).

STREAMING POTENTIAL (STREAMING POTENTIAL DIFFERENCE): A potential difference that occurs when a liquid under pressure is forced through a narrow opening or diaphragm. Streaming potential is one of four related electrokinetic phenomena which depend upon the presence of an electrical double layer at a solid-liquid interface. This electrical double layer is made up of ions of one charge type which are fixed to the surface of the solid and an equal number of mobile ions of the opposite charge which are distributed through the neighbouring region of the liquid phase. In such a system the movement of liquid over the surface of the solid produces an electric current, because the flow of liquid causes a displacement of the mobile counterions with respect to the fixed charges on the solid surface. The applied potential necessary to reduce the net flow of electricity to zero is the streaming potential. It can be measured between two electrodes at each end of a capillarity tube made of the same material as the diaphragm.

STREAMLINE (LINE OF FLOW): Paths followed by a particle whose velocity field is the given vector field. The vectors in a vector field are tangent to the flow lines.

STREAMLINE FLOW: A type of fluid flow in which no turbulence occurs and the particles of the fluid follow continuous paths at a velocity that remains constant or varies regularly with time.

STREAMLINING: Designing a body to offer least resistance through a medium such as air or water or the contouring of a body to reduce its resistance (drag) to motion through a fluid. For example, submarines, sharks and most modern cars have streamlined shapes.

STRENGTH: 1. The ability of a material to resist the effect of a force or a measure of how well a substance can resist being deformed or fractured. 2. A measure of the magnitude of a force or an effect. 3. The concentration of a solute in a solution. 4. The ability of flour to produce yeasted dough capable of retaining carbon dioxide bubble until the protein in the bubble walls becomes relatively rigid. It happens at about 75 °C (348 K).

STRENGTHENING TISSUE (MECHANICAL TISSUE; SUPPORTING TISSUE): Any tissue consisting of cells with thickened cell walls, such as collenchyma and sclerenchyma.

STREPTOBACILLUS: A rod-shaped bacterium that grows in culture as rods in chains. It often causes diseases such as the Haverhill fever form of rat bite fever characterised by fever, rash, chills, headache, vomiting, muscle pain, arthritis and bacteremia and by weight loss and diarrhoea in children.

STREPTOCARPUS: A genus of herbaceous flowering plants in the family Gesneriaceae, closely related to the genus *Saintpaulia*. It is native to subtropical regions and generally has only one large leaf. It has tubular and brightly coloured flowers. About 155 species of *Streptocarpus* are currently recognised, the first described being *S. rexii*. They are found growing on the ground, rock crevices and almost anywhere the seed can germinate and grow.

STREPTOCOCCUS: A genus of spherical chiefly non-motile and parasitic gram-positive bacteria occurring widely in nature, typically as chains or pairs of cells. Many are saprotrophic and exist as usually harmless commensals inhabiting the skin, mucous membranes and intestine of human beings and animals. Some are pathogens of humans and domestic animals, causing diseases such as scarlet fever, strep throat, strep pneumonia, scarlet fever and rheumatic fever.

STREPTODORNASE: An enzyme derived from haemolytic streptococci. It is used medically, often in combination with streptokinase, to dissolve purulent or fibrinous secretions from infections.

STREPTOKINASE: An enzyme produced by the bacterium streptococci, which can bind and activate human plasminogen. It is used in dissolving blood clots in some cases of myocardial infarction and pulmonary embolism.

STREPTOLYSIN: A substance produced by streptococci which breaks down red blood cells. Types of streptolysins are Streptolysin O (SLO), which is oxygen-labile and Streptolysin S (SLS), which is oxygen-stable.

STREPTOMYCIN: A broad spectrum of antibiotic, which is active against both gram-positive and gram-negative bacteria including species resistant to other antibiotics. Streptomycin acts by inhibiting protein synthesis and damaging cell membranes in susceptible microorganisms. It is used in the treatment of bacterial infections such as tuberculosis and against many types of infections especially some streptococci, penicillin-resistant staphylococci and bacteria of the genera *Proteus* and *Pseudomonas*. It is produced from the soil bacterium *Streptomyces griseus*.

STRESS: 1. In biology, it is a stimulus or succession of stimuli of such magnitude as to tend to disrupt the homeostasis of the organism. 2. In physics, it is the force or system of forces per unit area on a body that tends to cause it to deform. It is a measure of the internal forces in a body between particles of the material of which it consists. They resist separation, compression or sliding in response to externally applied forces. Tensile and compressive stresses are axial forces per unit area applied to a body that tend to extend it or compress it linearly whiles shear stress is a tangential force per unit area that tends to shear it. The unit of stress is the pascal and its symbol is Pa. 1 pascal is equal to 1 N/m^{-2} [equivalent to one newton (N) per square metre (m^2)]. 3. An abbreviation in computing which means **structural engineering solver**. It is a computer language designed for use in solving structural analysis problems in civil engineering. 4. A state of anxiety or strain. It may result in high blood pressure or depression.

STRESS GENES: Genes that are prompted to function by exposure to DNA-damaging agencies such as radiation or oxygen-free radicals.

STRESS GRAPHITISATION: The solid-state transformation of non-graphitic carbon into graphite by heat treatment combined with application of mechanical stress, resulting in a defined degree of graphitisation being obtained at a lower temperature and/or after a shorter time of heat treatment than in the absence of applied stress.

STRESS TENSOR: The second order symmetric tensor, σ, such that the stress vector, $\mathbf{t}$, at a point on a surface is given by $\sigma\mathbf{n}$, where $\mathbf{n}$ is the unit outward normal to the surface at the point.

STRESS VECTOR: The contact force density of a body.

STRESS-ASSISTED TRANSITION: A transition that takes place when an applied stress assists the transition to the new phase.

STRESS-POWER: The difference between the rate of change of the kinetic energy and the power of a sub-body, that is, the integral $\int tr(\sigma\Sigma)dv$ over the volume of the current configuration of the sub-body, where σ is the stress tensor and Σ is the Eulerian strain rate. The quantity $tr(\sigma\Sigma)$ is the stress-power per unit volume.

STRESSED COMMUNITY: A community that is disturbed by human activities such as road building or pollution. Some species become superabundant while others disappear.

STRETCH RECEPTOR: A specialised cell or group of cells found in muscles or tendons that is sensitive to mechanical stress or stretch. When muscles or tendons are stretched the stress is registered by special sensory nerve cells and converted into nerve impulses which are transmitted into the central nervous system.

STRETCH REFLEX (MYOTATIC REFLEX): reflex contraction of a muscle when an attached tendon is stretched; an example is the knee-jerk reflex. It is important in maintaining erect posture. Stretching of the muscle causes impulses to be generated in the muscle spindles. These impulses are transmitted by sensory neurons to the spinal cord where the sensory neurons synapse with motor neurons.

STRETCHING: A homothetic transformation of the form characterised by an invariant line and a scale factor called **one-way stretch** or two invariant lines and corresponding scale factors, called, **two-way stretch**. it is of the form $x' = kx$, $y' = ky$, where $k > 1$.

STRIATE: Marked with fine, usually parallel lines or grooves. **Stria** is a fine line or groove.

STRIATED MUSCLE (VOLUNTARY MUSCLE; STRIPED MUSCLE; SKELETAL MUSCLE): Muscle under voluntary control attached to bone, which produces all the movements of body parts in relation to each other. Its name is derived from its multinucleated fibres that are long and thin and are crossed with a regular pattern of fine red and white lines, giving the muscle its distinctive appearance. The muscle fibres may be of two main types: **fast twitch fibres**, adapted to produce quick, powerful movements; and **slow twitch fibres**, adapted to slower, endurance type movements. Examples of voluntary muscles are muscles of the chest, neck and abdomen. Involuntary muscles include muscles of the digestive system, smooth muscles, etc.

STRIATIONS: 1. Parallel scratches on the Earth's surface caused by rocks, boulders, gravel, and pebbles dragged by a glacier. The scratches point in the direction of the glacial movement. 2. Linear furrows, or linear marks, generated from fault movement. The striation's direction reveal the movement directions in the fault plane.

STRICT: 1. In mathematics, of a relation, etc., distinguished from some other relation of the same name by the fact that it applies more restrictively. For example, a strict inequality such as $x < = y$ is not strict, whereas $x < y$ is. 2. Distinguished from a relation of the same name that is not the subject of formal study. For example, strict identity is governed by a set of axioms and is distinguished from various uses in ordinary language.

STRICT IMPLICATION: The connective of modal logic usually defined in terms of the possibility of the truth of its antecedent at the same time as the falsehood of its consequent, i.e., $P \Rightarrow Q \equiv -\Diamond(P \& -Q)$, where $\Diamond$ is the possibility operator. It is the relation that holds between two sentences precisely when one is validly deducible from the other and is not truth-functional.

STRIDULATION: The production of sound by insects by rubbing one part of the body against another. The parts involved vary from species to species. Stridulation is typical of the Orthoptera such as grasshoppers, mantises, locusts and crickets but other animals such as a number of species of fishes, snakes and spiders also produce stridulation. The purpose of the sound is usually to bring sexes together, although they are also used in activities such as territorial behaviour and warning. **Stridulatory organs** are the parts of the body that produce sound when rubbed together.

STRIGA: A genus of 28 species of parasitic plants that occur naturally in parts of Africa, Asia and Australia. The genus is classified in the family Orobanchaceae, although older classifications place it in the Scrophulariaceae. It is attached to the roots of the host plant by which it obtains its water supply. It has green leaves and therefore photosynthesises. Three species: *Striga asiatica*, *S. gesnerioides* and *S. hermonthica* cause much damage to some crops and plants such as maize, cowpea, sorghum and pearl millet.

STRIGILLOSE (STRIGULOSE): Minutely strigose.

STRIGOSE: Bearing straight, stiff, sharp, appressed hairs.

STRIKE: The infestation of the flesh of a sheep by the larvae of a blowfly.

STRIKE-SLIP FAULT: A fault (a fracture or zone of fractures between two blocks of rock) in which the two blocks slide horizontally past one another.

STRING: 1. In mathematical logic, it is a sequence of symbols that are chosen from a set or alphabet. 2. In computer science, it is group of characteristers manipulated as a single object by a computer. 3. A fine chord for tying things, keeping things in place, etc. 4. Abbreviation for **S**earch **T**ool for the **R**etrieval of **I**nteracting **G**enes/Proteins (STRING). It is a database and web resource of known and predicted protein-protein interactions. The STRING database contains information from numerous sources, including experimental data, computational prediction methods and public text collections. It is freely accessible and it is regularly updated. Version 8.1 contains information about 2.5 million proteins from 630 species. 5. A one dimensional spatial object used in theories of elementary particles and in cosmology. It is unlike an elementary particle which is zero-dimensional or point-like. This theory replaces the idea of a point like elementary particle by a line or loop.

STRING THEORY: The large number, 10^{500} of possible states. In string theory, everything in the universe including all particles and forces and perhaps space-time itself consists of fantastically small strings under immense tension, vibrating and spinning in a multi-dimensional super space.

STRIP CROPPING (STRIP FARMING): A practice of growing different crops such as hay, wheat or other small grains with strips of row crops, such as corn, soybeans, cotton or sugar beets in narrow strips either at right angles to the direction of the prevailing wind or following the natural contours of the terrain to prevent wind and water erosion of the soil by creating natural dams for water, helping to preserve the strength of the soil.

STRIP CULTIVATION: A method of communal farming in which each family has a long piece of several strips.

STRIP CUP: A container into which the first drops of milk are drawn by hand from the teats of the cow before the milking machine is attached to the udders. To test the first milk of a cow for clots, a few streams are squirted onto the lid so that watery milk can be recognised and some is squirted through the mesh so that any clots present can be detected.

STRIP GRAZING: A system of grazing which allows animals access to a small part of the field. The rest of the field is protected by a temporary fence, usually electric.

STRIPPER HEADER: A machine which harvests a crop such as linseed and strips off the seedheads. The basic concept of the stripper header is that a rearwards rotating rotor fitted in the front of the header is fitted with 8 rows of stripping fingers that strip grain from the crop as the combine moves the head forwards while it spins backwards. The speed of the rotor can be varied according to crop conditions. After the grain has been stripped by the rotor, a series of deflectors within the header deflect the grain back into a conventional auger and pan.

STRIP INTERCROPPING: A type of crop cultivation in which different crops are simultaneously cultivated using alternate strips of land with uniform width and on the same field strips of land rather than narrow rows. This practice is suitable for mechanised commercial large-scale farming.

STRIP-TILLAGE SYSTEM (SHALLOW ZONE-TILLAGE): A modification to a direct drilling system where disturbance of less than one third of the total field area is cultivated. It involves tilling a narrow strip of soil, and leaving the majority of the surface undisturbed. Typically, strips of approximately 15 centimetres wide and 10 to 20 centimetres deep are tilled ahead of planting. It is well suited for areas where no-till is not well adapted for corn production due to cold, heavy, compacted and/or poorly drained soils.

STRIPPING: In chemistry, it is the process of removing solute(s) from a loaded solvent or extract. Generally this refers to the main solute(s) present.

STRIPPING FUNCTOR (FORGETFUL FUNCTOR; UNDERLYING FUNCTOR): A functor obtained by considering a category of algebraic forms such as groups, Abelian groups, modules, rings and vector spaces, as another category of sets with simpler objects. For example, the functor from the category of groups to the category of sets that preserves all mappings forgets the structure of the groups.

STRIPPING REACTION: 1. A nuclear or chemical reaction in which part of an incident particle combines with the target nucleus, while the rest of the incident particle proceeds with most of its original momentum in practically its original direction. It was first described in 1951 by Stuart Thomas Butler (1926-1982), an Australian nuclear physicist. A standard example of stripping reaction is a reaction in which a deuteron is the incident particle and a nucleus is the target particle with the deuteron combining with the target nucleus and the proton from the deuteron proceeding almost without disruption. 2. A chemical process, studied in a molecular beam, in which the reaction products are scattered forward with respect to the moving centre of mass of the system.

STRIPPING SOLUTION: A solution (usually aqueous, sometimes water alone) used for back-extracting the distribuend from the extract (usually organic).

STRIPPINGS: The last drops of milk from a cow's teats at the end of a milking session.

STROBILUS: A type of composite fruit that is formed from a complete inflorescence. It is characteristic of horsetail plants. It produces achenes enclosed in a bract and when mature is cone-shaped.

STROBOGRAMMATIC PRIME: A prime number that remains unchanged when rotated through 180°. An example is 619, which looks the same when read upside-down. It cannot contain digits other than 0, 1 and 8, which have a horizontal line of symmetry and 6 and 9, which are vertical reflections of each other. An **invertible prime** is one that yields a different prime when the digits are inverted.

STROBOSCOPE (STROBE): A device for making a moving body intermittently visible in order to make it appear stationary or slow moving. In its simplest form, it is a rotating disc with evenly-spaced holes placed in the line of sight between the observer and the moving object. The rotational speed of the disc is adjusted so that it becomes synchronised with the movement of the observed system, which seems to slow and stop. The illusion is caused by temporal aliasing or stroboscopic effect. In electronic versions, the perforated disc is replaced by a lamp capable of emitting brief and rapid flashes of light. The frequency of the flash is adjusted so that it is equal to or a unit fraction below or above the object's cyclic speed, at which point the object is seen to be either stationary or moving backward or forward, depending on the flash frequency. Stroboscopic photography is thus the taking of very short-exposure pictures of moving objects using an electronically controlled flash lamp.

STROKE (CEREBRAL INFARCTION; CEREBRAL HAEMORRHAGE): An interruption of the blood supply to any part of the brain. It can be caused by a blocked blood vessel carrying blood to the brain or by a blood clot. This is called an ischaemic stroke. When a blood vessel breaks open, causing blood to leak into the brain, it is called a **haemmorhagic stroke**. If blood flow is stopped for longer than a few seconds, the brain cannot get blood and oxygen. Brain cells can die, causing permanent damage.

STROKE VOLUME: The volume of blood ejected from one ventricle at each stroke of the heart during systole or the volume of blood pumped from one ventricle of the heart with each beat. It is calculated using measurements of ventricle volumes from an echocardiogram and subtracting the volume of the blood in the ventricle at the end of a beat called end-systolic volume from the volume of blood just prior to the beat called end-diastolic volume. A typical stroke volume of a normal 70 kg man at rest is 70 ml, which can increase to 120 ml during strenuous exercise. Stroke volume is an important determinant of **cardiac output**, which is the product of stroke volume and heart rate and is also used to calculate **ejection fraction**, which is stroke volume divided by end-diastolic volume.

STROMA (STROMATA): Tissues that form the supportive framework of a biological cell, tissue or organ and usually composed of connective tissue. An example is the tissue of the ovary that surrounds the reproductive cells or the gel-like matrix of chloroplast.

STROMATOLITE: A rocky cushion-like mass formed by the unchecked growth of millions of lime-secreting cyanobacteria. Stromatolites are found only in areas where other organisms that would normally keep down the bacteria number cannot survive, such as extremely salty bays. Such bacteria were abundant during the Proterozoic and Archean era. Stromatolites are the oldest macroscopic evidence of life on Earth, at least 2.5 billion years old and they are still forming in the seas.

STROMBUS: 1. A legume which is spirally coiled, as occurs in *Medicago*. 2. A genus of marine gastropods in which the shell is a brightly-coloured spiral with large outer lip. An example is a conch shell.

STRONG ACID: An acid that is completely dissociated in aqueous solution. Examples are mineral acids such as hydrochloric acid (HCl), sulfuric acid (H_2SO_4) or nitric acid (HNO_3).

STRONG BASE: A base that is completely dissociated into its component ions. Examples are sodium hydroxide (NaOH) and potassium hydroxide (KOH).

STRONG COMPLETENESS: The property of a logical theory that the addition to its axioms of any well-formed formula that is not a theorem yields an inconsistent theory.

STRONG DUALITY: The relationship between two constrained mathematical programs (P) $P = \inf_X f(x)$ and (D) $d = \sup_Y g(y)$ that if there exist feasible primal and dual solutions, then there exist feasible primal and dual solutions which have the same objective value. In other words, the primal and dual solutions are equivalent. In strong duality, there is no duality gap and strong duality holds if and only if the duality gap is equal to 0. In **weak duality**, the primal problem has optimal value greater than the dual problem or the duality gap is positive.

STRONG ELECTROLYTE: An electrolyte that is completely dissociated into its component ions in acqueous solution (NaCl). These ions are good conductors of electric current in the solution. On the other hand, a weak electrolyte is one which is incompletely dissociated. Strong acids, strong bases and soluble ionic salts that are not weak acids or weak bases are strong electrolytes.

STRONG ERGODIC THEOREM (BIRKHOFF ERGODIC THEOREM; POINTWISE ERGODIC THEOREM): The theorem that, for any measure-preserving transformation, T, on a measure space and an integrable f, the Cesaro means of $f(T^i x)$ converge almost everywhere to an invariant function f^* for which $f^*(Tx) = f^*(x)$; when the underlying space has finite measure, f and f^* have the same integrals.

STRONG FIELD LIGAND: A ligand that exerts a strong crystal or ligand electrical field and generally forms low spin complexes with metal ions when possible.

STRONG INTERACTION (STRONG FORCE; STRONG NUCLEAR FORCE): One of the four fundamental interactions of nature. The others are electromagnetism, the weak interaction and gravitation. It is a non-contact force and is responsible for the binding of protons and neutrons in the nucleus and interactions between quarks. At atomic scale, it is about 100 times stronger than electromagnetism.

STRONG LAW OF LARGE NUMBERS: A formulation of the law of large numbers that if a sequence of independent random variables has finite variance, σ_n such that $\sum_n \sigma_n{}^2/n^2$ is finite, then the sequence of averages of the given sequence converges almost everywhere. This is in contrast with the weak law of large numbers that concerns convergence in measure.

STRONG NUCLEAR FORCE (STRONG FORCE): One of the four fundamental interactions of nature. The others are electromagnetism, the weak interaction and gravitation. It is a short-range but powerful nuclear non-contact force that is responsible for the binding of protons and neutrons in the nucleus and interactions between quarks. At atomic scale, it is about 100 times stronger than electromagnetism.

STRONG TOPOLOGY: The topology on a normed space obtained from the given norm as contrasted with the associated weak topology. Thus, strong convergence means convergence in norm.

STRONTIANITE: A mineral form of strontium carbonate with the chemical formula $SrCO_3$. It is white, greenish or yellowish in colour, usually occurring in fibrous massive forms, but sometimes in prismatic crystals. It is an important raw material for the extraction of strontium.

STRONTIUM: A silvery-white metallic element in Group 2 (IIA) of the Periodic Table (alkali-earth group) with the symbol Sr, atomic number 38, density 2.64 g/cm^{-3}, melting point 777 °C (1050 K) and boiling point 1382° C (1655 K). The isotope, strontium-90, is present in radioactive fallout and can be metabolised. The human body absorbs strontium as if it were calcium, incorporating it into the skeleton. It can be obtained by roasting the oxide, followed by reduction with aluminium in a process called the Goldschmidt Process. Its compounds impart a bright red colour to flame and are used in flares and fireworks.

STRONTIUM CARBONATE: A white, odourless, non-flammable solid, which occurs naturally as the mineral strontianite with the chemical formula $SrCO_3$, molar mass 147.63 g/mol, density 3.74 g/cm^3, melting point 1494 °C and decomposes. It is prepared industrially by boiling strontium sulfate with ammonium carbonate. It can also be prepared by passing carbon dioxide over strontium oxide or hydroxide or by passing the gas through a solution of strontium salt. It is used to coat the glass of cathode-ray screens, for refining sugar and as a slagging agent in certain furnaces and in fireworks.

STRONTIUM CHLORIDE: A white compound having the chemical formula $SrCl_2$, which can be made by neutralising hydrochloric acid with strontium carbonate oxide or hydroxide. It is deliquescent and readily forms the hexahydrate. It is used in medicine, in pyrotechnics and to make strontium salts.

STRONTIUM HYDROGENCARBONATE (STRONTIUM BICARBONATE): A compound having the chemical formula $Sr(HCO_3)_2$ and is stable only in solution. It is formed by the action of carbon dioxide on a suspension of strontium carbonate in water. On heating the process is reversed.

STRONTIUM NITRATE: A colourless crystalline solid with the chemical formula $Sr(NO_3)_2$, density 2.986 g/cm^3 for anhydrous and 2.20 g/cm^3 for tetrahydrate, melting point 570 °C (843 K) for the anhydrous and 100 °C (273 K) for the tetrahydrate and a boiling point of 645 °C (918 K) and decomposes. Strontium nitrate is generated by the reaction of nitric acid on strontium carbonate. It is used in pyrotechnics, signals and flares, medicine, matches and as chemical intermediate.

STRONTIUM OXIDE (STRONTIA): A white or grey amorphous cubic crystal with the chemical formula SrO, molar mass 03.619 g/mol, density 4.70 g/cm^3, melting point 2,531 °C (2,804 K), boiling point 3,200 °C (3,470 K) and decomposes. It dissolves in water to form strontium hydroxide. It can be prepared by the decomposition of heated strontium carbonate, hydroxide or nitrate. It is a strongly basic oxide, which is used in the manufacture of other strontium salts, in pigments, soaps, greases and also in television picture tube glass.

STRONTIUM SULFATE (CELESTINE): A white orthorhombic crystal with the chemical formula $SrSO_4$, molar mass 183.68 g/mol, density 3.96 g/cm^3 and melting point 1,606 °C (1,879 K). It is made by dissolving strontium oxide, hydroxide or carbonate in sulfuric acid. It is insoluble in ethanol and alkalis and slightly soluble in acids. It can be used as ceramic glazes, in paper manufacture, to provide red colour in fireworks and as a pigment in paints.

STRONTIUM UNIT (S.U.): A measure of the concentration of the radio-isotope strontium-90 in substances such as bone, milk or soil relative to their calcium content. The human body absorbs strontium as if it were calcium, incorporating it into the skeleton. One strontium unit is equal to one picocurie from strontium-90 per gram of calcium (37 Becquerels per kilogram) in the subject's skeleton.

STROPHANTHUS: A genus of woody vines, shrubs or small trees in the dogbane family, Apocynaceae, with more than 40 species that are native to tropical Africa and Southeast Asia. The leaves are opposite or whorled, simple broad lanceolate, with an entire margin. The flower petals of some species are drawn out into long threads. The bark and seeds of some vinelike species contain toxic alkaloids called **strophanthins**, which are used as arrow poisons and in low dosages as cardiac and vascular stimulants. *Strophanthus hispidus*, also known as hairy strophanthus, brown strophanthus or arrow poison, is a liana growing up to 100 m long or a shrub or a small tree growing between 3 - 5m tall. It occurs from Senegal eastwards to the Central African Republic, DR Congo, Uganda and western Tanzania and south to northern Angola. It has white latex which becomes reddish clear and coagulates after exposure. The bark is dark-grey. The leaves are decussately opposite or rarely in whorls of three (ternate), glossy and dark green above, dull paler below, ovate, elliptic or obovate rounded or subcordate at the base, acuminate at the apex. The flowers are in one to multi-flowered inflorescences, lax or congested, with short stiff hairs in all parts. The seeds are highly toxic and therefore, not used in traditional medicine. It is used to treat skin diseases, leprosy and ulcers, malaria, dysentery, gonorrhoea, snakebite, heart failure and blood circulation disorders. *Strophanthus amboensis*, is a much branched shrub growing up to 4 m in height or a liana growing up to 30 m long, with shortly petioled or subsessile, ovate to broadly ovate leaves, occurring in bushveld and rocky places in South West Africa and southern Angola. It has an unusual and beautiful flowers with orange-yellow turning purple corolla tube, white-streaked on the inside, which makes it an ornamental plan. The plant is also used to treat rheumatism, enema of the root, venereal disease, etc. *Strophanthus sarmentosus* is a tropical liana that occurs in tropical Africa, South and Southeast Asia and grows as either a deciduous shrub or as a liana up to 40 m long, with a stem diameter of up to 15 cm. It is used as a source of the drug cortisone to treat different conditions such as allergic disorders, skin conditions, ulcerative colitis, arthritis, lupus, psoriasis or breathing disorders. Locally, the plant is used as arrow poison and for the treatment of joint pain, eye conditions and venereal disease.

STROPHIOLE: An appendage like a crest at the helium in some seeds; a caruncle.

STROPHOID: The locus of points, two on each line in a pencil of lines through a fixed point, whose distance from the intercept of the line with the y-axis is equal to the y-coordinate of that intercept.

STRUCK: In animal science, it is an acute and fatal disease of adult sheep, which is a form of enterotoxaemia. It manifests with abdominal pain, tenesmus and rapid death. It is caused by the rapid multiplication of Clostridium in the intestines and the consequent absorption of increased amounts of toxin.

STRUCTURAL DEFECT: Actual physical damages that manifest to the load-bearing elements of a building or structure caused by failure of such load-bearing elements that affects their load-bearing functions to the extent that the structure becomes unsafe.

STRUCTURAL DISORDER: Any deviation from the ideal three-dimensional regularity of the crystal structure.

STRUCTURAL DOMAIN: An element of protein tertiary structure that recurs in many structures or a part of protein sequence and structure that can evolve, function and exist independently of the rest of the protein chain. Many proteins consist of several structural domains. One domain may appear in a variety of evolutionarily related proteins. Domains vary in length from between about 25 amino acids up to 500 amino acids in length. The shortest domains such as zinc fingers are stabilised by metal ions or disulfide bridges. Domains often form functional units, such as the calcium-binding EF hand domain of calmodulin.

STRUCTURAL FORMULA: A chemical formula that shows the number and kinds of atoms in a molecule and how the atoms and bonds in the molecule are arranged. Structural formula identifies the location of chemical bonds between the atoms and molecules. It is an expanded molecular formula showing the arrangement within the molecule of atoms and of bonds or a chemical chemical formula showing the linkage of the atoms in a molecule diagrammatically. The atoms are represented by symbols and the structure is indicated by showing the relative positions of the atoms in space and the bonds between them. It consists of symbols for the atoms connected by short lines that represent the chemical bonds – one line for single bond, two for double and three lines standing for triple bonds. For example, H-O-O-H is the structural formula for hydrogen peroxide, "-" representing the bond. Structural formula is important for showing how compounds with identical kind and number of atoms differ. It is important to draw Structural Formulas for organic compounds because in most cases a molecular formula does not uniquely represent a single compound, as in the case of isomers. Also, there are many compounds where the slightest altering of the structure or choosing a different enantiomer can make the compound go from being helpful or harmless to being harmful or dangerous. For example, morphine, used to relieve pain is harmless while diamorphine (heroine or diacytylmorphine) is harmful and dangerous.

STRUCTURAL GENE: A gene encoding the amino acid sequence of a polypeptide chain that is not in regulation. It may code for a structural protein, an enzyme or an RNA molecule not involved in regulation.

STRUCTURAL ISOMERISM (CONSTITUTIONAL ISOMERISM): Isomerism of chemical compounds that have the same molecular formula but different structural formulae, physical and chemical properties. For example, propanal (C_2H_5CHO) and propanone (acetone), CH_3COCH_3, have the same molecular formulae, C_3H_6O, but different structures and are, therefore structural isomers.

STRUCTURAL PROTEIN: A fibrous protein forming part of the body or organelles of an organism. Measured in terms of their total mass, the largest class of proteins is the structural one. The most familiar of the fibrous proteins are probably the **keratins**, which form the protective covering of all land vertebrates: skin, fur, hair, wool, claws, nails, hooves, horns, scales, beaks and feathers. Some others are the **actin** and **myosin** proteins of muscle tissue. Another group of fibrous structural proteins are the **silks and insect fibres**. In addition, there are the **collagens** of tendons and hides, which form connective ligaments within the body and give extra support to the skin where needed.

STRUCTURAL TRANSITION: A reversible or irreversible transition that involves a change of the crystal structure. An example is the transition of NH_4Cl at 469 K from a CsCl-type structure to a NaCl-type structure.

STRUCTURE: 1. In biology, a construction and arrangement of tissues, parts or organs. 2. In chemistry, the arrangement of atoms in a molecule of a chemical compound, which is the structure of benzene. 3. In geology, the way in which a mineral, rock, rock mass or stratum, etc, is made up of its component parts. 4. In logic, an assignment to a first-order language of a nonempty set (the universe) of which the elements are individuals and of functions, predicates and constants in that universe to the corresponding symbols of the language with the exception of identity. A structure for a theory in which its non-logical axioms are true is a model for the theory. 5. A set endowed with some prescribed functions, predicates or relations, usually of an algebraic nature.

STRUCTURE FORMATION IN THE UNIVERSE: The non-uniform distribution of matter throughout the universe. Clumpiness occurs when the matter is not homogeneous, the gravitational attraction between particles of matter ceases to be uniform. This can occur as a result of quantum fluctuations in the early universe associated with inflation. There is some evidence for this hypothesis, based on measurements from Cosmic Background Explorer (COBE) a cosmology satellite. An alternative mechanism for structure formation is based on cosmic strings.

STRUCTURE INDEX: Any measurement of a soil's physical property, such as aggregation, porosity, permeability to air or water, or bulk density that denotes or indicates its structural condition.

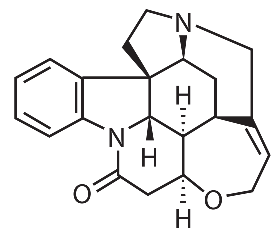
STRUCTURE-ACTIVITY RELATIONSHIP (SAR): Association between specific aspects of molecular structure and defined biological action.

STRUCTURE-BASED DESIGN: Structure-based design is a design strategy for new chemical entities based on the three-dimensional (*3D*) structure of the target obtained by X-ray or nuclear magnetic resonance (*NMR*) studies or from protein homology models.

STRUCTURE-PROPERTY CORRELATIONS (SPC): It refers to all statistical mathematical methods used to correlate any molecular property (intrinsic, chemical or biological) to any other property, using statistical regression or pattern recognition techniques.

STRUCTURED QUERY LANGUAGE (SQL): A high-level computer language used to access and modify information in a database. Some common SQL commands include "insert," "update," and "delete."

STRYCHNINE (VAQUELINE): A colourless or white crystalline insoluble alkaloid with the chemical formula $C_{21}H_{22}O_2N_2$, density 1.36 g/cm³, melting point 284-286°C (557-559 K) and boiling point 270 °C (543 K). It has a bitter taste and found in certain plants, such as Strychnous nux vomica, the strychnine tree. Strychnine is a terpene indole alkaloid belonging to the Strychnos family of Corynanthe alkaloids and it is derived from tryptamine and secologanin. It is one of the most powerful poisons known and used as a poison for rodents and other pests and topically in medicine as a stimulant for the central nervous system.



Chemical structure of Strychnine

STUBBLE: 1. The short stiff stems left on the ground after crops such as corn, wheat, oilseed rape, etc., have been cut. **Stubble mulch** is the stubble of crops or crop residues left essentially in place on the land as a surface cover, to prevent soil erosion from wind or water and conserve soil moisture. 2. The regrowth of shaven body hair on the face, legs, armpits or chest that is short and rough to the touch.

STUBBLE BURNING: A method of removing dry stubble by burning it before ploughing.

STUD FARM: A farm, where horses are kept for breeding. Many stud farms make male animals available for breeding to outside female animals that are not owned by the stud farm. This provides a source of revenue to a stud farm through the stud fees paid to obtain the services of the stud animal as well as contributing to the overall genetic diversity of the animal's offspring. It has an overall effect of improving the quality of animals throughout an area.

STUDENT'S T DISTRIBUTION (T-DISTRIBUTION): A probability distribution in statistics made up of the means of random samples from a collection of samples that has a normal distribution of unknown variance. It is used to test a hypothesis.

STUMP: The base of a tree trunk and its roots after the tree has been cut down or fallen.

STURM SEQUENCE: A finite sequence of polynomials, $p_0(x)$, ..., $p_k(x)$, defined for a given polynomial, p, as the sign variation in the sequence $p_0(x)$, ..., $p_k(x)$, where $p_0 = p$, $p_1 = p'$ and each $P_i; = -r_i$, where $r_2, r_3, \ldots, r_k$ are the successive remainders computed by Euclid's algorithm for the highest common factor of p and p'.

STURM'S THEOREM: A theorem in which the number and position of the real roots between given limits of an algebraic equation can be determined. It states that if a real polynomial is non-zero at the end points of an interval, the number of roots in that interval, counting multiplicity, is the difference between the numbers of sign changes of the Sturm sequence at the two end points.

STY (HORDEOLUM): 1. A structure for housing pigs usually with a small run next to it. 2. An acute infection of the secretory glands of the eyelids. This common infection results from blocked glands within the eyelid. When the gland is blocked, the oil produced by the gland occasionally backs up and extrudes through the wall of the gland, forming a lump (chalazion), which can be red, painful and nodular.

STYLE: 1. The usually slender part of a pistil, situated between the ovary and the stigma. The length of the style varies according to the plant species. 2. A slender, tubular or bristle-like process, for example, a cartilaginous style. 3. Also called **stylet** in medicine, it is a surgical probing instrument.

STYLOPODIUM: A disc-like expansion or swelling at the base of the style in plants of the umbel family.

STYPTICS: Agents or substances containing properties which stops bleeding or haemorrhages and heals wounds. Certain herbs can be effective in treating these problems; such herbs include marigold, pepper, lotus, oak bark and plantain. These are astringent herbs.

STYRENE (VINYL BENZENE; PHENYL ETHENE): A colourless oily liquid aromatic compound with the chemical formula $C_6H_5CH{=}CH_2$ that occurs in coal-tar. It polymerises slowly on standing, rapidly when exposed to sunlight and is used for making polymers such as polysterene. Low levels of styrene occur naturally in plants as well as a variety of foods such as fruits, vegetables, nuts, beverages and meats.

STYRENE-BUTADIENE RUBBER: Synthetic rubber made by the copolymerisation of butadiene with styrene. It is used for making vehicle tyres, footwear, adhesives and sealants.

SUB-: 1. A prefix denoting an element that is contained within a given structure a relatively small proportion and shares its structural properties. 2. In chemistry, it denotes a basic compound, for example, subnitrate. Also it is an element in a low oxidation state.

SUB-BASE: In mathematics, for a topology, it is a collection of open sets of which the finite intersections generate a base for the topology.

SUB-BODY: A subset of a body that is itself a body.

SUBADDITIVE: 1. In mathematics, for a function with a semi-group as its domain, having the property that the value at the sum of two elements of the domain is less than or equal to the sum of the function's values at each element, expressed as $f(x + y) \leq f(x) + f(y)$. A function, f, is superadditive if its additive inverse $-f$ is subadditive. 2. For a set function on a class, having the property that the value for a union of two elements in the domain is less than or equal to the sum of the function's values at each element, expressed as $S(A \cup B) \leq S(A) + S(B)$. A function possessing countable or finite subadditivity is **countably subadditive**, as in Lebesgue outer measure.

SUBALPINE: 1. Of or relating to regions at or near the foot of the Alps. 2. Growing in altitude, which is below the alpine zone that does not contain trees and above the montane zone (relatively moist cool upland slopes below timberline dominated by large coniferous trees).

SUBARACHNOID SPACE (SUBARACHNOID CAVITY): The space between the arachnoid membrane and the pia matter. It is filled with cerebrospinal fluid that bathes the brain and spinal cord, cushions the brain within the skull and serve as a shock absorber for the central nervous system and also circulates nutrients and chemicals filtered from the blood and removes waste products from the brain.

SUBATOMIC PARTICLE: The small particles that are smaller than an atom. Examples include the electron, proton, and neutron. They are the basic units of all matter and energy. There are two types of subatomic particles: **elementary particles** such as photons, which are not made of other particles; and **composite particles** such as a boson or fermion. Particle physics and nuclear physics study these particles and how they interact.

SUBCLAVIAN ARTERY: A paired artery that originates on the left from the aortic arch and on the right from the brachiocephalic and passes beneath the collar bone and branches to supply blood to the arm.

SUBCLAVIAN VEIN: The continuation of the axillary vein, which joins the internal jugular to form the brachiocephalic vein and carries blood from the capillaries toward the heart.

SUBCLAVIUS: A small muscle between the clavicle and the first rib and arises in a short thick tendon from the junction of the first rib and its cartilage under the clavicle. The subclavius helps to steady the clavicle in movements of the shoulder-girdle.

SUBCONTRARY: 1. In logic, for a pair of statements, having a relationship that both cannot be false at the same time, under the same circumstances or in the same interpretation. For example, x as non-negative and x is non-positive are subcontrary if x is restricted to the reals, since at least one must be true, but both statements are true when $x = 0$. For a single statement, not able to be false when a given statement is false. 2. A statement that is subcontrary to a given statement.

SUBCRITICAL: Describing a chain reaction in a nuclear reactor that is not self sustaining or having a mass of fissionable material that is less than that needed for a chain reaction.

SUBCULTURE: A microbiological culture made by transferring microorganisms from a previous culture to a fresh growth medium. This method is used to prolong the life of a particular strain of microorganism, where there is a tendency of degeneration in older cultures.

SUBCUTANEOUS: Located, living or made beneath the skin or pertaining to areas beneath the skin.

SUBCUTANEOUS TISSUE (ADIPOSE LAYER; ADIPOSE TISSUE; FATTY LAYER): A layer of cells found chiefly below the skin and around major organs such as the kidneys, heart and the gut made up of fat and oils which act as energy reserve and also providing insulation against heat loss.

SUBDESIGN: In mathematics, it is a block design in which the sets of blocks and varieties are subsets of those of a given design.

SUBDIAGONAL: In mathematics, for a square matrix, it is the set of elements lying immediately below the main diagonal. The entries are of the form $a_{i,\,i-1}$.

SUBDIAGONAL MATRIX: A matrix whose entries are zero except in the subdiagonal.

SUBDUCTION: The process in which one lithospheric plate collides with and is overridden by or descends under, an adjacent plate. The subduction zone is the place where two lithospheric plates come collide, one riding over the other. Often the upper plate is a continental plate and the lower one is an oceanic plate and subduction leads to the formation of an oceanic trench. Most volcanoes on land occur parallel to and inland from the boundary between the two plates.

SUBDWARF: A star that is smaller and less luminous by one to two magnitudes (2.5-6 times fainter) than a normal dwarf (main sequence) star of the same spectral type. Subdwarfs are mainly old, galactic halo population II objects of types F, G and K. They are denoted by the prefix "sd", as in sdK7 and are placed in luminosity class VI.

SUBELLIPTICAL EGG: An egg that is elongated and tapered towards its rounded ends.

SUBERIN: A mixture of waxy substance similar to cutin, present in the cell walls of cork and enderdomis. Suberin is highly hydrophobic or impermeable to water and its main function is to prevent water from penetrating the tissue.

SUBFIELD: A subring of a field or ring that is also a field.

SUBGEOMETRY: In mathematics, a geometry formed by all the points that have homogeneous coordinates that are linear combinations of those of a given set of points. For example, lines and planes are subgeometries of a three-dimensional Euclidean geometry.

SUBGIANT: A star that has stopped fusing hydrogen in its core and has begun to evolve off the main sequence, increasing in size and luminosity, on its way to becoming a red giant. Subgiants produce energy in a hydrogen-burning shell that surrounds the core. They are mainly of types F, G and K and are placed in luminosity class IV. Subgiants are of smaller radius and lower luminosity than normal giant stars and lie between the main sequence and giant region of the Hertzsprung-Russell diagram.

SUBGRADIENT: The set of linear functionals, defined in terms of a given convex function, g, at a point as $\partial g(x)=\{\varphi:\varphi(y-x)\}$.

SUBGRAPH: A graph whose sets of vertices and edges are subsets of those of a given graph and that contains all the vertices joined by any of its edges.

SUBGROUP: In mathematics, a subset of another group that is also a group under the same binary operation; for example, the integers are a subgroup of the reals under addition, but the integers modulo n are not a subgroup of these since the operations are differently defined.

SUBHARMONIC FUNCTION: A function of two real variables on a domain having the property that whenever it is dominated by an harmonic function on the boundary of a subdomain, it is dominated throughout the subdomain. If the function has continuous second partial derivatives, it is subharmonic precisely when it has a non-negative Laplacian throughout the given domain. A function, f, is **superharmonic** if its additive inverse, $-f$, is subharmonic.

SUBIRRIGATION METHOD: One of the ways of irrigating a field in which water is applied to open ditches or tile lines until the water table is high enough to wet the soil.

SUBITISE/SUBITIZE: In mathematics, to recognise instantaneously the number of objects in a set without consciously using any mental or mathematical processes.

SUBJACENT ORBITALS: The next-to-highest occupied molecular orbital (NHOMO) and the second-lower unoccupied molecular orbital (SLUMO). In certain cases, they play an important role in the interpretation of molecular interactions according to the frontier orbital approach.

SUBJECT OF THE FORMULA: A variable in the form of a letter solved for in a formula which equals the rest of the formula in an arrangement. The subject of a formula is the single variable, usually on the left of the "=" that everything else is equal to. For example, in the expression $s=ut+\frac{1}{2}at^2$, the subject of formula is s. The process of solving for the variable or letter is the changing of the subject by transformation of the formula. For example, $s=ut+\frac{1}{2}at^2$, if then in $a=\frac{2(s-ut)}{t^2}$, a is made the subject of formula by transforming the formula.

SUBLATION: A flotation process in which the material of interest, adsorbed on the surface of gas bubbles in a liquid, is collected on an upper layer of immiscible liquid. There is no liquid-phase mixing in the bulk of the system. As a result recoveries can approach 100%.

SUBLIMATE: 1. A deposit that has grown from a vapour rather than from a solution. Sublimates are often seen in blowpipe testing and some minerals around volcanoes form this way. 2. Also known as **sublime**, it is to change directly from a solid to a gas without passing through a liquid phase.

SUBLIMATION: The transition of a substance from the solid phase to the gas phase without passing through an intermediate liquid phase or change of state from the solid to gaseous state without passing through the liquid state. It is an endothermic phase transition that occurs at temperatures and pressures below a substance's triple point in its phase diagram. Examples of substances subjected to this process are carbon dioxide, iron(II) chloride and iodine.

SUBLIME NUMBER: A number such that both the sum of its divisors and the number of its divisors are perfect numbers. The smallest sublime number is 12. There are 6 divisors of 12: 1, 2, 3, 4, 6 and 12, the sum of which is 28. Both 6 and 28 are perfect. The second sublime number begins 60865..., ends ...91264 and has a total of 76 digits! It is not known if there are larger even sublime numbers, nor if there are any odd sublime numbers.

SUBLINEAR CONVERGENCE: Any rate of convergence worse than linear.

SUBLINEAR FUNCTION: 1. A Minkowski function. 2. A function whose domain is a vector space that is subadditive and positively homogeneous, as makes sense whenever the range is an ordered vector space. A function, f, is superlinear if its additive inverse, $-f$, is sublinear.

SUBLINGUAL GLANDS: The smallest of the salivary glands, the others are **parotid** and **submandibular** glands. Sublingual glands lie in the floor of the mouth underneath the tongue.

SUBLITTORAL: 1. Designating or occurring in the shallow water-zone of a sea over the continental shelf and below the tide mark or ranging in depth to about 200 metres. 2. Occurring in a lake below the littoral zone to a depth of 6-10 metres.

SUBMARINE: 1. A vessel capable of travelling and functioning under the sea, used in research and military operations or a boat built to operate and travel for long periods under the sea. 2. Taking place or growing under water, especially in the sea.

SUBMATRIX: A submatrix formed from a given matrix by deleting some of its rows or columns or both of the matrix.

SUBMERGENCE: A rise in the water level in relation to land, so that a former dry land become sinundated. It results either from a sinking of the land or from a rise of the water level. For example, submergent coastlines are stretches along the coast that have been inundated by the sea by a relative rise in sea levels.

SUBMILLIMETRE WAVES: The electromagnetic radiation with wavelengths below one millimetre (i.e. with frequencies greater than 300 GHz), extending to radiation of the far infrared. A medium pressure mercury lamp in quartz is a source of submillimetre radiation. Submillimetre waves can be detected by a Golay cell.

SUBMODULE: In mathematics, a module over a ring, R that is contained in another module, M over the same ring and has the same addition. In other words, if M is a left R-module, then the submodule of M is any subset of M that is a left R-module under the operations induced from M. The subset {0} is called the **trivial submodule** and is denoted by (0). The module M is a submodule of itself, an **improper submodule**.

SUBMUCOSA: A layer of dense irregular connective tissue or loose connective tissue that lies beneath the mucous membrane, for example, in the stomach and intestines. It contains blood vessels, nerves and lymphatics, vessels that carry lymph material and absorbs nutrients into the blood stream.

SUBNET MASK: A number that defines a range of IP addresses that can be used in a network. Subnet masks are used to designate subnetworks or subnets, which are typically local networks LANs that are connected to the Internet. Systems within the same subnet can communicate directly with each other, while systems on different subnets must communicate through a router. Therefore, subnetworks can be used to partition multiple networks and limit the traffic between them.

SUBPOPULATION: 1. In biology, it is a subdivision of a specific type of organism living in a given area and sharing one or more properties. For example, men are a subpopulation of humans. 2. In statistics, it is a subset of a given population that is subject to a statistical study.

SUBRING: A subset of a ring that is itself a ring under the same binary operations of addition and multiplication on the ring and are restricted to the subset and which shares the same multiplicative identity as the ring.

SUBSAMPLE: A subsample may be one of the following: (i) a portion of the sample obtained by selection or division; (ii) an individual unit of the lot taken as part of the sample; (iii) the final unit of multistage sampling. The term 'subsample' is used either in the sense of a 'sample of a sample' or as a synonym for 'unit'. In practice, the meaning is usually apparent from the context or is defined.

SUBSCRIBER IDENTITY MODULE (SUBSCRIBER IDENTIFICATION MODULE; SIM CARD): A small card that is used in cellular phones and sometimes other devices that contains subscriber information. One advantage of a SIM card is that it can be moved from phone to phone, making it easier to change phones if needed.

SUBSEQUENCE: In mathematics, a sequence that can be derived from another sequence by deleting some of its terms and retaining their order. For example, $\langle a_2, a_3 \rangle$ is a subsequence of $\langle a_1, a_2, a_3 \rangle$.

SUBSET: 1. A set is a subset of another set, if all its elements belong to the other set. In other words, set B is a subset of set A if every element of B is also an object of A. B is a subset of A is usually written as B ⊆ A and A is a subset of B is usually written as A ⊆ B. A **proper set** is strictly contained within a larger set, C as there are some elements in B that are not in C. 2. A programming language that contains all the features of a given language and has been expanded or enhanced to include other features as well.

SUBSHELL: Subdivision of an electron shell or orbitals in the electron shell of an atom.

SUBSIDENCE: The sudden sinking or gradual downward settling of the Earth's surface with little or no horizontal motion. This may be caused by geologic or anthropogenic processes, for example, the caving in of natural or man-made underground voids.

SUBSIDIARY CELL (ACCESSORY CELL): One of a group of morphologically differentiated epidermal cells immediately surrounding the guard cells. Stomatal complexes lacking subsidiary cells are termed **anomocytic**. The number, arrangement and development of subsidiary cells are often useful taxonomic characters.

SUBSISTENCE AGRICULTURE: Self-sufficiency farming in which farmers focus on growing enough food to feed their families. The typical subsistence farm has a range of crops and animals needed by the family to eat during the year. **Subsistence crops** are grown for consumption by the farmer and not for sale.

SUBSOIL (SUBSTRATA): Lower infertile layer of soil composed mainly of rock particles or layers of soil that lie beneath the top soil.

SUBSOILING: The breaking of compact subsoils, without inverting them, with a special knife-like instrument, which is pulled through the soil usually at depths of 30 to 60 cm and spacings of 60 to 150 cm. This is done to improve growth in all crops where soil compaction is a problem. A **subsoiler**, also called a **flat lifter** is a tractor-mounted farm implement used to loosen the soil at some depth below the surface without turning it over.

SUBSONIC SPEED: A speed relative to surrounding fluid less than that of the speed of sound in the same fluid. For aircraft speeds, which are very much less than the speed of sound, the aircraft is said to be subsonic.

SUBSPACE: In mathematics, it is a subset, A of a topological space X with the the same properties as X; the open set in A consists of the intersections of the open sets of X with A.

SUBSPECIES: A group of individuals within a species that breed more freely among themselves than with other members of the species. In biological classification, it is either a taxonomic rank subordinate to species or a taxonomic unit in that rank. For example, zebras are grouped into three species, each with a distinctive stripe pattern. They are Grévy's zebra, the mountain zebra and the plains zebra also known as Burchell's zebra.

SUBSTANCE: A sample of matter that can be identified by name and have specific chemical composition. An example of a chemical substance is pure water. It has the same properties and the same ratio of hydrogen to oxygen whether it is isolated from a river or made in a laboratory. Some other chemical substances are diamond, gold, salt (sodium chloride) and sugar (sucrose). Generally, chemical substances exist as a solid, liquid, gas or plasma and may change between these phases of matter with changes in temperature or pressure. Chemical reactions convert one chemical substance into another.

SUBSTANCE CONTENT: Amount-of-substance of a component divided by the mass of the system.

SUBSTANCE FLOW RATE: Amount-of-substance of a component crossing a surface divided by the time.

SUBSTANCE FRACTION: Ratio of the amount-of-substance of the component to the total amount-of-substance in the system containing the component.

SUBSTANCE P: A neuropeptide comprising 11 amino acid residue that is found widely in the brain, spinal cord and the peripheral nervous system. It mediates the effects of the parasympathetic nervous system by stimulating contraction of the gut, causing dilation of blood vessels and increasing the flow of saliva in the salivary glands.

SUBSTANTIVITY: 1. The affinity of a dye for its substrate. 2. The capacity of an oral antimicrobial agent to continue its therapeutic activity for a prolonged period of time.

SUBSTILSIN (PROTEASE): A group of enzymes that digest proteins. They are active within specific pH ranges and function in acidic medium.

SUBSTITUENT: 1. An atom or group that replaces another in a substitution reaction. 2. An atom or group of atoms substituted in place of a hydrogen atom on the parent chain of a hydrocarbon. In polymers, side chains extend from a backbone structure. In proteins side chains are attached to the alpha carbon atoms of the amino acid backbone.

SUBSTITUENT ATOM (GROUP): An atom (group) that replaces one or more hydrogen atoms attached to a parent structure or characteristic group except for hydrogen atoms attached to a chalcogen atom.

SUBSTITUTE: 1. To replace an atom or group of atoms in a molecule with another atom or group. 2. To replace one mathematical element with another of equal value or to replace a variable with a mathematical expression in a statement. For example, $x = 3y$ can be substituted in $2x - 4y = k$ to give $2y = k$.

SUBSTITUTION: 1. The replacement of one mathematical element with another of equal value. 2. The replacement of one logical expression with another or a replaced logical expression. 3. In chemistry, it is a reaction in which an atom or fragment within a molecule is replaced with another. 4. In genetics, a type of mutation due to replacement of one nucleotide in a DNA sequence by another nucleotide or replacement of one amino acid in a protein by another amino acid.

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SUBSTITUTION GROUP (PERMUTATION GROUP): In mathematics, a group whose elements are permutations of a given set and whose group operation is the composition of permutations in the group and multiplication is defined as successive permutation. The group of all permutations of n objects has $n!$ elements and determines the full symmetric group and the group of all even permutations of n objects is called the alternating group.

SUBSTITUTION INSTANCE: In logic, it is an expression derived from another by substitution of constants for variables.

SUBSTITUTION MUTATION (POINT MUTATION): It is a mutation in which one base pair in DNA molecule sequence is replaced by another. If the mutation occurs in noncoding sequences, it does not often result in an altered amino acid sequence during translation. If such mutation occurs in the promoter sequence of a gene, the effect may be apparent since the expression of the gene may change. Substitution mutations may be classified based on functionality as nonsense mutation, missense mutation andsilent mutation. Depending on the type of nitrogenous base involved, this type of mutation may be categorised as transition or transversion. An example of a disease caused by substitution mutation is sickle-cell anaemia, in which the codon GAG found in the normal Hb gene is mutated to GTG. This is called a base substitution mutation as adenine (A) is replaced by thymine (T).

SUBSTITUTION PRODUCT: A product obtained by replacing one atom or group in a molecule with another atom or group.

SUBSTITUTION REACTION: A reaction in which an atom or group of atoms replace an atom or group of atoms. For example, the free radical reaction between methane and chlorine, catalysed by an ultraviolet light, $CH_4 + Cl_2 \rightarrow CH_3Cl + HCl$. In the reaction, one chlorine atom replaces a hydrogen atom in CH_4 to form CH_3Cl.

SUBSTITUTION RULE: A method for evaluation of an integral by substitution, based on the identity between differentials, where u is a function of x as $du = u'dx$, which can be written as $\int f(u)u'\,dx = \int f(u)\,du$.

SUBSTITUTION THEOREM: In logic, the theorem that universally quantified statement implies any instance of it and that an existentially quantified statement is implied by any instance of it. Given the deduction theorem, this is equivalent to the instantiation rule for the universal quantifier and the introduction rule for the existentially quantifier.

SUBSTITUTIVE NAME: A name which indicates the exchange of one or more hydrogen atoms attached to a skeletal atom of a parent structure or to an atom of a characteristic group for another atom or group, which may be expressed by a suffix or by prefixes.

SUBSTITUTIVITY: The principle that terms with the same reference may be substituted for one another in a sentence without changing its truth value; thus since the morning star is the evening star, "if the morning star is visible in the morning" is true, then so is "the evening star is visible in the morning". This does not hold in opaque modal contexts or of de dicto modalities.

SUBSTOICHIOMETRIC EXTRACTION: Solvent extraction in which the amount of reagent used is lower than that dictated by stoichiometry. If the constants of formation and extraction of the complexes are high, the amount of extracted metal is dictated by the amount of extractant introduced.

SUBSTOICHIOMETRIC ISOTOPE DILUTION ANALYSIS: A kind of isotope dilution analysis in which the final isotopic abundance is estimated from the amount of the nuclide present in a known quantity of the relevant element separated from the test portion, where this quantity is smaller than the total amount of that element present in the test portion.

SUBSTRATE: 1. A compound or mixture of compounds acted on by an enzyme in a biochemical reaction or the substance that is affected by the action of a catalyst. 2. A material on which a sedentary organism such as barnacle lives and grows. In this case the substrate either acts as a support and/or provides nutrients to the organism. 3. The single crystal of a semiconductor used as the basis for an integrated circuit or transistor. 4. The substance on which some other substance is adsorbed or in which it is absorbed. For example, the porous solid absorbing a gas or the material to which a dye is attached.

SUBSTRATE FEEDERS: Animals such as earthworms or termites that eat the soil or wood through which they burrow, digesting organic materials to fertilise and aerate the soil; or animals that live in or on their food source and actually move by eating their way through the food. Substrate feeders include caterpillars, which eat their way through leafy greens, leaving a trail of faeces behind them.

SUBSTRATE-LEVEL PHOSPHORYLATION: The formation of adenosine triphosphate (ATP) by directly transferring a phosphate group to adenosine diphosphate (ADP) from an intermediate substrate in catabolism. An example of substrate-level phosphorylation occurs in glycolysis when phosphoenolpyruvate (PEP) reacts with adenosine diphosphate (ADP) to form ATP and pyruvate, a reaction catalysed by the enzyme pyruvate kinase. Substrate-level phosphorylation is one of the two ways in which a phosphate group is introduced into a molecule; the other one is oxidative phosphorylation.

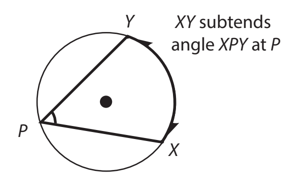
SUBTHRESHOLD STIMULUS: In biology, it is a stimulus that is not strong enough to trigger an action potential in excitable cells (such as neurons, muscle cells (skeletal, cardiac, smooth, etc.) that are electrically excited resulting in the generation of action potentials.

SUBSTRUCTURE: A structure whose domain is a subset of that of a given structure and has the same properties as that of the given structure. Examples are subgroup, subalgebra, subdesign, subfield, subgraph, sublattice, submodule and subring.

SUBSURFACE TILLAGE: Tillage with a special sweep-like plough or blade, which is drawn beneath the surface at depths of several centimetres and cuts plant roots and loosens the soil without inverting it or incorporating the surface cover.

SUBTANGENT: In mathematics, it is the projection of a tangent to a curve onto the given point and the coordinate axes or the x-axis in the two-dimensional Cartesian plane. In other words, it is a segment of the x-axis lying between the x-coordinate of the point at which a tangent is drawn to a curve and the intercept of the tangent with the x-axis.

SUBTEND: 1. In mathematics, if two points X and Y on a curve are joined to a point P then XY is said to have subtended the angle XPY at P. An angle subtended by an arc is one whose two rays pass through the endpoints of the arc. An example is the angle subtended by an arc of a circumference when the angle's vertex is a point on the circumference. A simple theorem of plane geometry states that arcs of equal lengths subtend equal angles in such a situation. 2. In botany, to extend beneath or close to, as between a leaf and its stem.



Subtend

SUBTERRANEAN: Existing or occurring below the earth's surface.

SUBTILISIN: Any group of protein-digesting enzymes produced by certain strains of a soil bacterium (*Bacillus amyloliquefaciens*). They have broad specificities and work over a wide range of pH values. Subtilisins belong to **subtilases**, a group of serine proteases that initiate the nucleophilic attack on the peptide (amide) bond through a serine residue at the active site. **Subtilin Carlsberg** is an alkaline protease, derived from bacillus lichen forms that is used widely in detergents. 30-40% of the peptide bonds in casein, a milk protein are hydrolysed by subtilin Carlsberg. Subtilisins are widely used in commercial products, for example, in laundry and dishwashing detergents, cosmetics, food processing, skin care ointments, contact lens cleaners and for research purposes in synthetic organic chemistry.

SUBTRACTION: A mathematical operation of deducting one number or quantity from another or a binary operation to find the remainder when part of a whole is removed. In other words, it is a mathematical operation to calculate the difference between two values. If the denominator in two fractions are different when one is being subtracted from another they have to be made similar to leave single fractional answer. To subtract is to calculate the difference between two values by substraction.

SUBTRACTION STRATEGIES: Various methods that help students to take away one number from another, for example, by forward counting, backward counting, place value understanding, manipulatives, etc.

SUBTRACTION WORD PROBLEMS: Mathematical problems that are expressed in words rather than in mathematical notations and require that the unknown quantity is found by using a subtraction operation.

SUBTRACTIVE: In mathematics, indicating or requiring subtraction; having the minus sign (−).

SUBTRACTIVE NAME: A name for a modified parent structure in which prefixes and/or suffixes indicate the removal of atoms or groups and, where required, replacement by an appropriate number of hydrogen atoms.

SUBTRAHEND: The number to be subtracted from another, called the **minuend**. For example, in 5 - 3, the minuend is 5 and the subtrahend is 3.

SUBTROPICAL: Distributed, related to or characteristic of geographic areas intermediate between tropical and temperate regions.

SUBUNIT: Individual polypeptide chains in a protein or one of the polypeptide chains composing an oligomeric protein.

SUCCESS: In statistics, in a binomial experiment, it is an outcome of an experiment, in contrast with failure. The probability of a given number of successes is described by a binominal distribution.

SUCCESSION (ECOLOGICAL SUCCESSION): In ecology, it is the sequence of ecological communities that develop in an area from the initial stages of colonisation until a stable mature climax community is reached or the series of changes that create a full-fledged plant and animal community, such as from the colonisation of a bare rock to the establishment of a forest. There are two types of ecological succession: **primary succession**, which is the ecological succession that occurs following an opening of uninhabited, barren habitat or that occurs on an environment that is devoid of vegetation and usually lacking topsoil; and **secondary succession**, which is the ecological succession that occurs on a pre-existing soil after the primary succession has been disrupted or destroyed due to a disturbance that reduced the population of the initial inhabitants. Numerous factors which may be brought about or by the colonisers affect the nature of the succession. For example, shrubs produce deep soil, which in turn produce trees, which may attract animals seeking to use them as shelter, whose droppings act as manure which results in the growth of other plants, etc.

SUCCESSIONAL CROPPING: 1. The growing of several crops one after the other during the same growing season. 2. The process of sowing a crop such as lettuce over a long period, so that harvesting takes place over a similarly long period.

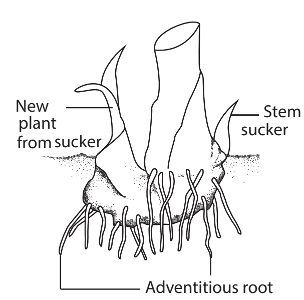
SUCCESSIVE: In mathematics, following one after the other. For example, 2 and 3 are successive whole numbers, 6 and 8 are successive even numbers. Also it is consecutive integers that follow each other in a patterned order, usually just one.

SUCCULENT: In botany, it is a thick fleshy plant such as cactus, agave and aloe plant that conserves water by storing it in fleshy leaves or stems to minimise evaporation. In succulents, the stomata close during the day and open at night in order to minimise transpiration. Succulent plants are usually found in regions of little rainfall and generally have long roots to absorb a maximum amount of water.

SUCCUS: A fluid, especially a secretion, of a plant or animal origin or fluid such as gastric juice or vegetable juice, contained in or secreted by living tissue.

SUCCUSENTERICUS: An alkaline mixture of water, mucoproteins and hydrogen carbonate ions produced by glands within the duodenum. They are responsible for most of the digestion that occur in the alimentary canal.

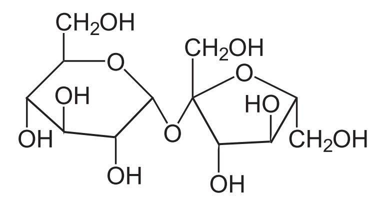
SUCKER: 1. A secondary shoot that arises from an underground root or stem and is often able to produce its own roots and grow into a new plant at the expense of the parent plant. 2. A muscular organ, found on the tentacles of octopuses and similar sea animals, used to cling to or hold things such as prey. 3. A young animal such as a young pig or whale that is still taking milk from its mother. 4. A bony bottom-feeding freshwater fish with a downward-facing sucking mouth without teeth. 5. The mouth of an animal such as the leech or lamprey that is adapted for sucking in food. 6. The piston or piston valve of a suction pump. 7. A pipe through which a liquid is drawn by means of suction.



Sucker of a Banana

SUCRASE: An enzyme that catalyses the hydrolysis of sucrose into its constituent molecules of glucose and fructose. Sucrase is secreted by the tips of the villi of the epithelium in the small intestine. Its levels are reduced in response to villi-blunting events such as celiac sprue and the inflammation associated with the disorder. The levels increase in pregnancy/lactation and diabetes as the villi hypertrophy.

SUCROSE (CANE SUGAR; BEET SUGAR; SACCHAROSE): A crystalline disaccharide comprising one molecule of glucose and one molecule of fructose with the chemical formula $C_{12}H_{22}O_{11}$. It is soluble in water, slightly soluble in alcohol and ether, crystallises in long, slender needles and is dextrorotatory, rotating the plane of polarised light to the right. It is found in many plants but extracted as ordinary sugar mainly from sugar cane and sugar beets. It is widely used as a sweetener or preservative and in the manufacture of plastics and soaps. When heated to temperatures above 180°C (453 K), sucrose becomes **caramel**, an amorphous, brown, syrupy substance.


Chemical Structure of Sucrose

SUCTION: The physical force created by a difference in pressure, such as that created by the intake side of a pump when a piston is moved up and down inside a cylinder or vacuum. The pressure gradient between this region and the ambient pressure will propel matter toward the low pressure area.

SUDAN STAINS: Any of various temporary aniline stains used to colour fats and waxes. They include (Sudan II, Sudan III, Sudan IV, Oil Red O and Sudan Black B. Sudan stains have high affinity to fats and are used to demonstrate triglycerides, lipids and lipoproteins.

SUDDEN INFANT DEATH SYNDROME, SIDS (CRIB/COT DEATH): A disorder resulting in the sudden and unexpected death during sleep of infants, usually between the ages of two weeks and one year. SIDS is more common in infants with low birth weights of 2.5 kg or less at birth. Other risk factors include sleeping on a soft surface, maternal smoking during pregnancy, overheating, late or no prenatal care and young maternal age.

SUDDEN POLARISATION: The occurrence of a large intramolecular charge separation in the singlet excited state of polyenes and their derivatives twisted about a double bond. Non-symmetrical substitution or geometrical distortion is effective in polarising the system. An example is the stabilisation of the zwitterionic structure of 90° twisted ethene with one methylene group pyramidalised.

SUDORIFICS (DIAPHORECTICS): Agents or substances containing properties that produce sweat. Sudorifics include herbs such as chilli pepper, sage, thyme, ginger.

SUFFICIENT CONDITION: A condition, P, that, if satisfied, guarantees a consequence Q, is obtained.

SUFFICIENT STATISTIC: For a parameter, it is a static, T(X) with the property that the conditional distribution of X given T(X) does not depend upon the given parameter and is the same for all members of the family. If the distribution of a sample is known, it contains all the useful information or statistics necessary to estimate the parameters, without further reference to the data.

SUFFOCATION: Impairment of breathing, resulting in a lack of oxygen and a build-up of carbon dioxide and other waste products in the blood and body tissues. Suffocation results from an obstruction in the air passages to the lungs, which may be caused either by a foreign object or by swelling of tissues in and around the throat. Suffocation also occurs when the air breathed contains too little oxygen. Drowning, severe pneumonia and congestive heart failure create excess fluid in the lungs, causing the lungs to stop functioning. Prolonged suffocation from any cause leads to death.

SUFFRUTESCENT HABIT: A growth form in which the plant is woody at the base but has herbaceous branches. It is seen in alpine willows.

SUFFRUTICOSE: Of or relating to a plant with a woody stem at the base, but shrubby higher up.

SUGAR (SACCHARIDE): A type of solid crystalline carbohydrate that dissolves easily in water and has more or less a sweet taste. Sugars fall into three groups: the monosaccharides, disaccharides and trisaccharides. The monosaccharides are the simple sugars; they include fructose and glucose. The disaccharides are formed by the union of two monosaccharides with the loss of one molecule of water. Disaccharides include lactose, maltose and sucrose. Sugars, which are widely distributed in nature are manufactured by plants during the process of photosynthesis and are found in many animal tissues. Ribose, a monosaccharide sugar containing five carbon atoms in its molecule, is a constituent of the nuclei of all animal cells. Five-carbon sugars are known as **pentoses**.

SUGAR ACID: An oxidised derivative of a monosaccharide. There are two biologically important types of sugar acid, the aldonic acids and the uronic acids. In aldonic acids, the aldehyde group of a monosaccharide is oxidised to a carboxylic acid. The aldonic acid of glucose, gluconic acid, is an intermediate in the pentose phosphate pathway. Ascorbic acid (vitamin C) is also an aldonic acid.

SUGAR ALCOHOL: A monosaccharide in which the aldehyde group is reduced to an alcohol. Thus, D-glucose yields the sugar alcohol sorbitol, while the alcohol from mannose is mannitol. Other important sugar alcohols are **glycerol**, a central compound in lipid metabolism and **inositol**, an important intermediate in cell wall polysaccharide synthesis. Sugar alcohols, other than glycerol, are limited in nature to the plant kingdom.

SUGAR CANE: A tall perennial monocotyledonous plant of the genus Saccharum of the grass family, Poaceae that is native to warm temperate to tropical regions. The most widely cultivated species is *Saccharum officinarum*. It measures between 2 - 6 metres in height and about 2.5 – 7.5 cm in diameter. It has stout jointed fibrous stalks that resemble bamboo. After harvesting, the stalks regenerate, ensuring another harvest. The leaves of the plant grow from the nodes of the stem, arranged in two rows on either side of the stem. Its inflorescence is a terminal panicle with two spikelets and seeds protected by husks (glumes) covered in silky hair. Two flowers are produced on the inflorescence, one sterile and the other bisexual. Its main product is sucrose which is in the sap that accumulates in the stalk internodes. The sucrose is extracted from the sap and purified in specialised mill factories and used as a raw material in human food industries such as for producing sugar. Additionally, it could be fermented to produce ethanol for alcoholic beverages or gasohol as fuel for cars. Bagasse, the pulpy residue left after the juice has been extracted can be burnt to produce steam, to generate electricity. Its stems and byproducts are used for feeding livestock. Brazil is the largest producer of sugar cane.

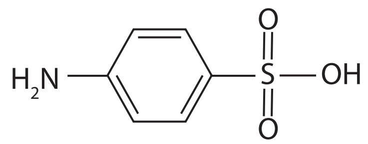
SUITCASE FARMER: A farmer who lives some distance (more than 48km) from his or her holding; or a grower of grain or other crops who lives outside the community except during the ploughing, seeding and harvesting seasons and often has a farm without buildings and does much of the farming by hired migratory crews.

SULCATE: In botany, having narrow, deep longitudinal furrows or grooves, as a stem or tissue.

SULFAMIC ACID: A colourless crystalline solid with the chemical formula NH_2SO_2OH, which is extremely soluble in water and normally exists as the zwitterions, $H_3N^+.SO_3^-$. It decomposes at 205 °C (478 K). It is a strong acid, readily forming sulfamate salts. It is used in electroplating, hard-water scale removers, herbicides and artificial sweetners.

SULFANES: Compounds of hydrogen and sulfur containing chains of sulfur atoms, having the general formula H_2S_n.

SULFANILIC ACID (4-AMINOBENZENE SULFONIC ACID): A colourless crystalline solid made by prolonged heating of phenylamine (aniline) sulfate with the chemical formula $H_6NC_6H_4SO_2OH$, molar mass 173.19, density 1.485 and melting point 288 °C (561 K). It readily forms diazo compounds and used to make dyes and sulphur drugs.



Chemical Structure of Sulfanilic Acid

SULFATE: Salt or ester derived from sulfuric(VI) acid or an ion and an acid radical with the chemical formula SO_4^{2-}. Most sulfates are water-soluble and require a very high temperature to decompose them. Sulfate salts contain the ion SO_4^{2-} and organic sulfates have the formula R_2SO_4 where R is an organic group. Sulfate can be prepared by dissolving a metal in sulfuric acid; reacting sulfuric acid with a metal hydroxide or oxide; or oxidising metal sulfides or sulfites.

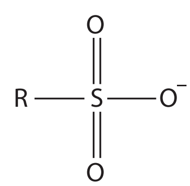
SULFATION: Introduction of sulfate (SO_4) into a compound. Biochemical sulfation is a phase II enzyme reaction. Sulfation can also be posttranslational modification of protein. The target amino acid is tyrosine. It is called tyrosine sulfation.

SULFIDE: A compound of sulfur with another element in which sulfur is the more electronegative element. They are produced by reaction with sulfur or hydrogen sulfide. Sulfides occur in a number of minerals. Some of the more volatile sulfides have extremely unpleasant odours. For example, hydrogen Sulfide smells of rotten egg. Compounds of sulfur with non-metals are covalent compounds (H_2S). Organic sulfides are named from the linking groups (CH_3SCH_3 – dimethyl sulfide).

SULFINATE (DITHIONITE; HYPOSULFITE): A salt that contains the negative ion $S_2O_4^{2-}$, usually formed by the reduction of sulfites with excess SO_2. Solutions are not very stable and decompose to give thiosulfate and hydrogensulfite ions.

SULFINIC ACID (DITHIONOUS ACID; HYPOSULFUROUS ACID): An unstable acid known in the form of its salts (sulfinates) with the chemical formula $H_2S_2O_4$. An example of a well-studied sulfinic acid is phenylsulfinic acid. A commercially important sulfinic acid is thiourea dioxide

SULFONAMIDES: Organic compounds containing the group $-SO_2NH_2$. They are amides of sulfonic acid and are members of a class of synthetic and anti-bacterial drugs which act by enzyme inhibition. Many have antibacterial action and are also known as **sulphur drugs**, including sulfadiazine ($NH_2C_6H_4SO_2NHC_4H_3N_2$), sulfathiazole and several others. They act by preventing the bacteria from reproducing and are used to treat a lot of bacterial infections, especially of the gut and urinary system.



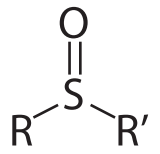
Chemical Structure of Sulfonate

SULFONATE: A salt or ester of sulfonic acid. It contains the functional group $R\text{-}SO_2O^-$.

SULFONATION: A substitution reaction that involves the replacement of a hydrogen atom by the sulfonic acid group or a type of reaction in which sulfonic acid group is substituted on a benzene ring to form a sulfonic acid. It is an example of electrophilic substitution in which the electrphile is a sulfurtrioxide molecule.

SULFONIC ACID: Compounds that contain the group $-SO_3H$ bonded directly to carbon. Organic sulfonic acids are used in the manufacture of dyes, detergents and drugs. These acids are formed by the reaction of aromatic hydrocarbons with concentrated sulfuric acid. Sulfonic acids are strong acids which ionise completely in solution to form the sulfonate ion $-SO_2O^-$.

SULFONIUM COMPOUND: Compounds containing the ion R_3S^+ (sulfonium ion) or of the general formula R_3SX, where R is an organic radical and X is an electronegative element or radical. These compounds can be formed by reacting organic sulfides with halogen compounds. Diethyl sulfide, for example, reacts with chloromethane to give diethylmethylsulfonium chloride.



Chemical Structure of Sulfoxides

SULFOXIDES: Organic compounds containing the group =S=O (sulfoxide group) linked to two other groups. They are of the general formula RSOR′, where R and R′ are organic radicals. An example is dimethyl sulfoxide (DMSO).

SULFUR: A brittle, yellow, non-metallic element with the symbol S that is a member of Group 16 (VIA) of the Periodic Table, which occurs alone in nature or combined in sulfide and sulfate minerals. Sulfur burns in air to form sulfur dioxide and it is used to make sulfuric acid, matches, fungicides and gunpowder. The two allotropic forms of sulfur are rhombic sulfur and monoclinic sulfur. It is an essential element in living organisms especially in many proteins. It is also found in various cell metabolites such as coenzyme A. It is also absorbed by plants from the soil as the sulfate ion.

SULFUR BACTERIA: Any of various groups of unrelated bacteria that utilises sulfur, sulfide or sulfate in their metabolism and important in the sulfur cycle in nature. The anaerobic photoautotrophic green sulfur bacteria and purple sulfur bacteria can use hydrogen sulfide as a source of electron in photosynthesis, producing sulfur as a by product. *Thiobacillus*, which is found widely in marine and terrestrial habitats, oxidises sulfur to produce sulfates that are useful to plants; in deep ground deposits it generates sulfuric acid, which dissolves metals in mines but also corrodes concrete and steel. *Desulfovibrio desulficans* reduces sulfates in waterlogged soils and sewage to hydrogen sulfide, a gas with the rotten egg odour so common to such places.

SULFUR CYCLE: Cycling of sulfur between the living and non-living components of the environment. Sulfate derived from weathering and oxidation of rocks is absorbed by plants and incorporated into sulfur-containing proteins, then passed along the food chains to animals. Decomposition of dead organic matter and faeces by anaerobic sulfate-reducing bacteria returning sulfur to the non-living or abiotic environment in the form of hydrogen sulfide which can be converted back to sulfate or to the element (sulfur) by the action of photosynthetic and sulfide-oxidising bacteria. The elemental sulfur becomes incorporated into rocks and the process begins again and continues.

SULFUR DICHLORIDE DIOXIDE (SULFURYL CHLORIDE): A colourless liquid with the chemical formula SO_2Cl_2, relative density 1.67, melting point -54 °C (219 K) and boiling point 69 °C (342 K). It decomposes in water as well as soluble in benzene. The compound is formed by the action of chlorine on sulfur dioxide in the presence of sunlight or iron(III) chloride as a catalyst. It is used as a chlorinating agent and a source of the related fluoride (SO_2F_2).

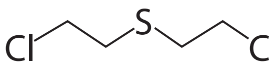
SULFUR DICHLORIDE OXIDE (THIONYL CHLORIDE): A colourless fuming liquid with the chemical formula $SOCl_2$, molar mass 18.97 g/mol, density 1.638 g/cm^3 (liquid), melting point -104.5 °C (168.7 K) and boiling point 74.6 °C (347.8 K) and decomposes at 80 °C (353 K). It hydrolyses rapidly in water as well as being soluble in benzene. It may be prepared by the direct action or by the reaction of phosphorous(V) chloride with sulfur dioxide. It is used as a chlorinating agent in synthetic organic chemistry in a reaction which replaces –OH groups with Cl.

SULFUR DUST: Powdered sulfur, used as an insecticide, especially against mites and scale insects and as a fungicide, particularly in the control of mildew. It gives off sulfur dioxide when ignited and is thus often used as a fumigant. It is also mixed with soil to kill soil-borne pathogens. Elemental sulfur, in combination with calcium hydroxide and water, forms lime sulfur, an orange-coloured liquid used to kill various mites and to control peach leaf curl, scab and powdery mildew.

SULFUR DYE: Dye made by heating certain organic compounds with sulfur or alkali polysulfides. The dyes are absorbed by cotton from a bath containing sodium sulfide or sodium hydrosulfite and are made insoluble within the fibre by oxidation. During this process, the dyes form larger complex molecules which are the basis of their good wash-fastness. The dyes have good all round fastness except to chlorine. Due to the highly polluting nature of the dye-bath effluent, sulfur dyes are being phased out. Sulfur dyes are primarily used for dark colours such as black, brown and dark blue. The deep indigo blues of denim blue jeans are a product of sulfur dyes.

SULFUR HEXAFLUORIDE: An inorganic, colourless, odourless, non-flammable gaseous compound. Its chemical formula is SF_6 with a molar mass of 146.06 g/mol, a density of 6.17 g/l, melting point of 209 k (−64 °c), and a boiling point of 222.3 (−50.8 °C). It is composed of an inorganic fluorinated inert gas comprised of six fluoride atoms bound to one sulfur atom. It is poorly soluble in water but quite soluble in nonpolar organic solvents. It is generally transported as a liquefied compressed gas. It is an extremely potent greenhouse gas, and an excellent electrical insulator of high-voltage equipment. It is of the six greenhouse gases to be curbed under the Kyoto Protocol. Its Global Warming Potential is 23,900.

SULFUR MUSTARD: A chemical warfare agent with the chemical formula $C_4H_8Cl_2S$, relative density 1.27, melting point 14.4 °C (287.5 K) and boiling point 217 °C (490 K). It is a class of related cytotoxic, vesicant chemical warfare agent with the ability to form large blisters on exposed skin.Pure sulfur mustards are colourless and viscous liquids at room temperature. However, when used in impure form, such as warfare agents, they are usually yellow-brown in colour and have an odour resembling mustard plants, garlic or horseradish.



Chemical Structure of Sulfur Mustard

SULFUR(IV) OXIDE (SULFUR DIOXIDE): The International Union of Pure and Applied Chemistry's (IUPAC) name for **sulfur dioxide**. It is a choking, acrid, pungent, irritating, colourless, poisonous gas with the chemical formula SO_2, molar mass 64.066 g mol^{-1}, density 2.6288 kg m^{-3}, melting point –72 °C (201 K) and boiling point –10 °C (263 K). It is made by burning sulfur in air or oxygen. It occurs in industrial flue gases and is a major cause of acid rain. In the laboratory, sulfur(IV) oxide is prepared by the action of dilute hydrochloric acid on sodium trioxosulfate(IV) such as Na_2SO_3. Crystals of Na_2SO_3 are placed in a flask, dilute HCl is added and the mixture is heated until it is hot. During the process, effervescence occurs and the gas produced is dried by concentrated H_2SO_4 and collected by downward delivery by displacement of air, since SO_2 is very soluble in water and much denser than air. The reaction is

$Na_2SO_{3\,(s)} + 2HCl_{(aq)} \rightarrow 2NaCl_{(aq)} + SO_{2(g)} + H_2O_{(l)}$.

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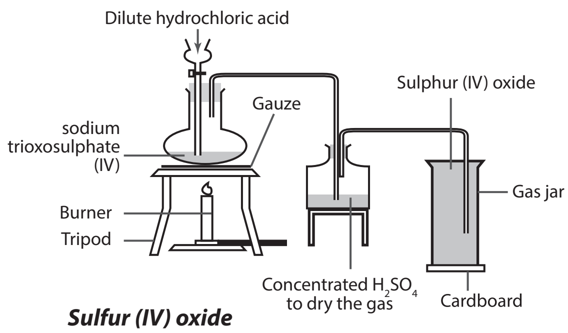

Sulfur (IV) oxide

SULFUR(VI) OXIDE: The International Union of Pure and Applied Chemistry's (IUPAC) name for **sulfur trioxide**, with the chemical formula SO_3. It is a volatile solid with a sharp odour. It is produced in the contact process by the combination of sulfur dioxide and oxygen over a catalyst (vanadium pentoxide). The oxide is acidic and it reacts violently with water to form sulfuric acid.

SULFURIC ACID (TETRAOXOSULFATE(VI) ACID): A dense, viscous, colourless liquid with the chemical formula H_2SO_4, relative density 1.84, melting point 10.36 °C (283.5 K) and boiling point 338 °C (611 K). It is extremely corrosive. The pure acid is rarely used. It is commonly available as a 96-98% solution with a melting point of 3.0°C (276 K). It is made by the contact process and is used extensively in refining petrol, in the manufacture of fertilisers, paints, detergents, explosives, dyes and fibres. It forms the acid component of car batteries.

SULFUROUS ACID (TETRAOXOSULFATE(IV) ACID): An acid with the chemical formula H_2SO_3 which only exists in aqueous solution, formed by passing sulfur dioxide in water. It may also be referred to as a solution of sulfur dioxide (SO_2) in water. It is a weak acid as well as a reducing agent. It is used as a disinfectant, as mild bleaches and for materials which may be damaged by chlorine-containing bleaches.

SULFURYL GROUP: The group $=SO_2$, as in sulfur dichloride oxide. The group is a functional group in inorganic chemistry consisting of a sulfur atom covalently bonded to two oxygen atoms (O_2SX_2). It occurs in compounds such as sulfuryl chloride (SO_2Cl_2) and sulfuryl fluoride (SO_2F_2).

SUM: 1. The result obtained from an arithmetic operation involving addition of numbers or algebraic expressions, with the symbol '+' or the total amount resulting when two or more numbers or quantities are added together. For example, 4 is the sum of adding 2 + 2 or 3 + 1. 2. The limit of the sequence of partial sums of the first n terms of infinite series, as n tends to infinity. For example, the series $\sum_{n=1}^{\infty}\frac{1}{2^n}=1+\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\ldots$ has sum 2, since that is the limit of the sequence of partial sums $1, 1\frac{1}{2}, 1\frac{3}{4}, 1\frac{7}{8}, \ldots$

SUM OF AN INFINITE GEOMETRIC SERIES: The sum of a geometric series is given by $S_n=\frac{a(1-r^n)}{1-r}$, for $|r|< 1$. For very large values of n, the sum where r^n approaches infinity, the sum, S, of an infinite geometric series for which $|r|< 1$ is given by the relation $S = \frac{a}{1-r}$, where S is the sum of the series, a is the first term and $|r|$ is the common ratio.

SUM OF AN INFINITE SERIES: If S_n is the sum of n terms of a series and S is a number such that $S > S_n$ for all n and $S - S_n$ approaches zero as n increases without limit, then the sum of the infinite series is S: $\lim_{x\to\infty} S_n = S$.

SUM OF SQUARES: In statistics, a method used in regression analysis to determine how well a function fits a set of data by squaring the distance between each data point and the line of best fit and all of the squares are summed up. The line of best fit will minimise this value.

SUM OF SQUARES THEOREM: The result that if the matrix of the quadratic form of a sum of squares of normal random variables is idempotent of rank r, then the sum of squares is distributed proportionally to a chi square distribution with T degrees of freedom. If the matrix associated with a second sum of squares is orthogonal to the first, then the two sums of squares are distributed independently.

SUMMABLE: Able to be added or integrated. The term is also used for infinite series, especially a divergent one capable of having a sum assigned to it by a method other than the usual one of taking the limit of successive partial sums.

SUMMAND: A number or value to be added to others, it forms one term of a series.

SUMMATION: 1. The process of adding or integrating a series of numbers or expressions up to find the total or the aggregate. The symbol is given by Σ (sigma), meaning "sum of". 2. The combined effort of changes in electric potential elicited in one or more postsynaptic membranes by the transmission of impulses at synapses that is sufficient to trigger an action potential in the postsynaptic neuron. It may consist of the effect of two or more potentials evoked simultaneously at different synapses on the same neuron or in rapid succession of the same synapse.

SUMMATION CONVENTION (DUMMY SUFFIX CONVENTION): A shorthand notation used in manupulating components of vectors and tensors, in accordance with which the summation symbol $\sum$ is omitted and the sum indicated by repetition of an index. For example, the scalar product $\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3$ may be more compactly written as a_ib_i.

SUMMATION CURVE (OF PARTICLE SIZES): A curve showing the accumulative percentage by weight of particles within increasing or decreasing size limits as a function of diameter; the percentage by weight of each size fraction is plotted accumulatively on the ordinate as a function of the total range of diameters represented in the sample plotted on the abscissa.

SUMMER: In temperate regions, it is the season following spring and before autumn when the weather is warmest, the Sun is highest in the sky and most plants flower and set seed. It consists of the months of June, July and August in the Northern Hemisphere or, as calculated astronomically, extending from the summer solstice to the autumnal equinox. In the Nouthern Hemisphere it occurs from December to February.

SUMMER ANNUAL: A plant that is planted and completes its growth between Spring and late Summer or early fall; or a plant that completes its growth in one growing season and dies.

SUMMER FALLOW: Land that has been ploughed but has purposely not been planted through the summer. This is done to let the land air out and store rainwater during the summer, as well as to break down organic matter and provide nutrients to the soil.

SUMMER FEEDING: The feeding of cattle on permanent pasture in the summer months.

SUMMER MASTITIS: An infection of the udder thought to be spread by biting flies. Cows become very ill, lameness may occur and milk is watery and later becomes bloody.

SUN: The star at the centre of our solar system. It is composed of about 70% hydrogen and 30% helium, with other elements making less than 1%. The Sun converts about 4.5 million tons of matter into energy every second by nuclear fusion reactions in its core, producing neutrinos and solar radiation. The small amount of this energy that penetrates the Earth's atmosphere provides the light and heat that support life. It is a sphere of luminous gas 1,392,000 km in diameter and a mass of about 330,000 times that of Earth. Its core temperature is close to 15,000,000 K and its surface temperature about 6,000 K. It rotates at different rates at different latitudes; one rotation takes 36 days at the poles but only 25 days at the equator. The visible surface or photosphere, is in constant motion, with the number and position of sunspots changing in a regular solar cycle. External phenomena include magnetic activity extending into the chromosphere and corona, solar flares, solar prominences and the solar wind. Effects on Earth include auroras and disruption of radio communications and power-transmission lines. Despite its activity, the Sun appears to have remained relatively unchanged for billions of years.

SUN PLANT: A plant that is able to flourish only in conditions of high light intensity. Sun plants usually have wide epidermal and palisade layers and numerous stomata on the lower surface of the leaf. Examples are the dominant trees of many woodland communities.

SUNBURN: Burning effect on the skin following prolonged exposure to ultraviolet radiation from the Sun, common in travellers from temperate zones to hot climates. First-degree burns may occur but usually only a delayed erythema is seen with extreme skin sensitivity. Systemic disturbance occurs in extreme cases. Fair-skinned persons are most susceptible.

SUNDIAL: A device for showing apparent solar time by the position of a shadow cast by an indicator called a gnomon. The gnomon slants upwards away from the dial at an angle determined by the latitude. It points due north in northern latitudes and due south in southern latitudes. Sundials were first used in the Near East about 5,000 years ago.

SUNFLOWER: Common name for annual and perennial herbs belonging to the genus *Helianthus* of the family Asteraceae. It is a large-headed flower that has yellow petals with a dark centre. The daily orientation of the flower to the Sun is a direct result of differential growth of the stem. A plant-growth regulator or auxin, accumulates on the shaded side of a plant when conditions of unequal light prevail. Because of this accumulation, the darker side grows faster than the sunlit side. Thus, the stem bends toward the Sun. It is grown commercially for its edible seeds and the oil extracted from them. Refined sunflower-seed oil is edible and considered by many equal in quality to olive oil. Cruder sunflower oil is used for making soap and candles. The oil cakes (solid residues after oil is expressed) are used as cattle feed. The raw seeds are used in poultry mixes and are consumed by humans as well.

SUNSHINE RECORDER: An instrument designed to record the duration of sunshine without regard to intensity at a given location. Sunshine recorders may be classified in two groups according to the method by which the time scale is obtained. In one group the time scale is obtained from the motion of the Sun in the manner of a sundial. In the second group the time scale is supplied by a chronograph.

SUNSPOT: A dark patch on the surface of the Sun, which is an area of cooler gas thought to be caused by strong magnetic fields that block the outward flow of heat to the Sun's surface or a dark patch in the Sun's photosphere resulting from a localised fall in temperature to about 4273ºC (4,000 K). It possesses a powerful magnetic field.

SUNSPOT CYCLE: The variation of the size and number of sunspots in an 11-year cycle which is shared by all other forms of solar activity. During a given sunspot cycle, the leading sunspots in groups in the northern solar hemisphere all tend to have the same polarity, while the same is true of sunspots in the Southern Hemisphere, except that the common polarity is reversed. During the next sunspot cycle, the leading spots in each hemisphere flip to the opposite polarity.

SUNSTROKE (HEATSTROKE): A rise in body temperature and failure of sweating in hot climates, often following exertion. Delirium, coma and convulsions may develop suddenly and rapid cooling should be effected.

SUPER-: Prefix meaning above, over or superimpose; an example is superscript or superfluid.

SUPER EXTENDED GRAPHICS ARRAY (SXGA): In computer science, it is a standard monitor that provides true colour (24-bit colour depth, giving over 16 million shades) at a resolution of 1280 x 1024 with an aspect ratio of 5:4. A widescreen format, SXGA+, provides 1400 x 1050 pixels.

SUPER HIGH FREQUENCY, SHF (CENTIMETRE BAND; CENTIMETRE WAVE): A radio frequency in the range of 3 and 30 GHz with wavelengths of 10 to 1 centimetres (100 mm to 10 mm). This frequency is used for microwave devices, WLAN, most modern radars, satellite uplinks/downlinks and terrestrial high-speed data links.

SUPER INPUT/OUTPUT (SIO; SUPER I/O): An integrated circuit on a computer motherboard that handles the slower and less prominent input/output devices. Super I/O communicates through the Southbridge and is still used with computers in order to support older legacy devices.

SUPER VIDEO GRAPHICS ARRAY (SVGA): In computer science, it is a graphics display monitor or interface providing a higher resolution than Video Graphics Array (VGA). SVGA monitors have resolutions of either 800 x 600 or 1,024 x 768 pixels.

SUPERACID: An acid that has a proton-donating ability equal to or greater than that of 100% pure anhydrous sulfuric acid. The strongest superacids are prepared by the combination of two components, a strong Lewis acid and a strong Brønsted acid. The strongest known superacid is fluoroantimonic acid, $HSbF_6$. It is formed by mixing hydrogen fluoride (HF) and antimony pentafluoride (SbF_5). Commercially available superacids include trifluoromethanesulfonic acid (CF_3SO_3H), also known as triflic acid and fluorosulfonic acid (FSO_3H), both of which are about a thousand times stronger or have more negative H_0 values than sulfuric acid.

SUPERACTINIDE: Any of the theoretical series of superheavy, radioactive elements, starting from unbiunium with atomic number 112 to unpenttrium with atomic number 153 that follow the transactinide series in the periodic table. None of these elements has yet been synthesised or discovered in nature. It is postulated that this series are highly unstable with respect to radioactive decay and have extremely short half lives, although element 114 is hypothesised to be within an "island of stability" that is resistant to fission but not to alpha decay has a group of elements.

SUPERAXILLARY (SUPRA-AXILLARY): Attached above the axillary.

SUPERBUG: Infectious bacterium that has developed resistance to most or all known antibiotics.

SUPERCLUSTER: Large groups of smaller galaxy groups and clusters that are among the largest structures of the cosmos, believed to have formed in the early universe when matter clumped together under the influence of gravity. They range in size from 100 million to at least 300 million light-years across and typically contain tens of thousands of galaxies. Because of their enormous sizes they are not gravitationally bound and, consequently, partake in the Hubble expansion. The largest known supercluster is the Shapley Concentration. Our Galaxy lies within the Local Supercluster.

SUPERCOILING: A type of DNA in which the double helix is further twisted about itself, to form a tightly coiled structure. This is the form adopted by DNA in nature and enables it to condense sufficiently to be packaged into living cells. In negative supercoiling, the DNA is twisted about an axis in a direction opposite that of the clockwise turn of the double helix. This decreases the number of turns of the helix around the other. In positive supercoiling, the twist of the supercoil is in the same direction as that of the double helix increasing the number of turns of the helix around the other.

SUPERCOMPUTER: An extremely fast and expensive complex computer with a very large memory capacity and processors capable of operating at several billion operations per second. Most supercomputers are really a series of computers that perform parallel processing. In the year 2014, the fastest supercomputer in the world was Tianhe-2, developed by China's National University of Defence Technology with a performance of 33.86 petaflop/s. The first supercomputer known as Cray-1 was made in 1976 by Roger Cray (1925-1996), an US engineer. Supercomputers find applications in specialised fields that require immense amounts of mathematical calculations and are used in various fields including weather forecasting, climate research, oil and gas exploration, electronic design, molecular modelling and quantum mechanics.

SUPERCONDUCTING TRANSITION: A transition at the critical temperature, below which the resistance of electrical conductors becomes zero and magnetic flux is excluded.

SUPERCONDUCTIVITY: A phenomenon occurring in many electrical conductors, in which the electrons responsible for conduction undergo a collective transition into an ordered state with many unique and remarkable properties. These include absence of resistance to the flow of electric current, the appearance of a large diamagnetism and other unusual magnetic effects, substantial alteration of many thermal properties and the occurrence of quantum effects otherwise observable only at the atomic and subatomic level. It is known to occur in some 26 metallic elements and many compounds and alloys. The temperature below which a substance becomes super-conducting is called the transition or critical temperature.

SUPERCONDUCTOR: A conductor that has no electrical resistance when cooled to a very low temperature usually below 23 K (-250 °C). Most of the physical properties of superconductors vary from material to material, such as the heat capacity and the critical temperature, critical field and critical current density at which superconductivity is destroyed. On the other hand, there is a class of properties that are independent of the underlying material. For instance, all superconductors have exactly zero resistivity to low applied currents when there is no magnetic field present or if the applied field does not exceed a critical value. Conventional superconductors usually have critical temperatures ranging from around 20 K (-253 °C) to less than 1 K (-272 °C). Solid mercury, for example, has a critical temperature of 4.2 K (-268.8 °C).

SUPERCOOLING: Cooling of a substance below the temperature at which a change of state would ordinarily take place without such a change of state occurring, for example, the cooling of a liquid below its freezing point without freezing taking place. This results in a metastable state and cooling of a vapour to make it super saturated until a disturbance causes condensation to occur.

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SUPERCRITICAL: 1. A chain reaction of a nuclear reactor which is above critical and is uncontrollable. 2. Having an effective multiplication constant greater than 1, so that the rate of reaction rises rapidly in a nuclear reactor. 3. Property of a gas which is above its critical pressure and temperature.

SUPERCRITICAL FLUID (SF): A fluid at temperatures and pressures above those at which the liquid and gaseous states of the given substance would have the same density or a fluid which is above or close to the critical state. SFs have characteristics which are intermediate to those of gases and liquids. SFs can effuse through solids like a gas and dissolve materials like a liquid. In addition, close to the critical point, small changes in pressure or temperature result in large changes in density. SFs are used in supercritical fluid chromatography and are suitable as a substitute for organic solvents in a range of industrial and laboratory processes. Carbon dioxide and water are the most commonly used supercritical fluids, being used for decaffeination and power generation, respectively.

SUPERCRITICAL FLUID CHROMATOGRAPHY (SFC): A separation technique in which the mobile phase is a fluid above and relatively close to its critical temperature and pressure. In general, the terms and definitions used in gas or liquid chromatography are equally applicable to supercritical fluid chromatography.

SUPERDEFORMED NUCLEUS: A nucleus that has an ellipsoidal shape, where the ratio of the long to short axis is considerably larger than 1.3 to 1. Such nuclei can be formed by fusing high-energy heavy ions with other nuclei. They lose energy by emitting a characteristic pattern of gamma rays as they revert to a more normal shape.

SUPERDIAGONAL (SECOND DIAGONAL): In a matrix, the line of elements lying immediately above the diagonal, of the form a_{i+1}. A **superdiagonal matrix** is a matrix in which all the entries are zero except in the superdiagonal.

SUPEREGO: In Freudian psychology, it is the the part of the human mind concerned with ethics and social standards of behaviour, learned from parents and teachers.

SUPEREQUIVALENT ADSORPTION: This occurs when the specifically adsorbed amount of charge in the inner Helmholtz plane is higher than the charge on the metal phase, taken with the reverse sign.

SUPEREXCHANGE INTERACTION: Electronic interaction between two molecular entities mediated by one or more different molecules or ions.

SUPERFICIAL CLEAVAGE: A meroblastic cleavage in which mitosis occurs without cytokinesis, resulting in many nuclei, which migrate to the periphery of a centrolecithal egg. It is found in insects that have centrolecithal eggs or the yolk is located in the centre of the egg cell.

SUPERFICIAL EXPANSIVITY (EXPANSIVITY): The increment in area of a solid surface per unit of area for a rise in temperature of 1°C (274 K) at constant pressure.

SUPERFICIAL FASCIA (HYPODERMIS; TELA SUBCUTANEA): The thin layer of loose fatty connective tissue underlying the skin and binding it to the parts beneath. In the scalp, the back of the neck, the palms of the hands and the soles of the feet, its attachment to the skin is very firm. In all other parts it is loose enough to allow the skin to move freely around; and its elasticity enables it to bring the skin back into place again. The thickness of the superficial fascia depends upon the quantity of fat in its meshes and therefore varies greatly in the bodies of different individuals and in different parts of the same body.

SUPERFLUID: A fluid that flows without viscosity or friction and has a very high thermal conductivity. **Superfluidity** is the unusual property of **superfluids** such as liquid helium below 2.186K, exhibiting apparently frictionless flow. At such low temperatures, helium exhibits an enormous rise in heat conductivity and rapid flow through capillaries or over the rim of its container. The temperature at which superfluidity occurs is called the **lambda point**. There is a fundamental connection between superfluidity and super conductivity; hence a super conductor is sometimes called a charged superfluid.

SUPERGENE (SUPER GENE; COMPLEX GENE): A group of tightly linked genes that affects the same trait and inherited apparently as a single unit. They lie close together on a chromosome, function as a unit and are rarely separated. Supergenes are noted to determine shell colour and pattern in terrestrial snails and wing shape and colouring essential to mimicry in certain butterflies. This tight linkage is advantageous since it ensures that the appropriate combination of allele for the character concerned is virtually always transmitted intact to the animal's offspring.

SUPERGIANT: Rarely seen stars, which are the largest and most luminous, with a diameter of up to 1,000 times that of the Sun and absolute magnitude (the brightness of a star as it would be seen at a distance of 32.6 light-years) between -5 and -9.

SUPERGRAVITY (SUPERGRAVITY THEORY): A unified-field theory that combines the principles of supersymmetry and general relativity. It is most naturally formulated as a Kaluza-Klein theory in eleven dimensions but it is believed that to obtain a consistent quantum theory of gravity one has to abandon quantum field theories since they deal with point objects.

SUPERHEATING: Heating a liquid to above its normal boiling point by increasing the pressure without boiling. It is caused by an additional force, the surface tension, which suppresses the growth of bubbles. Superheating is achieved by heating a homogeneous substance in a clean container, free of nucleation sites, while taking care not to disturb the liquid. An example is heating a cup of water undisturbed in a microwave oven.

SUPERHETERODYNE RECEIVER: A type of radio in which the incoming radio-frequency signal is mixed with an internally generated signal from a local oscillator. The output of the mixer has a carrier frequency equal to the difference between the transmitted frequency and the locally generated frequency, still retains the transmitted modulation and is called the intermediate frequency (IF). The IF signal is amplified and demodulated before being passed to the audio-frequency amplifier. This system enables the IF signal to be amplified with less distortion, greater gain, better selectivity and easier elimination of noise than can be achieved by amplifying the radio-frequency signal.

SUPERIONIC CONDUCTOR: A crystal or solid electrolyte in which some ions move so freely that the material conducts electricity as well as a molten salt does. A superionic crystal has a transition on heating to high ionic conductivity. The crystal remains rigid, but conducts electricity well, as though only one ionic species has 'melted' while the others stayed fixed in the crystal lattice. Such materials, for example, yttria-stabilised zirconias are used in fuel cells and other clean energy technologies. High conductivity generally only occurs at high temperature and this compromises performance.

SUPERIOR: A structure that is positioned above or higher than another structure in the body. For example, in flowering plants the ovary is described as superior if it stands above the petals.

SUPERIOR VENA CAVA (PRECAVA): One of the largest veins in the blood system formed by the left and right brachiocephalic veins or innominate veins that brings de-oxygenated blood from the upper limbs, head and neck, behind the lower border of the first right costal cartilage to the right atrium of the heart. In animals, superior vena cava is also known as the **cranial vena cava**. No valve separates the superior vena cava from the right atrium.

SUPERLATTICE: 1. A periodic structure of alternating layers of two different semiconductor materials, each several nanometres thick. 2. Also known as **artificial crystal**, **artificially layered structure** or **superstructure**, it is an ordered arrangement of atoms in a solid solution which forms a lattice superimposed on the normal solid solution lattice.

SUPERLUMINAL MOTION: An apparent optical illusion in which an object appears to be travelling faster than the speed of light as seen in some radio galaxies.

SUPERMEMBRANE THEORY: A unified theory of the fundamental interactions involving supersymmetry, in which the basic entities are two-dimensional extended objects (super membranes). They have the same length scale as superstrings, i.e. 10^{-35}m. Presently, there is no experimental evidence for super membranes.

SUPERMOON (PERIGEE FULL MOON): The Moon when seen from the Earth at the biggest and brightest that it can possibly appear. During a supermoon occurrence, the Moon appears up to 14% bigger and 30% brighter than when it is furthest from the Earth. Supermoons happen when the Moon is both full (i.e. on the opposite side of the Earth from the Sun) and also at a minimum distance from the Earth in its elliptical orbit, when it is 356,400 km away, compared to its mean distance of 384,000 km.

SUPERNATANT LIQUID: Clear liquid that lies above a sediment or precipitate. An example is a paint can, in which the liquid layer rests upon a layer of heavier pigments and other matter in the bottom of the can.

SUPERNORMAL STIMULUS (SUPER STIMULUS): A stimulus that produces a more vigorous response than the stimulus for which it evolved. For example, a bird would prefer brooding a giant egg in preference to its own egg which is smaller.

SUPERNOVA: The death explosion of certain types of star, resulting in a sudden, vast increase in brightness followed by a gradual fading. At peak light output, a supernova can outshine an entire galaxy. Most of the star is blown into interstellar space at speeds of several percent of the speed of light, eventually forming a supernova remnant.

SUPERNOVA REMNANT (SNR): The outwardly expanding remnants of a star that exploded as a supernova and the expanding gaseous shell swept up by a supernova shock wave. They heat up the interstellar medium, distribute heavy elements throughout the galaxy, and accelerate cosmic rays.

SUPERNUMERARY CHROMOSOME (ACCESSORY CHROMOSOME; B-CHROMOSOME): Any of a number of small chromosomes present in some organisms in addition to the fixed number of stable chromosomes characteristic of the species (the A-chromosomes). They are of widespread occurrence and have been reported in over 80 genera and 20 families of plants. The numbers found may differ between members of a species and even within a single individual, with gametic nuclei often having larger numbers. Supernumerary chromosomes are composed largely of heterochromatin and in many cases appear to have little effect. However, when present in large numbers they may be deleterious.

SUPERORGANISM: A collection or community of highly interdependent life forms which behaves as if it were a single organism of a higher order. Examples include the colonies of social insects such as ants. These appear to display an organisational ability and collective will that is not evident in the individuals of which it is composed.

SUPEROVULATION: The process in animal reproduction of injecting hormones to increase the number of eggs released by the ovaries or to produce mature ova at an accelerated rate or in a large number at one time.

SUPEROXIDE: 1. A compound that yields the free radical O_2^- which is highly toxic to living cells. They are formed in significant quantities only for potassium, sodium, rubidium and caecium. They are very powerful oxidising agents and react vigorously with water to give oxygen gas and OH- ions. 2. Oxide that yields both hydrogen peroxide and oxygen on treatment with an acid.

SUPEROXIDE DISMUTASE (SOD): An enzyme found in all living cells that removes the superoxide radical O^-_2 to form molecular oxygen and hydrogen peroxide during metabolism. It is one of the body's primary internal anti-oxidant defences and helps break down potentially harmful oxygen molecules in cells and plays a critical role in the defence of cells against the toxic effects of oxygen radicals. Its supplements are used to protect against the following conditions: Alzheimer's disease, cataracts, gout, inflammation, interstitial cystitis, osteoarthritis, Parkinson's disease, rheumatoid arthritis and scleroderma.

SUPERPARTNERS: Particles whose spins differ by 1/2 unit and that are paired by supersymmetry.

SUPERPHOSPHATE: A commercial phosphate mixture consisting mainly of monocalcium phosphate. It is the principal carrier of phosphate, the form of phosphorus usable by plants and is one of the world's most important fertilisers. Single-superphosphate is made by treating phosphate rock with sulfuric acid whiles triple-superphosphate is made by using phosphoric(V) acid in place of sulfuric acid.

SUPERPHYLUM: A taxonomic category lying between a kingdom and a phylum.

SUPERPLASTICITY: The ability of some metals and alloys to stretch uniformly by several thousand percent at high temperatures, unlike hot normal alloys, which fail after being stretched 100% or less. Alternatively, it may be described as an alloy that is capable of being easily shaped and moulded at high temperatures without fracturing. Many alloys and ceramics have been shown to possess this property. For superplasticity to occur the metal grain must be small and rounded and the alloy must have a slow rate of deformation.

SUPERPOSABILITY: The ability to bring two particular stereochemical formulae (or models) into coincidence (or to be exactly superposable in space and for the corresponding molecular entities or objects to become exact replicas of each other) by no more than translation and rigid rotation.

SUPERPOSE: 1. To move one geometric figure, such as a triangle, so that its edges coincide exactly with another figure without deforming either figure. 2. To set or place (one thing) over or above something else.

SUPERPOSITION PRINCIPLE (SUPERPOSITION PROPERTY): It states that, for all linear systems, the net response at a given place and time caused by two or more stimuli is the sum of the responses which would have been caused by each stimulus individually. This is a general principle of linear systems that when applied to wave phenomena asserts that the combined effect of any number of interacting waves at a point may be obtained by the algebraic summation of the amplitudes of all the waves at the point. For example, the superposition of two oscillations, x_1 and x_2, both of frequency v, produces a disturbance of the same frequency. The amplitude and phase angle of the resulting disturbance are functions of the component amplitudes.

SUPERRADIANCE: Spontaneous emission amplified by a single pass through a population inverted medium. It is distinguished from true laser action by its lack of coherence. The term superradiance is frequently used in laser technology.

SUPERRADIENT: Denoting a laser in which the efficiency of the transition of stimulated emission is sufficiently large for the radiation to be produced by a single pulse of electromagnetic radiation without using mirrors.

SUPERSATURATED SOLUTION: Unstable solution that contains more solute than a saturated solution would contain at the same temperature and pressure. It easily changes to a saturated solution when the excess solute is made to crystallise.

SUPERSATURATION: 1. In chemistry, it is an unstable system which has a greater concentration of a material in solution than would exist at equilibrium at a given temperature. 2. In meteorology, the condition which occurs in the atmosphere when the relative humidity is greater than 100%. When the air over a cloud droplet is supersaturated, the vapour pressure of the air is greater than the saturation vapour pressure on the droplet's surface. In such an environment, water droplets grow by vapour deposition; more water vapour molecules are condensing to the droplet than are evaporating from it. Greater supersaturation occurs in stronger storm updrafts because the process of vapour deposition cannot keep up with the decompressional cooling with ascent. Ice crystals in the presence of abundant cloud droplets are said to be supersaturated with respect to ice even though relative humidity may only be 100 percent. This is because the saturation vapour pressure over ice is less than that over water, allowing vapour deposition onto the ice.

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SUPERSCALAR: Microprocessor architecture capable of processing more than one instruction in every clock cycle. A superscalar architecture includes parallel execution units, which can execute instructions simultaneously.

SUPERSET: In mathematics, it is a set, A containing a subset or a smaller set, B. It is written as A ⊃ B. If A is a proper superset of B, this is written A ⊇ B.

SUPERSONIC SPEED: Speed greater than that at which sound travels through air. In dry air at 20°C (293 K), the threshold value required for an object to travel at a supersonic speed is approximately 343 m/s or 1,236 km/h. It is measured in Mach numbers. For example, speeds greater than 5 times the speed of sound (Mach 5) are often referred to as **hypersonic**.

SUPERSPACE: A space, A in which some other space, B is a subspace, so that they have the same structure.

SUPERSTRING THEORY: A mathematical theory developed in the 1980s to explain the properties of elementary particles and the forces between them, in particular gravity and the nuclear forces, in a way that combines relativity and quantum theory.

SUPERSYMMETRY (SUSY): An extension of the Standard Model theory that hypothesizes the existence of a complete set of additional particles which complement those that are known to exist. It relates the two classes of elementary particles: the fermions (electrons, protons and neutrons) and the bosons (alpha and photon particles) of one spin to other particles that differ by half a unit of spin known as super partners.

SUPERTASK: Any infinite sequence of operations that occur sequentially within a finite interval of time. When the number of operations is uncountably infinite they are called **hypertasks**; when each individual task must be completed in the same amount of time they are called **equisupertasks**.

SUPERTRANSURANICS (SUPERHEAVY ELEMENTS): A set of relatively stable elements with atomic numbers around 114 and mass numbers of about 298 predicted to exist beyond the transuranic elements. It is believed that these elements have not been noticed because of the difficulties in synthesising them rather than their instability. The existence of supertransuranic elements is hypothesised because of the stability associated with the closing of a proton or neutron shell that occurs for magic numbers. The number 114 is predicted to be a proton magic number and the number 184 is predicted to be a neutron magic number.

SUPINATION: 1. Rotation of the lower forearm in such a way that the hand faces forwards or upwards with the radius and ulna parallel. 2. A combination of plantar flexion, inversion and adduction of the foot during walking and running. The rear part of the foot inverts to some extent when the heel strikes the ground. Plantar flexion and adduction occurs as the foot rolls forward and the forefoot makes contact with the ground.

SUPPLEMENTAL CHORDS: In a circle, they are chords that join a point on the circumference to the two extremities of a diameter, they join the endpoints of supplementary arcs.

SUPPLEMENTARY ANGLE: An angle, when added to another angle sums up to 180°.

SUPPLEMENTARY ARC: An arc of a circle that together with another arc, form a semicircle, so that they sit on supplemental chords.

SUPPLEMENTAL PLUMAGE: A third set of feathers found in birds that have three different plumages in their annual cycle of moults.

SUPPLEMENTARY RATION: A type of concentrate that is fed to livestock to supplement feeds of hay and roots.

SUPPLEMENTARY UNITS: Dimensionless units that are used along with base units to form derived units in the S.I. system. This class contains only two, purely geometrical units: the radian and the steradian.

SUPPLEMENTS: 1. An arc or angle that, when added to another, makes a semicircle or 180°. 2. A substance with a specific nutritional value taken to make up for a supposed or real deficiency in diet.

SUPPORT: 1. In mathematics, the closure of the set of arguments for which a real- or complex-valued function has non-zero value. 2. Also called **kernel**, with respect to a regular Borel measure, the unique smallest closed set of which the complement has measure zero. 3. In computer science, it is to design in order to allow something, such as a specific type of computer device, software or programming language, to operate with it.

SUPPORT ENERGY: The total energy expenditure necessary for the production of plant and animal agricultural foodstuffs.

SUPPORT ENVIRONMENT: In computer science, it is a collection of programmes (software) used to help people design and write other programs.

SUPPORT FUNCTION: The Minkowski function of a convex set C in a real normed space, denoted $\delta^*_C(f)$, $s_C(f>$ or $s(f, C)$ and defined by $\delta^*_C(f) = \sup\{f(x) : x \in C\}$ on the dual space or, analogously, on the original space if an inner product is available. The function is everywhere finite if the set is bounded and it will produce a norm if the set is a symmetric convex body. The support function δ^*_C is conjugate in the sense of Fenchel to the indicator function δ_C.

SUPPORT POINT: In mathematics, a point of a convex set at which there exists a non-zero linear functional that is sometimes required to be continuous and takes its supremum over the set. Geometrically, this corresponds to the existence of a (closed) support or boundary hyperplane that contains the point and has the set entirely in one supporting half-space.

SUPPORT THEOREM: The consequence of the separation theorem of Mazur that a closed convex body is the intersection of its c!osed supporting half-spaces and every boundary point of a convex set with non-empty interior in normed space is a support point.

SUPPORTING ELECTROLYTE: An electrolyte solution, whose constituents are not electroactive in the range of applied potentials being studied and whose ionic strength (and, therefore, contribution to the conductivity) is usually much larger than the concentration of an electroactive substance to be dissolved in it.

SUPRAGLACIAL: Of, relating to, situated or occurring at the surface of a glacier. A **supraglacial deposition** is sedimentary material consisting of particles such as debris, sand, soil and rocks that is laid down on top of a glacier, which may form lateral and medial moraines. A **supraglacial stream** flows over the surface of the glacier.

SUPRARENAL GLANDS (ADRENAL GLANDS): A pair of two small triangular endocrine glands, weighing about 5 grammes in humans which are situated immediately above the kidneys. It lies anterior to the mammalian kidney. The inner portions of the adrenals secrete the hormone adrenaline, which causes increased heart beat on breathing, in response to conditions of stress.

SUPPRESSIVE SOIL (PATHOGEN SUPPRESSIVE SOIL): Soil in which a pathogen does not persist, or causes no damage. Suppresiveness can occur through substances such as certain chemicals present in the soil that inhibit or kill potential pathogens, competition for nutrients, parasitism and predation.

SUPPRESSOR: 1. A device that reduces unwanted interference or current in a circuit. 2. A substance which reduces emission, absorption or light scattering by an interferent, thus removing or lowering spectral interference. 3. A device attached to or part of the barrel of a firearm to reduce the amount of noise and flash generated by firing the weapon.

SUPPRESSOR GENE: A gene that can reverse the phenotype of a mutation in another gene or a gene which inhibits the expression of another gene.

SUPPRESSOR GRID: A wire grid in a pentode thermionic valve placed between the screen grid and the anode to prevent electrons produced by secondary emission from the anode from reaching the screen grid.

SUPPRESSOR MUTATION: A mutation that restores a function lost through an earlier mutation and that is located at a site different from the initial mutation or a mutation that alters the anticodon in a tRNA so that it is complementary to a termination codon, thus suppressing termination of the amino acid chain.

SUPPRESSOR T CELL (REGULATORY T CELLS): A cell that reduces or suppresses activation of the immune system and thereby maintains immune system homeostasis and tolerance to self-antigens.

SUPRACHIASMIC NUCLEUS (SCN): A region of the hypothalamus that controls internal cycles of endocrine secretion.

SUPRAMOLECULAR CHEMISTRY: Chemical research involving the formation and properties of large assemblies of molecules held together by intramolecular forces such as Van der Waal's forces and hydrogen bonds. A special characteristic of this field of study is that of self-assembly or self-organisation, which involves the spontaneous formation of the structure as a result of the nature of the molecules. Another feature involves the study of very large molecules which are able to be used in complex chemical reactions in a manner similar to the actions of naturally occurring haemoglobin and nucleic acid molecules. It also involves host-guest chemistry which is concerned with molecules specifically designed to accept other molecules.

SUPRAMOLECULE: A system of two or more molecular entities held together and organised by means of intermolecular (noncovalent) binding interactions.

SUPREMAL: Relating to or constituting a supremum.

SUPREMUM (LEAST UPPER BOUND): The unique smallest element of the set of upper bounds for a set that is greater than or equal to all elements of the set, if such an element exists. For example, the sequence 1/2, 2/3, 3/4, ..., has as upper bound every real number greater than or equal to 1; it has no maximum, but its supremum is 1.

SURCULOSE: Producing suckers or runners from the base or from rootstocks. **Surculose-proliferous** describes plants which reproduce by suckers or runners.

SURCURRENT: In botany, it refers to extending upward from the point of insertion, for example, a leaf base that extends up along the stem.

SURD: A quantity that cannot be expressed as a rational number or an irrational root or irrational number or an expression containing one or the other. It consists of the root of an arithmetic number to be exactly determined or the sum or difference of such roots. An example of a **simple surd** is $\sqrt{3}$ and that of a **compound surd** is $2 + \sqrt{3}$ or $2 - \sqrt{3}$. The conjugate surd of the compound surd $2 + \sqrt{3}$ is $2 - \sqrt{3}$. Conjugate surds can be used to express a fraction which has a compound surd as its denominator with a rational denominator. An **entire surd** is a surd that has no rational factors or terms; if it has it is called a **mixed surd**. A **pure surd** is a surd whose every term is an entire or mixed surd.

SURFACE: In mathematics, it refers to the boundary of a three-dimensional figure; or the two-dimensional locus of points located in three-dimensional space; or the portion of space having length and breadth but no thickness. A sphere, a torus, a pseudosphere and a Klein bottle are examples of different types of surface.

SURFACE AMOUNT: Amount of substance adsorbed on a surface.

SURFACE ANATOMY: The study of the external features of the body that can be studied by sight, without dissecting an organism.

SURFACE AREA: The area of the outer surface of a three dimensional shape or solid. The surface area of a cube is the area of the six squares that cover it. The area of one of them is a x a or a^2. Since these are all the same, one of them is multiplied by 6, so the surface area of a cube is 6 times the area of the square. The surface area of a rectangular prism is equal to $2ab + 2bc + 2ac$, where a, b and c are the lengths of the 3 sides. The surface area of a sphere $= 4\pi r^2$. The surface area of a cylinder $= 2\pi r^2 + 2\pi rh$, where h is the height of the cylinder and r is the radius of the cylinder. Since there is both a top and a bottom, it is multiplied by two.

SURFACE BARRIER SEMICONDUCTOR DETECTOR: A semiconductor detector utilising a junction due to a surface inversion layer.

SURFACE BRIGHTNESS: The measure of the amount of light that an object, especially a galaxy, emits per area of the sky. A galaxy that has a high total luminosity can be hard to see if it has a low surface brightness.

SURFACE CHARGE DENSITY: Electric charge on a surface divided by the surface area.

SURFACE CONCENTRATION: Amount of substance adsorbed on a surface divided by the surface area.

SURFACE CONTAMINATION: Material in the experimental surface which is either not characteristic of the sample or which would not be present if the sample had been prepared in an absolute vacuum by methods not contacting other substances to the sample.

SURFACE COVERAGE: Number of adsorbed molecules on a surface divided by the number of molecules in a filled monolayer on that surface.

SURFACE DIPOLE LAYER: Particles in the surface region of a phase are subjected to orienting forces as a result of the anisotropic force field. Polar molecules, for example, permanent dipoles, may thus be preferentially oriented in the surface region, while polarisable molecules may be polarised (induced dipoles). The array of oriented polar and/or polarised molecules is called the surface dipole layer with which an electric potential drop is associated, called the surface potential of the phase.

SURFACE DRAINAGE: The removal of surplus water from an area of land by means of ditches and channels or natural or artificial removal of excess groundwater.

SURFACE ENERGY: The amount of energy needed to create 1 m^2 of new surface in solid or liquid. To calculate the energy, let N represent the number of molecules per m^2 of cross-section; if the diameter of one molecule is a_0, then N is $1/a_0^2$ per m^2.

SURFACE GRAFTING: A process in which a polymer surface is chemically modified by grafting or by the generation of active sites that can lead to the initiation of a graft polymerisation.

SURFACE GRAVITY: The local gravitational field strength at the surface of an astronomical body. It determines, for example, how much a person would weigh if they were to stand on that object.

SURFACE HEAT CAPACITY (CS): Energy required to raise the temperature of a unit area of a planet's surface by 1 K.

SURFACE INTEGRAL: In calculus, the integral of a function of several variables calculated over a surface. For functions of a single variable, definite integrals are calculated over intervals on the x-axis and result in areas. For functions of two variables, the simplest double integrals are calculated over rectangular regions and result in volumes. More generally, an integral calculated over a plane or curved surface results in a surface integral representing a volume, though it also has many nongeometric applications.

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SURFACE IONISATION: Ionisation that takes place when an atom or molecule is ionised when it interacts with a solid surface. Ionisation occurs only when the work function of the surface, the temperature of the surface and the ionisation energy of the atom or molecule have an appropriate relationship.

SURFACE IONS: Ions that are constituents of the surface or which have a particularly high affinity for the surface or surface sites.

SURFACE IRRIGATION: A system involving water flowing over soil surface to crops.

SURFACE LAYER (INTERFACIAL LAYER): The region of space comprising and adjoining the phase boundary within which the properties of matter are significantly different from the values in the adjoining bulk phases.

SURFACE OF A PHASE (FREE SURFACE): The plane ideally marking the boundary between a phase and its environment

SURFACE OF REVOLUTION: The surface of revolution in Euclidean space created by rotating a curve, called the generatrix around a straight line in its plane (the axis). If the arc of the curve $y = f(x)$ between $x = a$ and $x = b$ is rotated around the x-axis, the area of the resulting surface is $2\pi\int_a^b y\sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx$. Some examples of surfaces of revolution include the apple, cone (excluding the base), conical frustum (excluding the ends), cylinder (excluding the ends), hyperboloid, lemon, oblate spheroid, paraboloid, prolate spheroid, pseudosphere, sphere and spheroid.

SURFACE PRESSURE: 1. The change of interfacial tension caused by addition of a given species to a base solution. It is the difference in surface tension measured between a clean subphase and a monolayer-covered subphase. Surface pressure greatly influences monolayer properties. 2. Atmospheric pressure at a location on the Earth's surface. It is directly proportional to the mass of air over that location.

SURFACE REGION: The tridimensional region, extending from the free surface of a condensed phase towards the interior, where the properties differ from the bulk.

SURFACE RUNOFF: Water flow that occurs when soil is infiltrated to full capacity or excess water from rain, meltwater or other sources that flow over land. This is a major component of the hydrologic cycle. Runoff that occurs on surfaces before reaching a channel is called a **non-point source**. If a non-point source contains man-made contaminants, the runoff is called **non-point source pollution**. A land area which produces runoff that drains to a common point is called a **watershed**. When runoff flows along the ground, it can pick up soil contaminants such as petroleum, pesticides (in particular herbicides and insecticides) or fertilisers that become discharged or non-point source pollution.

SURFACE SEALING: The orientation and packing of dispersed soil particles in the immediate surface layer of the soil to render the surface fairly impermeable to water. The dispersed soil particles have disintegrated from the soil aggregates due to the impact of rain drops. Surface seals are formed at the very surface of the soil, rendering it relatively impermeable to water.

SURFACE SOIL (TOPSOIL): Referred to as **A horizon**, it is the upper, outermost layer of soil with a depth of between 7.5 and 25 cm and the layer of soil that is moved in cultivation. It has the highest concentration of organic matter (humus) and microorganisms and is where most of the biological soil activity occurs. Its composition also includes mineral particles, organic matter, water, and air.

SURFACE STATES: Energy levels localised in the surface region of semiconductors, which do not bear any direct relation to the bulk energy distribution, but which can exchange electrons with the bulk.

SURFACE TENSION: The property of a liquid which makes the surface to behave as if it were a stretched elastic skin or the property of liquids that gives their surfaces a slightly elastic quality and enables them to form into separate drops. A drop of water, for example, tends to assume the shape of a sphere. It results from forces of attraction between the molecules in the liquid or from the interaction of molecules at or near the surface that tend to cohere and contract the surface into the smallest possible area.

SURFACE WATER: Water that flows across the surface of the soil as a stream, after rain drains into rivers rather than seeping into the soil itself. In a broader sense, surface water includes water in lakes, marshes, glaciers and reservoirs as well as that flowing in streams. In the broadest sense, surface water is all the water on the surface of the Earth and thus includes the water of the oceans.

SURFACE WAVE: A seismic wave that travels along the surface of a medium and that has a stronger effect near the surface of the medium than it has in the interior.

SURFACE-AREA-TO-VOLUME RATIO (SURFACE-TO-VOLUME RATIO): 1. The amount of surface area per unit volume of an object or collection of objects. The surface area to volume ratio is measured in units of inverse distance. A cube with sides of length a will have a surface area of $6a^2$ and a volume of a^3. The surface to volume ratio for a cube is thus $\frac{6}{a}$. 2. In biology, it applies to the ratio of the surface area of a cell organ or organism to its internal volume. This is high for small animals but low for large animals such as elephants. The higher the ratio, the greater the possibility for gaseous exchange and absorption of dissolved minerals. In higher plants, especially the possession of numerous flat leaves, serves to increase the surface area/volume ratio.

SURFACE-EXTENDED X-RAY ABSORPTION FINE-STRUCTURE SPECTROSCOPY (SEXAFS): It refers to a plant whose roots are shallow in the soil exploiting water and easily available nutrients. Where conditions are close to optimum in the surface layers of soil, the growth of surface roots is encouraged and they commonly become the dominant roots.

SURFACTANT (SURFACE ACTIVE AGENT): A substance that markedly affect the surface characteristics of a system, such as increasing its spreading or wetting properties by reducing its surface tension. In a two-phase system, for example, liquid-liquid or solid-liquid, a surfactant tends to locate at the interface of the two phases, where it introduces a degree of continuity between the two different materials. Soaps and detergents are classic examples of surfactants due to their dual (amphipathic) character.

SURFICIAL: In botany, it refers to growing near the ground or spreading over the surface of the ground.

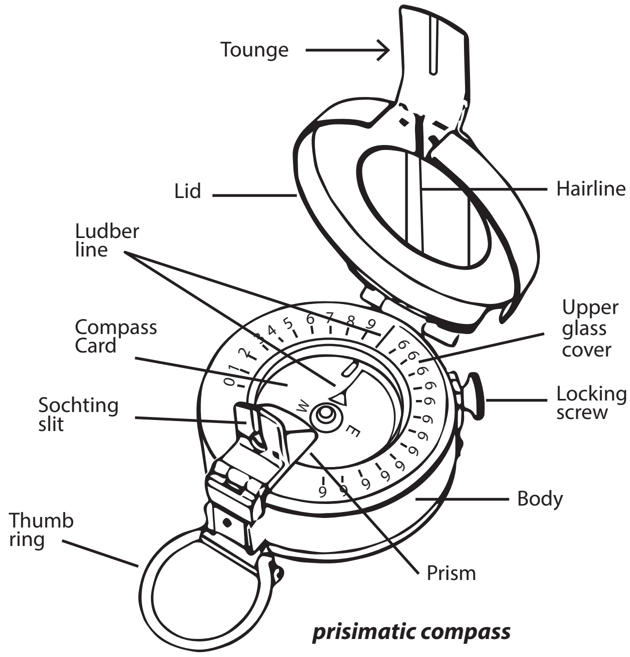
SURFING: In computer science, it is to spend time visiting a lot of websites from one link to another without a definite objective, unlike browsing.

SURGE SUPPRESSOR (SURGE PROTECTOR): A plug-in device that is designed to help protect the computer from surges and spikes. Many devices can be plugged in at one time and it protects each connected device from power surges. It prevents surges in electrical current by sending the excess current to the earth or ground through the earth wire or ground wire. If the surge is extra high, such as from a lightning strike, a fuse in the surge protector blows and the current is prevented from reaching any of the devices plugged into the surge protector.

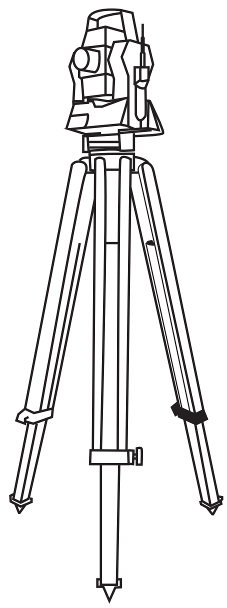
SURGICAL SPIRIT (METHYLATED SPIRIT): A denatured alcohol consisting of ethanol to which a small amount of methanol has been added to render it unfit to drink. It is used to sterilise surfaces and to clean skin abrasions and sores.

SURJECTIVE MAPPING (ONTO MAPPING; SURJECTION): A mapping f: $S \rightarrow T$ is subjective if every element of the codomain T is the image under f of at least one element of the domain S. So f is surjective if the image or range $f(S)$ is the whole of T.

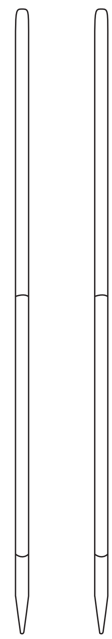
SURVEYING: Technique and science involving accurately determining the terrestrial or three-dimensional position of points and the distances and angles between them by measuring and recording the relative altitudes, angles and distances of features on, above or below a land surface the result of which are used to plot maps and plans. The methods used involve triangulation, trilateration level, plane tabling and transversing. Survey is necessary to locate and measure property lines and buildings for ownership. Examples of surveying instruments are ranging poles, theodolite and prismatic compass, as shown in the diagrams.



prisimatic compass



theodolite



Ranging pole

Examples of surveying tools

SURVIVAL: The act of living through something, especially under adverse or unusual life threatening circumstances such as drought, flood, famine or disease. The term "survival of the fittest" originated from Darwinian evolutionary theory as a way of describing the mechanism of natural selection.

SURVIVAL CURVE: 1. A graph that shows the relationship between survival and the age of a population of a particular species. Survival can be represented as a percentage of individuals of the original population that have survived after a specific time. Some types of organisms, notably those that produce many offspring have a low survival rate in the early stage of development and vice versa. 2. The curve obtained by plotting the number or percentage of organisms surviving at a given time against the dose of radiation or the number surviving at different intervals after a particular dose of radiation.

SUSCEPTANCE: 1. An expression of the ease with which alternating current (AC) passes through a capacitance or inductance. It is the reciprocal of the reactance of a circuit and thus the imaginary part of its admittance. It is measured in siemens and its symbol is B. 2. A dimensionless quantity referring to a dielectric equal to $p/\varepsilon_o E$, where p is the electric polarisation, E is the electric intensity producing it and ε_o is the electric constant. The symbol is B.

SUSCEPTIBILITY: 1. The ratio of the intensity of magnetisation produced in a material to the intensity of the magnetic field to which the material is exposed. It measures the extent to which a material is magnetised by an applied magnetic field. 2. The dimensionless quantity describing the contribution made by a substance when subjected to a magnetic field to the total magnetic flux density present. Its symbol is χ_m. 3. In botany, it is the extent to which a plant, vegetation complex or ecological community would suffer from a pathogen if exposed, without regard to the likelihood of exposure. This is not the same as vulnerability, which takes into account both the effect of exposure and the likelihood of exposure.

SUSLIN OPERATION: In mathematics, denoted by A, it is an operation that constructs a set from a collection of sets indexed by finite sequences of positive integers. If M is the family of closed subsets of a topological space, then the elements of $A(M)$ are called **Suslin sets**.

SUSPENDED MATTER: All particulate materials which persist in the atmosphere or in a flue gas stream for lengthy periods because the particles are too small in size to have an appreciable falling velocity.

SUSPENSION: A heterogeneous mixture consisting of small solid particles dispersed in a liquid or gas, which will settle on standing or a mixture made from small insoluble solid particles shaken with a liquid. An example is milk of magnesia, which is a suspension of magnesium hydroxide in water.

SUSPENSION CULTURE: A type of culture in which intact cells are maintained in liquid medium.

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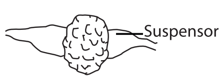
SUSPENSION EFFECT: An effect which occurs when ion-selective electrodes are used in concentrated, space-filled suspensions while the external reference electrode remains in the supernatant (suspension-free) solution. The suspensions are specifically solvent-swollen ion exchangers or other materials, like soils and clays that concentrate ions by adsorption and absorption. Space-filled, gravity-packed suspensions act like a second phase and form apparently an interfacial potential difference (PD) with respect to the supernatant. The measured ion activity in the suspension differs from the value in the supernatant by the interfacial PD and corresponds to a higher value approximating the activity inside the ion exchanger gel. The effect nearly disappears when the outer reference electrode is placed in the same region of the suspension as the sensor electrode. There are some changes in the junction potential differences of the reference electrode, between suspension and supernatant.

SUSPENSION-FEEDER: An aquatic animal, such as a clam, barnacle, sponge or baleen whale that feeds by filtering particulate organic material such as plankton from the water. It usually has various structural modifications for straining out its food from the water.

SUSPENSOR: 1. A lateral outgrowth of adjacent hyphae that develops into structure that supports the gametagium. 2. A structure made up of a chain of cells that support a plant embryo sac.



Suspensor of hyphae

SUSPENSORY LIGAMENT: Any ligament that supports a body part, especially an organ. They include suspensory ligament of the penis, suspensory ligament of the ovary, suspensory ligament of the lens of the eye and suspensory ligament of the breast.

SUSTAINABLE AGRICULTURE: Environmentally friendly methods of farming that allow the production of crops or livestock without damage to the ecosystem. It is an integrated system of plant and animal production practices having a site-specific application that will, over the long term satisfy human food and fibre needs as well as make the most efficient use of nonrenewable resources and on-farm resources. It also integrates, where appropriate, natural biological cycles and controls; sustain the economic viability of farm operations and enhance the quality of life for farmers and society as a whole.

SUSTAINABLE FOOD CHAIN: A food chain from producer to consumer which is environmentally responsible and sustainable at all stages.

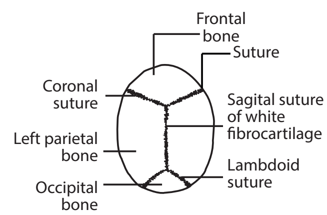
SUSTAINABLE PRODUCTION PROCESSES: Agricultural production methods which do not damage or deplete natural resources.

SUSTAINABLE YIELD: The greatest productivity that can be derived from a renewable resource without depleting the supply in a specific area or the ecological yield that can be extracted without reducing the base of capital itself. It is the surplus required to maintain nature's services at the same or increasing level over time.

SUSTAINED YIELD: 1. The removal of surplus individuals from a population of organisms so that the population maintains a constant size. 2. The continuing yield of a biological resource, such as timber from a forest, by controlled periodic harvesting. 3. The quantity of a resource harvested in this manner.

SUSTENTACULAR CELLS: Cells that provide structural support to Sertoli cell or a cell of the olfactory epithelium.

SUTURE: 1. The line marking the junction of two body structures or a seam or line at which two edges have been joined. 2. A joint, especially in the skull, in which the bones are tightly bound together by fibrous connective tissue so as to prevent movement between them. 3. A distinguishable line at the junction of adjacent structures for example, between the exoskeletal plates of an insect. 4. A line along which a seed pod or fruit will split to release its seeds. 5. The line formed where a wound has been closed or tissues have been joined or a piece of material used to close a wound or connect tissues, for example, catgut, thread or wire. 6. A major fault zone through an orogen or mountain range. Sutures separate terranes or tectonic units that have different plate tectonic, metamorphic and paleogeographic histories.



Suture: Fixed (immovable) joints of a skull

SV40 (SIMIAN VACUOLATING VIRUS 40; SIMIAN VIRUS 40): An infectious DNA-containing virus belonging to the agent of the genus Lentivirus in the family Retroviridae of the papovavirus group that is much used in cancer research for its ability to induce transformation in cells. The virus was originally isolated from monkeys and infects primates of the infraorder Simiiformes (hence its name), which includes the so-called anthropoids: apes, monkeys and humans. The virus attacks and integrates itself into our DNA which can lead to cancerous cell activity. **Simian immunodeficiency viruses** (**SIVs**) are retroviruses that naturally infect a wide range of African nonhuman primates and are the source of the human immunodeficiency viruses (HIV-1 and HIV-2).

SVGA (SUPER VIDEO GRAPHICS ARRAY): In computer science, it is a graphic display standard providing higher resolution than Video Graphics Array (VGA). SVGA screens have resolutions of either 800 x 600 or 1,024 x 768 pixels.

SWALLOW HOLE: A hole often found in limestone areas, through which a surface stream disappears underground or an opening that occurs occasionally at the bottom of a sinkhole which permits direct drainage from the surface into an underground channel.

SWAMP (SWAMP LAND): A region of low-lying land that is permanently saturated with water and usually overgrown with vegetation or freshwater wetland ecosystem characterised by poorly drained mineral soils and plant life dominated by trees. Swamps have a sufficient water supply to keep the ground waterlogged and the water has a high-enough mineral content to stimulate decay of organisms and to prevent the accumulation of organic materials. They are found throughout the world.

SWAP: In computer science, it is to move segments of data in and out of memory. Swapping is a useful technique that enables a computer to execute programs and manipulate data files larger than main memory. The operating system copies as much data as possible into main memory and leaves the rest on the disk. When the operating system needs data from the disk, it exchanges a portion of data (called a page or segment) in main memory with a portion of data on the disk.

SWARD: A cover of grasses and clovers which makes a pasture. **Swardsman** is a farm worker who looks after or grows pasture.

SWARM: Behaviour of an aggregate of animals of similar size and body orientation, often moving together or migrating in the same direction. Swarming is a general term that can be applied to any animal that swarms. The term is applied particularly to insects, but can also be applied to birds, fish, various microorganisms such as bacteria and people.

SWARM CELL: An amoeba-like cell that is produced on germination of a spore in cell of myxomycetes. Swarm cells have two anterior whiplash flagella and lack a cell wall. Usually one, but occasionally up to four such cells emerge from one spore. They ingest food, reproduce by division and then become gametes, in some instances becoming myxamoebae first. The gametes fuse in pairs to form a zygote that develops into the plasmodium. In adverse conditions, the swarm cells can transform into microcysts.

SWASH: A surge of turbulent seawater that rushes up the shore after a wave breaks. It runs back down the slope as backwash. The swash can carry materials such as driftwood, seashells and seaweed.

SWATH: 1. A row of grass or other plants lying on the ground after being cut. 2. A row of potatoes which have been lifted and left lying on the ground.

SWATHTURNER: A hay making machine used to move individual swath sideways and turn them over at the same time, so making the drying process faster. It is also used in wet conditions to scatter a swath to dry it more quickly.

SWAYBACK DISEASE: An often fatal disease of lambs caused by copper deficiency in the ewe's diet. Lambs become unsteady and unable to walk. The disease is often a problem when there has been no snow during the winter.

SWEAT: Fluid secreted by the sweat glands in the mammalian skin containing water, salts and some urea or the salty fluid secreted by the sweat glands onto the surface of the skin. Excess body heat is used to evaporate sweat, resulting in cooling of the skin surface.

SWEAT GLAND: A small gland, *Brassica rutebage* in the dermis of mammalian skin that secretes sweat or a small tube-shaped gland in the skin of most parts of the body from which sweat is released. Human sweat glands are of two types, eccrine or merocrine and apocrine. The **eccrine glands**, are usually absent from the hairy skin and limited to friction surfaces, are vital to the regulation of body temperature. The **apocrine glands** are scent glands, which exude a sticky fluid quite different from the watery sweat of the eccrines. They usually develop in association with hair follicles and open into them. In human beings, apocrine glands are concentrated in the underarm and in genital regions; the glands are inactive until they are stimulated by hormonal changes in puberty. In other mammals, apocrine glands are more numerous. They function in response to stress or sexual stimulation, playing no part in temperature regulation.

SWEEPERBULL: A bull used to serve cows that have not been artificially inseminated.

SWEETBREADS: The elongated thymus glands called neck or throat sweetbreads and pancreas called heart, chest or stomach sweetbreads of young animals, usually calf or lamb and sometimes pig used as food.

SWEETENER: Any chemical, either natural or synthetic that gives sweetness to food. Artificial sweetener is any synthetically produced, intensely sweet substance developed for use in non-nutritive or dietetic foods and beverages. Unlike natural sugars, artificial sweeteners do not promote tooth decay and they contribute few, if any calories to the foods they sweeten. Major artificial sweeteners include **aspartame** (NutraSweet) produced from aspartic acid; **saccharin** (375 times as sweet as sugar) that is a crystalline powder with the chemical formula $C_6H_4CONHSO_2$; **cyclamates** are salts or esters of cyclamic acid, especially sodium cyclamate; and **sucralose**, which is created from sugar by replacing three hydroxyl groups with three chlorine atoms is 600 times sweeter. Natural sweeteners are sugars, of which the commonest is sucrose, with the chemical formula $C_{12}H_{22}0_{11}$. It belongs to the group of carbohydrates known as disaccharides. Some other natural sugars include **fructose** or **fruit sugar**, a monosaccharide with the chemical formula $C_6H_{12}O_6$ that occurs with glucose in sweet fruits and fruit juices. It is 173% as sweet as sucrose; **glucose** or **grape sugar** (74% as sweet as sucrose) is a monosaccharide sugar with the chemical formula $C_6H_{12}O_6$, found in honey and the juices of many fruits; **maltose** (33% as sweet as sucrose) is a sugar with the chemical formula $C_{12}H_{22}O_{11}$, formed by the action of the enzyme diastase on starch; **lactose** (16% as sweet as sucrose) is sugar with the chemical formula $C_{12}H_{22}O_{11}$, present in milk; and **honey** is a mixture of glucose and fructose.

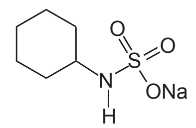


Fig. I: sodium cyclamate

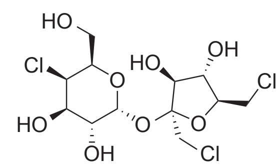


Fig. II: sucralose

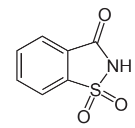


Fig. III: saccharin

Examples of sweetener

SWELLING: The increase in volume of a gel or solid associated with the uptake of a liquid or gas.

SWELLING AGENT: Fluid used to swell a gel, network or solid.

SWELLING PRESSURE: That pressure difference which has to be established between a gel and its equilibrium liquid, to prevent further swelling of the gel.

SWIM BLADDER (AIR BLADDER): A hydrostatic organ of thin-walled air sac, containing gas, usually oxygen, lying above the alimentary canal in bony fishes that enables them to adjust their density and thereby control their buoyancy and also as a resonating chamber to produce or receive sound; in certain primitive fishes it functions as a lung or respiratory aid. It is missing in some bottom-dwelling and deep-sea bony fish (teleosts) and in all cartilaginous fish such as sharks, skates and rays.

SWINE FLU (H_1N_1 FLU): Virulent, highly contagious form of influenza, infecting pigs that can be transmitted to humans. It is a respiratory disease caused by infection with swine influenza A virus (SIV). General symptoms include coughing, sore throat, runny or stuffy nose, watery red eyes, body aches, headache, fatigue, diarrhoea, nausea and vomiting.

SWIPE: In computer science, it refers to quickly moving a finger across a touchscreen or trackpad. For example, with a smartphone you may need to slide or swipe your finger from left-to-right, right-to-left or from low-to-high in order to unlock the phone. It is used primarily with touchscreen devices, such as smartphones and tablets, also by some laptops with trackpads and desktop computers with trackpad input.

SWITCH: 1. A device that can interrupt the current flow in a circuit or an electrical device that closes or opens a circuit by operating a switch to cause current to stop or start flowing or change its path. 2. A tuft of hair at the end of the tail of a crow or other animal. 3. A programmed technique for indicating which alternative path to take at a decision point in a program's logic.

S

SWITCHING TRANSITION: A transition in semiconductive glasses in which, beyond a critical applied voltage, there is an avalanche breakdown of conducting electrons that causes local melting and hence local crystallisation that gives metallic conductivity.

SXGA (SUPER EXTENDED GRAPHICS ARRAY): In computer science, it is a standard monitor that provides true colour (24-bit colour depth, giving over 16 million shades) at a resolution of 1280 x 1024 with an aspect ratio of 5:4. A widescreen format, SXGA+, provides 1400 x 1050 pixels.

SYBYL LINE NOTATION (SLN): A type of line notation, similar to SMILES but with some additional features. In SLN, hydrogens are not omitted so ethane is CH_3CH_3. SLN differs from SMILES in several significant ways. SLN can specify molecules, molecular queries and reactions in a single line notation, whereas SMILES handles these through language extensions.

SYCONUS (SYCONIUM): A type of composite fruit formed from a hollow fleshy inflourescence stalk inside which tiny flowers develop. An example is the fig.

SYENITE: Coarse-grained igneous rock, similar in appearance and composition to granite. Unlike granite, it contains little or no quartz. The chief minerals in syenite are the feldspars, with mica, hornblende and pyroxene. They are occasionally substituted for granites as building stones.

SYLEPTIC SHOOTS: Abnormal shoots that develop from lateral buds before they have reached maturity.

SYLLOGISM: 1. A deductive inference in which a conclusion is drawn from two given or assumed premises or propositions with one of the premises, called the **major premise** containing the **major term** that is the predicate of the conclusion. The premise in which the subject of the conclusion, called the **minor term** occurs is the **minor premise**. The minor term occurs in both premises but not the conclusion. Syllogism is of the form "all A is C; all B is A; therefore all B is C." 2. A deductive reasoning of certain other forms with two premises, for example, hypothetical syllogism.

SYLOW SUBGROUP (SYLOW P-SUBGROUP): A subgroup of a given finite group, G of order a maximal prime power; that is, if p divides |G| and α is the largest integer such that p^α divides |G|, then a Sylow p-subgroup is a subgroup, H, of order $|H| = p^\alpha$.

S

SYLOW'S THEOREMS: A group of theorems that describe the number of subgroups of fixed order that a given finite group contains. It is named after Ludwig Sylow (1832-1918), Norwegian mathematician. Sylow' first theorem states that if p divides |G|, G has a Sylow p-subgroup; whence every finite group possesses subgroups of order p^s for any prime p for which p^s divides the order of the group. The second theorem states that in a finite group, for some fixed p, all the Sylow p-subgroups are conjugate and so isomorphic. The third theorem states that the number of Sylow p-subgroups for a given p is congruent to 1 mod p; consequently, any maximal p-subgroup of a finite group is a Sylow p-subgroup.

SYLVESTER'S LAW OF INERTIA: In mathematics, the theorem that if A is a symmetric matrix that defines the quadratic form and S is any invertible matrix such that $D = SAS^T$ is diagonal, then the number of negative elements in the diagonal of D is always the same, for all such S; and the same goes for the number of positive elements. It is named after James Joseph Sylvester (1814-97) English mathematician.

SYLVESTER'S THEOREM: The result that given a finite set of non-collinear points in the plane, there is a line passing through exactly two of the points.

SYLVITE (SYLVINE): A mineral form of potassium chloride with the chemical symbol KCl. It forms crystals in the isometric system very similar to normal rock salt, halite (NaCl). Sylvite is colourless to white with shades of yellow and red due to inclusions. It has a Mohs Scale hardness of 2.5 and a relative density of 1.99. It has a refractive index of 1.490. Sylvite is soluble in water and has a bitter, salty taste. It is found in deposits formed by the evaporation of seawater and salty lakes. Sylvite is also occasionally found deposited directly from volcanic fumes. It is the principal source of potassium compounds and potash fertilisers. It is used by people on low sodium diets.

SYM-: An affix used in names to denote symmetrical.

SYMBIONTS: A plant or animal living in close and often mutually beneficial association with another of a different species or an organism that is a partner in a symbiotic relationship. **Symbiosis** is a close and lasting relationship between two organisms of different species, which includes commensalism, mutualism and parasitism.

SYMBIOTIC NITROGEN FIXATION: The biological processes of converting nitrogen (N_2) from the atmosphere to inorganic nitrogen by microorganisms that live in nodules on the roots of soybean plants. It is symbiotic because it involves interaction between two different organisms living in close physical association (host plant and rhizobia). It is a mutualistic relationship for both symbiotic partners; the host plant provides the rhizobia with carbon and source of energy for growth and functions while the rhizobia fix atmospheric N_2 and provide the plant with a source of reduced nitrogen in the form of ammonium.

SYMBOL: 1. Anything that represents something else by association, resemblance or convention, especially a material object used to represent something invisible. 2. A printed or written sign used to represent an operation, element, quantity, quality or relation, as in mathematics or music. 3. An object or image that an individual unconsciously uses to represent repressed thoughts, feelings or impulses, for example, phallic symbol.

SYMBOLIC ADDRESS: In computer science, it is a symbol used in assembly language programming to represent the binary address of a memory location.

SYMBOLIC LOGIC (FORMAL LOGIC): The study of deductive argument and of the structure and relations of statements, which includes primitive symbols, combinations of these symbols, axioms and rules of inference.

SYMBOLIC MANIPULATION: In computer science, it is the use of computer programs or languages that allow one to manipulate quantities symbolically rather than merely numerically. An example of such a program is as MACSYMA (MAC's SYmbolic MAnipulator).

SYMBOLIC PROCESSOR: A computer that is built to run symbol-manipulation programs rather than programs involving a great deal of numerical computation. They exist principally for the artificial intelligence language LISP, although some have also been built to run PROLOG.

SYMMEDIAN: Reflection of a median of a triangle about the corresponding angle bisector. They are three particular geometrical lines (symmedian point, Lemoine point or Grebe point) associated with every triangle. They are constructed by taking a median of the triangle (a line connecting a vertex with the midpoint of the opposite side) and reflecting the line over the corresponding angle bisector (the line through the same vertex that divides the angle of the triangle in two equal parts).

SYMMETRIC (SYMMETRICAL): 1. In botany, being divisible into two similar parts by more than one plane passing through the centre; actinomorphic. 2. For a flower, having the same number of parts in each whorl. For example, the flowers of roses, magnolias, buttercups and water lilies are **radially symmetric**; the petals are identical in design and project from a central point or focus. The flowers of orchids and peas are **bilaterally symmetric**; its two sides are mirror images of each other. 3. In chemistry, having a structure that exhibits a regular repeated pattern of the component parts. 4. Of a benzene derivative, having three substitutions occurring at alternate carbon atoms. 5. In mathematics, for a figure or configuration, being identical with its own reflection in an axis of symmetry or centre of symmetry; having pairs of points identically placed, except for sense, with respect to some line, point, plane, etc. 6. For a relation, holding between a pair of arguments x and y only when it also holds between y and x. In other words, an element x in relation to a second element y implies the y is in relation to y. 7. For a **square matrix**, being equal to its transpose. Its entries are symmetric around its main diagonal and is equal to its transpose 8. For a dyad, being equal to its conjugate. 9. For a binary operator, having the property that the order of arguments is immaterial. 10. A **symmetric function** is a function of several variables that is unchanged by any permutation of its variables. 11. A **symmetric rotation** is a rotation of a regular polygon or polyhedron that is congruent to the original.

SYMMETRIC DESIGN: In mathematics, it is a block design in which the number of blocks is equal to the number of varieties or points. Also, block design in which the number of points belonging to each block is equal to the number of blocks in which each point is contained.

SYMMETRIC DIGITAL SUBSCRIBER LINE (SDSL): A technology, which is used for transferring data over existing copper telephone lines. It has a transfer speed of 1.544 Mbps for both the upstream and downstream data transmissions. SDSL is called symmetric because it supports the same data rates for upstream and downstream traffic. A similar technology that supports different data rates for upstream and downstream data is called **asymmetric digital subscriber line** (**ADSL**).

SYMMETRIC DIFFERENCE: For two given sets, A and B, the set of elements that is in one set but not in both of them. It is the union of their relative complements, the relative complement of their intersection in their union. The symmetric difference of A and B is denoted by A $\ominus$ B, A $\oplus$ B or A ⍰ B. For example, the symmetric difference of the sets {1, 2, 3} and {3, 4} is {1, 2, 4}.

SYMMETRIC FUNCTION (SYMMETRIC POLYNOMIAL): 1. A function that is symmetric about the origin; an even function. 2. A function of two or more variables that is unchanged by any permutation of its variables with the expression $f(y_1, y_2, ..., y_n) = f(x_1, x_2, ..., x_n)$, where $y_i = x_{\pi(i)}$ and π is an arbitrary permutation of the indices 1, 2, ..., n.

SYMMETRIC GROUP: In mathematics, the group whose elements are all the permutation operations of a given set that is finite of order n. The composition of such permutation operations are defined as bijective functions from the set of symbols to itself. There are $n!$ possible permutation operations that can be performed on a tuple composed of n symbols and the order (the number of elements) of the symmetric group is n!.

SYMMETRY GROUP OF AN ELASTIC BODY: In relation to a given reference configuration of a body, it is the set of transformations to a new reference configuration, such that the response function of the body is invariant.

SYMMETRICAL FILMS: The properties of fluid films depend on the nature of the film phase and that of each of the two neighbouring bulk phases. These films should be described, where appropriate, by three capital letters such as A for air, W for water, O for oil and S for solid, separated by solidi, the middle letter indicating the film phase. For symmetrical films the first and last symbols are the same, for eample, A/W/A: water film in air or W/O/W: oil film in water, whereas for unsymmetrical films these are different, for eample, W/O/ A: oil film between water and air.

SYMMETRY: 1. Exact likeness in shape about a given line, axis, point or plane. A figure is said to have symmetry if one half can be rotated and/or reflected onto the other or if a change in a system leaves the essential features of the system unchanged. It preserves the length, angle but not necessarily orientation. 2. The set of invariances of a system. Upon application of a symmetry operation of a system, the system is unchanged. 3. In biology, it is the balanced distribution of duplicate body parts or shapes. The body plans of most multicellular organisms exhibit some form of symmetry, either radial symmetry or bilateral symmetry or spherical symmetry. A small minority, which exhibits no symmetry, are asymmetric. In nature and biology, symmetry is approximate. For example, plant leaves, while considered symmetric, will rarely match up exactly when folded in half.

SYMMETRY CONCENTRATION: The quantity of cations or anions equivalent to the exchange capacity of a soil. For example, if the cation exchange capacity of a soil is 10 meq/100 g of soil, then 1 symmetry concentration is 10 meq of any cation.

SYMMETRY NUMBER: The symmetry number of a molecule is obtained by imagining all identical atoms to be labelled and then counting the number of different but equivalent arrangements that can be obtained by rotating (but not reflecting) the molecule. In the statistical-mechanical treatment of chemical equilibrium, the partition function for each molecular species must be divided by its symmetry number.

SYMMETRY VALUE: The percentage of the adsorbed ion released when one symmetry concentration of another ion is added.

SYMMETRY-CONSERVING TRANSITION: A transition in which the cell dimensions and/or angles of the one phase differ from those in the other phase, but where the space-group symmetry is conserved. For example, the transition of face-centred cubic Ce, upon cooling, to a face-centred cubic phase that is 10% denser. Upon cooling, enough contraction takes place to allow an overlap of the fsp^2 configuration and the change from an isolated non-bonding magnetic f electron to a bonding non-magnetic electron pair.

SYMMORPHOSIS: A theory that biological systems adhere to an economy of design given a close match between their various structural and functional parameters. Thus no single parameter in the system has an unnecessary excess capacity beyond the requirement of the system.

SYMPATHETIC NERVOUS SYSTEM: Part of the autonomic nervous system, which is the opposite of the parasympathetic nervous system's functions. Its nerve endings release noradrenaline or adrenaline as a neurotransmitter and its action tends to antagonise those of the parasympathetic nervous system, thus achieving a balance in the organ it serves. For example, it decreases salivary gland secretion, increases heart beat rate and constricts blood vessels.

SYMPATHETIC TONE: The condition of the muscle when the tone is maintained predominantly by impulse from the sympathetic nervous system.

SYMPATRIC: A term describing groups of similar organisms that although in close proximity and theoretically capable of interbreeding, do not interbreed because of differences in behaviour such as flowering time, etc.

SYMPATRIC SPECIATION: A type of speciation, in which new species evolve from a single ancestral species while inhabiting the same geographic region. It occurs, when one individual develops an abnormal number of chromosomes, either extra chromosomes (polyploidy) or fewer, such that viable interbreeding can no longer occur. An example is parent plants, which produce offspring that are polyploid. Hence, the offspring live in the same environment as their parents but are reproductively isolated.

SYMPETALAE (ASTERIDAE): A subclass of the dicotyledons containing mostly herbaceous plants having flowers with fused petals. They have few stamens and usually only two carpels and the seeds are surrounded by only one integument (unitegmic).

SYMPETALOUS (GAMOPETALOUS): With the petals united or wholly fused, at least near the base, forming a corolla shaped like a tube or funnel.

SYMPHYSIS: 1. A joint that is only slightly movable. The bones at a symphysis articulate by means of smooth layers of cartilage and string fibres. Examples are the joints between the vertebrae of the vertebral column and that between the two pubic bones in the pelvic girdle. 2. The fusion or coalescence of like parts, as occurs in sympetalous corolla.

SYMPLAST: In plants, the continuum of cytoplasm connected by plasmodesmata between cells. This effectively forms a continuous system of cytoplasm bounded by the plasma membrane of the cells. The movement of water through the symplast is known as the **symplast pathway**. It is one of the routes by which water travels by the root cortex from the root hair and is the only means by which water crosses the endodermis.

SYMPLASTIC GROWTH: A type of growth in which adjacent cells grow at the same rate so that the symplast is not disrupted.

SYMPLESIOMORPHY (SYMPLESIOMORPHIC CHARACTER): An ancestral trait that is shared by two or more modern groups because of their ancient origin. Symplesiomorphies are not usually helpful in assessing more recent evolutionary relationships within a larger group. For example, simple leaves of some modern flowering plants are probably inherited from simple-leave ancestors, yet it does not make all simple leaves relative.

SYMPODIAL: Branching without a main axis but with many, more or less, equal laterals.

SYMPODIAL BRANCHING: A type of growth seen in some plants, for example, elm and lime, in which the apical bud withers at the end of the growing season and growth is continued the following season by the lateral bud immediately below.

SYMPODIUM: The composite primary axis of growth in such plants as lime and horse chestnut. After each season's growth the shoot tip of the main stem stops growing. Growth is continued by the tip of one or more of the lateral buds.

SYMPTOM: An indication of a disease or disorder or the visible change to the host caused by the presence of a disease or abnormality. For example, the symptoms of the common flu or influenza are chills, fever, headaches, muscle aches and fatigue.

SYN (SYM): A prefix denoting united. For example, in botany, **synandrous** means anthers which are united.

SYNANGIUM: A compound fruiting unit developed from the lateral fusion of individual sporangia, as seen in some ferns, for example, Marattia and gymnosperms such as Welwitschia.

SYNANTHEROUS: Having leaves and flowers appearing at same time.

SYNAPOMORPHY: A character or trait that is shared by two or more taxonomic groups that evolved in an ancestor common to all species on one branch of a fork in a cladogram, but not common to species on the other branch.

S

SYNAPSE: The junction between two adjacent nerve cells or between a nerve cell and a muscle across which nerve impulses are transmitted and received or a site of transmission of electric nerve impulses between two nerve cells or between a nerve cell and a gland or muscle cell.

SYNAPSIS: The pairing of homologous chromosomes, seen during the first meiotic prophase or the side-by-side association of homologous paternal and maternal chromosomes during the first prophase of meiosis.

SYNAPTIC CLEFT: The space between the end of a neuron and an adjacent cell where neurotransmitters from presynaptic cell, for example, acetylcholine or noradrenaline are released into by exocytosis and diffuses across to bind with the receptors in the cell membrane of a postsynaptic cell. This gap is only a minute space allowing the concentration of neurotransmitters to be raised and lowered rapidly.

SYNAPTIC KNOB: The relay point at the tip of a transmitting neuron's axon, where signals are sent to another neuron or to an effector.

SYNAPTIC PLASTICITY: The change in the efficacy of connections of the junctions between neurons in the nervous system. It is a crucial process that underlies modification of an animal's behaviour during development and in response to previous activity or experience, including learning and memory.

SYNAPTIC TERMINAL: A bulb at the end of an axon in which neurotransmitter molecules are stored and released.

SYNAPTIC VESICLES (NEUROTRANSMITTER VESICLES): Vesicles at the synapse end of an axon that contain neurotransmitters that are released at the synapse. The release is regulated by a voltage-dependent calcium channel. Vesicles are essential for propagating nerve impulses between neurons and are constantly recreated by the cell.

SYNAPTONEMAL COMPLEX: A ribbon-like complex of protein that binds the members of each homologous pair of chromosomes together during pairing in prophase of the first division of meiosis. Formation of the complex begins at the telomeres.

SYNC (SYNCHRONISE; SYNCHRONIZE): A term used to describe the process of connecting a peripheral via a USB or wireless Bluetooth to a computer such as a cell phone and updating any information that differs on either the computer or cell phone. Often, this is done to help keep a peripheral's data backed up and up-to-date.

SYNCARP: An aggregate or multiple fruit such as pineapple and mulberry, formed from two or more carpels of one flower or the aggregated fruits of several flowers. **Syncarpous** describes a situation in which the carpels are united or syncarpous.

SYNCARPY: The situation in which the female reproductive parts (carpels) of a flower are joined to each other. Examples are found in the primrose.

SYNCHROCYCLOTRON: A form of cyclotron in which the frequency of the accelerating potential is synchronised with the increasing period of revolution of a group of the accelerated particles, resulting from their relativistic increase in mass as they reach relativistic speeds. The accelerator is used with protons, deuterons and alpha-particles.

SYNCHRONISATION: Fixing of different events to take place at the same point in time. In television, for example, synchronisation is essential in order that the electron beams of receiver picture tubes are at exactly the same spot on the screen at each instant as is the beam in the television camera tube at the transmitter.

SYNCHRONOUS: Taking place at the same time, at the same rate or with the same period. A synchronous computer is a computer in which all operations are controlled by one master clock. Most synchronous events occur in physical science and engineering.

SYNCHRONOUS DYNAMIC RANDOM ACCESS MEMORY, SDRAM (SDR-RAM): A type of dual in-line memory module (DIMM) that synchronises itself with the computer's system clock to provide synchronisation between the memory and the computer processor. DIMM is a small circuit board that holds semiconductor memory chips with two independent rows of input/output contacts. Synchronisation allows the memory to run at higher speeds than previous memory types.

SYNCHRONOUS MOTOR: An alternating current (ac) motor which operates at a fixed synchronous speed proportional to the frequency of the applied alternating current (ac) power. Unlike an induction motor, which must slip in order to produce torque, synchronous motor does not rely on slip under usual operating conditions and as a result, produces torque at synchronous speed. A synchronous machine may operate as a generator, motor or capacitor depending only on its applied shaft torque (whether positive, negative or zero) and its excitation. Synchronous motors are used where constant speed is required. It is a highly efficient means of converting alternating current (ac) energy to work and can operate at leading or unity power factor and thereby providing power-factor correction.

SYNCHRONOUS ORBIT (GEOSYNCHRONOUS ORBIT): An orbit in which an orbiting body, usually a satellite has a period equal to the average rotational period of the body being orbited, usually a planet and in the same direction of rotation as that body. If the orbit, however, lies in the planet's equatorial plane and is circular, the satellite will appear to be stationary. This is called a **stationary orbit** or **geostationary orbit**.

SYNCHRONOUS ROTATION: The rotation of a natural satellite in which the period of rotation is equal to its orbital period. In other words, it is a body orbiting another body, where the orbiting body takes as long to rotate on its axis as it does to make one orbit. It therefore, always keeps the same hemisphere pointed at the body it is orbiting. In other words, from the surface of the satellite, the main planet appears to be locked in place in the sky as it slowly rotates. An example is the Moon which is in synchronous rotation about the Earth and, therefore, always presents the same face to the Earth.

SYNCHRONOUSLY EXCITED (FLUORESCENCE, PHOSPHORESCENCE) SPECTRUM: A two-dimensional spectrum obtained by varying both the excitation and emission wavelengths simultaneously and which corresponds to the curve where a plane, parallel to the z-axis, intersects the excitation-emission spectrum (EES).

SYNCHROTRON: A cyclic particle accelerator used to impart energy to electrons and protons in order to carry out experiments in particle physics and some cases to make use of the synchrotron radiation produced. The particles are accelerated in closed orbits, often circular, by radio-frequency fields. Magnets are spaced around the orbit to bend the trajectory of the particles and separate focusing magnets are used to keep the particles in a narrow beam. The radio-frequency accelerating cavities are interspersed between the magnets. The particle energy increases giving rise to an increase in the field strength. This causes the motion of the particles to synchronise automatically with the magnetic field.

SYNCHROTRON RADIATION (MAGNETOBREMSSTRAHLUNG): The electromagnetic radiation emitted by charged particles moving at relativistic speeds in a circular orbit in a magnetic field. It is so called because it is produced by high-speed particles in a synchrotron. Such radiation is highly polarised and continuous. Its intensity and frequency depend on the strength of the magnetic field that alters the path of the particles, as well as on the energy of those particles. Synchrotron radiation is emitted by a variety of astronomical objects, from planets to supernova remnants to quasars. At radio frequencies it is emitted by high-energy electrons as they spiral through magnetic fields in space, such as those around Jupiter.

SYNCLASTIC: Having a surface such as that of a sphere that is curved toward the same side in all directions with all points bending away from a tangent plane toward the same side, as opposed to anticlastic.

SYNCLINE: A fold in the rocks of the Earth's crust in which the layers or beds dip inwards thus forming a trough-like structure with the beds arching upwards in an anticline.

SYNCOTYLY: Coalesced cotyledons, forming a funnel or trumpet.

SYNCYTIUM: A large cell-like structure filled with cytoplasm containing many nuclei. For example, the cells of striated muscle form a syncytium.

SYNDET: A synthetic detergent or a detergent other than soap.

SYNDETOCHEILIC: A term used to describe a gymnosperm stomatal complex in which the subsidiary cells are derived from the same initial as the guard cells, as occurred in the Bennettitales.

SYNDIOTACTIC MACROMOLECULE: A tactic macromolecule, essentially comprising alternating enantiomeric configurational base units, which have chiral or prochiral atoms in the main chain in a unique arrangement with respect to their adjacent constitutional units.

SYNDIPLOIDY: A doubling of the chromosome number through a fault in the mitotic process. Commonly, there is either a failure of spindle formation at anaphase or a failure to form a dividing wall between the two daughter cells. If such cells continue to divide normally an autopolyploid segment of tissue may arise in the plant. If this develops flowers, autopolyploid seeds may result.

SYNDROME: The group or recognisable pattern of symptoms or abnormalities that indicate a particular trait or disease.

SYNECOLOGY: The study of the ecological interrelationships between different species that form a community and their interactions with the surrounding environment. This study involves both biotic and abiotic factors.

SYNEMA: The column composed of united filaments in a flower with monadelphous stamens.

SYNERESIS: The spontaneous separation of the solid and liquid components of a gel owing to further coagulation or the process by which a liquid is separated from a gel owing to further coagulation. An example is when lymph drains from a contracting clot of blood or the collection of whey on the surface of yogurt.

SYNERGIDS (SYNERGIDAE): The two nuclei in the embryo sac of flowering plants, which are closely associated with the oosphere or egg cell, to form the egg structure.

SYNERGY: The co-operative action of two or more drugs or body parts such as muscles, nerves or organs so that their combined effect is greater than the sum of their individual effects. Synergism applies in medicine, especially to enhance the efficacy of a set of drugs.

SYNGAMY: The fusion of gametes resulting in the formation of a zygote, which develops into a new organism.

SYNGENESIOUS: United by the anthers or having an androecium in which the anthers are fused, as in composite plants.

SYNODIC: In astronomy, of or relating to a conjunction between two celestial objects.

SYNODIC MONTH (LUNATION): The average period of revolution of the Moon about the Earth with respect to the Sun. It is the time averaging 29.53059 days or 29 days 12 hours 44 minutes 2.8 seconds, required for the Moon to pass from a particular phase, such as full moon, back to the same phase. This is sometimes called the "month of the phases", because it extends from the new moon to the next new moon.

SYNODIC PERIOD: The mean time taken to return to the same position for a planet, moon or any object in the solar system in its orbit, as seen from the Earth or the mean time taken by an object in the solar system to travel between successive returns to the same position, relative to the Sun, as seen from the Earth. The synodic period of the Moon, which is called the **lunar month** or **lunation**, is 29.53059 days or 29 days 12 hours 44 minutes 2.8 seconds. The **sidereal month** is the time required to make one revolution from a particular star back to the same star by the Moon, as estimated from the centre of the Earth averaging 27.32166 days.

SYNOECIOUS: Having male and female reproductive organs or staminate and pistillate flowers together in the same head.

SYNOPTIC METEOROLOGY: A branch of meteorology dealing with collective meteorological observations made more or less simultaneously at a number of points over a large geographical area, so as to obtain an overall view of weather conditions.

SYNOPTIC SCALE: In meteorology, it is the size or scale of ordinary weather systems or cyclones; typically horizontally.

SYNOVARIOUS: Having ovaries of adjacent carpels completely fused, styles and stigmas separate.

SYNOVIAL CAPSULE: A membrane structure between two bones of a movable joint. It is shaped like a bag and contains a fluid that lubricates the joints.

SYNOVIAL FLUID: A viscous yellow fluid that bathes movable joints between the bones of vertebrates. It nourishes and lubricates the cartilage at the end of each bone to reduce friction.

SYNOVIAL MEMBRANE (SYNOVIUM): The soft tissue that lines the non-cartilaginous surfaces within joints with cavities or the synovial capsule.

SYNSEPALOUS (GAMOPETALOUS): Having united or fused sepals.

SYNSTYLOVARIOUS: Having ovaries and styles of adjacent carpels completely fused, with stigmas separate.

SYNTACTIC: 1. Relating to or determined by the rules of syntax. 2. In logic, able to described wholly in terms of the grammatical structure of an expression or the rules of well-formedness for a formal theory, without regard to their sense.

SYNTAX: 1. In logic, the set of rules usually in the form of an algorithm, stated in terms of the structure of formal languages or formal systems without regard to any interpretation or meaning given to them. 2. A term used in programming language or command to refer to a set of rules that are associated with the language or command. **Syntax error** is an error that is encountered when the programmer or individual who wrote the code has not followed the rules of the language, causing the program to fail.

SYNTECTIC REACTION: A reversible reaction that involves the conversion of two liquid phases 1′ + 1″, into a solid phase α on cooling: 1′ + 1″ $\leftrightharpoons$ α. The maximum temperature at which this reaction can occur is the congruent melting point of the solid phase. An example is the conversion of co-existing K-rich and Zn-rich phases in the K-Zn system to form an intermediate solid phase KZn_{13}.

SYNTENY: 1. The conservation of chromosomal segment containing a group of homologous genes in the same order and orientation among different species. Chromosomal regions containing such genes are termed synthenic blocks. 2. The physical colocalisation of genetic loci on the same chromosome within individual or species.

SYNTHESIS: 1. A combination of two or more entities that together form something new. 2. In chemistry, it is the formation of a substance or compound from two or more elementary substances, often through chemical reactions. **Synthesis reaction** is one of the most common types of chemical reactions, usually of the form A + B → AB. An example is the formation of potassium chloride from potassium and chlorine gas: $2K_{(s)} + Cl_{2(g)} \rightarrow 2KCl_{(s)}$. 3. In biology, **biosynthesis** is the production of an organic compound in a living thing, usually by the help of enzymes to catalyse the reaction and energy source such as adenosine triphosphate (ATP). Examples of biosynthesis include photosynthesis, chemosynthesis, amino acid synthesis, nucleic acid synthesis and ATP synthesis. 4. The production of sounds or speech by using electronic equipment. To **synthesise** (**synthesise**) is to combine two or more entities to form something new; or to form a substance or a compound from two or more elementary substances, often through chemical reaction.

SYNTHESIS GAS: 1. A mixture of one part carbon monoxide and two parts hydrogen used in the manufacture of organic chemicals such as hydrocarbons and methanol. 2. A mixture of gases made as feedstock, especially a fuel produced by controlled combustion of coal in the presence of water vapour.

S

SYNTHESIS PHASE (S-PHASE): In the cell cycle, the phase in which the DNA of the chromosomes is replicated and DNA-associated proteins, such as histones, are synthesised. It occurs in between two Gap phases (Gap 1 and Gap 2). By the end of Gap 1 phase, the cell will decide if it will proceed to the Synthesis phase (S phase) or if it will enter a period of dormancy called Gap 0 phase. The transition of the cell from Gap 1 to S phase is determined by certain factors, for example, levels of nutrients, growth factors and energy.

SYNTHESISER (SYNTHESIZER): An electronic instrument, often played with a keyboard that combines simple waveforms to produce more complex sounds, such as those of various other instruments. Most synthesisers can be attached to computers and sequencers using Musical Instrument Digital Interface (MIDI).

SYNTHETIC: 1. Any material made from chemicals or a substance made artificially that does not come from a natural source. An example is nylon, which is synthesised by polymerising adipic acid and hexamethylenediamine. The principal goals of synthetic chemistry are to create new chemical substances and to develop better, less-expensive methods for the synthesis of known substances. 2. In logic, of a noncontradictory proposition, neither true nor false by virtue of meaning alone, because the predicate is not included in or entailed by, the subject.

SYNTHETIC DIVISION (HORNER'S RULE): A rule for polynomial division in which an algorithm is used to simplify the process of evaluating a polynomial $f(x)$ at a certain value $x = x_0$ by dividing the polynomial into monomials or polynomials of the first degree. Each monomial involves a maximum of one multiplication and one addition processes. The result obtained from one monomial is added to the result obtained from the next monomial and so forth in an accumulative addition fashion.

SYNTHETIC FIBRE: Fibre made by chemical processes. Most synthetic fibres are now derived from organic polymers, materials consisting of large organic molecules. Most of them are thermoplastic and are softened by heat. Examples are nylon, polyester, acrylic and polyolefin.

SYNTHETIC GRAPHITE: A material consisting of graphitic carbon which has been obtained by graphitizing of non-graphitic carbon by chemical vapour deposition (CVD) from hydrobcarbons at temperatures above 2500 K, by decomposition of thermally unstable carbides or by crystallising from metal melts supersaturated with carbon.

SYNTHETIC GEOMETRY (PURE GEOMETRY): The study of geometry, usually projective geometry, by the synthetic method without the use of coordinates.

SYNTHETIC METAL: A substance such as conducting polymers that is not a metal but has free electrons that can contribute to electrical conductivity.

SYNTHETIC NATURAL GAS (SNG): A mixture of gaseous hydrocarbons produced from coal, petroleum, etc., and used as a fuel. It is clean-burning, high-quality fuels used in motor vehicles, for cooking, for commercial, industrial and other fuel consuming facilities.

SYNTHETIC NUTRIENTS: Nutrients manufactured from chemicals. Synthetic nutrients are extracted using various chemicals and often lack the important cofactors and accessory nutrients required for the proper absorption and digestion of vitamins and minerals. A natural source contains co-factors that come with the nutrient in nature. The body recognises and uses the whole food supplement and all its phytonutrients as designed by nature.

SYNTHETIC PROOF (SYNTHETIC METHOD): The deduction of the properties of some entity from a set of axioms, as opposed to analytic and algebraic geometries, which use analysis and algebraic techniques to obtain geometric results.

SYNTHETIC RUBBER: A synthetic compound with a structure resembling that of natural rubber usually produced by the polymerisation or copolymerisation of petroleum-derived olefinic or other unsaturated compounds. It consists of long chain polymer molecules that are built of monomers such as acetylene (ethyne), isoprene, styrene and vinyl compounds.

SYNTHON: A fragment obtained by disconnection in retrosynthetic analysis and defined as a structural unit within a molecule which is related to a possible synthetic operation. By analogy, units building up structures by self-assembly in supermolecular chemistry are sometimes known as **supermolecular synthons**.

SYNTROPHY: Interaction between two different species of microorganisms that are mutually dependent on each other for growth and existence, or one species lives off the products of another.

SYPHILIS: A serious infectious disease, caused by a spirally twisted bacterium *Treponema pallidum* and sexually transmitted by direct contact sexual intercourse, kissing or passed from mother to child in utero. It affects many body organs and parts, including the genitals, brain, skin and nervous tissue. In the final stage of syphilis, known as **tertiary syphilis**, the disease spreads throughout the body, affecting the brain, spinal cord, heart, skin, bones and joints.

SYRINGE: A simple piston pump consisting of a plunger that fits tightly in a tube. The plunger can be pulled and pushed along inside a cylindrical tube (the barrel), allowing the syringe to take in and expel a liquid or gas through an orifice at the open end of the tube. The open end of the syringe may be fitted with a hypodermic needle, a nozzle or tubing to help direct the flow into and out of the barrel. Syringes are often used to administer injections, apply compounds such as glue or lubricant and measure liquids.

SYRINX: 1. The sound-producing organ of a bird, situated at the lower end of the trachea where it splits into the bronchi, producing sounds without the vocal cords of mammals. It has a complex structure with a number of vibrating membranes. The syrinx enables some species of birds such as parrots, crows and mynas to mimic human speech. Unlike the larynx of mammals, the syrinx is located where the trachea forks into the lungs and because of this some songbirds can produce more than one sound at a time. 2. A rare, fluid-filled neuroglial cavity within the spinal cord (syringomyelia) or in the brain stems (syringobulbia). A syrinx results when a watery, protective substance known as cerebrospinal fluid, that normally flows around the spinal cord and brain, transporting nutrients and waste products, collects in a small area of the spinal cord and forms a cyst.

SYRUP: A thick sweet liquid made from sugar cane juice or by boiling sugar with water. It varies in density and strength and it may be used in conveying oral medicine.

SYSTEM: 1. A group of things or parts working together in a regular relation. 2. A set of abstract entities having a structure by a set of axioms and regarded as uninterpreted calculus, such as groups, rings, fields, Boolean algebras, etc. 3. A set of equations or inequalities that are to be solved simultaneously or treated together. 4. An assembly of computer software, hardware and peripherals functioning together. 5. An assembly of electronic or mechanical parts which function together as a unit. 6. An assembly of substances in chemical or physical equilibrium. 7. A division of rocks larger than a series but smaller than a stage. It is used to distinguish formations of a specific era or period. 8. A division used in classifying minerals according to their crystal structures. 9. A group of astronomical objects or other gravitationally linked objects.

SYSTEM ANALYST: An individual responsible for examining a computer or computer equipment, verifying that it works properly without errors or is user friendly.

SYSTEM CRASH (CRASH): A software program error or hardware problem that is encountered without warning. A crash is often caused by either a software error where either the programming has errors or with conflictions encountered with other programs. A crash may also occur because of a hardware problem, such as a failing component. For example, when a hard drive dies it is often referred to as a hard drive crash.

SYSTEM FLOW CHART: A flow chart used to describe the flow of data through a particular computer system.

SYSTEM REQUIREMENTS: In computer science, it is the minimum specification necessary in order to use a piece of hardware or software efficiently.

SYSTEM SOFTWARE: Files and programs that make up a computer's operating system. System files include libraries of functions, system services, drivers for printers and other hardware, system preferences and other configuration files. The programs that are part of the system software include assemblers, compilers, file management tools, system utilities and debuggers.

SYSTEM X: In communications, it is a modular, computer-controlled, digital switching system used in telephone exchanges.

SYSTEMATIC ACQUIRED RESISTANCE (SAR): A type of induced resistance in plants caused by localised infection but which spreads throughout the plant to offer strong protection against infection.

SYSTEMATIC ERROR: An error that causes the results of an instrument to vary from the correct value in a predictable manner, which often can be identified and corrected. Its causes include problem with an instrument or its data handling system or the instrument is not used wrongly correctly. Two types of systematic error can occur with instruments having a linear response: (i) Offset or zero setting error in which the instrument does not read zero when the quantity to be measured is zero. (ii) Multiplier or scale factor error in which the instrument consistently reads changes in the quantity to be measured greater or less than the actual changes.

SYSTEMATIC NOMENCLATURE: A system for naming chemical compounds according to a set of rules, usually those developed by the International Union of Pure and Applied Chemistry (IUPAC).

SYSTEMATICS: The study of the diversity of organisms and their natural relationships. It is sometimes used as a synonym for taxonomy.

SYSTÉME INTERNATIONAL D'UNITÉS (SI UNITS): The official French name for SI units, which is a unit adopted for international use in science and technology. The seven fundamental units under this system are the metre, kilogram, second, ampere, Kelvin, candela and mole.

SYSTEMIC (TRANSLOCATED): In botany, it is a term used to refer a chemical that is absorbed by a plant and transported throughout the tissues. Systemic fungicides and insecticides are applied to render plant tissues toxic to fungi or insects. Their effects are longer lasting than those of contact pesticides. Systemic herbicides, such as 2,4-D, 2,4,5-T and MCPA, are used as selective weedkillers, killing most broad-leaved plants but leaving grasses and cereals unaffected. Such herbicides are also used to kill deeply rooted weeds.

SYSTEMIC ACQUIRED RESISTANCE (SAR): A generalised state of enhanced immunity to infection demonstrated by plants following an initial localised injury. SAR is important for plants to resist disease, to fight prevailing diseases. SAR can be induced by a wide range of pathogens, especially (but not only) those that cause tissue necrosis. The resistance observed following induction of SAR is effective against a wide range of pathogens.

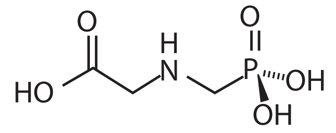
S

SYSTEMIC ARCH: A paired blood vessel in the embryos of tetrapods responsible for transporting blood from the aorta to the trunk and hind limbs. It is derived from the fourth aortic arch. Adult amphibians and reptiles retain both arches; but birds and mammals have only one.

SYSTEMIC CIRCULATION: Circulation of oxygenated blood throughout the body from the left ventricle to the arteries, the capillaries, the organs and the various tissues and returns deoxygenated blood from the tissues to the right atrium of the heart.

SYSTEMIC EFFECT: A consequence that is either of a generalised nature or that occurs at a site distant from the point of entry of a substance.

SYSTEMIC HERBICIDE: A herbicide which is translocated through the plant, either from foliar application down to the roots or from soil application up to the leaves. They are capable of controlling perennial plants and may be slower acting but ultimately more effective than contact herbicides. An example is glyphosate whose structure is shown in the diagram.



Chemical Structure of glyphosate, an example of a systemic herbicide

SYSTEMIC SIGNALLING: The dissemination of chemical signals throughout a plant in response to injury or infection of part of the plant. It triggers a systemic defence response that helps the plant to combat the spread of infection or restrict further injury.

SYSTEMS ANALYSIS: The investigation of a business activity or clerical procedure with a view of deciding if and how it can be computerised or the use of mathematics to determine how a set of interconnected components whose individual characteristics are known will behave in response to a given input or set of inputs.

SYSTEMS APPLICATION ARCHITECTURE (SAA): In computer science, it is the IBM standards for computer software, introduced in 1987 to achieve consistency and some amount of program portability among IBM's range of computer systems. It uses CUA (common user access) standards to ensure that commands and keystrokes are used consistently in different applications.

SYSTEMS BIOLOGY: A biology-based interdisciplinary study approach that seeks to study organisms as complete systems or complex interactions in biological systems-networks of interacting genes, biomolecules and biochemical reactions. It attempts to integrate all relevant structural and functional information rather than focusing on just one particular gene or protein at a time.

SYSTEMS DESIGN: The detailed design of an application package or determination in detail of the exact operational requirements of a system, resolution of these into file structures and input/output formats and relating of each to management tasks and information requirements. Alternatively, it is the preparation of an assembly of methods, procedures or techniques united by regulated interaction to form an organised whole making use of available components to perform in a specified manner. Systems design is most effective when more than one solution can be proposed.

SYSTEMS ECOLOGY: The use of mathematical models and relationships in the study of ecology or a branch of ecosystem ecology that attempts to clarify the structure and function of ecosystems by means of applied mathematics, mathematical models and computer programs. It includes the study of energy budgets, biogeochemical cycles and feeding and behavioural aspects of ecological communities.

SYSTEMS ENGINEERING: The design and implementation of production systems that require the integration of diverse and complex tasks. In other words, it is the design of a complex interrelation of many elements (a system) to maximise an agreed-upon measure of system performance, taking into consideration all of the elements related in any way to the system, including utilisation of worker power as well as the characteristics of each of the system's components. An example is an automobile assembly line.

SYSTEMS MANAGEMENT SERVER (SMS): A tool developed by Microsoft that helps administrators perform diagnostics, maintain and create an inventory of computers that are connected to the Network and are running SMS.

SYSTEMS NETWORK ARCHITECTURE (SNA): The IBM architecture that defines the logical structure, formats, protocols and operational sequences for transmitting information units through and controlling the configuration and operation of, networks. The layered structure of SNA allows the ultimate origins and destinations of information (the users) to be independent of and unaffected by the specific SNA network services and facilities that are used for information exchange.

SYSTEMS PROGRAM: A program that performs a task related to the operation and performance of the computer system itself. It is a product of the computer manufacturer to aid the user in easily and productively operating the system.

SYSTEMS SOFTWARE (APPLICATIONS SOFTWARE): The operating system and utility programs used to operate and maintain a computer system and provide resources for application programs or software, including spreadsheets and word processors.

SYSTOLE: The phase of the heartbeat during which the ventricles of the heart contract to pump blood into the arteries. The opposite of systole is the **diastole**, which is the phase of a heartbeat that occurs between two contractions of the heart. In this case the muscles of the heart relax and the ventricles are filled with blood. **Systolic pressure** is the pressure in an artery during the ventricular contraction phase of the heart cycle. Systolic and diastolic pressures are measured in millimetres of mercury (abbreviated mm Hg) using an instrument called a **sphygmomanometer**. In a healthy adult, the normal blood pressure is 120/80 mm Hg (systolic/diastolic). At this level, there is a much lower risk of heart disease or stroke.

SYSTRAY (SYSTEM TRAY): A section of the taskbars in the Microsoft Windows desktop user interface located on the right side of the Windows toolbar. In other words, it is the collection of small icons on the opposite side of the Start Menu. The volume control and the date and time are default items in the systray and many more can be added. Systray allows quick and easy access to programs and control settings. Most systray icons will open a control panel or program when double-clicked.

SYZYGY: The situation in which the Sun, the Moon or a planet and the Earth are in a straight line or the straight-line conjunction or opposition of three astronomical objects such as the Sun, Earth and Moon or the alignment of three celestial objects within a solar system or within any other system of objects in orbit about a star.

SZILARD-CHALMERS EFFECT: An effect that is used to separate radioactive products in a nuclear reaction involving the absorption of a neutron and the emission of gamma rays or the breaking of a chemical bond between a radioactive atom formed in a nuclear reaction and the molecule to which the atom belonged. It is used in the separation of isotopes by chemical means.

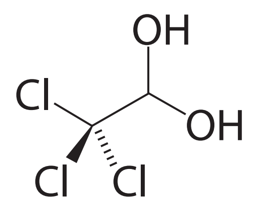
T

1,1,1-TRICHLOROETHANE (METHYL CHLOROFORM; TRICHLOROETHANE): A volatile colourless non-flammable liquid with a sweet odour and the chemical formula $C_2H_3Cl_3$, density 1.32 g/cm^3, melting point -33°C (240 K) and boiling point 74 °C (347 K). It is used as solvents for adhesives, pesticides and lubricants and in industrial cleaning solutions.

1,1,2-TRICHLOROETHENE (ACETYLENE TRICHLORIDE; 1,1-DICHLORO-2-CHLOROETHYLENE; 1-CHLORO-2,2-DICHLOROETHYLENE; TRICHLOROETHENE;): A colourless, toxic and non-flammable liquid with the chemical formula C_2HCl_3, density 1.46 g/cm^3 (20 °C), melting point -73 °C (200 K) and boiling point 87.2 °C (360 K). It has a smell resembling that of chloroform (trichloromethane). It is widely used as a solvent in dry cleaning and degreasing. It is also used to extract oils from nuts and fruit, as an anaesthetic and fire extinguisher.

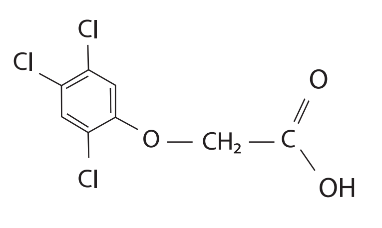
1,2,3-TRINITROXYPROPANE (GLYCERYL TRINITRATE; NITROGLYCERINE; TRINITROGLYCERIN; TRINITROGLYCERINE): A heavy, colourless, oily explosive liquid obtained by nitrating glycerol. Its chemical formula is $C_3H_5N_3O_9$, density 1.6 g cm^{-3} (at 15°C, 288 K), melting point 14 °C (287 K) and boiling point 50-60 °C (323-333 K). It is the active ingredient in the manufacture of explosives (dynamite). It has been used by the military as an active ingredient and a gellatinizer for nitrocellulose in some solid propellant, such as cordite and ballistite. In medicine, it is used as vasodilator to treat heart conditions such as angina and chronic heart failure.

2,2,2-TRICHLOROETHANEDIOL (CHLORAL HYDRATE): A colourless crystalline solid with the chemical formula $CCl_3CH(OH)_2$, relative density 1.91, melting point 57 °C (330 K) and boiling point 96.3 °C (369.45 K). It is made by the hydrolysis of trichloroethanal. It is unusual in having two –OH groups on the same carbon atom. It is soluble in both water and oil; and used as a sedative and hypnotic drug.



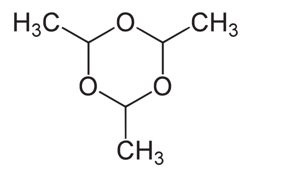
Chemical Structure of 2,2,2-trichloroethanediol

2,4,5-TRICHLOROPHENOXYACETIC ACID (2,4,5-T; 2,4,5-TRICHLOROPHENOXYETHANOIC ACID): A synthetic auxin or a plant hormone with the chemical formula $C_6H_2Cl_3OCH_2CO_2H$ and a melting point of 152 °C (425 K). It is soluble in alcohol and insoluble in water. It is used as a defoliant, plant hormone and herbicide. At low levels, it can promote plant growth, but at higher levels it is herbicidal. It is banned in a number of countries, because it tends to become contaminated with a toxic chemical called dioxin.



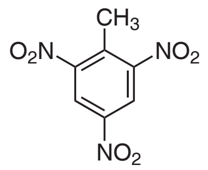
Chemical Structure of 2,4,5-Trichlorophenoxyacetic Acid

2,4,6-TRIMETHYL-1,3,5-TRIOXANE (ETHANAL TRIMER; PARALDEHYDE): A colourless flammable liquid with a pungent odour having the chemical formula $C_6H_{12}O_3$, molar mass 132.16 g mol^{-1}, melting point 12 °C (285 K) and boiling point 124 °C (397 K). It is produced by treating acetaldehyde with a small amount of sulfuric acid. It is slightly soluble in water and highly soluble in alcohol. It is used as a sedative and hypnotic drug (the safest hypnotic drug); in the manufacture of synthetic resins; as a preservative; and in preparing leather.



Chemical Structure of Paraldehyde

2,4,6-TRINITROTOLUENE (TRINITROTOLUENE, TNT; TRILITE; TOLITE; TRINOL; TROTYL): A yellow highly explosive crystalline solid with the chemical formula $CH_3C_6H_2(NO_2)_3$. It is made by nitrating toluene (methylbenzene). It is an explosive material with convenient handling properties unaffected by ordinary shocks and jarring and must be set off by a detonator. It is sometimes used as a reagent in chemical synthesis and also used to generate charge transfer salts. The explosive yield of TNT is considered to be the standard measure of strength of bombs and other explosives.



Chemical Structure of TNT

***t*:** 1. A symbol denoting the independent real variable of a function of time. 2. Symbol for an independent variable in parametric equations, often non-angular as opposed to the angular parameter, θ.

T: 1. Symbol for the transpose of a matrix., usually written as a superscript (^{T}). 2. In logic, it is usually used in truth-tables to represent "truth". It is also written 1 as opposed to 0 for falsehood. If it is inverted, it is the symbol for falsehood. It is also written as F or 0. 3. It is an abbreviation used in symbols for multiples of the physical units of the Systeme International (S.I.). 4. Symbol for orthogonal, perpendicular or singular (of a measure). 5. Symbol for tesla, the S.I. unit of magnetic flux density.

T CELL (T LYMPHOCYTE): Any of a population of lymphocytes that are the principal agents of cell-mediated immunity. T cells are derived from the bone marrow but migrate to the thymus to mature. T-cells attack other body cells that are infected by some bacteria, a virus or another pathogen. Some T lymphocytes, called cytotoxic (cell-poisoning) or killer T lymphocytes, generate cell-mediated immune responses, directly destroying cells that have specific antigens on their surface that are recognised by the killer T cells. Helper T lymphocytes, a second kind of T lymphocyte, regulate the immune system by controlling the strength and quality of all immune responses. The HIV/AIDS virus destroys a type of T-cell, leading to the syndrome characterised by a defective immune system. T-cell counts are used as a diagnostic test to indicate the strength of the immune system in AIDS patients.

T CELL RECEPTOR (T-CELL RECEPTOR): A protein molecule found attached to the plasma membrane of a T cell that recognises foreign antigen displayed on the surface of host cells and activates an immune response by the T cell. A single T cell typically has about 30,000 such receptors.

T DISTRIBUTION (T-DISTRIBUTION; STUDENT'S T DISTRIBUTION): A probability distribution in statistics made up of the means of random samples from a collection of samples that has a